EXAM-READY REVIEW · ROBBINS STANDARD

Pathology:
All-in-One

Special Pathology · 4th Year MBBS

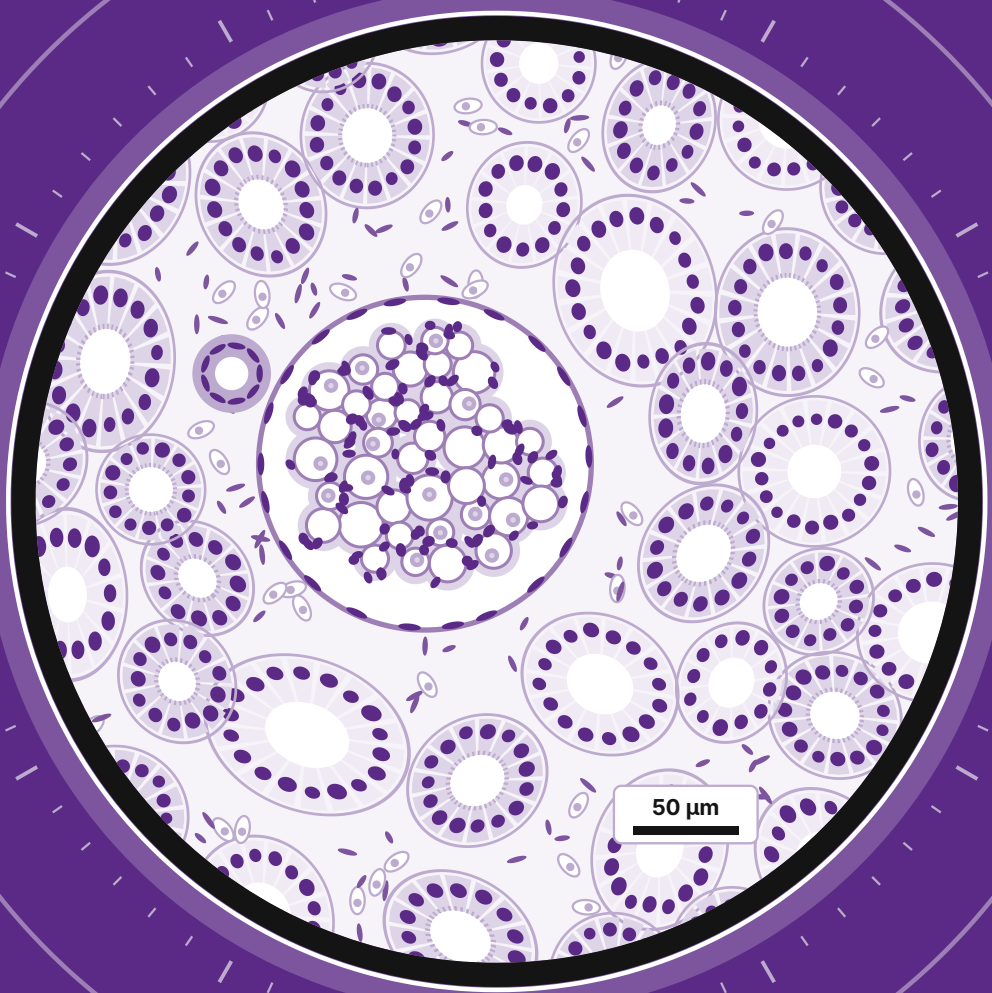
31
syllabus-mapped chapters

Robbins
standard concepts

Flowcharts
tables & mind maps

Solved
SEQs · BCQs · Viva · OSPE

Night-before
revision pages



MODULES 10 · 11 · 12

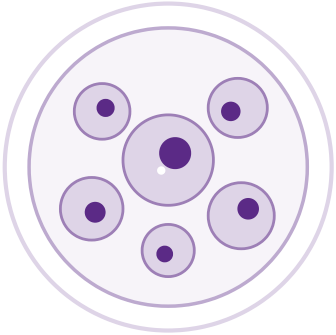
SYLLABUS-MAPPED · EXAM-READY

COMPILED BY

FIRST EDITION 2026

4TH YEAR MBBS · SPECIAL PATHOLOGY

Pathology: All-in-One



Special Pathology for 4th Year MBBS · Modules 10–12

BASED ON THE ROBBINS STANDARD · SYLLABUS-MAPPED · EXAM-READY

ELEVEN PARTS · THIRTY-ONE CHAPTERS

A Gastrointestinal Tract
B Liver, Gallbladder & Pancreas
C Lung
D Blood & Lymphoid System
E Kidney & Urinary Tract
F Reproductive System & Breast
G Endocrine System
H Cardiovascular System
I Musculoskeletal System
J Nervous System
K Skin

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FIRST EDITION 2026

Pathology: All-in-One

Special Pathology for 4th Year MBBS · Modules 10–12 · First edition 2026

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REFERENCE STANDARD AND INDEPENDENCE

Definitions, classifications and morphological descriptions in this book follow the standard set by *Robbins Basic Pathology* and *Robbins & Cotran Pathologic Basis of Disease* (10th editions), the standard pathology texts in Pakistani medical colleges. Students are encouraged to read Robbins alongside this book. *Pathology: All-in-One* is an independent review written in its own words, with original diagrams, tables and questions. It is not affiliated with, sponsored by or endorsed by Elsevier or by the authors or editors of Robbins; the name is used only to identify the reference text. Examination questions in this book are original practice questions written in the style of university papers; they are not reproductions of any past paper.

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How to use this book

Every chapter is built the same way, so you always know where to look, what to learn and how to test yourself.

Anatomy of a chapter

Each of the 31 chapters covers the lectures, interactive sessions and labs of one block of the syllabus (see the Syllabus Map). Six parts appear in the same order every time.



The six parts of every chapter. Learn the layout once and you can find anything in seconds.

- **Opener.** The objectives tell you what an examiner expects you to do. The **Exam Radar** rates the topic for each format: ★★★ a core topic examined often; ★★☆ examined regularly; ★☆☆ asked now and then. The **Robbins link** names the matching Robbins chapter for further reading.
- **Mind map.** Look at it before you start and redraw it when you finish. If you can rebuild the map, you know the chapter.
- **Concept sections** (N.1, N.2…). Every disease follows one rhythm: **definition → aetiology → pathogenesis → morphology → clinical features → complications**. It is also the skeleton of a full-mark SEQ answer.
- **Morphology panels.** Gross on the left, microscopy on the right, clinical note below: the exact words examiners look for in SEQs and the OSPE.
- **Exam Zone.** Solved SEQs tagged **Prof pattern** (annual examination style), **Module pattern** (module examination style) or **Classic** (a theme that keeps returning). All questions are original, written in the university style, with a model answer plan that shows where the marks are. Then 12 BCQs with reasons, 12 viva questions and OSPE stations.
- **Night-before page.** Fifteen one-liners, memory hooks, fill-the-gaps, two draw-from-memory prompts and a revision tracker: the page to read on exam morning.

The 3-pass method

Reading a chapter once feels like learning; recalling it with the book shut is what actually makes it stay. Take every chapter through three passes.

LEARN 1

In the week of the lecture (the Syllabus Map tells you which chapter). Read the concept sections, think through every Concept Clear box and copy each flowchart once by hand.

Tracker: tick "Read"

COMPRESS 2

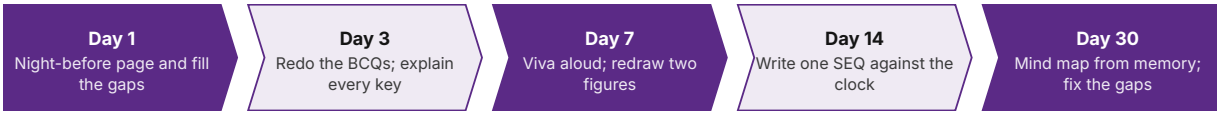
Within two days. Squeeze each disease into the six headings on one side of paper, fill the comparison tables from memory and add your own notes to the Night-before page.

Tracker: tick "Compressed"

RECALL 3

Book closed. Do the BCQs, answer the viva aloud, fill the gaps and draw the two figures from memory. Then mark, and reread only what you got wrong.

Tracker: tick "BCQs done"



Spaced revision. Five short reviews, each 15–30 minutes, spread over a month. Tick them on the tracker at the end of each chapter.

In the text: bold marks the words that earn marks · **keyword** is a term to use in your answers · **3.5 g/day** is a number worth quoting · highlighted is the one line to remember · "see Chapter 24" sends you to the chapter that owns a topic.

The boxes: a legend

Ten kinds of box recur throughout the book. Each has one job; once you know the job, you know how to use the box.

DEFINITION

The wording that opens an SEQ answer and earns the definition mark.

Barrett oesophagus: intestinal metaplasia (columnar epithelium with goblet cells) replacing the squamous lining of the distal oesophagus, as a complication of long-standing GERD.

CONCEPT CLEAR

The why behind a fact, so you can reason instead of memorising.

Antral *H. pylori* gastritis lowers somatostatin, so gastrin and acid output rise and the **duodenum** ulcerates. Autoimmune gastritis destroys parietal cells instead: acid falls and the risk becomes **B12 deficiency**, not ulcer.

PAKISTAN FOCUS

Local burden, risk factors and practice that examiners in Pakistan expect.

Smokeless tobacco (naswar, gutka) and betel quid with areca nut make **oral cancer** one of the commonest cancers in Pakistani men. Areca nut also causes **oral submucous fibrosis**.

MEMORY HOOK

An accurate mnemonic worth carrying into the exam hall.

CRAB

Multiple myeloma: hyper**C**alcaemia, **R**enal failure, **A**naemia, **B**one (lytic) lesions.

EXAM TRAP

Look-alikes, classic BCQ traps, and the Robbins version where teachers differ.

A raised **AFP** rules out a **pure seminoma**. Seminoma may secrete some hCG (from syncytiotrophoblastic cells), but a raised AFP means a non-seminomatous (yolk sac) component.

LAB VALUES

Reference ranges, cut-offs and how to read a test.

Microcytic MCV < 80 fL · **Macrocytic** MCV > 100 fL

Iron deficiency ferritin ↓, TIBC ↑, saturation ↓

Chronic disease ferritin normal or ↑, TIBC ↓

CLINICAL CASE

Vignette → reasoning → diagnosis: how BCQ stems and OSPE scenarios are written.

A 58-year-old smoker has haematuria, a loin mass and haemoglobin 18.5 g/dL. **Reasoning:** renal mass plus secondary polycythaemia (tumour erythropoietin). **Diagnosis:** renal cell carcinoma, clear cell type.

CLINICAL CORRELATION

Links the lesion to the bedside: signs, tests, complications.

Why does a patient with cirrhosis bleed? Less synthesis of factors II, VII, IX and X (prolonged PT), low platelets from hypersplenism, and varices from portal hypertension.

MOLECULAR KEY

The one gene, translocation or marker examiners ask, and what it does.

CML: t(9;22) Philadelphia chromosome → **BCR–ABL1** fusion gene → constitutively active tyrosine kinase → the target of imatinib.

LAB SPOTTER

How to name the slide or specimen in ten seconds.

Papillary carcinoma thyroid: branching papillae with fibrovascular cores; crowded, clear "Orphan Annie eye" nuclei with grooves and inclusions; psammoma bodies.

MORPHOLOGY PANEL · EXAMPLE: HASHIMOTO THYROIDITIS

GROSS

- Thyroid **diffusely, symmetrically enlarged**; capsule intact
- Cut surface **pale grey-tan**, firm, somewhat friable
- Gland well demarcated from neighbouring structures

MICROSCOPY

- Dense **lymphoplasmacytic infiltrate** with **germinal centres**
- Atrophic follicles; **Hürthle cells** (abundant pink granular cytoplasm)
- Increased interstitial fibrous tissue

How to read a panel: gross = size, shape, colour, consistency, cut surface; microscopy = architecture → cells → special features; the bottom strip adds the clinical link (here: painless goitre, slow hypothyroidism, anti-TPO antibodies, risk of B-cell lymphoma).

Syllabus map

Every lecture, interactive session and lab of the Special Pathology syllabus, matched to the chapter that teaches it.

Find the week you are in, read across to the chapter number, and study that chapter alongside the lecture. Chapters follow organ systems, so a chapter may collect topics taught in different weeks (for example, Chapter 24 gathers the pituitary and diabetes lectures of Module 11 and the adrenal lectures of Module 12).

Module 10
Chapters 1–13
+ Chapter 14 (Lymphoma I)
GIT, liver, gallbladder, pancreas, lung, anaemias, leukaemias

Module 11
Chapters 14–24
Lymphoma II, bleeding, kidney, male and female genital tract, breast, thyroid, pituitary, diabetes

Module 12
Chapters 25–31
+ Chapter 24 (adrenal)
Vessels, heart, bone, joints, muscle, CNS, skin

ROBBINS STANDARD Definitions, classifications, morphology and figures in every chapter follow *Robbins Basic Pathology* and *Robbins & Cotran Pathologic Basis of Disease* (10th editions), the texts your teachers and examiners use; each chapter opener names the matching Robbins chapter.

M10 GIT, Liver, Lung & Blood I

LECTURES

Week	Lecture	Ch
W1	Introduction to special pathology	1
	Diseases of salivary glands	1
W2	Gastritis / acid peptic disease	2
	Diseases of oesophagus	1
W3	Gastritis / acid peptic disease (contd)	2
W4	Gastric carcinoma	2
	H. pylori gastritis	2
	Malabsorption	3
W5	Inflammatory bowel disease	3
	Intestinal polyps	4
W6	Colorectal carcinoma	4
	Jaundice	5
W8	Haemochromatosis	6
	Hepatocellular carcinoma	6
W9	Diseases of gallbladder	7
	Anaemias I	11
W10	Interstitial lung diseases	9
	Anaemias II	11 12
W11	Thalassaemias	12
	Lymphoma I	14
W12	Leukaemia I & II	13

INTERACTIVE SESSIONS

Topic	Ch
Oral pre-malignant lesions	1
Barrett oesophagus & GERD	1
Chronic pancreatitis	7
Viral hepatitis	5
Cirrhosis	6
Pneumonia	8
COPD I	9
Lung tumours	10

LABS

Topic	Ch
H. pylori gastritis	2
Chronic pancreatitis	7
Chronic hepatitis	5
Cirrhosis	6
Hepatocellular carcinoma	6
Pneumonia	8
Pulmonary tuberculosis	8
Squamous cell carcinoma lung	10
Anaemias (blood smears)	11 12
Leukaemias (smears, marrow)	13
Gross specimens	1–13

M11 Blood II, Kidney, Reproduction, Breast & Endocrine

LECTURES

Week	Lecture	Ch
W2	Bleeding disorders	15
	Lymphoma II	14
W3	Glomerular diseases I & II	16
W4	Nephrotic syndrome I & II	16
W5	Tubulointerstitial nephritis I & II	17
W6	Renal cell carcinoma	18
	Urolithiasis	18
W7	Diseases of prostate	19
	Testicular tumours	19
W8	Diseases of vulva & cervix	20
	Ovarian tumours	21
W9	Diseases of endometrium	20
	Endometrial cancer	20
W10	Breast carcinoma	22
	Non-neoplastic diseases of thyroid	23
W11	Neoplastic diseases of thyroid	23
W12	Diabetes mellitus	24
	Diseases of pituitary	24

INTERACTIVE SESSIONS

Topic	Ch
Cystic diseases of kidney	17
Acute tubular necrosis	17
RFTs & acid–base balance	17
TCC & congenital testicular diseases	18 19
Benign epithelial lesions of breast	22
Diseases of parathyroid	23

LABS

Topic	Ch
Renal cell carcinoma	18
Transitional cell carcinoma	18
Seminoma	19
Benign prostatic hyperplasia	19
Adenomyosis & endometrial carcinoma	20
Mucinous cystadenoma ovary	21
Trophoblastic disease	21
Fibroadenoma breast	22
Breast carcinoma	22
Hashimoto thyroiditis	23
Papillary carcinoma thyroid	23
Gross specimens	14–24

M12 CVS, Adrenal, Bone & Joint, CNS & Skin

LECTURES

Week	Lecture	Ch
W1	Atherosclerosis	25
W2	Ischaemic heart disease	26
	Adrenal disorders I	24
W3	Adrenal disorders II	24
	Vasculitis	25
W4	Aortic aneurysm & dissection	25
	Vascular tumours	25
W5	Osteoporosis	27
	Bone tumours	27
W6	Arthritis	28
	Gout & gouty arthritis	28
W7	Cerebrovascular diseases	29
	Demyelinating disorders	29
W8	Neurodegenerative diseases	29
	Brain tumours I	30
W9	Introduction to skin pathology	31
	Brain tumours II	30
W10	Melanocytic naevi & melanoma	31
	Inflammatory dermatoses	31
W11	Bullous disorders	31
	Skin cancers	31

INTERACTIVE SESSIONS

Topic	Ch
Hypertensive & valvular heart disease	26
Cardiomyopathies	26
Rickets, osteomalacia & osteomyelitis	27
Muscular dystrophies & myopathies	28
Meningitis & brain abscess	29
Benign epithelial tumours of skin	31

LABS

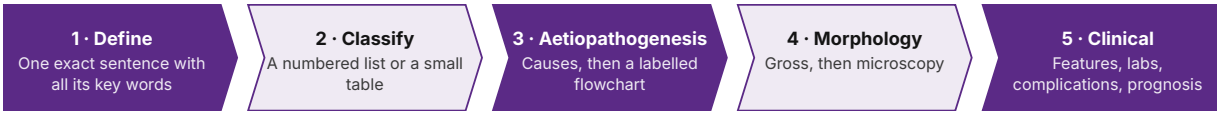
Topic	Ch
Osteochondroma	27
Osteosarcoma	27
Giant cell tumour of bone	27
Bone tumours (other)	27
Schwannoma	30
Brain tumours	30
Malignant melanoma	31

Exam craft for pathology

Knowing pathology and scoring in pathology are different skills. These pages teach the second: how examiners award marks, and how to collect them.

1 Writing a 5-mark SEQ

Examiners mark against a key: a list of points, each worth a fraction of a mark. An answer laid out in a predictable order lets the examiner find those points in seconds. For any disease, use the same five-part frame and leave out the parts the question does not ask for.



The SEQ frame. The same order Robbins uses for every disease, and the order of every chapter in this book.

TIME PER MARK

- Before you start: **minutes per mark = minutes in the paper ÷ total marks**. For example, 45 marks in 120 minutes is about 2.5 minutes a mark: roughly **12 minutes per 5-mark SEQ**, with 10 minutes left to review.
- Split time by marks inside a question: in a **2 + 3** question the 2-mark part gets two-fifths of the time, however much you know about it.
- Attempt every part. The first marks of a question are the easiest to earn; the last mark of a long answer is the hardest.

PRESENTATION THAT EARNS MARKS

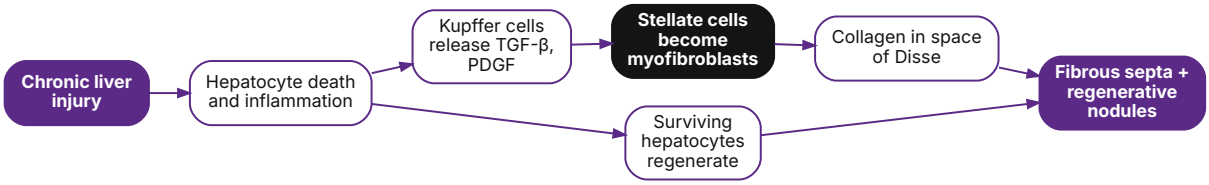
- Use the question's own parts as headings: **a)**, **b)**, **c)**.
- Write numbered points, not paragraphs; **underline the key words**.
- **Draw the pathogenesis as a flowchart** and give it a title. A 1-minute diagram often carries a full mark.
- "Differentiate" means a **table** with a "Feature" column; label every table and diagram.
- Write morphology under two sub-headings, **Gross** and **Microscopy**.

WORKED EXAMPLE 1 Define cirrhosis. Describe its pathogenesis and morphology. **1 + 2 + 2 marks**

HOW A FULL-MARK ANSWER IS LAID OUT

a) Definition (1). A **diffuse** process of the whole liver in which **fibrous septa** divide the parenchyma into **regenerative nodules**, with loss of normal architecture and vascular shunting.

b) Pathogenesis (2). Causes first (chronic HCV/HBV, alcohol, fatty liver disease, biliary disease, haemochromatosis, Wilson), then the flowchart:



Pathogenesis of cirrhosis, as you would draw it in the exam.

c) Morphology (2). Gross: shrunken, firm liver with a nodular surface; **micronodular** (nodules < 3 mm, e.g. alcohol) or **macronodular** (> 3 mm, e.g. viral). **Microscopy:** bridging fibrous septa, regenerative nodules without normal lobular arrangement, ductular reaction, variable inflammation.

Where the marks are: the three key words of the definition (diffuse, fibrosis, nodules); the stellate cell in the flowchart; the nodule size cut-off. Each is a mark-earning point, so make each one easy to see.

2 Describing a gross specimen

At a gross station you have about a minute. Describe in a fixed order, out loud or on paper, and give the diagnosis **last**: a good description earns marks even when the diagnosis is wrong.

1. **Organ** — which organ, which part; whole, bisected or resected
2. **Cut surface** — or outer surface, whichever is shown
3. **Size** — estimate in cm, against the jar or a ruler
4. **Colour** — uniform or variegated; haemorrhage, necrosis, pigment
5. **Consistency** — as it looks: firm, soft, friable, gritty, cystic
6. **Lesion** — number, site, shape: nodule, ulcer, cyst, mass, cavity
7. **Borders** — circumscribed or infiltrative; capsule; spread
8. **Diagnosis** — last; if unsure, add a two-item differential

WORKED EXAMPLE 2 · GROSS SPECIMEN

Specimen: half of a kidney in a jar.

> SAY *This is a bisected nephrectomy specimen. At the upper pole there is a rounded tumour about 6 cm across. Its cut surface is variegated: golden-yellow areas with grey-white, red-brown haemorrhagic and cystic areas. It looks soft, is well circumscribed and pushes the kidney aside; a plug of tumour fills the renal vein. The diagnosis is renal cell carcinoma, clear cell type.*

Why yellow? Lipid and glycogen in the clear cells.

GROSS CLUES THAT NAME THE LESION

Look	Think of
Nutmeg cut surface, liver	Chronic passive congestion
Leather-bottle stomach	Diffuse gastric carcinoma (linitis plastica)
Cobblestone mucosa, skip lesions	Crohn disease
Honeycomb lung, lower lobes	Usual interstitial pneumonia
Chocolate-filled ovarian cyst	Endometriotic cyst
Fish-flesh cut surface	Lymphoma, sarcoma
Flea-bitten kidney	Malignant hypertension

3 Identifying a slide in the OSPE

Slides are recognised by **pattern** at low power, then confirmed by **cells** at high power. Work through the same five steps every time, starting at the lowest power.

THE 10-SECOND CHECKLIST

1. **Tissue** — low power first: which organ? Find a landmark: villi, glomeruli, alveoli, follicles.
2. **Architecture** — is the normal structure kept or lost? Glands, sheets, papillae, nodules?
3. **Cells** — a mixture suggests inflammation; one monotonous or pleomorphic population suggests a neoplasm.
4. **Special feature** — look for one clue from the table.
5. **Diagnosis** — commit: name it, point to two features you can see, add one related fact.

WORKED EXAMPLE 3 · SLIDE

Slide: lung. Alveolar walls are intact, but every alveolus is packed with neutrophils, strands of fibrin and red cells.

Answer: lobar pneumonia, **red hepatisation**. Features: neutrophil–fibrin exudate filling alveoli; preserved alveolar walls. Related fact: commonest cause *Streptococcus pneumoniae*.

SPECIAL FEATURES THAT NAME THE SLIDE

You see	Think of
Caseating granulomas, Langhans giant cells	Tuberculosis
Keratin pearls, intercellular bridges	Squamous cell carcinoma
Signet-ring cells	Diffuse gastric carcinoma
Starry-sky pattern	Burkitt lymphoma
Reed–Sternberg cells in a mixed background	Hodgkin lymphoma
Psammoma bodies	Papillary thyroid, serous ovarian tumour, meningioma
Whorls of cells	Meningioma
Peripheral palisading of basaloid cells	Basal cell carcinoma
Pseudopalisading necrosis	Glioblastoma
Verocay bodies, Antoni A and B areas	Schwannoma
Cartilage cap over bony stalk	Osteochondroma
Even sheet of osteoclast-like giant cells	Giant cell tumour of bone

4 Data interpretation stations

Every data station is solved the same way: **mark each value** as high, low or normal; **find the pattern**; **name** the disorder; suggest **one confirmatory test**. Reference ranges vary between laboratories, so always use the ranges printed on the station when they are given.

LIVER FUNCTION TESTS: THREE PATTERNS OF JAUNDICE

Test	Pre-hepatic	Hepatic	Obstructive
Bilirubin	↑ unconjugated	↑ both	↑ conjugated
Urine bilirubin	Absent	Present	Present
Urobilinogen	↑	N or ↑	↓ or absent
ALT / AST	Normal	↑↑	N or mild ↑
ALP / GGT	Normal	Mild ↑	↑↑
Stool	Normal	Normal	Pale

Normal: pH 7.35–7.45 · $PaCO_2$ 35–45 mmHg · HCO_3^- 22–26 mmol/L · anion gap = $Na^+ - (Cl^- + HCO_3^-)$ ≈ 8–12 mmol/L. **High gap:** ketoacidosis, lactic acidosis, renal failure, toxins. **Normal gap:** diarrhoea, renal tubular acidosis.

FOUR PRIMARY ACID–BASE DISORDERS

Disorder	pH	Primary	Compensation
Metabolic acidosis	↓	HCO_3 ↓	$PaCO_2$ ↓ ($1.5 \times HCO_3 + 8 \pm 2$)
Metabolic alkalosis	↑	HCO_3 ↑	$PaCO_2$ ↑
Respiratory acidosis	↓	$PaCO_2$ ↑	HCO_3 ↑ (kidney, days)
Respiratory alkalosis	↑	$PaCO_2$ ↓	HCO_3 ↓

THYROID PROFILE

TSH	Free T4	Interpretation
↓	↑	Primary hyperthyroidism (Graves, toxic nodule)
↑	↓	Primary hypothyroidism (Hashimoto)
↑	Normal	Subclinical hypothyroidism
↓	Normal	Subclinical hyperthyroidism or T3 toxicosis (check T3)
N or ↓	↓	Secondary (pituitary) hypothyroidism

RENAL FUNCTION: PRERENAL FAILURE VS ATN

Test	Prerenal	ATN
BUN : creatinine	> 20 : 1	≈ 10–15 : 1
Urine Na^+	< 20 mmol/L	> 40 mmol/L
FENa	< 1%	> 2%
Urine osmolality	> 500 mOsm/kg	< 350 mOsm/kg
Sediment	Bland, hyaline casts	Muddy-brown granular casts

WORKED EXAMPLE 4 · DATA STATION — 2 minutes

Diabetic aged 19, drowsy, deep sighing breaths. pH 7.18, $PaCO_2$ 22 mmHg, HCO_3^- 9, Na^+ 136, Cl^- 100 mmol/L.

STEP BY STEP

pH ↓ with HCO_3 ↓ = **metabolic acidosis**. Gap = 136 − 109 = **27**: high. Expected $PaCO_2 = 1.5 \times 9 + 8 = 21.5 \pm 2$; measured 22, so compensation is appropriate (Kussmaul breathing). **Diabetic ketoacidosis**: confirm with blood glucose and ketones.

CSF IN MENINGITIS

Finding	Pyogenic	Viral	Tuberculous
Appearance	Turbid	Clear	Clear; cobweb clot on standing
Cells	Neutrophils, often thousands	Lymphocytes	Lymphocytes (mixed early)
Protein	↑↑	N or mild ↑	↑↑ (often marked)
Glucose	↓↓	Normal	↓

Normal CSF: clear; 0–5 lymphocytes/µL; protein 15–45 mg/dL; glucose about two-thirds of blood glucose.

COAGULATION SCREEN

Pattern	Think of
Platelets ↓, BT ↑; PT, aPTT normal	ITP, TTP, marrow failure
BT ↑; count normal	Platelet function defect, aspirin, von Willebrand
aPTT ↑ alone	Haemophilia A or B, heparin
PT ↑ alone	Early warfarin or vitamin K deficiency (factor VII)
PT ↑, aPTT ↑	Vitamin K deficiency, liver disease
PT, aPTT ↑; platelets ↓; fibrinogen ↓; D-dimer ↑	DIC

COMPLETE BLOOD COUNT AND SMEAR: THE CLUE AND WHAT IT MEANS

Clue	Think of	Clue	Think of
Microcytic, ferritin ↓, RDW ↑	Iron deficiency	Bite cells; Heinz bodies on supravital stain	G6PD deficiency
Microcytic, RBC count high, HbA_2 > 3.5%	β-thalassaemia trait	Blasts with Auer rods	Acute myeloid leukaemia
Macrocytic, hypersegmented neutrophils	B12 or folate deficiency	Smudge cells, small mature lymphocytes	Chronic lymphocytic leukaemia
Spherocytes, reticulocytes ↑	Hereditary spherocytosis, warm AIHA	Whole granulocyte series, basophils ↑	Chronic myeloid leukaemia
Schistocytes, platelets ↓	DIC, TTP, HUS	Target cells	Thalassaemia, liver disease

5 BCQs: one best answer

1. **Cover the options.** Read the stem and last line; answer in your head first.
2. **Find the keywords**: age, sex, time course, site, exposure, the one abnormal test.
3. **Uncover and match.** If your answer is there, it is usually right; if not, eliminate and choose between the last two.
4. **Read the qualifier**: *most common*, *most specific*, *earliest*, *except*.
5. **Do not change** an answer without a clear reason; know the paper's negative-marking rule.

LOOK-ALIKES EXAMINERS LOVE

Pair	The one fact that separates them
Crohn vs UC	Transmural, skip lesions, granulomas vs continuous from rectum, mucosal, crypt abscesses
Pemphigus vs pemphigoid	Suprabasal acantholysis, fishnet IgG vs subepidermal blister, linear IgG/C3
Gout vs pseudogout	Negatively birefringent needles vs positively birefringent rhomboids
Haemophilia vs von Willebrand	aPTT ↑, BT normal vs BT ↑ (± aPTT ↑)
Leukaemoid reaction vs CML	No Philadelphia chromosome, LAP score high vs t(9;22), LAP low
Seminoma vs yolk sac tumour	AFP normal vs AFP raised; Schiller–Duval bodies

WORKED EXAMPLE 5 · BCQ TECHNIQUE

A 45-year-old woman has tiredness, weight gain and a **firm, painless, diffusely enlarged** thyroid. Which finding is most likely on fine-needle aspiration?

A. Clear, grooved nuclei with inclusions
B. Lymphocytes and Hürthle cells
C. Granulomas with multinucleate giant cells
D. Spindle cells in an amyloid stroma
E. Crowded follicular cells in microfollicles

Cover, then predict: middle-aged woman + hypothyroid features + painless diffuse goitre = **Hashimoto thyroiditis**, so expect lymphocytes and Hürthle cells. **Uncover:** B matches. **Answer B.** Every distractor is the right answer to another question: A papillary carcinoma, C de Quervain (painful) thyroiditis, D medullary carcinoma, E follicular neoplasm.

6 The viva

Examiners judge clarity as much as knowledge: give a short, structured answer, then stop and let the examiner lead.

THE 1-1-1 ANSWER *Definition, one cause, one feature.* **"What is a granuloma?"** — *A focus of chronic inflammation made of activated macrophages that have become epithelioid cells, often with giant cells and a rim of lymphocytes. The commonest cause in Pakistan is tuberculosis, and its typical feature there is central caseous necrosis.*

GOOD HABITS

- Listen to the whole question; a one-second pause is fine.
- For a specimen, begin: "This is a specimen of… showing…".
- Give numbers with units; offer to draw a flowchart if it helps.

WHEN YOU DO NOT KNOW

- **Ask** for the question to be repeated or rephrased.
- **Bridge** from what you know: "I am not sure of the exact figure, but I know that…".
- **Never invent**: an honest "I don't remember" costs less than a confident wrong answer.
- **Accept correction** politely and move on; never argue.

Study plans

Three timetables for three situations, all built on chapter numbers. Pick the one that matches the time you have, then tick your progress on the master tracker overleaf.

A Six weeks before a module examination

Weeks 1–5 take each chapter through Pass 1 and Pass 2 on weekdays and Pass 3 (recall) at the weekend. Week 6 is for recall only: no new reading.

Week	Module 10	Module 11	Module 12
1	1 2 3 Oesophagus, stomach, malabsorption, IBD	14 15 Lymphomas, bleeding disorders	25 Atherosclerosis, aneurysms, vasculitis
2	4 5 6 Colorectal, jaundice, hepatitis, cirrhosis	16 17 Glomerular, tubular and cystic disease	26 24 Heart; adrenal (Chapter 24, adrenal part)
3	7 8 9 Gallbladder, pancreas, pneumonia, COPD	18 19 Renal tumours, stones, prostate, testis	27 Metabolic bone disease, bone tumours
4	10 11 12 Lung tumours, anaemias I and II	20 21 Cervix, endometrium, ovary, GTD	28 29 Joints, muscle, stroke, CNS infection
5	13 14 Leukaemias; lymphoma (Hodgkin, Lymphoma I)	22 23 24 Breast, thyroid, pituitary, diabetes	30 31 Brain tumours, skin
6	**Recall week:** Night-before pages 1–14, redo all BCQs, OSPE labs	**Recall week:** Night-before pages 14–24, redo all BCQs, OSPE labs	**Recall week:** Night-before pages 24–31, redo all BCQs, OSPE labs

B Four-week prof plan

Week	Days 1–6: learn and compress	Day 7: recall
1	1 – 8 GIT, liver, pancreas, pneumonia, TB	BCQs and viva 1–8
2	9 – 16 Lung, blood, glomerulus	BCQs 9–16; Day-7 review of 1–8
3	17 – 24 Kidney, reproductive, breast, endocrine	BCQs 17–24; Day-14 review of 1–8
4	25 – 31 on days 1–5 Days 6–7: grand mock paper, OSPE revision	All Night-before pages

Each day: about 1½ chapters. Morning: Pass 1 of a new chapter. Afternoon: Pass 2 and its BCQs. Evening: 30 minutes of due spaced reviews from the tracker.

C Ten-day crash plan

Day	Chapters	System
1	1 – 4	Gastrointestinal tract
2	5 – 7	Liver, gallbladder, pancreas
3	8 – 10	Lung
4	11 – 13	Anaemias, leukaemias
5	14 – 17	Lymphoma, bleeding, kidney I
6	18 – 21	Urinary tract, genital tracts
7	22 – 24	Breast, endocrine
8	25 – 28	Vessels, heart, bone, joints
9	29 – 31	Nervous system, skin
10	Grand mock paper; every Night-before page	

Crash mode: read only the mind map, the tables, the ★★★ topics of the Exam Radar, the Exam Zone and the Night-before page.

Master chapter tracker

Tick a box when the step is done. Pass 1 = **Read**, Pass 2 = **Compressed**, Pass 3 = **BCQs**; then the five spaced reviews, counted from the day you finished Pass 3.

Ch	Chapter	Module	Read	Compressed	BCQs	D1	D3	D7	D14	D30
MODULE 10										
1	Oral cavity, salivary glands & oesophagus	M10	☐	☐	☐	☐	☐	☐	☐	☐
2	Stomach: gastritis, peptic ulcer & tumours	M10	☐	☐	☐	☐	☐	☐	☐	☐
3	Malabsorption & inflammatory bowel disease	M10	☐	☐	☐	☐	☐	☐	☐	☐
4	Intestinal polyps & colorectal carcinoma	M10	☐	☐	☐	☐	☐	☐	☐	☐
5	Jaundice & viral hepatitis	M10	☐	☐	☐	☐	☐	☐	☐	☐
6	Cirrhosis, haemochromatosis & HCC	M10	☐	☐	☐	☐	☐	☐	☐	☐
7	Gallbladder & pancreas	M10	☐	☐	☐	☐	☐	☐	☐	☐
8	Pneumonia & tuberculosis	M10	☐	☐	☐	☐	☐	☐	☐	☐
9	COPD, asthma & interstitial lung disease	M10	☐	☐	☐	☐	☐	☐	☐	☐
10	Lung tumours	M10	☐	☐	☐	☐	☐	☐	☐	☐
11	Anaemias I: approach, iron, megaloblastic, aplastic	M10	☐	☐	☐	☐	☐	☐	☐	☐
12	Anaemias II: haemolytic & thalassaemias	M10	☐	☐	☐	☐	☐	☐	☐	☐
13	Leukaemias & myeloproliferative neoplasms	M10	☐	☐	☐	☐	☐	☐	☐	☐
14	Lymphomas	M10–11	☐	☐	☐	☐	☐	☐	☐	☐
MODULE 11										
15	Bleeding disorders	M11	☐	☐	☐	☐	☐	☐	☐	☐
16	Glomerular diseases	M11	☐	☐	☐	☐	☐	☐	☐	☐
17	Tubulointerstitial, cystic kidney & renal function	M11	☐	☐	☐	☐	☐	☐	☐	☐
18	Renal tumours, urothelial carcinoma & stones	M11	☐	☐	☐	☐	☐	☐	☐	☐
19	Prostate & testis	M11	☐	☐	☐	☐	☐	☐	☐	☐
20	Vulva, cervix & endometrium	M11	☐	☐	☐	☐	☐	☐	☐	☐
21	Ovarian tumours & trophoblastic disease	M11	☐	☐	☐	☐	☐	☐	☐	☐
22	Breast	M11	☐	☐	☐	☐	☐	☐	☐	☐
23	Thyroid & parathyroid	M11	☐	☐	☐	☐	☐	☐	☐	☐
24	Pituitary, adrenal & diabetes mellitus	M11–12	☐	☐	☐	☐	☐	☐	☐	☐
MODULE 12										
25	Blood vessels	M12	☐	☐	☐	☐	☐	☐	☐	☐
26	Heart	M12	☐	☐	☐	☐	☐	☐	☐	☐
27	Bone	M12	☐	☐	☐	☐	☐	☐	☐	☐
28	Joints & skeletal muscle	M12	☐	☐	☐	☐	☐	☐	☐	☐
29	CNS I: vascular, infection, demyelination, degeneration	M12	☐	☐	☐	☐	☐	☐	☐	☐
30	CNS II: brain & nerve sheath tumours	M12	☐	☐	☐	☐	☐	☐	☐	☐
31	Skin	M12	☐	☐	☐	☐	☐	☐	☐	☐

MODULE 10 EXAM
Date: _____

MODULE 11 EXAM
Date: _____

MODULE 12 EXAM
Date: _____

PROFESSIONAL EXAM
Date: _____

Notes

01

Oral Cavity, Salivary Glands & Oesophagus

The upper gut keeps a record of how we live. Naswar and paan turn the cheek white, then red, then malignant; acid that escapes the stomach rebuilds the lower oesophagus in the image of the intestine. Read the lining and you can read the risk.

BY THE END OF THIS CHAPTER YOU CAN

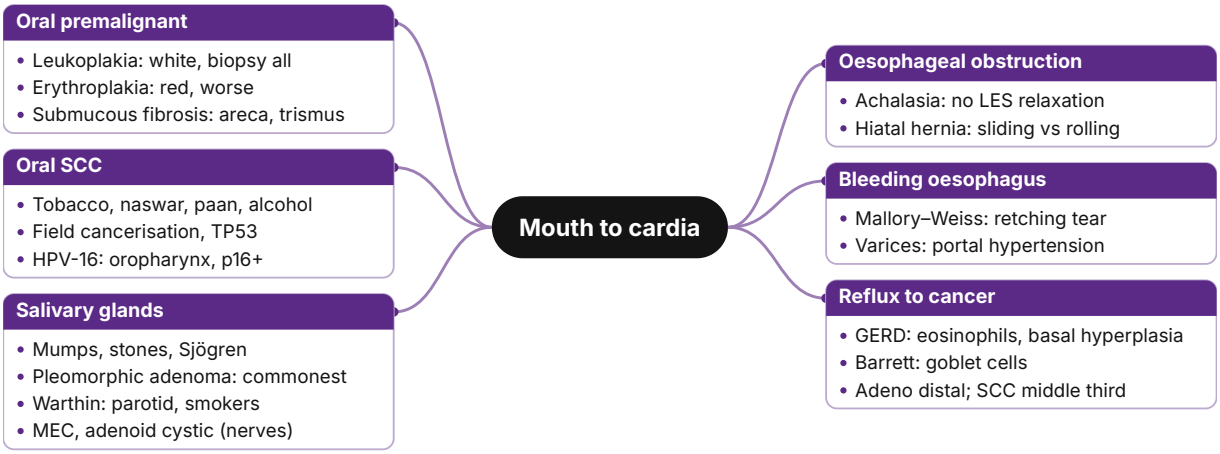
1. Define leukoplakia and erythroplakia, grade their risk and describe oral submucous fibrosis
2. Explain the two pathways of oral squamous cell carcinoma and its local risk factors
3. Classify sialadenitis and describe the four salivary tumours examiners ask most
4. Explain achalasia, hiatal hernia, Mallory–Weiss tears and oesophageal varices
5. Differentiate reflux, eosinophilic and infectious oesophagitis
6. Describe Barrett oesophagus and its progression to adenocarcinoma
7. Compare squamous cell carcinoma and adenocarcinoma of the oesophagus

EXAM RADAR

Theory SEQs	★★★
BCQs / MCQs	★★★
Viva voce	★★★
OSPE / slides	★★☆

Leukoplakia vs erythroplakia, pleomorphic adenoma, Barrett oesophagus and SCC vs adenocarcinoma of the oesophagus are the repeat favourites.

ROBBINS LINK Robbins Basic Pathology — *Oral Cavity and Gastrointestinal Tract*: oral lesions, salivary glands, oesophagus (obstruction, oesophagitis, Barrett, tumours).



Chapter at a glance. Six ideas carry the whole chapter.

INTRODUCTION TO SPECIAL PATHOLOGY

General pathology taught the processes (cell injury, inflammation, repair, haemodynamics, immunity, neoplasia); **special (systemic) pathology** applies them organ by organ. Every disease here, and every long answer you write, follows one frame: **definition → aetiology → pathogenesis → morphology (gross, microscopy) → clinical features and tests → complications and prognosis**.

1.1 Oral premalignant lesions

Most of the mouth is lined by **non-keratinised stratified squamous epithelium**. Chronic irritation by tobacco, areca nut or alcohol makes it thicken and keratinise, which looks **white**; or it becomes thin, dysplastic and vascular, which looks **red**. The colour is a rough guide to the risk.

DEFINITION

Leukoplakia is a white patch or plaque of the oral mucosa that **cannot be scraped off** and cannot be characterised clinically or microscopically as any other disease (a clinical label, not a diagnosis). **Erythroplakia** is a red, velvety, often eroded area, flat or slightly depressed, that cannot be explained by any other cause. A mixed lesion is **speckled leukoplakia (erythroleukoplakia)**.

LEUKOPLAKIA VS ERYTHROPLAKIA

Feature	Leukoplakia	Erythroplakia
Frequency	**Common** (about 3% of adults worldwide); men 40–70 years	Much less common
Microscopy	**Hyperkeratosis**, acanthosis; from no dysplasia to carcinoma in situ	Usually **severe dysplasia**, CIS or early invasion; little keratin
Malignant risk	About **5–25%** are dysplastic and may progress	**Much higher** than leukoplakia

MEMORY HOOK

White: watch. Red: real danger.

Both are **premalignant until a biopsy** proves otherwise. The usual march: hyperkeratosis → dysplasia → carcinoma in situ → invasive SCC.

EXAM TRAP

White, but not leukoplakia: candidiasis (scrapes off), **lichen planus** (lacy striae), frictional keratosis, and **oral hairy leukoplakia**: EBV-driven corrugated plaques on the **lateral tongue** in HIV; **not premalignant**.

Oral submucous fibrosis (OSMF)

A chronic, progressive **scarring** disease of the oral mucosa, seen almost entirely in South Asia, caused by chewing **areca nut (supari)** in paan, gutka and paan masala. It is a recognised **premalignant condition**.

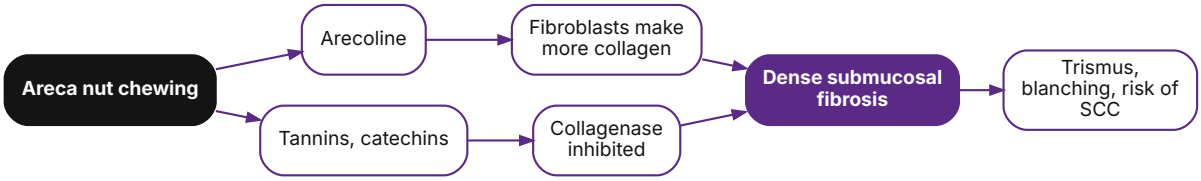


Figure 1.1 · Pathogenesis of oral submucous fibrosis. More collagen made, less collagen removed.

- **Clinical:** burning with spicy food, **blanched marble-white** mucosa, **vertical fibrous bands**, progressive **trismus**
- **Microscopy: atrophic** (sometimes dysplastic) epithelium over dense, **hyalinised, poorly vascular collagen** reaching muscle

1.2 Oral squamous cell carcinoma

About **95%** of oral cancers are squamous cell carcinomas (SCC). Two separate roads lead to them, and the road decides the site, the patient and the prognosis.

ROAD 1: CARCINOGEN-DRIVEN (ORAL CAVITY)

- **Tobacco** smoked or chewed (naswar, gutka), **betel quid with areca**, alcohol
- **Field cancerisation:** the whole exposed mucosa is mutated, so **second primaries** are common
- TP53 (commonest), loss of CDKN2A (p16), NOTCH1
- South Asia: **buccal mucosa, gingivobuccal sulcus** (where the quid rests); West: tongue, floor of mouth

ROAD 2: HPV-DRIVEN (OROPHARYNX)

- **HPV-16**; tonsillar crypts and base of tongue
- E7 inactivates RB, so cells over-express **p16**: the surrogate immunostain
- Younger patients, often non-smokers; non-keratinising
- **Better prognosis**

GROSS

- Early: raised, firm, **pearly plaque** or verrucous patch
- Later: **ulcer with raised, rolled, indurated edges** or a fungating mass, fixed to deeper tissue

MICROSCOPY

- Nests and cords of atypical squamous cells **invading** the stroma
- Well differentiated: **keratin pearls**, single-cell keratinisation, **intercellular bridges**

Spread: local, then **cervical lymph nodes** (often at diagnosis); lungs, liver and bone late. Early lesions do well, advanced ones poorly. **Verrucous carcinoma**: warty, well differentiated, pushing border, rarely metastasises.

PAKISTAN FOCUS

Oral cavity cancer is one of the most common cancers in Pakistan, at or near the top in men in the Karachi cancer registry. The drivers are **naswar**, **gutka** and **mainpuri** (smokeless tobacco), **paan with supari** and smoking; about **19%** of adults use tobacco in some form. Areca nut also causes OSMF, so a young chewer with trismus already carries a premalignant mucosa.

Source: Global Adult Tobacco Survey (GATS) Pakistan, 2014.

1.3 Salivary glands

NON-NEOPLASTIC SALIVARY DISEASE: A VIRUS, A STONE OR AUTOIMMUNITY

Disease	Cause and setting	Key points
Mumps	Paramyxovirus; children; commonest viral sialadenitis	**Bilateral** painful parotitis; interstitial oedema, mononuclear infiltrate; raised serum amylase; adults risk **orchitis**, pancreatitis, meningoencephalitis
Sialolithiasis and bacterial sialadenitis	Stones mostly in the **submandibular** gland (long uphill duct, thick saliva); *S. aureus*, *Strep. viridans* after obstruction or dehydration	Painful swelling **at meal times**; infection gives a tender gland with pus at the duct; chronic cases fibrose
Mucocele	Commonest salivary lesion; trauma ruptures a minor-gland duct, usually **lower lip**	Mucin pool with foamy macrophages and granulation tissue; **no epithelial lining**. Floor of mouth: **ranula**
Sjögren syndrome	Autoimmune; women; anti-**SS-A (Ro)**, anti-**SS-B (La)**; often with RA	Dry eyes + dry mouth; periductal lymphocytic infiltrate (lip biopsy); risk of **MALT lymphoma**

Salivary gland tumours

CONCEPT CLEAR

The smaller the gland, the more likely the tumour is malignant. About 65–80% of tumours arise in the parotid, but only 15–30% of these are malignant; about 40% of submandibular, half of minor-gland and 70–90% of sublingual tumours are malignant. A parotid lump with **facial palsy, pain or fixation** is malignant until proven otherwise.

Pleomorphic adenoma (benign mixed tumour)

The **commonest salivary tumour** (about 60% of parotid tumours). "Mixed" because neoplastic **epithelial and myoepithelial cells** lie in a myxoid, hyaline or chondroid stroma that the myoepithelial cells themselves make. Driven by PLAG1 or HMGA2 rearrangement.

GROSS

- Rounded, well-demarcated, rarely over 6 cm
- Capsule **incomplete**: tongue-like **protrusions** into the gland
- Grey-white, myxoid, with **bluish translucent chondroid** areas

MICROSCOPY

- **Ducts, acini, cords and sheets** of epithelial and myoepithelial cells
- **Myxoid and chondroid** matrix; cartilage, rarely bone
- No atypia

Clinical: painless, slow, mobile lump at the angle of the jaw. **Enucleation** leaves protrusions behind (recurrence up to about 25%; about 4% after parotidectomy). **Carcinoma ex pleomorphic adenoma**: about 2% if present < 5 years, about 10% if > 15 years; highly aggressive.

Warthin tumour (papillary cystadenoma lymphomatosum)

Second commonest benign tumour, practically confined to the **parotid**; 5th–7th decades; **smokers** have about eight times the risk; about **10% bilateral**. **Gross:** encapsulated, 2–5 cm, pale grey with **slit-like cysts** of mucoid fluid. **Microscopy:** cysts and papillae lined by a **double layer of oncocytic cells** (granular pink cytoplasm) over a dense **lymphoid stroma with germinal centres**.

THE TWO SALIVARY CARCINOMAS EXAMINERS ASK

Feature	Mucoepidermoid carcinoma	Adenoid cystic carcinoma
Frequency, site	**Commonest primary malignant** salivary tumour; parotid (60–70%)	Uncommon; about half in **minor glands** (palate)
Cells	**Squamous**, **mucus-secreting** and **intermediate** cells; mucin cysts	Small dark cells in **cribriform ("Swiss cheese")**, tubular or solid patterns around hyaline basement-membrane material
Gene	t(11;19) CRTC1–MAML2	t(6;9) MYB–NFIB
Hallmark	Behaviour follows **grade**: low grade, 5-year survival > 90%; high grade about 50%	**Perineural invasion** → pain, facial palsy; relentless recurrence, late lung/bone metastases

LAB SPOTTER

Salivary slide in 10 seconds: blue cartilage-like matrix with ducts = **pleomorphic adenoma**. Pink double-layered epithelium over lymphoid follicles = **Warthin**. Squamous cells mixed with mucous cells = **mucoepidermoid**. Sieve-like nests of small dark cells round nerves = **adenoid cystic**.

1.4 Oesophagus: obstruction and bleeding

CONCEPT CLEAR

Read the dysphagia. A **mechanical** block (stricture, carcinoma) narrows the lumen gradually, so **solids stick first** and liquids follow later. A **motility** disorder (achalasia) fails to propel anything, so **solids and liquids** are difficult from the start and undigested food is regurgitated.

Achalasia

DEFINITION

Achalasia is a motility disorder with a triad: **incomplete relaxation of the lower oesophageal sphincter (LES)**, **increased LES tone** and **aperistalsis** of the oesophageal body.

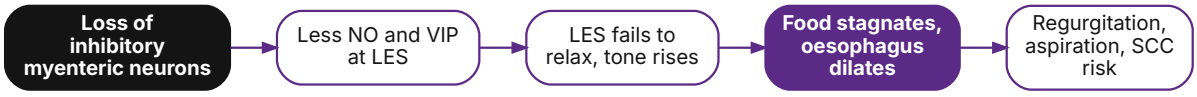


Figure 1.2 · Pathogenesis of achalasia.

- **Primary** (most): degeneration of inhibitory ganglion cells; autoimmune or post-viral damage suspected
- **Secondary: Chagas disease** (*T. cruzi*), diabetic neuropathy, amyloid, sarcoidosis, malignancy ("pseudo-achalasia")
- **Morphology:** dilated body above a narrow LES; few or no myenteric ganglion cells; stasis oesophagitis, candidiasis
- **Diagnosis:** barium **"bird-beak"**; manometry. **Complications:** aspiration pneumonia, diverticula, **SCC**

Hiatal hernia

The diaphragmatic crura separate and part of the stomach rises into the thorax; often silent.

- **Sliding (axial), about 95%:** GE junction and cardia slide up as a bell-shaped pouch; **GERD**, ulcers, bleeding

Mallory–Weiss tears

Severe **retching**, classically in **acute alcohol intoxication**, raises gastric pressure before the oesophagus relaxes, and the wall tears. The tears are **longitudinal, linear, mucosal** and cross the **GE junction**. They cause

- **Paraoesophageal (rolling):** GE junction stays put; fundus rolls up **beside** the oesophagus; risk of **strangulation**

haematemesis but usually heal by themselves. A **transmural** rupture (**Boerhaave syndrome**) is rare and catastrophic (mediastinitis).

Oesophageal varices



Figure 1.3 · How varices form and bleed. Portal blood bypasses the liver through the left gastric vein into the oesophageal plexus and on to the azygos vein.

- Develop in about **half of patients with cirrhosis**; schistosomiasis is a major cause worldwide
- **Morphology:** dilated, tortuous veins in the **submucosa of the distal oesophagus and proximal stomach**; collapse after death, so easily missed at autopsy
- **Rupture:** sudden massive haematemesis without warning; death from shock or hepatic coma; rebleeding common (see Chapter 6)

1.5 Oesophagitis, GERD and Barrett oesophagus

Reflux oesophagitis (gastro-oesophageal reflux disease, GERD)

The **commonest cause of oesophagitis**: refluxed acid, pepsin and bile injure the squamous lining when the LES fails. LES tone falls with **alcohol, tobacco, CNS depressants** and fatty meals; reflux rises with **obesity, pregnancy, hiatal hernia** and delayed gastric emptying.

ENDOSCOPY

- Often only **redness**; may look normal
- Severe: erosions, ulcers, stricture

MICROSCOPY

- **Intraepithelial eosinophils**; neutrophils if severe
- **Basal zone hyperplasia** > 20% of thickness
- **Elongated papillae** reaching the upper third

Clinical: heartburn, sour regurgitation, dysphagia; chest pain may mimic angina. **Complications:** ulcer and bleeding, **stricture**, **Barrett oesophagus**.

REFLUX VS EOSINOPHILIC OESOPHAGITIS: THE EOSINOPHIL COUNT SETTLES IT

Feature	Reflux oesophagitis	Eosinophilic oesophagitis
Cause, patient	Acid and pepsin; heartburn	**Allergic** (food antigens) in **atopic** patients; adults: **food impaction**, dysphagia
Endoscopy	Distal redness, erosions	**Rings** ("trachealised" oesophagus), furrows, white specks; proximal too
Eosinophils	Few, distal	**> 15 per high-power field**, clustered superficially
Treatment	Responds to proton-pump inhibitors	Elimination diet, topical steroids; poor PPI response

INFECTIOUS AND CHEMICAL OESOPHAGITIS HERPES AT THE EDGE, CMV AT THE BASE

- **Candida:** immunosuppressed, diabetics; adherent **grey-white pseudomembranes** of matted hyphae
- **Herpes simplex: punched-out ulcers**; inclusions and multinucleated **squamous cells at the edge**
- **CMV:** shallow ulcers; inclusions in **endothelial and stromal cells of the base**
- **Chemical:** lye, very hot drinks, lodged pills (doxycycline, bisphosphonates, iron, NSAIDs)

Barrett oesophagus

DEFINITION

Barrett oesophagus is a complication of chronic GERD in which distal oesophageal squamous mucosa is replaced by **intestinal-type columnar epithelium with goblet cells (intestinal metaplasia)**. It develops in about **10%** of people with symptomatic GERD, mostly white men aged 40–60, and is the main precursor of **oesophageal adenocarcinoma**. The metaplasia is an **adaptation** (columnar cells resist acid better), but the clone can acquire TP53 mutations and progress; most patients never do.

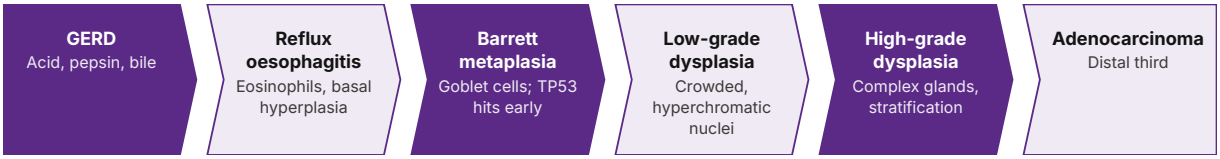


Figure 1.4 · From reflux to adenocarcinoma. Surveillance biopsies aim to catch the dysplasia steps.

ENDOSCOPY / GROSS

- **Tongues of red, velvety (salmon-pink) mucosa** extending up from the GE junction, between pale squamous mucosa
- **Long segment** ≥ 3 cm; **short** < 3 cm

MICROSCOPY

- Columnar epithelium replaces squamous
- **Goblet cells**: pale blue mucin vacuole, **wine-goblet** shape
- **Dysplasia** low or high grade; **intramucosal carcinoma** once lamina propria is invaded

Management: acid suppression and **endoscopic surveillance with biopsies**; high-grade dysplasia or intramucosal cancer → endoscopic resection or ablation, or oesophagectomy.

EXAM TRAP

Diagnosis needs two things (Robbins): columnar mucosa **above** the GE junction at endoscopy **and** goblet cells on biopsy. Some (e.g. British) guidelines accept columnar mucosa without goblet cells: give the Robbins definition and mention the variant.

1.6 Oesophageal carcinoma: SCC vs adenocarcinoma

SQUAMOUS CELL CARCINOMA VS ADENOCARCINOMA OF THE OESOPHAGUS: THE MOST-ASKED COMPARISON OF THE CHAPTER

Feature	Squamous cell carcinoma	Adenocarcinoma
Epidemiology	**Commonest worldwide**; Asian "oesophageal cancer belt"; over 45; M:F about 4:1	USA and Western Europe, rising; white men (about 7:1)
Precursor	Squamous dysplasia	**Barrett oesophagus** with dysplasia
Risk factors	**Tobacco, alcohol**, very **hot beverages**, poverty and poor diet, caustic injury, **achalasia**, **Plummer–Vinson syndrome**, radiation, HPV in high-risk areas	**GERD, Barrett**, tobacco, **obesity**, radiation; fruit and vegetables protect
Site	**Middle third** (about half)	**Distal third**, may invade the gastric cardia
Molecular	TP53 and CDKN2A loss, SOX2 amplification, cyclin D1	TP53 early; later ERBB2 (HER2) amplification
Gross	Grey-white plaque → polypoid, **ulcerated** or infiltrating mass	Flat or raised patch in Barrett mucosa → nodular or ulcerated mass
Microscopy	Keratinising SCC: **keratin pearls**, intercellular bridges	**Mucin-producing glands**, intestinal type
Local spread	Trachea (**tracheo-oesophageal fistula**, aspiration), aorta; nodes by level: upper → cervical, middle → mediastinal, lower → coeliac	Cardia; periesophageal and coeliac nodes

MEMORY HOOK

SCC = 3 S · Adeno = 2 A

Smoke, **S**pirits, **S**calding tea: middle third. **A**cid (Barrett), **A**diposity: lower third.

CLINICAL CORRELATION

Both present late: progressive dysphagia (solids, then liquids), odynophagia, marked **weight loss**, haematemesis, hoarseness, cough on swallowing (fistula). Overall survival is poor; mucosal tumours do well.

PAKISTAN FOCUS

An **"oesophageal cancer belt"** runs through Iran and Afghanistan into **Balochistan**, and in Pakistan **squamous cell carcinoma predominates**. Very **hot tea**, **naswar**, smoking and nutritional deficiencies are the local risk factors: a middle-aged patient from Quetta with progressive dysphagia should be scoped early.

EXAM ZONE Past-Paper Pattern SEQs — Solved

PROF PATTERN **a)** Define leukoplakia and erythroplakia. Which carries the greater risk of malignancy? **b)** Give the risk factors and morphology of squamous cell carcinoma of the oral cavity. **2 + 3 marks**

MODEL ANSWER PLAN

a) Leukoplakia: white plaque that cannot be scraped off or attributed to another disease; hyperkeratosis ± dysplasia (5–25% dysplastic). Erythroplakia: red velvety lesion, usually severe dysplasia/CIS: **much higher risk**. Both need biopsy. **b)** Risk: tobacco (naswar, gutka, smoking), betel quid/areca, alcohol, HPV-16 (oropharynx), OSMF. Gross: indurated ulcer with rolled edges; buccal mucosa in South Asia. Micro: invasive squamous nests, keratin pearls, intercellular bridges; cervical node spread.

Score booster: mention **field cancerisation** (second primaries) and list the steps from hyperkeratosis through dysplasia and CIS to invasion.

MODULE PATTERN A 45-year-old woman has a painless, mobile swelling in front of the right ear for 6 years. The facial nerve is intact. **a)** Most likely diagnosis? **b)** Describe its gross and microscopic features. **c)** Why does it recur after enucleation? **1 + 2 + 2 marks**

MODEL ANSWER PLAN

a) Pleomorphic adenoma of the parotid. **b)** Well-demarcated grey-white mass with bluish chondroid areas; epithelial and myoepithelial cells forming ducts and sheets in myxoid/chondroid stroma. **c)** Incomplete capsule with tongue-like protrusions left behind by enucleation (recurrence up to about 25%, about 4% after parotidectomy).

Score booster: add one line on **carcinoma ex pleomorphic adenoma**, whose risk rises with duration.

PROF PATTERN **a)** Define Barrett oesophagus and describe its morphology. **b)** Differentiate between squamous cell carcinoma and adenocarcinoma of the oesophagus. **2 + 3 marks**

MODEL ANSWER PLAN

a) Intestinal metaplasia (goblet cells) replacing distal squamous mucosa in chronic GERD (about 10%). Gross: salmon-pink velvety tongues above the GE junction. Micro: columnar epithelium with goblet cells ± dysplasia. **b)** Table: precursor, risk factors, site (middle vs distal third), histology (keratin pearls vs glands), geography.

BCQs One-Best-Answer Practice

1 A 50-year-old naswar user has a white patch on the buccal mucosa that cannot be wiped off. The most appropriate next step is:

A. Biopsy of the lesion
B. Antifungal treatment and review
C. Reassurance only
D. Serology for EBV
E. Wide surgical excision without biopsy

2 A 28-year-old supari chewer cannot open his mouth fully; the buccal mucosa is blanched with palpable vertical bands. The diagnosis is:

A. Leukoplakia
B. Sjögren syndrome
C. Oral submucous fibrosis
D. Lichen planus
E. Candidiasis

3 p16 immunostaining in an oropharyngeal squamous carcinoma is used as a marker of:

A. EBV infection
B. Tobacco exposure
C. TP53 mutation
D. HPV infection
E. Areca nut exposure

4 A parotid tumour shows a double layer of oncocytic epithelial cells lining cysts over a lymphoid stroma with germinal centres. The patient is a 60-year-old smoker. Diagnosis?

A. Pleomorphic adenoma
B. Mucoepidermoid carcinoma
C. Adenoid cystic carcinoma
D. Acinic cell carcinoma
E. Warthin tumour

5 The commonest primary malignant tumour of the salivary glands is:

A. Mucoepidermoid carcinoma
B. Adenoid cystic carcinoma
C. Carcinoma ex pleomorphic adenoma
D. Squamous cell carcinoma
E. Lymphoma

6 A palatal tumour with cribriform nests of small dark cells shows extensive perineural invasion. Which fusion gene is characteristic?

A. CRTC1–MAML2
B. MYB–NFIB
C. PLAG1 rearrangement
D. BCR–ABL
E. EWS–FLI1

7 A man has dysphagia to both solids and liquids, regurgitation of undigested food and a bird-beak narrowing on barium swallow. The basic defect is:

A. Mucosal ring at the squamocolumnar junction
B. Squamous cell carcinoma of the middle third
C. Loss of inhibitory neurons of the myenteric plexus
D. Eosinophilic infiltration of the mucosa
E. Herniation of the fundus beside the oesophagus

8 After a night of heavy drinking and repeated retching, a 25-year-old vomits fresh blood. Endoscopy will most likely show:

A. Dilated submucosal veins
B. Barrett mucosa
C. A transmural rupture with mediastinitis
D. Longitudinal mucosal tears at the GE junction
E. Punched-out ulcers

9 An atopic young man presents with food impaction. Endoscopy shows rings throughout the oesophagus; biopsy shows 40 eosinophils per high-power field. He did not improve on proton-pump inhibitors. Diagnosis?

A. Reflux oesophagitis
B. Candida oesophagitis
C. Barrett oesophagus
D. Achalasia
E. Eosinophilic oesophagitis

10 The histological feature required by Robbins to diagnose Barrett oesophagus is:

A. Basal zone hyperplasia
B. Intraepithelial eosinophils
C. Goblet cells in columnar epithelium
D. Keratin pearls
E. Ganglion cell loss

11 A 60-year-old man from Balochistan, a heavy smoker who drinks very hot tea, has progressive dysphagia. The most likely tumour and site are:

A. Adenocarcinoma, distal third
B. Squamous cell carcinoma, middle third
C. Adenocarcinoma, upper third
D. Leiomyoma, distal third
E. Squamous cell carcinoma of the gastric cardia

12 Inclusions within endothelial and stromal cells at the base of an oesophageal ulcer in a transplant recipient indicate:

A. Cytomegalovirus
B. Herpes simplex virus
C. Candida
D. Pill injury
E. Reflux

ANSWER KEY WITH ONE-LINE REASONS

1. **A** Leukoplakia is premalignant until histology excludes dysplasia.
2. **C** Areca nut → collagen deposition → trismus and blanching (OSMF).
3. **D** HPV E7 inactivates RB, causing p16 over-expression.

4. **E** Oncocytic double layer + lymphoid stroma = Warthin tumour.
5. **A** Squamous, mucous and intermediate cells; t(11;19).
6. **B** Adenoid cystic carcinoma: t(6;9) MYB–NFIB.
7. **C** Achalasia: loss of NO/VIP neurons → LES fails to relax.
8. **D** Mallory–Weiss: longitudinal mucosal tears; heal.
9. **E** > 15 eosinophils/HPF, rings, atopy, PPI failure.
10. **C** Barrett = intestinal metaplasia with goblet cells.
11. **B** Hot drinks + tobacco → SCC, mostly middle third.
12. **A** CMV: endothelium/stroma; HSV: squamous cells at the edge.

VIVA Rapid-Fire Viva Voce

Q Define leukoplakia.	White oral patch that cannot be scraped off or classified as another disease.
Q Is oral hairy leukoplakia premalignant?	No: EBV-related lateral-tongue plaque in HIV/immunosuppression.
Q What causes oral submucous fibrosis?	Areca nut: arecoline drives collagen synthesis; tannins block collagenase.
Q What is field cancerisation?	Carcinogens mutate the whole mucosa, so independent second primaries arise.
Q Commonest salivary gland tumour?	Pleomorphic adenoma, usually in the parotid.
Q Which salivary tumour is almost confined to the parotid and linked to smoking?	Warthin tumour.
Q Which salivary carcinoma is famous for perineural invasion?	Adenoid cystic carcinoma.
Q Where do salivary stones form most often, and why?	Submandibular: long uphill duct, thick mucous saliva.
Q Define achalasia.	Incomplete LES relaxation, raised LES tone, aperistalsis.
Q Name three histological features of reflux oesophagitis.	Intraepithelial eosinophils, basal hyperplasia > 20%, elongated papillae.
Q What is a goblet cell in Barrett oesophagus?	Columnar cell with a goblet-shaped blue mucin vacuole: intestinal metaplasia.
Q Name four risk factors for oesophageal SCC.	Tobacco, alcohol, very hot drinks, achalasia (also Plummer–Vinson, lye).

OSPE Practice Stations

STATION 1 · SLIDE — 2 minutes

TASK

Parotid lump: circumscribed tumour with ducts and sheets of epithelial and spindle cells in a pale blue myxoid matrix with cartilage-like islands. (a) Identify. (b) Two features. (c) Long-term risk if neglected?

ANSWER

(a) Pleomorphic adenoma. (b) Epithelial + myoepithelial cells; myxochondroid stroma. (c) Carcinoma ex pleomorphic adenoma.

STATION 2 · SLIDE — 2 minutes

TASK

Biopsy 2 cm above the GE junction, long-standing heartburn: columnar epithelium with scattered cells holding large blue mucin vacuoles. (a) Diagnosis. (b) Name the cells. (c) Cancer risk?

ANSWER

(a) Barrett oesophagus (intestinal metaplasia). (b) Goblet cells. (c) Adenocarcinoma of the distal oesophagus.

STATION 3 · GROSS SPECIMEN 1 minute

TASK

Oesophagectomy: ulcerated, grey-white, firm growth with everted edges in the middle third, narrowing the lumen. (a) Diagnosis. (b) Two risk factors. (c) One local complication.

ANSWER

(a) Squamous cell carcinoma. (b) Tobacco/naswar, very hot tea, alcohol, achalasia. (c) Tracheo-oesophageal fistula with aspiration pneumonia.

RAPID REVISION The Night-Before Page

15 ONE-LINERS

01 Leukoplakia: white, cannot scrape off; 5–25% dysplastic; biopsy all.

02 Erythroplakia: red, velvety; much higher malignant risk.

03 OSMF: areca nut, blanching, fibrous bands, trismus; premalignant.

04 Oral SCC: tobacco, naswar, paan; TP53; field cancerisation.

05 HPV-16 oropharyngeal SCC: p16+, younger, better outcome.

06 Smaller salivary gland = higher chance of malignancy.

07 Pleomorphic adenoma: myxochondroid stroma; recurs after enucleation.

08 Warthin: parotid, smokers, oncocytic double layer + lymphoid stroma.

09 Mucoepidermoid: commonest salivary cancer; CRTC1–MAML2.

10 Adenoid cystic: cribriform, perineural invasion, MYB–NFIB.

11 Achalasia: no LES relaxation; bird-beak; Chagas secondary.

12 Mallory–Weiss: longitudinal tears at GE junction after retching.

13 Varices: portal hypertension; submucosal veins of distal oesophagus.

14 Barrett: goblet cells above GE junction; precursor of adenocarcinoma.

15 SCC middle third (tobacco, hot tea); adeno distal third (Barrett).

MEMORY HOOKS

- **White: watch; Red: real danger** (leukoplakia vs erythroplakia)
- **Smaller gland, bigger danger** (salivary tumours)
- **Herpes at the Edge, CMV at the Base**
- **SCC = 3 S** (smoke, spirits, scalding tea); **Adeno = 2 A** (acid, adiposity)

TEST YOURSELF: FILL THE GAPS

1. Arecoline causes _____ of the oral mucosa.
2. Pleomorphic adenoma gene: _____ or HMGA2.
3. EoE needs more than _____ eosinophils per HPF.
4. Barrett long segment is _____ cm or more.
5. Secondary achalasia in South America: _____ disease.
6. Mucoepidermoid carcinoma translocation: t(_____).

Answers: submucous fibrosis · PLAG1 · 15 · 3 · Chagas · 11;19

DRAW FROM MEMORY

Pleomorphic adenoma under the microscope: label ducts, myoepithelial cells, myxoid and chondroid stroma, capsule with protrusion.

DRAW FROM MEMORY

The GERD → Barrett → dysplasia → adenocarcinoma ladder, with one histological marker at each step.

SPACED REVISION ☐ Day 1 ☐ Day 3 ☐ Day 7 ☐ Day 14 ☐ Day 30

02

Stomach: Gastritis, Peptic Ulcer & Gastric Tumours

The stomach digests a steak but not itself, because mucus, bicarbonate and prostaglandins hold the line. Tip the balance with H. pylori or a painkiller and the wall erodes; leave H. pylori for decades and the same inflammation can end in cancer or lymphoma.

BY THE END OF THIS CHAPTER YOU CAN

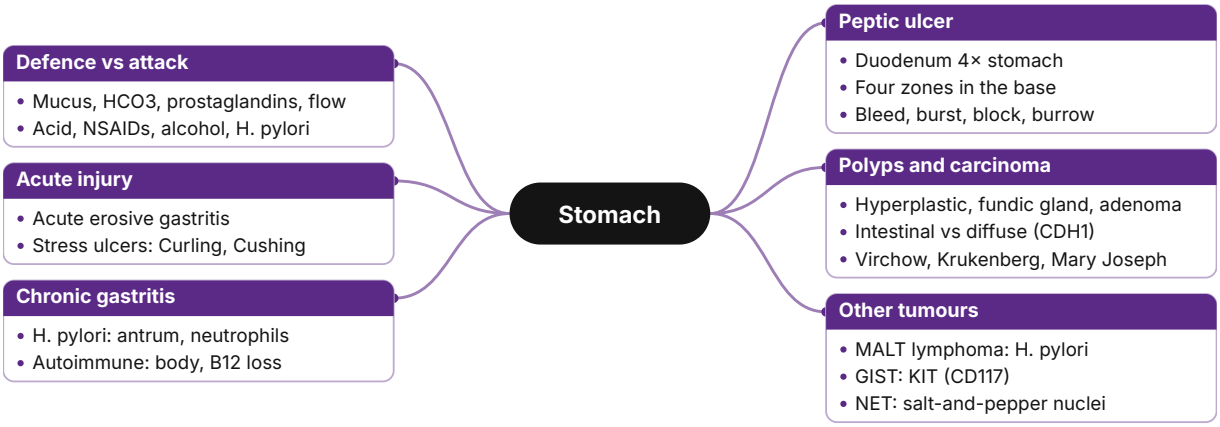
1. Explain the balance between mucosal defence and injury and apply it to acute gastritis and stress ulcers
2. Compare H. pylori gastritis with autoimmune gastritis
3. Describe the pathogenesis, morphology and complications of peptic ulcer disease
4. Differentiate a benign from a malignant gastric ulcer and classify gastric polyps
5. Classify gastric adenocarcinoma (Lauren) and describe its spread and eponymous signs
6. Outline MALT lymphoma, GIST and gastric neuroendocrine tumours

EXAM RADAR

Theory SEQs	★★★
BCQs / MCQs	★★★
Viva voce	★★★
OSPE / slides	★★★

Peptic ulcer (pathogenesis, morphology, complications), H. pylori gastritis and gastric carcinoma are perennial long questions; the H. pylori gastritis slide is a lab favourite.

ROBBINS LINK Robbins Basic Pathology — *Oral Cavity and Gastrointestinal Tract*: stomach (acute and chronic gastritis, peptic ulcer disease, polyps, adenocarcinoma, lymphoma, GIST, neuroendocrine tumours).



Chapter at a glance. Six ideas carry the whole chapter.

2.1 Mucosal defence, acute gastritis and stress ulcers

CONCEPT CLEAR

Every gastric lesion in this chapter is a lost balance. Acid (pH about 1) and pepsin are kept off the epithelium by a surface layer of **mucus** into which the cells pour **bicarbonate**; tight junctions and rapid restitution seal gaps; a rich **blood flow** carries away acid and brings oxygen; and **prostaglandins** (PGE_2, PGI_2) drive mucus, bicarbonate and blood flow. Anything that weakens defence or increases attack causes gastritis or ulcer.

DEFENCES WEAKENED BY

- **NSAIDs, aspirin**: block COX, so prostaglandins fall
- **Ischaemia, shock**, hypoxia (high altitude), ageing
- **Alcohol, smoking**, chemotherapy, radiation
- **Uraemia**: ammonia inhibits bicarbonate transport

ATTACK INCREASED BY

- **H. pylori** (antral infection → acid hypersecretion)
- **Gastrinoma** (Zollinger–Ellison syndrome)
- **Vagal overdrive** after head injury (Cushing ulcer)
- Bile reflux, swallowed acids or alkalis

Acute gastritis

A transient inflammation of the gastric mucosa. It may be silent or cause epigastric pain, nausea and vomiting; severe cases erode and bleed, giving **haematemesis or melaena**.

ENDOSCOPY / GROSS

- Mild: oedema and redness
- Severe: multiple **erosions** with dark punctate haemorrhages (**acute erosive haemorrhagic gastritis**)

MICROSCOPY

- Oedema, congestion; **neutrophils** above the basement membrane in contact with epithelium = **active** inflammation
- **Erosion**: loss of surface epithelium **not** crossing the muscularis mucosae, with fibrin and neutrophils

Stress-related mucosal disease (acute stress ulcers)

Seen in critically ill patients: shock, sepsis, major trauma or surgery, burns and intracranial injury. The main cause is **local ischaemia** from hypotension and splanchnic vasoconstriction; systemic acidosis and low prostaglandins add to it.

TWO EPONYMS

Curling ulcer: in the **proximal duodenum** after **severe burns** or trauma.

Cushing ulcer: in the stomach, duodenum or oesophagus with **intracranial disease** (head injury, tumour, surgery). Direct **vagal stimulation causes acid hypersecretion**; perforation is common.

MEMORY HOOK

Curling = Cooking (burns) · Cushing = Cranium

Burns → duodenum; brain → acid hypersecretion and perforation.

ACUTE STRESS ULCER VS CHRONIC PEPTIC ULCER: A CLASSIC "DIFFERENTIATE" QUESTION

Feature	Acute stress ulcer	Chronic peptic ulcer
Setting	Shock, burns, sepsis, head injury, ICU	H. pylori, NSAIDs
Number, size	**Multiple**, small (< 1 cm), round	Usually **solitary**, larger
Site	Anywhere in the stomach (Curling: duodenum)	First part of duodenum; lesser curvature of stomach
Depth, base	Shallow; base **brown-black** (acid-digested blood)	Deep, often into muscularis propria; base **clean** grey-white
Surroundings	Normal mucosa	Chronic gastritis; puckered mucosa
Scar, vessels	**No** scarring, no thickened vessels	**Fibrous scar**, thick-walled vessels
Outcome	Heal completely in days to weeks once the cause goes	Chronic, relapsing; heals with scar

2.2 Chronic gastritis

Chronic mucosal inflammation that, over years, leads to **atrophy** and **intestinal metaplasia**, usually without erosions. **H. pylori is by far the commonest cause**; autoimmune gastritis accounts for under 10%. Rarer causes: radiation, chronic bile reflux, Crohn disease, graft-versus-host disease.

H. pylori gastritis

Helicobacter pylori is a curved, flagellated, Gram-negative rod that lives in the **mucus layer over the antral epithelium**. It is acquired in **childhood**, spreads by the faeco-oral and oral–oral routes, and thrives where there is crowding, poverty and poor sanitation. Untreated, it persists for life.

FOUR VIRULENCE TOOLS

Flagella swim through mucus · **Urease** splits urea to ammonia, buffering acid around the bacterium (the basis of the urea breath test) · **Adhesins** (BabA) bind epithelial blood-group antigens · **Toxins**: CagA (injected into epithelial cells, pro-proliferative) and VacA. CagA-positive strains carry the higher risk of ulcer and cancer.

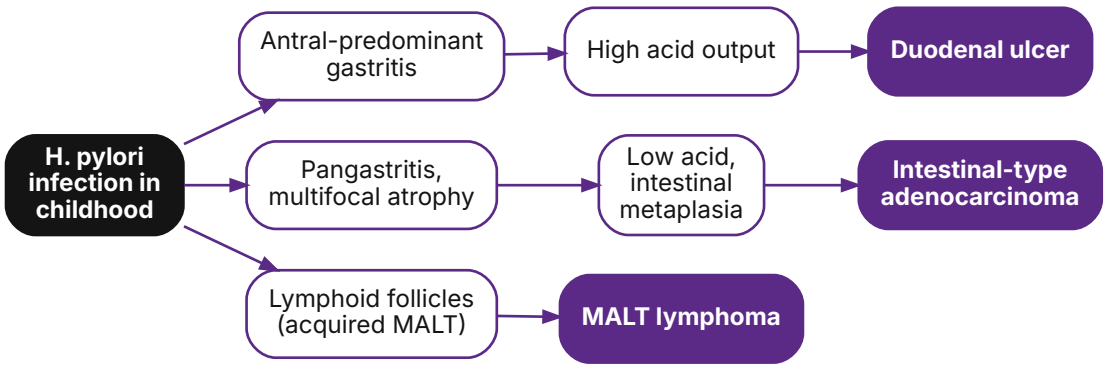


Figure 2.1 · Three outcomes of one infection. The distribution of gastritis decides the disease.

ENDOSCOPY

- **Antrum** reddened, coarse, nodular (lymphoid follicles)
- Long-standing: thin, flat mucosa (atrophy)

MICROSCOPY

- **Curved rods** in the surface mucus over the epithelium and in the pits (H&E; better with **Giemsa**, Warthin–Starry or immunostain)
- **Neutrophils** within the epithelium and in pit lumens (**pit abscesses**)
- Many **plasma cells**, lymphocytes, macrophages in superficial lamina propria
- **Lymphoid aggregates with germinal centres**
- Late: atrophy and **intestinal metaplasia** (goblet cells)

Tests: non-invasive: **urea breath test**, **stool antigen**, serology (shows exposure, not current infection). Invasive (endoscopic biopsy): histology, **rapid urease (CLO) test**, culture. **Treatment:** a proton-pump inhibitor plus two antibiotics; recurrence is usually reinfection.

LAB SPOTTER

H. pylori gastritis slide in 10 seconds: (1) Antral mucosa with a **plasma-cell-rich** lamina propria. (2) **Neutrophils** crawling into the pit epithelium. (3) Look at the **surface mucus** at high power for slender curved or S-shaped rods. (4) A **lymphoid follicle** with a germinal centre clinches chronic H. pylori gastritis.

Autoimmune gastritis

T cells (CD4+) destroy the **parietal cells** of the **body and fundus**, sparing the antrum. Antibodies to parietal cells (**H^+/K^+-ATPase**) and **intrinsic factor** are markers; the IF antibodies also block B12 binding. Patients are older (median about 60), slightly more often women, and often have other autoimmune diseases.

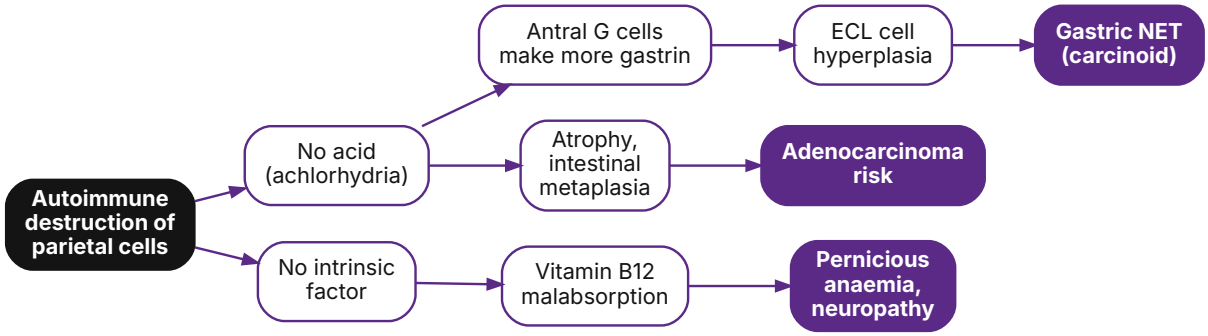


Figure 2.2 · Consequences of autoimmune gastritis. One lost cell, three diseases.

H. PYLORI GASTRITIS VS AUTOIMMUNE GASTRITIS (ROBBINS PATTERN)

Feature	H. pylori-associated	Autoimmune
Location	**Antrum**	**Body and fundus** (antrum spared)
Inflammation	Neutrophils; subepithelial plasma cells; superficial	Lymphocytes, macrophages; deep, around glands

Feature	H. pylori-associated	Autoimmune
Acid	Increased to slightly decreased	**Decreased** (achlorhydria)
Gastrin	Normal to decreased	**Increased**
Other lesions	Hyperplastic / inflammatory polyps	Neuroendocrine (ECL) hyperplasia
Serology	Antibodies to H. pylori	Anti-parietal cell, anti-intrinsic factor
Sequelae	**Peptic ulcer**, adenocarcinoma, **MALT lymphoma**	Atrophy, **pernicious anaemia**, adenocarcinoma, **carcinoid**
Associations	Low socio-economic status, crowding	Hashimoto thyroiditis, type 1 diabetes, Graves disease

MEMORY HOOK

Type A = four As, in the body

Autoimmune, **A**ntibodies, **A**chlorhydria, pernicious **A**naemia. Type **B** = **B**acterial (H. pylori), in the antrum.

EXAM TRAP

The letters mislead: type A (autoimmune) sits in the **body**, type B (bacterial) in the **antrum**. And acid goes **up** in antral H. pylori gastritis but **down** in autoimmune gastritis, so gastrin is high only in the latter.

PAKISTAN FOCUS

H. pylori infection is acquired in childhood and is highly prevalent in Pakistan, as in other developing countries where crowding and unsafe water favour spread. About **19%** of adults use tobacco in some form, and smoking delays ulcer healing. Treating H. pylori, not just the acid, is what stops ulcers from coming back.

Source: Global Adult Tobacco Survey (GATS) Pakistan, 2014.

2.3 Peptic ulcer disease

DEFINITION

Peptic ulcer is a chronic, usually solitary breach of the mucosa that extends **through the muscularis mucosae** into the submucosa or deeper, in any part of the gut exposed to acid-pepsin. Most are caused by **H. pylori** or **NSAIDs**.

- **Sites: first part of duodenum** (anterior wall, within a few cm of the pylorus) about **4×** commoner than stomach (**lesser curvature**, body–antrum junction); also lower oesophagus (GERD), gastro-jejunostomy margins, Meckel diverticulum
- **Frequency:** lifetime risk about 10% in men and 4% in women. Most ulcer patients carry H. pylori, yet only a small minority of infected people ever develop an ulcer

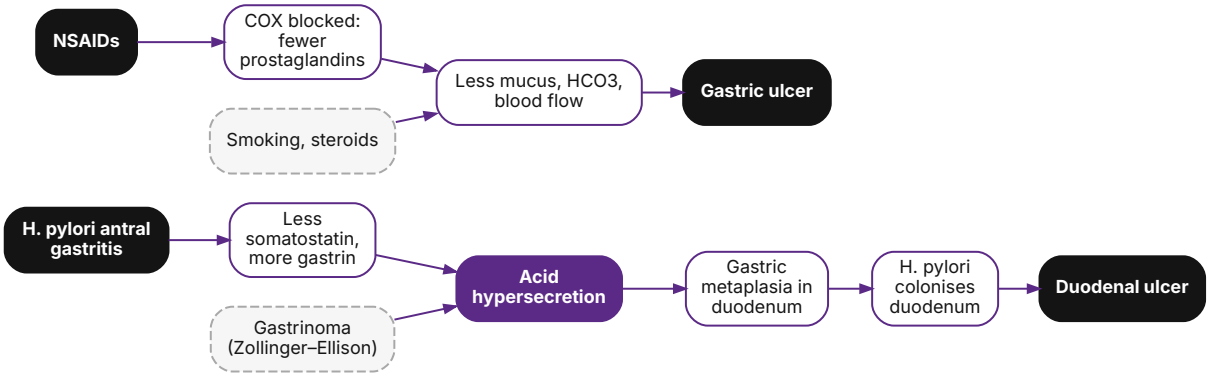


Figure 2.3 · Pathogenesis of peptic ulcer. More attack, less defence.

Other associations: cirrhosis, COPD, chronic kidney disease and hyperparathyroidism (calcium stimulates gastrin). **Zollinger–Ellison syndrome** (gastrinoma, often with MEN1) gives multiple, refractory ulcers in unusual sites such as the jejunum, with diarrhoea.

Morphology of a chronic peptic ulcer

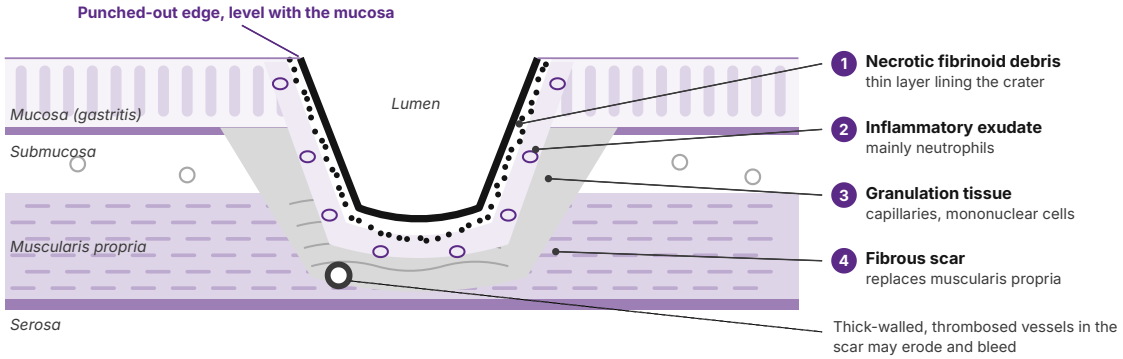


Figure 2.4 · Chronic peptic ulcer in section. The four zones of the base, from the lumen outward: draw this in every answer on peptic ulcer.

GROSS

- **Round to oval, sharply punched-out** defect with straight walls
- Margins **level** with the surrounding mucosa (heaped-up margins suggest cancer)
- Base **smooth and clean** (exudate digested); a thrombosed or open vessel may be visible
- **Mucosal folds radiate** from the crater like wheel spokes
- Depth limited by muscularis, or by adherent pancreas, omentum or liver

MICROSCOPY

- Base in four zones (Fig 2.4): **necrotic fibrinoid debris → neutrophilic exudate → granulation tissue → fibrous scar**
- Scar replaces the muscularis propria; vessels in it are **thick-walled**, sometimes thrombosed
- Surrounding mucosa: **chronic (H. pylori) gastritis**

CLINICAL FEATURES

- **Epigastric burning or gnawing pain** 1–3 hours after meals, worse at night, relieved by alkali or food
- Nausea, vomiting, bloating, belching, weight loss
- Some present first with **iron-deficiency anaemia**, bleeding or perforation

MEMORY HOOK

Bleed, Burst, Block, Burrow

The four complications of a peptic ulcer, in order of frequency of the first three, plus penetration into neighbouring organs.

COMPLICATIONS OF PEPTIC ULCER DISEASE

Complication	Frequency and importance	Pathology and picture
Haemorrhage (bleed)	**Commonest** (15–20% of patients); about a quarter of ulcer deaths	Erosion of a vessel in the base (posterior duodenal ulcer → gastroduodenal artery); haematemesis, melaena, anaemia
Perforation (burst)	About 5%; **two-thirds of ulcer deaths**	Anterior duodenal ulcer → acute peritonitis; **free gas under the diaphragm**; board-like rigidity
Obstruction (block)	About 2%	Oedema or scarring of a pyloric channel or duodenal ulcer → gastric outlet obstruction, vomiting, succussion splash
Penetration (burrow)	Posterior ulcers	Into pancreas (back pain, raised amylase), liver or omentum
Malignancy	**Never** in duodenal; extremely rare in gastric	Most "malignant ulcers" were ulcerated carcinomas from the start: **biopsy every gastric ulcer**

DUODENAL VS GASTRIC PEPTIC ULCER

Feature	Duodenal	Gastric
Frequency	About 4× commoner	Less common
Site	1st part, anterior wall	Lesser curvature
Acid	Normal or high	Normal or low
Pain (classic)	Relieved by food; nocturnal	Worse with food
Cancer risk	None	Exclude carcinoma

BENIGN VS MALIGNANT GASTRIC ULCER (GROSS)

Feature	Benign	Malignant
Shape	Round, punched-out	Irregular
Margins	Flat, level	**Heaped-up, everted**
Base	Smooth, clean	Necrotic, nodular
Folds	Radiate to the edge	Stop short, clubbed
Size	Unreliable: benign ulcers may exceed 4 cm	

CLINICAL CASE

A 42-year-old smoker with months of night-time epigastric pain relieved by milk takes ibuprofen for back pain. He develops sudden severe epigastric pain; the abdomen is rigid and silent; erect chest X-ray shows gas under the diaphragm.

Reasoning: classic duodenal ulcer pain + NSAID + smoking → **perforated anterior duodenal ulcer** with chemical, then bacterial, peritonitis. After surgical repair: test and treat H. pylori, stop NSAIDs and smoking.

2.4 Gastric polyps

THE THREE GASTRIC POLYPS TO KNOW

Feature	Inflammatory / hyperplastic	Fundic gland polyp	Gastric adenoma
Frequency	About **75%** of gastric polyps	Common	About 10%
Setting	Chronic gastritis (H. pylori); 50–60 years	Sporadic (**proton-pump inhibitor** use) or FAP	Chronic gastritis with atrophy and intestinal metaplasia; FAP
Microscopy	Elongated, tortuous, cystic foveolar glands; oedematous inflamed lamina propria; surface erosion	Cystically dilated glands lined by flattened **parietal and chief cells**	**Dysplastic** columnar epithelium, usually intestinal type
Cancer risk	Dysplasia rises with size (> 1.5 cm)	None (sporadic)	**Carcinoma in up to 30%**; high if > 2 cm

2.5 Gastric adenocarcinoma

Adenocarcinoma makes up **over 90%** of gastric cancers. Incidence is highest in **Japan, Korea, Chile and Eastern Europe**, and has fallen sharply in countries where H. pylori has declined and refrigeration replaced salting and smoking of food. Men are affected more than women.

RISK FACTORS

- **Infection: H. pylori** (chronic gastritis → atrophy → metaplasia); **EBV** (about 10% of cases)
- **Diet: nitrites/nitrosamines** in smoked, salted and pickled foods; lack of fresh fruit and vegetables
- **Host:** autoimmune gastritis, partial gastrectomy, gastric adenoma, smoking
- **Genetic:** germline CDH1 (hereditary diffuse gastric cancer), FAP, Lynch syndrome

MOLECULAR KEY

Diffuse type: loss of CDH1 (**E-cadherin**), the glue that holds cells together, so cells infiltrate singly; germline in familial cases, somatic in about half of sporadic ones. **Intestinal type:** Wnt pathway (APC/β-catenin), microsatellite instability, TP53. **EBV-positive** tumours: PIK3CA mutation, heavy DNA methylation. ERBB2 (HER2) amplification in a subset → trastuzumab.



Figure 2.5 · The intestinal-type cascade (Correa sequence). Diffuse-type cancer has no recognised precursor.

- **Location:** antrum and pylorus 50–60%, cardia about 25%; **lesser curvature** (about 40%) far more than greater curvature
- **Depth: early gastric cancer** is confined to the **mucosa and submucosa**, whatever the nodal status; **advanced** cancer reaches the muscularis propria or beyond
- **Growth patterns:** exophytic (polypoid, fungating), flat or depressed, and excavated (ulcerated)
- **Lauren histological types:** intestinal and diffuse (table below); mixed tumours exist

LAUREN CLASSIFICATION: INTESTINAL VS DIFFUSE GASTRIC CARCINOMA

Feature	Intestinal type	Diffuse type
Growth	**Cohesive** cells forming **glands**, like colonic adenocarcinoma	**Discohesive** cells infiltrating singly or in small clusters
Hallmark cell	Columnar cells; mucin inside gland lumens	**Signet-ring cells**: intracytoplasmic mucin pushes the nucleus to one side
Precursor	Intestinal metaplasia, dysplasia, adenoma	**None** recognised
Gene	APC/β-catenin, MSI, TP53	CDH1 (E-cadherin) loss
Patient	Older (mean about 55); M:F about 2:1; high-risk regions; H. pylori	Younger; M = F; incidence fairly constant worldwide
Gross	Bulky **exophytic** or ulcerated mass	**Linitis plastica**: diffusely thickened, rigid wall; flattened rugae
Trend	Falling with H. pylori control	Not falling; now a larger share of cases

GROSS

- Intestinal: polypoid or fungating mass, or an **ulcer with heaped-up, rolled margins** and necrotic base
- Diffuse: **"leather-bottle" stomach** (linitis plastica): thick, rigid, non-distensible wall, grey-white on cut section

MICROSCOPY

- Intestinal: malignant glands with nuclear stratification, mitoses, invasion of the wall
- Diffuse: sheets or single **signet-ring cells** (PAS/mucicarmine positive) in a **desmoplastic** stroma; pools of extracellular mucin

Spread and eponymous signs

Route	What examiners want
Local	Duodenum, pancreas, transverse colon, retroperitoneum
Lymphatic	Perigastric nodes; **Virchow node**: left supraclavicular sentinel node, sometimes the first sign of an occult cancer
Transcoelomic	Peritoneal seeding, ascites; **Krukenberg tumour**: bilateral ovarian metastases full of **signet-ring cells** (diffuse type); drop deposits in the pelvic pouch (Blumer shelf)
Skin of umbilicus	**Sister Mary Joseph nodule**: periumbilical subcutaneous metastasis
Blood	Liver (commonest distant organ), lungs, bone

CLINICAL CORRELATION

Clinical: early disease is silent or causes dyspepsia; by the time of **weight loss, anorexia, early satiety, vomiting** (antral obstruction), dysphagia (cardia) or iron-deficiency anaemia, it is usually advanced. Paraneoplastic clues: **acanthosis nigricans**, **Trousseau migratory thrombophlebitis**, sudden seborrhoeic keratoses (Leser–Trélat). **Prognosis** depends on depth and spread: 5-year survival of early gastric cancer is about 90% even with nodal metastasis, but under 20% for advanced disease. Diagnosis is by endoscopy and biopsy.

2.6 MALT lymphoma, GIST and neuroendocrine tumours

THREE NON-EPITHELIAL-LOOKING GASTRIC TUMOURS

Feature	MALT lymphoma	GIST	Neuroendocrine tumour (carcinoid)
Origin	Marginal-zone **B cells** of acquired MALT (extranodal marginal zone lymphoma)	**Interstitial cells of Cajal** (gut pacemaker cells)	Neuroendocrine (ECL) cells of the mucosa
Cause / gene	**H. pylori** gastritis; t(11;18) API2–MALT1, t(1;14) BCL10	Gain-of-function KIT (75–80%) or PDGFRA (about 8%); NF1; Carney triad	**Hypergastrinaemia**: autoimmune atrophic gastritis, Zollinger–Ellison (MEN1); or sporadic
Gross	Thickened mucosa, erosions, ulcers	Solitary, well-circumscribed **fleshy mass** in the wall; mucosa intact or ulcerated; can be very large	Small submucosal, polypoid nodules; **yellow-tan**, firm
Microscopy	Small B cells in lamina propria; **lymphoepithelial lesions** (tumour cells invading glands); reactive follicles	**Spindle** cells in fascicles or **epithelioid** cells	Nests, trabeculae of uniform cells; **salt-and-pepper** chromatin; pink granular cytoplasm
Markers	CD20+, CD5−, CD10−	**KIT (CD117)**, DOG1	**Chromogranin**, synaptophysin
Behaviour, treatment	**Regresses after H. pylori eradication** (not if t(11;18)); may transform to DLBCL	Risk by size, mitoses, site (gastric less aggressive); **imatinib** for unresectable disease	Gastrin-driven ones indolent; sporadic ones aggressive; carcinoid syndrome only with liver metastases

⚠ **EXAM TRAP**

GIST vs leiomyoma: both are spindle-cell wall masses, but GIST is **KIT (CD117)/DOG1-positive** while smooth-muscle tumours are desmin-positive and KIT-negative. **MALT lymphoma vs gastritis:** lymphoepithelial lesions and a monotonous B-cell sheet mean lymphoma. Remember: **antibiotics can cure a lymphoma** here, which is why H. pylori status is checked first.

EXAM ZONE Past-Paper Pattern SEQs — Solved

PROF PATTERN **a)** Differentiate between H. pylori gastritis and autoimmune gastritis. **b)** Describe the pathogenesis of peptic ulcer. **2 + 3 marks**

MODEL ANSWER PLAN

a) Table of 5 points: site (antrum vs body), infiltrate (neutrophils, plasma cells vs lymphocytes, deep), acid (normal/high vs low), gastrin (normal vs high), sequelae (ulcer, MALT lymphoma vs pernicious anaemia, carcinoid). **b)** Imbalance of defence and attack. H. pylori: antral gastritis → less somatostatin, more gastrin → acid hypersecretion → gastric metaplasia in duodenum → colonisation → ulcer; urease, CagA, VacA damage. NSAIDs: COX inhibition → less mucus, bicarbonate, blood flow. Others: ZES, smoking, steroids.

Score booster: draw Fig 2.3 and name one virulence factor per step.

PROF PATTERN A 45-year-old man has epigastric pain that wakes him at night and is relieved by food. **a)** Most likely diagnosis? **b)** Describe the gross and microscopic features of the lesion. **c)** List its complications. **1 + 2 + 2 marks**

MODEL ANSWER PLAN

a) Chronic duodenal peptic ulcer. **b)** Round, punched-out ulcer, level margins, clean base, radiating folds; micro four zones: necrotic debris, neutrophilic exudate, granulation tissue, fibrous scar with thick-walled vessels. **c)** Haemorrhage (commonest), perforation (most deaths), obstruction, penetration; no malignant change in duodenal ulcers.

Score booster: a labelled diagram of the four zones (Fig 2.4) earns the morphology marks quickly.

PROF PATTERN **a)** Enumerate the risk factors of gastric carcinoma. **b)** Compare intestinal and diffuse types of gastric carcinoma and describe their spread. **2 + 3 marks**

MODEL ANSWER PLAN

a) H. pylori, EBV, nitrosamine-rich diet, low fruit/vegetables, autoimmune gastritis, adenoma, partial gastrectomy, smoking, CDH1/FAP/Lynch. **b)** Table: cohesive glands vs discohesive signet-ring cells; precursor vs none; APC/MSI vs CDH1; older men vs younger, equal sex; exophytic vs linitis plastica. Spread: local, lymphatic (Virchow node), transcoelomic (Krukenberg, peritoneum), Sister Mary Joseph nodule, liver.

CLASSIC **a)** Differentiate an acute stress ulcer from a chronic peptic ulcer. **b)** What are Curling and Cushing ulcers? **3 + 2 marks**

MODEL ANSWER PLAN

a) Multiple, small, shallow, brown-black base, normal surroundings, no scar vs solitary, deep, clean base, scar with thick vessels, chronic gastritis. **b)** Curling: proximal duodenum after burns/trauma. Cushing: stomach/duodenum/oesophagus in intracranial disease; vagal acid hypersecretion; perforation-prone.

BCQs One-Best-Answer Practice

1 An antral biopsy shows neutrophils within pit epithelium, many plasma cells in the lamina propria and a lymphoid follicle with a germinal centre. The most likely cause is:

A. Helicobacter pylori
B. Autoimmune gastritis
C. NSAID gastropathy
D. Bile reflux
E. Cytomegalovirus

2 The H. pylori enzyme that forms the basis of the breath test is:

A. Catalase
B. Urease
C. Protease
D. Lipase
E. Phospholipase

3 A 60-year-old woman with vitiligo has megaloblastic anaemia and a raised serum gastrin. Gastric biopsy will most likely show:

A. Antral neutrophils with curved rods
B. Foveolar hyperplasia only
C. Loss of parietal cells in the body with antral sparing
D. Granulomas in the antrum
E. Signet-ring cells

4 A patient with 40% burns develops haematemesis on day 3. The lesion is most likely a:

A. Cushing ulcer in the stomach
B. Mallory–Weiss tear
C. Chronic gastric ulcer
D. Curling ulcer in the proximal duodenum
E. Dieulafoy lesion

5 The commonest complication of peptic ulcer disease is:

A. Perforation
B. Gastric outlet obstruction
C. Malignant change
D. Penetration into pancreas
E. Haemorrhage

6 The base of an active chronic peptic ulcer, from the lumen outward, shows:

A. Necrotic debris, neutrophils, granulation tissue, fibrous scar
B. Fibrous scar, granulation tissue, neutrophils, debris
C. Granulation tissue, debris, scar, neutrophils
D. Neutrophils, scar, debris, granulation tissue
E. Lymphoid follicles, debris, scar, mucosa

7 A gastric ulcer is most likely to be malignant if it has:

A. A smooth clean base
B. Folds radiating to its edge
C. Heaped-up, everted margins
D. A round punched-out shape
E. Location on the lesser curvature

8 A 35-year-old woman with a strong family history of gastric cancer has a rigid, thickened stomach full of signet-ring cells. The germline mutation is most likely in:

A. APC
B. CDH1
C. TP53
D. MLH1
E. KIT

9 A woman has bilateral solid ovarian masses composed of mucin-filled signet-ring cells. The most likely primary site is:

A. Breast
B. Colon
C. Endometrium
D. Stomach
E. Pancreas

10 Which feature defines early gastric cancer?

A. Size under 2 cm
B. Absence of lymph node metastasis
C. Intestinal histological type
D. Absence of ulceration
E. Confinement to mucosa and submucosa

11 A gastric MALT lymphoma that does not regress after H. pylori eradication most likely carries:

A. t(11;18) API2–MALT1
B. t(14;18) IGH–BCL2
C. t(8;14) MYC–IGH
D. t(9;22) BCR–ABL
E. t(11;14) CCND1–IGH

12 A gastric wall spindle-cell tumour is positive for CD117 and DOG1. The best targeted drug is:

A. Trastuzumab
B. Rituximab
C. Imatinib
D. Octreotide
E. Tamoxifen

13 Small gastric nodules showing uniform cells with salt-and-pepper chromatin, positive for chromogranin, arise in a patient with autoimmune gastritis. They are driven by:

A. H. pylori
B. KIT mutation
C. EBV
D. Hypergastrinaemia
E. CDH1 loss

14 A left supraclavicular lymph node enlarged by metastatic adenocarcinoma in a patient with weight loss is called:

A. Irish node
B. Sister Mary Joseph nodule
C. Krukenberg node
D. Blumer shelf
E. Virchow node

ANSWER KEY WITH ONE-LINE REASONS

1. **A** Active chronic antral gastritis with follicles = H. pylori.
2. **B** Urease splits labelled urea, releasing labelled CO2.
3. **C** Autoimmune gastritis destroys body/fundus parietal cells.
4. **D** Curling ulcers follow severe burns and affect the proximal duodenum.
5. **E** Bleeding occurs in 15–20%; perforation causes most deaths.
6. **A** Four zones from the surface: debris → exudate → granulation → scar.
7. **C** Raised, rolled margins and interrupted folds suggest carcinoma.
8. **B** Hereditary diffuse gastric cancer: E-cadherin (CDH1).
9. **D** Krukenberg tumour: classically metastatic diffuse gastric carcinoma.
10. **E** Depth alone defines it, regardless of nodal status.
11. **A** t(11;18) lymphomas are independent of H. pylori antigen drive.
12. **C** GIST: KIT tyrosine kinase inhibitor imatinib.
13. **D** Achlorhydria → high gastrin → ECL hyperplasia → NET.
14. **E** Virchow node may be the first sign of an occult gastric cancer.

VIVA Rapid-Fire Viva Voce

Q Name four gastric mucosal defences.	Mucus, bicarbonate, epithelial barrier with restitution, blood flow (all supported by prostaglandins).
Q How do NSAIDs cause ulcers?	COX inhibition lowers prostaglandins, so mucus, bicarbonate and blood flow fall; also direct injury.
Q What is an erosion?	Loss of surface epithelium that does not cross the muscularis mucosae.
Q Where do you find H. pylori in a biopsy?	In surface mucus over antral epithelium and in gastric pits; not invading tissue.
Q Which stains show H. pylori best?	Giemsa, Warthin–Starry silver, or immunohistochemistry.
Q Why is gastrin high in autoimmune gastritis?	Achlorhydria removes acid feedback, so antral G cells oversecrete gastrin.
Q What is the commonest site of peptic ulcer?	First part of the duodenum, anterior wall, within a few cm of the pylorus.

Q Which complication of peptic ulcer kills most patients?	Perforation (about two-thirds of ulcer deaths).
Q Define linitis plastica.	Diffuse-type gastric carcinoma making the wall thick, rigid and leather-bottle-like.
Q What is a signet-ring cell?	Tumour cell with intracytoplasmic mucin pushing the nucleus to the periphery.
Q What is a Sister Mary Joseph nodule?	Periumbilical subcutaneous metastasis of an abdominal or pelvic cancer, classically gastric.
Q Cell of origin of GIST?	Interstitial cells of Cajal; KIT (CD117) positive.
Q What is a lymphoepithelial lesion?	Neoplastic B cells infiltrating and destroying gastric glands: hallmark of MALT lymphoma.

OSPE Practice Stations

STATION 1 · SLIDE (LAB) — 2 minutes

TASK

Antral biopsy: lamina propria crowded with plasma cells and lymphocytes; a lymphoid follicle with a germinal centre; neutrophils in the pit epithelium; slender curved rods in the surface mucus. (a) Diagnosis. (b) Two features seen. (c) Name one non-invasive test and two long-term complications.

ANSWER

(a) Chronic active H. pylori gastritis. (b) Neutrophils in epithelium (pit abscesses), plasma-cell-rich lamina propria, lymphoid follicles, organisms in mucus. (c) Urea breath test or stool antigen; peptic ulcer, intestinal-type adenocarcinoma, MALT lymphoma.

STATION 2 · GROSS SPECIMEN — 1 minute

TASK

Partial gastrectomy: a 2 cm round, punched-out ulcer on the lesser curvature with flat margins, a clean base and mucosal folds radiating from it. (a) Diagnosis. (b) Two features favouring a benign lesion. (c) The commonest complication.

ANSWER

(a) Chronic peptic (gastric) ulcer. (b) Level, non-everted margins; clean base; radiating folds; round shape. (c) Haemorrhage.

STATION 3 · GROSS SPECIMEN + SLIDE — 2 minutes

TASK

A stomach with a uniformly thick, rigid, grey-white wall and flattened rugae. Microscopy shows single cells with clear mucin vacuoles pushing the nuclei aside, scattered in fibrous stroma. (a) Name the gross appearance and the cells. (b) Lauren type. (c) Name the ovarian metastasis it may cause.

ANSWER

(a) Linitis plastica; signet-ring cells. (b) Diffuse type (CDH1/E-cadherin loss). (c) Krukenberg tumour.

RAPID REVISION

The Night-Before Page

15 ONE-LINERS

01 Gastric injury = defence (mucus, HCO3, flow, PGs) losing to attack.

02 Erosion stops above muscularis mucosae; ulcer goes through it.

03 Curling: burns, duodenum. Cushing: head injury, vagal acid, perforates.

04 Stress ulcers: multiple, shallow, brown-black, no scar.

05 H. pylori: antrum, neutrophils in pits, plasma cells, follicles.

06 H. pylori tools: flagella, urease, adhesins, CagA, VacA.

07 Autoimmune gastritis: body, achlorhydria, high gastrin, B12 loss.

08 Peptic ulcer: duodenum 4× stomach; H. pylori or NSAIDs.

09 Ulcer base: debris → neutrophils → granulation → scar.

10 Bleeding commonest; perforation kills most; duodenal never malignant.

11 Gastric adenoma: carcinoma in up to 30%; fundic polyps with PPIs.

12 Intestinal type: glands, metaplasia precursor, older men.

13 Diffuse type: signet rings, CDH1, linitis plastica, younger.

14 Virchow node, Krukenberg tumour, Sister Mary Joseph nodule.

15 MALT: H. pylori, t(11;18); GIST: KIT, imatinib; NET: gastrin.

MEMORY HOOKS

- **Curling = Cooking** (burns); **Cushing = Cranium**
- **Type A = 4 As** (Autoimmune, Antibodies, Achlorhydria, Anaemia) in the body; **Type B = Bacterial** in the antrum
- **Bleed, Burst, Block, Burrow**: ulcer complications
- **Diffuse = Discohesive** (no E-cadherin)

TEST YOURSELF: FILL THE GAPS

1. Autoimmune gastritis antibodies target H^+/K^+-ATPase and _____.
2. Cushing ulcers result from _____ stimulation.
3. Perforation causes about _____ of ulcer deaths.
4. Early gastric cancer is limited to mucosa and _____.
5. GIST arises from interstitial cells of _____.
6. Gastric adenocarcinoma linked to EBV: about _____%.

Answers: intrinsic factor · vagal · two-thirds · submucosa · Cajal · 10

DRAW FROM MEMORY

A chronic peptic ulcer in section with its four zones and the scar replacing muscularis propria (Figure 2.4).

DRAW FROM MEMORY

The three outcomes of H. pylori infection: duodenal ulcer, intestinal-type carcinoma, MALT lymphoma (Figure 2.1).

SPACED REVISION ☐ Day 1 ☐ Day 3 ☐ Day 7 ☐ Day 14 ☐ Day 30

03

Malabsorption & Inflammatory Bowel Disease

The small intestine has the surface of a tennis court folded into a tube, and malabsorption begins whenever that surface is flattened, blocked or starved of enzymes. Inflammatory bowel disease is the opposite problem: a gut that cannot stop fighting its own bacteria.

BY THE END OF THIS CHAPTER YOU CAN

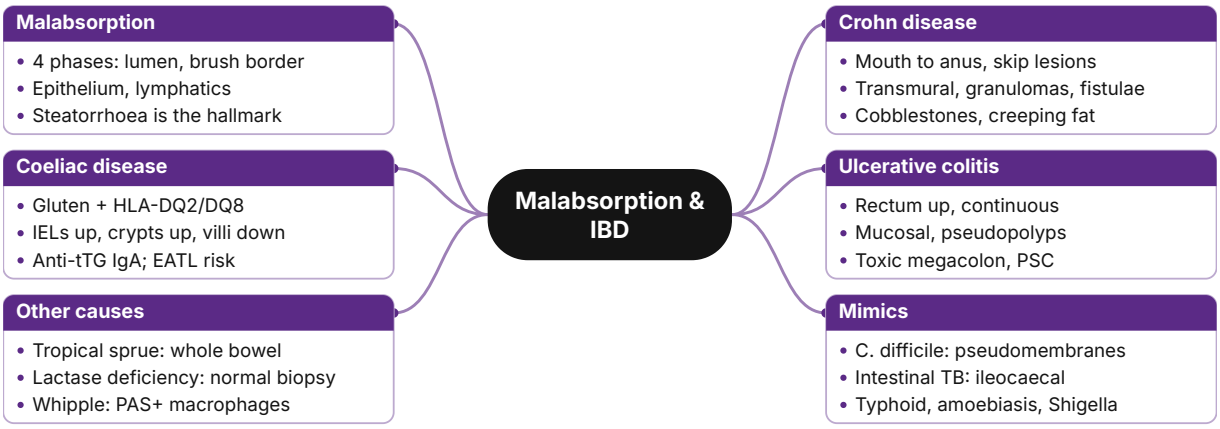
1. Classify malabsorption by the phase of absorption that fails and outline its investigation
2. Explain the immunopathogenesis, morphology, serology and complications of coeliac disease
3. Differentiate coeliac disease from tropical sprue, lactase deficiency, Whipple disease and abetalipoproteinaemia
4. Explain how genes, microbes, epithelium and T cells interact in inflammatory bowel disease
5. Compare Crohn disease and ulcerative colitis in gross, microscopic and clinical terms
6. State the risk factors for colitis-associated neoplasia
7. Separate IBD from infective, pseudomembranous and tuberculous colitis

EXAM RADAR

Theory SEQs	★★★
BCQs / MCQs	★★★
Viva voce	★★★
OSPE / slides	★★☆

Crohn vs ulcerative colitis is among the most-asked comparisons in the GIT paper; coeliac disease and the approach to malabsorption are frequent SEQs and BCQs.

ROBBINS LINK Robbins Basic Pathology — *Oral Cavity and Gastrointestinal Tract*: malabsorptive diarrhoea (coeliac disease, environmental/tropical enteropathy, lactase deficiency, abetalipoproteinaemia), infectious enterocolitis (pseudomembranous colitis, Whipple disease), inflammatory bowel disease, colitis-associated neoplasia.



Chapter at a glance. Six ideas carry the whole chapter.

3.1 Approach to malabsorption

DEFINITION

Malabsorption is defective absorption of fats, fat- and water-soluble vitamins, proteins, carbohydrates, electrolytes, minerals and water. It presents as **chronic diarrhoea**, and its hallmark is **steatorrhoea**: bulky, pale, greasy, foul-smelling stools that float. The commonest causes are **pancreatic insufficiency, coeliac disease and Crohn disease**.

CONCEPT CLEAR

Absorption is a four-step relay; malabsorption is a dropped baton. (1) **Intraluminal digestion**: pancreatic enzymes and bile salts break food down. (2) **Terminal digestion**: brush-border enzymes (disaccharidases, peptidases) finish the job. (3) **Transepithelial transport**: enterocytes take up the products and package fat into chylomicrons. (4) **Lymphatic transport**: lacteals carry chylomicrons away. Fat needs all four steps, so **fat is the first thing to escape** into the stool when any step fails.

WHICH PHASE FAILS? A WORKING CLASSIFICATION

Phase	Defect	Examples
Intraluminal digestion	No enzymes or bile salts	**Chronic pancreatitis**, cystic fibrosis; cholestasis, ileal resection, bacterial overgrowth (bile salts deconjugated)
Terminal (brush-border) digestion	Enzyme missing or surface flattened	**Lactase deficiency**; coeliac disease
Transepithelial transport	Enterocytes lost or cannot export fat	**Coeliac disease**, tropical sprue, Crohn disease, **abetalipoproteinaemia**
Lymphatic transport	Lacteals blocked	**Whipple disease**, lymphangiectasia, lymphoma

WHAT THE PATIENT LOSES, AND WHAT YOU SEE

- **Iron, folate, B12** → anaemia, glossitis, stomatitis
- **Vitamin K** → bruising, bleeding, long PT
- **Calcium, magnesium, vitamin D** → osteopenia, tetany
- **Vitamin A, B12** → night blindness, neuropathy
- **Protein, calories** → weight loss, wasting, oedema; amenorrhoea, infertility
- **Unabsorbed sugars** fermented → bloating, flatus, borborygmi

LAB VALUES

Faecal fat

> 7 g/day (or Sudan stain positive) = steatorrhoea

D-xylose test

low in mucosal disease; normal in pancreatic disease

Faecal elastase low in pancreatic insufficiency

Anti-tTG IgA

screens for coeliac disease (check total IgA)

Hydrogen breath test

rises after lactose in lactase deficiency

Duodenal biopsy the gold standard for mucosal disease

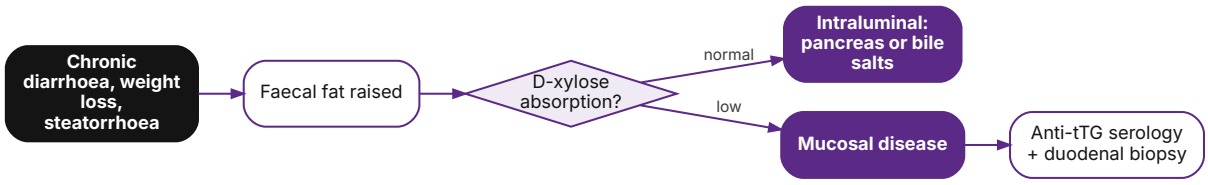


Figure 3.1 · Approach to a patient with suspected malabsorption.

3.2 Coeliac disease (gluten-sensitive enteropathy)

DEFINITION

Coeliac disease is an **immune-mediated enteropathy** triggered by eating **gluten**-containing cereals (**wheat, barley, rye**) in **genetically predisposed** people. It affects roughly 1% of Western populations and resolves when gluten is withdrawn.

Pathogenesis

Gluten's alcohol-soluble fraction, **gliadin**, contains a peptide that resists gastric, pancreatic and brush-border proteases. Two arms of immunity then attack the mucosa.

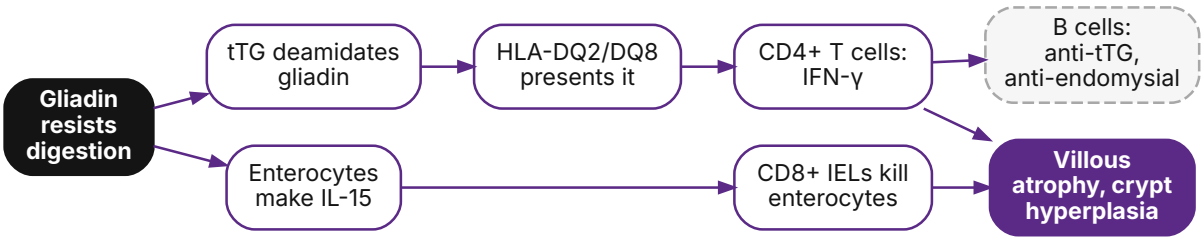


Figure 3.2 · Pathogenesis of coeliac disease. Adaptive CD4+ arm (top) and innate CD8+ arm (bottom) meet at the villus.

MOLECULAR KEY

Genes: almost every patient carries **HLA-DQ2** or **HLA-DQ8**, but so do about 40% of healthy people: the HLA type is **necessary but not sufficient**. Its absence practically rules coeliac disease out. **Associations:** type 1 diabetes, autoimmune thyroiditis, Sjögren syndrome, IgA nephropathy, Down and Turner syndromes, **selective IgA deficiency** (use IgG-based tests).

ENDOSCOPY

- Biopsy the **second part of duodenum** or proximal jejunum (most gluten exposure)
- Flattened mucosa, **scalloped** or reduced folds, mosaic pattern
- Changes are **proximal** and **reversible** on a gluten-free diet

MICROSCOPY

- **Intraepithelial lymphocytosis** (CD8+ T cells): the earliest change
- **Crypt hyperplasia** with increased mitoses
- **Villous atrophy**: partial to total flattening
- Lamina propria: more **plasma cells**, mast cells, eosinophils

MEMORY HOOK

Lymphs up, crypts up, villi down

The three microscopic features of coeliac disease in the order they appear. Gluten grains: **B**arley, **R**ye, **W**heat ("BRaW" bread).

CLINICAL FORMS

- **Children** (6–24 months, after gluten weaning): irritability, abdominal distension, chronic diarrhoea, **failure to thrive**, muscle wasting
- **Adults** (30–60 years): diarrhoea, bloating, fatigue, **iron-deficiency anaemia**; extra-intestinal: short stature, delayed puberty, aphthous ulcers, arthralgia, osteopenia
- **Silent**: positive serology + atrophy, no symptoms; **latent**: positive serology, no atrophy

- **Dermatitis herpetiformis** (about 10%): itchy blisters on extensor surfaces; IgA in dermal papillae; also responds to a gluten-free diet
- **Serology: IgA anti-tTG** (most sensitive), IgA/IgG anti-deamidated gliadin; **anti-endomysial** (most specific). Confirm by biopsy and response to diet
- **Complications: enteropathy-associated T-cell lymphoma** (EATL, the commonest coeliac-associated cancer, from IELs) and small-intestinal adenocarcinoma
- **Refractory coeliac disease:** symptoms and atrophy persist despite a strict diet; investigate for lymphoma

3.3 Other malabsorptive disorders

COELIAC DISEASE VS TROPICAL SPRUE

Feature	Coeliac disease	Tropical sprue (environmental enteropathy)
Who	Genetically predisposed (HLA-DQ2/DQ8); worldwide	Residents of or visitors to the **tropics** (South Asia, Caribbean); often after an acute diarrhoeal illness
Cause	Immune reaction to gluten	No single agent; bacterial overgrowth or infection suspected; poor sanitation
Distribution	Mainly **proximal** small bowel	**Whole small bowel**, including the ileum
Histology	IELs, crypt hyperplasia, villous atrophy	Similar but usually milder villous blunting and IELs

Feature	Coeliac disease	Tropical sprue (environmental enteropathy)
Key deficiency	Iron, folate	**Folate and B12** → megaloblastic anaemia
Treatment	**Gluten-free diet**	**Broad-spectrum antibiotics + folic acid**

Lactase (disaccharidase) deficiency

Lactase sits on the brush border. Undigested lactose stays in the lumen and holds water (**osmotic diarrhoea**); colonic bacteria ferment it to gas and acids, causing bloating, flatus and acidic stools after milk. The **biopsy is normal**; the defect is biochemical.

- **Congenital**: autosomal recessive, rare; explosive frothy diarrhoea with the first milk feeds
- **Acquired**: lactase gene switched off after childhood (common in Asian and African populations) or **transient after viral or bacterial enteritis**
- Diagnosis: **hydrogen breath test**; treatment: avoid lactose

Abetalipoproteinaemia

A rare autosomal recessive disease: mutation of **microsomal triglyceride transfer protein (MTTP)** means enterocytes cannot load lipid onto apolipoprotein B, so **no chylomicrons** leave the cell.

- Infancy: **failure to thrive**, diarrhoea, steatorrhoea
- Biopsy: enterocytes with **clear, lipid-vacuolated cytoplasm** (oil red O positive)
- Blood: **absent apoB lipoproteins**; **acanthocytes** (burr cells)
- Deficiency of fat-soluble vitamins (A, D, E, K): retinal and spinocerebellar degeneration

Whipple disease

A rare systemic infection by *Tropheryma whipplei*, a Gram-positive actinomycete, usually in middle-aged men. Bacteria-laden macrophages pack the lamina propria and block the lacteals, so absorbed fat cannot leave (a **lymphatic** cause of malabsorption).

GROSS

- Small-bowel mucosa thick and **shaggy**; yellow flecks
- Mesenteric nodes enlarged

MICROSCOPY

- Villi **expanded and blunted** by sheets of **foamy macrophages** in the lamina propria
- Macrophages full of **PAS-positive, diastase-resistant** granules (bacteria in lysosomes); rod-shaped bacilli on EM
- Dilated lacteals, fat droplets

Clinical: the triad of **diarrhoea (malabsorption), weight loss and arthralgia**, plus fever, lymphadenopathy, skin pigmentation, endocarditis and CNS disease (dementia). Similar macrophages fill joints, valves, nodes and brain. It responds to antibiotics.

MEMORY HOOK

Foamy Whipped cream in a CAN

Foamy PAS+ macrophages; **C**ardiac, **A**rthralgia, **N**eurological involvement.

EXAM TRAP

Whipple vs Mycobacterium avium complex (MAC) in AIDS: both fill the lamina propria with PAS-positive foamy macrophages. MAC bacilli are **acid-fast positive** (ZN); *T. whipplei* is **acid-fast negative**.

3.4 Inflammatory bowel disease

DEFINITION

Inflammatory bowel disease (IBD) is a chronic, relapsing condition caused by **inappropriate mucosal immune activation against the normal gut flora** in genetically susceptible people. Its two forms are **Crohn disease (CD)** and **ulcerative colitis (UC)**. Both usually begin in the teens and twenties; they are commonest in whites and in industrialised countries (the "hygiene hypothesis"), and rising wherever lifestyles westernise.

Pathogenesis

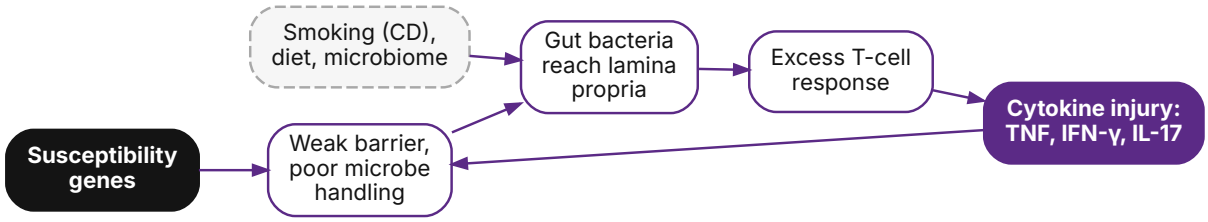


Figure 3.3 · Pathogenesis of IBD. A leaky barrier lets microbes in; an overeager immune response damages the barrier further, so the cycle sustains itself.

- **Genes (CD):** NOD2, an intracellular sensor of bacterial peptidoglycan; loss-of-function variants let microbes survive in the mucosa. ATG16L1 and IRGM impair **autophagy** and Paneth-cell defence
- **Immune response: TH1 and TH17** cells drive CD (anti-TNF drugs work); UC may involve TH2-type cytokines (IL-13). Defects of IL-10 or its receptor cause severe infant colitis
- **Epithelium:** leaky tight junctions; abnormal Paneth-cell antimicrobial granules
- **Smoking** raises the risk of CD but **lowers** that of UC; UC may first appear after stopping smoking

Crohn disease

Can affect **any part of the gut from mouth to anus**, most often the **terminal ileum, ileocaecal valve and caecum**. At diagnosis about 40% have small-bowel disease alone, 30% small bowel and colon, and the rest colon only. Lesions are separated by normal bowel (**skip lesions**) and inflammation is **transmural**.

GROSS

- Sharply demarcated **skip lesions**
- Earliest: **aphthous ulcers** → long **serpentine, linear ulcers** along the bowel axis
- **Cobblestone** mucosa: islands of oedematous mucosa between ulcers
- **Fissures** → **fistulae**, abscesses, perforation
- Thick, rubbery wall → **strictures** ("string sign" on barium)
- **Creeping fat**: mesenteric fat wraps the serosa

MICROSCOPY

- Neutrophils in crypts (**crypt abscesses**), crypt destruction, ulcers
- Chronicity: **distorted, branched crypts**; pyloric and Paneth-cell **metaplasia**
- **Transmural** inflammation with **lymphoid aggregates**
- **Non-caseating granulomas** in about 35% (any layer, nodes); their absence does not exclude CD
- Submucosal **fibrosis**, muscular hypertrophy

Clinical: intermittent mild diarrhoea, fever and abdominal pain; about a fifth present acutely with **right lower quadrant pain mimicking appendicitis**. Terminal ileal disease causes **B12 and bile-salt malabsorption** (gallstones, oxalate kidney stones), iron deficiency and hypoalbuminaemia. Fistulae run to bowel, bladder, vagina and **perianal skin**. Disease **recurs after resection**.

Ulcerative colitis

Confined to the **colon and rectum**. It **always involves the rectum** and spreads proximally in a **continuous** sheet (proctitis → left-sided colitis → pancolitis), and it is limited to the **mucosa and superficial submucosa**. Severe pancolitis may inflame the last few centimetres of ileum (**backwash ileitis**).

GROSS

- **Continuous** disease from the rectum; abrupt line where normal colon starts
- Red, **granular, friable** mucosa; **broad-based** ulcers
- **Pseudopolyps**: islands of regenerating mucosa; mucosal bridges
- Wall **not thickened**, serosa normal, **no strictures**
- Chronic: **flat, atrophic** mucosa without folds ("lead-pipe" colon)

MICROSCOPY

- **Crypt abscesses**, crypt distortion, Paneth-cell metaplasia
- Inflammation **limited to mucosa** and superficial submucosa
- **No granulomas**, no transmural inflammation
- Healed disease: mucosal atrophy, submucosal fibrosis

Clinical: relapsing attacks of **bloody diarrhoea with stringy mucus**, lower abdominal cramps relieved by defecation. **Toxic megacolon**: inflammation damages the muscularis propria and its nerves, the colon dilates and may perforate. **Colectomy cures the bowel disease** (but not all extra-intestinal disease).

CROHN DISEASE VS ULCERATIVE COLITIS: THE COMPARISON EVERY GIT PAPER ASKS

Feature	Crohn disease	Ulcerative colitis
Region	Ileum ± colon; mouth to anus	Colon only
Rectum	Sometimes involved	**Always** involved
Distribution	**Skip lesions**	**Continuous** (diffuse)
Depth of inflammation	**Transmural**	Mucosa and submucosa
Wall	**Thick**, strictures common	Thin; strictures rare
Ulcers	**Deep, knife-like**, linear; cobblestones	**Superficial, broad-based**
Pseudopolyps	Moderate	**Marked**
Fibrosis, serositis, lymphoid reaction	Marked	Mild or none
Granulomas	**Yes** (about 35%), non-caseating	No
Fistulae, perianal disease	**Yes**	No
Fat/vitamin malabsorption	Yes (ileal disease)	No
Toxic megacolon	No (rare)	**Yes**
Malignant potential	With colonic involvement	**Yes**
Recurrence after surgery	**Common**	No (colectomy cures)
Smoking	Increases risk	Appears protective

MEMORY HOOK

C-R-O-H-N

Cobblestones and creeping fat · **R**ecurs after resection (rectum often spared) · **O**bstruction from strictures · **H**oles right through the wall (transmural → fistulae) · **N**on-caseating granulomas. UC is simply **continuous from the rectum, and mucosal**.

EXTRA-INTESTINAL DISEASE (BOTH, BUT SOME FAVOUR ONE)

- Joints: **migratory polyarthritis**, sacroiliitis, ankylosing spondylitis
- Eyes: **uveitis**; skin: **erythema nodosum**, pyoderma gangrenosum; finger clubbing
- Liver: **primary sclerosing cholangitis**, commoner with **UC** (and p-ANCA)

Colitis-associated neoplasia

The most feared long-term complication of UC and colonic Crohn disease. Dysplasia arises in chronically inflamed mucosa, is often **flat and multifocal**, and tends to acquire TP53 mutations **early** (the reverse of the APC-first sporadic pathway; see Chapter 4).

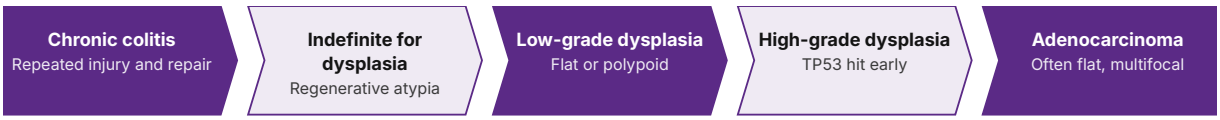


Figure 3.4 · Colitis-associated neoplasia. Surveillance colonoscopy with biopsies looks for the dysplasia steps.

CONCEPT CLEAR

Risk rises with four things: (1) **duration**: it climbs sharply 8–10 years after onset; (2) **extent**: pancolitis carries more risk than left-sided disease; (3) **severity** of active (neutrophilic) inflammation; and (4) coexisting **primary sclerosing cholangitis**. Hence surveillance colonoscopy starts about 8 years after diagnosis, or at diagnosis if PSC is present.

3.5 The mimics: infective, pseudomembranous and tuberculous colitis

CONCEPT CLEAR

Time is written in the crypts. Acute infective colitis inflames the mucosa but leaves the **crypt architecture intact**. IBD has been injuring and repairing the mucosa for months, so it shows **chronicity**: **branched, distorted crypts**, basal plasma cells and metaplasia. A biopsy can therefore separate a first attack of dysentery from IBD.

INFECTIVE COLITIS: ORGANISMS THAT IMITATE IBD

Organism	Site and lesion	Clue
Shigella	Left colon, rectum; mucosal ulcers, pseudomembranes	Bacillary dysentery; tiny infective dose; HUS with Shiga toxin
Salmonella typhi	Terminal ileum: **oval ulcers along the long axis over Peyer patches**	Typhoid fever; macrophage nodules; perforation in the 3rd week
Campylobacter, EHEC O157:H7	Colon; acute self-limited colitis	Guillain–Barré after Campylobacter; **HUS** after EHEC
Entamoeba histolytica	Caecum, ascending colon: **flask-shaped ulcers** (narrow neck, broad base)	Trophozoites with ingested red cells; amoebic liver abscess
Yersinia	Ileum, appendix, mesenteric nodes; granulomas	Mimics appendicitis and Crohn disease

Pseudomembranous colitis (*Clostridioides difficile*)

Antibiotics (classically third-generation cephalosporins, clindamycin, fluoroquinolones) wipe out the normal flora, and toxin-producing *C. difficile* overgrows. Its **toxins A and B** disrupt the epithelial cytoskeleton, causing barrier loss, cytokine release and cell death. Risk is highest in the elderly, hospitalised and immunosuppressed.

GROSS

- Colon studded with raised **yellow-white plaques** (pseudomembranes) that coalesce
- Mucosa between plaques is relatively normal early

MICROSCOPY

- Surface epithelium denuded; lamina propria full of **neutrophils**, fibrin thrombi in capillaries
- Damaged crypts distended by mucopurulent exudate that **erupts like a volcano** ("summit lesion") and spreads into a **pseudomembrane** of mucin, fibrin and neutrophils

Clinical: fever, leukocytosis, cramps, profuse **watery diarrhoea**, dehydration, hypoalbuminaemia. **Diagnosis:** detect **C. difficile toxin** (or toxin genes by PCR) in stool. **Treatment:** oral vancomycin or fidaxomicin (metronidazole in mild cases); faecal microbiota transplant for recurrence. Pseudomembranes are **not specific**: ischaemia and other necrotising infections can produce them.

Intestinal tuberculosis: the great Crohn mimic

Bacilli reach the gut in **swallowed sputum** from pulmonary TB, in **unpasteurised milk** (*M. bovis*) or by the blood. The **ileocaecal region** is the favourite site because of its abundant lymphoid tissue and relative stasis. Forms: **ulcerative**, **hypertrophic** (a fibrotic ileocaecal mass that mimics carcinoma) and **ulcerohypertrophic**. Ulcers follow the lymphatics, which run **around** the bowel, so they are **transverse (girdle) ulcers** that heal by fibrosis into **strictures**.

INTESTINAL TUBERCULOSIS VS CROHN DISEASE

Feature	Intestinal tuberculosis	Crohn disease
Ulcers	**Transverse, circumferential (girdle)**, undermined edges	**Longitudinal, linear**, serpentine; aphthous; cobblestones
Granulomas	**Large, confluent, caseating**, many; Langhans giant cells	**Small, discrete, non-caseating**, sparse
Organisms	**AFB on ZN stain** (often scanty); GeneXpert or culture positive	None
Mesenteric nodes	**Caseating**, may calcify	Reactive; non-caseating granulomas
Fistulae, perianal disease, skip lesions	Uncommon; short ileocaecal segment	**Common**
Other clues	Chest X-ray TB, ascites, fever, night sweats; responds to ATT	Extra-intestinal (uveitis, arthritis); relapsing course

PAKISTAN FOCUS

Pakistan is among the **top five high-TB-burden countries** (incidence about 266 per 100,000), and **abdominal tuberculosis is common**. Before labelling a young adult with ileocaecal ulcers as Crohn disease, **exclude TB** (biopsy for caseation and AFB, GeneXpert, chest X-ray): steroids and anti-TNF drugs given for presumed Crohn disease can make TB disseminate. Remember also **typhoid**, for which Pakistan faced an extensively drug-resistant (XDR) outbreak in Sindh from 2016 and became the first country to add typhoid conjugate vaccine to routine immunisation (2019). Typhoid ulcers are **oval and longitudinal** over Peyer patches; TB ulcers are **transverse**.

CLINICAL CASE

A 24-year-old man from Peshawar has 4 months of right lower abdominal pain, colicky episodes after meals, low-grade evening fever and weight loss. Colonoscopy: transverse ulcers and a narrowed ileocaecal valve. Biopsy: confluent granulomas with central caseation and Langhans giant cells; a few acid-fast bacilli.

Reasoning: ileocaecal disease fits both Crohn disease and TB, but **transverse ulcers + caseation + AFB** settle it: **ileocaecal tuberculosis** with early stricture. Start antituberculous therapy; steroids would have been dangerous.

EXAM ZONE Past-Paper Pattern SEQs — Solved

PROF PATTERN Differentiate between Crohn disease and ulcerative colitis under the headings of site, gross, microscopy and complications. **5 marks**

MODEL ANSWER PLAN

Table of at least 10 points: site (mouth to anus, terminal ileum vs colon with rectum always); distribution (skip vs continuous); depth (transmural vs mucosal); wall (thick, strictures vs thin); ulcers (deep linear, cobblestone vs broad superficial, pseudopolyps); granulomas (35% vs none); fistulae (yes vs no); creeping fat; malabsorption (yes vs no); toxic megacolon (no vs yes); cancer (colonic CD vs UC, higher); recurrence after surgery.

Score booster: end with two lines on colitis-associated neoplasia risk factors (duration, extent, activity, PSC).

MODULE PATTERN A 7-year-old girl has chronic diarrhoea, abdominal distension, short stature and iron-deficiency anaemia. IgA anti-tTG is strongly positive. **a)** Diagnosis? **b)** Explain the pathogenesis. **c)** Describe the duodenal biopsy findings. **1 + 2 + 2 marks**

MODEL ANSWER PLAN

a) Coeliac disease. **b)** Gliadin resists digestion → deamidated by tTG → presented on HLA-DQ2/DQ8 → CD4+ T cells (IFN-γ); epithelial IL-15 → CD8+ IELs (NKG2D) kill enterocytes; B cells make anti-tTG/endomysial antibodies. **c)** Intraepithelial lymphocytosis, crypt hyperplasia, villous atrophy, plasma cells in lamina propria; reversible on gluten-free diet.

Score booster: name the long-term risk (EATL) for an extra mark.

CLASSIC **a)** Classify malabsorption syndromes according to the phase of absorption affected. **b)** Write a short note on Whipple disease. **2 + 3 marks**

MODEL ANSWER PLAN

a) Intraluminal (chronic pancreatitis, cystic fibrosis, bile salt loss), terminal digestion (lactase deficiency), transepithelial (coeliac, tropical sprue, abetalipoproteinaemia), lymphatic (Whipple, lymphangiectasia). **b)** *T. whipplei*; foamy PAS+ diastase-resistant macrophages in lamina propria, blocked lacteals; diarrhoea, weight loss, arthralgia; heart, CNS; acid-fast negative (vs MAC); antibiotics.

PROF PATTERN **a)** Describe the pathogenesis and morphology of pseudomembranous colitis. **b)** How would you differentiate intestinal tuberculosis from Crohn disease? **2 + 3 marks**

MODEL ANSWER PLAN

a) Antibiotics → loss of flora → *C. difficile* toxins A/B → epithelial injury; yellow plaques; volcano-like crypt eruptions of neutrophils, mucin, fibrin; stool toxin test. **b)** Table: transverse vs longitudinal ulcers; caseating confluent vs small non-caseating granulomas; AFB/GeneXpert; caseous nodes; fistulae and skip lesions in Crohn; response to ATT.

BCQs One-Best-Answer Practice

1 A 30-year-old woman has bulky, foul-smelling floating stools. Faecal fat is high; the D-xylose test is normal. The defect most likely lies in:

A. Intraluminal digestion (pancreas or bile)
B. Brush-border lactase
C. Enterocyte chylomicron assembly
D. Lacteal drainage
E. Ileal B12 absorption

2 The HLA alleles almost always present in coeliac disease are:

A. HLA-B27 and HLA-B51
B. HLA-DQ2 and HLA-DQ8
C. HLA-DR3 and HLA-DR4 only
D. HLA-A3 and HLA-B14
E. HLA-DR2 and HLA-DR15

3 Which enzyme deamidates gliadin, increasing its affinity for HLA molecules?

A. Lactase
B. Trypsin
C. Tissue transglutaminase
D. Enterokinase
E. Pepsin

4 The earliest histological change in coeliac disease is:

A. Total villous atrophy
B. Crypt abscesses
C. Non-caseating granulomas
D. Intraepithelial lymphocytosis
E. Foamy macrophages

5 The malignancy most characteristic of long-standing coeliac disease is:

A. MALT lymphoma
B. Burkitt lymphoma
C. Gastric adenocarcinoma
D. Carcinoid tumour
E. Enteropathy-associated T-cell lymphoma

6 A small-bowel biopsy shows lamina propria packed with foamy macrophages containing PAS-positive, diastase-resistant granules that are acid-fast negative. The patient has arthralgia and weight loss. Diagnosis?

A. Mycobacterium avium complex
B. Abetalipoproteinaemia
C. Whipple disease
D. Coeliac disease
E. Tropical sprue

7 An infant with steatorrhoea has enterocytes with clear lipid vacuoles and acanthocytes in the blood. The defective protein is:

A. Microsomal triglyceride transfer protein
B. Lactase
C. CFTR
D. Tissue transglutaminase
E. NOD2

8 A man who returned from a long stay in the tropics has chronic diarrhoea and megaloblastic anaemia. Biopsies show villous blunting in jejunum and ileum. The best treatment is:

A. Gluten-free diet
B. Broad-spectrum antibiotic plus folic acid
C. Pancreatic enzyme supplements
D. Corticosteroids
E. Lactose-free diet

9 Which feature favours Crohn disease over ulcerative colitis?

A. Continuous involvement from the rectum
B. Toxic megacolon
C. Inflammation limited to the mucosa
D. Transmural inflammation with fistulae
E. Marked pseudopolyps

10 A colectomy specimen shows continuous disease from the rectum to the splenic flexure with broad shallow ulcers and pseudopolyps; the wall is thin. Which is true of this condition?

A. Granulomas are seen in about a third
B. Smoking is a strong risk factor
C. It recurs in the remaining bowel after colectomy
D. Perianal fistulae are typical
E. It is associated with primary sclerosing cholangitis

11 A gene whose loss-of-function variants impair sensing of bacterial peptidoglycan and predispose to Crohn disease is:

A. APC
B. NOD2
C. CDH1
D. HLA-B27
E. MTTP

12 After a course of cephalosporins, an elderly inpatient has watery diarrhoea and fever. Colonoscopy shows yellow-white plaques. Biopsy will most likely show:

A. Flask-shaped ulcers with trophozoites
B. Caseating granulomas
C. Volcano-like eruption of neutrophils, mucin and fibrin from damaged crypts
D. Transmural lymphoid aggregates
E. Foamy macrophages in the lamina propria

13 Ileocaecal ulcers that run transversely around the bowel, with confluent caseating granulomas, suggest:

A. Typhoid fever
B. Crohn disease
C. Amoebiasis
D. Yersiniosis
E. Intestinal tuberculosis

14 The risk of colitis-associated carcinoma is LEAST influenced by:

A. Duration of colitis
B. Extent of colitis
C. Coexisting primary sclerosing cholangitis
D. Patient's blood group
E. Severity of active inflammation

ANSWER KEY WITH ONE-LINE REASONS

1. **A** Normal D-xylose means the mucosa absorbs; the problem is digestion in the lumen.
2. **B** DQ2/DQ8 present gliadin to CD4+ T cells; absence virtually excludes the disease.
3. **C** tTG is also the target of the most sensitive serological test.
4. **D** IEL increase precedes crypt hyperplasia and villous atrophy.
5. **E** EATL arises from intraepithelial T lymphocytes.
6. **C** T. whipplei; MAC would be acid-fast positive.
7. **A** Abetalipoproteinaemia: MTTP mutation, no apoB lipoproteins.
8. **B** Tropical sprue involves the whole small bowel and responds to antibiotics + folate.
9. **D** Transmural disease causes fissures, fistulae and strictures.
10. **E** Ulcerative colitis: PSC association; colectomy cures the bowel disease.
11. **B** NOD2 activates NF-κB in response to muramyl dipeptide.
12. **C** C. difficile pseudomembranous colitis.
13. **E** TB ulcers follow circumferential lymphatics; typhoid ulcers are longitudinal.
14. **D** Duration, extent, activity and PSC drive the risk.

VIVA Rapid-Fire Viva Voce

Q Define steatorrhoea.	Excess fat in stool (over about 7 g/day): pale, bulky, greasy, foul, floating stools.
Q Name the four phases of absorption.	Intraluminal digestion, terminal (brush-border) digestion, transepithelial transport, lymphatic transport.
Q Which cereals contain toxic gluten?	Wheat, barley and rye.
Q Most sensitive serological test for coeliac disease?	IgA anti-tissue transglutaminase (check total IgA first).
Q Where is the biopsy taken in coeliac disease?	Second part of duodenum or proximal jejunum.
Q Skin disease associated with coeliac disease?	Dermatitis herpetiformis.
Q Why is the biopsy normal in lactase deficiency?	The defect is biochemical in the brush-border enzyme, not structural.
Q Causative organism of Whipple disease?	Tropheryma whipplei, a Gram-positive actinomycete.

Q What is creeping fat?	Mesenteric fat wrapping around the serosa of bowel affected by Crohn disease.
Q What is a pseudopolyp?	An island of regenerating mucosa between ulcers, typical of ulcerative colitis.
Q What is toxic megacolon?	Acute colonic dilatation in severe colitis from muscle and nerve damage; risk of perforation.
Q Which ulcer is longitudinal: typhoid or TB?	Typhoid (over Peyer patches); TB ulcers are transverse.
Q When does surveillance colonoscopy start in UC?	About 8 years after onset; at diagnosis if PSC coexists.

OSPE Practice Stations

STATION 1 · SLIDE — 2 minutes

TASK

Duodenal biopsy: flat mucosa with no villi, elongated hyperplastic crypts with many mitoses, numerous lymphocytes within the surface epithelium and plasma cells in the lamina propria. (a) Diagnosis. (b) Three features. (c) Two serological tests.

ANSWER

(a) Coeliac disease. (b) Villous atrophy, crypt hyperplasia, intraepithelial lymphocytosis. (c) IgA anti-tTG; anti-endomysial antibody (or anti-deamidated gliadin).

STATION 2 · GROSS SPECIMENS — 2 minutes

TASK

Specimen A: segment of terminal ileum with a thick, rubbery wall, narrowed lumen, cobblestone mucosa and fat wrapping the serosa. Specimen B: colon with continuous red granular mucosa from the rectum upward and numerous small polypoid projections. (a) Identify A and B. (b) One complication specific to each.

ANSWER

(a) A: Crohn disease (stricture, cobblestones, creeping fat). B: ulcerative colitis with pseudopolyps. (b) A: fistula or stricture with obstruction; B: toxic megacolon or colitis-associated carcinoma.

STATION 3 · SLIDE — 2 minutes

TASK

Ileocaecal biopsy from a 22-year-old with fever and weight loss: large confluent granulomas with central amorphous pink necrosis, epithelioid cells and Langhans giant cells. (a) Diagnosis. (b) Special stain to confirm. (c) One disease it must be distinguished from, and why the distinction matters.

ANSWER

(a) Intestinal tuberculosis (caseating granulomas). (b) Ziehl–Neelsen for acid-fast bacilli (or GeneXpert). (c) Crohn disease; steroids or anti-TNF therapy would disseminate TB.

RAPID REVISION The Night-Before Page

15 ONE-LINERS

01 Malabsorption hallmark: steatorrhoea (faecal fat > 7 g/day).

02 Four phases: lumen, brush border, enterocyte, lymphatics.

03 Normal D-xylose = intraluminal cause (pancreas, bile).

04 Coeliac: gluten + HLA-DQ2/DQ8; tTG deamidates gliadin.

05 Coeliac biopsy: IELs up, crypts up, villi down; proximal.

06 Anti-tTG IgA most sensitive; anti-endomysial most specific.

07 Coeliac cancers: EATL, small-bowel adenocarcinoma.

08 Tropical sprue: whole small bowel; B12 + folate; antibiotics.

09 Lactase deficiency: osmotic diarrhoea, normal biopsy, breath test.

10 Abetalipoproteinaemia: MTTP, vacuolated enterocytes, acanthocytes.

11 Whipple: PAS+ foamy macrophages; diarrhoea, weight loss, arthralgia.

12 Crohn: skip, transmural, granulomas, fistulae, creeping fat.

13 UC: rectum, continuous, mucosal, pseudopolyps, toxic megacolon.

14 Colitis cancer risk: duration, extent, activity, PSC.

15 TB ulcers transverse, caseating; typhoid ulcers longitudinal.

MEMORY HOOKS

- **Lymphs up, crypts up, villi down**: coeliac biopsy
- **BRaW**: Barley, Rye, Wheat
- **Foamy Whipped cream in a CAN**: Cardiac, Arthralgia, Neurological
- **C-R-O-H-N**: Cobblestones, Recurs, Obstruction, Holes (fistulae), Non-caseating granulomas

TEST YOURSELF: FILL THE GAPS

1. Coeliac biopsy site: _____ part of duodenum.
2. Granulomas occur in about _____% of Crohn disease.
3. Whipple macrophages are PAS-positive, diastase-_____.
4. UC always involves the _____.
5. C. difficile pseudomembranes erupt like a _____.
6. NOD2 senses bacterial _____.

Answers: second · 35 · resistant · rectum · volcano · peptidoglycan

DRAW FROM MEMORY

The colon and ileum twice: shade Crohn disease (skip, ileocaecal, transmural) and ulcerative colitis (continuous from rectum, mucosal).

DRAW FROM MEMORY

The coeliac pathogenesis chain from gliadin to villous atrophy, with both the CD4+ and CD8+ arms (Figure 3.2).

SPACED REVISION ☐ Day 1 ☐ Day 3 ☐ Day 7 ☐ Day 14 ☐ Day 30

04

Intestinal Polyps & Colorectal Carcinoma

Most colorectal cancers start life as a small, harmless-looking polyp and take ten years or more to turn malignant. That long, visible runway is why colonoscopy saves lives, and why examiners never tire of the adenoma–carcinoma sequence.

BY THE END OF THIS CHAPTER YOU CAN

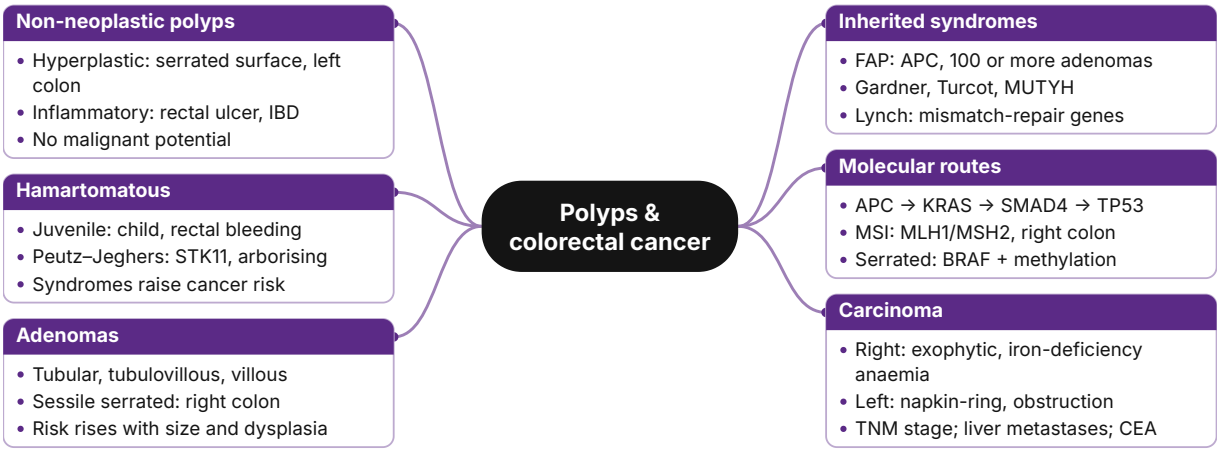
1. Classify intestinal polyps into non-neoplastic, hamartomatous and neoplastic types
2. Describe the morphology of hyperplastic polyps, adenomas and sessile serrated lesions and list the features that predict malignant change
3. Compare the inherited polyposis syndromes (FAP, Gardner, Turcot, Peutz–Jeghers, juvenile polyposis) with Lynch syndrome
4. Explain the APC/β-catenin and mismatch-repair (MSI) pathways of colorectal carcinogenesis
5. Describe the gross and microscopic morphology of colorectal carcinoma and contrast right- with left-sided tumours
6. Stage colorectal carcinoma (TNM, Dukes) and explain its spread, prognosis and the use of CEA

EXAM RADAR

Theory SEQs	★★★
BCQs / MCQs	★★★
Viva voce	★★☆
OSPE / slides	★★☆

Adenoma–carcinoma sequence, FAP vs Lynch and right- vs left-sided carcinoma are classic long questions; polyp types and their genes are favourite BCQs.

ROBBINS LINK Robbins Basic Pathology — *Oral Cavity and Gastrointestinal Tract*: colonic polyps, hamartomatous and polyposis syndromes, colorectal adenocarcinoma.



Chapter at a glance. From a harmless bump to a staged cancer.

4.1 Polyps: what they are and how to classify them

A polyp is any mass that projects above the mucosal surface into the lumen. It may be **sessile** (broad-based, no stalk) or **pedunculated** (on a stalk). "Polyp" describes a **shape**, not a disease: the pathologist must decide whether it is reactive, hamartomatous or neoplastic, because only the neoplastic ones are on the road to cancer.

CLASSIFICATION OF INTESTINAL POLYPS

Group	Types	Cancer risk
Non-neoplastic	Hyperplastic polyp; inflammatory polyp (solitary rectal ulcer syndrome, pseudopolyps of IBD)	None
Hamartomatous	Juvenile polyp; Peutz–Jeghers polyp; Cowden-type polyps	Sporadic: none. Syndromic: **raised** (in and outside the gut)
Neoplastic: conventional adenomas	Tubular, tubulovillous, villous	**Yes**: precursor of most colorectal cancers

Group	Types	Cancer risk
Neoplastic: serrated	Sessile serrated lesion (adenoma); traditional serrated adenoma	**Yes**: via the methylation/MSI route

Hyperplastic polyps

Common in the **6th–7th decades**; **small (< 5 mm)**, often multiple, mostly in the **left colon and rectum**. **Delayed shedding** of surface cells makes them crowd and fold, so the glands look **serrated (saw-tooth, star-shaped)**, but only in the **upper crypt**. No dysplasia and no meaningful malignant potential; the look-alike to exclude is the sessile serrated lesion.

Inflammatory polyps

In **solitary rectal ulcer syndrome** a poorly relaxing anorectal sphincter makes the rectal mucosa prolapse, ulcerate and heal repeatedly, producing a polyp of inflamed, regenerating mucosa with **fibromuscular hyperplasia of the lamina propria**. IBD pseudopolyps: see Chapter 3.

Hamartomatous polyps

A hamartoma is a disorganised overgrowth of tissues native to the site. Sporadic juvenile polyps are harmless, but the **polyposis syndromes** that produce hamartomas carry a real cancer risk.

JUVENILE POLYP VS PEUTZ–JEGHERS POLYP

Feature	Juvenile polyp	Peutz–Jeghers polyp
Patient	Child **under 5 years** with **rectal bleeding**; may prolapse or auto-amputate	Median age about 11; **mucocutaneous pigmentation** of lips, buccal mucosa, palms, perianal skin
Syndrome, gene	Juvenile polyposis: AD, **SMAD4** or **BMPR1A** (TGF-β pathway)	Peutz–Jeghers syndrome: AD, **STK11 (LKB1)**
Gross	Rectum; smooth, red, pedunculated, < 3 cm; **cystic** cut surface	Mainly **small intestine**; large, pedunculated, **lobulated**
Microscopy	**Cystically dilated glands** full of mucin and debris in an oedematous, inflamed lamina propria	**Arborising (tree-like) bands of smooth muscle** carrying normal-looking glands
Risk	Sporadic: none. Polyposis: colonic adenocarcinoma	**Intussusception**; cancers of colon, pancreas, breast, lung, ovary, uterus, testis

4.2 Adenomas: the true precursors

DEFINITION

Adenoma is a benign neoplastic polyp of glandular epithelium whose defining feature is **epithelial dysplasia** (intraepithelial neoplasia). Colonic adenomas are the precursor of most colorectal adenocarcinomas; about 30% of adults in Western countries have at least one by the age of 60.

CONCEPT CLEAR

Why dysplasia starts at the surface. In normal crypts, cells divide at the base and mature as they climb, stopping division before they reach the surface. In an adenoma the cells **fail to mature** as they rise, so the surface is lined by cells that still look like crypt-base stem cells: tall, crowded, with **enlarged, elongated, hyperchromatic, stratified ("pencil") nuclei** and little mucin. That failed maturation is what the pathologist calls dysplasia.

HISTOLOGICAL TYPES OF CONVENTIONAL ADENOMA

Type	Architecture	Typical gross form
Tubular (commonest)	Small rounded or tubular glands (villous part < 25%)	Small, **pedunculated**
Tubulovillous	Mixture (villous 25–75%)	Intermediate

Type	Architecture	Typical gross form
Villous	Long, slender, **finger-like villi** (villous > 75%)	Large, **sessile**, velvety; invasive foci more often (they are larger)

FEATURES THAT PREDICT MALIGNANT CHANGE

- **Size**: the most important. Cancer is very rare in adenomas < 1 cm; nearly 40% of those > 4 cm contain carcinoma
- **High-grade dysplasia**
- Villous architecture (largely a reflection of size)
- Number of adenomas (multiple = higher risk)

CLINICAL FEATURES

- Most are silent, found at screening colonoscopy
- **Occult bleeding** → iron-deficiency anaemia
- Rarely, a large **villous adenoma** secretes mucus rich in protein and potassium → **hypoproteinaemia, hypokalaemia**

Sessile serrated lesion (sessile serrated adenoma)

Looks like a hyperplastic polyp but behaves like an adenoma. Mostly in the **right colon**, flat and easily missed, and usually **without classic cytological dysplasia**. The clue is architectural: **serration runs down to the crypt base**, where crypts dilate and grow sideways (**boot- or L-shaped**). Malignant potential is similar to adenomas, via **BRAF** mutation and DNA methylation.

HYPERPLASTIC POLYP VS SESSILE SERRATED LESION VS TUBULAR ADENOMA

Feature	Hyperplastic polyp	Sessile serrated lesion	Tubular adenoma
Site	Left colon, rectum	**Right colon**	Anywhere
Serration	**Upper crypt** only	**Whole crypt**, dilated base	None
Dysplasia	None	Usually absent	**Present** (defining)
Main gene	—	BRAF, CpG methylation	APC
Cancer risk	None	**Yes**	**Yes**

> ⚠ **EXAM TRAP**
>
> **Intramucosal carcinoma and pseudo-invasion.** Dysplastic cells confined to the lamina propria or muscularis mucosae (**intramucosal carcinoma**) almost never metastasise, because the mucosa has few lymphatics; invasion counts only once the **submucosa** is reached. Glands pushed into the stalk of a twisted polyp, with haemorrhage and haemosiderin, are **pseudo-invasion**, not carcinoma.

4.3 Familial polyposis and cancer syndromes

Two syndromes illustrate the two molecular routes to cancer: **FAP** (loss of a tumour suppressor, APC) and **Lynch syndrome** (loss of DNA mismatch repair).

Familial adenomatous polyposis (FAP) and its variants

Autosomal dominant germline mutation of APC (chromosome **5q21**). Hundreds to thousands of adenomas carpet the colon from adolescence; **at least 100** are needed for a diagnosis of classic FAP. Each polyp looks like an ordinary tubular adenoma, but with so many, **colorectal adenocarcinoma develops in 100% of untreated patients**, often before 30 years. Hence **prophylactic colectomy**. Other lesions: gastric and **duodenal (ampullary) adenomas**, fundic gland polyps and congenital hypertrophy of the retinal pigment epithelium (CHRPE).

VARIANTS OF ADENOMATOUS POLYPOSIS

Syndrome	Gene / inheritance	Distinguishing features
Gardner syndrome	APC; AD	FAP + **osteomas** (mandible, skull, long bones), epidermoid cysts, **desmoid tumours**, thyroid tumours, dental abnormalities (unerupted or extra teeth)
Turcot syndrome	APC (about two-thirds) or mismatch-repair genes	Adenomas + **CNS tumours**: **medulloblastoma** with APC; **glioblastoma** with mismatch-repair mutations

Syndrome	Gene / inheritance	Distinguishing features
Attenuated FAP	APC (mutations at the ends of the gene); AD	Fewer polyps (about 30), later onset
MUTYH-associated polyposis	MUTYH (base-excision repair); **autosomal recessive**	Resembles attenuated FAP; no affected parent

Lynch syndrome (hereditary non-polyposis colorectal cancer, HNPCC)

Autosomal dominant germline mutation of a **DNA mismatch-repair gene**: most often MSH2 or MLH1, less often MSH6 or PMS2. It accounts for about **2–4%** of colorectal cancers. Patients do not get hundreds of polyps ("non-polyposis"); instead, their few adenomas progress to cancer unusually fast. Cancers appear at a **younger age** than sporadic cancer, favour the **right colon** and are often **mucinous**. Extracolonic cancers: **endometrium (commonest)**, ovary, stomach, small bowel, hepatobiliary tract, **ureter and renal pelvis**, brain and skin (sebaceous tumours).

FAP VS LYNCH SYNDROME: A FAVOURITE "DIFFERENTIATE" QUESTION

Feature	FAP	Lynch syndrome (HNPCC)
Gene	**APC** (5q21)	**MSH2, MLH1**, MSH6, PMS2
Gene role	Tumour suppressor: the "**gatekeeper**"	DNA repair: the "**caretaker**"
Inheritance	Autosomal dominant	Autosomal dominant
Polyps	**≥ 100**, from adolescence	Few adenomas
Cancer risk	**100%** if untreated, often < 30 years	High, at a younger age than sporadic cancer
Site and type	Throughout colon; usual adenocarcinoma	**Right colon**; mucinous, lymphocyte-rich
Molecular pathway	Chromosomal instability	**Microsatellite instability**
Extracolonic	Duodenal adenoma, osteomas, desmoids, CHRPE	**Endometrium**, ovary, stomach, urinary tract
Prevention	Prophylactic colectomy	Frequent colonoscopy from early adulthood

MEMORY HOOK

Lynch "3–2–1"

3 relatives with a Lynch-type cancer, **2** successive generations, **1** diagnosed before 50 (FAP excluded). And: **Lynch hits the Lining of the uterus**.

4.4 How colorectal cancer develops

The APC/β-catenin pathway (adenoma–carcinoma sequence)

About **80%** of sporadic cancers follow this chromosomal-instability route. It starts with loss of APC and proceeds by the stepwise addition of mutations, which we can see under the microscope as normal mucosa → adenoma → carcinoma.

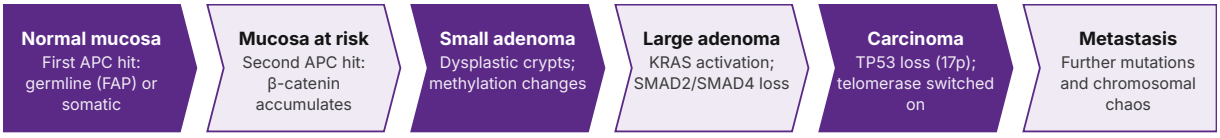


Figure 4.1 · The adenoma–carcinoma sequence. Each morphological step is matched by a genetic hit; the order matters more than any single gene.

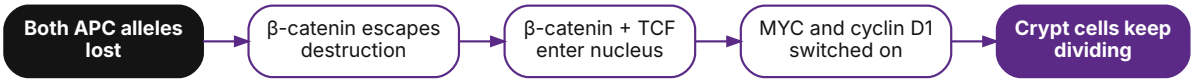


Figure 4.2 · Why losing APC makes crypts grow. APC normally destroys β-catenin, the messenger of Wnt signalling.

MOLECULAR KEY

APC (5q21): gatekeeper in the β-catenin destruction complex. **KRAS**: mutated in < 10% of adenomas under 1 cm but about half of larger ones; keeps growth signals on. **SMAD2/4** (18q): carry growth-inhibitory TGF-β signals. **TP53** (17p): lost in 70–80% of cancers but rarely in adenomas, so it marks invasion.

The mismatch-repair (MSI) and serrated pathways

Mismatch-repair enzymes proofread DNA after replication. When they fail, errors accumulate fastest in microsatellites (short tandem repeats such as CACACA), which lengthen or shorten: **microsatellite instability (MSI)**. Genes that contain such repeats in their coding sequence, such as **TGFBR2** (TGF-β receptor) and **BAX** (pro-apoptotic), are then knocked out. The defect is germline in Lynch syndrome, but in sporadic cancers it arises from **methylation of the MLH1 promoter**, usually in a sessile serrated lesion carrying a **BRAF** mutation (the CpG island methylator phenotype, CIMP).

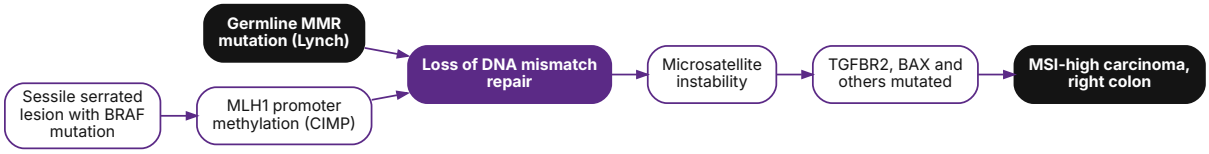


Figure 4.3 · Two entries into the mismatch-repair pathway.

CHROMOSOMAL INSTABILITY VS MICROSATELLITE INSTABILITY PATHWAYS

Feature	APC/β-catenin (CIN)	Mismatch repair (MSI)
Share of sporadic cancers	About 80%	About 10–15%
Precursor	Tubular / villous adenoma	Sessile serrated lesion (sporadic)
Key genes	APC, KRAS, SMAD4, TP53	MLH1, MSH2; **BRAF** only in sporadic cases (Lynch tumours lack it)
Site	Mostly **left** colon	Mostly **right** colon
Chromosomes	Aneuploid	Near-diploid
Histology	Usual gland-forming adenocarcinoma	**Mucinous**, poorly differentiated, many tumour-infiltrating lymphocytes
Prognosis (stage for stage)	Usual	**Better**

4.5 Colorectal adenocarcinoma

Adenocarcinoma is the **commonest malignancy of the gastrointestinal tract**. Incidence peaks at **60–70 years**; fewer than 20% of cases occur before 50. A cancer in a young patient should make you think of FAP, Lynch syndrome or long-standing colitis.

Risk factors: diet low in fibre and high in refined carbohydrate, fat and red or processed meat; obesity, physical inactivity, smoking; adenomas, inherited syndromes and inflammatory bowel disease (see Chapter 3). **Protective:** aspirin and other NSAIDs, which inhibit **COX-2**, an enzyme over-expressed in most adenomas and cancers.

Morphology

Tumours occur along the whole colon. The **shape depends on the site**: the wide caecum and ascending colon carry liquid stool, so tumours there grow outwards into the lumen; the narrow distal colon carries solid stool, so an encircling tumour soon obstructs.

GROSS

- **Proximal (right)**: **polypoid, exophytic, fungating** masses along one wall; ulcerate and bleed; rarely obstruct
- **Distal (left)**: **annular, encircling** growth with heaped-up edges: "**napkin-ring**" constriction ("apple-core" on barium enema)
- Cut surface grey-white, firm; ulcerated centre

MICROSCOPY

- **Tall columnar cells** resembling adenoma epithelium, forming irregular, back-to-back glands
- Invasion through the muscularis mucosae into the wall with a strong **desmoplastic** stroma (makes the tumour firm)
- Gland lumens contain necrotic "**dirty necrosis**"
- Variants: **mucinous** (> 50% extracellular mucin), **signet-ring cell**: both poor prognosis

RIGHT-SIDED VS LEFT-SIDED COLORECTAL CARCINOMA

Feature	Right (caecum, ascending)	Left (descending, sigmoid, rectum)
Gross form	**Exophytic, polypoid**	**Annular, napkin-ring**
Presentation	**Fatigue, weakness, iron-deficiency anaemia**; mass in right iliac fossa	**Change in bowel habit**, obstruction, bright-red bleeding; tenesmus (rectum)
Obstruction	Rare, late	**Common**
Molecular link	More often MSI / serrated; mucinous	More often CIN pathway

CLINICAL CORRELATION

Two bedside rules. (1) An older man or post-menopausal woman with **iron-deficiency anaemia** has GI cancer until proven otherwise. (2) ***Streptococcus gallolyticus* (*S. bovis*)** bacteraemia or endocarditis is linked to colorectal cancer: do a colonoscopy.

PAKISTAN FOCUS

Ileocaecal **tuberculosis** is common in Pakistan and can present as a right iliac fossa mass with weight loss and anaemia, mimicking caecal carcinoma (and Crohn disease). Biopsy settles it: caseating granulomas in TB, invasive dysplastic glands in carcinoma.

Spread, staging and prognosis

ROUTES OF SPREAD

- **Direct**: through the wall into pericolic fat, bladder, uterus, vagina
- **Lymphatic**: regional (pericolic, mesenteric) nodes
- **Haematogenous**: portal vein → **liver** (commonest distant site); then lungs and bone. Low rectal tumours drain to systemic veins and can reach the lungs directly
- **Transcoelomic**: peritoneal seeding

PROGNOSTIC FACTORS

- The **two most important**: **depth of invasion** and **lymph node metastasis**
- Distant metastasis
- Poor grade, mucinous or signet-ring type; lymphovascular invasion; involved margins
- MSI-high tumours do better stage for stage
- Most stage I patients are cured; few with distant metastases survive 5 years

TNM STAGING OF COLORECTAL CARCINOMA, WITH THE DUKES EQUIVALENT (FOR VIVA)

Stage	TNM	Meaning	Dukes
0	Tis N0 M0	Carcinoma in situ / intramucosal (lamina propria only)	—
I	T1–T2 N0 M0	T1 invades **submucosa**; T2 invades **muscularis propria**	**A**: confined to the bowel wall
II	T3–T4 N0 M0	T3 into pericolic tissue; T4a through visceral peritoneum; T4b into other organs	**B**: through the wall, nodes negative
III	Any T, N1–N2, M0	N1: 1–3 regional nodes; N2: 4 or more	**C**: lymph nodes involved
IV	Any T, any N, M1	Distant metastasis (liver, lung, peritoneum)	**D**: distant spread (later addition)

LAB VALUES

CEA glycoprotein made by many CRCs

Use baseline; **follow-up for recurrence**

Not for screening

insensitive; raised in smokers, IBD, cirrhosis

Screening FOBT/FIT, colonoscopy from 45–50 y

LAB SPOTTER

Colon slides in 10 seconds. (1) **Serration** only at the top = hyperplastic; down to a dilated crypt base = sessile serrated lesion. (2) Surface nuclei **pencil-shaped, crowded, dark** = adenoma (tubules = tubular, fronds = villous). (3) Glands past the **muscularis mucosae** in **desmoplastic** stroma with dirty necrosis = adenocarcinoma. (4) Branching **smooth muscle** = Peutz–Jeghers; **cystic glands** = juvenile polyp.

EXAM ZONE Past-Paper Pattern SEQs — Solved

PROF PATTERN **a)** Classify polyps of the large intestine. **b)** Differentiate between familial adenomatous polyposis and Lynch syndrome. **2 + 3 marks**

MODEL ANSWER PLAN

a) Non-neoplastic (hyperplastic, inflammatory), hamartomatous (juvenile, Peutz–Jeghers), neoplastic (tubular, tubulovillous, villous adenomas; sessile serrated lesion), malignant. **b)** Table: APC vs MMR genes; ≥ 100 polyps vs few; 100% cancer vs high risk when young; whole colon vs right-sided mucinous; CIN vs MSI; osteomas/desmoids vs endometrium/ovary/urinary tract; colectomy vs surveillance.

CLASSIC Describe the molecular pathogenesis of colorectal carcinoma. **5 marks**

MODEL ANSWER PLAN

APC/β-catenin (about 80%): two APC hits → β-catenin → MYC/cyclin D1 → adenoma; KRAS; SMAD2/4 loss; TP53 loss → carcinoma; telomerase; metastasis. **Mismatch repair (MSI, 10–15%)**: germline MMR mutation (Lynch) or MLH1 methylation in BRAF-mutant serrated lesions → MSI → TGFBR2, BAX hit → right-sided, mucinous, better prognosis.

Score booster: draw Figure 4.1 with the gene written under each step.

MODULE PATTERN A 64-year-old man has weakness, Hb 8 g/dL with microcytic hypochromic indices and a mass in the right iliac fossa. **a)** Give the most likely diagnosis. **b)** Describe its gross and microscopic features. **c)** How will you stage it, and which factors decide prognosis? **1 + 2 + 2 marks**

MODEL ANSWER PLAN

a) Adenocarcinoma of caecum/ascending colon. **b)** Gross: polypoid, exophytic, ulcerated mass (contrast left-sided napkin-ring). Micro: dysplastic columnar cells in irregular glands invading beyond the muscularis mucosae, desmoplasia, dirty necrosis. **c)** TNM (T by depth; N1 1–3, N2 ≥ 4 nodes; M); Dukes A–D. Prognosis: depth and nodal status (most important), metastasis, grade, lymphovascular invasion, MSI.

BCQs One-Best-Answer Practice

1 The commonest site of distant haematogenous metastasis from colonic carcinoma is the:

A. Liver
B. Lung
C. Bone
D. Brain
E. Adrenal

2 Serum CEA in colorectal carcinoma is most useful for:

A. Population screening
B. Diagnosing early cancer
C. Distinguishing adenoma from carcinoma
D. Identifying Lynch syndrome
E. Detecting recurrence after resection

3 A 15-year-old boy has hundreds of colonic adenomas at colonoscopy. His father died of colon cancer at 34. Which gene is most likely mutated?

A. MLH1
B. APC
C. STK11
D. SMAD4
E. MUTYH

4 A girl of 10 has dark macules on her lips and an intussusception caused by a polyp with branching bands of smooth muscle. The mutated gene is:

A. STK11
B. PTEN
C. APC
D. BMPR1A
E. MSH2

5 Which feature of a colonic adenoma correlates best with the risk that it contains invasive carcinoma?

A. Presence of a stalk
B. Tubular architecture
C. Size
D. Location in the rectum
E. Age of the patient

6 A 44-year-old woman has a mucinous carcinoma of the ascending colon with microsatellite instability. Her mother had endometrial carcinoma at 47. The germline defect most likely involves:

A. APC
B. TP53
C. KRAS
D. MSH2
E. BRAF

7 The earliest genetic event in the classic adenoma–carcinoma sequence is:

A. KRAS activation
B. TP53 loss
C. SMAD4 loss
D. Telomerase activation
E. Loss of APC

8 A 70-year-old woman with a large rectal villous adenoma has watery mucous diarrhoea and a serum potassium of 2.8 mmol/L. The best explanation is:

A. Secretion of mucus rich in protein and potassium
B. Chronic bleeding from the tumour
C. Partial obstruction with overflow
D. Aldosterone secretion by the tumour
E. Bacterial overgrowth

9 A flat 1.5 cm right-colon polyp shows serrated crypts extending to the crypt base, with dilated, boot-shaped crypts and no cytological dysplasia. It is best classified as:

A. Hyperplastic polyp with no malignant potential
B. Juvenile polyp
C. Sessile serrated lesion
D. Tubular adenoma
E. Inflammatory pseudopolyp

10 A 72-year-old man has fatigue and iron-deficiency anaemia but no change in bowel habit. A colonic carcinoma is most likely located in the:

A. Rectum
B. Caecum and ascending colon
C. Sigmoid colon
D. Descending colon
E. Anal canal

11 A 30-year-old man has multiple colonic adenomas, an osteoma of the mandible and a desmoid tumour of the abdominal wall. The diagnosis is:

A. Turcot syndrome
B. Gardner syndrome
C. Peutz–Jeghers syndrome
D. Cowden syndrome
E. Lynch syndrome

12 The most important prognostic factor in a resected colorectal carcinoma is:

A. Tumour diameter
B. Histological grade alone
C. Site within the colon
D. Depth of invasion and lymph node status
E. Pre-operative CEA

ANSWER KEY WITH ONE-LINE REASONS

1. **A** Venous drainage via the portal vein delivers tumour cells to the liver first.
2. **E** CEA lacks sensitivity and specificity for diagnosis; a rising level after surgery signals recurrence.
3. **B** Autosomal dominant inheritance with ≥ 100 adenomas = classic FAP (APC, 5q21).
4. **A** Peutz–Jeghers syndrome: STK11 (LKB1); arborising smooth muscle.
5. **C** Size is the most important; nearly 40% of adenomas over 4 cm contain cancer.
6. **D** Lynch syndrome: mismatch-repair genes (MSH2, MLH1); endometrium is the commonest extracolonic site.
7. **E** APC loss starts the sequence; TP53 loss marks the step to invasion.
8. **A** Large villous adenomas can secrete K^+- and protein-rich mucus.
9. **C** Full-length serration with a dilated, laterally growing base = sessile serrated lesion; premalignant.
10. **B** Right-sided tumours are exophytic, bleed occultly and rarely obstruct.
11. **B** Gardner = FAP + osteomas, epidermoid cysts, desmoids, dental anomalies.
12. **D** Stage (depth of invasion and nodal metastasis) outweighs all other factors.

VIVA Rapid-Fire Viva Voce

Q Define a polyp.	A mass projecting above the mucosal surface into the lumen; sessile or pedunculated.
Q Which serrated polyp has no malignant potential?	Hyperplastic polyp: left colon, serration confined to the upper crypt, no dysplasia.
Q What is the hallmark of an adenoma?	Epithelial dysplasia: crowded, elongated, hyperchromatic, stratified nuclei with loss of mucin.
Q How many adenomas define classic FAP?	At least 100.
Q Why is APC called the gatekeeper?	It controls β-catenin; its loss is the first step that lets crypt cells proliferate.
Q Name the genes of Lynch syndrome.	DNA mismatch-repair genes: MSH2 and MLH1 mostly; MSH6 and PMS2.
Q Commonest extracolonic cancer in Lynch syndrome?	Endometrial carcinoma.
Q What is a napkin-ring carcinoma?	Annular, encircling carcinoma of the left colon that narrows the lumen and obstructs.
Q What is pseudo-invasion?	Adenomatous glands displaced into the stalk of a twisted polyp; mimics invasive carcinoma.
Q What does Dukes C mean?	Lymph node metastasis, whatever the depth (TNM stage III).
Q Which bacteraemia should prompt a colonoscopy?	Streptococcus gallolyticus (S. bovis) bacteraemia or endocarditis.
Q Is CEA a screening test?	No. It is insensitive for early cancer and raised in smokers, IBD and cirrhosis; used for follow-up.

OSPE Practice Stations

STATION 1 · SLIDE — 2 minutes

TASK

Colon wall: normal mucosa gives way to irregular, back-to-back glands of tall columnar cells with elongated, hyperchromatic, stratified nuclei, lying in a desmoplastic stroma deep in the muscularis propria; lumens hold necrotic debris. (a) Identify. (b) Two diagnostic features. (c) Commonest site of distant spread and the follow-up marker.

ANSWER

(a) Adenocarcinoma of the colon (moderately differentiated). (b) Invasion beyond the muscularis mucosae with desmoplasia; dysplastic glands with dirty necrosis. (c) Liver; serum CEA.

STATION 2 · SLIDE — 2 minutes

TASK

A stalked polyp is covered by crowded tubular glands with pencil-shaped, hyperchromatic, stratified nuclei and little mucin; nothing crosses the muscularis mucosae. (a) Identify. (b) Two features that raise its malignant risk. (c) Gene usually lost first?

ANSWER

(a) Tubular adenoma (low-grade dysplasia). (b) Size > 1 cm (especially > 4 cm), high-grade dysplasia, villous component. (c) APC.

STATION 3 · GROSS SPECIMEN 1 minute

TASK

Colectomy from a 22-year-old: mucosa carpeted with hundreds of small polyps; one ulcerated, encircling growth. (a) Diagnosis? (b) Gene and inheritance. (c) Why remove the colon?

ANSWER

(a) Familial adenomatous polyposis with carcinoma. (b) APC, autosomal dominant. (c) Cancer develops in 100% of untreated patients, often before 30: prophylactic colectomy.

RAPID REVISION The Night-Before Page

15 ONE-LINERS

01 Polyp = a shape; only adenomas and serrated lesions are premalignant.

02 Hyperplastic: small, left-sided, serration in the upper crypt, no dysplasia.

03 Sessile serrated lesion: right colon, serration to crypt base, BRAF.

04 Juvenile polyp: child, rectal bleeding, cystic glands; polyposis = SMAD4/BMPR1A.

05 Peutz–Jeghers: STK11, lip pigmentation, arborising smooth muscle.

06 Adenoma hallmark = dysplasia; size is the strongest predictor of cancer.

07 Villous adenoma: large, sessile; can cause hypokalaemia.

08 Invasion counts only after the muscularis mucosae is crossed.

09 FAP: APC 5q21, ≥ 100 adenomas, 100% cancer if untreated.

10 Gardner = FAP + osteomas + desmoids; Turcot = polyps + brain tumours.

11 Lynch: MSH2/MLH1, right-sided mucinous cancer, endometrium.

12 Sequence: APC → KRAS → SMAD2/4 → TP53.

13 MSI tumours: right, mucinous, lymphocyte-rich, better prognosis.

14 Right CRC = anaemia; left CRC = napkin-ring obstruction.

15 Depth + nodes = prognosis; liver = first metastasis; CEA = follow-up.

MEMORY HOOKS

- **Gatekeeper vs caretaker**: APC guards the door to proliferation; MMR genes look after the DNA
- **3–2–1**: Lynch family history (3 relatives, 2 generations, 1 under 50)
- **Right bleeds, left blocks**: right-sided cancer gives anaemia, left-sided gives obstruction
- **Size is king**: < 1 cm rarely malignant, > 4 cm nearly 40%

TEST YOURSELF: FILL THE GAPS

1. APC lies on chromosome _____.
2. Peutz–Jeghers gene: _____.
3. Commonest extracolonic cancer in Lynch: _____.
4. The late gene for invasion: _____.
5. Sporadic MSI cancers carry a _____ mutation.
6. Dukes B = through the wall, nodes _____.

Answers: 5q21 · STK11 · endometrium · TP53 · BRAF · negative

DRAW FROM MEMORY

The adenoma–carcinoma sequence with the gene under each step (Figure 4.1).

DRAW FROM MEMORY

Right- and left-sided colon cancer side by side: polypoid mass vs napkin-ring constriction.

SPACED REVISION ☐ Day 1 ☐ Day 3 ☐ Day 7 ☐ Day 14 ☐ Day 30

05

Jaundice & Viral Hepatitis

A yellow sclera is the liver's alarm bell, but the fault may lie before, inside or after the liver. Follow bilirubin on its journey and the lab report names the culprit; learn the timeline of the hepatitis B markers and every serology panel becomes a puzzle you can solve.

BY THE END OF THIS CHAPTER YOU CAN

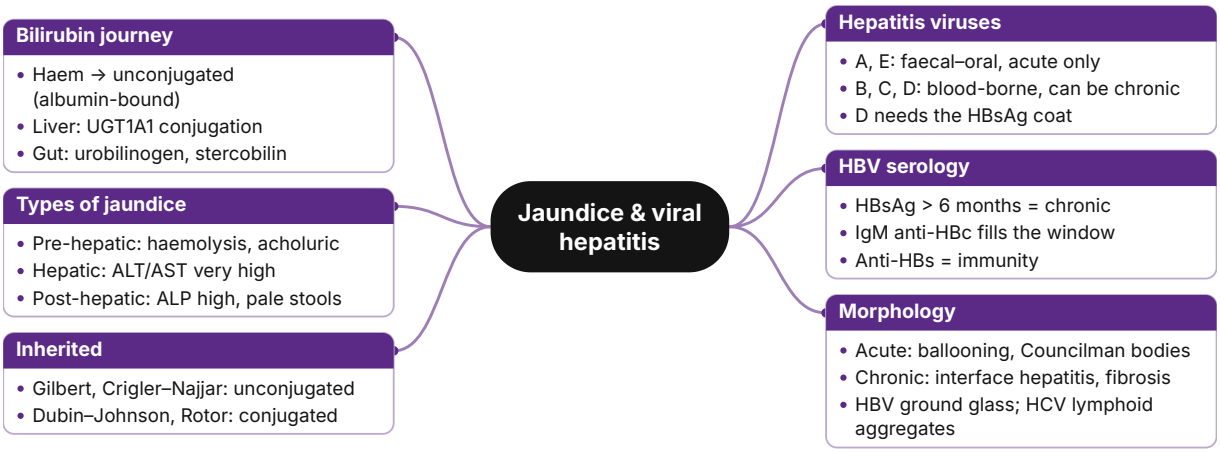
1. Trace the metabolism of bilirubin and explain why conjugated, but not unconjugated, bilirubin appears in urine
2. Classify jaundice and differentiate pre-hepatic, hepatic and post-hepatic jaundice by laboratory findings
3. Compare the hereditary hyperbilirubinaemias and outline neonatal jaundice
4. Compare hepatitis viruses A to E by structure, transmission and outcome
5. Interpret hepatitis B serological markers in acute, chronic, resolved and vaccinated states
6. Describe the morphology of acute and chronic viral hepatitis, its grading and staging, and fulminant hepatitis

EXAM RADAR

Theory SEQs	★★★
BCQs / MCQs	★★★
Viva voce	★★★
OSPE / slides	★★★

Types of jaundice with their lab differences and HBV serology are perennial SEQ and data-station themes; chronic hepatitis is a lab slide.

ROBBINS LINK Robbins Basic Pathology — *Liver and Gallbladder*: jaundice and cholestasis, hereditary hyperbilirubinaemias, viral hepatitis, acute liver failure.



Chapter at a glance. One pigment, five viruses.

5.1 Bilirubin metabolism and jaundice

DEFINITION

Jaundice (icterus) is yellow discolouration of the skin and sclerae caused by retention of bilirubin. It becomes visible when serum bilirubin exceeds about **2.0–2.5 mg/dL** (normal adult level < 1.2 mg/dL). **Cholestasis** is the wider failure of bile flow, in which **bile salts and cholesterol** are retained as well as bilirubin: it adds pruritus, xanthomas, raised alkaline phosphatase and malabsorption of fat-soluble vitamins.

Bilirubin is the end product of haem breakdown; about 0.2–0.3 g is made each day. Roughly **85%** comes from **senescent red cells** destroyed by macrophages in the spleen, liver and marrow; the rest comes from turnover of haem proteins (such as cytochrome P-450) and from red-cell precursors destroyed in the marrow (ineffective erythropoiesis).

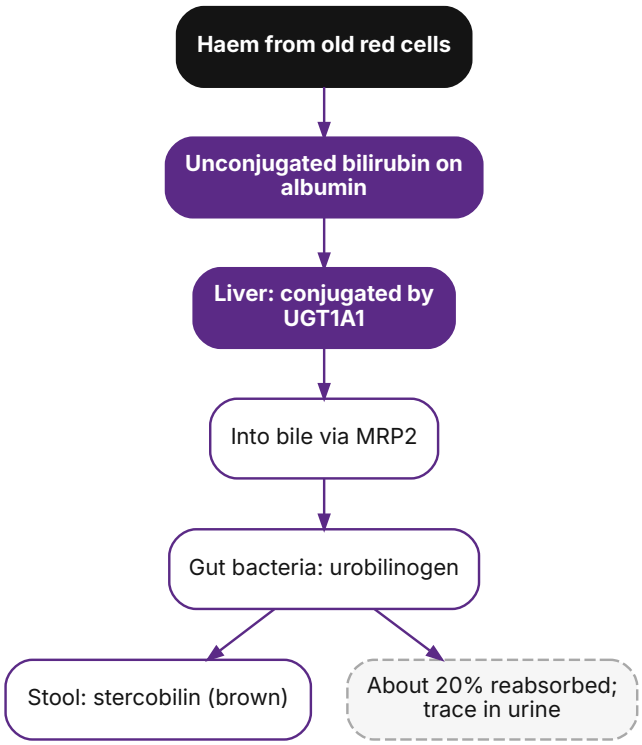


Figure 5.1 · The journey of bilirubin.

1. **Haem oxygenase** opens haem to biliverdin; biliverdin reductase turns it into **unconjugated bilirubin**, which is water-insoluble.
2. In blood it is **tightly bound to albumin**, so it is not filtered by the glomerulus.
3. Hepatocytes take it up across the sinusoidal membrane; in the endoplasmic reticulum **UGT1A1** (bilirubin uridine diphosphate-glucuronosyltransferase) adds one or two glucuronic acids, making it **water-soluble**.
4. Bilirubin glucuronides are pumped into the canaliculus by **MRP2**, an energy-dependent step.
5. In the colon, bacteria deconjugate and reduce it to colourless **urobilinogens**. Most leave in the stool (oxidised to brown **stercobilin**); about 20% is reabsorbed and re-excreted by the liver, and a trace reaches the urine.

CONCEPT CLEAR

Solubility explains the urine and stools. Unconjugated bilirubin is insoluble and albumin-bound, so it **never enters urine**: haemolytic jaundice is "**acholuric**". But free unconjugated bilirubin is lipid-soluble and can cross the immature blood–brain barrier of a newborn, causing **kernicterus**. Conjugated bilirubin is water-soluble and only loosely bound, so when it backs up it spills into urine (**dark, tea-coloured urine**). If bile cannot reach the gut, no stercobilin forms (**pale, clay-coloured stools**) and no urobilinogen is reabsorbed (**urine urobilinogen absent**).

5.2 Classifying jaundice

Robbins sorts jaundice by the **type of bilirubin** that rises; clinicians sort it by **site** (pre-hepatic, hepatic, post-hepatic). Both lead to the same causes.

CAUSES OF JAUNDICE BY THE TYPE OF BILIRUBIN

Mechanism	Examples
Predominantly unconjugated hyperbilirubinaemia	
Excess production	**Haemolytic anaemias** (hereditary spherocytosis, G6PD deficiency, sickle cell, autoimmune, malaria); **ineffective erythropoiesis** (thalassaemia, megaloblastic anaemia); resorption of a large haematoma
Reduced uptake	Drugs (e.g. rifampicin); Gilbert syndrome
Impaired conjugation	**Physiological jaundice of the newborn**; Crigler–Najjar types I and II; Gilbert syndrome; diffuse hepatocellular disease
Predominantly conjugated hyperbilirubinaemia	
Impaired canalicular excretion	Dubin–Johnson and Rotor syndromes
Hepatocellular damage, intrahepatic cholestasis	**Viral or drug hepatitis**, alcoholic hepatitis, cirrhosis, sepsis, primary biliary cholangitis, total parenteral nutrition
Extrahepatic obstruction	**Gallstones** in the common bile duct, **carcinoma of the head of pancreas**, cholangiocarcinoma, strictures, biliary atresia

PRE-HEPATIC VS HEPATIC VS POST-HEPATIC JAUNDICE: THE CLASSIC LAB TABLE

Finding	Pre-hepatic (haemolytic)	Hepatic (hepatocellular)	Post-hepatic (obstructive)
Serum bilirubin	**Unconjugated** ↑ (rarely > 5 mg/dL)	Both ↑	**Conjugated** ↑↑
Urine bilirubin	**Absent** (acholuric)	Present	Present, **dark urine**
Urine urobilinogen	**Increased**	Normal or ↑ (falls if cholestatic)	**Absent**
Stool colour	Normal or dark	Normal or pale	**Pale, clay-coloured**
ALT / AST	Normal	**Markedly ↑** (often > 10 × normal)	Mildly ↑
ALP / GGT	Normal	Mildly ↑ (< 3 × normal)	**Markedly ↑** (> 3 × normal)
Prothrombin time	Normal	Prolonged; **not corrected** by vitamin K	Prolonged; **corrected** by parenteral vitamin K
Other clues	Anaemia, reticulocytosis, ↑ LDH, ↓ haptoglobin, splenomegaly	Tender hepatomegaly; low albumin if chronic	**Pruritus**, steatorrhoea, ↑ cholesterol, palpable gallbladder (tumour)

LIVER FUNCTION TESTS: WHAT EACH ONE TELLS YOU

Function tested	Test (usual adult range; varies by lab)	Pattern
Hepatocyte integrity	**ALT** (about 7–56 U/L), **AST** (about 10–40 U/L)	Very high in hepatitis; ALT is more liver-specific; AST:ALT > 2 suggests alcohol
Biliary excretion	Bilirubin (total 0.3–1.2, direct < 0.3 mg/dL); **ALP** (about 40–130 U/L); **GGT**; urine bilirubin	ALP with GGT high = cholestasis (ALP alone may be bone)
Synthetic function	**Albumin** (3.5–5.0 g/dL); **PT/INR** (factors II, V, VII, IX, X, fibrinogen)	Low albumin = chronic damage; long PT = severe or acute failure
Metabolic function	Blood ammonia	High in hepatic encephalopathy

CLINICAL CASE

A 62-year-old man has painless, deepening jaundice for 6 weeks with severe itching, tea-coloured urine and clay-coloured stools; a non-tender gallbladder is palpable. Bilirubin 14 mg/dL (conjugated 11), ALP 820 U/L, GGT 610 U/L, ALT 90 U/L; PT 17 s, normal after intramuscular vitamin K. **Reasoning:** conjugated bilirubin + ALP/GGT far above ALT + pale stools + PT corrected by vitamin K → **post-hepatic (obstructive) jaundice**. Painless progressive jaundice with a palpable gallbladder points to a **carcinoma of the head of the pancreas** rather than a stone (see Chapter 7).

5.3 Hereditary hyperbilirubinaemias and neonatal jaundice

Each inherited syndrome is a fault at one step of Figure 5.1: two at **conjugation** (unconjugated rise), two at **excretion** (conjugated rise).

HEREDITARY HYPERBILIRUBINAEMIAS

Syndrome	Inheritance	Defect	Bilirubin	Key features
Gilbert	Usually AR	**Reduced UGT1A1** activity (promoter variant)	Unconjugated, mild, fluctuating	About 6% of people; rises with **fasting**, stress, infection; harmless
Crigler–Najjar I	AR	**Absent** UGT1A1	Unconjugated, severe	**Kernicterus**; fatal in infancy without liver transplant
Crigler–Najjar II	AD, variable penetrance	Severely reduced UGT1A1	Unconjugated	Milder, rarely kernicterus; **phenobarbital** induces the enzyme
Dubin–Johnson	AR	Absent **MRP2** (canalicular export)	Conjugated	**Black liver**: coarse dark granules in hepatocytes (not bilirubin); benign
Rotor	AR	Defective hepatic uptake and storage	Conjugated	Liver **not pigmented**; benign

MEMORY HOOK

Ask the DR for Direct bilirubin

Dubin–Johnson and **R**otor raise **direct (conjugated)** bilirubin; Dubin–Johnson gives a **Dark** liver. Gilbert and Crigler–Najjar raise **indirect** bilirubin: **Gilbert is gentle**, Crigler–Najjar I is catastrophic.

PHYSIOLOGICAL JAUNDICE OF THE NEWBORN

- **Immature UGT1A1** + high red-cell turnover after birth → transient unconjugated hyperbilirubinaemia
- Appears **after 24 hours**, peaks around days 3–5, settles within about 2 weeks in term babies
- Exaggerated in breast-fed babies: milk β-glucuronidases deconjugate bilirubin in the gut and it is reabsorbed

PATHOLOGICAL: INVESTIGATE IF

- Appears in the **first 24 hours** (haemolytic disease: Rh, ABO, G6PD deficiency)
- Very high or rapidly rising levels (risk of **kernicterus**: basal ganglia staining)
- **Conjugated** or lasting > 2 weeks: **neonatal cholestasis** from **biliary atresia** or neonatal hepatitis

5.4 Viral hepatitis: the five hepatotropic viruses

"Hepatitis viruses" are the five viruses whose main target is the hepatocyte; EBV, CMV, herpes simplex and yellow fever can also inflame the liver during systemic infection. Any hepatitis virus can cause: **acute asymptomatic** infection; **acute symptomatic** hepatitis (icteric or anicteric) with recovery; **chronic hepatitis** (disease lasting **> 6 months**), with or without cirrhosis; a **carrier state**; or **fulminant hepatitis** (acute liver failure).

HEPATITIS VIRUSES A TO E

Feature	HAV	HBV	HCV	HDV	HEV
Virus, genome	Picornavirus; ssRNA; no envelope	Hepadnavirus; **partially double-stranded DNA**; enveloped	Flavivirus; ssRNA; enveloped	**Defective** circular RNA; coat of **HBsAg**	Hepevirus; ssRNA; no envelope
Transmission	**Faecal–oral** (water, food, shellfish)	**Parenteral, sexual, perinatal**	**Parenteral** (blood, needles); sexual and vertical less often	Parenteral (only with HBV)	**Faecal–oral**, waterborne epidemics
Incubation	2–6 weeks	2–26 weeks (mean about 8)	4–26 weeks (mean about 9)	As for HBV	2–8 weeks
Chronic infection	**Never**	**5–10%** of adults; about **90%** of neonates	**Most (about 80%)**	Co-infection < 5%; **superinfection about 80%**	No (except immunosuppressed)
Fulminant	Rare	About 1% of acute cases	Rare	More frequent, esp. superinfection	**Pregnancy: mortality up to about 20%**
Hepatocellular carcinoma	No	**Yes** (even without cirrhosis)	**Yes** (mostly via cirrhosis)	Yes (with HBV)	No
Diagnosis	**IgM anti-HAV**	HBsAg, **IgM anti-HBc**, HBV DNA	Anti-HCV (screen); **HCV RNA** (active infection)	IgM anti-HDV, HDV RNA	IgM anti-HEV, HEV RNA
Vaccine	Yes	Yes (recombinant HBsAg)	**None**	HBV vaccine protects	Not widely available

MEMORY HOOK

"Vowels hit the bowels"

Hepatitis **A** and **E** spread by the faecal–oral route and do not become chronic in healthy people. **B**, **C** and **D** need **Blood**: **C** is the **Chronic** one, and **D** is **Dependent** on B for its coat.

CONCEPT CLEAR

The immune system, not the virus, injures the liver. HBV is not directly cytopathic: hepatocytes die when **CD8+ cytotoxic T cells** recognise viral antigens on their surface. A vigorous T-cell response gives florid acute hepatitis but clears the virus; a weak, tolerant response lets the virus persist. That is why **90% of infected neonates** (immature, tolerant immunity) become chronic carriers, but only **5–10% of adults** do.



Figure 5.2 · Clinical course of acute viral hepatitis.

Hepatitis A

Benign and self-limited; most childhood infections are silent. Virus is shed in stool for 2–3 weeks **before** and about a week after jaundice appears, so spread has happened by the time the patient turns yellow. **IgM anti-HAV** = acute infection; IgG anti-HAV = lifelong immunity. No carrier state.

5.5 Hepatitis B and its serology

The complete virion (**Dane particle**) has an envelope and a core. Its genome is a small, partially double-stranded circular DNA that replicates through an RNA intermediate using a **reverse transcriptase**, and can **integrate** into host DNA.

- **HBsAg**: envelope protein; made in excess and released into blood
- **HBcAg**: core protein; stays inside hepatocytes (not in serum)
- **HBeAg**: secreted precore protein: marker of **active replication and high infectivity**
- **HBx**: transactivating protein; important in hepatocellular carcinoma

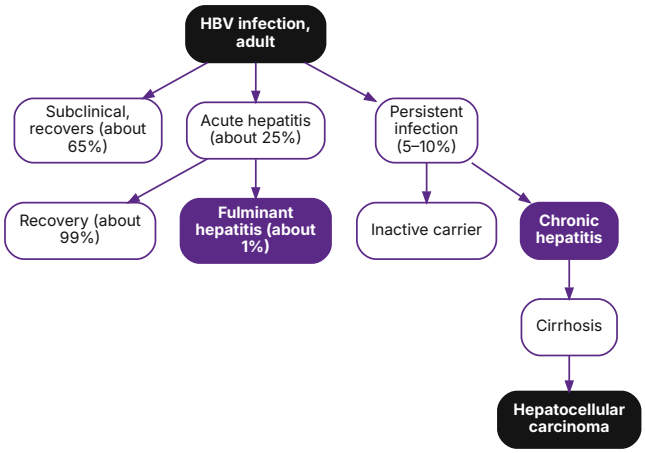


Figure 5.3 · **Outcomes of HBV infection in adults** (approximate).

The serological markers in time order

1. **HBsAg** — appears **before symptoms**, peaks during the illness and disappears within **3–6 months**. Persistence beyond 6 months = **chronic infection**.
2. **HBeAg, HBV DNA, DNA polymerase** — appear soon after HBsAg; all mean **active replication**. Persistent HBeAg predicts chronicity.
3. **IgM anti-HBc** — rises just before symptoms, with the ALT; over months it is replaced by **IgG anti-HBc**, which stays for life.
4. **Anti-HBe** — appears as HBeAg vanishes: the infection has peaked.
5. **Anti-HBs** — rises only after the disease is over, weeks to months after HBsAg has gone; persists for life and is **protective**. It is the **only** marker after vaccination.

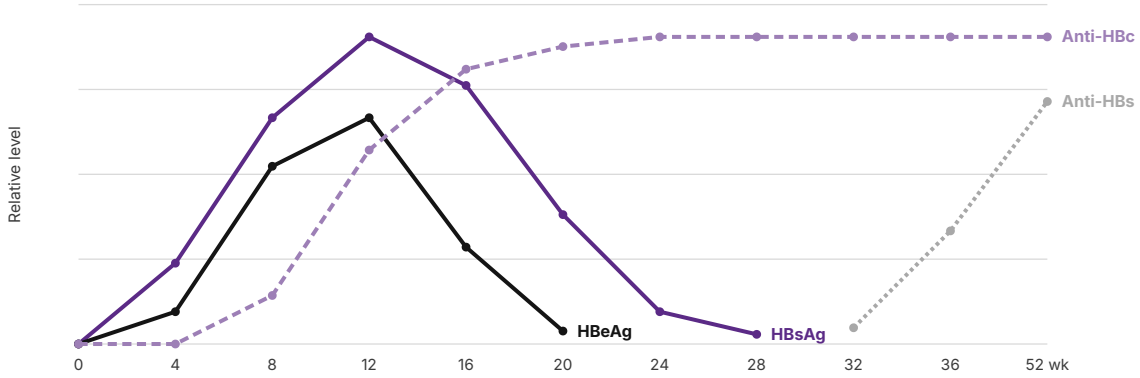


Figure 5.4 · Serological course of acute hepatitis B that resolves (weeks after exposure; relative levels). The gap between the fall of HBsAg and the rise of anti-HBs is the **window period**: IgM anti-HBc is then the diagnostic marker.

READING A HEPATITIS B SEROLOGY PANEL (A FAVOURITE DATA STATION)

HBsAg	HBeAg	IgM anti-HBc	IgG anti-HBc	Anti-HBs	Interpretation
+	+	+	–	–	**Acute hepatitis B**, highly infectious
–	–	+	–	–	**Window period** of acute infection (anti-HBe may be +)
–	–	–	+	+	**Recovered**: immune after natural infection
–	–	–	–	+	**Vaccinated**: immune
+	+	–	+	–	**Chronic hepatitis B**, active replication (infectious)
+	–	–	+	–	Chronic infection with anti-HBe: **inactive carrier** if HBV DNA low; **precore mutant** if HBV DNA high
–	–	–	–	–	Never infected, not immune: **vaccinate**

⚠ **EXAM TRAP**

Three serology traps. (1) A negative HBsAg does **not** exclude acute HBV: in the window period **IgM anti-HBc** is the diagnostic marker (anti-HBe may also be present). (2) Anti-HBc (any class) means **infection**, never vaccination; the vaccine contains only HBsAg, so a vaccinated person has anti-HBs alone. (3) HBeAg-negative is not always "safe": **precore mutants** cannot make HBeAg yet replicate actively (high HBV DNA).

5.6 Hepatitis C, D and E

Hepatitis C

HCV is an enveloped RNA virus whose polymerase makes many copying errors. Its envelope proteins keep changing, so each patient carries a swarm of variants (**quasispecies**) that stay one step ahead of neutralising antibodies. This explains its three hallmarks: **persistence** (most infections become chronic), **no vaccine**, and possible **re-infection**.

- Acute infection is usually **mild or silent**; many are found only when they present with chronic disease
- Chronic HCV: **episodic rises of ALT** with normal spells between them
- **Anti-HCV** screens but does not protect; **HCV RNA** proves active infection
- About one in five develop **cirrhosis** over about 20 years; risk of **HCC**. Direct-acting antivirals now cure most patients

CLINICAL CORRELATION

Extrahepatic HCV. Mixed **cryoglobulinaemia** with membranoproliferative GN (see Chapter 16), porphyria cutanea tarda, lichen planus, B-cell lymphoma and insulin resistance. Think HCV when a patient has purpura, arthralgia and proteinuria with low complement.

PAKISTAN FOCUS

Viral hepatitis is a major public-health problem in Pakistan, which carries one of the **largest hepatitis C burdens in the world**. The national survey of 2007–08 found **anti-HCV in about 4.8%** and **HBsAg in about 2.5%** of the general population. The main routes are **unsafe injections and reuse of syringes**, unscreened blood, **barbers' razors** and dental procedures. **HDV co-infection** is relatively common in HBV carriers in some regions. Every blood bag must be screened and every needle used once.

Source: PMRC national survey of hepatitis B and C, 2007–08.

Hepatitis D (delta)

HDV is a **defective** RNA virus: it can replicate only inside a cell already infected by HBV, because it borrows **HBsAg** as its envelope. Preventing HBV (vaccination) therefore prevents HDV.

HDV CO-INFECTION VS SUPERINFECTION

Feature	Co-infection	Superinfection
Setting	HBV and HDV acquired **together**	HDV infects a **chronic HBV carrier**
Illness	Acute hepatitis, usually self-limited (may be more severe)	Severe acute hepatitis on top of chronic disease
Chronicity	**< 5%**	**About 80%**; accelerates cirrhosis
Serology	IgM anti-HBc + IgM anti-HDV	HBsAg, **IgG anti-HBc** + anti-HDV

Hepatitis E

A waterborne, faecal–oral infection that causes **epidemics after contamination of drinking water** and sporadic cases in South Asia. Like HAV it is self-limited and does not become chronic in healthy people (chronic HEV occurs in transplant recipients). Its notorious feature is the high mortality, up to about **20%**, in **pregnant women**, from fulminant hepatitis.

5.7 Morphology of viral hepatitis

The **time course**, not the virus, decides the pattern: **acute** injury is scattered through the **lobule**; **chronic** inflammation is centred on **portal tracts**, gnaws into the parenchyma and lays down fibrosis.

Acute viral hepatitis

GROSS

- Liver normal-sized or slightly enlarged, swollen and soft; tense capsule
- May be greenish if cholestatic
- Severe necrosis: shrunken, soft liver (see fulminant hepatitis)

MICROSCOPY

- **Ballooning degeneration**: swollen hepatocytes with pale, wispy cytoplasm
- **Acidophil (Councilman) bodies**: apoptotic hepatocytes, shrunken and deeply eosinophilic with pyknotic or absent nuclei
- **Lobular disarray**; spotty necrosis with lymphocytes in the lobule
- **Kupffer cell** hypertrophy, laden with debris; mild portal inflammation
- Canalicular **bile plugs**; in severe cases **bridging necrosis**

Chronic viral hepatitis

GROSS

- Early: liver looks normal
- Late: firm, finely or coarsely nodular as **cirrhosis** develops (Chapter 6)

MICROSCOPY

- **Portal tracts expanded** by lymphocytes, macrophages and plasma cells
- **Interface hepatitis** (formerly piecemeal necrosis): inflammation crosses the **limiting plate** and kills periportal hepatocytes
- Lobular inflammation, apoptosis; **bridging necrosis** when active
- **Fibrosis**: portal → periportal → **bridging septa** → cirrhosis

HBV clue: ground-glass hepatocytes (finely granular, pale-pink cytoplasm because the ER is packed with **HBsAg**; confirmed by orcein stain or immunostaining); "sanded" nuclei hold HBcAg. **HCV clue:** dense **portal lymphoid aggregates** or follicles, **bile-duct injury** and **steatosis** (fatty change).

GRADING AND STAGING OF CHRONIC HEPATITIS

	Grade	Stage
Measures	**Activity**: interface, lobular and portal necroinflammation; reversible	**Fibrosis**: architectural damage
Scale (METAVIR)	A0 (none) to A3 (severe)	**F0** none · **F1** portal fibrosis · **F2** few septa · **F3** many bridging septa · **F4 cirrhosis** (Ishak uses 0–6)

LAB SPOTTER

Chronic hepatitis slide in 10 seconds. Look at the **portal tracts first**: swollen with lymphocytes that spill into the surrounding liver cells (**interface hepatitis**) = chronic hepatitis. A **lymphoid follicle** + fat droplets = **HCV**; glassy pink hepatocytes = **HBV**; septa joining tracts = advanced stage. In **acute** hepatitis the tracts are quiet and ballooning and Councilman bodies are scattered through the lobule.

Fulminant hepatitis (acute liver failure)

Acute liver illness with **encephalopathy and coagulopathy** within **26 weeks** of the initial injury, without pre-existing liver disease. **Paracetamol overdose** leads in Western countries; in South Asia **viral hepatitis** (HEV, HBV ± HDV, HAV) is a leading cause. Others: drugs, autoimmune hepatitis, Wilson disease, *Amanita* mushrooms, ischaemia.

GROSS

- **Massive hepatic necrosis**: liver shrunken (can fall to 500–700 g), soft, limp, red-brown
- **Wrinkled** capsule

MICROSCOPY

- Whole lobules destroyed; **collapsed reticulin** framework
- Only portal tracts remain, with a **ductular reaction**
- Survivors regenerate; extensive collapse heals with coarse scars

Clinical: jaundice, **encephalopathy**, high INR, hypoglycaemia, cerebral oedema, renal failure, sepsis; high mortality without transplantation (see Chapter 6).

EXAM ZONE **Past-Paper Pattern SEQs — Solved**

PROF PATTERN **a)** Describe the normal metabolism of bilirubin. **b)** Differentiate between haemolytic, hepatocellular and obstructive jaundice on the basis of laboratory findings. **2 + 3 marks**

MODEL ANSWER PLAN

a) Haem (85% from senescent RBCs) → biliverdin → unconjugated bilirubin bound to albumin → hepatocyte uptake → conjugation by UGT1A1 → excretion into bile by MRP2 → gut bacteria form urobilinogen → stercobilin in stool; about 20% reabsorbed, trace in urine. **b)** Table: unconjugated vs mixed vs conjugated; urine bilirubin absent vs present vs present; urobilinogen ↑ vs normal vs absent; stools normal vs normal/pale vs clay; ALT normal vs very high vs mild; ALP normal vs mild vs very high; PT and its response to vitamin K.

Score booster: add one cause of each type (hereditary spherocytosis; viral hepatitis; carcinoma head of pancreas) and the word "acholuric".

MODULE PATTERN A 24-year-old man has jaundice for a week. ALT 1650 U/L; HBsAg positive, HBeAg positive, IgM anti-HBc positive, anti-HBs negative. **a)** What is the diagnosis? **b)** Describe, with a labelled diagram, the sequence of serological markers in hepatitis B. **c)** List the possible outcomes of this infection. **1 + 2 + 2 marks**

MODEL ANSWER PLAN

a) Acute hepatitis B, highly infectious (HBeAg +). **b)** Draw Figure 5.4: HBsAg before symptoms, gone by 3–6 months; HBeAg/HBV DNA = replication; IgM anti-HBc with symptoms, replaced by IgG; anti-HBe after HBeAg; anti-HBs after recovery, lifelong; label the window period. **c)** Recovery (most), fulminant hepatitis (about 1%), chronic infection (5–10% of adults) as inactive carrier or chronic hepatitis → cirrhosis → hepatocellular carcinoma.

Score booster: state that HBsAg persisting beyond 6 months defines chronic infection.

PROF PATTERN Describe the morphological features of acute and chronic viral hepatitis. How are grading and staging done? **5 marks**

MODEL ANSWER PLAN

Acute: ballooning degeneration, Councilman bodies, lobular disarray with lymphocytes, Kupffer cell hyperplasia, cholestasis, bridging necrosis if severe. Chronic: portal mononuclear infiltrate, interface hepatitis, bridging necrosis, progressive fibrosis to cirrhosis; HBV ground-glass hepatocytes; HCV lymphoid aggregates, duct injury, steatosis. Grade = necroinflammatory activity (A0–A3); stage = fibrosis (F0–F4).

CLASSIC **a)** Compare hepatitis A, B and C in a table. **b)** Differentiate between HDV co-infection and superinfection. **3 + 2 marks**

MODEL ANSWER PLAN

a) Rows: genome (RNA, DNA, RNA), route (faecal–oral, parenteral/sexual/perinatal, parenteral), incubation, chronicity (never, 5–10% adults, about 80%), HCC (no, yes, yes), diagnosis (IgM anti-HAV; HBsAg + IgM anti-HBc; anti-HCV + HCV RNA), vaccine (yes, yes, none). **b)** Simultaneous vs on a chronic carrier; usually self-limited vs severe; chronicity < 5% vs about 80%; IgM vs IgG anti-HBc.

BCQs One-Best-Answer Practice

1 A child with hereditary spherocytosis is jaundiced. Which urine finding is expected?

A. Bilirubin present, urobilinogen absent
B. Bilirubin absent, urobilinogen increased
C. Bilirubin present, urobilinogen increased
D. Bilirubin absent, urobilinogen absent
E. Haemoglobinuria with bilirubinuria

2 A 21-year-old student becomes mildly yellow after fasting during Ramadan. Total bilirubin 2.8 mg/dL, almost all unconjugated; Hb, reticulocytes and liver enzymes are normal. The underlying defect is:

A. Absent MRP2
B. Haemolysis
C. Reduced UGT1A1 activity
D. Bile duct obstruction
E. Absent UGT1A1

3 A liver biopsy from a young woman with mild conjugated hyperbilirubinaemia is jet black, with coarse dark granules in hepatocytes. The defective protein is:

A. UGT1A1
B. ATP7B
C. HFE
D. MRP2
E. Ferroportin

4 In obstructive jaundice, which finding best distinguishes it from hepatocellular jaundice?

A. Prolonged PT that corrects with parenteral vitamin K
B. Raised conjugated bilirubin
C. Bilirubin in urine
D. Raised ALT
E. Low serum albumin

5 A patient recovering from acute hepatitis B is negative for HBsAg and anti-HBs. The marker that confirms acute infection now is:

A. HBeAg
B. IgG anti-HBc alone
C. HBV DNA
D. Anti-HBs
E. IgM anti-HBc

6 A nursing student's serology shows anti-HBs positive; HBsAg and anti-HBc negative. She is:

A. Immune after vaccination
B. In the window period
C. A chronic carrier
D. Recovered from natural infection
E. Acutely infected

7 Chronic hepatitis B is defined by persistence of which marker for more than 6 months?

A. Anti-HBs
B. HBsAg
C. IgM anti-HBc
D. Anti-HBe
E. IgG anti-HAV

8 Ground-glass hepatocytes in a liver biopsy owe their appearance to:

A. Glycogen
B. Fat droplets
C. Endoplasmic reticulum packed with HBsAg
D. Copper
E. Mallory–Denk bodies

9 A liver biopsy shows portal lymphoid follicles, bile-duct injury and scattered macrovesicular fat. The most likely cause is:

A. Hepatitis A
B. Hepatitis E
C. Alcohol
D. Hepatitis C
E. Gilbert syndrome

10 During a waterborne epidemic, a woman in her third trimester develops fulminant hepatitis. The likely virus is:

A. HAV
B. HCV
C. HDV
D. HBV
E. HEV

11 Acidophil (Councilman) bodies in acute viral hepatitis represent:

A. Viral inclusions
B. Apoptotic hepatocytes
C. Hyaline in alcoholic injury
D. Bile plugs
E. Kupffer cells with debris

12 The morphological hallmark that separates chronic from acute viral hepatitis is:

A. Interface hepatitis with portal-based fibrosis
B. Ballooning degeneration
C. Cholestasis
D. Kupffer cell hyperplasia
E. Lobular disarray

ANSWER KEY WITH ONE-LINE REASONS

1. **B** Unconjugated bilirubin cannot enter urine (acholuric); extra pigment reaching the gut raises urobilinogen.
2. **C** Gilbert syndrome: mild, fasting-induced unconjugated hyperbilirubinaemia.
3. **D** Dubin–Johnson syndrome: failure of canalicular export of bilirubin glucuronides.
4. **A** Fat-soluble vitamin K is malabsorbed but the liver can still use it; a damaged liver cannot.
5. **E** Window period: HBsAg has gone, anti-HBs not yet detectable.
6. **A** Vaccine contains only HBsAg, so only anti-HBs appears.
7. **B** HBsAg beyond 6 months means chronic infection.
8. **C** Finely granular cytoplasm of HBsAg-laden ER; stains with orcein.
9. **D** The classic triad of chronic HCV.
10. **E** HEV mortality approaches 20% in pregnant women.
11. **B** Shrunken, eosinophilic hepatocytes dying by apoptosis.
12. **A** Chronic hepatitis is portal-centred, erodes the limiting plate and lays down fibrosis.

VIVA Rapid-Fire Viva Voce

Q At what level is jaundice clinically visible?	Serum bilirubin above about 2.0–2.5 mg/dL; first seen in the sclerae.
Q Which enzyme conjugates bilirubin?	UGT1A1 (bilirubin uridine diphosphate-glucuronosyltransferase) in the hepatocyte ER.
Q Why is haemolytic jaundice acholuric?	Unconjugated bilirubin is albumin-bound and water-insoluble, so it is not filtered into urine.
Q What causes kernicterus?	Free unconjugated bilirubin crossing the immature blood–brain barrier and staining basal ganglia.
Q Why are stools pale in obstruction?	No bilirubin reaches the gut, so no stercobilin forms.
Q Which hereditary jaundice gives a black liver?	Dubin–Johnson syndrome (MRP2 defect).
Q Which hepatitis virus has a DNA genome?	Hepatitis B (partially double-stranded DNA, hepadnavirus).
Q Marker of HBV replication and infectivity?	HBeAg (and HBV DNA).
Q What is the window period?	Interval after HBsAg disappears and before anti-HBs appears; IgM anti-HBc is the diagnostic marker.
Q Why is there no vaccine for HCV?	Its envelope proteins mutate constantly (quasispecies), escaping neutralising antibodies.
Q What is interface hepatitis?	Portal inflammation crossing the limiting plate and destroying periportal hepatocytes; hallmark of chronic hepatitis.
Q Grade vs stage?	Grade = necroinflammatory activity; stage = extent of fibrosis up to cirrhosis.

OSPE Practice Stations

STATION 1 · SLIDE (LAB SLIDE) — 2 minutes

TASK

Liver biopsy: portal tracts are expanded by dense lymphocytes, one forming a lymphoid follicle. The infiltrate spills across the limiting plate and surrounds periportal hepatocytes. Some hepatocytes contain large fat vacuoles; thin fibrous septa extend from portal tracts. (a) Identify. (b) Two features supporting it. (c) Which virus is most likely, and what do “grade” and “stage” mean?

ANSWER

(a) Chronic viral hepatitis. (b) Interface hepatitis; portal-based fibrosis with septa. (c) HCV (lymphoid aggregates + steatosis). Grade = necroinflammatory activity; stage = fibrosis.

STATION 2 · HEPATITIS SEROLOGY — 2 minutes

TASK

A 35-year-old blood donor, well, ALT normal. HBsAg positive (also positive 1 year ago); HBeAg negative; anti-HBe positive; IgG anti-HBc positive; IgM anti-HBc negative; anti-HBs negative; HBV DNA low. (a) Interpret. (b) Can he donate blood? (c) Name two long-term risks.

ANSWER

(a) Chronic HBV infection, low replication: inactive carrier. (b) No: HBsAg-positive blood is infectious. (c) Reactivation/chronic hepatitis, cirrhosis, hepatocellular carcinoma; HDV superinfection.

STATION 3 · LAB DATA 2 minutes

TASK

A 12-year-old boy: Hb 8.6 g/dL, reticulocytes 9%, total bilirubin 4.0 mg/dL (direct 0.4), ALT 30 U/L, ALP normal; urine: bilirubin nil, urobilinogen increased. Spleen palpable. (a) Type of jaundice? (b) Explain the urine findings. (c) Two possible causes.

ANSWER

(a) Pre-hepatic (haemolytic), unconjugated. (b) Unconjugated bilirubin is not filtered (acholuric); more pigment reaches the gut, so more urobilinogen. (c) Hereditary spherocytosis, thalassaemia, G6PD deficiency.

RAPID REVISION The Night-Before Page

15 ONE-LINERS

01 Jaundice is visible above about 2–2.5 mg/dL bilirubin.

02 Unconjugated bilirubin: albumin-bound, not in urine, causes kernicterus.

03 UGT1A1 conjugates; MRP2 exports into bile.

04 Haemolytic: unconjugated, acholuric, urobilinogen ↑.

05 Hepatocellular: ALT/AST very high, mixed bilirubin.

06 Obstructive: ALP/GGT very high, pale stools, itching, PT corrects with vitamin K.

07 Gilbert: mild, fasting-induced, benign; Crigler–Najjar I: fatal kernicterus.

08 Dubin–Johnson: MRP2, conjugated, black liver.

09 A and E: faecal–oral, no chronicity; HEV kills pregnant women.

10 HBV: DNA virus; 90% of neonates, 5–10% of adults become chronic.

11 HBsAg > 6 months = chronic; HBeAg = infectivity.

12 Window period: IgM anti-HBc is the key marker; vaccine gives anti-HBs only.

13 HCV: most become chronic; quasispecies; no vaccine.

14 Acute hepatitis: ballooning, Councilman bodies, lobular disarray.

15 Chronic: interface hepatitis + fibrosis; HBV ground glass, HCV lymphoid aggregates.

MEMORY HOOKS

- **Vowels hit the bowels**: A and E are faecal–oral
- **Ask the DR for Direct**: Dubin–Johnson, Rotor = conjugated
- **C is Chronic, D is Dependent** on B's coat
- **S for Surface, S for Shield**: anti-HBs protects; anti-HBc only says "infected"

TEST YOURSELF: FILL THE GAPS

1. Canalicular bilirubin pump: _____.
2. Stool pigment: _____.
3. HBV marker of infectivity: _____.
4. Chronic HBV: HBsAg beyond _____ months.
5. HCV triad: lymphoid aggregates, duct injury, _____.
6. Apoptotic hepatocyte: _____ body.

Answers: MRP2 · stercobilin · HBeAg · 6 · steatosis · Councilman (acidophil)

DRAW FROM MEMORY

The HBV serology curve with the window period labelled (Figure 5.4).

DRAW FROM MEMORY

Bilirubin's path from haem to stercobilin and urinary urobilinogen, marking where each type of jaundice arises.

SPACED REVISION ☐ Day 1 ☐ Day 3 ☐ Day 7 ☐ Day 14 ☐ Day 30

06

Cirrhosis, Haemochromatosis & Hepatocellular Carcinoma

Cirrhosis is the scar a liver is left with after years of injury, whatever the cause. That scar blocks portal blood flow, starves the body of liver function and seeds cancer, so one lesion explains ascites, vomited blood, coma and a hard liver mass.

BY THE END OF THIS CHAPTER YOU CAN

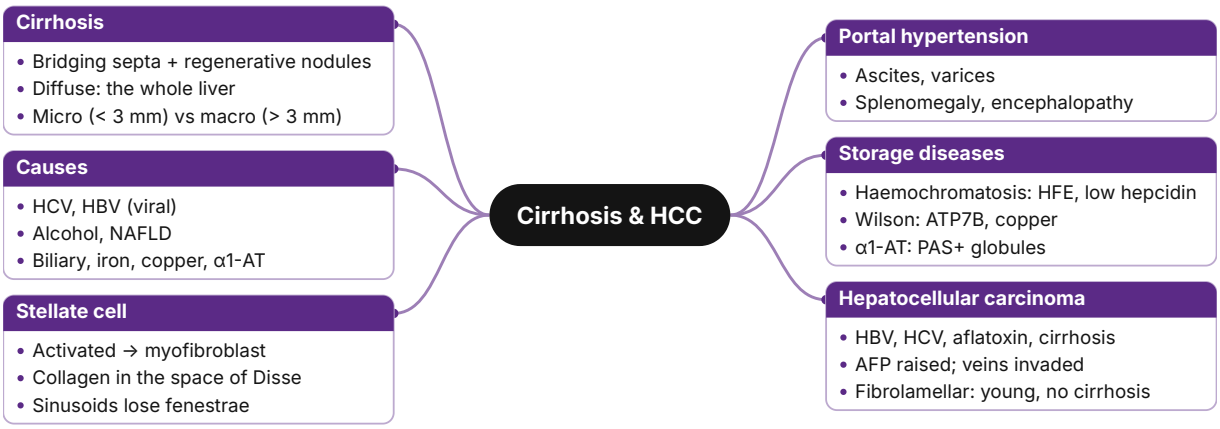
1. Define cirrhosis, list its causes and explain how hepatic stellate cells produce it
2. Describe the gross and microscopic morphology of cirrhosis
3. Explain the causes and consequences of portal hypertension and the clinical features of hepatic failure
4. Outline alcoholic and non-alcoholic fatty liver disease
5. Describe the pathogenesis, morphology and clinical features of hereditary haemochromatosis and compare it with Wilson disease and α1-antitrypsin deficiency
6. Describe the risk factors, pathogenesis, morphology and markers of hepatocellular carcinoma, including the fibrolamellar variant

EXAM RADAR

Theory SEQs	★★★
BCQs / MCQs	★★★
Viva voce	★★★
OSPE / slides	★★★

Cirrhosis with portal hypertension and hepatocellular carcinoma are classic long questions and lab slides; haemochromatosis is a favourite short note.

ROBBINS LINK Robbins Basic Pathology — *Liver and Gallbladder*: cirrhosis, portal hypertension, liver failure, alcoholic and non-alcoholic fatty liver disease, haemochromatosis, Wilson disease, α1-antitrypsin deficiency, hepatocellular carcinoma.



Chapter at a glance. Injury, scar, pressure, failure, cancer.

6.1 Cirrhosis

DEFINITION

Cirrhosis is a **diffuse** process in which the normal architecture of the **whole liver** is replaced by **regenerative parenchymal nodules** surrounded by **bridging fibrous septa**. It is the end stage of many chronic liver diseases.

CONCEPT CLEAR

All three features are needed. (1) **Bridging fibrous septa**, delicate bands or broad scars linking portal tracts to each other and to central veins; (2) **parenchymal nodules** of regenerating hepatocytes trapped by the septa; (3) **diffuse** involvement of the entire liver. A single scar is not cirrhosis; nodules without fibrosis are not cirrhosis.

CAUSES

- **Chronic viral hepatitis**: HCV, HBV (± HDV)
- **Alcoholic liver disease**; **non-alcoholic fatty liver disease** (NAFLD)
- Biliary disease: primary biliary cholangitis, primary sclerosing cholangitis, long-standing duct obstruction
- Metabolic: **haemochromatosis**, **Wilson disease**, **α1-antitrypsin deficiency**
- Autoimmune hepatitis; drugs (e.g. methotrexate)
- Chronic right heart failure (cardiac fibrosis)
- **Cryptogenic**: no cause found (many are burnt-out NAFLD)

MEMORY HOOK

HEPATIC

Hepatitis viruses B, C (± D) · **E**thanol and Excess fat (NAFLD) · **P**rimary biliary cholangitis / PSC · **A**utoimmune hepatitis and α1-antitrypsin · **T**oxins and drugs · **I**ron and copper (haemochromatosis, Wilson) · **C**ardiac and Cryptogenic

PAKISTAN FOCUS

Hepatitis C is the leading cause of cirrhosis and hepatocellular carcinoma in Pakistan, with hepatitis B second (prevalence figures in Chapter 5). With about one in four adults diabetic, NAFLD is a growing contributor.

Pathogenesis: the stellate cell is the scar-maker

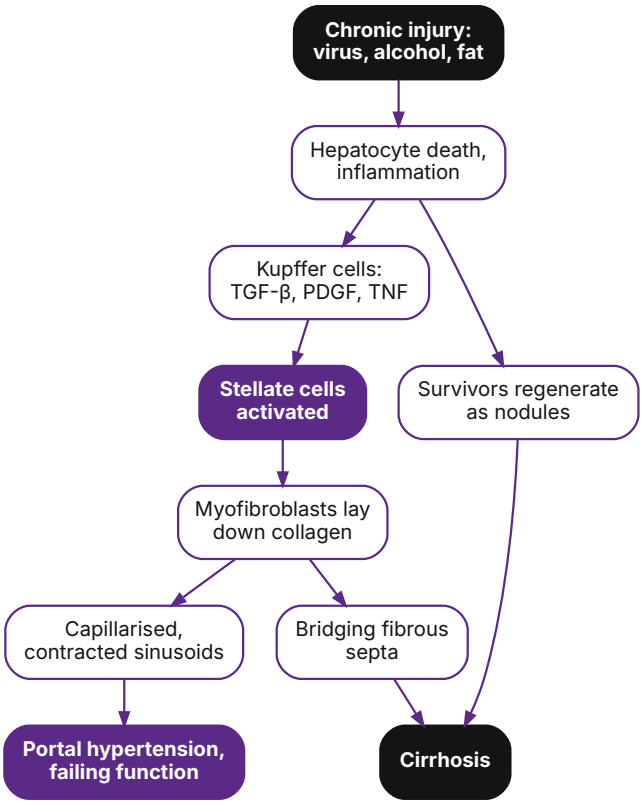


Figure 6.1 · Pathogenesis of cirrhosis.

Hepatic stellate cells (Ito cells) lie in the **space of Disse** and normally store **vitamin A**. Injury activates them through cytokines from Kupffer cells (**TGF-β**, PDGF, TNF), reactive oxygen species and damaged matrix. They lose their vitamin A and become **contractile myofibroblasts** that lay down **type I and III collagen**.

- **Collagen in the space of Disse** and loss of endothelial fenestrae ("**capillarisation**" of sinusoids) block exchange between plasma and hepatocytes → **liver failure**
- **Contraction** of myofibroblasts and septal scarring raise sinusoidal resistance → **portal hypertension**
- Vascular **shunts** in the septa divert blood past hepatocytes
- Surviving hepatocytes regenerate as **nodules**, helped by progenitor cells (**ductular reaction**)
- Fibrosis can slowly **regress** if the cause is removed (e.g. HCV cured)

GROSS

- Liver usually **shrunken**, firm, with a **nodular surface**
- **Micronodular**: uniform nodules < 3 mm (alcohol, haemochromatosis, biliary)
- **Macronodular**: nodules > 3 mm, varying sizes, broad scars (post-viral)
- Cut surface: nodules separated by **grey-white fibrous bands**; nodules yellow (fat), green (bile) or rusty brown (iron)

MICROSCOPY

- **Regenerative nodules** of hepatocytes without normal lobular architecture or central veins
- Nodules encircled by **fibrous septa** joining portal tracts and central veins
- Septa contain lymphocytes and **ductular reaction**
- Clues to the cause: Mallory–Denk bodies (alcohol, NAFLD), ground-glass cells (HBV), lymphoid aggregates (HCV), iron (Prussian blue), PAS+ globules (α1-AT)

Outcome: many patients are silent for years (compensated). Decompensation brings anorexia, weight loss, jaundice, ascites, variceal bleeding and encephalopathy. Death comes from **progressive liver failure**, a **complication of portal hypertension**, or **hepatocellular carcinoma**.

EXAM TRAP

Micro- vs macronodular is a weak classification. One cause can produce both: alcoholic cirrhosis begins micronodular and becomes macronodular as nodules keep growing. Robbins therefore classifies cirrhosis by **cause**; quote the size cut-off (3 mm) but give the aetiology.

6.2 Portal hypertension

Portal pressure rises when **resistance** to portal flow increases or, less often, when **flow** increases. In cirrhosis both happen: the scarred, contracted sinusoids resist flow, and nitric oxide-mediated dilatation of splanchnic arteries pours more blood into the portal system (a hyperdynamic circulation).

CAUSES OF PORTAL HYPERTENSION BY SITE

Site	Causes
Pre-hepatic	Portal vein thrombosis (sepsis, hypercoagulable state, tumour), narrowing of the portal vein before the liver, massive splenomegaly
Intrahepatic	**Cirrhosis** (commonest), schistosomiasis, massive fatty change, granulomatous disease (sarcoidosis), nodular regenerative hyperplasia
Post-hepatic	Severe right heart failure, constrictive pericarditis, **hepatic vein outflow obstruction** (Budd–Chiari syndrome)

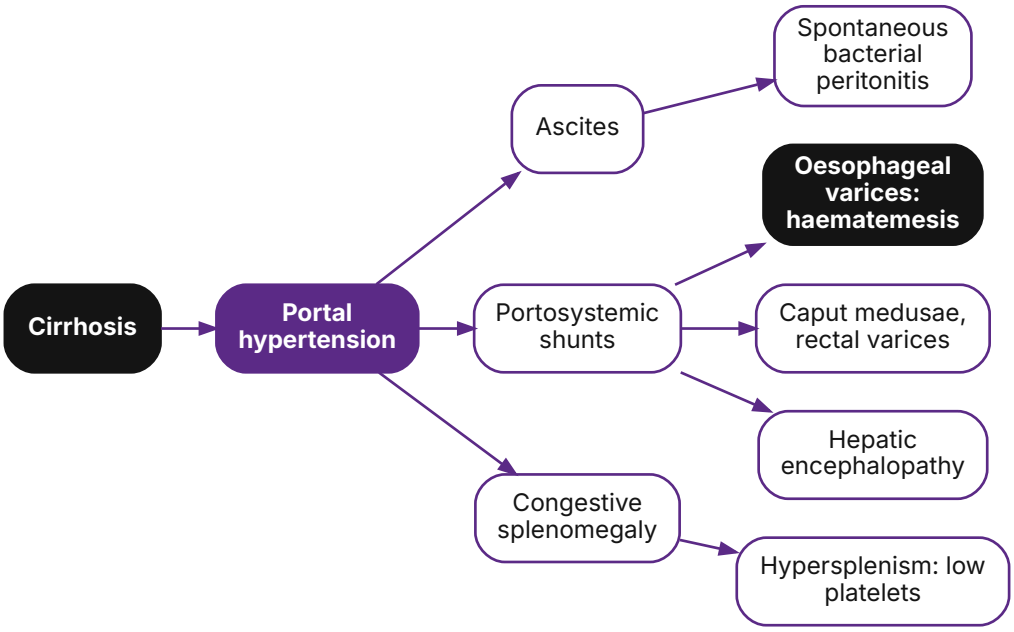


Figure 6.2 · Consequences of portal hypertension. Robbins' four: ascites, portosystemic shunts, congestive splenomegaly, hepatic encephalopathy.

WHY ASCITES FORMS

1. **Sinusoidal hypertension** pushes fluid into the space of Disse; hepatic lymph flow overwhelms the thoracic duct and lymph weeps into the peritoneum.
2. **Splanchnic vasodilatation** lowers effective arterial volume → RAAS and ADH → **sodium and water retention**.
3. **Hypoalbuminaemia** lowers plasma oncotic pressure.

Ascitic fluid is serous, low in protein; infection without an obvious source is **spontaneous bacterial peritonitis**.

PORTOSYSTEMIC SHUNT SITES

Site	Result
Oesophagogastric junction (left gastric → azygos)	**Oesophageal varices**: rupture → massive **haematemesis**, a leading cause of death
Rectum (superior → middle and inferior rectal veins)	Rectal varices
Umbilicus (paraumbilical veins)	**Caput medusae**
Retroperitoneum	Retroperitoneal collaterals

LAB VALUES

SAAG serum albumin − ascitic albumin

< 1.1 g/dL

peritoneal disease: tuberculosis, carcinomatosis, pancreatitis

≥ 1.1 g/dL

portal hypertension (cirrhosis, heart failure, Budd–Chiari)

SBP ascitic neutrophils ≥ 250/mm³

6.3 Hepatic failure

About **80–90%** of hepatic function must be lost before the liver fails. Failure may be **acute** (massive necrosis, Chapter 5), **chronic** (cirrhosis, the commonest route), **acute-on-chronic** (a new insult to a cirrhotic liver), or occur **without overt necrosis** when hepatocytes are poisoned (microvesicular fat in Reye syndrome and acute fatty liver of pregnancy).

CLINICAL FEATURES OF HEPATIC FAILURE AND THEIR MECHANISMS

Feature	Mechanism
Jaundice, cholestasis	Failure of bilirubin uptake, conjugation and excretion
Oedema, ascites	Hypoalbuminaemia (plus portal hypertension)
Bleeding, prolonged PT	Low factors II, VII, IX, X, V and fibrinogen; thrombocytopenia from hypersplenism
Palmar erythema, spider angiomas, gynaecomastia, testicular atrophy	**Hyperoestrogenism**: the failing liver cannot metabolise oestrogens
Hepatic encephalopathy: confusion, asterixis (flapping tremor), coma	Ammonia and other gut-derived toxins bypass the liver (failure + portosystemic shunts) and disturb neurotransmission
Fetor hepaticus	Musty breath from mercaptans
Hepatorenal syndrome	Splanchnic vasodilatation → intense renal vasoconstriction; kidneys structurally normal; oliguria, very low urine sodium
Hepatopulmonary syndrome; portopulmonary hypertension	Intrapulmonary vascular dilatation → hypoxaemia; or pulmonary arterial hypertension

6.4 Alcoholic and non-alcoholic fatty liver disease

Alcohol and the metabolic syndrome injure the liver in the same three overlapping stages. Only a minority progress: about **10–15%** of heavy drinkers develop cirrhosis.



Figure 6.3 · The fatty liver spectrum (alcoholic and non-alcoholic).

HOW ALCOHOL INJURES HEPATOCYTES

- **Acetaldehyde** (from alcohol dehydrogenase, CYP2E1) forms toxic protein adducts
- **CYP2E1** generates reactive oxygen species
- Excess **NADH** diverts metabolism to fat synthesis; lipoprotein export is impaired
- Gut **endotoxin** triggers TNF release from Kupffer cells

MOLECULAR KEY

Mallory–Denk bodies are tangled **keratin 8 and 18** intermediate filaments forming eosinophilic, ropy cytoplasmic inclusions in ballooned hepatocytes. Typical of alcoholic hepatitis but **not specific**: also NAFLD, Wilson disease, chronic cholestasis and HCC.

ALCOHOLIC VS NON-ALCOHOLIC FATTY LIVER DISEASE

Feature	Alcoholic liver disease	NAFLD / NASH
Cause	Chronic heavy drinking	**Insulin resistance**: obesity, type 2 diabetes, dyslipidaemia
Definition	—	Fat in > 5% of hepatocytes without significant alcohol use; NASH adds inflammation and ballooning
AST : ALT	**> 2 : 1**	Usually < 1 (ALT higher)
Morphology	Neutrophils and Mallory–Denk bodies prominent	Steatosis prominent; inflammation milder
Outcome	Alcoholic hepatitis, cirrhosis, HCC	Commonest chronic liver disease in many countries; cirrhosis, HCC

6.5 Hereditary haemochromatosis

DEFINITION

Haemochromatosis is excessive accumulation of body iron, most of it deposited in **parenchymal cells** of the liver, pancreas, heart, joints and endocrine glands. The **hereditary** form is an autosomal recessive disorder of **excess intestinal iron absorption**; the **secondary** form (haemosiderosis) follows parenteral iron, mainly **repeated transfusions** (β-thalassaemia major, Chapter 12), or ineffective erythropoiesis.

Total body iron is normally 2–6 g, and we absorb only 1–2 mg a day; there is no route for excreting excess iron. In hereditary haemochromatosis iron builds up by about 0.5–1 g a year and may exceed **50 g**. Disease usually appears in the **40s–50s**, after about 20 g has collected, and men are affected far more often (women lose iron through menstruation and pregnancy).

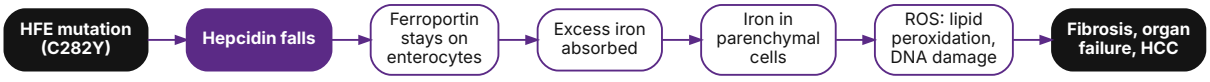


Figure 6.4 · Pathogenesis of hereditary haemochromatosis. Hepcidin is the brake on iron absorption; HFE mutation releases the brake.

MOLECULAR KEY

HFE lies on chromosome **6p** close to the HLA-A locus; most patients are homozygous for the **C282Y** mutation (cysteine → tyrosine at position 282). **Hepcidin**, made by the liver, binds the iron exporter **ferroportin** and makes the cell degrade it; less hepcidin means ferroportin keeps pumping iron out of enterocytes and macrophages into blood. Rarer forms: mutations of hepcidin itself or **haemojuvelin** (severe **juvenile haemochromatosis**), transferrin receptor 2, ferroportin.

GROSS

- **Liver**: dark rusty-brown, finely nodular (**micronodular pigment cirrhosis**)
- **Pancreas**: deeply pigmented, fibrotic
- **Heart**: enlarged, brown myocardium
- **Skin**: slate-grey to bronze

MICROSCOPY

- **Haemosiderin** granules (golden-brown) first in **periportal hepatocytes**, then the whole lobule, bile ducts and Kupffer cells
- **Prussian blue (Perls) stain** turns iron blue
- Little inflammation; fibrosis leads to cirrhosis
- Pancreas: iron in acini and **islets** → fibrosis; heart: iron in cardiac myocytes

Skin pigmentation is mainly **increased melanin** (with some dermal haemosiderin).

MEMORY HOOK

Haemochromatosis: the seven Cs

Cirrhosis · **C**arcinoma (HCC, risk about 200-fold) · **C**ardiomyopathy (failure, arrhythmia) · **C**arbohydrate intolerance (diabetes) · **C**olour (bronze skin) · **C**alcium pyrophosphate arthropathy · **C**astration (hypogonadism)

CLINICAL AND LAB PICTURE

- Classic but late triad: **micronodular cirrhosis, diabetes mellitus, skin pigmentation** ("**bronze diabetes**")
- **Transferrin saturation** high (the earliest clue), **ferritin** high, serum iron high; HFE genotyping; liver iron on biopsy or MRI
- **Treatment: repeated phlebotomy**. Started before cirrhosis, life expectancy is normal; screen first-degree relatives

EXAM TRAP

Hereditary vs secondary iron overload. In **hereditary** haemochromatosis iron enters through the gut and lands first in **parenchymal cells** (periportal hepatocytes). In **transfusional** overload, iron from destroyed red cells lands first in **macrophages (Kupffer cells)** and spleen, spilling into hepatocytes only when they are saturated.

6.6 Wilson disease and α1-antitrypsin deficiency

Two more inherited metabolic diseases end in cirrhosis. Learn them side by side with haemochromatosis: examiners love the comparison.

INHERITED METABOLIC LIVER DISEASES COMPARED

Feature	Haemochromatosis	Wilson disease	α1-antitrypsin deficiency
Inheritance, gene	AR; **HFE** (chr 6)	AR; **ATP7B** (chr 13)	AR (alleles codominant); **SERPINA1** (chr 14), **PiZZ**
Basic defect	Low hepcidin → excess iron absorption	Copper not excreted into bile nor built into **ceruloplasmin**	Misfolded Z protein **trapped in hepatocyte ER**; little reaches blood
Usual age	40s–50s, men	Children and young adults (5–40 years)	Neonates (cholestasis) to adults
Liver	Pigment cirrhosis; HCC	Fatty change, acute or chronic hepatitis, massive necrosis, cirrhosis	Neonatal hepatitis, chronic hepatitis, cirrhosis; HCC
Outside the liver	Pancreas, heart, skin, joints, gonads	**Brain** (basal ganglia, putamen): movement and psychiatric disorders; **Kayser–Fleischer rings** (copper in Descemet membrane)	**Panacinar emphysema** (unopposed neutrophil elastase), worse in smokers (Chapter 9)
Stain	**Prussian blue** (iron)	Rhodanine or orcein (copper)	**PAS-positive, diastase-resistant** globules in periportal hepatocytes
Key lab test	↑ Transferrin saturation, ↑ ferritin	**↓ Serum ceruloplasmin**, ↑ urinary copper; ↑ **hepatic copper** (most sensitive)	↓ Serum α1-AT; Pi typing
Treatment	Phlebotomy	Copper chelation (penicillamine), zinc; transplant	Liver transplant; stop smoking

6.7 Hepatocellular carcinoma

The commonest malignant tumours found in the liver are **metastases** (from colon, breast, lung, pancreas), which form multiple nodules that are often umbilicated. Among primary malignancies, **hepatocellular carcinoma (HCC)** dominates. Over 85% of cases arise in countries with high rates of chronic HBV infection (Asia, sub-Saharan Africa), where infection in infancy can produce HCC in young adults. Men are affected several times more often than women.

RISK FACTORS

- **Chronic HBV**: viral DNA integrates; **HBx** protein disturbs growth control; HCC can arise **without cirrhosis**
- **Chronic HCV**: mostly through cirrhosis
- **Alcohol**; **NAFLD**, obesity and diabetes
- **Aflatoxin B1** from *Aspergillus flavus* on badly stored grain and groundnuts: causes a characteristic **TP53 codon 249** mutation
- Haemochromatosis, α1-AT deficiency, tyrosinaemia
- **Cirrhosis of any cause**: most HCCs arise in cirrhotic livers

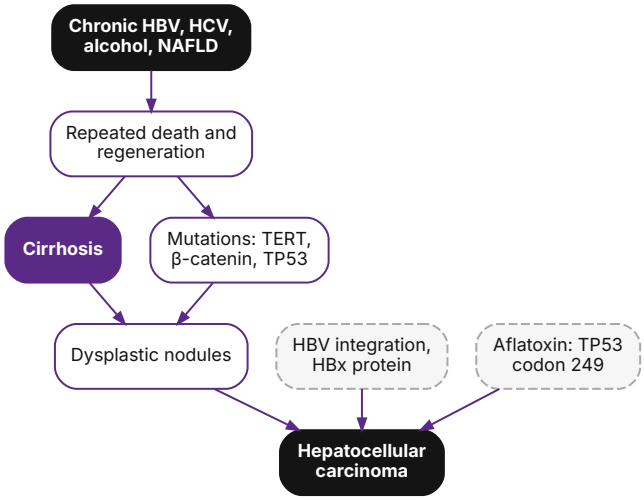


Figure 6.5 · Roads to hepatocellular carcinoma.

Precursors: in cirrhotic livers, **large-cell and small-cell change** (dysplasia) and **dysplastic nodules** (low- and high-grade) precede HCC. Common mutations activate **TERT** (telomerase) and **β-catenin** or inactivate **TP53**.

GROSS

- Three patterns: **unifocal** (often large) mass, **multifocal** nodules, or **diffusely infiltrative**
- Paler than the surrounding liver; soft; may be **bile-stained green**, haemorrhagic or necrotic
- Usually on a background of **cirrhosis**
- Strong tendency to **invade veins**: portal vein and hepatic veins/IVC, sometimes reaching the right atrium

MICROSCOPY

- **Trabecular** (cords several cells thick), **pseudoglandular** or solid sheets
- Polygonal cells resembling hepatocytes: **eosinophilic cytoplasm**, central nuclei, **prominent nucleoli**; pleomorphism and giant cells when poorly differentiated
- **Bile** in canaliculi or cells proves hepatocellular origin
- **No portal tracts**; endothelium-lined sinusoid-like vessels
- IHC: HepPar-1, arginase-1, glypican-3

LAB SPOTTER

Cirrhosis vs HCC in 10 seconds. Cirrhosis: round islands of liver cells, each ringed by pink-blue fibrous bands carrying lymphocytes and little ducts; no normal central veins. **HCC**: no fibrous rings and **no portal tracts**, just thick trabeculae of atypical liver-like cells with big nucleoli, separated by blood spaces; look for **green bile** and vascular invasion. Both together on one slide (HCC beside cirrhotic nodules) is a favourite.

Fibrolamellar carcinoma

A distinct variant of **young adults (20–40 years)**, with equal sex incidence and **no link to HBV, cirrhosis or other risk factors**. It forms a single, large, **hard, scirrhous** tumour with fibrous bands. Microscopy: large polygonal **oncocytic** cells (abundant pink, mitochondria-rich cytoplasm) with prominent nucleoli, separated by parallel **lamellae of dense collagen**. It carries a **DNAJB1–PRKACA** fusion gene; AFP is usually normal, and the outlook is better because the tumour is often resectable from a healthy liver.

CLINICAL FEATURES AND SPREAD

- Vague upper abdominal pain, malaise, weight loss; **hard, irregular hepatomegaly**
- In a cirrhotic: sudden decompensation, bloody ascites, variceal bleed
- Paraneoplastic: hypoglycaemia, erythrocytosis, hypercalcaemia
- Spread within the liver and along veins; distant spread mainly to **lungs**, late
- Death from cachexia, GI or variceal bleeding, liver failure with coma, or tumour rupture with haemorrhage

LAB VALUES

α-Fetoprotein (AFP) is raised in about half or more of patients with HCC, but it is **neither sensitive nor specific**: it also rises in yolk sac tumours, cirrhosis, chronic hepatitis, massive necrosis, normal pregnancy and fetal neural tube defects. Cirrhotic patients are screened with **ultrasound (± AFP) every 6 months**.

OTHER LIVER MASSES TO RECOGNISE

Cavernous haemangioma	Commonest benign liver tumour; red-blue, soft; must not be needle-biopsied blindly
Hepatocellular adenoma	Young women on **oral contraceptives**, anabolic steroids; can rupture and bleed; β-catenin-activated type may become malignant
Focal nodular hyperplasia	Well-demarcated nodule with a **central stellate scar**; not neoplastic
Cholangiocarcinoma	Bile-duct adenocarcinoma with desmoplasia; liver flukes (*Opisthorchis*, *Clonorchis*), primary sclerosing cholangitis
Hepatoblastoma	Commonest liver tumour of early childhood

CLINICAL CASE

A 58-year-old man with HCV cirrhosis known for 6 years now has weight loss, increasing abdominal girth and right upper quadrant pain. The liver is hard and irregular. AFP 1,200 ng/mL. Ultrasound: a 7 cm mass in the right lobe with thrombus in the portal vein.

Reasoning: cirrhosis + sudden decompensation + hard mass + very high AFP + portal vein invasion → **hepatocellular carcinoma** arising in HCV cirrhosis. The portal vein thrombus is tumour invading the vein and worsens his ascites and portal hypertension.

EXAM ZONE Past-Paper Pattern SEQs — Solved

PROF PATTERN **a)** Define cirrhosis and enumerate its causes. **b)** Describe its pathogenesis and morphology. **2 + 3 marks**

MODEL ANSWER PLAN

a) Diffuse process: bridging fibrous septa + regenerative nodules replacing the architecture of the whole liver. Causes: HBV/HCV, alcohol, NAFLD, biliary (PBC, PSC), haemochromatosis, Wilson, α1-AT deficiency, autoimmune, drugs, cardiac, cryptogenic. **b)** Injury → Kupffer cell cytokines (TGF-β, PDGF) → stellate cells become myofibroblasts → collagen I/III in space of Disse, capillarisation, septa; regeneration → nodules. Gross: small, firm, nodular; micro- (< 3 mm) vs macronodular. Micro: nodules ringed by fibrous septa with inflammation and ductular reaction.

Score booster: draw Figure 6.1 and name the three causes of death (liver failure, portal hypertension, HCC).

CLASSIC **a)** Classify the causes of portal hypertension. **b)** Describe its consequences, with the mechanism of ascites. **2 + 3 marks**

MODEL ANSWER PLAN

a) Pre-hepatic (portal vein thrombosis), intrahepatic (cirrhosis, schistosomiasis), post-hepatic (right heart failure, Budd–Chiari). **b)** Four consequences: ascites (sinusoidal hypertension and lymph leak, splanchnic vasodilatation → RAAS, hypoalbuminaemia), portosystemic shunts (oesophageal varices → haematemesis, caput medusae, rectal varices), congestive splenomegaly → hypersplenism, hepatic encephalopathy.

MODULE PATTERN A 50-year-old man has diabetes, darkening of the skin, hepatomegaly and pain in the knuckles. Serum ferritin 2,100 ng/mL; transferrin saturation 86%. **a)** What is the likely diagnosis? **b)** Explain its pathogenesis. **c)** Describe the liver morphology and list two other complications. **1 + 2 + 2 marks**

MODEL ANSWER PLAN

a) Hereditary haemochromatosis ("bronze diabetes"). **b)** AR, HFE C282Y → low hepcidin → ferroportin active → excess iron absorption → parenchymal deposition → ROS, lipid peroxidation, stellate cell activation → fibrosis. **c)** Rusty-brown micronodular cirrhosis; haemosiderin in periportal hepatocytes, Prussian blue positive. Complications: HCC (about 200-fold), cardiomyopathy, hypogonadism, pseudogout.

Score booster: name the treatment (phlebotomy) and advise screening of siblings.

PROF PATTERN **a)** Enumerate the risk factors for hepatocellular carcinoma. **b)** Describe its morphology and the value of serum AFP. **2 + 3 marks**

MODEL ANSWER PLAN

a) HBV (integration, HBx; even without cirrhosis), HCV, alcohol, NAFLD, aflatoxin B1 (TP53 codon 249), haemochromatosis, α1-AT, tyrosinaemia; cirrhosis of any cause. **b)** Gross: unifocal, multifocal or diffuse; pale or bile-green, haemorrhagic; portal vein invasion. Micro: trabecular/pseudoglandular hepatocyte-like cells with prominent nucleoli, bile, no portal tracts. Fibrolamellar variant: young, no cirrhosis, oncocytic cells in collagen lamellae. AFP raised in about half or more; false positives (yolk sac tumour, pregnancy, cirrhosis); used for screening with ultrasound.

BCQs One-Best-Answer Practice

1 The cell chiefly responsible for collagen deposition in cirrhosis is the:

A. Kupffer cell
B. Hepatic stellate cell
C. Sinusoidal endothelial cell
D. Cholangiocyte
E. Hepatocyte

2 A cirrhotic man vomits a large amount of fresh blood. The most likely source is:

A. Oesophageal varices
B. Mallory–Weiss tear
C. Duodenal ulcer
D. Gastric carcinoma
E. Rectal varices

3 Spider angiomas, palmar erythema and gynaecomastia in a cirrhotic man are due to:

A. Hypoalbuminaemia
B. Portal hypertension
C. Hyperammonaemia
D. Hyperoestrogenism
E. Vitamin K deficiency

4 A cirrhotic patient becomes oliguric. Urine sodium is under 10 mmol/L and urine sediment is bland; renal biopsy is normal. The diagnosis is:

A. Acute tubular necrosis
B. Membranoproliferative GN
C. Hepatorenal syndrome
D. Renal vein thrombosis
E. Pyelonephritis

5 Ascitic fluid in a patient with abdominal distension gives a serum–ascites albumin gradient of 1.8 g/dL. The most likely cause is:

A. Tuberculous peritonitis
B. Peritoneal carcinomatosis
C. Pancreatic ascites
D. Nephrotic syndrome
E. Cirrhosis with portal hypertension

6 A liver biopsy shows ballooned hepatocytes with ropy eosinophilic cytoplasmic inclusions surrounded by neutrophils. The inclusions are composed of:

A. HBsAg
B. Keratin 8 and 18
C. α1-antitrypsin
D. Copper-binding protein
E. Lipofuscin

7 Hereditary haemochromatosis caused by HFE mutation results in excess iron absorption because of:

A. Increased transferrin synthesis
B. Loss of ferroportin
C. Decreased hepcidin
D. Increased erythropoietin
E. Defective ceruloplasmin

8 The stain that best demonstrates haemosiderin in a liver biopsy is:

A. PAS with diastase
B. Rhodanine
C. Congo red
D. Prussian blue
E. Ziehl–Neelsen

9 A 15-year-old with tremor and chronic hepatitis has a golden-brown corneal ring. Which finding is expected?

A. Low serum ceruloplasmin
B. High transferrin saturation
C. PAS-positive globules in hepatocytes
D. Raised α-fetoprotein
E. Anti-mitochondrial antibody

10 A man with emphysema and cirrhosis has round PAS-positive, diastase-resistant globules in periportal hepatocytes. The disease is:

A. Haemochromatosis
B. α1-antitrypsin deficiency
C. Wilson disease
D. Glycogen storage disease
E. Alcoholic hepatitis

11 A specific TP53 codon 249 mutation in hepatocellular carcinoma points to exposure to:

A. Vinyl chloride
B. Thorotrast
C. Oral contraceptives
D. Alcohol
E. Aflatoxin B1

12 A 24-year-old woman has a large, hard liver mass in a non-cirrhotic liver; AFP is normal. Microscopy shows large eosinophilic cells separated by parallel bands of collagen. The diagnosis is:

A. Fibrolamellar carcinoma
B. Hepatoblastoma
C. Hepatocellular adenoma
D. Focal nodular hyperplasia
E. Cholangiocarcinoma

ANSWER KEY WITH ONE-LINE REASONS

1. **B** Activated stellate (Ito) cells become collagen-secreting myofibroblasts.
2. **A** Portosystemic shunting at the oesophagogastric junction; rupture causes massive haematemesis.
3. **D** The failing liver cannot metabolise oestrogens.
4. **C** Functional renal failure from renal vasoconstriction; the kidneys are structurally normal.
5. **E** SAAG ≥ 1.1 g/dL indicates portal hypertension.
6. **B** Mallory–Denk bodies: aggregated intermediate filaments, typical of alcoholic hepatitis.
7. **C** Low hepcidin leaves ferroportin active on enterocytes.
8. **D** Perls Prussian blue turns ferric iron blue.
9. **A** Wilson disease: Kayser–Fleischer ring, low ceruloplasmin, high hepatic copper.
10. **B** Misfolded PiZ protein retained in hepatocyte ER; low serum α1-AT causes emphysema.
11. **E** Aflatoxin from Aspergillus flavus on stored grain and nuts.
12. **A** Young adult, no cirrhosis, oncocytic cells in collagen lamellae; DNAJB1–PRKACA fusion.

VIVA Rapid-Fire Viva Voce

Q Define cirrhosis.	Diffuse replacement of liver architecture by regenerative nodules surrounded by bridging fibrous septa.
Q Size cut-off for micronodular cirrhosis?	Nodules under 3 mm; macronodular above 3 mm.
Q What are stellate cells?	Vitamin A-storing cells in the space of Disse that become collagen-producing myofibroblasts in injury.
Q What is capillarisation of sinusoids?	Loss of endothelial fenestrae with collagen in the space of Disse, blocking plasma–hepatocyte exchange.
Q Four consequences of portal hypertension?	Ascites, portosystemic shunts (varices), congestive splenomegaly, hepatic encephalopathy.
Q What is asterixis?	Flapping tremor of outstretched hands in hepatic encephalopathy.
Q Why gynaecomastia in cirrhosis?	Hyperoestrogenism: the failing liver cannot inactivate oestrogens.

Q Gene and chromosome in hereditary haemochromatosis?	HFE on chromosome 6p; C282Y mutation.
Q What is bronze diabetes?	Haemochromatosis: skin pigmentation plus diabetes from pancreatic iron, with cirrhosis.
Q Kayser–Fleischer ring?	Copper deposited in Descemet membrane of the cornea in Wilson disease.
Q Tumour marker for HCC?	α-Fetoprotein; raised in about half or more, but not specific.
Q Most specific microscopic proof that a tumour is hepatocellular?	Bile production by tumour cells (bile in canaliculi).

OSPE Practice Stations

STATION 1 · SLIDE (LAB SLIDE) — 2 minutes

TASK

Liver: normal lobules are lost. Rounded islands of hepatocytes with no central veins are completely encircled by fibrous bands that contain lymphocytes and small proliferating ductules. (a) Identify. (b) Two features. (c) Name the cell that makes the fibrous tissue and one complication.

ANSWER

(a) Cirrhosis of the liver. (b) Regenerative nodules; bridging fibrous septa with ductular reaction. (c) Hepatic stellate cell; portal hypertension / HCC.

STATION 2 · SLIDE (LAB SLIDE) — 2 minutes

TASK

Liver: thick trabeculae and pseudoglands of polygonal cells with pink cytoplasm, high nuclear–cytoplasmic ratio and prominent nucleoli, separated by vascular spaces. No portal tracts; green bile plugs in some lumens; adjacent liver shows fibrous septa. (a) Identify. (b) Two features. (c) Serum marker and one risk factor.

ANSWER

(a) Hepatocellular carcinoma on a background of cirrhosis. (b) Trabecular pattern of atypical hepatocyte-like cells; bile production; no portal tracts. (c) AFP; HBV/HCV/aflatoxin.

STATION 3 · GROSS SPECIMEN — 1 minute

TASK

A small, firm liver with a nodular outer surface. The cut surface shows nodules separated by grey-white bands, and one large, soft, pale-green, haemorrhagic mass; a tumour plug fills a branch of the portal vein. (a) Diagnosis? (b) Two gross features. (c) Most likely cause in Pakistan?

ANSWER

(a) HCC arising in a cirrhotic liver. (b) Nodular cirrhotic background; bile-stained haemorrhagic mass with portal vein invasion. (c) Chronic hepatitis C (HBV second).

STATION 4 · LAB DATA — 2 minutes

TASK

A man with shifting dullness. Serum albumin 2.9 g/dL; ascitic albumin 1.0 g/dL; ascitic neutrophils 40/mm³; platelets 78,000/mm³; PT prolonged; spleen palpable. (a) Calculate SAAG and interpret. (b) Explain the low platelets. (c) Is there infection?

ANSWER

(a) 1.9 g/dL (≥ 1.1): portal hypertension, most likely cirrhosis. (b) Congestive splenomegaly with hypersplenism. (c) No: spontaneous bacterial peritonitis needs ≥ 250 neutrophils/mm³.

RAPID REVISION

The Night-Before Page

15 ONE-LINERS

01 Cirrhosis = diffuse bridging fibrosis + regenerative nodules.

02 Micronodular < 3 mm, macronodular > 3 mm; classify by cause.

03 Stellate cells → myofibroblasts → collagen in the space of Disse.

04 Death in cirrhosis: liver failure, portal hypertension, HCC.

05 Portal hypertension: ascites, varices, splenomegaly, encephalopathy.

06 SAAG ≥ 1.1 g/dL = portal hypertension.

07 Hyperoestrogenism: spider angiomas, palmar erythema, gynaecomastia.

08 Hepatorenal syndrome: normal kidneys, very low urine sodium.

09 Mallory–Denk bodies = keratin 8/18; alcohol, NASH, Wilson.

10 AST:ALT > 2 suggests alcohol.

11 Haemochromatosis: HFE C282Y, low hepcidin, Prussian blue.

12 Bronze diabetes = cirrhosis + diabetes + pigmentation.

13 Wilson: ATP7B, low ceruloplasmin, K–F rings, basal ganglia.

14 α1-AT: PiZZ, PAS+ diastase-resistant globules, emphysema.

15 HCC: HBV, HCV, aflatoxin; AFP; vein invasion; fibrolamellar in the young.

MEMORY HOOKS

- **HEPATIC**: causes of cirrhosis
- **Seven Cs** of haemochromatosis: cirrhosis, carcinoma, cardiomyopathy, carbohydrate, colour, calcium pyrophosphate, castration
- **Iron, copper, protein**: chromosomes 6, 13, 14 (HFE, ATP7B, SERPINA1)
- **Blue, brown-gold, pink**: Prussian blue iron, K–F ring copper, PAS globules

TEST YOURSELF: FILL THE GAPS

1. Stellate cells normally store vitamin _____
2. Paraumbilical shunts cause _____.
3. Hepcidin degrades _____.
4. Wilson gene: _____.
5. Aflatoxin mutates _____ codon 249.
6. Fibrolamellar fusion gene: DNAJB1–_____.

Answers: A · caput medusae · ferroportin · ATP7B · TP53 · PRKACA

DRAW FROM MEMORY

The portal hypertension flowchart with its four consequences (Figure 6.2).

DRAW FROM MEMORY

A cirrhotic nodule ringed by a septum, with a stellate cell laying collagen in the space of Disse.

SPACED REVISION ☐ Day 1 ☐ Day 3 ☐ Day 7 ☐ Day 14 ☐ Day 30

07

Gallbladder & Pancreas: Gallstones, Cholecystitis & Pancreatitis

A stone no bigger than a pea can sit quietly in the gallbladder for decades, or slip down to the ampulla one evening and set the pancreas digesting itself. Follow the stone and most of this chapter falls into place.

BY THE END OF THIS CHAPTER YOU CAN

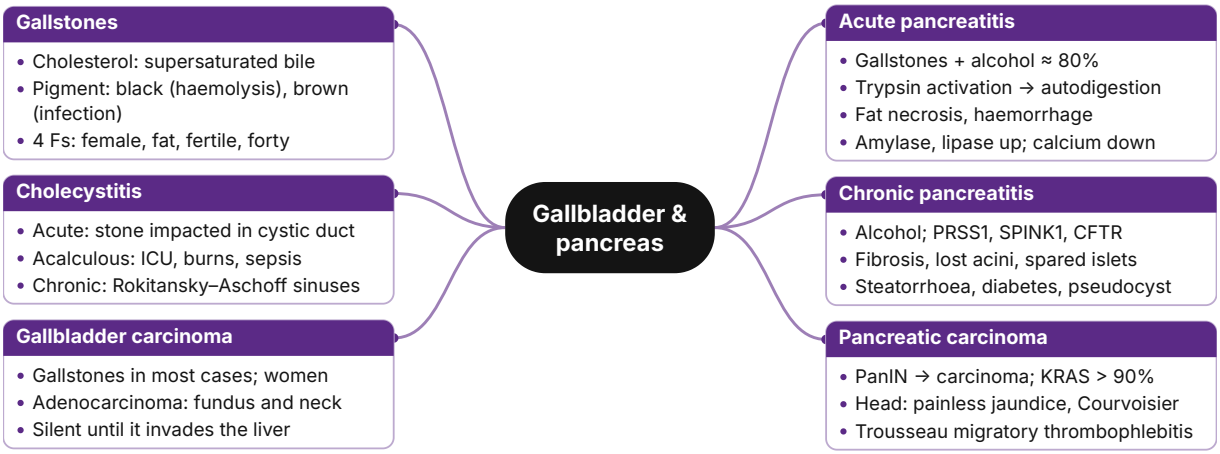
1. Classify gallstones and explain how cholesterol and pigment stones form
2. Describe the pathogenesis and morphology of acute and chronic cholecystitis and list the complications of gallstones
3. Recognise carcinoma of the gallbladder and its link with gallstones
4. Explain the pathogenesis of acute pancreatitis from enzyme activation to fat necrosis, haemorrhage and shock
5. Describe the causes, morphology and complications of chronic pancreatitis and pseudocyst
6. Outline pancreatic carcinoma: precursor lesions, KRAS, Trousseau sign and Courvoisier law

EXAM RADAR

Theory SEQs	★★★
BCQs / MCQs	★★★
Viva voce	★★☆
OSPE / slides	★★☆

Acute pancreatitis (causes, pathogenesis, morphology) and gallstones are classic long questions; chronic pancreatitis is a lab slide.

ROBBINS LINK Robbins Basic Pathology — *Liver and Gallbladder*: cholelithiasis, cholecystitis, gallbladder carcinoma; *Pancreas*: acute and chronic pancreatitis, pseudocysts, pancreatic carcinoma.



Chapter at a glance. One stone, two organs, six ideas.

7.1 Gallstones (cholelithiasis)

Bile is a clever solvent. Water-insoluble **cholesterol** is kept in solution only because **bile salts** and **lecithin** wrap it into micelles and vesicles. Tip the balance (too much cholesterol, too little bile salt) and cholesterol crystallises. Pigment stones follow the same logic with insoluble **unconjugated bilirubin**. About 10–20% of adults in high-income countries carry gallstones, most without knowing it.

CHOLESTEROL VS PIGMENT GALLSTONES

Feature	Cholesterol stones	Pigment stones
Frequency	Most stones in Western populations (about 80%)	Commoner in Asia and in rural populations
Composition	Cholesterol monohydrate ("mixed" stones add calcium salts); form only in the **gallbladder**	Calcium salts of **unconjugated bilirubin**; form anywhere in the biliary tree
Gross	Pale yellow, hard, round–ovoid; often multiple and **faceted**; cut surface shows **radiating crystals**	**Black**: small, numerous, crumble easily. **Brown**: few, soft, greasy, soap-like

Feature	Cholesterol stones	Pigment stones
X-ray	Mostly **radiolucent** (up to 20% opaque from calcium carbonate)	Black: 50–75% **radiopaque**; brown: radiolucent
Main causes	Hormones, obesity, age, stasis (below)	Black: **chronic haemolysis** (spherocytosis, sickle cell), cirrhosis, ileal disease. Brown: **infected ducts** (E. coli, Ascaris, liver flukes)

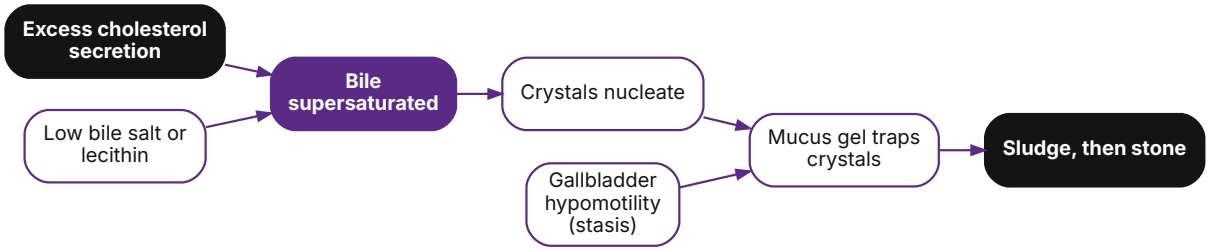


Figure 7.1 · How a cholesterol stone forms. Four conditions must coincide.

RISK FACTORS FOR CHOLESTEROL STONES

- **Female sex hormones:** female sex, pregnancy, oral contraceptives
- **Obesity**, insulin resistance, **rapid weight loss**
- **Advancing age**; Native and Mexican Americans, northern Europeans
- **Stasis:** pregnancy, parenteral nutrition, spinal cord injury
- Hyperlipidaemia, clofibrate; bile acid defects; **ABCG8** variants

MEMORY HOOK

The 4 Fs: Female, Fat, Fertile, Forty

Some teachers add **Fair** and **Flatulent** (6 Fs). Flatulent dyspepsia is a symptom, not a cause; in an answer, list the 4 Fs and then the risk factors alongside.

Clinical features and complications of gallstones

About 70–80% of people with stones remain asymptomatic for life. Pain begins when a stone obstructs the outlet: **biliary colic**, a severe, steady right-upper-quadrant or epigastric pain, often after a fatty meal, radiating to the right shoulder.

- **Acute and chronic cholecystitis**; empyema, gangrene, perforation
- **Choledocholithiasis** → obstructive jaundice, **ascending cholangitis**
- **Acute pancreatitis** (stone at the ampulla)
- **Mucocele (hydrops):** obstructed gallbladder fills with clear mucus
- **Gallstone ileus:** a large stone erodes into the bowel (cholecystoenteric fistula) and impacts at the ileocaecal valve
- **Carcinoma of the gallbladder** (long-term risk)

7.2 Cholecystitis

Acute calculous cholecystitis

Acute inflammation of a gallbladder that contains stones, triggered in about 90% of cases by **obstruction of the neck or cystic duct**. It is the commonest complication of gallstones and the commonest reason for emergency cholecystectomy.

CONCEPT CLEAR

Chemical first, bacterial later. Trapped bile is the first irritant: mucosal phospholipases turn lecithin into toxic **lysolecithin**, the mucus barrier is stripped and prostaglandins inflame the wall. The distended gallbladder squeezes its own blood supply, and only then do gut bacteria (E. coli, enterococci) invade. Early attacks may be sterile; neglected ones become gangrenous.

Acute acalculous cholecystitis

About **5–12%** of gallbladders removed for acute cholecystitis contain no stones. It strikes **critically ill** patients: major non-biliary surgery, severe trauma, **burns**, sepsis, prolonged parenteral nutrition, the postpartum state. The mechanism is **ischaemia** (the cystic artery is an end artery) plus stasis. Gangrene and perforation are commoner than in calculous disease, and the diagnosis is easily missed in a sedated patient.

Chronic cholecystitis

May follow repeated acute attacks but often arises without them; **stones are present in over 90%**. Bacteria grow in only about a third, so supersaturated bile itself seems to drive the inflammation.

ACUTE VS CHRONIC CHOLECYSTITIS: MORPHOLOGY

Feature	Acute cholecystitis	Chronic cholecystitis
Size, serosa	**Enlarged, tense**, bright red or blotchy violet-green-black (subserosal haemorrhage); **fibrinous to fibrinopurulent** serosal exudate	Contracted, normal or enlarged; serosa smooth or with fibrous adhesions
Wall	Oedematous, thick; **green-black necrotic** if gangrenous. Lumen: turbid bile or pus (**empyema**); stone impacted in neck	**Thick, grey-white, fibrotic**; may calcify (**porcelain gallbladder**)
Microscopy	Oedema, congestion, **neutrophils**, mucosal ulceration, necrosis	**Lymphocytes, plasma cells, macrophages**; subserosal fibrosis; **Rokitansky–Aschoff sinuses** (mucosal outpouchings through the muscle coat)

Clinical: acute attacks give progressive RUQ pain, fever, vomiting, leukocytosis and a positive Murphy sign (jaundice suggests a duct stone); chronic disease gives recurrent colic or nothing. **Variants: porcelain gallbladder** (dystrophic calcification of the wall; raised cancer risk) · **xanthogranulomatous cholecystitis** (nodular wall packed with lipid-laden macrophages; mimics carcinoma) · **hydrops** (atrophic, obstructed gallbladder filled with clear secretions).

7.3 Carcinoma of the gallbladder

The commonest malignancy of the extrahepatic biliary tract, typically in older women. The key risk factor is **gallstones**, present in the great majority (up to about 95%): years of stone-induced inflammation drive metaplasia, dysplasia and carcinoma. **Porcelain gallbladder** and chronic biliary infection add risk.

GROSS

- Commonest in the **fundus and neck**
- **Infiltrating** (commoner): diffuse, firm, grey-white thickening of the wall
- **Exophytic**: cauliflower-like mass growing into the lumen

MICROSCOPY

- **Adenocarcinoma** in most (papillary to poorly differentiated glands in desmoplastic stroma); few adenosquamous or squamous
- Direct invasion of the **liver**, cystic duct and portal nodes is usual by diagnosis

Clinical: insidious, mimicking gallstone disease (RUQ pain, jaundice, anorexia). Prognosis is poor; the lucky few are found incidentally at cholecystectomy.

PAKISTAN FOCUS

Gallbladder carcinoma is relatively common in Pakistan and northern India, as part of a regional belt of high incidence. It is strongly gallstone-associated and more frequent in women. Because early cancers are often found unexpectedly, every gallbladder removed for "stones" should be sent for histopathology.

7.4 Acute pancreatitis

DEFINITION

Acute pancreatitis is a **reversible** inflammatory disorder of the pancreas whose severity ranges from interstitial oedema with focal fat necrosis to widespread **haemorrhagic parenchymal necrosis**. Its basis is **autodigestion** of the gland by inappropriately activated pancreatic enzymes.

Biliary tract disease (**gallstones**) and **alcohol** together cause about **80%** of cases; about 5% of people with gallstones develop pancreatitis, and 10–20% of cases are idiopathic. Robbins groups the rest as **metabolic** (hypertriglyceridaemia, hypercalcaemia, drugs), **genetic** (PRSS1 cationic trypsinogen, gain of function; SPINK1 trypsin inhibitor, loss of function; CFTR), **mechanical** (trauma, ERCP, pancreas divisum, periampullary tumour, Ascaris), **vascular** (shock, atheroembolism, vasculitis) and **infectious** (mumps, coxsackievirus).

MEMORY HOOK

GET SMASHED

Gallstones · **E**thanol · **T**rauma · **S**teroids · **M**umps · **A**utoimmune · **S**corpion venom · **H**ypercalcaemia, **H**ypertriglyceridaemia · **E**RCP · **D**rugs (azathioprine, thiazides, valproate)

CONCEPT CLEAR

Trypsin is the master switch. The pancreas normally protects itself by making enzymes as inactive zymogens, packaging them in granules away from lysosomes, and carrying a trypsin inhibitor (SPINK1). Once trypsinogen is activated inside the gland, trypsin activates everything else: prophospholipase, proelastase, kallikrein, the clotting and complement cascades. A single misplaced activation therefore becomes a chain reaction. This is also why a gain-of-function **PRSS1** mutation (trypsin that resists inactivation) or loss of **SPINK1** causes hereditary pancreatitis.

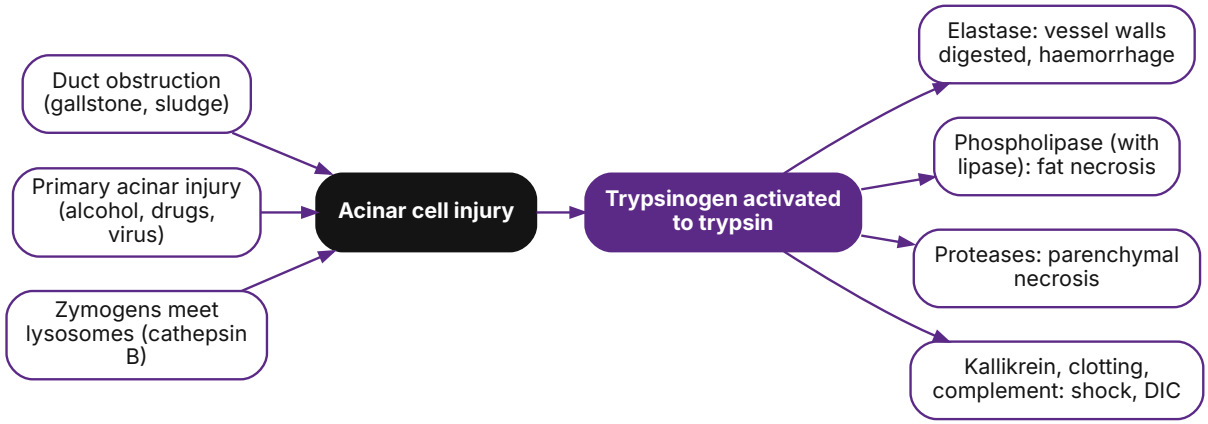


Figure 7.2 · Pathogenesis of acute pancreatitis. Three ways in, one chain reaction, four kinds of damage.

Alcohol contracts the sphincter of Oddi, is toxic to acinar cells and forms **protein plugs** in small ducts. Duct obstruction acts through oedema, enzyme leak and ischaemia.

GROSS

- **Mild (interstitial):** oedematous gland with focal fat necrosis
- **Severe (necrotising/haemorrhagic): red-black haemorrhage** mixed with **chalky yellow-white foci of fat necrosis**
- Fat necrosis also in **omentum, mesentery**, even subcutaneous fat
- Peritoneum: turbid, brown-tinged fluid with fat globules

MICROSCOPY

- Microvascular leakage → **interstitial oedema**
- **Enzymatic fat necrosis**: ghost outlines of fat cells with **basophilic calcium soap** deposits (saponification)
- Acute inflammatory reaction (neutrophils) around necrosis
- Proteolytic destruction of acini, ducts and islets; necrosis of vessel walls → **haemorrhage**

CLINICAL FEATURES AND LABORATORY FINDINGS

- **Severe, constant upper abdominal pain radiating to the back**, often after a heavy meal or drinking bout
- Severe cases: acute abdomen, **shock** (fluid sequestration, haemorrhage, cytokines, endotoxaemia); flank (Grey Turner) or periumbilical (Cullen) bruising

COMPLICATIONS

- **Systemic:** shock, **DIC**, **ARDS**, acute tubular necrosis, SIRS
- **Local:** infected necrosis and **pancreatic abscess** (gram-negative bacteria), **pseudocyst**, duodenal obstruction

- **Serum amylase** rises within 24 h and falls in a few days; **lipase** rises a little later, stays high longer and is more specific
- **Hypocalcaemia** (calcium trapped in fat necrosis), **hyperglycaemia**, leukocytosis, raised bilirubin
- **Diagnosis:** typical pain plus amylase or lipase > 3 × upper limit, or contrast CT showing a swollen, necrotic gland
- Mild attacks settle within days; severe necrotising disease still carries a significant mortality

Pancreatic pseudocyst

A collection of necrotic, haemorrhagic, enzyme-rich fluid walled off by **granulation and fibrous tissue without an epithelial lining** (hence "pseudo"), usually in the lesser sac after acute (especially alcoholic) pancreatitis or trauma. Pseudocysts make up about three-quarters of pancreatic cysts. They may resolve, or become **infected**, bleed, rupture or compress neighbouring organs.

7.5 Chronic pancreatitis

DEFINITION

Chronic pancreatitis is long-standing inflammation with **irreversible destruction of the exocrine parenchyma**, **fibrosis** and, in late stages, loss of endocrine tissue. Irreversibility is what separates it from acute pancreatitis.

- **Long-term alcohol abuse**: commonest; middle-aged men
- Long-standing **duct obstruction**: stones, tumours, pancreas divisum
- **Hereditary:** PRSS1, SPINK1; **CFTR** mutations
- **Tropical pancreatitis** (Asia, Africa; some SPINK1): young, large ductal stones
- **Autoimmune** (IgG4-related) pancreatitis
- A sizeable fraction are **idiopathic**

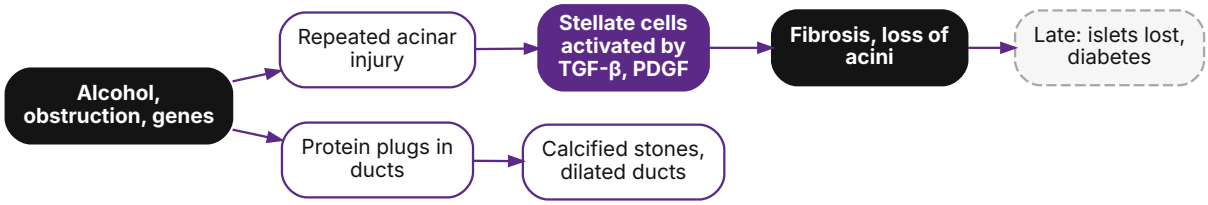


Figure 7.3 · The necrosis–fibrosis sequence of chronic pancreatitis.

GROSS

- Gland **hard, shrunken**, nodular
- **Dilated ducts** with visible **calcified concretions**
- Pseudocysts may be attached

MICROSCOPY

- **Parenchymal fibrosis**; acini reduced in number and size
- **Relative sparing of islets**, which cluster and may look enlarged within the scar
- Variably **dilated ducts** with eosinophilic **protein plugs** or stones; ductal epithelium atrophic, hyperplastic or squamous
- Chronic inflammatory cells around lobules and ducts

Autoimmune (IgG4) pancreatitis: duct-centred lymphoplasmacytic infiltrate, storiform fibrosis, venulitis; steroid-responsive; can mimic carcinoma.

CLINICAL FEATURES AND COMPLICATIONS

- Recurrent or persistent abdominal pain; some are silent
- **Exocrine insufficiency** → malabsorption, **steatorrhoea**, weight loss, fat-soluble vitamin deficiency
- **Diabetes mellitus** (late, once islets are destroyed)
- **Pseudocysts** (about 10%); raised risk of **carcinoma**
- Calcification on X-ray or CT is characteristic; amylase may be normal in a "burnt-out" gland

LAB SPOTTER

Chronic pancreatitis slide in 10 seconds: (1) Lobules are **shrunken** and separated by broad bands of **pink fibrous tissue**. (2) **Acini are scanty**; what survives are **islets of Langerhans**, pale islands stranded in the scar. (3) **Ducts are dilated**, some with pink inspissated plugs. (4) Lymphocytes and plasma cells sprinkle the fibrosis. No neutrophil-rich necrosis: that would be acute.

ACUTE VS CHRONIC PANCREATITIS

Feature	Acute pancreatitis	Chronic pancreatitis
Nature	Reversible if patient survives	**Irreversible** loss of function
Commonest causes	Gallstones and alcohol	Alcohol; duct obstruction; genetic
Morphology	Oedema, **fat necrosis**, haemorrhage, neutrophils	**Fibrosis**, acinar loss, spared islets, ductal calculi
Amylase / lipase	**Markedly raised**; low serum calcium	Mildly raised or normal; calcification on imaging
Key complications	Shock, DIC, ARDS, abscess, pseudocyst	Malabsorption, diabetes, pseudocyst, carcinoma

7.6 Carcinoma of the pancreas

Pancreatic ductal adenocarcinoma is silent until advanced, and fewer than about one in ten patients survive five years. It arises from **precursor lesions**: usually microscopic **pancreatic intraepithelial neoplasia (PanIN)**, less often cystic precursors (IPMN, mucinous cystic neoplasm).

MOLECULAR KEY

The "big four" genes: KRAS (activating mutation, > 90%; the commonest oncogene) · CDKN2A/p16 (lost in most) · TP53 · SMAD4, accumulating stepwise from low-grade PanIN to invasive carcinoma. Inherited risk: **BRCA2**, PRSS1, CDKN2A (familial melanoma), Peutz–Jeghers. Strongest environmental risk: **smoking**; also chronic pancreatitis, diabetes, obesity.

- **Site:** about 60% **head**, 15% body, 5% tail, 20% diffuse
- **Gross:** hard, **stellate, grey-white**, poorly defined mass
- **Micro:** infiltrating glands, intense **desmoplasia**, **perineural** and vascular invasion
- **Head tumours:** block the distal bile duct → **painless progressive jaundice**, weight loss
- **Body/tail tumours:** silent; late back pain and liver metastases
- **Trousseau sign: migratory thrombophlebitis** from tumour procoagulants
- **CA19-9:** raised in many; not specific, not for screening; used to follow treatment

EXAM TRAP

Courvoisier law. In a jaundiced patient, a **palpable, non-tender gallbladder** is unlikely to be due to stones, because a stone-bearing gallbladder is chronically inflamed, fibrotic and cannot distend. It points to **malignant obstruction** of the distal bile duct, classically **carcinoma of the head of the pancreas**. Do not confuse it with the **Trousseau sign** of malignancy (migratory thrombophlebitis).

EXAM ZONE Past-Paper Pattern SEQs — Solved

PROF PATTERN **a)** Enumerate the causes of acute pancreatitis. **b)** Describe its pathogenesis and morphology. **2 + 3 marks**

MODEL ANSWER PLAN

a) Gallstones and alcohol (about 80%); metabolic (hypertriglyceridaemia, hypercalcaemia, drugs); genetic (PRSS1, SPINK1, CFTR); mechanical (trauma, ERCP); vascular (shock, vasculitis); infections (mumps); idiopathic 10–20%.

b) Duct obstruction, acinar injury or misrouted zymogens → trypsin activated → phospholipase, elastase, kallikrein, clotting, complement → fat necrosis, haemorrhage, parenchymal necrosis, shock/DIC. Morphology: oedema and focal fat necrosis (mild); red-black haemorrhage with chalky fat necrosis (severe); micro: calcium soaps, neutrophils, necrotic vessels.

Score booster: draw Figure 7.2 and add one line on why calcium falls (soaps). Examiners reward the link between enzyme and lesion.

MODULE PATTERN A 40-year-old obese mother of five has had right upper abdominal pain after fatty meals for months. She now has fever, vomiting and RUQ tenderness. Ultrasound: multiple stones and a thickened gallbladder wall. **a)** Diagnosis? **b)** Classify gallstones and give the risk factors for her type. **c)** List four complications of gallstones. **1 + 2 + 2 marks**

MODEL ANSWER PLAN

a) Acute calculous cholecystitis with cholelithiasis. **b)** Cholesterol (pure/mixed) vs pigment (black, brown); risk: female hormones, obesity, multiparity, age, rapid weight loss, stasis. **c)** Empyema/perforation, duct stone with cholangitis and jaundice, acute pancreatitis, gallstone ileus, mucocele, carcinoma.

PROF PATTERN **a)** Define chronic pancreatitis and list its causes. **b)** Describe its morphology and complications. **2 + 3 marks**

MODEL ANSWER PLAN

a) Irreversible exocrine destruction with fibrosis and late endocrine loss. Causes: alcohol, duct obstruction, hereditary (PRSS1, SPINK1, CFTR), tropical, autoimmune (IgG4), idiopathic. **b)** Hard gland, dilated ducts, calculi; micro fibrosis, loss of acini, spared islets, protein plugs, chronic inflammation. Complications: steatorrhoea, diabetes, pseudocyst, pain, carcinoma.

Score booster: "relative sparing of islets" and "calcified concretions" are the words that earn the morphology mark.

CLASSIC Write short notes on: **a)** Pancreatic pseudocyst. **b)** Carcinoma of the head of the pancreas. **2 + 3 marks**

MODEL ANSWER PLAN

a) Enzyme-rich necrotic collection with fibrous/granulation wall, no epithelium; after acute (alcoholic) pancreatitis or trauma; lesser sac; complications: infection, haemorrhage, rupture, compression. **b)** Risk: smoking, chronic pancreatitis, diabetes, BRCA2; PanIN precursor; KRAS, CDKN2A, TP53, SMAD4; hard grey-white mass with desmoplasia and perineural invasion; painless jaundice, palpable gallbladder (Courvoisier), Trousseau sign, CA19-9; poor prognosis.

BCQs One-Best-Answer Practice

1 A 42-year-old obese mother of four has recurrent right upper quadrant pain after fatty meals. Ultrasound shows many faceted stones. These stones are most likely composed mainly of:

A. Calcium bilirubinate
B. Cholesterol monohydrate
C. Calcium oxalate
D. Uric acid
E. Magnesium ammonium phosphate

2 Black pigment gallstones are most typically associated with:

A. Ascaris in the bile ducts
B. Rapid weight loss
C. Oral contraceptive use
D. Chronic haemolysis, e.g. hereditary spherocytosis
E. Obesity

3 Rokitansky–Aschoff sinuses in a cholecystectomy specimen are a feature of:

A. Chronic cholecystitis
B. Acute acalculous cholecystitis
C. Choledochal cyst
D. Mucocele only
E. Gallbladder adenocarcinoma

4 A patient with 40% burns develops fever and RUQ tenderness on day 6 in ICU. Ultrasound shows a distended gallbladder without stones. The main mechanism is:

A. Stone impacted in the cystic duct
B. Salmonella carriage
C. Ischaemia and stasis of the gallbladder
D. IgG4-mediated inflammation
E. Tumour obstructing the neck

5 The single most important risk factor for carcinoma of the gallbladder is:

A. Hepatitis B infection
B. Alcohol abuse
C. Cigarette smoking
D. Haemochromatosis
E. Gallstones

6 In acute pancreatitis, the enzyme whose premature activation sets off activation of the other proenzymes is:

A. Lipase
B. Trypsin
C. Amylase
D. Elastase
E. Phospholipase A2

7 At autopsy of a patient who died of acute pancreatitis, chalky white foci are seen in the omentum. They are due to:

A. Caseous necrosis with dystrophic calcification
B. Coagulative necrosis
C. Amyloid deposits
D. Calcium soaps formed in enzymatic fat necrosis
E. Metastatic calcification from hypercalcaemia

8 A 50-year-old with severe acute pancreatitis has serum calcium of 7.0 mg/dL. This finding:

A. Reflects calcium trapped in fat necrosis and is a poor prognostic sign
B. Indicates primary hyperparathyroidism
C. Means the attack is mild
D. Is caused by vomiting alone
E. Suggests pancreatic carcinoma

9 Five weeks after alcoholic acute pancreatitis a man has an epigastric mass; CT shows a 7-cm fluid collection beside the pancreas. Its wall would show:

A. Mucinous columnar epithelium
B. Stratified squamous epithelium
C. Cuboidal epithelium with ovarian-type stroma
D. Glycogen-rich serous epithelium
E. Granulation and fibrous tissue without epithelium

10 Which histological picture is characteristic of chronic pancreatitis?

A. Haemorrhagic necrosis of acini with neutrophils
B. Fat necrosis with calcium soaps only
C. Loss of acini, fibrosis and relative sparing of islets
D. Selective destruction of islets with insulitis
E. Sheets of malignant glands with perineural invasion

11 The gene most frequently mutated in pancreatic ductal adenocarcinoma is:

A. TP53
B. KRAS
C. SMAD4
D. BRCA2
E. APC

12 A 65-year-old smoker has painless progressive jaundice, weight loss and a palpable non-tender gallbladder. The most likely cause is:

A. Stone in the common bile duct
B. Acute viral hepatitis
C. Chronic calculous cholecystitis
D. Carcinoma of the head of the pancreas
E. Gilbert syndrome

ANSWER KEY WITH ONE-LINE REASONS

1. **B** Cholesterol stones: female, fat, fertile, forty; faceted and multiple.
2. **D** Excess unconjugated bilirubin from haemolysis precipitates in sterile gallbladder bile.
3. **A** Mucosal outpouchings through the muscle wall reflect chronic inflammation and raised pressure.
4. **C** Acalculous cholecystitis: the cystic artery is an end artery; ischaemia leads to gangrene.
5. **E** Gallstones accompany the great majority of gallbladder carcinomas.
6. **B** Trypsin activates proelastase, prophospholipase, kallikrein, clotting and complement.
7. **D** Lipases release fatty acids that combine with calcium (saponification).
8. **A** Persistent hypocalcaemia signals extensive fat necrosis and a severe attack.
9. **E** Pseudocyst: no epithelial lining, by definition.
10. **C** Exocrine tissue is lost first; islets survive in the scar until late.
11. **B** Activating KRAS mutations occur in more than 90% of cases.
12. **D** Courvoisier law: a distended gallbladder with jaundice suggests malignant obstruction.

VIVA Rapid-Fire Viva Voce

Q What are the 4 Fs of gallstones?	Female, fat, fertile, forty: classic risk factors for cholesterol stones.
Q Which four conditions produce a cholesterol stone?	Cholesterol supersaturation, gallbladder hypomotility, crystal nucleation, and mucus hypersecretion trapping the crystals.
Q Black vs brown pigment stones?	Black: sterile gallbladder bile, haemolysis, often radiopaque. Brown: infected ducts, soft and greasy, radiolucent.
Q What are Rokitansky–Aschoff sinuses?	Outpouchings of gallbladder mucosa through the muscle coat, seen in chronic cholecystitis.
Q What is a porcelain gallbladder?	Chronic cholecystitis with calcification of the wall; it carries an increased risk of carcinoma.
Q What is gallstone ileus?	A large stone passes through a cholecystoenteric fistula and obstructs the small bowel, usually at the ileocaecal valve.
Q Why does fat necrosis look chalky white?	Released fatty acids bind calcium to form insoluble soaps (saponification).
Q Amylase or lipase: which is more specific?	Lipase: it stays raised longer and has fewer non-pancreatic sources.
Q Why is hypocalcaemia a bad sign in acute pancreatitis?	It reflects extensive fat necrosis trapping calcium, so the attack is severe.
Q Why is a pseudocyst called "pseudo"?	It has no epithelial lining; the wall is granulation and fibrous tissue.
Q Genes in hereditary pancreatitis?	PRSS1 (cationic trypsinogen, gain of function) and SPINK1 (trypsin inhibitor, loss of function).
Q What is Trousseau sign of malignancy?	Migratory thrombophlebitis caused by tumour procoagulants, classically with pancreatic carcinoma.

OSPE Practice Stations

STATION 1 · SLIDE — 2 minutes

TASK

Section of pancreas: lobules are shrunken and separated by dense fibrous bands containing lymphocytes and plasma cells. Few acini remain; islets of Langerhans lie in the fibrous tissue. Ducts are dilated and some contain pink inspissated material. (a) Identify the lesion. (b) Give two causes. (c) Name two complications.

ANSWER

(a) Chronic pancreatitis. (b) Chronic alcohol abuse; duct obstruction or hereditary (PRSS1/SPINK1/CFTR). (c) Malabsorption with steatorrhoea; diabetes mellitus (also pseudocyst, carcinoma).

STATION 2 · GROSS SPECIMEN — 1 minute

TASK

An opened gallbladder with a thick, grey-white, fibrotic wall and flattened mucosa. The lumen contains many pale yellow, faceted stones. (a) Identify. (b) What type of stones? (c) Two complications.

ANSWER

(a) Chronic calculous cholecystitis with cholelithiasis. (b) Cholesterol (mixed) stones. (c) Acute cholecystitis/empyema; obstructive jaundice or pancreatitis from a migrating stone (also carcinoma).

STATION 3 · LAB DATA 2 minutes

TASK

A 48-year-old woman with known gallstones has severe epigastric pain radiating to the back. Serum amylase 1,450 U/L (ref 30–110), lipase 2,100 U/L (ref < 160), calcium 7.4 mg/dL (ref 8.5–10.5), glucose 190 mg/dL, WBC 16 × 10^9/L. (a) Diagnosis. (b) Explain the calcium. (c) Two systemic complications to watch for.

ANSWER

(a) Acute (gallstone) pancreatitis. (b) Calcium precipitates as soaps in fat necrosis; persistent hypocalcaemia indicates severity. (c) Shock, ARDS, DIC or acute tubular necrosis.

RAPID REVISION The Night-Before Page

15 ONE-LINERS

01 Cholesterol stones: supersaturation + stasis + nucleation + mucus.

02 Cholesterol stones form only in the gallbladder; mostly radiolucent.

03 Black pigment stones: haemolysis; brown: infected ducts.

04 Acute cholecystitis: stone in neck/cystic duct; chemical then bacterial.

05 Acalculous cholecystitis: ICU, burns, sepsis; ischaemia; gangrene.

06 Chronic cholecystitis: fibrosis, mononuclear cells, Rokitansky–Aschoff sinuses.

07 Porcelain gallbladder: calcified wall; cancer risk.

08 Gallbladder carcinoma: gallstones, women, fundus/neck, liver invasion.

09 Acute pancreatitis: gallstones + alcohol ≈ 80%.

10 Trypsin activation is the key event in autodigestion.

11 Fat necrosis + calcium = chalky soaps; hypocalcaemia is a bad sign.

12 Pseudocyst: no epithelial lining; follows acute pancreatitis.

13 Chronic pancreatitis: fibrosis, acinar loss, spared islets, calculi.

14 Pancreatic cancer: PanIN precursor; KRAS > 90%; smoking.

15 Painless jaundice + palpable gallbladder = think pancreatic head cancer.

MEMORY HOOKS

- **4 Fs**: female, fat, fertile, forty (cholesterol stones)
- **GET SMASHED**: causes of acute pancreatitis
- **"Stones shrink, cancer swells"**: a stone-scarred gallbladder cannot distend; a palpable one with jaundice suggests cancer (Courvoisier)
- **Big four of pancreatic cancer**: KRAS, CDKN2A, TP53, SMAD4

TEST YOURSELF: FILL THE GAPS

1. Brown pigment stones form in _____ ducts.
2. Mucosal outpouchings in chronic cholecystitis: _____ sinuses.
3. Fat necrosis is due to the action of _____.
4. Hereditary pancreatitis gene for cationic trypsinogen: _____.
5. In chronic pancreatitis the _____ are relatively spared.
6. Migratory thrombophlebitis with cancer: _____ sign.

Answers: infected · Rokitansky–Aschoff · lipase and phospholipase · PRSS1 · islets · Trousseau

DRAW FROM MEMORY

The pathogenesis of acute pancreatitis: three routes in, trypsin, and the four enzyme effects (Figure 7.2).

DRAW FROM MEMORY

The chronic pancreatitis slide: fibrous bands, scanty acini, spared islets, dilated duct with a plug.

SPACED REVISION ☐ Day 1 ☐ Day 3 ☐ Day 7 ☐ Day 14 ☐ Day 30

08

Pneumonia & Tuberculosis

Every breath carries microbes into the lung. Pneumonia is what happens when the defences are briefly outnumbered; tuberculosis is what happens when the body fights a germ it cannot kill, and the fight itself hollows out the lung.

BY THE END OF THIS CHAPTER YOU CAN

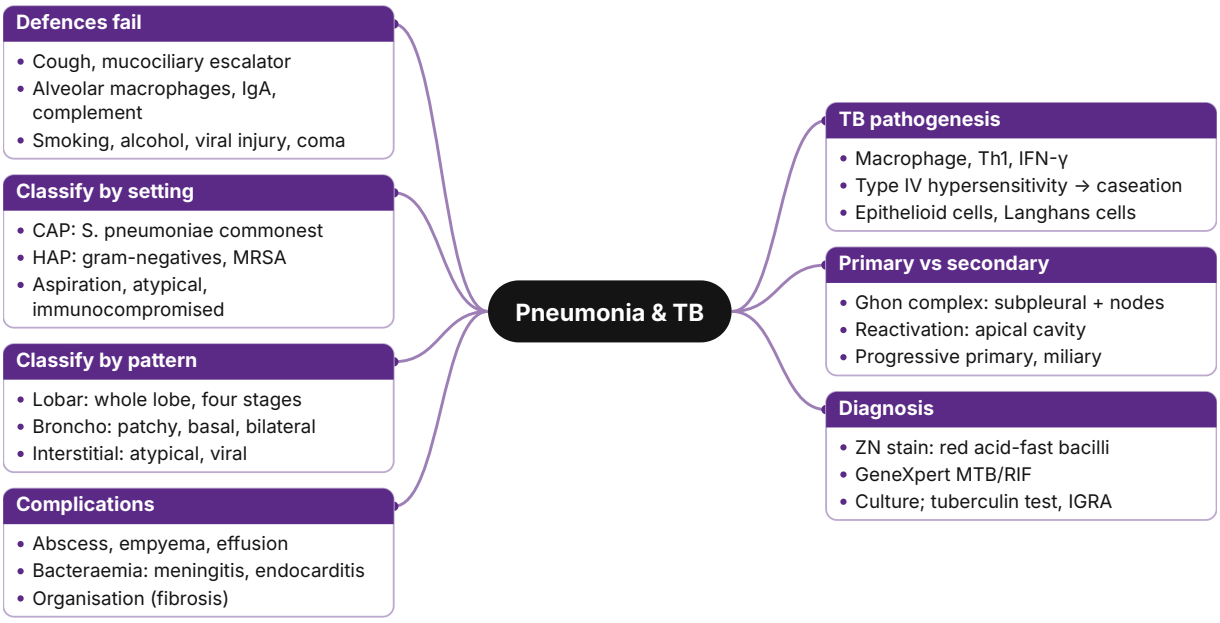
1. List the defences of the lung and explain how their failure invites particular organisms
2. Classify pneumonia by clinical setting and by anatomical pattern
3. Describe the four stages of lobar pneumonia and contrast lobar pneumonia with bronchopneumonia
4. Describe atypical, hospital-acquired and aspiration pneumonia, lung abscess and the complications of pneumonia
5. Explain the pathogenesis of tuberculosis, including the Th1 response and type IV hypersensitivity
6. Differentiate primary from secondary tuberculosis and describe the tubercle, cavitation and miliary spread
7. Interpret the laboratory diagnosis of TB: ZN stain, culture, GeneXpert, tuberculin test and IGRA

EXAM RADAR

Theory SEQs	★★★
BCQs / MCQs	★★★
Viva voce	★★★
OSPE / slides	★★★

Stages of lobar pneumonia, pathogenesis of TB and primary vs secondary TB are perennial long questions; the pneumonia and TB slides are lab staples.

ROBBINS LINK Robbins Basic Pathology — *The Lung*: pulmonary infections (community-acquired, hospital-acquired, aspiration and atypical pneumonia, lung abscess, tuberculosis); *Inflammation and Repair*: granulomatous inflammation.



Chapter at a glance. Seven ideas carry the whole chapter.

8.1 Defences of the lung and classification of pneumonia

Pneumonia is any infection of the lung parenchyma. The lung is exposed to microbes with every breath and to oral flora with every unnoticed aspiration in sleep, yet it stays sterile below the larynx because its defences are layered. Pneumonia means one or more layers have failed, and **which layer fails predicts which organism invades**.

DEFENCES

- **Nose and upper airway** filter large particles; **cough** and glottic closure
- **Mucociliary escalator** sweeps trapped particles upwards

WHAT BREAKS THEM

- **Lost cough reflex**: coma, anaesthesia, alcohol, drugs, neuromuscular disease → aspiration
- **Damaged cilia**: cigarette smoke, viral infection (influenza), hot or corrosive gases, immotile cilia

- **Alveolar macrophages**, surfactant, IgA, complement, neutrophil recruitment
- Adaptive immunity: **antibody** (opsonises capsulated bacteria), **T cells** (intracellular microbes)
- **Retained secretions**: cystic fibrosis, bronchial obstruction (tumour)
- **Impaired macrophages**: alcohol, smoke, hypoxia; **pulmonary oedema**

CONCEPT CLEAR

The defect predicts the bug. Weak **antibody** or no **spleen** (sickle cell disease, splenectomy) invites **capsulated bacteria** such as pneumococcus and H. influenzae. Defective **neutrophils or complement** invite pyogenic bacteria and Aspergillus. Weak **T-cell** immunity (HIV, transplant) invites intracellular organisms: **mycobacteria**, Pneumocystis, CMV and herpesviruses. A preceding **influenza** infection strips the cilia and opens the door to **S. aureus** and pneumococcus.

CLASSIFICATION OF PNEUMONIA BY CLINICAL SETTING (WITH THE ORGANISMS EXAMINERS EXPECT)

Setting	Typical organisms	Clues
Community-acquired, acute (bacterial)	**S. pneumoniae** (commonest), H. influenzae, Moraxella catarrhalis, S. aureus (after influenza), **Klebsiella**, Legionella	Abrupt fever, rigors, productive cough, consolidation
Community-acquired, atypical	**Mycoplasma pneumoniae**, Chlamydia, Coxiella (Q fever); viruses: influenza, RSV, adenovirus, SARS-CoV-2	Dry cough, few chest signs, interstitial pattern
Hospital-acquired (nosocomial)	Gram-negative rods (Klebsiella, E. coli, Serratia, **Pseudomonas**), **MRSA**	Starts ≥ 48 h after admission; ventilators, ICU
Aspiration	Oral **anaerobes** (Bacteroides, Prevotella, Fusobacterium, Peptostreptococcus) mixed with aerobes	Stroke, alcohol, seizures, vomiting; abscess
Chronic	**M. tuberculosis**, atypical mycobacteria, Nocardia, Actinomyces, fungi (Histoplasma)	Granulomas, cavitation, weeks–months
Immunocompromised host	**Pneumocystis jirovecii**, CMV, Aspergillus, Candida, M. avium complex, usual bacteria	HIV (CD4 < 200/µL), transplant, chemotherapy

Pathologists also classify by **anatomical pattern**: **lobar pneumonia** (consolidation of a large part or all of a lobe), **bronchopneumonia** (patchy consolidation centred on bronchioles) and **interstitial pneumonia** (inflammation of alveolar walls, typical of atypical and viral infections).

8.2 Lobar pneumonia and bronchopneumonia

Streptococcus pneumoniae (pneumococcus), a gram-positive lancet-shaped diplococcus, is the commonest cause of community-acquired pneumonia and of lobar pneumonia. Sputum Gram stain shows neutrophils with diplococci; blood cultures are positive in a fraction of cases. Pneumococcal vaccines protect high-risk groups. **Klebsiella pneumoniae**, the commonest gram-negative cause, strikes debilitated and malnourished people and chronic alcoholics, with thick, gelatinous ("red-currant jelly") sputum from its viscid capsule.

CONCEPT CLEAR

Why lobar pneumonia fills a whole lobe. Pneumococci multiply in the alveoli and provoke a flood of protein-rich oedema fluid. Bacteria ride this fluid from alveolus to alveolus through the **pores of Kohn**, so the infection spreads like a stain until it reaches the **pleura of the fissure**, which stops it. Bronchopneumonia starts instead in many separate bronchioles, so it produces many separate patches.



Figure 8.1 · The four stages of lobar pneumonia. Classic teaching timings are approximate; antibiotics shorten every stage.

- **Pleura:** a **fibrinous pleuritis** overlies the affected lobe in the early stages (pleuritic pain, friction rub). It may resolve or organise into fibrous thickening and adhesions.
- **If resolution fails**, fibroblasts grow into the exudate and **organise** it into solid fibrous tissue.
- **Clinical:** abrupt high fever, shaking chills, cough with mucopurulent or **rusty** sputum, pleuritic chest pain; X-ray: a dense, well-demarcated lobar opacity.

> ⚠ **EXAM TRAP**
>
> **Days are teaching aids, not Robbins facts.** Robbins describes the four stages without fixed days; the timings above are classic approximations. In grey hepatisation the lobe is grey because the **red cells have broken down** while the fibrin and neutrophils remain; it is not a sign of healing.

Bronchopneumonia

Patchy foci of acute suppurative inflammation, centred on bronchioles, that may be confined to one lobe but are usually **multilobar, bilateral and basal** (secretions gravitate to the lower lobes). Typical patients are infants, the elderly, the bedridden and the terminally ill; organisms include staphylococci, streptococci, H. influenzae, Pseudomonas and coliforms.

LOBAR PNEUMONIA VS BRONCHOPNEUMONIA

Feature	Lobar pneumonia	Bronchopneumonia
Distribution	Large part or whole of **one lobe**	**Patchy**, multilobar, often bilateral and basal
Usual host	Previously healthy adult	Extremes of age, debilitated, bedridden
Organisms	Pneumococcus (most); Klebsiella	Staphylococci, streptococci, H. influenzae, Pseudomonas, coliforms
Gross	Uniform consolidation passing through four stages	Grey-red to yellow, dry, granular foci 3–4 cm, poorly delimited
Microscopy	Alveoli uniformly filled with exudate at the same stage	Neutrophil-rich exudate filling **bronchi, bronchioles** and adjacent alveoli
X-ray	Dense lobar opacity, air bronchogram	Multiple patchy opacities

COMPLICATIONS: LOCAL

- **Lung abscess** from tissue necrosis (type 3 pneumococcus, Klebsiella, S. aureus)
- Spread to the pleura: **parapneumonic effusion**, fibrinosuppurative pleuritis, **empyema**
- **Organisation** of unresolved exudate → fibrous scarring

COMPLICATIONS: SYSTEMIC (BACTERAEMIA)

- **Meningitis**, **infective endocarditis**, pericarditis
- Suppurative arthritis; metastatic abscesses in brain, kidney, spleen
- **Sepsis** and septic shock

LAB SPOTTER

Lobar pneumonia slide in 10 seconds: (1) **Alveolar walls are intact** and thin; the lung architecture is preserved. (2) Every alveolus is **stuffed with exudate**: neutrophils and pink **fibrin strands** (plus red cells in red hepatisation, few red cells in grey). (3) All fields look alike, because the whole lobe is at one stage. (4) The pleura may carry a fibrin layer. Intact walls separate it from an abscess, where the walls are destroyed.

8.3 Atypical, hospital-acquired and aspiration pneumonia; lung abscess

Atypical (interstitial) pneumonia

Called "atypical" because sputum is scanty, chest signs of consolidation are absent, the white count is only modestly raised and, histologically, the alveoli contain **no exudate**. **Mycoplasma pneumoniae** is the commonest cause in children and young adults; viruses (influenza, RSV, adenovirus, SARS-CoV-2) are the others. Viruses damage the respiratory epithelium and cilia, which also explains **secondary bacterial pneumonia** after influenza.

GROSS

- Patchy or lobar; areas are red-blue, congested and subcrepitant
- Pleuritis is unusual

MICROSCOPY

- **Interstitial** inflammation: alveolar septa widened, oedematous, infiltrated by **lymphocytes, macrophages**, plasma cells
- Alveolar spaces largely free of cellular exudate
- Severe cases: **hyaline membranes** (diffuse alveolar damage)

HOSPITAL-ACQUIRED PNEUMONIA

Pneumonia starting **48 hours or more after admission**, common in ICU and in **ventilated** patients, the immunosuppressed and those on prolonged antibiotics. Gram-negative rods and **MRSA** dominate; mortality is high.

ASPIRATION PNEUMONIA

Debilitated or unconscious patients (stroke, alcohol intoxication, seizures, repeated vomiting) aspirate gastric contents and oral flora. Injury is partly **chemical** (acid) and partly bacterial (anaerobes). It is often necrotising, frequently fatal, and the commonest route to lung abscess.

PNEUMONIA IN THE IMMUNOCOMPROMISED

- **Pneumocystis jirovecii**: HIV with CD4 < 200/μL; alveoli filled with **foamy, frothy pink exudate**; cup-shaped cysts on **silver (GMS)** stain
- **CMV**: enlarged cells with **owl-eye** intranuclear inclusions
- **Invasive aspergillosis**: neutropenia; septate hyphae branching at acute angles invade vessels → haemorrhagic infarcts
- Mycobacteria (typical and atypical) and the usual bacteria

Lung abscess

DEFINITION

Lung abscess is a localised area of **suppurative necrosis** of lung parenchyma that forms one or more cavities. Anaerobes are found in almost all, usually mixed with aerobes.

ROUTES

- **Aspiration** of infected material (commonest): usually **single, right-sided** (right main bronchus is wider and more vertical)
- **Post-pneumonic**: S. aureus, Klebsiella, type 3 pneumococcus
- **Septic emboli** (right-sided endocarditis, IV drug use): multiple
- **Behind an obstructing tumour**: always exclude carcinoma in an older smoker
- Penetrating trauma; spread from a subphrenic abscess

MORPHOLOGY AND COURSE

- A few mm to 5–6 cm; cavity of suppurative debris; **air–fluid level** if it drains into a bronchus
- Micro: suppurative destruction of lung tissue; with time, a **fibrous wall**
- Copious, **foul-smelling** purulent sputum, fever, weight loss, **clubbing**
- Complications: **empyema**, haemorrhage, **brain abscess** or meningitis, secondary (**AA**) amyloidosis

8.4 Tuberculosis: the organism and pathogenesis

DEFINITION

Tuberculosis is a chronic communicable **granulomatous** disease caused by **Mycobacterium tuberculosis**. It usually involves the lungs but can affect any organ, and its hallmark lesion is the **caseating granuloma**.

M. tuberculosis is a slender, aerobic, non-motile rod with a waxy, **mycolic acid**-rich wall that holds carbol fuchsin against acid-alcohol: it is **acid-fast**. It grows slowly (weeks in culture). Spread is by **droplet nuclei** coughed out by a patient with open (cavitary) disease. **M. bovis** in unpasteurised milk causes oropharyngeal and intestinal TB. **Infection**

(bacilli present, usually latent and symptomless) must be distinguished from **disease** (clinically active TB); most infected people never develop disease.

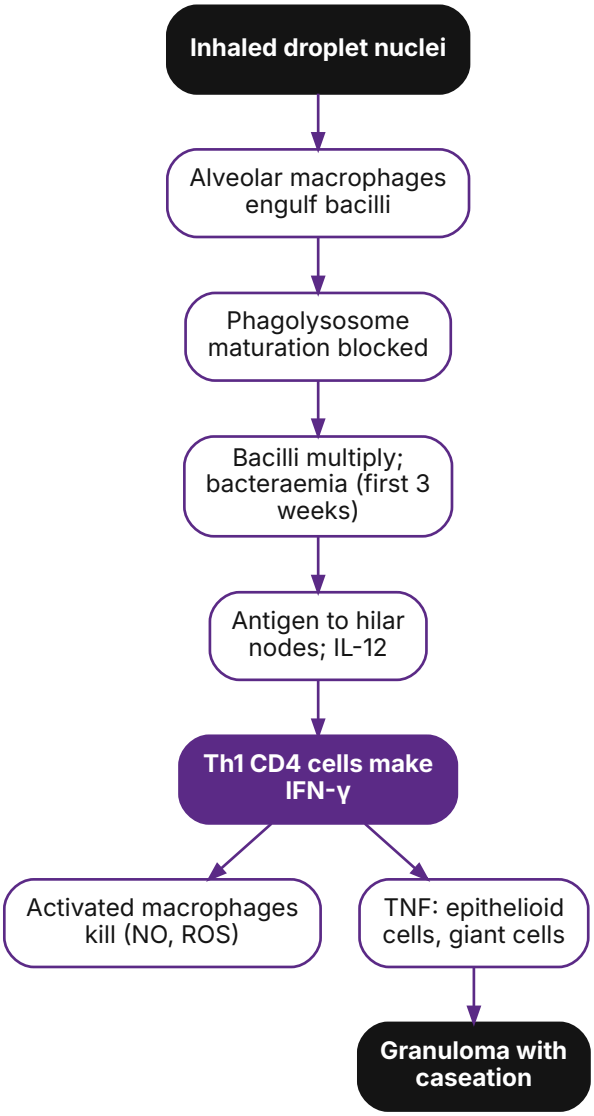


Figure 8.2 · Pathogenesis of tuberculosis. The same Th1 response gives immunity and tissue damage.

1. **Entry:** alveolar macrophages take up bacilli through mannose and complement receptors.
2. **Replication:** the bacillus blocks phagolysosome maturation and multiplies unchecked for about 3 weeks; bacteraemia seeds many organs silently.
3. **Th1 response:** dendritic cells carry antigen to hilar nodes; **IL-12** drives CD4+ Th1 cells that secrete **IFN-γ**.
4. **Macrophage activation:** IFN-γ enables phagolysosome fusion, **nitric oxide**, reactive oxygen species and autophagy: bacilli are killed or contained.
5. **Granuloma and damage: TNF** and chemokines recruit monocytes that become **epithelioid cells** and fuse into **Langhans giant cells**; **type IV hypersensitivity** produces caseous necrosis.

CONCEPT CLEAR

Immunity and hypersensitivity are the same response. The Th1/IFN-γ reaction that lets macrophages kill bacilli also kills host tissue: caseation, cavitation and fibrosis are all products of it. Hence: (1) a **positive tuberculin test** shows T-cell sensitisation (past or present infection, or BCG), not active disease; (2) defects in **IL-12/IFN-γ receptor** genes or **anti-TNF drugs** (infliximab) cause disseminated or reactivated mycobacterial disease; (3) in advanced **HIV**, granulomas are poorly formed, bacilli abound, cavitation is rare and the tuberculin test is often falsely negative.

TUBERCLE: GROSS

- **Caseous necrosis**: soft, dry, crumbly, **yellow-white, cheese-like** material
- Healing: grey-white fibrosis and chalky **calcification**

TUBERCLE: MICROSCOPY

- Centre: **caseation**, an amorphous, granular, pink area with no cell outlines
- **Epithelioid histiocytes**: pale-pink cytoplasm, elongated slipper-shaped nuclei
- **Langhans giant cells**: nuclei arranged in a **horseshoe** at the periphery
- Cuff of **lymphocytes**, then fibroblasts and collagen
- **ZN stain**: red acid-fast bacilli, most often in or near the caseous centre

8.5 Primary and secondary tuberculosis

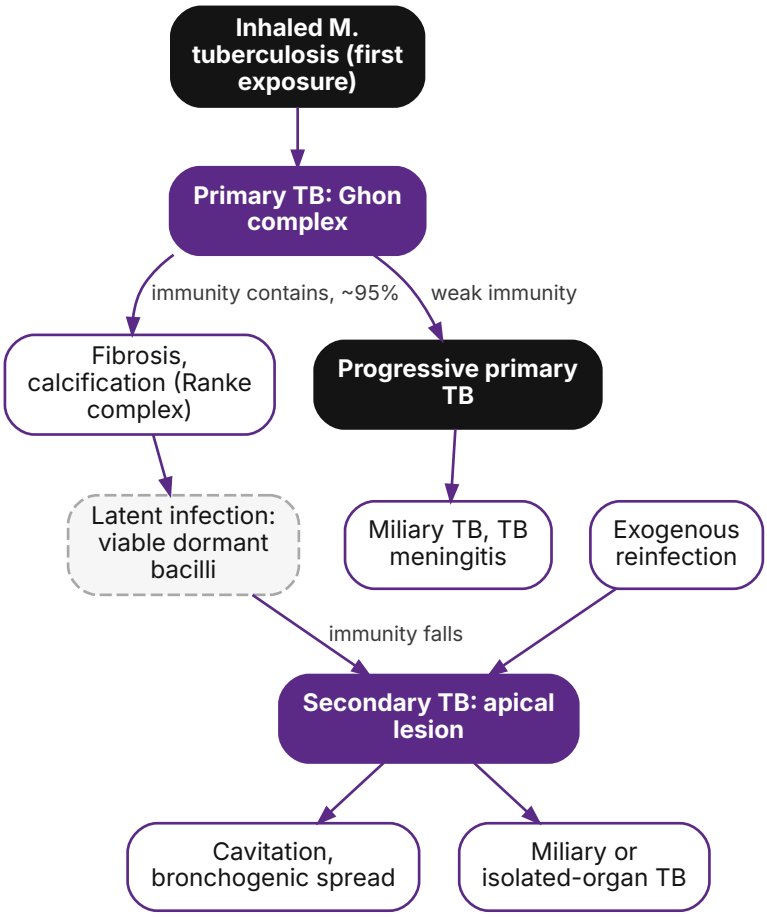


Figure 8.3 · Natural history of tuberculosis. The host's immunity decides every branch.

Primary tuberculosis

The form that develops in a previously **unexposed, unsensitised** person. Inhaled bacilli lodge in the **subpleural** air spaces of the **lower part of the upper lobe or upper part of the lower lobe**, near the interlobar fissure. A **Ghon focus** (1–1.5 cm, grey-white, caseating) forms; bacilli drain to the **hilar lymph nodes**, which also caseate. Focus + nodes = **Ghon complex**. In about 95% of people, cell-mediated immunity controls the infection: the complex fibroses and calcifies (**Ranke complex**), but viable bacilli may persist for decades.

Progressive primary TB occurs when immunity is weak (HIV, malnutrition, infancy). It mimics bacterial pneumonia: consolidation of lower and middle lobes, hilar lymphadenopathy, pleural effusion, little cavitation, and lymphohaematogenous spread to **meninges** and the whole body (**miliary TB**).

Secondary (post-primary, reactivation) tuberculosis

Disease in a **previously sensitised** host, usually **reactivation** of a dormant focus when resistance falls, sometimes **exogenous reinfection** in high-prevalence areas. It is classically localised to the **apex of one or both upper lobes**, probably because of the high oxygen tension there. Because the host is already sensitised, the tissue reaction is prompt and intense: the lesion is walled off (so nodes are less involved) but caseation and **cavitation** are frequent. A cavity that erodes into a bronchus discharges bacilli into the sputum, making the patient **infectious**.

GROSS

- Early: firm, grey-white to yellow focus < 2 cm within 1–2 cm of the **apical pleura**, central caseation, peripheral fibrosis
- Healing: **fibrocalcific scar** that puckers the pleura; pleural adhesions
- Progressive: caseation expands and erodes a bronchus → **ragged cavity** lined by caseous material, poorly walled off by fibrous tissue (**fibrocavitary TB**)

MICROSCOPY

- Coalescent tubercles with central **caseation**, epithelioid cells, **Langhans giant cells**, lymphocytes
- Fibrous wall around older lesions
- Acid-fast bacilli on ZN, abundant in cavity walls

Miliary TB: bacilli reach the blood (lymphatics → thoracic duct → right heart → lungs, or pulmonary veins → systemic arteries) and seed countless small (**about 2 mm**) **yellow-white foci** resembling millet seeds in the lungs or in liver, spleen, marrow, adrenals, meninges, kidneys, fallopian tubes and epididymis.

PRIMARY VS SECONDARY TUBERCULOSIS

Feature	Primary TB	Secondary TB
Host	Unsensitised; often a child	**Sensitised**; usually an adult
Origin	First infection	**Reactivation** of latent focus (or reinfection)
Site	**Subpleural**, lower part of upper lobe / upper part of lower lobe	**Apex** of upper lobe(s)
Lymph nodes	**Prominent** hilar node caseation (Ghon complex)	Little nodal involvement
Cavitation	Rare	**Common**
Usual outcome	Heals by fibrosis and calcification; latent infection	Fibrocaseous, cavitary, chronic; infectious
Spread	Lymphohaematogenous when progressive	Bronchial (endobronchial, laryngeal, intestinal), haematogenous

SPREAD OF SECONDARY TB

- **Bronchogenic**: tuberculous bronchopneumonia, endobronchial and **laryngeal** TB
- **Swallowed sputum**: **intestinal TB** (ileocaecal)
- **Pleura**: effusion, empyema, fibrous obliteration
- **Blood**: miliary TB; **isolated-organ TB**: meninges, kidneys, adrenals (Addison disease), bone (**Pott disease** of the spine), fallopian tubes, epididymis
- **Lymphadenitis**, commonest extrapulmonary form, usually cervical ("scrofula")

CLINICAL FEATURES

- Insidious **low-grade evening fever**, **night sweats**, anorexia, **weight loss**, malaise
- Cough, purulent sputum, **haemoptysis**, pleuritic pain
- Massive haemoptysis may come from a **Rasmussen aneurysm** (pulmonary artery branch in a cavity wall)
- Long-standing disease: **secondary (AA) amyloidosis**, cor pulmonale

8.6 Diagnosis of tuberculosis

TESTS FOR TUBERCULOSIS AND WHAT EACH ONE TELLS YOU

Test	What it shows	Points to remember
Ziehl–Neelsen (ZN) smear of sputum	**Red, slender, beaded rods** on a blue background	Quick and cheap; needs many bacilli, so a negative smear does not exclude TB. Fluorescent (auramine) stain is more sensitive
Culture (Löwenstein–Jensen; liquid culture)	Growth of M. tuberculosis	Gold standard; allows drug-susceptibility testing; takes weeks (liquid culture is faster)
GeneXpert MTB/RIF	M. tuberculosis DNA + **rifampicin resistance** (rpoB mutations)	Cartridge PCR, result in about 2 hours; recommended first test
Tuberculin skin test (Mantoux)	Induration at 48–72 h = T-cell hypersensitivity	Infection or BCG, not active disease. False negative: HIV, malnutrition, miliary TB, sarcoidosis, measles

Test	What it shows	Points to remember
IGRA (interferon-γ release assay)	IFN-γ from T cells exposed to TB-specific antigens	Not affected by BCG; still cannot separate latent from active TB
Histopathology / FNAC	Caseating granulomas ± acid-fast bacilli	Lymph node, pleura, bowel and other tissues

MOLECULAR KEY

Drug resistance: rifampicin resistance = **rpoB** mutations (detected by GeneXpert); isoniazid resistance = **katG** or **inhA**. **MDR-TB** resists at least isoniazid and rifampicin. Rifampicin resistance is a good surrogate for MDR-TB.

PAKISTAN FOCUS

Pakistan is among the **top five high-TB-burden countries**, with an incidence of roughly **266 per 100,000** population. **Drug-resistant TB** is a growing concern, and **GeneXpert MTB/RIF** is now widely used. BCG at birth protects children mainly against miliary TB and TB meningitis.

Source: WHO estimates (approximate).

LAB SPOTTER

Pulmonary TB slide in 10 seconds: (1) Look at low power for **round nodules** (tubercles), often merging. (2) Each has a **pink, amorphous, granular centre** without cell outlines: caseation. (3) Around it, pale **epithelioid cells** and **Langhans giant cells** with horseshoe nuclei. (4) A rim of **lymphocytes** and fibrosis. Name it "caseating granulomatous inflammation consistent with tuberculosis" and ask for ZN stain.

EXAM ZONE Past-Paper Pattern SEQs — Solved

PROF PATTERN **a)** Classify pneumonia. **b)** Describe the morphology of lobar pneumonia. **2 + 3 marks**

MODEL ANSWER PLAN

a) By setting: community-acquired (typical, atypical), hospital-acquired, aspiration, chronic, immunocompromised, with one or two organisms each. By pattern: lobar, bronchopneumonia, interstitial. **b)** Four stages with gross and micro: congestion (heavy red lung; fluid, bacteria); red hepatisation (firm red liver-like lobe; neutrophils, RBCs, fibrin); grey hepatisation (grey-brown dry; RBCs lysed, fibrinosuppurative exudate); resolution (enzymatic digestion, macrophages). Add fibrinous pleuritis and organisation if resolution fails.

Score booster: draw the four chevrons of Figure 8.1; it earns the morphology marks at a glance.

PROF PATTERN **a)** Describe the pathogenesis of pulmonary tuberculosis. **b)** Differentiate between primary and secondary tuberculosis. **3 + 2 marks**

MODEL ANSWER PLAN

a) Droplet nuclei → alveolar macrophages → phagolysosome maturation blocked → replication and bacteraemia (3 weeks) → antigen to nodes, IL-12 → Th1 cells secrete IFN-γ → macrophage activation (NO, ROS) → TNF-driven granuloma with epithelioid and Langhans cells → type IV hypersensitivity causes caseation. **b)** Table: host (unsensitised vs sensitised), site (subpleural vs apical), nodes (prominent vs minimal), cavitation (rare vs common), outcome (healing/latency vs fibrocavitary).

MODULE PATTERN A 30-year-old man has cough for 2 months, evening fever, night sweats, weight loss and blood-streaked sputum. Chest X-ray shows a cavity in the right apex. **a)** Most likely diagnosis? **b)** Describe the gross and microscopic features of the lesion. **c)** Name two tests to confirm the diagnosis and one to detect drug resistance. **1 + 2 + 2 marks**

MODEL ANSWER PLAN

a) Secondary (reactivation) pulmonary TB with cavitation. **b)** Apical fibrocaseous lesion; ragged cavity lined by caseous material, fibrous wall, pleural puckering; micro: caseating granulomas, epithelioid cells, Langhans giant cells, lymphocytes, AFB. **c)** Sputum ZN smear, culture (or GeneXpert); GeneXpert MTB/RIF for rifampicin resistance.

CLASSIC **a)** Differentiate lobar pneumonia from bronchopneumonia. **b)** List the complications of pneumonia. **2 + 3 marks**

MODEL ANSWER PLAN

a) Distribution (lobe vs patchy, bilateral, basal), host, organisms, gross, micro (alveoli at one stage vs bronchiolocentric exudate), X-ray. **b)** Abscess, effusion and empyema, organisation; bacteraemia → meningitis, endocarditis, pericarditis, arthritis, metastatic abscesses, septic shock.

CLASSIC Write a note on lung abscess: causes, morphology and complications. **5 marks**

MODEL ANSWER PLAN

Definition (localised suppurative necrosis with cavity; anaerobes). Causes: aspiration (commonest, right side), post-pneumonic, septic emboli, obstructing tumour, trauma. Morphology: mm to 6 cm cavity with pus, air–fluid level, fibrous wall. Complications: empyema, haemorrhage, brain abscess, AA amyloidosis.

Score booster: "Exclude an obstructing carcinoma in an older smoker" shows clinical maturity.

BCQs One-Best-Answer Practice

1 A 60-year-old man has sudden fever, rigors, pleuritic pain and rusty sputum. X-ray shows consolidation of the whole right lower lobe. The most likely organism is:

A. Mycoplasma pneumoniae
B. Streptococcus pneumoniae
C. Pneumocystis jirovecii
D. Legionella pneumophila
E. Influenza virus

2 At autopsy a lobe is grey-brown, dry and firm. Alveoli contain fibrin and neutrophils with disintegrating red cells. The stage is:

A. Congestion
B. Red hepatisation
C. Grey hepatisation
D. Resolution
E. Organisation

3 A malnourished chronic alcoholic has lobar consolidation and thick, mucoid "red-currant jelly" sputum. The likely organism is:

A. Staphylococcus aureus
B. Haemophilus influenzae
C. Pseudomonas aeruginosa
D. Klebsiella pneumoniae
E. Moraxella catarrhalis

4 In Mycoplasma pneumonia, the inflammation is mainly located in the:

A. Alveolar septa, with lymphocytes and macrophages
B. Alveolar spaces, with neutrophils
C. Bronchial lumen, with eosinophils
D. Pleural cavity
E. Hilar lymph nodes, with caseation

5 Aspiration lung abscess is commoner in the right lung mainly because:

A. The right lung is larger
B. The right lung has three lobes
C. The right lung has a better blood supply
D. The left main bronchus is wider
E. The right main bronchus is wider and more vertical

6 Within alveolar macrophages, M. tuberculosis survives early infection mainly by:

A. Producing a powerful exotoxin
B. Blocking phagolysosome maturation
C. Lysing complement components
D. Antigenic variation of pili
E. Forming spores

7 The cytokine most essential for activating macrophages to kill M. tuberculosis is:

A. IL-4
B. IL-5
C. IL-10
D. TGF-β
E. Interferon-γ

8 The Ghon complex consists of:

A. An apical cavity and pleural effusion
B. Miliary nodules and meningitis
C. A subpleural parenchymal focus and caseous hilar lymph nodes
D. Bronchiectasis and fibrosis
E. An apical scar and calcified pleura

9 A man with rheumatoid arthritis develops active TB three months after starting infliximab. The best explanation is that:

A. TNF is required to form and maintain granulomas
B. Infliximab raises IgE levels
C. Infliximab is contaminated with mycobacteria
D. Infliximab causes agranulocytosis
E. Infliximab destroys BCG immunity

10 Secondary tuberculosis characteristically involves the lung apex. The usual explanation is:

A. Better lymphatic drainage
B. Highest oxygen tension
C. Greatest number of macrophages
D. Proximity to hilar nodes
E. Thinnest pleura

11 A positive tuberculin (Mantoux) test in an adult indicates:

A. Active pulmonary tuberculosis
B. Complete protection against TB
C. Sputum-positive infectious disease
D. Cell-mediated sensitisation from past infection or BCG
E. Humoral immunity to mycobacteria

12 GeneXpert MTB/RIF performed on sputum detects:

A. Acid-fast bacilli by fluorescence
B. Antibodies to mycobacterial antigens
C. Interferon-γ release by T cells
D. M. tuberculosis DNA and rpoB mutations for rifampicin resistance
E. Isoniazid resistance through katG mutations

ANSWER KEY WITH ONE-LINE REASONS

1. **B** Pneumococcus is the commonest cause of community-acquired and lobar pneumonia.
2. **C** Red cells break down while the fibrinosuppurative exudate persists.
3. **D** Klebsiella's viscid capsule gives gelatinous sputum in debilitated alcoholics.
4. **A** Atypical pneumonia is interstitial; alveolar spaces are largely clear.
5. **E** Aspirated material follows the straighter right main bronchus.
6. **B** It arrests the phagosome before fusion with lysosomes and multiplies inside.
7. **E** Th1 cells secrete IFN-γ, which drives phagolysosome fusion and nitric oxide.
8. **C** Ghon focus plus involved hilar nodes = primary complex.
9. **A** TNF blockade breaks down granulomas and reactivates latent TB.
10. **B** Robbins attributes apical localisation probably to high oxygen tension.
11. **D** It shows T-cell hypersensitivity; it does not separate infection from disease.
12. **D** Cartridge PCR for TB DNA and rifampicin resistance in about 2 hours.

VIVA Rapid-Fire Viva Voce

Q Define pneumonia.	Infection of the lung parenchyma, producing alveolar exudate (consolidation) or interstitial inflammation.
Q Commonest cause of community-acquired pneumonia?	Streptococcus pneumoniae.
Q Name the stages of lobar pneumonia.	Congestion, red hepatisation, grey hepatisation and resolution.
Q Why is it called hepatisation?	The consolidated lobe is solid, airless and firm like liver.
Q Why is bronchopneumonia basal and bilateral?	Secretions gravitate to the lower lobes and infection starts in many bronchioles at once.
Q Why is atypical pneumonia 'atypical'?	Scanty sputum, no consolidation signs, modest WBC; inflammation lies in alveolar septa, not alveolar spaces.
Q What is a Ghon complex?	The subpleural Ghon focus of primary TB together with caseous hilar lymph nodes.
Q What is a Ranke complex?	A healed Ghon complex that has fibrosed and calcified.

Q Which cells form a tubercle?	Epithelioid histiocytes, Langhans giant cells, lymphocytes and fibroblasts around caseous necrosis.
Q Why are mycobacteria acid-fast?	Their mycolic acid-rich waxy wall retains carbol fuchsin despite decolorising with acid-alcohol.
Q What is miliary TB?	Haematogenous spread producing countless yellow-white foci about 2 mm across, like millet seeds, in lungs or other organs.
Q What is a Rasmussen aneurysm?	Dilated pulmonary artery branch in a TB cavity wall; rupture causes massive haemoptysis.

OSPE Practice Stations

STATION 1 · SLIDE — 2 minutes

TASK

Section of lung: alveolar walls are intact and thin. Every alveolus is filled with neutrophils and pink fibrin strands; red cells are scanty. The pleura is covered by a layer of fibrin. (a) Identify the lesion and its stage. (b) Name the commonest organism. (c) Give two complications.

ANSWER

(a) Lobar pneumonia, grey hepatisation. (b) Streptococcus pneumoniae. (c) Lung abscess, empyema (also meningitis, endocarditis, organisation).

STATION 2 · SLIDE — 2 minutes

TASK

Section of lung: several rounded nodules, each with a central pink, amorphous, granular area lacking cell outlines, surrounded by pale elongated cells, multinucleated cells with peripheral horseshoe nuclei and a rim of lymphocytes. (a) Identify. (b) Name the multinucleated cell. (c) Which stain confirms the organism, and how does it appear?

ANSWER

(a) Caseating granulomatous inflammation: pulmonary tuberculosis. (b) Langhans giant cell. (c) Ziehl–Neelsen: slender red (magenta) acid-fast bacilli on a blue background.

STATION 3 · GROSS SPECIMEN — 1 minute

TASK

A lung with a ragged cavity at the apex of the upper lobe, lined by yellow-white cheesy material, with surrounding grey-white fibrosis and thickened, puckered pleura. (a) Diagnosis? (b) Two features you see. (c) One complication.

ANSWER

(a) Secondary (fibrocavitary) pulmonary tuberculosis. (b) Apical cavity with caseous lining; fibrosis with pleural puckering. (c) Haemoptysis (Rasmussen aneurysm), bronchogenic or miliary spread, empyema, AA amyloidosis.

STATION 4 · LAB DATA — 2 minutes

TASK

A 35-year-old woman with 6 weeks of cough and weight loss. Sputum ZN: acid-fast bacilli 2+. GeneXpert MTB/RIF: *MTB detected; rifampicin resistance detected.* (a) Interpret. (b) What does rifampicin resistance usually indicate? (c) What further test is needed?

ANSWER

(a) Active, infectious pulmonary TB, rifampicin-resistant. (b) Likely MDR-TB (resistance to isoniazid and rifampicin). (c) Culture with full drug-susceptibility testing (including isoniazid and second-line drugs).

RAPID REVISION The Night-Before Page

15 ONE-LINERS

01 The defect predicts the bug: antibody/spleen → capsulated; T cell → intracellular.

02 Commonest CAP and lobar pneumonia organism: S. pneumoniae.

03 Klebsiella: alcoholics, red-currant jelly sputum.

04 Lobar stages: congestion → red → grey hepatisation → resolution.

05 Grey hepatisation: red cells lysed; fibrin and neutrophils persist.

06 Bronchopneumonia: patchy, basal, bilateral; extremes of age.

07 Atypical pneumonia: interstitial mononuclear infiltrate; Mycoplasma.

08 HAP: ≥ 48 h after admission; gram-negatives, MRSA.

09 Lung abscess: aspiration, anaerobes, right side, air–fluid level.

10 TB survives by blocking phagolysosome maturation.

11 Th1 → IFN-γ → macrophage activation; TNF keeps granulomas.

12 Primary TB: Ghon focus + hilar nodes; heals as Ranke complex.

13 Secondary TB: apex, caseation, cavity, infectious.

14 Miliary TB: about 2 mm millet-seed foci by haematogenous spread.

15 GeneXpert: TB DNA + rifampicin resistance (rpoB) in 2 hours.

MEMORY HOOKS

- **"Congested Red Grey Rose"**: Congestion, Red, Grey, Resolution
- **Ghon goes low, reactivation goes high**: primary subpleural mid-lung; secondary apical
- **"Immunity bites back"**: the Th1 response that kills TB also caseates the lung
- **Tubercle = C-E-L-L**: Caseation, Epithelioid cells, Langhans cells, Lymphocytes

TEST YOURSELF: FILL THE GAPS

1. Bacteria spread between alveoli through pores of _____.
2. Pneumocystis is shown best by _____ stain.
3. TB macrophages are activated by _____.
4. Healed calcified primary complex: _____ complex.
5. Secondary TB favours the apex due to high _____.
6. Rifampicin resistance gene: _____.

Answers: Kohn · silver (GMS) · IFN-γ · Ranke · oxygen tension · rpoB

DRAW FROM MEMORY

The four stages of lobar pneumonia with one gross and one microscopic feature each (Figure 8.1).

DRAW FROM MEMORY

A tubercle: caseous centre, epithelioid cells, a Langhans giant cell, lymphocyte cuff and fibrosis.

SPACED REVISION ☐ Day 1 ☐ Day 3 ☐ Day 7 ☐ Day 14 ☐ Day 30

09

COPD, Asthma & Interstitial Lung Disease

Chronic lung disease comes in two shapes: either air cannot get out, or the lung cannot stretch to let it in. Ask that one question of the spirometer and you have half the diagnosis before anyone looks down a microscope.

BY THE END OF THIS CHAPTER YOU CAN

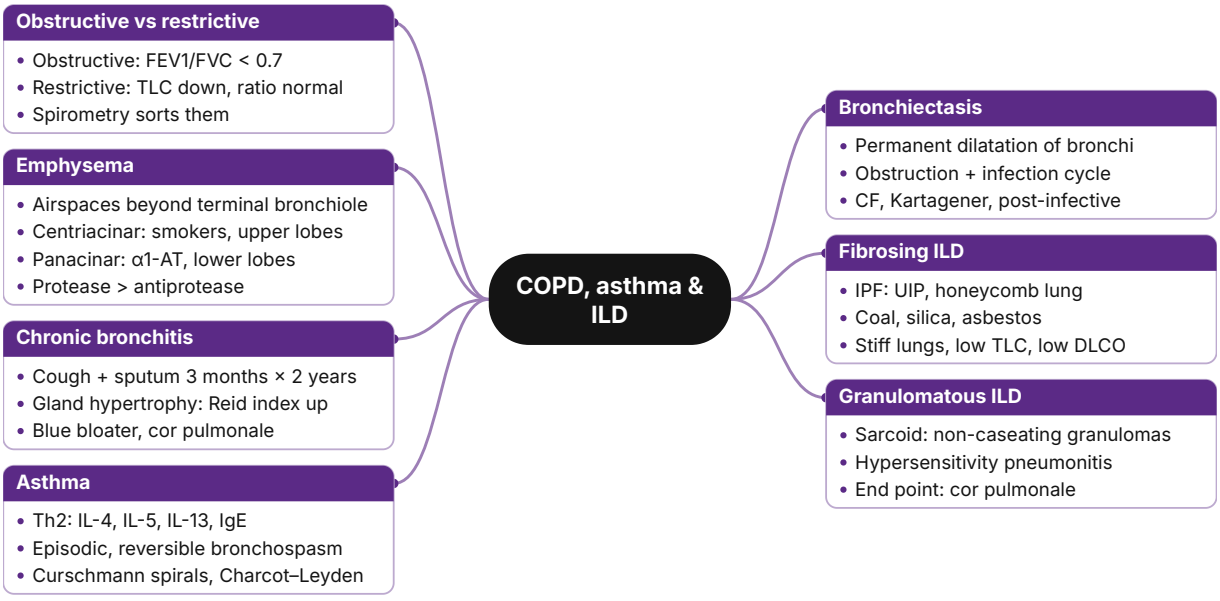
1. Differentiate obstructive from restrictive lung disease using spirometry
2. Define emphysema, describe its types and explain the protease–antiprotease and oxidant mechanisms
3. Describe chronic bronchitis (Reid index) and contrast the 'pink puffer' with the 'blue bloater'
4. Explain the pathogenesis and morphology of atopic and non-atopic asthma
5. Describe the causes, pathogenesis and morphology of bronchiectasis
6. Describe idiopathic pulmonary fibrosis, the pneumoconioses, sarcoidosis and hypersensitivity pneumonitis
7. Explain how chronic lung disease leads to cor pulmonale

EXAM RADAR

Theory SEQs	★★★
BCQs / MCQs	★★★
Viva voce	★★★
OSPE / slides	★★☆

Emphysema (types, pathogenesis), asthma, bronchiectasis and obstructive vs restrictive are favourite long questions; silica, asbestos and sarcoidosis are BCQ staples.

ROBBINS LINK Robbins Basic Pathology — *The Lung*: obstructive lung diseases (emphysema, chronic bronchitis, asthma, bronchiectasis), chronic interstitial (restrictive) diseases, pneumoconioses, granulomatous diseases; *The Heart*: cor pulmonale.



Chapter at a glance. Two families of disease and the heart they finally strain.

9.1 Obstructive vs restrictive lung disease

Obstructive diseases raise resistance to airflow anywhere from trachea to respiratory bronchiole, so air is trapped and expiration is slow. **Restrictive** diseases reduce expansion of the lung, either because the parenchyma is stiff (interstitial fibrosis) or because the chest wall cannot move; lung volumes shrink but the air that is inhaled comes out fast.

OBSTRUCTIVE VS RESTRICTIVE LUNG DISEASE: THE SPIROMETRY QUESTION

Feature	Obstructive	Restrictive
Basic problem	Air cannot get **out** (airway resistance up)	Air cannot get **in** (lung or chest wall stiff)
FEV1	**Markedly reduced**	Reduced in proportion to FVC
FVC	Normal or mildly reduced	**Markedly reduced**
FEV1/FVC	**Low (< 0.7)**	**Normal or high**
Lung volumes	TLC normal or high; **RV high** (air trapping)	**TLC low**
Examples	Emphysema and chronic bronchitis (acinus, bronchus), asthma, bronchiectasis, small airway disease (bronchiolitis)	**Chest wall:** obesity, kyphoscoliosis, pleural disease, neuromuscular disease. **Lung:** IPF, pneumoconioses, sarcoidosis, ARDS

LAB VALUES

Normal FEV1/FVC about 0.75–0.80

Obstruction post-bronchodilator FEV1/FVC < 0.70

Reversibility

FEV1 up ≥ 12% and ≥ 200 mL after bronchodilator → asthma

DLCO

low in emphysema and ILD; normal in asthma, chronic bronchitis

9.2 Emphysema

COPD is persistent, poorly reversible airflow limitation caused mainly by smoking; its two components, **emphysema** and **chronic bronchitis**, usually coexist. Only a minority of smokers develop significant COPD, so genes and other exposures matter.

DEFINITION

Emphysema is **irreversible enlargement of the airspaces distal to the terminal bronchiole**, accompanied by **destruction of their walls** without significant fibrosis. It is a morphological definition; chronic bronchitis is a clinical one.

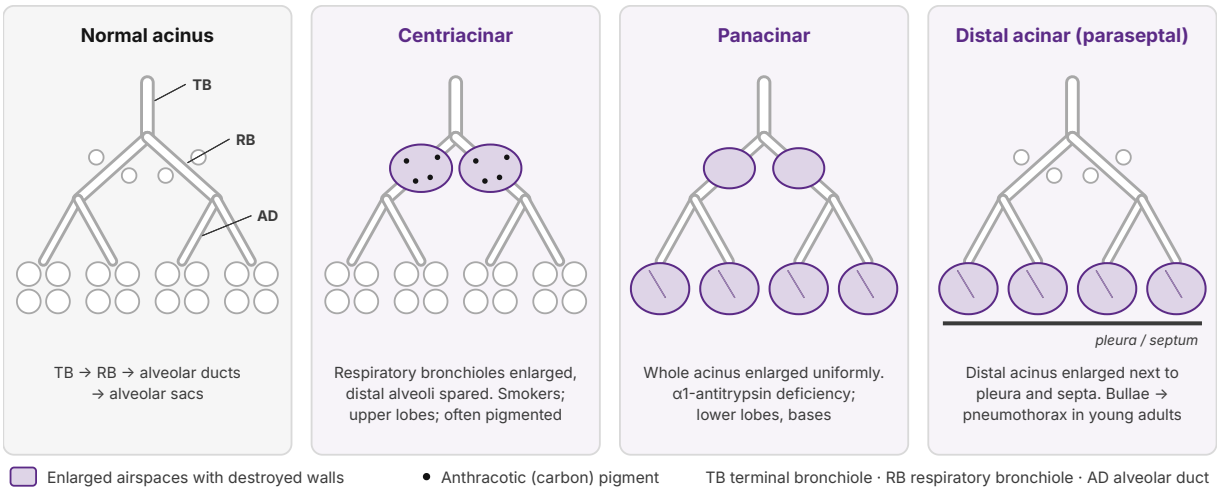


Figure 9.1 · Types of emphysema by where they sit in the acinus. The acinus is the unit beyond the terminal bronchiole: respiratory bronchioles, alveolar ducts and alveoli.

- **Centriacinar: respiratory bronchioles** enlarged, distal alveoli spared; **upper lobes** (apical segments); black pigment in walls; **heavy smokers**, often with chronic bronchitis. More than 95% of clinically significant emphysema.
- **Panacinar:** whole acinus enlarged **uniformly**; **lower zones**, worst at the bases; **α1-antitrypsin deficiency**.

- **Distal acinar (paraseptal):** next to **pleura** and septa, upper lung, near scars; forms **bullae → spontaneous pneumothorax** in young adults.
- **Irregular** (airspace enlargement with fibrosis): around old scars; common but clinically insignificant.

Pathogenesis

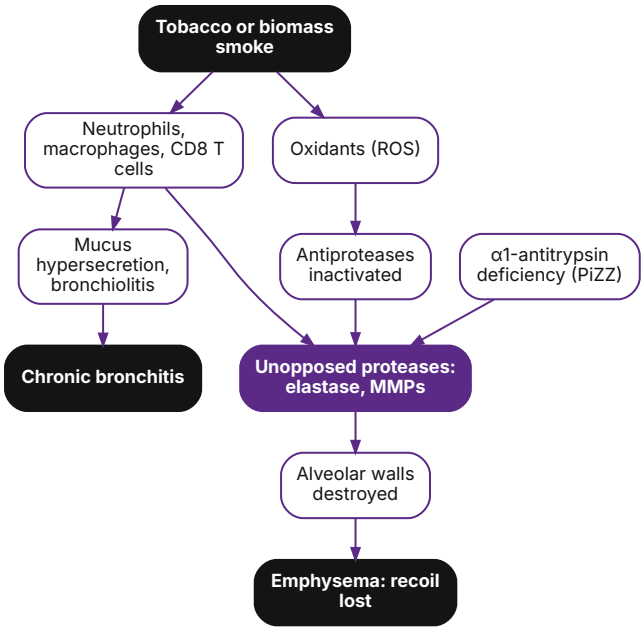


Figure 9.2 · Pathogenesis of COPD. Too much protease, too little antiprotease.

- **Inflammation:** neutrophils, macrophages, CD4/CD8 T cells; LTB4, IL-8, TNF.
- **Protease–antiprotease imbalance:** neutrophil **elastase** and macrophage **MMPs** digest elastic tissue faster than antiproteases neutralise them.
- **Oxidant–antioxidant imbalance:** smoke free radicals injure cells and **inactivate α1-antitrypsin**. Infection drives exacerbations.

MOLECULAR KEY

α1-antitrypsin (gene SERPINA1, chromosome 14) is the main serum inhibitor of neutrophil elastase. **PiZZ** homozygotes have very low levels and develop **panacinar emphysema** early, much earlier if they smoke, plus liver disease (PAS-positive globules in hepatocytes).

CONCEPT CLEAR

Why emphysema obstructs although the airways are open. Small airways have no cartilage; alveolar walls hold them open like guy-ropes on a tent. Destroy the walls and the lung loses **elastic recoil**, and bronchioles **collapse during expiration**, trapping air. The lungs hyperinflate and the patient breathes out through pursed lips to splint the airways.

GROSS

- Advanced panacinar: **pale, voluminous lungs** that cover the heart when the chest is opened
- Centriacinar: changes worse in the **upper two-thirds**
- Bullae (large subpleural blebs) in distal acinar and irregular types

MICROSCOPY

- **Thinned, destroyed alveolar walls**; adjacent alveoli fuse into large airspaces
- Free-floating fragments of septa; reduced capillary bed
- Bronchioles deformed and narrowed; bronchiolar inflammation and submucosal fibrosis

Clinical: dyspnoea is the first symptom, insidious and steadily progressive; cough and wheeze; weight loss; **barrel chest**; sits hunched forward, breathing through **pursed lips**. Death from respiratory acidosis and coma, **right heart failure** (cor pulmonale) or pneumothorax.

Not true emphysema: compensatory hyperinflation (after lobectomy), obstructive overinflation (ball-valve tumour or foreign body), bullous emphysema (bullae > 1 cm), interstitial (mediastinal) emphysema (air in stroma after an alveolar tear).

9.3 Chronic bronchitis

DEFINITION

Chronic bronchitis is defined clinically: a **persistent productive cough for at least 3 consecutive months in at least 2 consecutive years**, in the absence of another identifiable cause.

The defining abnormality is **mucus hypersecretion**, starting in large airways. **Tobacco smoke**, polluted air and dusts (grain, cotton, silica) cause **hypertrophy of submucosal mucous glands** and more **goblet cells** in small airways. Inflammation is by macrophages, neutrophils and CD8 T cells, not eosinophils (unlike asthma). Infection maintains it and triggers exacerbations.

GROSS

- Mucosa of large airways **hyperaemic, oedematous**, coated with mucopurulent secretions
- Small bronchi and bronchioles filled with similar secretions

MICROSCOPY

- **Enlarged mucous glands**: **Reid index** (gland layer ÷ wall between epithelium and cartilage) above the normal **0.4**
- Goblet cell metaplasia, mucus plugs, bronchiolar inflammation and fibrosis (**small airway disease**)
- **Squamous metaplasia** and dysplasia of bronchial epithelium

Clinical: years of cough and sputum, then dyspnoea, **hypoxaemia, hypercapnia, cyanosis**, recurrent infections and **cor pulmonale**.

EMPHYSEMA ("PINK PUFFER") VS CHRONIC BRONCHITIS ("BLUE BLOATER")

Feature	Predominant emphysema	Predominant chronic bronchitis
Dyspnoea	**Early, severe**	Late, milder
Cough, sputum	Late, scanty	**Early, copious**
Blood gases	Near-normal at rest (hyperventilates)	**Hypoxaemia, hypercapnia**
Cor pulmonale	Rare, terminal	**Common**
Chest X-ray	**Hyperinflation**, small heart	Prominent vessels, large heart
Appearance	Thin, pink, pursed-lip breathing	Cyanosed, oedematous, often overweight

9.4 Asthma

DEFINITION

Asthma is a chronic inflammatory disorder of the airways causing recurrent episodes of **wheezing, breathlessness, chest tightness and cough**, particularly at night or early morning. Its hallmarks are **intermittent, reversible airway obstruction**, chronic bronchial inflammation with **eosinophils**, smooth muscle **hypertrophy and hyperreactivity**, and excess mucus. **Status asthmaticus** is a severe attack unresponsive to therapy that can be fatal.

ATOPIC VS NON-ATOPIC ASTHMA (AND TWO SPECIAL TYPES)

Feature	Atopic (extrinsic)	Non-atopic (intrinsic)
Frequency, onset	**Commonest**; usually begins in childhood	Often adult onset
Mechanism	**Type I (IgE) hypersensitivity**, Th2 response	No allergen sensitisation; hyperreactive airways
Triggers	Allergens: dust mite, pollen, cockroach, animal dander, foods; family history of atopy	**Viral infections** (rhinovirus, parainfluenza), pollutants (SO_2, ozone), cold air, exercise, stress
Skin test, IgE	**Positive**; raised serum IgE	Negative; IgE normal

Drug-induced: aspirin-sensitive asthma (asthma + **nasal polyps** + urticaria): COX block shunts arachidonic acid to **leukotrienes**.
Occupational: fumes (epoxy resins), dusts (wood, cotton, platinum), toluene, formaldehyde; tiny repeated exposures suffice.

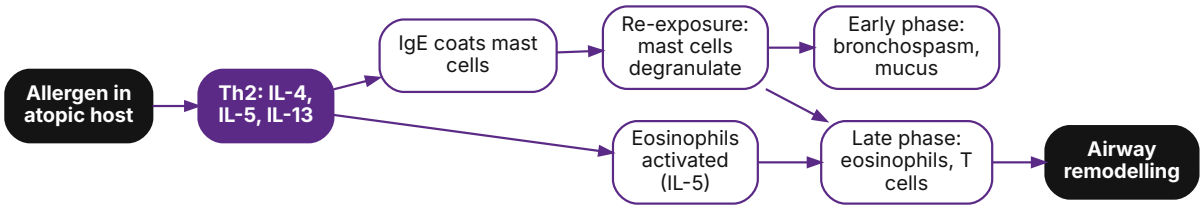


Figure 9.3 · Pathogenesis of atopic asthma. Two waves: minutes and hours.

- **IL-4**: IgE switch; **IL-5**: eosinophils; **IL-13**: mucus, IgE. Epithelial **IL-33, IL-25, TSLP** amplify Th2
- **Early phase** (minutes): histamine, PGD2, **leukotrienes C4, D4, E4** → bronchospasm, mucus, leak; vagal reflexes
- **Late phase** (hours): eotaxin recruits **eosinophils**; major basic protein injures epithelium
- **Genes:** 5q cytokine cluster, ADAM33, β2-receptor and IL-4R variants; "hygiene hypothesis"
- **Drug targets:** leukotrienes → montelukast · ACh → tiotropium · IgE → omalizumab · IL-5 → mepolizumab

GROSS

- Lungs **overdistended** (overinflation) with small areas of atelectasis
- Bronchi and bronchioles **occluded by thick, tenacious mucus plugs** (the striking finding in fatal asthma)

MICROSCOPY

- Mucus plugs with **Curschmann spirals** (whorls of shed epithelium)
- Many **eosinophils** and **Charcot–Leyden crystals** (eosinophil protein galectin-10)
- **Airway remodelling:** thickened **sub-basement membrane** (collagen I and III), enlarged submucosal glands, more goblet cells, **smooth muscle hypertrophy/hyperplasia**, increased vascularity
- **No alveolar wall destruction** (unlike emphysema)

Clinical: attacks of wheeze, dyspnoea and cough with prolonged expiration lasting hours, near-normal between attacks; eosinophilia; raised IgE (atopic); obstruction **reversible** with bronchodilator.

9.5 Bronchiectasis

DEFINITION

Bronchiectasis is **permanent dilatation of bronchi and bronchioles** caused by destruction of their muscle and elastic supporting tissue, resulting from or associated with **chronic necrotising infection**. It is not a primary disease but the end result of persistent infection or obstruction.

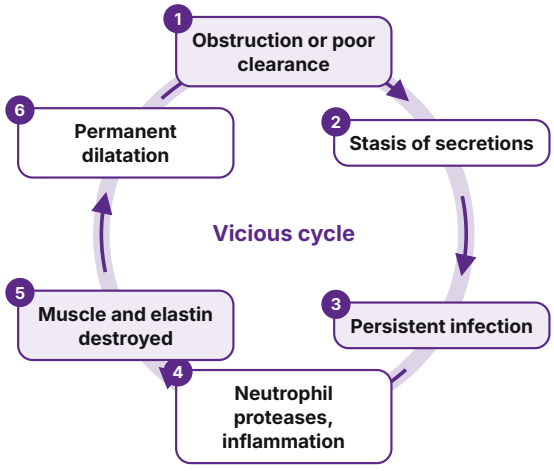


Figure 9.4 · The vicious cycle of bronchiectasis. Obstruction and infection feed each other.

CAUSES

- **Hereditary: cystic fibrosis**, **primary ciliary dyskinesia** (Kartagener), immunodeficiency, sequestration
- **Post-infectious:** necrotising pneumonia (**TB**, S. aureus, H. influenzae, Pseudomonas), adenovirus, influenza, HIV, **Aspergillus** (ABPA in asthmatics)
- **Obstruction:** tumour, foreign body, mucus impaction (one segment)
- **Other:** RA, SLE, IBD, lung transplant rejection; many idiopathic

EXAM TRAP

Kartagener syndrome: defective ciliary **dynein arms** (autosomal recessive) → **situs inversus, chronic sinusitis, bronchiectasis**; men infertile (immotile sperm).

GROSS

- Usually **lower lobes, bilateral**, most vertical airways; localised to one segment if due to tumour or foreign body
- Bronchi dilated up to **four times** normal; can be **followed out to the pleura** (normally impossible)
- Cylindrical, fusiform or **saccular**; cut surface: cyst-like spaces full of **mucopus**

MICROSCOPY

- Active disease: intense **acute and chronic inflammation** of bronchial walls; ulceration of mucosa
- Remaining epithelium pseudostratified or **squamous metaplastic**
- Destruction of muscle and elastic tissue; necrosis may form an **abscess**
- Chronic: **peribronchial fibrosis**, obliteration of bronchioles

Clinical: persistent cough with **copious, foul, mucopurulent sputum**, haemoptysis, **clubbing**, flares with infections. **Complications: cor pulmonale**, **brain abscess**, **AA amyloidosis**.

9.6 Interstitial lung disease: IPF and the pneumoconioses

Chronic interstitial lung diseases scar the **alveolar walls**, not the airways. The lung becomes **stiff** (more work to breathe) and thick walls block gas exchange (early **hypoxaemia**): **restrictive** spirometry, low DLCO, nodular or reticular shadows, and finally a **honeycomb lung** with cor pulmonale. Groups: **fibrosing** (IPF, NSIP, organising pneumonia, connective tissue disease, **pneumoconioses**, bleomycin, amiodarone, radiation), **granulomatous** (**sarcoidosis**, **hypersensitivity pneumonitis**), eosinophilic, smoking-related and others.

Idiopathic pulmonary fibrosis (IPF)

A progressive, patchy, bilateral interstitial fibrosis of unknown cause, in older adults (more often men and smokers). Its histological and radiological pattern is **usual interstitial pneumonia (UIP)**; UIP is required for the diagnosis but can also be seen in asbestosis, connective tissue disease and chronic hypersensitivity pneumonitis.

CONCEPT CLEAR

IPF is failed repair, not runaway inflammation. Repeated injury to alveolar epithelium (smoking, reflux, environment) in a susceptible lung is repaired badly: injured cells release **TGF-β**, driving **fibroblastic foci** that lay down collagen. Susceptibility: **short telomeres** (TERT, TERC) and a MUC5B promoter variant. Hence steroids fail; antifibrotics (pirfenidone, nintedanib) only slow it.

GROSS

- Pleura **cobblestoned** by retracting scars along septa
- Firm, rubbery white fibrosis, worst in the **lower lobes, subpleural** regions and along interlobular septa
- Cystic **honeycombing** late

MICROSCOPY

- **Patchy** interstitial fibrosis alternating with near-normal lung
- **Fibroblastic foci** (early) next to dense collagen (old): **temporal heterogeneity**, the hallmark of UIP
- **Honeycomb fibrosis**: cystic spaces lined by bronchiolar epithelium or type II pneumocytes
- Mild interstitial inflammation; pulmonary hypertensive vascular changes

Clinical: exertional dyspnoea, dry cough, **"velcro" crackles**, clubbing; mean survival about 3 years or less; transplant is the only cure. **NSIP**: uniform (same-age) fibrosis, better prognosis.

Pneumoconioses

Lung reactions to inhaled mineral dusts at work. Harm depends on particle **size** (1–5 μm particles lodge at distal airway bifurcations), shape, solubility and reactivity. Macrophages engulf the dust; silica and asbestos activate the **NLRP3 inflammasome** (IL-1) and growth factors (TGF-β, PDGF) drive fibrosis. Smoking worsens every mineral dust, asbestos most.

COAL, SILICA AND ASBESTOS: THE THREE PNEUMOCONIOSES EXAMINERS COMPARE

Feature	Coal workers' pneumoconiosis	Silicosis	Asbestosis and asbestos disease
Exposure	Coal mining	Sandblasting, hard-rock mining, **stone cutting and crushing**, foundries, glass	Insulation, shipbuilding, roofing, brake linings, construction
Zone	**Upper** lobes	**Upper** zones	**Lower** lobes, subpleural
Hallmark lesion	Anthracosis → simple CWP: **coal macules** (1–2 mm, carbon-laden macrophages) and nodules → **PMF**: black scars > 2 cm	**Silicotic nodule**: **whorled hyalinised collagen**, weakly **birefringent** silica; nodules coalesce → PMF	Plaques, effusions, interstitial fibrosis → honeycomb; **asbestos (ferruginous) bodies**: golden-brown beaded rods, **Prussian blue** positive
Special features	Centrilobular emphysema; **Caplan syndrome** (with RA)	**Eggshell calcification** of hilar nodes; progresses after exposure stops	**Pleural plaques**: commonest lesion; dense collagen, often calcified, parietal pleura and diaphragm domes
Cancer / TB	Lung cancer not clearly increased	**TB risk raised** (impaired macrophages); crystalline silica is a carcinogen	**Lung carcinoma** (about fivefold; multiplied by smoking); **mesothelioma** (not affected by smoking)

CONCEPT CLEAR

Why amphiboles are worse. Curly **serpentine** chrysotile (most used) is cleared more easily; straight, stiff **amphiboles** (crocidolite, amosite) align with airflow, penetrate deep to the pleura and persist, so they cause most **mesotheliomas**. Asbestos is both tumour initiator and promoter, and carries tobacco carcinogens on its surface.

PAKISTAN FOCUS

Silicosis is a documented hazard of **stone-crushing workers** in Khyber Pakhtunkhwa and Punjab; screen them for **silicotuberculosis**. About 19% of adults use tobacco (men about 32%), and **biomass fuel** smoke in rural kitchens causes COPD in women who never smoked.

Tobacco figures: GATS Pakistan, 2014.

9.7 Granulomatous ILD and cor pulmonale

Sarcoidosis

A multisystem disease of unknown cause with **non-caseating granulomas**, usually presenting as **bilateral hilar lymphadenopathy** or lung shadows on X-ray, before age 40; commoner in **non-smokers**. A **diagnosis of exclusion** (TB, fungi, berylliosis). **Pathogenesis:** an exaggerated **CD4+ Th1** response (IL-2, IFN-γ) to an unknown antigen in a predisposed host (HLA); lavage shows a high **CD4:CD8 ratio**, yet skin tests show **anergy**; polyclonal hypergammaglobulinaemia.

ORGANS

- **Lungs** (about 90%): interstitial granulomas along bronchovascular bundles, septa, pleura; 5–15% fibrose
- **Hilar/paratracheal nodes** (75–90%): firm, rubbery, **non-matted**
- **Skin** (about a quarter): **erythema nodosum**, nodules
- **Eyes**: uveitis; lacrimal and salivary glands (**Mikulicz syndrome**); spleen, liver, heart

MICROSCOPY

- Well-formed, tightly packed **non-necrotising epithelioid granulomas** with Langhans or foreign-body giant cells
- Thin rim of lymphocytes; later concentric fibrosis and hyaline scars
- **Schaumann bodies** (laminated calcium–protein concretions) and **asteroid bodies** (stellate inclusions in giant cells): typical but not specific

Labs and course: hypercalcaemia (macrophages make calcitriol), raised serum ACE. About 65–70% recover; about 20% keep lung or visual damage; 10–15% die of pulmonary fibrosis and cor pulmonale.

EXAM TRAP

Sarcoidosis vs tuberculosis. Sarcoid: **non-caseating**, tight "naked" granulomas, no AFB, **discrete** nodes, tuberculin **negative**. TB: **caseation**, **matted** nodes, sinuses, AFB on ZN, tuberculin positive. In Pakistan always exclude TB before calling a granuloma "sarcoid".

Hypersensitivity pneumonitis (extrinsic allergic alveolitis)

An immunologically mediated, mainly **interstitial** disease caused by intense or prolonged inhalation of **organic antigens**. Unlike asthma, the damage is **alveolar** (restrictive). Examples: **farmer's lung** (actinomycetes in mouldy hay), **pigeon breeder's lung**, humidifier lung, **bagassosis** (sugarcane). Immune basis: precipitating antibodies and complement in vessel walls (**type III**) plus granulomas (**type IV**).

Micro: patchy **bronchiolocentric** interstitial lymphocytes, plasma cells, macrophages; **loose non-necrotising granulomas** in about two-thirds. **Acute** attacks (fever, cough, dyspnoea **4–6 h after exposure**) resolve if exposure stops; **chronic** disease leaves irreversible upper-lobe fibrosis.

Cor pulmonale

Cor pulmonale is **right ventricular hypertrophy and dilatation**, often with failure, caused by **pulmonary hypertension** from disease of the lung, pulmonary vessels or chest wall (left heart and congenital disease excluded).

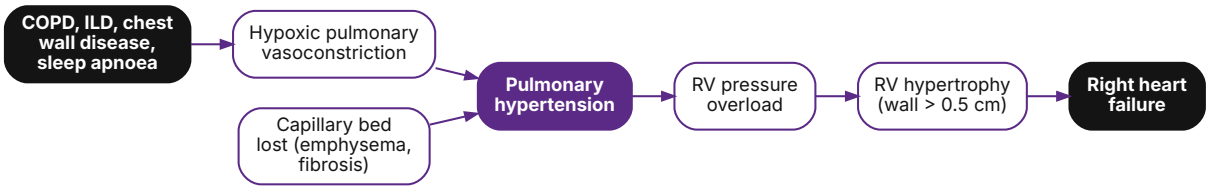


Figure 9.5 · How chronic lung disease strains the right heart.

Acute: massive pulmonary embolism (RV dilates, no hypertrophy). **Chronic**: RV wall thickens from 0.3–0.5 cm to 1.0 cm or more; pulmonary arteries show medial hypertrophy and intimal fibrosis. COPD is the commonest cause; raised JVP, hepatomegaly, pitting oedema.

EXAM ZONE Past-Paper Pattern SEQs — Solved

PROF PATTERN **a)** Define emphysema and classify it. **b)** Describe its pathogenesis. **2 + 3 marks**

MODEL ANSWER PLAN

a) Irreversible enlargement of airspaces distal to the terminal bronchiole with wall destruction, no significant fibrosis. Types: centriacinar (smokers, upper lobes), panacinar (α1-AT, lower lobes), distal acinar (paraseptal; bullae, pneumothorax), irregular. **b)** Smoke → inflammatory cells → elastase and MMPs; oxidants inactivate α1-AT; genetic α1-AT deficiency (PiZZ) → protease–antiprotease imbalance → wall destruction → loss of elastic recoil and expiratory collapse of bronchioles.

Score booster: sketch the four acini of Figure 9.1; one picture answers "classify".

MODULE PATTERN An 8-year-old boy with eczema has recurrent night-time cough and wheeze, worse in spring; his mother has hay fever. **a)** Diagnosis and type? **b)** Explain the pathogenesis. **c)** Describe the morphology of the lungs in a fatal attack. **1 + 2 + 2 marks**

MODEL ANSWER PLAN

a) Atopic (extrinsic) bronchial asthma. **b)** Th2 response: IL-4 (IgE), IL-5 (eosinophils), IL-13 (mucus); IgE on mast cells → re-exposure → early phase (histamine, leukotrienes: bronchospasm, mucus, oedema) → late phase (eosinophils, neutrophils) → remodelling. **c)** Overinflated lungs, mucus plugs; Curschmann spirals, Charcot–Leyden crystals, eosinophils, thick sub-basement membrane, gland and smooth muscle hypertrophy; no alveolar destruction.

PROF PATTERN **a)** Define bronchiectasis and list its causes. **b)** Describe its pathogenesis and morphology. **2 + 3 marks**

MODEL ANSWER PLAN

a) Permanent dilatation of bronchi from destruction of muscle and elastic tissue with chronic necrotising infection. Causes: CF, Kartagener/PCD, immunodeficiency; post-infectious (TB, staph, adenovirus, Aspergillus); obstruction (tumour, foreign body); RA, SLE. **b)** Obstruction ⇄ infection cycle (Figure 9.4). Lower lobes, bilateral; bronchi dilated to pleura, filled with pus; acute and chronic inflammation, ulceration, squamous metaplasia, peribronchial fibrosis.

CLASSIC **a)** Differentiate obstructive from restrictive lung disease. **b)** Enumerate asbestos-related diseases and describe asbestosis. **2 + 3 marks**

MODEL ANSWER PLAN

a) FEV1/FVC low vs normal/high; TLC normal/high vs low; examples of each. **b)** Pleural plaques, effusion, asbestosis, lung carcinoma, mesothelioma, laryngeal and other cancers. Asbestosis: lower-lobe subpleural interstitial fibrosis → honeycomb lung, asbestos bodies (iron-coated, Prussian blue), dyspnoea 10–20 years after exposure, cor pulmonale.

BCQs One-Best-Answer Practice

1 A 58-year-old smoker has progressive dyspnoea. FEV1/FVC is 0.52, TLC is increased and DLCO is reduced. The most likely diagnosis is:

A. Asthma
B. Emphysema
C. Idiopathic pulmonary fibrosis
D. Sarcoidosis
E. Kyphoscoliosis

2 A 34-year-old non-smoker has basal panacinar emphysema and hepatic cirrhosis. The deficient protein normally inhibits:

A. Thrombin
B. Neutrophil elastase
C. Plasmin
D. Angiotensin-converting enzyme
E. Pancreatic lipase

3 The Reid index is the ratio of:

A. Mucous gland layer thickness to wall thickness between epithelium and cartilage
B. Goblet cells to ciliated cells
C. Airway lumen to wall thickness
D. Smooth muscle to cartilage
E. Alveolar to capillary surface area

4 Charcot–Leyden crystals in the sputum of an asthmatic are derived from:

A. Neutrophils
B. Mast cells
C. Eosinophils
D. Shed epithelial cells
E. Mucus glycoproteins

5 Curschmann spirals seen in asthma are:

A. Crystals of eosinophil protein
B. Fungal hyphae
C. Asbestos fibres coated with iron
D. Whorls of shed epithelium in mucus plugs
E. Laminated calcium concretions

6 A woman with nasal polyps develops severe bronchospasm after taking aspirin. The mechanism is:

A. IgE antibody against aspirin
B. Cyclooxygenase block shunting arachidonic acid to leukotrienes
C. Complement activation by aspirin
D. Direct histamine release from basophils
E. β2-receptor blockade

7 A man with dextrocardia, chronic sinusitis, bronchiectasis and infertility most likely has a defect in:

A. CFTR chloride channel
B. α1-antitrypsin
C. Surfactant protein C
D. Fibrillin-1
E. Dynein arms of cilia

8 The histological hallmark of usual interstitial pneumonia (UIP) is:

A. Non-caseating granulomas along lymphatics
B. Uniform interstitial inflammation of the same age
C. Patchy fibrosis with fibroblastic foci beside old collagen
D. Intra-alveolar foamy exudate
E. Hyaline membranes lining alveoli

9 A stone crusher has upper-zone nodules and eggshell calcification of hilar nodes. Biopsy shows whorled hyalinised collagen nodules. The diagnosis is:

A. Asbestosis
B. Coal workers' pneumoconiosis
C. Sarcoidosis
D. Silicosis
E. Hypersensitivity pneumonitis

10 A shipyard worker who smokes has lower-lobe fibrosis with ferruginous bodies and calcified pleural plaques. Which malignancy is most strongly potentiated by his smoking?

A. Bronchogenic carcinoma
B. Malignant mesothelioma
C. Laryngeal papilloma
D. Hepatic angiosarcoma
E. Bladder carcinoma

11 A 30-year-old woman has bilateral hilar lymphadenopathy, erythema nodosum and hypercalcaemia; biopsy shows non-caseating granulomas. The hypercalcaemia is due to:

A. PTH-related peptide from granulomas
B. Osteolytic bone deposits
C. Calcitriol made by activated granuloma macrophages
D. Primary hyperparathyroidism
E. Milk-alkali syndrome

12 A farmer develops fever, cough and breathlessness 4–6 hours after handling mouldy hay. Biopsy shows bronchiolocentric interstitial lymphocytes with loose non-necrotising granulomas. The diagnosis is:

A. Atopic asthma
B. Silicosis
C. Sarcoidosis
D. Byssinosis
E. Hypersensitivity pneumonitis

ANSWER KEY WITH ONE-LINE REASONS

1. **B** Obstruction with hyperinflation and low DLCO (lost capillary bed) = emphysema.
2. **B** α1-antitrypsin (PiZZ) is the main inhibitor of neutrophil elastase.
3. **A** Normal about 0.4; raised in chronic bronchitis.
4. **C** They are crystalloids of the eosinophil protein galectin-10.
5. **D** Spiral casts of shed epithelium within mucus plugs.
6. **B** Aspirin-sensitive asthma: more leukotrienes, fewer bronchodilator prostaglandins.
7. **E** Kartagener syndrome: primary ciliary dyskinesia.
8. **C** Temporal heterogeneity with honeycombing defines UIP/IPF.
9. **D** Silicotic nodules and eggshell calcification are classic; TB risk is raised.
10. **A** Asbestos and smoking multiply lung cancer risk; smoking does not raise mesothelioma risk.
11. **C** Macrophage 1α-hydroxylase converts vitamin D to active calcitriol.
12. **E** Farmer's lung: thermophilic actinomycetes; type III and IV reactions in alveoli.

VIVA Rapid-Fire Viva Voce

Q Define emphysema.	Irreversible enlargement of airspaces distal to the terminal bronchiole with destruction of their walls, without significant fibrosis.
Q Which emphysema occurs in smokers?	Centriacinar: respiratory bronchioles enlarged, distal alveoli spared, upper lobes.
Q Which genotype causes α1-antitrypsin deficiency?	PiZZ (SERPINA1, chromosome 14): panacinar emphysema of the lower lobes plus liver disease.
Q Define chronic bronchitis.	Persistent productive cough for at least 3 consecutive months in at least 2 consecutive years.
Q What is the normal Reid index?	About 0.4; it rises in chronic bronchitis as mucous glands enlarge.
Q Pink puffer vs blue bloater?	Emphysema: thin, breathless, well oxygenated. Chronic bronchitis: cyanosed, hypercapnic, oedematous, cor pulmonale.
Q What are Curschmann spirals?	Whorls of shed epithelium in mucus plugs, seen in asthma.
Q Name the Th2 cytokines in asthma and their roles.	IL-4: IgE; IL-5: eosinophils; IL-13: mucus secretion and IgE.
Q What is Kartagener syndrome?	Primary ciliary dyskinesia with situs inversus, chronic sinusitis and bronchiectasis; male infertility.
Q What is a honeycomb lung?	End-stage fibrosis with cystic spaces lined by bronchiolar or type II epithelium.

Q What is an asbestos body?	An asbestos fibre coated with iron-containing protein: golden-brown, beaded, Prussian blue positive.
Q Define cor pulmonale.	Right ventricular hypertrophy and dilatation from pulmonary hypertension caused by lung, pulmonary vascular or chest wall disease.

OSPE Practice Stations

STATION 1 · DATA (SPIROMETRY) — 2 minutes

TASK

Patient A: FEV1 40% predicted, FVC 82%, FEV1/FVC 0.42, TLC 125%, DLCO low, FEV1 rises 4% after salbutamol. **Patient B:** FEV1 58%, FVC 55%, FEV1/FVC 0.88, TLC 60%, DLCO low. (a) Name each pattern. (b) Give the likely disease in each. (c) What would a 20% rise in FEV1 after salbutamol suggest?

ANSWER

(a) A: obstructive with air trapping; B: restrictive. (b) A: emphysema (irreversible, low DLCO); B: interstitial fibrosis such as IPF or a pneumoconiosis. (c) Reversible obstruction: asthma.

STATION 2 · GROSS SPECIMEN — 1 minute

TASK

Lower lobe of lung: bronchi are markedly dilated, some saccular, and can be traced to just beneath the pleura; they contain thick yellow pus. The surrounding lung is fibrotic. (a) Diagnosis? (b) Two causes. (c) Two complications.

ANSWER

(a) Bronchiectasis. (b) Cystic fibrosis, post-infective (TB, staphylococcal pneumonia), Kartagener syndrome, obstruction. (c) Lung or brain abscess, cor pulmonale, AA amyloidosis, haemoptysis.

STATION 3 · SLIDE — 2 minutes

TASK

Section of bronchus: the lumen is filled with a mucus plug containing spiral whorls of epithelial cells. The wall shows many eosinophils, a thickened pink band beneath the epithelium, enlarged submucosal glands and a thick smooth muscle layer. (a) Identify. (b) Name the spiral structures and the crystals you may find. (c) Name one mediator of the early phase.

ANSWER

(a) Bronchial asthma with airway remodelling. (b) Curschmann spirals; Charcot–Leyden crystals (galectin-10). (c) Histamine, leukotriene C4/D4/E4 or prostaglandin D2.

RAPID REVISION The Night-Before Page

15 ONE-LINERS

01 Obstructive: FEV1/FVC < 0.7; restrictive: TLC low, ratio normal or high.

02 Emphysema: airspaces beyond terminal bronchiole enlarged, walls destroyed.

03 Centriacinar: smokers, upper lobes; panacinar: α1-AT, lower lobes.

04 Paraseptal emphysema: bullae, spontaneous pneumothorax.

05 COPD = proteases + oxidants beat antiproteases.

06 Chronic bronchitis: cough 3 months × 2 years; Reid index > 0.4.

07 Pink puffer = emphysema; blue bloater = chronic bronchitis.

08 Atopic asthma: Th2, IgE, eosinophils; early and late phases.

09 Asthma morphology: mucus plugs, Curschmann, Charcot–Leyden, thick BM.

10 Bronchiectasis: obstruction + infection; bronchi traced to pleura.

11 Kartagener: dynein defect, situs inversus, sinusitis, bronchiectasis.

12 IPF = UIP: patchy fibrosis, fibroblastic foci, honeycombing.

13 Silica: upper zone, whorled nodules, eggshell nodes, TB risk.

14 Asbestos: lower lobes, ferruginous bodies, plaques, mesothelioma.

15 Sarcoid: non-caseating granulomas, BHL, high calcium, anergy.

MEMORY HOOKS

- **"Smokers go Centri and Up; Antitrypsin goes Pan and Down"**
- **Blue Bloater = Bronchitis**; **Pink Puffer = emPhysema**
- **Silica Sits up, Asbestos Aims low** (upper vs lower zones)
- **Asthma IL-4-5-13**: IgE (4), eosinophils (5), mucus (13)

TEST YOURSELF: FILL THE GAPS

1. α1-antitrypsin gene: _____ on chromosome 14.
2. Normal Reid index: about _____
3. Charcot–Leyden crystals are made of _____.
4. IPF hallmark: temporal _____.
5. Hilar node calcification in silicosis: _____.
6. Sarcoid hypercalcaemia: macrophages make _____.

Answers: SERPINA1 · 0.4 · galectin-10 · heterogeneity · eggshell · calcitriol

DRAW FROM MEMORY

Normal, centriacinar, panacinar and paraseptal acini side by side (Figure 9.1).

DRAW FROM MEMORY

The early and late phases of atopic asthma, from allergen to airway remodelling.

SPACED REVISION ☐ Day 1 ☐ Day 3 ☐ Day 7 ☐ Day 14 ☐ Day 30

10 Lung Tumours

Close to nine in ten lung cancers begin with a cigarette. Where the tumour sits, what it secretes and which gene drives it decide how the patient presents, how it spreads and which drug can help.

BY THE END OF THIS CHAPTER YOU CAN

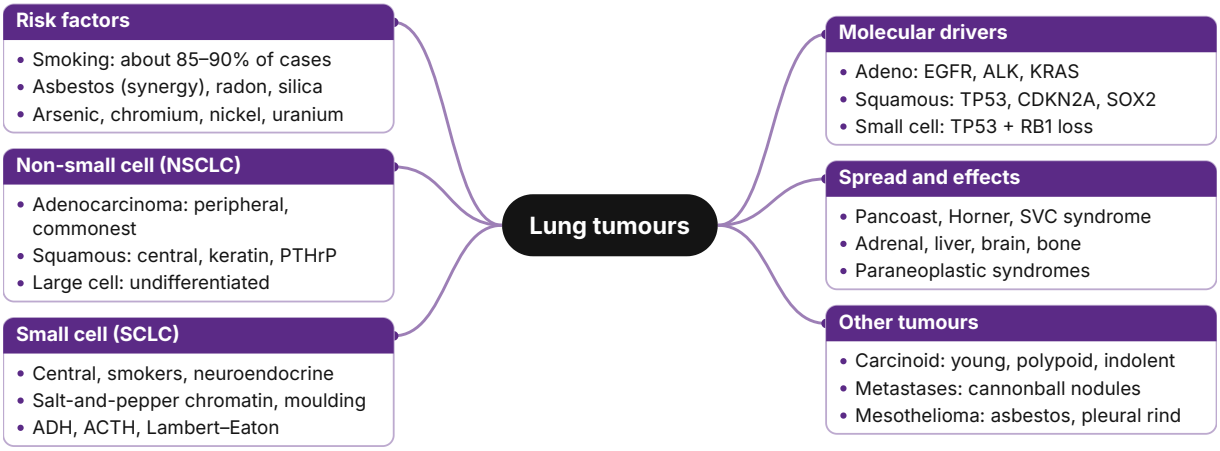
1. Classify lung tumours and contrast non-small cell with small cell carcinoma
2. Relate smoking and other carcinogens to the stepwise molecular changes of lung cancer
3. Describe the gross and microscopic features of squamous cell, adeno-, small cell and large cell carcinoma
4. Explain the local effects of spread, including Pancoast tumour, Horner syndrome and SVC syndrome
5. Tabulate the paraneoplastic syndromes of lung cancer with their mediators
6. Describe bronchial carcinoid, metastatic tumours of the lung and malignant mesothelioma

EXAM RADAR

Theory SEQs	★★★
BCQs / MCQs	★★★
Viva voce	★★☆
OSPE / slides	★★★

Classification, SCC morphology and paraneoplastic syndromes are the usual long questions; the SCC lung slide is a fixed lab station.

ROBBINS LINK Robbins Basic Pathology — *The Lung*: lung carcinomas, carcinoid tumours, metastatic tumours, pleural tumours (malignant mesothelioma).



Chapter at a glance. Six branches cover every lung tumour question.

10.1 Classification and the big divide

Lung cancer is the most frequently diagnosed major cancer and the **commonest cause of cancer death** worldwide. Most patients are in their **50s to 70s**; it is uncommon before 40. About **95%** of primary lung tumours are **carcinomas** (bronchogenic carcinoma); the rest are mainly carcinoids, mesenchymal tumours and lymphomas.

WHO TYPES OF LUNG CARCINOMA: SITE, SMOKING LINK, HISTOLOGY AND MARKERS

Type	Usual site	Smoking / patient	Key microscopy	Markers and drivers
Adenocarcinoma (commonest)	**Peripheral**; may arise in scars	Commonest type in **women and non-smokers** (most patients still smoke)	Glands, papillae, mucin; lepidic spread along alveolar walls	TTF-1, napsin A; **EGFR, ALK, KRAS**
Squamous cell carcinoma	**Central** (main, lobar bronchi)	Strong link; men	**Keratin pearls**, intercellular bridges	p40, p63, CK5/6; TP53, CDKN2A, SOX2
Small cell carcinoma (about 15%)	**Central**, hilar	**Strongest** link: almost never in non-smokers	Small cells, scant cytoplasm, **salt-and-pepper** chromatin, moulding	Chromogranin, synaptophysin, CD56, TTF-1; **TP53 + RB1**

Type	Usual site	Smoking / patient	Key microscopy	Markers and drivers
Large cell carcinoma	Peripheral or central	Smokers	Sheets of large cells, big nucleoli; no glands, keratin or NE features	Diagnosis of exclusion
Carcinoid tumour	Central bronchus	Not smoking-related; under 40	Nests of uniform cells	Chromogranin, synaptophysin

CONCEPT CLEAR

Why we still say "small cell versus non-small cell". The split is about **treatment**. A **non-small cell** carcinoma found before it has spread can be **cured by surgery**, and many adenocarcinomas carry a driver mutation that a tablet can switch off. A **small cell** carcinoma has almost always **spread by the time it is found**, so it gets chemotherapy and radiotherapy, not surgery: it melts with the first course, then returns.

10.2 Aetiology and pathogenesis

Tobacco and other carcinogens

- **Cigarette smoking:** about **85–90%** of cases. Risk rises with **pack-years**: about **10-fold** for average and up to **60-fold** for heavy smokers. Stopping lowers risk but never to baseline; passive smoking adds risk.
- Carcinogens: **polycyclic aromatic hydrocarbons** (benzo[a]pyrene), **nitrosamines**, activated by **CYP1A1**. Only about 1 in 10 heavy smokers gets lung cancer, so susceptibility matters.
- **Asbestos:** raises risk on its own; with smoking the effects **multiply** (synergy).
- **Radon** and uranium (miners), **arsenic, chromium, nickel**, vinyl chloride, mustard gas, **crystalline silica**, ionising radiation.
- **Air pollution**; **scars** (old infarcts, healed TB) may host a peripheral "scar carcinoma".

PAKISTAN FOCUS

About **one in five Pakistani adults** uses tobacco (GATS 2014: **19.1%**; men 31.8%, women 5.8%). Silicosis is a documented hazard among **stone-crushing workers** in KP and Punjab, and silica is itself a lung carcinogen. Take a **smoking and occupational history** from every adult with a chest mass.

Source: Global Adult Tobacco Survey (GATS) Pakistan, 2014.

Stepwise molecular pathogenesis

Lung cancer develops through a **stepwise accumulation** of mutations. The earliest changes (loss of chromosome **3p**, then **TP53** mutation) are found even in **normal-looking** bronchial epithelium of smokers. The whole airway lining has been exposed, so smokers carry a **field effect** and second primary tumours are common.

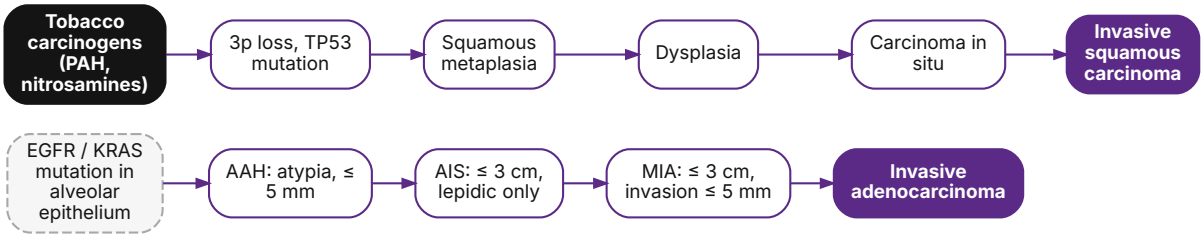


Figure 10.1 · From smoke to squamous carcinoma (top) and the peripheral adenocarcinoma sequence (bottom). Each step is a visible lesion.

AAH = atypical adenomatous hyperplasia; AIS = adenocarcinoma in situ (old "bronchioloalveolar carcinoma"); MIA = minimally invasive adenocarcinoma. AIS and MIA are cured by resection.

MOLECULAR KEY

Driver genes you must know (driver mutations are usually mutually exclusive in a given tumour):

Gene	Alteration	Typical tumour / patient	Why it matters
EGFR	Activating mutation of the tyrosine kinase	Adenocarcinoma; women, non-smokers, East Asians	Responds to **EGFR tyrosine-kinase inhibitors** (gefitinib, erlotinib, osimertinib)
ALK (also ROS1, RET)	Gene fusion, e.g. **EML4–ALK**	Adenocarcinoma; younger non-smokers	Responds to ALK inhibitors (crizotinib)
KRAS	Activating mutation	Adenocarcinoma in **smokers**	Worse outlook; few targeted options
TP53, CDKN2A, SOX2, FGFR1	Loss / amplification	Squamous cell carcinoma	Few targeted drugs
TP53 + RB1	Loss of function in almost all	Small cell carcinoma	Also MYC amplification

10.3 Morphology of the carcinomas

Why the type predicts the site: squamous and small cell carcinomas arise in the **large bronchi**, where smoke particles land first and neuroendocrine cells sit, so they are **central** and block a bronchus early (cough, haemoptysis, collapse, pneumonia). Adenocarcinoma arises from **peripheral** airway and alveolar epithelium and stays silent until it reaches the pleura or spreads. All types start as small, firm, **grey-white** mucosal thickenings. They may grow into the bronchial lumen, infiltrate along the wall or push a bulky mass into the lung. Large tumours show **necrosis, cavitation** and haemorrhage, then reach the pleura, pericardium and chest wall.

Squamous cell carcinoma

GROSS

- **Central** (hilar) mass arising in a main, lobar or segmental bronchus
- Grey-white, firm, gritty cut surface; may **cavitate** from central necrosis
- Bronchial obstruction: distal collapse, bronchiectasis, pneumonia, abscess

MICROSCOPY

- Islands and sheets of **polygonal cells** with abundant pink cytoplasm and distinct borders
- **Keratin pearls**, individual cell keratinisation, **intercellular bridges** (prickles)
- Pleomorphic hyperchromatic nuclei, mitoses; desmoplastic stroma
- Adjacent mucosa: squamous metaplasia, dysplasia, carcinoma in situ
- IHC: **p40/p63**, CK5/6 positive; TTF-1 negative

Associations: men, heavy smokers; **hypercalcaemia** from PTHrP; the type that most often cavitates.

LAB SPOTTER

SCC lung slide in 10 seconds: look for pink, glassy, sharply outlined cells in nests, then find a **whorled pink keratin pearl**. At high power, tiny spines join neighbouring cells (**intercellular bridges**). Bronchial cartilage or respiratory mucosa nearby tells you the organ is lung.

Adenocarcinoma

GROSS

- **Peripheral**, usually smaller than central tumours
- Grey-white, may be glistening with mucin; **pleural puckering** over it
- Mucinous type can look like consolidation (pneumonia-like)

MICROSCOPY

- Malignant **glands** (acinar), papillae, micropapillae or solid sheets with mucin
- **Lepidic** pattern: tumour cells creep along intact alveolar walls (AIS, MIA)
- Mucin on PAS-diastase or mucicarmine
- IHC: **TTF-1** and napsin A positive (non-mucinous types)

Behaviour: grows more slowly than squamous carcinoma but tends to **metastasise widely and early**. Test every advanced adenocarcinoma for EGFR, ALK, ROS1 and other drivers.

Small cell carcinoma (SCLC)

GROSS

- **Central, perihilar** grey-white mass
- Early, bulky spread to **hilar and mediastinal nodes**; may encase bronchi and vessels

MICROSCOPY

- Small cells (less than **3 resting lymphocytes** across), round to spindle, **scant cytoplasm**
- Finely granular **"salt-and-pepper"** chromatin; nucleoli absent or inconspicuous
- **Nuclear moulding**, very high mitotic count, extensive **necrosis**, **crush artefact**
- Basophilic DNA coating vessel walls (Azzopardi effect)
- EM: **dense-core neurosecretory granules**; IHC chromogranin, synaptophysin, CD56, TTF-1

Behaviour: the most aggressive type. Staged as **limited** (one hemithorax, one radiation field) or **extensive**. Very chemosensitive but nearly always recurs; median survival is about a year even with treatment. Source of most endocrine paraneoplastic syndromes.

Large cell carcinoma is an undifferentiated malignant epithelial tumour: sheets of **large cells** with vesicular nuclei, **prominent nucleoli** and moderate cytoplasm, showing **no** glandular, squamous or small-cell features on light microscopy and IHC. It is a **diagnosis of exclusion**. **Large cell neuroendocrine carcinoma** (organoid nests, rosettes, NE markers) behaves like SCLC.

EXAM TRAP

Old names in old papers. "**Oat cell**" carcinoma = SCLC. "**Bronchioloalveolar carcinoma**" is no longer used: Robbins splits it into adenocarcinoma in situ (non-mucinous) and invasive mucinous adenocarcinoma. A peripheral mass in a non-smoking woman is adenocarcinoma until proved otherwise.

10.4 Clinical course, spread and paraneoplastic syndromes

Lung cancer is **silent and insidious**; many tumours have spread before the first symptom. Common complaints are **cough, weight loss, chest pain, dyspnoea** and **haemoptysis**. Sometimes a metastasis (brain, bone, adrenal) is the first sign.

LOCAL EFFECTS OF SPREAD WITHIN THE CHEST

Clinical effect	Cause	Clinical effect	Cause
Pneumonia, abscess, collapse	Tumour blocks a bronchus	Raised hemidiaphragm	Phrenic nerve palsy
Pleural effusion (often bloody)	Spread to pleura	Rib destruction, chest pain	Chest wall invasion
Hoarseness	Recurrent laryngeal nerve (usually left)	**SVC syndrome**	Compression of superior vena cava

Clinical effect	Cause	Clinical effect	Cause
Dysphagia	Oesophageal invasion	**Horner syndrome**	Cervical sympathetic chain invasion
Pericarditis, tamponade	Pericardial spread	Brachial plexus pain	Apical (Pancoast) tumour

THREE NAMED SYNDROMES

Pancoast (superior sulcus) tumour: an **apical** tumour that invades the **brachial plexus** (C8, T1, T2), causing severe pain along the **ulnar nerve** distribution and wasting of hand muscles, often with rib destruction.

Horner syndrome: invasion of the cervical sympathetic chain gives ipsilateral **ptosis, miosis, anhidrosis** and apparent enophthalmos. With a Pancoast tumour it forms the **Pancoast syndrome**.

Superior vena cava syndrome: a central tumour or nodal mass compresses the SVC: **facial and arm swelling**, dusky cyanosis, distended neck and chest-wall veins, headache worse on bending. SCLC is the classic cause.

Distant spread. Lymphatic spread goes to **hilar**, then **mediastinal** and **supraclavicular (scalene)** nodes. Blood-borne metastases favour the **adrenals** (more than half of cases), **liver** (30–50%), **brain** (about 20%) and **bone** (about 20%). Adrenal deposits rarely cause adrenal failure.

Paraneoplastic syndromes

Effects **not** explained by local or metastatic tumour, caused by hormones, cytokines or antibodies; they may be the **first sign** of an occult cancer. Most endocrine syndromes come from **small cell**, hypercalcaemia from **squamous**, clotting problems from **adeno**carcinoma.

PARANEOPLASTIC SYNDROMES OF LUNG CANCER

Syndrome	Mediator / mechanism	Classic tumour
Hyponatraemia (SIADH)	Ectopic **ADH**	Small cell
Cushing syndrome	Ectopic **ACTH**	Small cell; carcinoid
Hypercalcaemia	**PTH-related peptide** (also IL-1, TNF, TGF-α, prostaglandin E)	**Squamous cell**
Lambert–Eaton myasthenic syndrome	Antibodies to presynaptic **voltage-gated calcium channels**: proximal weakness that improves with repeated effort	Small cell
Sensory neuropathy, cerebellar degeneration, encephalitis	Anti-neuronal (anti-Hu) antibodies	Small cell
Hypertrophic pulmonary osteoarthropathy, clubbing	Uncertain	Non-small cell
Migratory thrombophlebitis (Trousseau), NBTE, DIC	Procoagulants from tumour	Adenocarcinoma
Others: carcinoid syndrome, hypocalcaemia, gynaecomastia, acanthosis nigricans	Serotonin, calcitonin, gonadotropins, growth factors	Various

MEMORY HOOK

Squamous = 4 Cs · Small cell = 3 As

Squamous: Central, **C**avitation, **C**alcium high (PTHrP), **C**igarettes. **Small cell: A**DH (low sodium), **A**CTH (Cushing), **A**ntibodies (Lambert–Eaton, anti-Hu).

Staging and prognosis. NSCLC is staged by **TNM**; tumours found before nodal or distant spread are resected (lobectomy or pneumonectomy) with a real chance of cure. Overall, **fewer than one in five** patients survive 5 years, because most present late.

10.5 Carcinoid and benign lung tumours

Bronchial carcinoid tumour

A **low-grade malignant neuroendocrine** tumour, 1–5% of lung tumours. Patients are usually **under 40**; there is **no link with smoking**. A few occur with MEN1. It sits at the mild end of a spectrum whose aggressive end is SCLC.

GROSS

- Main bronchus: **polypoid, spherical, intraluminal** mass up to 3–4 cm, covered by intact mucosa
- Or a **"collar-button"** lesion: a small endobronchial knob with a larger part pushing through the wall into the lung

MICROSCOPY

- **Organoid nests**, trabeculae, ribbons of **uniform** cells
- Regular round nuclei, **salt-and-pepper** chromatin, little pleomorphism, rare mitoses
- Neurosecretory granules; chromogranin, synaptophysin positive

Clinical: cough, haemoptysis, recurrent pneumonia or collapse distal to the block. **Carcinoid syndrome** (flushing, diarrhoea, cyanosis) is uncommon. Typical carcinoids are cured by resection in most cases.

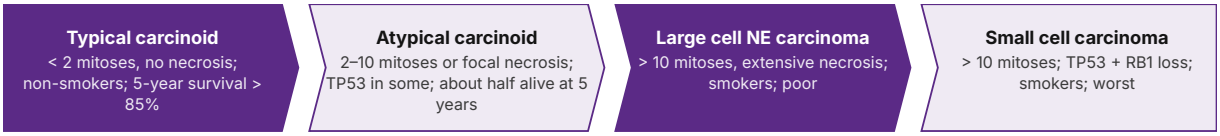


Figure 10.2 · The neuroendocrine spectrum of lung tumours. Mitoses are counted per 2 mm²; aggression rises from left to right.

Pulmonary hamartoma, the commonest **benign** lung tumour, is a round **"coin lesion"** under 3–4 cm: nodules of **mature hyaline cartilage** with fat, fibrous tissue and clefts lined by respiratory epithelium; clonal **HMGA2** (12q14–15) rearrangements make it a true neoplasm. Other solitary nodules: healed **tuberculous granuloma**, carcinoma, carcinoid, metastasis.

10.6 Metastatic tumours and malignant mesothelioma

Metastatic tumours of the lung

The lung is the **commonest site of metastases**. It receives the whole venous return and lymph through the thoracic duct, so its capillary bed is the first filter for tumour emboli from almost anywhere. Both carcinomas and sarcomas spread here.

- **Multiple discrete nodules** scattered through all lobes, mostly peripheral: the **"cannonball"** pattern (sarcomas, renal cell carcinoma, choriocarcinoma and testicular tumours, colon, breast)
- **Lymphangitic carcinomatosis**: tumour fills peribronchial and perivascular lymphatics (breast, stomach, pancreas)
- Solitary nodule, endobronchial deposit or **malignant pleural effusion**
- Lepidic spread along alveolar walls, mimicking primary adenocarcinoma

Malignant mesothelioma

DEFINITION

Malignant mesothelioma is a rare, highly malignant tumour of the **mesothelial cells** lining the pleura (less often peritoneum, pericardium or tunica vaginalis), strongly linked to **occupational asbestos exposure**.

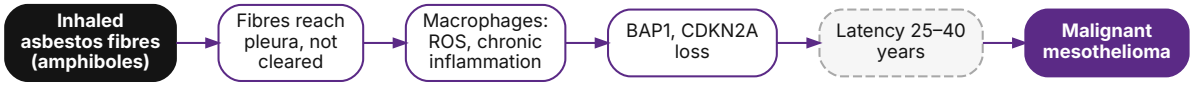


Figure 10.3 · Pathogenesis of mesothelioma. Note the long latency.

- **Amphibole** fibres (crocidolite: long, straight, stiff) are far more dangerous than serpentine chrysotile: they penetrate deep and stay.
- Lifetime risk in heavily exposed workers is as high as about **7–10%**.
- **Smoking does not increase** the risk of mesothelioma, unlike asbestos-related lung carcinoma.
- Germline **BAP1** mutations cause familial mesothelioma (with uveal melanoma).
- Markers of exposure: **pleural plaques** and **asbestos (ferruginous) bodies**: golden-brown beaded rods with an iron-protein coat, Prussian blue positive.

GROSS

- Starts as small pleural nodules, then spreads over the surface
- Lung **encased by a thick, yellow-white, firm, sometimes gelatinous rind** that obliterates the pleural space
- Often a pleural effusion; invasion of chest wall, fissures and diaphragm; distant spread is late

MICROSCOPY

- **Epithelioid** (commonest): cuboidal cells forming tubules and papillae, resembling adenocarcinoma
- **Sarcomatoid**: malignant spindle cells
- **Biphasic**: both patterns
- EM: **long, slender, branching microvilli**

Clinical: chest pain, dyspnoea, recurrent effusions. Prognosis is grim: about half die within a year of diagnosis and few survive 2 years.

EPITHELIOID MESOTHELIOMA VS PLEURAL ADENOCARCINOMA: A FAVOURITE "DIFFERENTIATE" QUESTION

Feature	Mesothelioma	Adenocarcinoma (lung)
Growth pattern	Diffuse pleural rind	Mass in lung, pleural spread secondary
Positive IHC	**Calretinin**, WT1, CK5/6, D2-40 (podoplanin)	**TTF-1**, napsin A, CEA, MOC-31, BerEP4, claudin-4
Mucin (PAS-diastase)	Negative	Often positive
Electron microscopy	Long, slender microvilli	Short, stubby microvilli
Smoking	No effect	Major risk

EXAM ZONE Past-Paper Pattern SEQs — Solved

PROF PATTERN **a)** Classify malignant tumours of the lung. **b)** Describe the gross and microscopic features of squamous cell carcinoma of the lung. **2 + 3 marks**

MODEL ANSWER PLAN

a) Carcinomas (95%): non-small cell (adenocarcinoma, squamous cell, large cell) and small cell carcinoma; neuroendocrine tumours (typical and atypical carcinoid, LCNEC); others (lymphoma, sarcoma); metastatic tumours. One line on the NSCLC vs SCLC treatment logic earns the second mark.

b) Gross: central hilar grey-white firm mass from a large bronchus, necrosis/cavitation, distal collapse or pneumonia. Micro: nests of polygonal cells with pink cytoplasm, keratin pearls, intercellular bridges, pleomorphism, mitoses, stromal invasion; adjacent metaplasia/dysplasia/CIS; p40 positive.

Score booster: add "associated with hypercalcaemia due to PTHrP"; examiners hold a mark for one clinical link.

MODULE PATTERN A 60-year-old smoker has weakness, confusion and a serum sodium of 122 mmol/L with low serum and high urine osmolality. CT chest shows a hilar mass. **a)** Name the likely tumour. **b)** Explain the electrolyte disturbance. **c)** Describe its microscopy. **1 + 2 + 2 marks**

MODEL ANSWER PLAN

a) Small cell carcinoma of the lung. **b)** Ectopic ADH → water retention → dilutional hyponatraemia, low serum osmolality, inappropriately concentrated urine (SIADH). **c)** Small cells with scant cytoplasm, salt-and-pepper chromatin, absent nucleoli, nuclear moulding, high mitoses, necrosis, crush artefact; NE markers and dense-core granules.

Score booster: name two other SCLC paraneoplastic syndromes (Cushing, Lambert–Eaton).

CLASSIC Enumerate the paraneoplastic syndromes associated with bronchogenic carcinoma, giving the mediator and the tumour type for each. **5 marks**

MODEL ANSWER PLAN

Define paraneoplastic syndrome (one line). Then a table of at least 6: SIADH/ADH/SCLC; Cushing/ACTH/SCLC; hypercalcaemia/PTHrP/SCC; Lambert–Eaton/anti-VGCC antibodies/SCLC; hypertrophic pulmonary osteoarthropathy with clubbing/NSCLC; Trousseau syndrome and NBTE/procoagulants/adenocarcinoma; carcinoid syndrome; anti-Hu neuropathy.

BCQs One-Best-Answer Practice

1 A 58-year-old heavy smoker has a cavitating central lung mass and a serum calcium of 13.2 mg/dL with low PTH. The hypercalcaemia is best explained by tumour secretion of:

A. Parathyroid hormone-related peptide
B. Calcitonin
C. Vitamin D binding protein
D. ACTH
E. Antidiuretic hormone

2 A 45-year-old woman who has never smoked has a peripheral lung adenocarcinoma. Which molecular change is she most likely to have, with a tablet treatment available?

A. RB1 loss
B. KRAS mutation
C. Activating EGFR mutation
D. SOX2 amplification
E. MYC amplification

3 Which microscopic feature best identifies small cell carcinoma of the lung?

A. Keratin pearls
B. Prominent cherry-red nucleoli
C. Mucin-filled glands
D. Nuclear moulding with finely granular chromatin
E. Psammoma bodies

4 A smoker has proximal leg weakness that improves after repeated contraction. The antibodies are directed against:

A. Acetylcholine receptors
B. Presynaptic voltage-gated calcium channels
C. Muscle-specific kinase
D. Myelin basic protein
E. Thyroid peroxidase

5 A man with an apical lung tumour has pain along the inner arm and ptosis with a small pupil on the same side. The eye signs result from invasion of the:

A. Phrenic nerve
B. Recurrent laryngeal nerve
C. Vagus nerve
D. Superior vena cava
E. Cervical sympathetic chain

6 The organ most frequently involved by blood-borne metastasis from lung carcinoma is the:

A. Adrenal gland
B. Spleen
C. Kidney
D. Thyroid
E. Skin

7 A 30-year-old non-smoker has recurrent right lower lobe pneumonia. Bronchoscopy shows a smooth polypoid mass in the bronchus; biopsy shows nests of uniform cells with salt-and-pepper chromatin and rare mitoses. Diagnosis:

A. Small cell carcinoma
B. Squamous cell carcinoma
C. Bronchial carcinoid tumour
D. Pulmonary hamartoma
E. Adenocarcinoma in situ

8 Which pair of tumour suppressor genes is inactivated in almost all small cell lung carcinomas?

A. APC and KRAS
B. BRCA1 and BRCA2
C. VHL and PTEN
D. TP53 and RB1
E. NF1 and NF2

9 A pleural tumour shows tubulopapillary epithelioid cells that stain for calretinin and WT1 but not TTF-1. The diagnosis is:

A. Metastatic adenocarcinoma
B. Solitary fibrous tumour
C. Pleural lymphoma
D. Squamous cell carcinoma
E. Malignant mesothelioma

10 Regarding asbestos exposure, which statement is correct?

A. Smoking multiplies the risk of mesothelioma
B. Smoking multiplies the risk of lung carcinoma but not of mesothelioma
C. Chrysotile is more carcinogenic than crocidolite
D. Mesothelioma usually follows within 5 years of exposure
E. Pleural plaques are premalignant

11 A 2.5 cm peripheral adenocarcinoma grows purely along intact alveolar walls with no stromal invasion. It is classified as:

A. Atypical adenomatous hyperplasia
B. Adenocarcinoma in situ
C. Minimally invasive adenocarcinoma
D. Invasive mucinous adenocarcinoma
E. Large cell carcinoma

12 A chest X-ray of a 70-year-old shows multiple round nodules of different sizes in both lungs, mostly peripheral. The most likely diagnosis is:

A. Metastatic tumour deposits
B. Multiple hamartomas
C. Bronchial carcinoid
D. Mesothelioma
E. Lobar pneumonia

ANSWER KEY WITH ONE-LINE REASONS

1. **A** SCC secretes PTHrP; native PTH is suppressed.
2. **C** EGFR mutants (women, non-smokers) respond to EGFR TKIs.
3. **D** SCLC: scant cytoplasm, moulding, crush artefact.
4. **B** Lambert–Eaton syndrome, classically with SCLC.
5. **E** Pancoast tumour causing Horner syndrome.
6. **A** Adrenals in over half; then liver, brain, bone.
7. **C** Young non-smoker, polyp, uniform NE cells: carcinoid.
8. **D** TP53 + RB1 loss is the signature of SCLC.
9. **E** Calretinin, WT1, CK5/6, D2-40 = mesothelioma.
10. **B** Synergy is for carcinoma only; latency 25–40 years.
11. **B** AIS: 3 cm or less, pure lepidic growth, no invasion.
12. **A** Cannonball metastases: commonest lung tumours.

VIVA Rapid-Fire Viva Voce

Q Commonest histological type of lung cancer?	Adenocarcinoma (also commonest in non-smokers).
Q Which type has the strongest link with smoking?	Small cell carcinoma, then squamous cell carcinoma.
Q Name two central lung carcinomas.	Squamous cell carcinoma and small cell carcinoma.
Q Hallmarks of SCC on a slide?	Keratin pearls and intercellular bridges.
Q IHC to prove SCLC?	Chromogranin, synaptophysin, CD56 (TTF-1).
Q What is a Pancoast tumour?	Apical tumour: brachial plexus pain plus Horner syndrome.
Q Features of Horner syndrome?	Ipsilateral ptosis, miosis, anhidrosis and apparent enophthalmos.
Q Commonest paraneoplastic syndrome of SCC lung?	Hypercalcaemia due to PTH-related peptide.
Q What is Lambert–Eaton syndrome?	Anti-calcium-channel weakness easing with use; SCLC.
Q Typical vs atypical carcinoid?	Atypical: 2–10 mitoses/2 mm² or necrosis; worse.
Q What are asbestos bodies?	Iron-coated fibres: golden-brown beaded rods.
Q Two markers for mesothelioma?	Calretinin and WT1 (also CK5/6, D2-40).

OSPE Practice Stations

STATION 1 · SLIDE 2 minutes

TASK

Lung section near a bronchus: infiltrating nests of large polygonal cells with abundant eosinophilic cytoplasm, distinct borders and pleomorphic nuclei. Some nests have concentric pink lamellated centres. (a) Identify. (b) Two diagnostic features. (c) One paraneoplastic syndrome.

ANSWER

(a) Well-differentiated squamous cell carcinoma, lung. (b) Keratin pearls; intercellular bridges (also individual cell keratinisation). (c) Hypercalcaemia from PTHrP.

STATION 2 · GROSS SPECIMEN 1 minute

TASK

Lung with the pleural cavity obliterated by a thick, firm, grey-white to yellow rind of tissue encasing the whole lobe and extending into the fissures. (a) Diagnosis. (b) Main risk factor. (c) One microscopic subtype and one IHC marker.

ANSWER

(a) Malignant pleural mesothelioma. (b) Occupational asbestos (amphibole) exposure, latency 25–40 years. (c) Epithelioid (or sarcomatoid, biphasic); calretinin.

RAPID REVISION The Night-Before Page

15 ONE-LINERS

01 Lung cancer: top cancer killer; 85–90% smokers.
02 Adeno: commonest, peripheral, non-smokers.
03 SCC: central, keratin pearls, cavitation, PTHrP.
04 SCLC: central, moulding, crush, NE granules.
05 Large cell carcinoma: diagnosis of exclusion.
06 Smokers: 3p loss → TP53 → metaplasia → CIS.
07 AAH → AIS (≤ 3 cm) → MIA (≤ 5 mm) → invasive.
08 EGFR, ALK: tablet-treatable adeno drivers.
09 SCLC: TP53 + RB1 loss; chemo, not surgery.
10 Pancoast: brachial plexus pain + Horner.
11 SVC syndrome: swollen face, arms, neck veins.
12 Metastases: adrenal > liver > brain, bone.
13 Carcinoid: young non-smoker, bronchial polyp.
14 Lung: commonest site of metastases (cannonballs).
15 Mesothelioma: asbestos, rind, calretinin.

MEMORY HOOKS

- **Squamous = 4 Cs**: central, cavitation, calcium, cigarettes
- **Small cell = 3 As**: ADH, ACTH, antibodies
- **"Adeno is Away"**: peripheral, away from the hilum
- **Meso = Calretinin; Adeno = TTF-1**

TEST YOURSELF: FILL THE GAPS

1. Hypercalcaemia in SCC lung is due to _____.
2. Lambert–Eaton antibodies target _____ channels.
3. Commonest site of distant metastasis: _____.
4. ALK fusion partner in lung adeno: _____.
5. Latency of mesothelioma: _____ years.
6. Commonest benign lung tumour: _____.

Answers: PTHrP · voltage-gated calcium · adrenal · EML4 · 25–40 · hamartoma

DRAW FROM MEMORY

A chest outline showing a central SCC/SCLC and a peripheral adenocarcinoma, with arrows to SVC, recurrent laryngeal nerve, sympathetic chain and pleura.

DRAW FROM MEMORY

The smoke-to-SCC sequence and the AAH → AIS → MIA → invasive adenocarcinoma ladder.

SPACED REVISION ☐ Day 1 ☐ Day 3 ☐ Day 7 ☐ Day 14 ☐ Day 30

11

Anaemias I: Approach, Iron Deficiency, Megaloblastic & Aplastic Anaemia

Every anaemia answers two questions: are red cells being lost, destroyed or under-produced, and what do they look like? The MCV and the reticulocyte count point to the answer before any special test is ordered.

BY THE END OF THIS CHAPTER YOU CAN

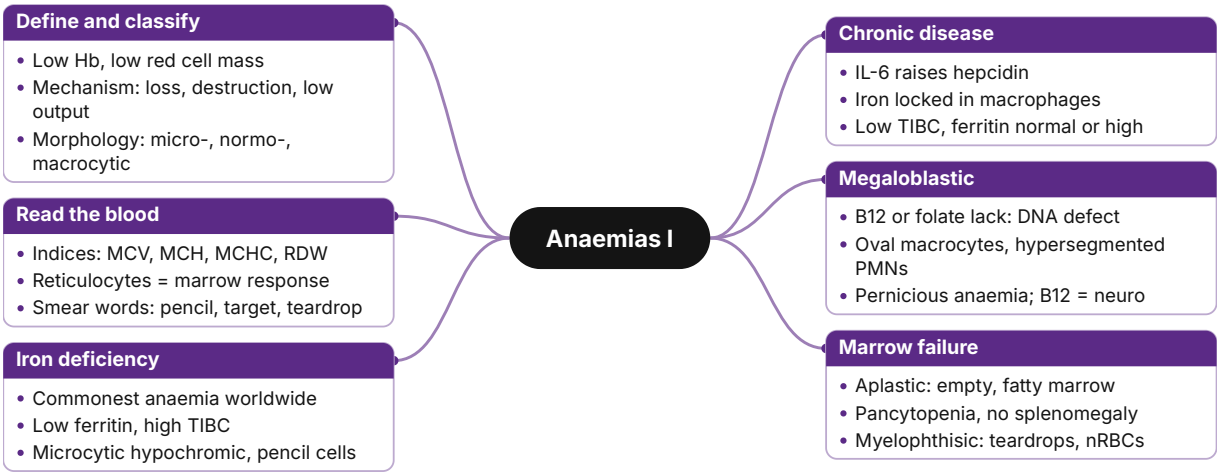
1. Define anaemia and classify it by mechanism and by red cell morphology
2. Interpret red cell indices, the reticulocyte count and the vocabulary of the peripheral smear
3. Explain iron metabolism and the role of hepcidin
4. Describe the causes, stages, morphology and laboratory diagnosis of iron deficiency anaemia
5. Differentiate iron deficiency, anaemia of chronic disease and thalassaemia trait from iron studies
6. Compare vitamin B12 and folate deficiency, including the pathogenesis of pernicious anaemia
7. Describe aplastic and myelophthisic anaemia

EXAM RADAR

Theory SEQs	★★★
BCQs / MCQs	★★★
Viva voce	★★★
OSPE / slides	★★★

Iron deficiency (causes, labs), pernicious anaemia and B12 vs folate are long-question regulars; peripheral smear and iron-study stations are fixed in the lab exam.

ROBBINS LINK Robbins Basic Pathology — *Hematopoietic and Lymphoid Systems*: anaemias (classification, blood loss), anaemias of diminished erythropoiesis: iron deficiency, anaemia of chronic inflammation, megaloblastic anaemias, aplastic and myelophthisic anaemia.



Chapter at a glance. Size the cell, count the reticulocytes, then find the cause.

11.1 Anaemia: definition, effects and classification

DEFINITION

Anaemia is a reduction in the **oxygen-carrying capacity** of the blood, usually because the total **red cell mass** has fallen below normal. In practice it is diagnosed when the **haemoglobin** (or haematocrit) is below the reference range for age and sex. WHO cut-offs: Hb **< 13 g/dL** in men, **< 12 g/dL** in non-pregnant women, **< 11 g/dL** in pregnancy and in children aged 6 months to 5 years.

WHY PATIENTS FEEL WHAT THEY FEEL

- **Tissue hypoxia:** pallor, fatigue, weakness, dyspnoea and palpitations on exertion, headache, faintness;

BLOOD LOSS ANAEMIA

- **Acute:** the danger is **hypovolaemia** and shock, not anaemia. The Hb is normal at first and falls over 2–3

angina in those with coronary disease.

- **Compensation:** raised cardiac output (flow murmurs, high-output failure if severe), more 2,3-BPG so Hb releases O_2 more easily, and renal **erythropoietin** drives marrow erythroid hyperplasia.
- Severe chronic hypoxia: **fatty change** of liver, kidney and heart ("tigered" myocardium).

days as tissue fluid dilutes the blood. Reticulocytes rise to about **10–15% by day 7**.

- **Chronic:** small repeated losses deplete iron, so it presents as **iron deficiency anaemia**.

CLASSIFICATION OF ANAEMIA BY MECHANISM (ROBBINS)

Mechanism	Main groups	Examples
Blood loss	Acute; chronic	Trauma; GI bleeding, menorrhagia, hookworm
Increased destruction (haemolysis, see Chapter 12)	Intrinsic (intracorpuscular) defects	Spherocytosis, G6PD deficiency, sickle cell, thalassaemia, PNH
	Extrinsic (extracorpuscular) factors	Autoimmune, transfusion, microangiopathic, malaria, hypersplenism
Decreased production	Nutritional deficiency	**Iron** (Hb synthesis); **B12, folate** (DNA synthesis)
	Marrow failure / replacement	**Aplastic** anaemia, pure red cell aplasia; **myelophthisis**, leukaemia, MDS
	Low erythropoietin / inflammation	Chronic kidney disease; **anaemia of chronic disease**
	Other	Parvovirus B19, hypothyroidism, liver disease

11.2 Reading the blood: indices, reticulocytes and the smear

CONCEPT CLEAR

Two numbers do most of the work. The **MCV** tells you *what kind* of factory fault exists: too little haemoglobin to fill the cell (small, pale cells) or too little DNA synthesis to let the cell divide on time (big cells). The **reticulocyte count** tells you *whether the marrow is answering*: high in bleeding and haemolysis (the factory works overtime), low when the factory itself is short of raw materials or damaged.

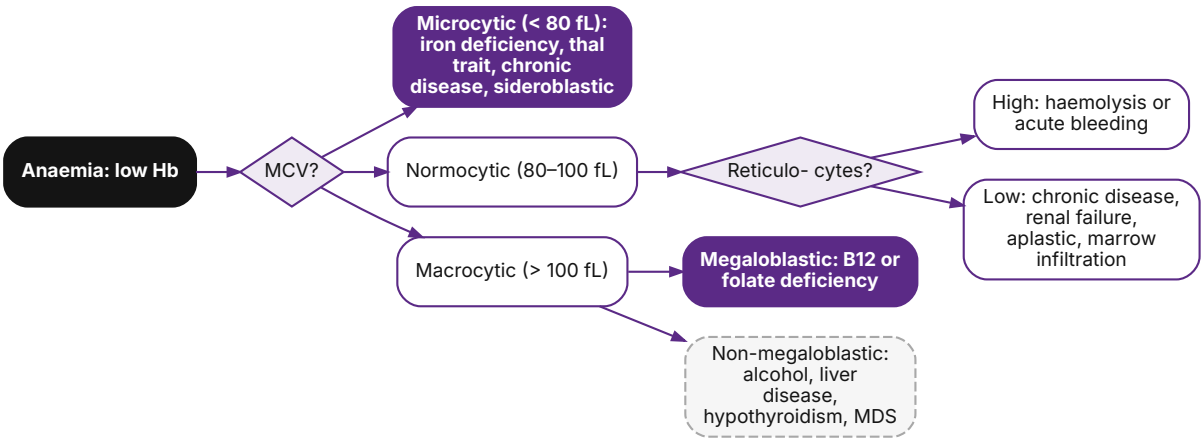


Figure 11.1 · Approach to anaemia by MCV. Size first, then the reticulocyte count or the marrow.

LAB VALUES

Hb men 13.6–17.2; women 12.0–15.0 g/dL

MCV = Hct ÷ RBC count: **82–96 fL**

MCH = Hb ÷ RBC count: 27–33 pg

MCHC = Hb ÷ Hct: 33–37 g/dL

RDW (size variation): 11.5–14.5%

Reticulocytes 0.5–1.5% of red cells

Corrected retics = retic % × (patient Hct ÷ 45)

Mentzer index = MCV ÷ RBC (millions/µL)

PERIPHERAL SMEAR VOCABULARY: WHAT YOU SEE AND WHAT IT SUGGESTS

Finding	Think of	Finding	Think of
Hypochromia: central pallor > ⅓ of cell	Iron deficiency, thalassaemia	**Spherocytes**: small, dense, no pallor	Hereditary spherocytosis, warm AIHA
Pencil cells: thin elongated cells	Iron deficiency	**Schistocytes**: helmet and triangle fragments	DIC, TTP/HUS, prosthetic valves
Target cells: central dot of Hb	Thalassaemia, HbC, liver disease, post-splenectomy	**Bite cells, Heinz bodies**	G6PD deficiency
Macro-ovalocytes	Megaloblastic anaemia	**Sickle cells**	Sickle cell disease
Hypersegmented neutrophils	B12 / folate deficiency	**Teardrop cells** (dacrocytes)	Myelofibrosis, myelophthisis
Howell–Jolly bodies: nuclear remnant	Asplenia, megaloblastic anaemia	**Basophilic stippling**	Lead poisoning, thalassaemia
Polychromasia: blue-grey large cells	Reticulocytosis: haemolysis, bleeding	**Nucleated RBCs**	Severe haemolysis, marrow infiltration
Rouleaux: stacked coins	Myeloma, high globulins	**Anisocytosis / poikilocytosis**	Variation in size / shape (raised RDW)

11.3 Iron metabolism and iron deficiency anaemia

Iron balance and hepcidin

The body holds about **2.5 g** of iron in women and **3.5 g** in men. Roughly 80% is **functional** (haemoglobin, myoglobin, cytochromes, catalase); the rest is **stored** as ferritin and haemosiderin in liver, spleen and marrow macrophages. There is **no regulated route of excretion**: we lose only 1–2 mg a day in shed skin and gut cells (more with menstruation). Iron balance is therefore controlled entirely at the point of **absorption**, in the **duodenum**. Haem iron (meat) is absorbed far better than non-haem iron (cereals, vegetables).

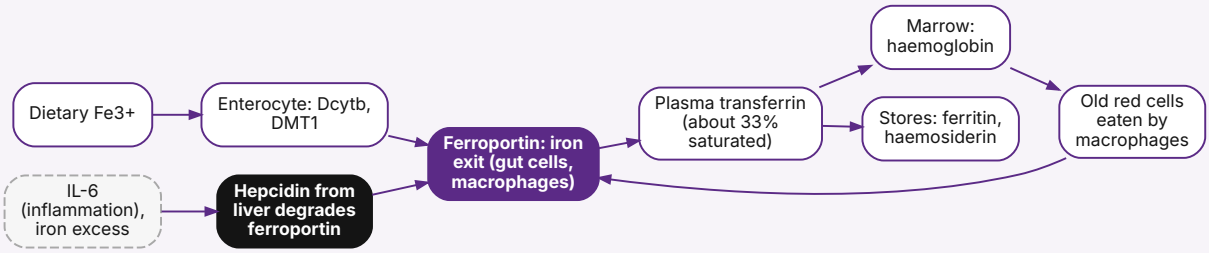


Figure 11.2 · Iron absorption, recycling and its master switch, hepcidin. Hepcidin shuts the same exit door (ferroportin) on gut cells and macrophages; iron lack, hypoxia and active erythropoiesis switch hepcidin off.

CONCEPT CLEAR

Hepcidin is the iron thermostat. When iron is plentiful or infection is present, the liver releases hepcidin, which binds **ferroportin** and causes its destruction: absorption stops and macrophages hold on to their iron. When iron is short, or the marrow is working hard, hepcidin falls and the doors open. **Too little hepcidin** → iron overload (haemochromatosis, Chapter 6; thalassaemia, Chapter 12). **Too much hepcidin** → anaemia of chronic disease.

Iron deficiency anaemia (IDA)

Iron deficiency is the **commonest nutritional disorder** in the world and the **commonest cause of anaemia**. It develops whenever iron loss or need outruns intake and absorption.

- **Low intake:** infants on prolonged milk feeds, the poor, the elderly, strict vegetarians.
- **Poor absorption:** coeliac disease, gastrectomy, achlorhydria, chronic diarrhoea.
- **Increased need:** growing infants and children, adolescents, **pregnancy** and lactation.
- **Chronic blood loss** (the commonest cause in adults): **menorrhagia**; GI bleeding from peptic ulcer, NSAIDs, piles, **colorectal carcinoma**; **hookworm** infestation in developing countries.

> **EXAM TRAP**
> In an **adult man or post-menopausal woman**, iron deficiency means **GI blood loss until proved otherwise**: look for an occult colonic or gastric cancer.

> **PAKISTAN FOCUS**
> Anaemia is very common in Pakistan: the National Nutrition Survey 2018 found it in about **half of children under five** (54%) and about **two-fifths of women of reproductive age** (42%). **Iron deficiency** is the main cause. Dietary **B12** deficiency is seen in vegetarians and **folate** deficiency in pregnancy.
> Source: National Nutrition Survey Pakistan, 2018.



Figure 11.3 · Stages of iron deficiency. Stores empty first; the blood count is the last thing to change.

PERIPHERAL SMEAR

- **Microcytic, hypochromic** red cells: Hb only a thin rim, central pallor enlarged
- **Anisocytosis** (high RDW) and poikilocytosis with **pencil cells**
- Reticulocytes low for the degree of anaemia; platelets often raised

BONE MARROW

- Mild to moderate **erythroid hyperplasia**; small erythroblasts with ragged cytoplasm (micronormoblasts)
- **Absent stainable iron** in macrophages on **Prussian blue** (Perls) stain: the most reliable morphological proof

Clinical: general symptoms of anaemia plus **koilonychia** (spoon nails), angular cheilitis, **atrophic glossitis**, alopecia, **pica** (craving for clay, ice). **Plummer–Vinson syndrome:** iron deficiency + atrophic glossitis + **oesophageal webs** (dysphagia); premalignant for post-cricoid carcinoma. In children, iron lack impairs growth and learning.

LAB DIAGNOSIS OF IDA

CBC low Hb, **MCV, MCH, MCHC** ; high **RDW**	**Serum ferritin** **low** (below about 12–15 µg/L): the best single test
Serum iron low; **TIBC high** (more transferrin)	**Transferrin saturation** low (< 15%)
Marrow no stainable iron (rarely needed)	**Proof** reticulocyte rise, then Hb rise on oral iron

11.4 Anaemia of chronic disease (inflammation)

The **commonest anaemia in hospitalised patients**. It accompanies **chronic infections** (tuberculosis, osteomyelitis, lung abscess, infective endocarditis), **chronic immune disorders** (rheumatoid arthritis, inflammatory bowel disease) and **cancers** (Hodgkin lymphoma, carcinoma of lung and breast). It is mild to moderate and does not progress beyond a certain level.

PATHOGENESIS

1. **IL-6** from inflammation stimulates the liver to make **hepcidin**.
2. Hepcidin degrades ferroportin: iron is **trapped in macrophages** and gut absorption falls. Serum iron drops although **stores are full**.
3. Erythropoietin output is **inappropriately low**, and IL-1, TNF and IFN-γ suppress red cell progenitors.

> **CONCEPT CLEAR**
>
> **Why hide iron?** Bacteria need iron to grow. Locking it inside macrophages starves invading microbes, a defence called **nutritional immunity**. The price is an iron-starved marrow despite plentiful stores.

Labs: normocytic normochromic (sometimes mildly microcytic) cells; **low serum iron**, **low TIBC**, **normal or high ferritin**; increased iron in marrow macrophages. Treat the underlying disease; erythropoietin helps some patients.

MICROCYTIC ANAEMIAS: IRON DEFICIENCY VS CHRONIC DISEASE VS β-THALASSAEMIA TRAIT (THE MOST USEFUL DATA TABLE IN THIS CHAPTER)

Test	Iron deficiency	Chronic disease	β-thalassaemia trait
Hb	Low, may be very low	Mild to moderately low	Normal or slightly low
MCV	Low	Normal or mildly low	**Very low** for the Hb
RBC count	Low	Low	**Normal or high** (often > 5 million/μL)
RDW	**High**	Normal	Normal or slightly raised
Mentzer index (MCV ÷ RBC)	**> 13**	Not useful	**< 13**
Serum ferritin	**Low**	**Normal or high**	Normal
Serum iron	Low	Low	Normal
TIBC (transferrin)	**High**	**Low**	Normal
Transferrin saturation	Low	Low or normal	Normal
Marrow iron (Perls)	Absent	**Increased** in macrophages	Present
Hepcidin	Low	High	Normal or low
Smear / special test	Pencil cells	Non-specific	Target cells, stippling; **HbA2 > 3.5%** on electrophoresis

> **EXAM TRAP**
>
> **Ferritin is an acute-phase protein.** Infection, inflammation, liver disease and cancer raise it, so a "normal" ferritin in a sick patient does not exclude iron deficiency; a **low** ferritin, however, always means low stores. Do not give long-term iron to a thalassaemia carrier: the Mentzer index and Hb electrophoresis prevent this common error (see Chapter 12).

11.5 Megaloblastic anaemias

The two causes are **vitamin B12 (cobalamin)** deficiency and **folate** deficiency. Both vitamins are needed to make **thymidine** for DNA.

> **CONCEPT CLEAR**
>
> **The nucleus lags behind the cytoplasm.** Without enough DNA precursors, cells cannot complete S phase on time, but RNA and protein synthesis carry on. The cytoplasm keeps growing and fills with haemoglobin while the nucleus stays large and immature: **nuclear–cytoplasmic asynchrony**. The result is giant precursors (**megaloblasts**), many of which die in the marrow (**ineffective erythropoiesis**: raised LDH and indirect bilirubin). Every dividing cell line is hit, so there is **pancytopenia**, and rapidly dividing epithelium suffers too (sore, smooth tongue).

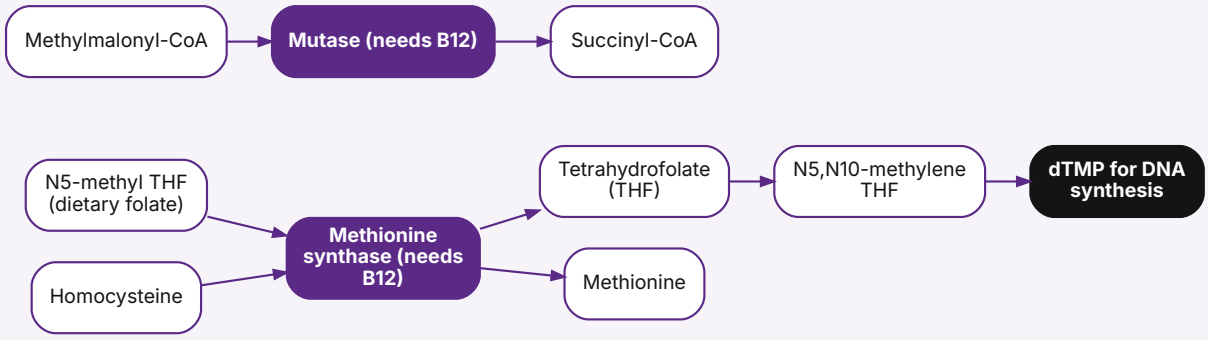


Figure 11.4 · Why B12 lack looks exactly like folate lack. Without B12, folate is trapped as N5-methyl THF (the "methyl-folate trap"), so DNA synthesis fails as in folate deficiency; the second B12 reaction explains the raised methylmalonic acid.

PERIPHERAL SMEAR

- **Macro-ovalocytes** (MCV > 100 fL), well filled with Hb, no central pallor; MCHC normal
- Marked anisopoikilocytosis; occasional nucleated RBCs, Howell–Jolly bodies
- **Hypersegmented neutrophils** (≥ 6 lobes, or > 5% with 5 lobes): an early clue
- Low reticulocytes; leucopenia and thrombocytopenia (**pancytopenia**)

BONE MARROW

- **Hypercellular**, with marked erythroid hyperplasia
- **Megaloblasts**: large cells with fine, open, "sieve-like" chromatin; Hb-rich cytoplasm around an immature nucleus
- **Giant metamyelocytes** and band forms
- Large megakaryocytes with bizarre multilobed nuclei

VITAMIN B12 VS FOLATE DEFICIENCY

Feature	Vitamin B12 (cobalamin)	Folate
Source	Animal foods only (meat, eggs, milk)	Green vegetables, fruit, liver; destroyed by prolonged cooking
Absorption	With **intrinsic factor** in the **terminal ileum**	**Proximal jejunum**
Body stores	Large: deficiency takes **years**	Small: deficiency in **weeks to months**
Main causes	**Pernicious anaemia**, gastrectomy, ileal resection or Crohn disease, fish tapeworm, blind-loop bacteria, strict vegans	Poor diet (alcoholics, elderly), **pregnancy**, haemolytic anaemias, malabsorption (coeliac, tropical sprue), **drugs**: methotrexate, phenytoin, trimethoprim
Neurological damage	**Yes**: subacute combined degeneration, neuropathy	**No** (but neural tube defects in the fetus)
Serum homocysteine	Raised	Raised
Methylmalonic acid	**Raised**	Normal

Vitamin B12 absorption

1. **Stomach** — pepsin frees B12 from food; it binds haptocorrin (R-binder) from saliva
2. **Duodenum** — pancreatic proteases release B12, which binds **intrinsic factor** from parietal cells
3. **Terminal ileum** — the IF–B12 complex is taken up by the **cubilin** receptor; transcobalamin II carries B12 to liver and marrow

Pernicious anaemia

DEFINITION

Pernicious anaemia is a megaloblastic anaemia caused by an **autoimmune atrophic gastritis** that destroys parietal cells, so there is no **intrinsic factor** and dietary B12 cannot be absorbed. It is the commonest cause of B12 deficiency.

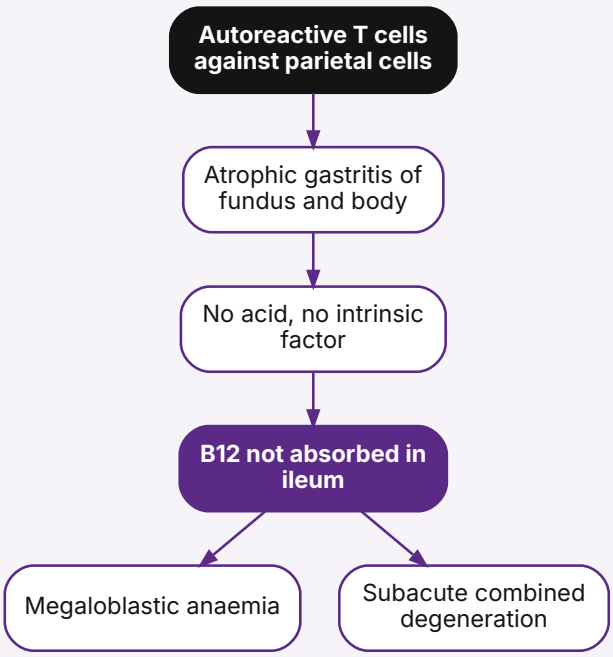


Figure 11.5 · Pathogenesis of pernicious anaemia.

AUTOANTIBODIES

- **Type I** (blocking): stop B12 binding to IF; in about 75%; highly specific.
- **Type II** (binding): stop the IF–B12 complex binding its ileal receptor.
- **Type III**: against the parietal cell proton pump (**H^+,K^+-ATPase**); in 85–90%, less specific.

Associated autoimmune diseases: Hashimoto thyroiditis, Addison disease, type 1 diabetes. Usually older adults (median about 60 years).

STOMACH AND TONGUE

- Thin, flat mucosa of fundus and body (**atrophic gastritis**); antrum spared
- Smooth, red, shiny "beefy" tongue (**atrophic glossitis**)

MICROSCOPY

- Loss of parietal and chief cells; lymphocytes and plasma cells in mucosa
- **Intestinal metaplasia**; megaloblastic change in epithelial cells
- **Spinal cord:** demyelination of **dorsal and lateral columns**

Clinical: insidious anaemia with mild jaundice ("lemon-yellow" pallor), glossitis, and neurological signs: symmetric tingling and numbness of feet and hands, loss of vibration and position sense, ataxia. Nerve damage can occur **without anaemia** and may not reverse. Risk of **gastric carcinoma** and carcinoid tumour is raised (Chapter 2). **Diagnosis:** low serum B12, raised MMA and homocysteine, anti-IF antibodies, and a brisk reticulocyte response within days of parenteral B12.

EXAM TRAP

Never treat a macrocytic anaemia with folate alone until B12 deficiency is excluded. Folate bypasses the methyl-folate trap and corrects the blood picture, but the **neurological damage progresses**. Also remember that not every macrocytosis is megaloblastic: **alcohol**, liver disease, hypothyroidism, a high reticulocyte count and MDS raise the MCV without megaloblasts.

B12 = "Brain and Blood"; Folate = "Foetus"

B12 lack damages **brain/cord and blood** (MMA up). Folate lack damages **blood and the foetal neural tube** (MMA normal). Both raise homocysteine.

11.6 Aplastic and myelophthisic anaemia

DEFINITION

Aplastic anaemia is a failure of multipotent myeloid stem cells, causing **pancytopenia** (anaemia, neutropenia, thrombocytopenia) with a **hypocellular, fat-filled marrow**, in the absence of marrow infiltration or fibrosis.

CAUSES

- **Idiopathic:** more than half of all cases.
- **Drugs and chemicals:** dose-related (alkylating agents, antimetabolites, **benzene**) or idiosyncratic (**chloramphenicol**, gold salts, carbamazepine, phenylbutazone).
- **Whole-body radiation.**
- **Viruses:** seronegative (non-A to E) hepatitis, EBV, CMV, HIV.
- **Inherited: Fanconi anaemia** (defective DNA repair; short stature, absent thumbs or radii, renal anomalies); **telomerase** gene defects.
- Associated: **PNH** may emerge from an aplastic marrow (Chapter 12).

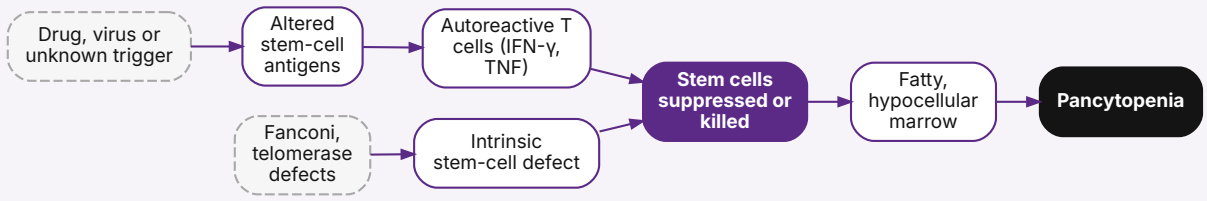


Figure 11.6 · Two routes to an empty marrow. The immune route explains why most patients respond to antithymocyte globulin + ciclosporin; stem cell transplantation cures many young patients.

BONE MARROW

- Aspirate often a **"dry tap"**: a **trephine biopsy** is needed
- Markedly **hypocellular**: more than 90% of the space is **fat**
- Scattered lymphocytes and plasma cells; few or no haematopoietic cells

BLOOD AND OTHER ORGANS

- Pancytopenia; red cells normocytic or slightly macrocytic; **reticulocytopenia**
- No blasts, no abnormal cells
- Secondary changes: fatty liver, haemorrhages, infections

Clinical: any age; insidious **weakness and pallor** (anaemia), **petechiae and bruising** (thrombocytopenia), and **recurrent infections**, fever and mouth ulcers (neutropenia). **Splenomegaly is characteristically absent**; if the spleen is big, look for another diagnosis (leukaemia, lymphoma, myelofibrosis, hypersplenism).

Pure red cell aplasia affects only the erythroid line: **parvovirus B19** infects red cell progenitors (a transient "aplastic crisis" in haemolytic anaemias), and an acquired form accompanies **thymoma** and large granular lymphocytic leukaemia.

Myelophthisic anaemia

Here the marrow is not empty but **replaced** by a space-occupying lesion: **metastatic carcinoma** (breast, lung, prostate), myeloma, lymphoma, leukaemia, granulomas (tuberculosis), myelofibrosis or storage diseases. The distorted marrow architecture lets immature cells escape early, giving a **leucoerythroblastic picture**.

APLASTIC VS MYELOPHTHISIC ANAEMIA

Feature	Aplastic anaemia	Myelophthisic anaemia
Marrow	Empty, fatty	Infiltrated by tumour, granulomas or fibrosis
Blood film	Pancytopenia; no abnormal cells	**Leucoerythroblastic**: nucleated RBCs + immature granulocytes; **teardrop cells**
Spleen	Not enlarged	Often enlarged (extramedullary haematopoiesis)
Diagnosis	Trephine biopsy: hypocellular	Trephine biopsy shows the infiltrate

CLINICAL CASE

A 64-year-old woman with hypothyroidism has tiredness, a sore tongue and tingling feet. Hb 7.8 g/dL, MCV 118 fL, WBC 3.2 × 10^9/L, platelets 110 × 10^9/L; oval macrocytes and six-lobed neutrophils; LDH and indirect bilirubin raised.

Reasoning: macrocytic pancytopenia + hypersegmented neutrophils + ineffective erythropoiesis = **megaloblastic anaemia**; neuropathy and an autoimmune background point to **pernicious anaemia**. Confirm with serum B12, MMA and anti-IF antibody; give lifelong parenteral B12, never folate alone.

EXAM ZONE Past-Paper Pattern SEQs — Solved

PROF PATTERN **a)** Classify anaemia on morphological and aetiological basis. **b)** Describe the laboratory diagnosis of iron deficiency anaemia. **2 + 3 marks**

MODEL ANSWER PLAN

a) Morphological: microcytic hypochromic (IDA, thal, ACD, sideroblastic), normocytic normochromic (acute bleed, haemolysis, ACD, aplastic, renal), macrocytic (megaloblastic: B12/folate; non-megaloblastic). Aetiological: blood loss, increased destruction, decreased production (with 2 examples each).

b) CBC: low Hb, MCV, MCH, MCHC; high RDW. Smear: microcytic hypochromic cells, pencil cells. Iron studies: low ferritin (best test), low serum iron, high TIBC, low saturation. Marrow: absent Perls iron. Find the cause (stool occult blood, endoscopy, worm ova). Response to iron.

Score booster: draw the 4-stage ladder (Fig 11.3); it shows you know ferritin falls first.

MODULE PATTERN A 30-year-old woman with heavy periods has fatigue and brittle, spoon-shaped nails. Hb 8.2 g/dL, MCV 65 fL, MCH 19 pg, RDW 19%, platelets 480 × 10^9/L. **a)** Most likely diagnosis? **b)** Describe the expected blood film. **c)** Give the expected iron studies and explain the stages of iron depletion. **1 + 2 + 2 marks**

MODEL ANSWER PLAN

a) Iron deficiency anaemia due to menorrhagia. **b)** Microcytic hypochromic cells with a thin rim of Hb, anisopoikilocytosis, pencil cells; reactive thrombocytosis. **c)** Ferritin low, serum iron low, TIBC high, saturation low. Stages: negative balance → storage depletion (ferritin falls) → iron-deficient erythropoiesis (iron and saturation fall) → anaemia.

PROF PATTERN **a)** Discuss the pathogenesis and morphological features of pernicious anaemia. **b)** How will you differentiate vitamin B12 from folate deficiency? **3 + 2 marks**

MODEL ANSWER PLAN

a) Autoimmune gastritis (T cells; anti-IF blocking and binding antibodies; anti-H^+,K^+-ATPase) → no IF → B12 malabsorption → impaired DNA synthesis → megaloblasts. Morphology: macro-ovalocytes, hypersegmented neutrophils, hypercellular megaloblastic marrow, giant metamyelocytes; atrophic fundic gastritis, beefy tongue; dorsal/lateral column demyelination. **b)** Neuro signs, raised MMA and anti-IF antibodies only in B12 deficiency; stores, absorption site and causes differ (table).

Score booster: one line on the methyl-folate trap explains why the blood picture is identical.

CLASSIC **a)** Define aplastic anaemia. **b)** Enumerate its causes. **c)** Describe the marrow and blood findings. **1 + 2 + 2 marks**

MODEL ANSWER PLAN

a) Stem cell failure with pancytopenia and hypocellular marrow, without infiltration. **b)** Idiopathic (most), drugs (chloramphenicol, alkylating agents, benzene), radiation, viruses (seronegative hepatitis), Fanconi anaemia, telomerase defects. **c)** Trephine: > 90% fat, scattered lymphocytes and plasma cells; blood: pancytopenia, reticulocytopenia, normocytic or macrocytic cells; no splenomegaly.

BCQs One-Best-Answer Practice

1 A 26-year-old woman with menorrhagia has Hb 8.4 g/dL and MCV 63 fL. Which result would confirm depleted iron stores most reliably?

A. Raised serum ferritin
B. Low serum ferritin
C. Low TIBC
D. Raised reticulocyte count
E. Raised HbA2

2 A 66-year-old man has iron deficiency anaemia but no visible bleeding and a normal diet. The most important next step is:

A. Start oral iron and review in a year
B. Bone marrow aspiration
C. Haemoglobin electrophoresis
D. Serum B12 assay
E. Endoscopy and colonoscopy for occult GI cancer

3 Hepcidin lowers plasma iron by:

A. Binding ferroportin and causing its degradation
B. Blocking DMT1 in the ileum
C. Increasing ferritin excretion in urine
D. Inhibiting transferrin synthesis
E. Stimulating erythropoietin release

4 A woman with active rheumatoid arthritis has Hb 9.8 g/dL, MCV 84 fL, low serum iron, low TIBC and high ferritin. The anaemia is due mainly to:

A. Occult GI bleeding
B. Folate deficiency
C. Hepcidin-mediated iron sequestration
D. Autoimmune haemolysis
E. β-thalassaemia trait

5 A 20-year-old has Hb 11.2 g/dL, MCV 61 fL, RBC 6.1 million/µL, normal RDW and normal ferritin. The best next test is:

A. Bone marrow iron stain
B. Serum B12
C. Coombs test
D. Haemoglobin electrophoresis for HbA2
E. Osmotic fragility

6 A 68-year-old woman with MCV 116 fL has loss of vibration sense in both feet. Her spinal cord lesion involves the:

A. Anterior horn cells
B. Dorsal and lateral columns
C. Spinothalamic tracts only
D. Cerebellar Purkinje cells
E. Dorsal root ganglia only

7 Which test best separates vitamin B12 deficiency from folate deficiency?

A. Serum homocysteine
B. MCV
C. Hypersegmented neutrophils
D. Serum methylmalonic acid
E. Serum LDH

8 A patient on long-term methotrexate develops macrocytic anaemia. The mechanism is:

A. Inhibition of dihydrofolate reductase
B. Loss of intrinsic factor
C. Ileal malabsorption of B12
D. Autoimmune haemolysis
E. Marrow infiltration

9 A peripheral smear shows oval macrocytes and neutrophils with six nuclear lobes. The marrow is expected to show:

A. Hypocellular fatty marrow
B. Absent erythroid precursors
C. Ring sideroblasts
D. Sheets of lymphoblasts
E. Megaloblasts and giant metamyelocytes

10 A 19-year-old has pancytopenia, a dry tap on aspiration and a trephine showing 95% fat. Which finding would argue AGAINST aplastic anaemia?

A. Petechiae
B. Recurrent infections
C. Massive splenomegaly
D. Low reticulocyte count
E. History of chloramphenicol use

11 A woman treated for breast cancer has anaemia, teardrop cells, nucleated red cells and myelocytes in her blood. The most likely diagnosis is:

A. Myelophthisic anaemia
B. Iron deficiency anaemia
C. Aplastic anaemia
D. Hereditary spherocytosis
E. Anaemia of chronic disease

12 A child has pancytopenia, short stature, absent thumbs and a renal anomaly. The underlying defect is in:

A. Globin chain synthesis
B. Red cell membrane spectrin
C. DNA repair
D. GPI anchor synthesis
E. Intrinsic factor production

ANSWER KEY WITH ONE-LINE REASONS

1. **B** Low ferritin reliably means low stores; TIBC is high in IDA.
2. **E** In older adults, IDA is GI blood loss (e.g. colon cancer) until proved otherwise.
3. **A** Ferroportin loss traps iron in enterocytes and macrophages.
4. **C** Anaemia of chronic disease: IL-6 → hepcidin → iron locked in macrophages.
5. **D** Mentzer index 10 (< 13) with normal ferritin suggests β-thalassaemia trait.
6. **B** Subacute combined degeneration in B12 deficiency.
7. **D** MMA rises only in B12 deficiency; homocysteine rises in both.
8. **A** Methotrexate is a folate antagonist that blocks THF regeneration.
9. **E** Megaloblastic marrow: N/C asynchrony, giant metamyelocytes.
10. **C** Splenomegaly is characteristically absent in aplastic anaemia.

11. **A** Leucoerythroblastic picture + teardrops = marrow infiltration by metastases.

12. **C** Fanconi anaemia: inherited DNA repair defect causing marrow failure.

VIVA Rapid-Fire Viva Voce

Q Define anaemia.	Fall in O2-carrying capacity of blood, usually from reduced red cell mass; Hb below range.
Q Normal MCV and its formula?	82–96 fL; Hct divided by RBC count.
Q What is RDW and when is it raised?	Variation in red cell size; raised in iron deficiency and megaloblastic anaemia.
Q First test to become abnormal in iron deficiency?	Serum ferritin, as stores empty first.
Q What are pencil cells?	Thin, elongated hypochromic red cells typical of iron deficiency.
Q Plummer–Vinson syndrome?	Iron deficiency, atrophic glossitis and oesophageal webs.
Q What does hepcidin do?	Degrades ferroportin, blocking gut absorption and macrophage iron release.
Q Why is ferritin high in anaemia of chronic disease?	Iron is trapped in macrophages and ferritin is an acute-phase protein.
Q Where and how is B12 absorbed?	Terminal ileum, bound to intrinsic factor, via cubilin receptors.
Q Define a hypersegmented neutrophil.	Neutrophil with six or more lobes, or over 5% with five lobes.
Q Why no splenomegaly in aplastic anaemia?	Marrow failure without infiltration or extramedullary haematopoiesis; a big spleen suggests another diagnosis.
Q What is a leucoerythroblastic picture?	Nucleated red cells plus immature granulocytes in blood: marrow infiltration.

OSPE Practice Stations

STATION 1 · PERIPHERAL SMEAR — 2 minutes

TASK

Leishman-stained film: most red cells are smaller than a lymphocyte nucleus, with only a thin rim of pink and a wide central pallor; many are elongated and cigar-shaped. Platelets are increased. (a) Describe the picture in one term. (b) Diagnosis. (c) Two confirmatory tests.

ANSWER

(a) Microcytic hypochromic anaemia with pencil cells. (b) Iron deficiency anaemia. (c) Serum ferritin (low); serum iron/TIBC (low/high); search for blood loss.

STATION 2 · PERIPHERAL SMEAR — 2 minutes

TASK

Film: large oval red cells without central pallor, marked variation in shape, and neutrophils with 6–7 lobes. CBC: Hb 7 g/dL, MCV 121 fL, WBC and platelets low. (a) Diagnosis. (b) Name the neutrophil change. (c) Two causes.

ANSWER

(a) Megaloblastic anaemia. (b) Hypersegmented neutrophils. (c) Vitamin B12 deficiency (pernicious anaemia) or folate deficiency (diet, pregnancy, methotrexate).

STATION 3 · LAB DATA — 2 minutes

TASK

Man with pulmonary TB for 8 months: Hb 9.6 g/dL, MCV 82 fL, serum iron low, TIBC low, transferrin saturation 14%, serum ferritin 410 µg/L. (a) Type of anaemia. (b) Explain the ferritin. (c) Key mediator.

ANSWER

(a) Anaemia of chronic disease (inflammation). (b) Stores are full; iron is trapped in macrophages, and ferritin is an acute-phase protein. (c) Hepcidin, induced by IL-6.

STATION 4 · MARROW TREPHINE — 1 minute

TASK

Bone marrow trephine from a young man with pancytopenia: bony trabeculae with marrow spaces almost entirely filled by fat cells; only a few lymphocytes and plasma cells; no tumour cells or fibrosis. (a) Diagnosis. (b) Two causes. (c) One clinical sign that should be absent.

ANSWER

(a) Aplastic anaemia. (b) Idiopathic; drugs (chloramphenicol), benzene, radiation, viral hepatitis, Fanconi anaemia. (c) Splenomegaly.

RAPID REVISION — The Night-Before Page

15 ONE-LINERS

01 WHO anaemia: Hb < 13 men, < 12 women.
02 MCV 82–96 fL; RDW high in IDA, not thal.
03 High retics: bleeding, haemolysis; low: output.
04 Iron is controlled at absorption only.
05 Hepcidin degrades ferroportin; IL-6 raises it.
06 IDA: commonest; adults = chronic blood loss.
07 Ferritin falls first; Hb falls last.
08 IDA: low ferritin, high TIBC, pencil cells.
09 ACD: low iron, low TIBC, high ferritin.
10 Mentzer < 13 + high RBC = thal trait.
11 Megaloblast: nucleus lags behind cytoplasm.
12 Hypersegmented PMNs + macro-ovalocytes.
13 MMA raised only in B12 deficiency.
14 Pernicious: anti-IF, atrophic fundus, cord.
15 Aplastic: fatty marrow, no splenomegaly.

MEMORY HOOKS

- **B12 = Brain and Blood**; **Folate = Foetus**
- **Ferritin First**: stores empty before Hb falls
- **TIBC: High in Hunger** (IDA), low in inflammation
- **Big spleen? Not aplastic**

TEST YOURSELF: FILL THE GAPS

1. Iron is exported from cells by _____.
2. Best single test for iron stores: _____.
3. B12–IF complex is absorbed in the _____.
4. Folate lack in pregnancy causes _____ defects.
5. Spinal tracts hit by B12 lack: _____ columns.
6. Teardrop cells + nRBCs suggest _____ anaemia.

Answers: ferroportin · serum ferritin · terminal ileum · neural tube · dorsal and lateral · myelophthisic

DRAW FROM MEMORY

The MCV approach flowchart (Fig 11.1) with two causes in each box.

DRAW FROM MEMORY

The iron cycle: duodenum → ferroportin → transferrin → marrow → macrophages, with hepcidin blocking ferroportin.

SPACED REVISION ☐ Day 1 ☐ Day 3 ☐ Day 7 ☐ Day 14 ☐ Day 30

12

Anaemias II: Haemolytic Anaemias & Thalassaemias

In haemolysis the marrow is working flat out and still losing. Find out whether the red cell is faulty from within or attacked from outside, and whether it dies in the bloodstream or in the spleen, and the diagnosis is half made.

BY THE END OF THIS CHAPTER YOU CAN

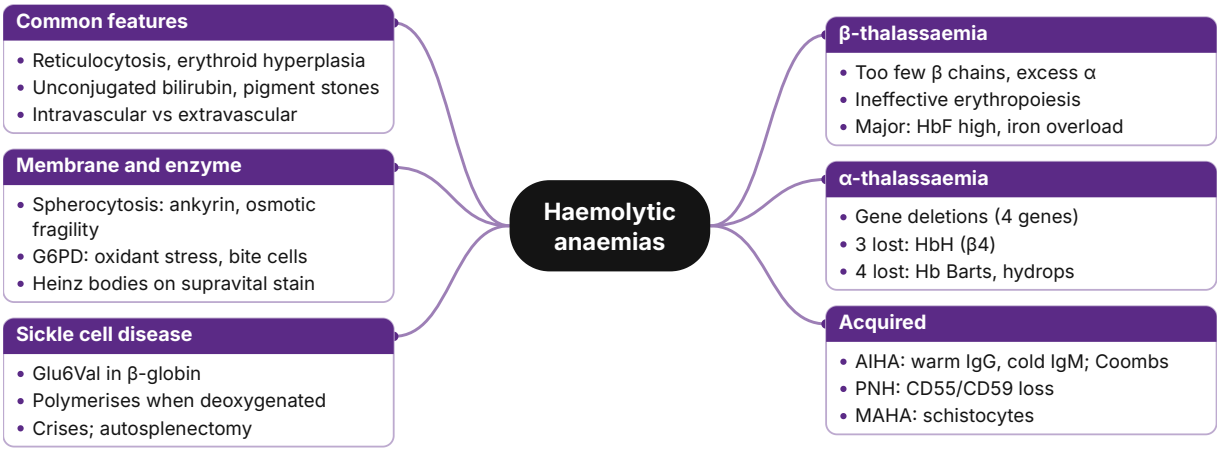
1. List the general features of haemolytic anaemia and classify the haemolytic anaemias
2. Differentiate intravascular from extravascular haemolysis clinically and in the laboratory
3. Describe the pathogenesis and laboratory diagnosis of hereditary spherocytosis and G6PD deficiency
4. Explain the pathogenesis, morphology and crises of sickle cell disease
5. Explain the genetics and pathogenesis of α- and β-thalassaemia and interpret haemoglobin electrophoresis
6. Classify autoimmune haemolytic anaemias and interpret the Coombs test
7. Outline PNH and microangiopathic haemolytic anaemia

EXAM RADAR

Theory SEQs	★★★
BCQs / MCQs	★★★
Viva voce	★★★
OSPE / slides	★★★

β-thalassaemia major (pathogenesis, blood picture, complications) and sickle cell anaemia are favourite long questions; intra- vs extravascular haemolysis and HS are common short notes.

ROBBINS LINK Robbins Basic Pathology — *Hematopoietic and Lymphoid Systems*: haemolytic anaemias (hereditary spherocytosis, G6PD deficiency, sickle cell disease, thalassaemias, PNH, immunohaemolytic and mechanical haemolytic anaemias).



Chapter at a glance. Faulty cell or hostile environment; spleen or bloodstream.

12.1 Haemolytic anaemias: common ground

DEFINITION

Haemolytic anaemia is anaemia caused by **premature destruction of red cells** (life span shorter than the normal **120 days**), with a compensatory increase in erythropoiesis.

FEATURES SHARED BY ALL

- **Raised erythropoietin → erythroid hyperplasia** of marrow and **reticulocytosis** (polychromasia); extramedullary haematopoiesis in severe cases.
- **Hb breakdown products accumulate:** unconjugated (acholuric) **jaundice**, **pigment gallstones**, raised LDH, low **haptoglobin**.

CLASSIFICATION

Intrinsic (intracorpuscular)	Extrinsic (extracorpuscular)
Membrane: hereditary spherocytosis, elliptocytosis	**Immune:** autoimmune, transfusion reaction, haemolytic disease of newborn

- Chronic marrow expansion in childhood deforms bone (**crew-cut skull**).
- Iron is not lost (except in intravascular haemolysis), so iron overload can follow transfusions.

Intrinsic (intracorpuscular)	Extrinsic (extracorpuscular)
Enzyme: G6PD, pyruvate kinase deficiency	**Mechanical:** MAHA, prosthetic valves
Haemoglobin: sickle cell disease, thalassaemias	**Infection:** malaria, clostridial sepsis
Acquired membrane defect: PNH	**Toxins, hypersplenism**

Rule: intrinsic defects are inherited (except PNH); extrinsic causes are acquired.

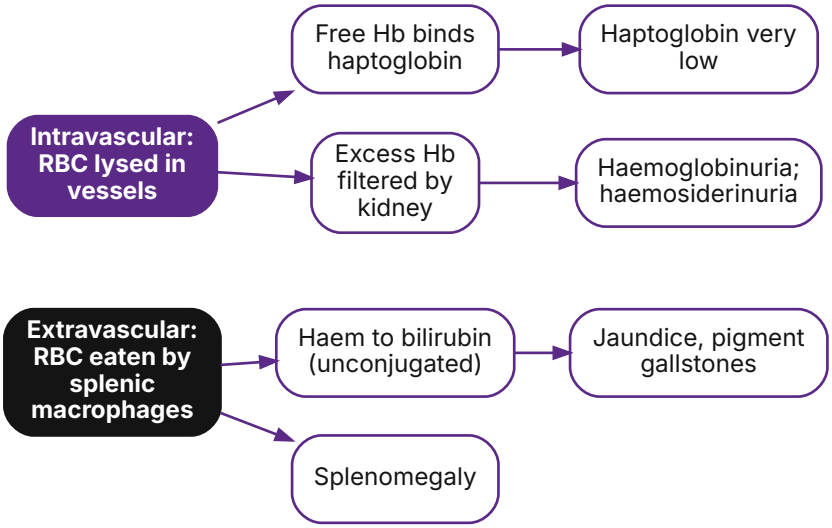


Figure 12.1 · Where the red cell dies decides what you find. Extravascular (black): macrophages in spleen and liver. Intravascular (purple): lysis inside vessels.

INTRAVASCULAR VS EXTRAVASCULAR HAEMOLYSIS

Feature	Intravascular	Extravascular
Site of destruction	Inside blood vessels	Macrophages of **spleen** (and liver)
Mechanism	Mechanical trauma, complement fixation, toxins, parasites	Cells made **less deformable** or coated with antibody
Examples	Mismatched transfusion, **G6PD** attack, **PNH**, MAHA, valves, falciparum malaria	**HS**, sickle cell, thalassaemia, **warm AIHA**
Haemoglobinaemia, haemoglobinuria	**Present** (red-brown urine)	Absent
Urine haemosiderin	**Present** (tubular cells shed iron)	Absent
Serum haptoglobin	**Absent or very low**	Reduced
Splenomegaly	Usually absent	**Present**
Iron loss	Yes: iron deficiency can follow	No: iron is recycled
Jaundice, LDH	Both raised	Both raised

12.2 Hereditary spherocytosis (HS)

An inherited defect of the red cell **membrane skeleton** that makes cells spherical, rigid and easy prey for the spleen. It is commonest in northern Europeans; about **75%** of cases are **autosomal dominant**. Mutations hit **ankyrin** (commonest), band 3, **spectrin** or band 4.2.

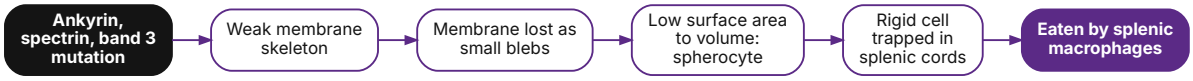


Figure 12.2 · Pathogenesis of hereditary spherocytosis. Splenectomy removes the trap, not the defect.

CONCEPT CLEAR

Why the spleen kills spherocytes. To pass from the splenic cords into the sinusoids, a 7–8 µm red cell must squeeze through slits only 2–3 µm wide. A normal biconcave disc has spare membrane and folds easily; a **sphere has no spare membrane**, so it gets stuck, loses more membrane in the hostile, glucose-poor cords and is eaten. Remove the spleen and the spherocytes live almost normally.

PERIPHERAL SMEAR

- **Spherocytes**: small, round, **dark-staining** cells with **no central pallor**
- Polychromasia (reticulocytosis)
- After splenectomy: **Howell–Jolly bodies**

ORGANS

- **Spleen**: moderately enlarged (500–1000 g), congested cords, prominent macrophages
- Marrow erythroid hyperplasia
- **Pigment gallstones** in 40–50% of adults

Clinical: anaemia, splenomegaly and jaundice, often from childhood. **Aplastic crisis** when **parvovirus B19** shuts down red cell production; haemolytic crisis with infections. **Labs:** raised **MCHC** (dehydrated cells), **increased osmotic fragility** (cells burst in mildly hypotonic saline), reduced eosin-5-maleimide (EMA) binding on flow cytometry, **direct Coombs negative**. **Treatment:** splenectomy in severe cases (vaccinate first).

EXAM TRAP

Spherocytes are not specific to HS. Warm AIHA also produces spherocytes, because macrophages nibble antibody-coated membrane. The **direct Coombs test** separates them: **negative in HS**, **positive in AIHA**. A family history and raised MCHC favour HS.

12.3 Glucose-6-phosphate dehydrogenase (G6PD) deficiency

G6PD drives the **hexose monophosphate shunt**, the red cell's only source of **NADPH**, which keeps **glutathione** in its reduced form to neutralise oxidants. The gene is **X-linked**, so hemizygous males are affected. It is common in malaria-endemic regions of Africa, the Mediterranean, the Middle East and Asia. In the **A⁻** variant (Africa) the enzyme is unstable and only **older** red cells are deficient, so attacks are mild and self-limited; the **Mediterranean** variant is more severe.

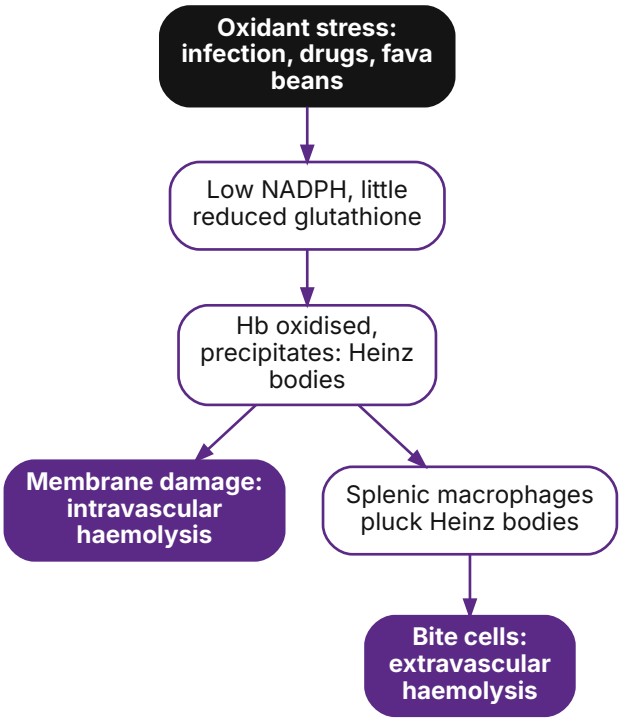


Figure 12.3 · Oxidant haemolysis in G6PD deficiency.

TRIGGERS

- **Infections** (the commonest): oxidants from activated phagocytes.
- **Drugs:** antimalarials (**primaquine**, chloroquine), **sulfonamides**, nitrofurantoin, dapsone.
- **Fava beans** (favism), mainly Mediterranean variant.

PICTURE

- Episodic, acute **intravascular** haemolysis 2–3 days after exposure: pallor, jaundice, dark urine (**haemoglobinuria**); also neonatal jaundice.
- Smear: **bite cells** (degmacytes), blister cells, a few spherocytes; **Heinz bodies** only on **supravital stain** (crystal violet).
- Between attacks the blood is normal.

> ⚠ **EXAM TRAP**
>
> **A normal G6PD level during an attack can be false.** The oldest, most deficient cells have just been destroyed and young reticulocytes carry more enzyme. Repeat the assay a few weeks after recovery.

12.4 Sickle cell disease

A **point mutation** in codon 6 of the β-globin gene replaces **glutamic acid with valine**, producing **HbS**. Homozygotes (HbSS) have **sickle cell anaemia**; heterozygotes (**sickle trait**, about 40% HbS) are usually well. The trait protects against **falciparum malaria**, which explains its high frequency in Africa, the Middle East and parts of India.

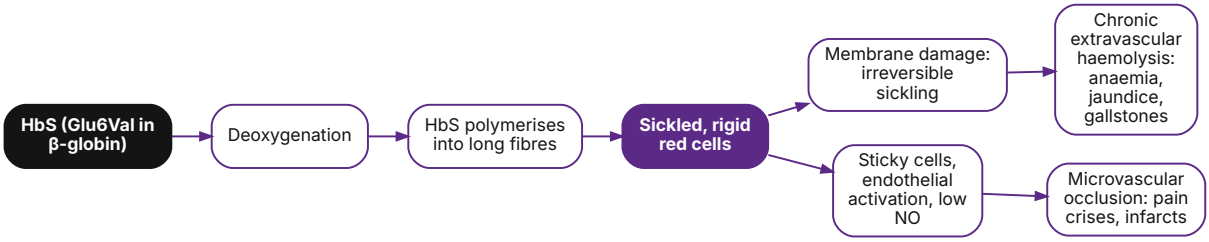


Figure 12.4 · Pathogenesis of sickle cell disease. Two consequences: chronic haemolysis and vaso-occlusion.

WHAT MAKES CELLS SICKLE MORE (OR LESS)

- **HbF** inhibits polymerisation: infants are well until **5–6 months**; **hydroxyurea** raises HbF.
- **Other Hb:** HbA in trait protects; **HbC** co-polymerises (HbSC disease, milder than SS).
- **MCHC:** dehydration increases sickling; coexistent α-thalassaemia lowers MCHC and eases disease.
- **Low pH** and low O_2; **slow transit** (spleen, marrow, inflamed tissue).

DIAGNOSIS

- Smear: sickled cells, target cells, polychromasia; **Howell–Jolly bodies** once the spleen is lost.
- **Sickling test:** sodium metabisulfite deoxygenates blood and induces sickling.
- **Hb electrophoresis / HPLC:** HbS with no HbA (SS); HbS about 40% (trait); prenatal DNA testing.

ORGANS

- **Spleen:** enlarged in childhood (congested red pulp), then repeated infarcts and fibrosis shrink it to a tiny nubbin by adolescence: **autosplenectomy**
- **Bone:** marrow hyperplasia (crew-cut skull, prominent cheekbones), **infarcts**, avascular necrosis of femoral head, **Salmonella osteomyelitis**
- **Kidney:** papillary necrosis, inability to concentrate urine

OTHER EFFECTS

- **Lung:** acute chest syndrome
- **Brain:** stroke; **eye:** retinopathy
- Leg ulcers, priapism, pigment gallstones
- Extramedullary haematopoiesis; secondary haemosiderosis

CRISES AND COMPLICATIONS OF SICKLE CELL ANAEMIA

Crisis	What happens	Key point
Vaso-occlusive (pain)	Ischaemia of bone, chest, abdomen; **hand–foot syndrome** (dactylitis) in infants	Commonest; precipitated by infection, dehydration, cold, acidosis
Acute chest syndrome	Fever, cough, chest pain, new lung infiltrates; vicious cycle of hypoxia and sickling	A leading cause of death
Sequestration	Massive pooling of blood in the spleen (or liver): sudden pallor, big spleen, shock	Young children with intact spleens
Aplastic	**Parvovirus B19** stops red cell production; reticulocytes vanish	Transient but dangerous
Infections	**S. pneumoniae**, H. influenzae sepsis and meningitis; Salmonella osteomyelitis	Due to **functional asplenia**: vaccinate, penicillin prophylaxis

12.5 The thalassaemias

CONCEPT CLEAR

Wrong amount, not wrong shape. Sickle cell disease is a **qualitative** defect: the globin is abnormal. Thalassaemia is a **quantitative** defect: normal globin, too little of it. The missing chain causes small, pale cells; the **partner chain left unpaired** causes the real damage. Normal adult Hb: **HbA** (α2β2) about 97%, **HbA2** (α2δ2) 2–3.5%, **HbF** (α2γ2) under 1%. β-globin is coded on **chromosome 11** (2 genes per cell); α-globin on **chromosome 16** (4 genes per cell).

β-thalassaemia

Caused mostly by **point mutations**: **β^0** alleles make no β chain, **β^+** alleles make a reduced amount. **Splicing mutations** are the commonest (β^0 or β^+); **promoter** mutations reduce transcription (β^+); **chain-terminator** (nonsense, frameshift) mutations give β^0.

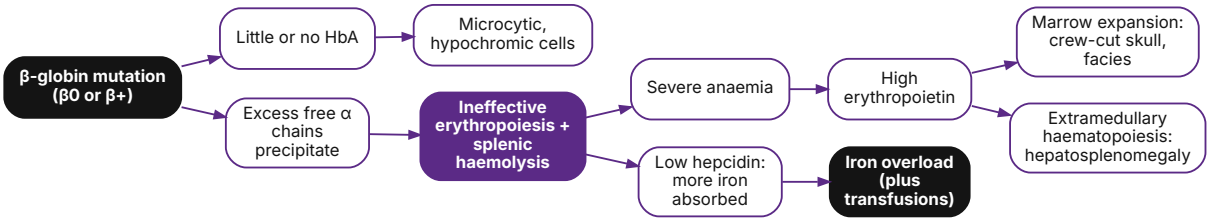


Figure 12.5 · Pathogenesis of β-thalassaemia major. Learn this chain: it answers the SEQ on pathogenesis, clinical features and complications.

CLINICAL FORMS OF β-THALASSAEMIA

Feature	β-thal major	β-thal intermedia	β-thal minor (trait)
Genotype	β^0/β^0, β^0/β^+, β^+/β^+ (severe)	Milder β^+ combinations	β/β^0 or β/β^+ (one gene)
Onset, Hb	**6–9 months** (as HbF falls); Hb 3–6 g/dL	Later; moderate anaemia	Asymptomatic; mild anaemia or none
Transfusion	**Lifelong, regular**	Occasional	Not needed

Feature	β-thal major	β-thal intermedia	β-thal minor (trait)
Smear	Severe microcytosis, hypochromia, target cells, stippling, fragments, many **nucleated RBCs**	Similar, milder	Microcytosis, target cells, basophilic stippling
Hb electrophoresis	**HbF markedly raised**; HbA absent (β^0) or low; HbA2 variable	HbF raised	**HbA2 raised (4–8%)**; HbF normal or slightly up
Importance	Iron overload, growth failure	Iron overload	Carrier detection; must not be mistaken for IDA

β-THAL MAJOR: BODY

- **Marrow expansion:** "crew-cut" (hair-on-end) skull X-ray, frontal bossing, prominent maxilla and cheekbones (**chipmunk facies**), thin cortices
- **Hepatosplenomegaly** from extramedullary haematopoiesis; spleen may exceed 1500 g
- Growth retardation, delayed puberty, cachexia

IRON OVERLOAD (HAEMOSIDEROSIS)

- Prussian blue-positive iron in **heart, liver, pancreas** and endocrine glands
- **Cardiomyopathy** and arrhythmias: the main cause of death
- Liver fibrosis/cirrhosis, diabetes, hypogonadism, hypothyroidism

Management: regular transfusion with iron **chelation** (desferrioxamine, deferasirox) lets children survive into adulthood. **Haematopoietic stem cell transplantation** is the only cure. Transfused children also risk transfusion-transmitted infections (HBV, HCV).

α-thalassaemia

Usually caused by **deletion** of α-globin genes. Severity rises with each gene lost. Free β chains (as **HbH**, β4) and γ chains (as **Hb Barts**, γ4) form **soluble tetramers**, so precipitation and ineffective erythropoiesis are milder than in β-thalassaemia; but these tetramers bind O_2 so tightly that they release almost none to tissues.

α-THALASSAEMIA BY NUMBER OF GENES DELETED

Genes lost	Genotype	Syndrome	Findings
1	−α/αα	Silent carrier	Normal blood count
2	−−/αα (**cis**, Asian) or −α/−α (trans, African)	α-thal trait	Mild microcytosis like β-thal minor, but **HbA2 normal**; cis carriers can have hydropic babies
3	−−/−α	**HbH disease**	Moderate to severe anaemia (like thal intermedia); HbH (β4) forms **"golf-ball" inclusions** on supravital stain
4	−−/−−	**Hydrops fetalis**	**Hb Barts** (γ4); severe hypoxia, oedema, hepatosplenomegaly; death in utero unless transfused

HB ELECTROPHORESIS / HPLC: PATTERNS TO RECOGNISE

Normal adult HbA ~97%, HbA2 2–3.5%, HbF < 1%

β-thal major **HbF very high** , HbA absent or low

Sickle cell anaemia HbS ~90%, HbF rest, **no HbA**

β-thal trait **HbA2 > 3.5%** (4–8%), HbF normal or slightly up

α-thal trait normal pattern (diagnosis by DNA)

Sickle trait HbA ~60%, **HbS ~40%**

PAKISTAN FOCUS

In Pakistan the β-thalassaemia **carrier rate is about 5–7%**, and an estimated **5,000 or more** children are born with thalassaemia major every year. **Cousin marriage** is common, which raises the chance that both parents carry the gene. Several provinces have passed **thalassaemia prevention laws** encouraging premarital or carrier screening, and **cascade screening** of relatives of every affected child is recommended. Screening is simple: CBC (low MCV/MCH, Mentzer index) then HPLC for HbA2; carrier couples can be offered prenatal diagnosis.

MEMORY HOOK

Thal major: "HbF Holds the Fort"

No β chains → HbA gone → the fetal haemoglobin (α2γ2) **Holds the Fort**, so HbF is the main Hb. In the **trait**, the δ chain steps up: **HbA2 > 3.5%** is the carrier's fingerprint.

12.6 Acquired haemolytic anaemias

Autoimmune haemolytic anaemia (AIHA)

Antibodies against the patient's own red cells. Diagnosis rests on the **Coombs (antiglobulin) test**. **Direct Coombs (DAT):** the patient's red cells are mixed with anti-human IgG or anti-C3; agglutination shows the cells are already coated. **Indirect Coombs:** the patient's serum is tested against reagent red cells to detect free antibody (cross-matching, antenatal screening).

WARM VS COLD AUTOIMMUNE HAEMOLYTIC ANAEMIA

Feature	Warm antibody type	Cold agglutinin type
Antibody	**IgG**, active at **37 °C**	**IgM**, binds below about 30 °C (ears, fingers, toes)
Frequency	**Commonest** form	Less common
Mechanism	Splenic macrophages nibble IgG-coated cells → **spherocytes** → extravascular haemolysis	IgM agglutinates cells and fixes **C3b**; cells cleared by liver macrophages; some intravascular lysis
Causes	Idiopathic; **SLE**, **CLL** and lymphomas; **drugs**: α-methyldopa (autoantibody), penicillin (hapten)	Acute: **Mycoplasma pneumoniae**, infectious mononucleosis; chronic: B-cell lymphomas
Clinical clues	Splenomegaly, spherocytes	**Raynaud phenomenon**, acrocyanosis, agglutination on the smear
DAT	IgG (± C3)	C3 only

Cold haemolysin type (paroxysmal cold haemoglobinuria): an IgG Donath–Landsteiner antibody binds at low temperature, fixes complement and lyses cells on warming; mostly children after viral infections.

Paroxysmal nocturnal haemoglobinuria (PNH)

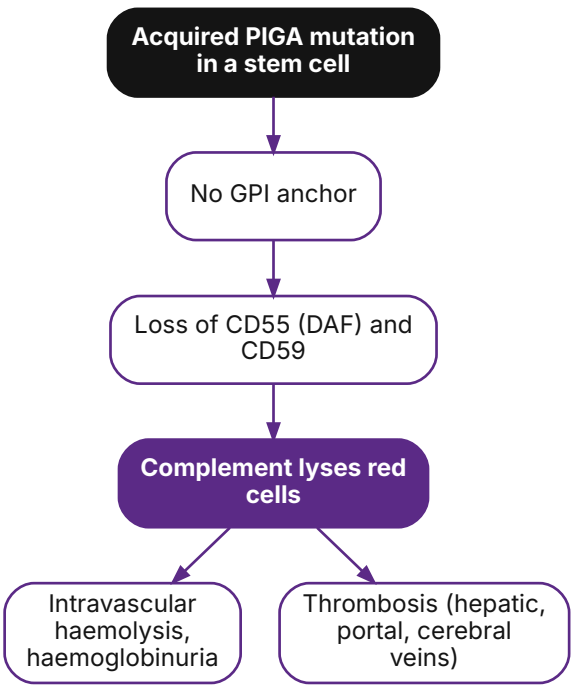


Figure 12.6 · PNH: a red cell without its shield.

- The only **acquired intrinsic** red cell defect; PIGA is X-linked, so one hit is enough.
- Only a minority have classic night-time haemoglobinuria.
- **Venous thrombosis** (Budd–Chiari) is the leading cause of death; iron deficiency from urinary loss.
- Linked to **aplastic anaemia**; may evolve to AML or MDS.
- Diagnosis: **flow cytometry** for CD55/CD59-deficient cells. Treatment: **eculizumab** (anti-C5).

Microangiopathic haemolytic anaemia

Red cells are cut by fibrin strands or damaged endothelium in small vessels, or by prosthetic valves in large ones: intravascular haemolysis with **schistocytes** (helmet, triangle and burr cells). Causes: **DIC, TTP, HUS**, malignant hypertension, SLE, disseminated cancer (see Chapter 15).

CLINICAL CASE

A 2-year-old boy from a cousin marriage has been pale since 8 months of age. He has frontal bossing, a protuberant abdomen with liver 4 cm and spleen 6 cm. Hb 5.1 g/dL, MCV 58 fL; the film shows target cells, basophilic stippling and many nucleated red cells. HPLC: HbF 92%, HbA2 3%, no HbA.

Reasoning: onset after HbF switch-off + severe microcytic anaemia + marrow expansion + hepatosplenomegaly + HbF dominant with no HbA = **β^0-thalassaemia major**. Plan: regular transfusion with chelation, HSCT if a matched sibling exists, and carrier screening of the parents and relatives with counselling for future pregnancies.

EXAM ZONE Past-Paper Pattern SEQs — Solved

PROF PATTERN **a)** Classify haemolytic anaemias. **b)** Differentiate between intravascular and extravascular haemolysis. **2 + 3 marks**

MODEL ANSWER PLAN

a) Intrinsic: membrane (HS), enzyme (G6PD, PK), haemoglobin (sickle, thalassaemia), acquired PNH. Extrinsic: immune (AIHA, transfusion, HDN), mechanical (MAHA, valves), infections (malaria), toxins, hypersplenism.

b) Table of 6 points: site, examples, haemoglobinuria/haemosiderinuria, haptoglobin, splenomegaly, iron loss. Draw Fig 12.1 if time allows.

MODULE PATTERN A 2-year-old child of first-cousin parents has pallor since infancy, frontal bossing and hepatosplenomegaly. Hb 5 g/dL, MCV 60 fL, many nucleated red cells. **a)** Most likely diagnosis? **b)** Explain its pathogenesis. **c)** Give the expected Hb electrophoresis and two complications of treatment. **1 + 2 + 2 marks**

MODEL ANSWER PLAN

a) β-thalassaemia major. **b)** β-globin mutation → little HbA (microcytic hypochromic cells) + excess α chains precipitate → ineffective erythropoiesis and splenic haemolysis → anaemia → EPO → marrow expansion and extramedullary haematopoiesis; low hepcidin → iron overload. **c)** HbF markedly raised, HbA absent/low. Complications: iron overload (cardiomyopathy, cirrhosis, diabetes, hypogonadism), transfusion-transmitted HBV/HCV, alloimmunisation.

Score booster: add "onset at 6–9 months because HbF synthesis falls"; examiners love the timing.

PROF PATTERN **a)** Describe the molecular basis and pathogenesis of sickle cell anaemia. **b)** List its complications. **2 + 3 marks**

MODEL ANSWER PLAN

a) Glu→Val at position 6 of β-globin → HbS polymerises on deoxygenation → sickled rigid cells → chronic extravascular haemolysis and microvascular occlusion; modifiers: HbF, MCHC, pH, transit time. **b)** Pain crises/dactylitis, acute chest syndrome, stroke, sequestration crisis, aplastic crisis (parvovirus B19), autosplenectomy with pneumococcal sepsis, Salmonella osteomyelitis, papillary necrosis, leg ulcers, priapism, gallstones.

CLASSIC Write short notes on: **a)** hereditary spherocytosis; **b)** G6PD deficiency. **2.5 + 2.5 marks**

MODEL ANSWER PLAN

a) AD, ankyrin/spectrin/band 3 → membrane loss → spherocytes trapped in spleen; anaemia, jaundice, splenomegaly, gallstones, aplastic crisis; raised MCHC, increased osmotic fragility, DAT negative; splenectomy. **b)** X-linked; low NADPH/glutathione; triggers: infection, primaquine, sulfonamides, fava beans; Heinz bodies, bite cells; episodic intravascular haemolysis; assay after recovery.

BCQs One-Best-Answer Practice

1 A patient develops fever, back pain and red-brown urine minutes into a blood transfusion. Which finding is expected?

A. Raised haptoglobin
B. Splenomegaly
C. Absent serum haptoglobin with haemoglobinuria
D. Spherocytes with negative Coombs test
E. Low reticulocytes and fatty marrow

2 A 10-year-old with jaundice and splenomegaly has small dark red cells without central pallor, raised MCHC and a negative direct Coombs test. The most useful confirmatory test is:

A. Osmotic fragility (or EMA binding)
B. Hb electrophoresis
C. G6PD assay
D. Sickling test
E. Bone marrow biopsy

3 Three days after starting primaquine, a man passes dark urine. Inclusions seen with crystal violet supravital stain are:

A. Howell–Jolly bodies
B. Pappenheimer bodies
C. Cabot rings
D. Heinz bodies
E. Basophilic stippling

4 The molecular defect in sickle cell anaemia is:

A. Deletion of both β-globin genes
B. Valine replacing glutamic acid at position 6 of β-globin
C. Lysine replacing glutamic acid at position 6 of β-globin
D. Splicing mutation in α-globin
E. Deletion of three α-globin genes

5 A 3-year-old with sickle cell anaemia suddenly becomes pale and shocked; the spleen has enlarged to the umbilicus. This is:

A. Aplastic crisis
B. Vaso-occlusive crisis
C. Acute chest syndrome
D. Hyperhaemolytic crisis
E. Splenic sequestration crisis

6 An adult with sickle cell anaemia has Howell–Jolly bodies. He is most at risk of severe infection by:

A. Streptococcus pneumoniae
B. Mycobacterium tuberculosis
C. Candida albicans
D. Pneumocystis jirovecii
E. Cytomegalovirus

7 A healthy 22-year-old has Hb 11.8 g/dL, MCV 64 fL, RBC 5.9 million/µL and normal ferritin. The finding that confirms β-thalassaemia trait is:

A. HbF 90%
B. HbH inclusions
C. HbA2 5.2%
D. HbS 40%
E. Hb Barts 80%

8 Symptoms of β-thalassaemia major usually begin at 6–9 months of age because:

A. Maternal antibodies disappear
B. Iron stores from birth are exhausted
C. The spleen matures
D. γ-chain (HbF) production falls and β chains are needed
E. Erythropoietin production starts

9 Iron overload occurs in β-thalassaemia intermedia even without transfusion mainly because:

A. Haemolysis releases iron into urine
B. Ineffective erythropoiesis suppresses hepcidin
C. Ferroportin is mutated
D. Transferrin is absent
E. Dietary iron intake is high

10 A stillborn fetus has massive oedema and hepatosplenomegaly; most of its haemoglobin is γ4. The genotype is:

A. −α/αα
B. −−/αα
C. −−/−α
D. β0/β0
E. −−/−−

11 Two weeks after Mycoplasma pneumonia, a student has anaemia and blue fingertips in the cold. The antibody is most likely:

A. IgG warm autoantibody
B. IgE against red cells
C. IgA against band 3
D. IgM cold agglutinin fixing complement
E. Donath–Landsteiner IgG haemolysin

12 A man with haemoglobinuria and hepatic vein thrombosis has red cells lacking CD55 and CD59. The mutated gene is:

A. PIGA
B. JAK2
C. HFE
D. ANK1
E. G6PD

ANSWER KEY WITH ONE-LINE REASONS

1. **C** ABO mismatch causes complement-mediated intravascular haemolysis.
2. **A** Hereditary spherocytosis: increased osmotic fragility, reduced EMA binding.
3. **D** Oxidised, precipitated Hb in G6PD deficiency forms Heinz bodies.
4. **B** HbS: Glu6Val; lysine at 6 gives HbC.

5. **E** Young children with intact spleens can pool much of their blood in the spleen.
6. **A** Autosplenectomy causes susceptibility to encapsulated bacteria.
7. **C** Raised HbA2 (4–8%) is the hallmark of β-thal minor.
8. **D** The switch from γ to β chain synthesis exposes the β defect.
9. **B** Low hepcidin increases gut iron absorption.
10. **E** Loss of all four α genes: Hb Barts hydrops fetalis.
11. **D** Cold agglutinin disease: IgM (anti-I), C3 on cells, acrocyanosis.
12. **A** PNH: acquired PIGA mutation blocks GPI anchor synthesis.

VIVA Rapid-Fire Viva Voce

Q Normal red cell life span?	About 120 days.
Q Two signs of intravascular haemolysis?	Haemoglobinuria and haemosiderinuria, with absent haptoglobin.
Q Commonest protein defect in HS?	Ankyrin (also spectrin, band 3, band 4.2).
Q Why is MCHC raised in HS?	Membrane loss and cell dehydration concentrate the haemoglobin.
Q What are bite cells?	Red cells from which splenic macrophages removed Heinz bodies; G6PD deficiency.
Q What is favism?	Acute haemolysis after eating fava beans in G6PD deficiency.
Q Why are sickle cell infants well at first?	High HbF inhibits HbS polymerisation until about 5–6 months.
Q What is autosplenectomy?	Spleen shrunk to a fibrous nubbin by repeated infarcts in sickle cell disease.
Q Cause of aplastic crisis?	Parvovirus B19 infecting red cell precursors.
Q Crew-cut skull: why?	Marrow expansion widens diploë; new bone forms perpendicular spicules.
Q What is HbH?	β4 tetramer in three-gene-deletion α-thalassaemia; golf-ball inclusions.
Q Direct vs indirect Coombs?	Direct: antibody on patient's cells. Indirect: free antibody in patient's serum.

OSPE Practice Stations

STATION 1 · PERIPHERAL SMEAR — 2 minutes

TASK

Child, Hb 5.4 g/dL. Film: marked anisopoikilocytosis, small pale cells, many target cells, basophilic stippling, fragments and numerous nucleated red cells. (a) Diagnosis. (b) Expected Hb electrophoresis. (c) Cause of death if treated by transfusion alone.

ANSWER

(a) β-thalassaemia major. (b) HbF markedly raised, HbA absent or low. (c) Cardiac failure from iron overload (haemosiderosis).

STATION 2 · PERIPHERAL SMEAR — 2 minutes

TASK

Film from a 7-year-old: crescent-shaped red cells with pointed ends, target cells, polychromasia and Howell–Jolly bodies. (a) Diagnosis. (b) Bedside confirmatory test. (c) What do the Howell–Jolly bodies indicate?

ANSWER

(a) Sickle cell anaemia. (b) Sickling test with sodium metabisulfite; Hb electrophoresis/HPLC. (c) Functional asplenia (autosplenectomy).

STATION 3 · LAB DATA 2 minutes

TASK

Engaged cousins, both well. Man: Hb 12.9, MCV 63, RBC 6.2 million, HbA2 5.4%. Woman: Hb 11.6, MCV 66, RBC 5.6 million, HbA2 4.9%. Ferritin normal in both. (a) Interpret. (b) Risk to each pregnancy. (c) Advice.

ANSWER

(a) Both are β-thalassaemia carriers (Mentzer < 13, raised HbA2). (b) 25% thal major, 50% carrier, 25% normal. (c) Genetic counselling; prenatal diagnosis (CVS DNA) in each pregnancy.

STATION 4 · LAB DATA 1 minute

TASK

Woman with SLE: Hb 8.1 g/dL, reticulocytes 9%, unconjugated bilirubin raised, LDH high, spherocytes on film, direct Coombs positive for IgG at 37 °C. (a) Diagnosis. (b) Site of red cell destruction. (c) One drug that can cause the same picture.

ANSWER

(a) Warm autoimmune haemolytic anaemia (secondary to SLE). (b) Spleen (extravascular). (c) α-methyldopa (or penicillin).

RAPID REVISION The Night-Before Page

15 ONE-LINERS

01 Haemolysis: retics up, bilirubin up, haptoglobin down.
02 Intravascular: haemoglobinuria, haemosiderinuria.
03 Extravascular: splenomegaly, spherocytes, jaundice.
04 HS: ankyrin; MCHC up; osmotic fragility up.
05 HS vs warm AIHA: Coombs negative vs positive.
06 G6PD: X-linked; infection, primaquine, fava beans.
07 G6PD smear: bite cells; Heinz bodies (supravital).
08 HbS: Glu6Val; polymerises when deoxygenated.
09 HbF protects; hydroxyurea raises HbF.
10 Sickle: autosplenectomy → pneumococcal sepsis.
11 β-thal major: 6–9 months, HbF high, nRBCs.
12 β-thal trait: HbA2 > 3.5%, Mentzer < 13.
13 Low hepcidin + transfusion = iron overload.
14 α-thal: 3 genes HbH (β4); 4 genes Barts (γ4).
15 PNH: PIGA, CD55/CD59, thrombosis, eculizumab.

MEMORY HOOKS

- **HbF Holds the Fort** in thal major
- **Warm = IgG = Spleen**; **Cold = IgM = C3**
- **Bite cells Betray G6PD**
- **Barts = all four α Bars down**

TEST YOURSELF: FILL THE GAPS

1. Commonest mutated protein in HS: _____.
2. NADPH in red cells comes from the _____ shunt.
3. Aplastic crisis is caused by _____.
4. HbA2 in β-thal trait: _____ %.
5. Cold agglutinins follow _____ pneumonia.
6. PNH red cells lack CD55 and _____.

Answers: ankyrin · hexose monophosphate · parvovirus B19 · 4–8 · Mycoplasma · CD59

DRAW FROM MEMORY

The β-thalassaemia pathogenesis chain (Fig 12.5) from mutation to iron overload.

DRAW FROM MEMORY

Extravascular vs intravascular haemolysis: spleen arm and bloodstream arm with their lab findings.

SPACED REVISION ☐ Day 1 ☐ Day 3 ☐ Day 7 ☐ Day 14 ☐ Day 30

13

Leukaemias & Myeloproliferative Neoplasms

Every leukaemia is a marrow that has lost a brake. Acute leukaemias cannot mature, so blasts pile up and the marrow fails within weeks; chronic leukaemias mature but cannot stop, so the blood fills with cells that look almost normal.

BY THE END OF THIS CHAPTER YOU CAN

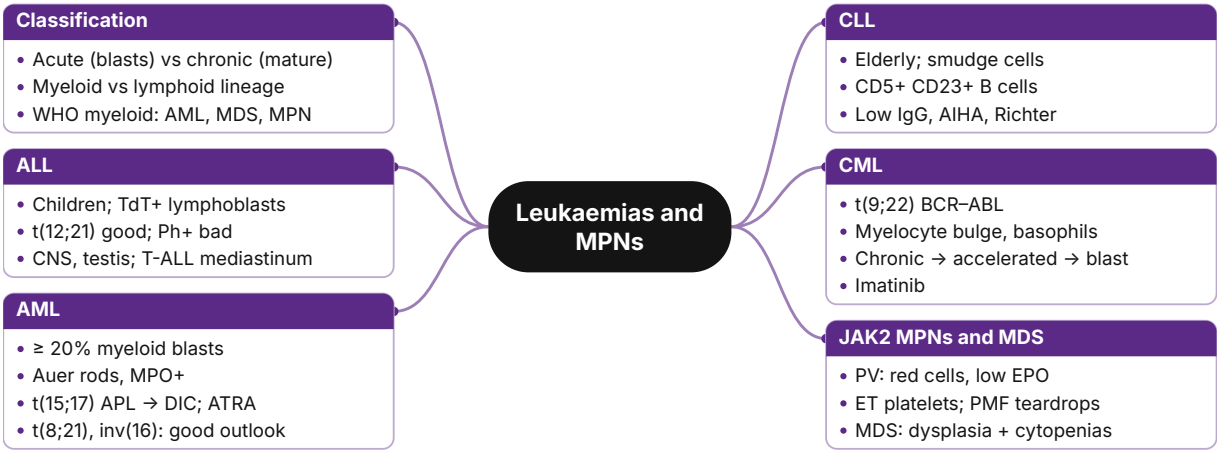
1. Classify leukaemias and myeloid neoplasms, and explain why acute and chronic leukaemias behave so differently
2. Differentiate ALL from AML by morphology, cytochemistry, immunophenotype and genetics
3. Explain why acute promyelocytic leukaemia causes DIC and why it responds to retinoic acid
4. Describe CLL and CML: smear findings, key genetic lesions, phases and complications
5. Explain how JAK2-pathway mutations produce polycythaemia vera, essential thrombocythaemia and primary myelofibrosis
6. Distinguish CML from a leukaemoid reaction and outline the myelodysplastic syndromes

EXAM RADAR

Theory SEQs	★★★
BCQs / MCQs	★★★
Viva voce	★★★
OSPE / slides	★★★

AML vs ALL, CML vs leukaemoid reaction and the translocations are favourite long and short questions; leukaemia smears are a standard OSPE station.

ROBBINS LINK Robbins Basic Pathology — *Hematopoietic and Lymphoid Systems* (PBD: *Diseases of White Blood Cells, Lymph Nodes, Spleen, and Thymus*): ALL, CLL, AML, MDS, MPN, leukaemoid reaction.



Chapter at a glance. Two questions (which lineage? how mature?) sort every leukaemia.

13.1 Classifying leukaemias

White-cell neoplasms are named by **where they live** and **how far they mature**. A **leukaemia** arises in the marrow and spills into the blood; a **lymphoma** forms masses in lymph nodes or other tissues (Chapter 14). The line is blurred: one tumour can present either way (ALL vs lymphoblastic lymphoma; CLL vs SLL).

> **DEFINITION**
> **Leukaemia** is a malignant, clonal proliferation of haematopoietic stem or progenitor cells that replaces the normal bone marrow and usually appears in large numbers in the peripheral blood.

Two questions name a leukaemia: **which lineage** (myeloid or lymphoid) and **how mature** (acute = blasts that cannot mature; chronic = cells that mature but proliferate to excess). Together they give the classic four: **ALL, AML, CLL, CML**. The WHO sorts myeloid neoplasms by what happens to maturation: **AML** (blocked; blasts ≥ 20%), **MDS** (disordered and ineffective → cytopenias) and **MPN** (normal, but production excessive → high counts: CML, PV, ET, PMF). Lymphoid neoplasms are covered in Chapter 14.

ACUTE VS CHRONIC LEUKAEMIA

Feature	Acute leukaemia	Chronic leukaemia
Dominant cell	**Blasts** (immature)	**Mature or maturing** cells
Onset	Abrupt: days to weeks	Insidious; often found on a routine count
Usual age	ALL: children; AML: adults	Adults (CLL: elderly)
WBC count	High, normal or even low	Usually very high
Marrow failure	**Early and severe**: anaemia, infection, bleeding	Late
Untreated survival	Weeks to months	Years

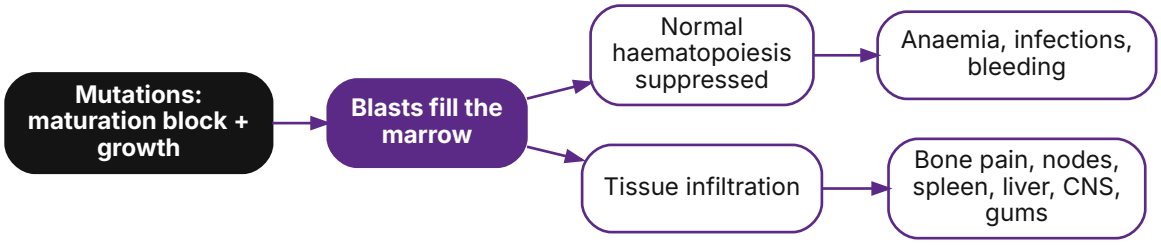


Figure 13.1 · How an acute leukaemia makes the patient ill. Symptoms come from what is missing, and from where blasts settle.

RISK FACTORS

- **Inherited:** Down syndrome, Fanconi anaemia, Bloom syndrome, ataxia–telangiectasia
- **Radiation**, **benzene**, smoking
- **Chemotherapy:** alkylating agents, topoisomerase II inhibitors → therapy-related AML/MDS
- **Viruses:** HTLV-1 → adult T-cell leukaemia/lymphoma
- **Prior MDS or MPN** → AML

CONCEPT CLEAR

Why does a leukaemic child bleed with a "normal" WBC? Symptoms come from what is **missing**. Blasts crowd out normal stem cells, so neutrophils, red cells and platelets fall. A normal count made of blasts protects nobody: read the smear.

13.2 Acute leukaemias: ALL and AML

An acute leukaemia needs two classes of mutation working together. One hits **transcription factors** that drive differentiation, so cells stay as blasts. The other switches on **growth signals** (tyrosine kinases, RAS), so blasts keep dividing. Cells that cannot mature and will not stop accumulate, and the marrow fails.

Acute lymphoblastic leukaemia (ALL)

DEFINITION

ALL is a neoplasm of precursor B or T cells (**lymphoblasts**) that mainly involves the marrow and blood. When the same cells form a mass (usually thymic) with little marrow disease, it is called **lymphoblastic lymphoma**.

WHO AND WHY

- **Commonest cancer of childhood**; peak about 3 years; about **85% are B-ALL**
- **T-ALL:** adolescent boys with a **mediastinal (thymic) mass**
- B-ALL: loss-of-function mutations in B-cell transcription factors (*PAX5*, *E2A*, *EBF*); T-ALL: activating *NOTCH1* mutations (over 70%)
- **Hyperdiploidy** and **t(12;21) *ETV6–RUNX1*** in children; **t(9;22) *BCR–ABL*** (Ph+) in about a quarter of adults

CLINICAL PICTURE

- Abrupt **fatigue, fever, bleeding** (marrow failure)
- **Bone pain and sternal tenderness** (marrow expansion, subperiosteal infiltration)
- Generalised lymphadenopathy, splenomegaly, hepatomegaly, testicular swelling
- **CNS spread** (headache, vomiting, cranial nerve palsies): commoner than in AML
- T-ALL: mediastinal mass compresses SVC or airway

PERIPHERAL SMEAR

- WBC high, normal or low; anaemia, thrombocytopenia
- **Lymphoblasts**: small to medium, high N:C ratio
- **Condensed** chromatin, **inconspicuous nucleoli**
- Scant blue **agranular** cytoplasm; **no Auer rods**

MARROW AND MARKERS

- Marrow hypercellular, **packed with lymphoblasts**
- **TdT positive** (> 95%); MPO negative; PAS coarse block positivity
- B-ALL: CD19, PAX5, **CD10** (CALLA); T-ALL: CD3, CD7, CD1

Prognosis: good with age 2–10 years, low WBC, hyperdiploidy (trisomy 4, 7, 10), t(12;21) and rapid blast clearance; **poor** with age under 2 (often *KMT2A*/11q23 rearranged), adolescence or adulthood, WBC over 100,000/µL, t(9;22), hypodiploidy and minimal residual disease after induction. About **95% of children** remit and **75–85% are cured**; adults fare much worse. Drugs reach the CNS and testes poorly ("sanctuary sites"), so CNS-directed therapy is standard.

Acute myeloid leukaemia (AML)

DEFINITION

AML is a neoplasm of myeloid progenitors in which **at least 20% of the cells in the marrow (or blood) are myeloid blasts**. Maturation is arrested, so blasts accumulate and normal haematopoiesis fails.

Mainly adults; incidence rises with age, peaking after 60. Mutations that **block differentiation** (*RUNX1*, *CBFB*, *RARA* fusions, *NPM1*) combine with ones that **drive growth** (*FLT3*, *KIT*, *RAS*) and epigenetic ones (*DNMT3A*, *TET2*, *IDH1/2*).

PERIPHERAL SMEAR

- **Myeloblasts**: larger, **delicate (fine) chromatin**, **3–5 nucleoli**
- Fine azurophilic granules; **Auer rods**: red, needle-like, specific for myeloid lineage
- Monoblasts: folded, lobulated nuclei; no Auer rods
- **Aleukaemic leukaemia:** few or no blasts in the blood

MARROW AND MARKERS

- Hypercellular; **≥ 20% blasts**
- **MPO** and **Sudan black B** positive; non-specific esterase (NSE) in monocytic types
- CD13, CD33, CD117 (KIT), CD34; CD14, CD64 in monocytic types
- **Myeloid sarcoma (chloroma):** a tissue mass of myeloblasts

WHO CLASSIFICATION OF AML, WITH THE FAB LABELS STILL ASKED IN VIVA

Category	Genetics / example	FAB	Point to remember
AML with recurrent genetic abnormalities	**t(8;21)** *RUNX1–RUNX1T1*	M2	Good prognosis; Auer rods
	inv(16) *CBFB–MYH11*	M4Eo	Abnormal marrow eosinophils; good
	t(15;17) *PML–RARA*	M3	APL: DIC; responds to ATRA and arsenic
	11q23 *KMT2A*; *NPM1* **mutation**	M4/M5	KMT2A poor; NPM1 good (no FLT3-ITD)
AML with myelodysplasia-related changes	Follows MDS; −5, −7, complex	—	Older patients; poor
Therapy-related AML	Alkylators, radiation; topoisomerase II inhibitors	—	Very poor
AML, not otherwise specified	By lineage and maturation	M0–M7	M5: gums and skin; M7: Down syndrome, marrow fibrosis

MEMORY HOOK: FAB M0–M7

M0 minimally differentiated · M1 without maturation · M2 with maturation · **M3 promyelocytic** · M4 myelomonocytic · M5 monocytic · M6 erythroid · M7 megakaryoblastic.

"3 bleeds, 4–5 chew, 7 has Down": M3 = DIC; M4/M5 = gum infiltration; M7 = Down syndrome.

CLINICAL FEATURES OF AML

- **Marrow failure:** anaemia, infections (bacterial and fungal), bleeding
- **DIC** in APL
- Monocytic types: **gum hypertrophy**, skin infiltration (leukaemia cutis)
- Very high counts: **leukostasis** in lungs and brain; hyperuricaemia

- Most adults remit, but only about **15–30%** stay disease-free for 5 years; APL, t(8;21), inv(16) do best

ACUTE PROMYELOCYTIC LEUKAEMIA (APL): THE LEUKAEMIA THAT BLEEDS

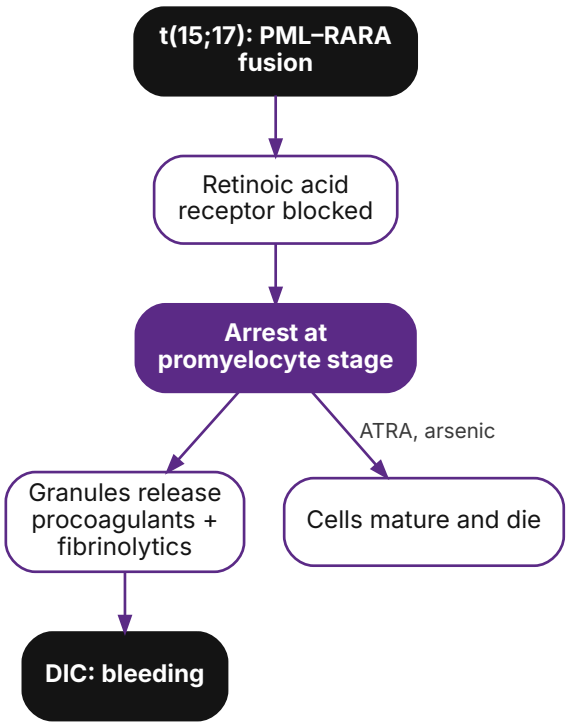


Figure 13.2 · APL: one fusion gene, two consequences.

- **t(15;17)** fuses PML with the retinoic acid receptor-α gene (RARA). The fusion protein represses genes needed for maturation, so cells stop at the **promyelocyte** stage.
- Smear/marrow: abnormal promyelocytes with heavy azurophilic granules, often bilobed nuclei, and **bundles of Auer rods ("faggot cells")**.
- The granules release **tissue factor-like procoagulants and fibrinolytic activators → DIC**, often present at diagnosis. Haemorrhage is the main early cause of death.
- **All-trans retinoic acid (ATRA)** in high doses overrides the block and forces maturation; **arsenic trioxide** degrades PML–RARA. Most patients are now cured.

ALL VS AML: THE MOST-ASKED COMPARISON IN LEUKAEMIA

Feature	ALL	AML
Age	Children (peak about 3 years)	Adults; rises with age
Chromatin, nucleoli	Condensed; nucleoli inconspicuous	Fine, delicate; **3–5 nucleoli**
Cytoplasm	Scant, agranular	More; azurophilic granules
Auer rods	Absent	**Present** in many
MPO / Sudan black B	Negative	**Positive**
PAS	**Coarse block** positivity	Negative or fine diffuse (erythroid type positive)
Non-specific esterase	Negative	Positive in monocytic types (M4, M5)
TdT	**Positive** (> 95%)	Usually negative
Surface markers	B: CD19, CD10, PAX5 · T: CD3, CD7	CD13, CD33, CD117, CD34
Special features	CNS, testis, mediastinum (T-ALL)	Gums, skin (monocytic); **DIC in APL**
Outcome	Most children cured	Poor overall; APL does well

⚠ **EXAM TRAP**

Three classic traps. (1) A **normal or low WBC does not exclude acute leukaemia** (aleukaemic leukaemia): examine the marrow. (2) Auer rods prove a myeloid blast, but monoblasts lack them, so their **absence does not exclude AML.** (3) **Down syndrome** raises the risk of both: acute megakaryoblastic leukaemia (M7) in children under 5, ALL in older children.

13.3 Chronic lymphocytic leukaemia (CLL)

DEFINITION

CLL is a neoplasm of small, mature-looking B cells involving the blood, marrow and lymphoid organs, diagnosed when the absolute count of clonal B lymphocytes in the blood exceeds **5000/µL**. When the same tumour lives mainly in lymph nodes it is called **small lymphocytic lymphoma (SLL)**: one disease, two addresses.

WHO AND WHY

- Commonest leukaemia of adults in Western countries; less common in Asia
- Median age about **60**; men twice as often as women
- Deletion of **13q14** removes miR-15/16, which normally restrain **BCL2** → BCL2 overexpression; also del(11q) (*ATM*), del(17p) (*TP53*), trisomy 12
- Survival signals flow through the B-cell receptor via **BTK** (the target of ibrutinib)

CLINICAL PICTURE

- Often **asymptomatic**; found on a routine count
- Fatigue, weight loss; generalised painless lymphadenopathy, hepatosplenomegaly
- **Hypogammaglobulinaemia** → bacterial infections
- **Warm autoimmune haemolytic anaemia** and immune thrombocytopenia
- **Richter transformation** to diffuse large B-cell lymphoma: a rapidly enlarging mass, poor outlook

PERIPHERAL SMEAR

- Lymphocytosis (> 5000/µL, often far higher)
- **Small round lymphocytes** with dense clumped chromatin and scant cytoplasm
- Many **smudge (basket) cells**: fragile cells crushed during smearing
- A few larger prolymphocytes

NODE AND MARROW

- Node diffusely effaced by small lymphocytes
- Pale **proliferation centres** of prolymphocytes and paraimmunoblasts: pathognomonic
- Marrow, splenic pulp and hepatic portal tracts infiltrated

CONCEPT CLEAR

CLL accumulates; it does not race. BCL2 keeps the cells alive long after they should have died, but they divide slowly: hence the indolent course (many patients live over 10 years) and the success of the BCL2 inhibitor venetoclax.

MOLECULAR KEY

Immunophenotype: CD19+, CD20 dim, **CD5+**, **CD23+**, dim surface Ig with one light chain. Mantle cell lymphoma is CD5+ but **CD23− and cyclin D1+**. **Bad markers:** del(17p)/*TP53*, del(11q), unmutated *IGHV*, ZAP-70, CD38.

13.4 Myeloproliferative neoplasms (MPN)

CONCEPT CLEAR

The MPN idea. The stem cell carries a **constitutively active tyrosine kinase**, so it overproduces without waiting for growth factors, but **maturation is normal**: the blood fills with **mature** cells, the opposite of AML. Four shared features follow: (1) raised counts of one or more lineages; (2) **extramedullary haematopoiesis** → hepatosplenomegaly; (3) a late **"spent phase"** of marrow fibrosis with falling counts; (4) transformation to **acute leukaemia** (common in CML, uncommon in the rest). High cell turnover adds **hyperuricaemia and gout**.

THE FOUR CLASSIC MPNS AND THEIR DRIVERS

MPN	Driver lesion	Main lineage expanded	Hallmark
CML	**BCR–ABL** [t(9;22)] in all cases	Granulocytes	Ph chromosome, basophilia, massive spleen
Polycythaemia vera	**JAK2 V617F** in > 95%	Red cells (plus WBC, platelets)	High haematocrit, **low EPO**, itching
Essential thrombocythaemia	JAK2 in about half; CALR or MPL	Megakaryocytes	Platelets persistently > 450,000/µL

MPN	Driver lesion	Main lineage expanded	Hallmark
Primary myelofibrosis	JAK2 in about half; CALR or MPL	Atypical megakaryocytes → fibrosis	Teardrops, leukoerythroblastosis, dry tap

Chronic myeloid leukaemia (CML)

DEFINITION

CML is an MPN of the pluripotent haematopoietic stem cell defined by the **BCR–ABL fusion gene**, usually created by the balanced translocation **t(9;22)(q34;q11)**. The shortened chromosome 22 is the **Philadelphia (Ph) chromosome**.

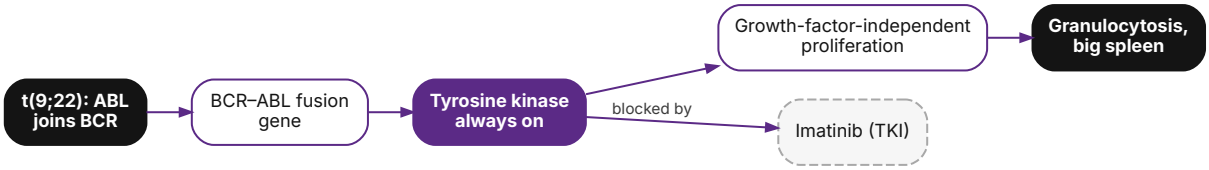


Figure 13.3 · Pathogenesis of CML. A kinase stuck in the 'on' position; imatinib switches it off.

Adults aged 25–60 (peak in the 40s and 50s). BCR–ABL drives growth **without blocking maturation**, so the blood contains every stage of granulocyte. A later hit that does block maturation produces blast crisis. Onset is insidious: fatigue, weight loss, anorexia and a **dragging sensation** from a massive spleen; sudden left-upper-quadrant pain means **splenic infarction**.

PERIPHERAL SMEAR

- WBC usually **> 100,000/µL**
- **Full spectrum** of granulocytes; **myelocytes and neutrophils** dominate ("myelocyte bulge")
- Blasts usually < 10% (chronic phase)
- **Basophilia** and eosinophilia; platelets often high
- Mild normocytic anaemia; **LAP score low**

MARROW AND SPLEEN

- Marrow **almost 100% cellular**: granulocytic and megakaryocytic hyperplasia, high M:E ratio
- Small ("dwarf") megakaryocytes; **sea-blue histiocytes** (pseudo-Gaucher cells)
- Spleen **massive**: red-pulp extramedullary haematopoiesis, infarcts
- Reticulin fibrosis increases with time

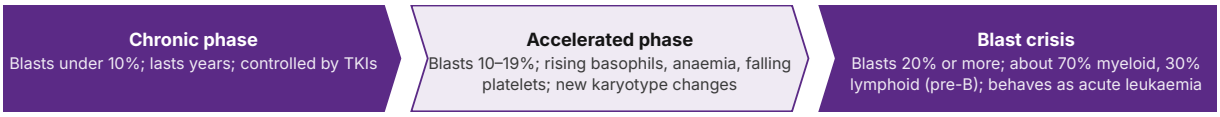


Figure 13.4 · Natural history of CML. Untreated, progression comes after a median of about 3 years.

About half of untreated patients pass through an accelerated phase; the rest enter blast crisis abruptly. **Imatinib** and later BCR–ABL tyrosine kinase inhibitors give lasting remissions in most patients but do not eradicate the stem cell; resistance can arise from ABL kinase mutations (e.g. T315I). Allogeneic stem-cell transplantation can cure.

A **leukaemoid reaction** is a **reactive** leukocytosis so high (usually > 50,000/µL) or immature that it mimics leukaemia: sepsis, tuberculosis, burns, severe haemolysis or haemorrhage, marrow infiltration by cancer.

LEUKAEMOID REACTION VS CML: A FAVOURITE "DIFFERENTIATE BETWEEN"

Feature	Leukaemoid reaction	CML
Nature, cause	Reactive: severe infection, burns, haemolysis	Clonal neoplasm: t(9;22) BCR–ABL
WBC	Usually 50,000–100,000/µL	Often > 100,000/µL
Differential	Mostly **bands and metamyelocytes**	Full spectrum; **myelocyte bulge**
Toxic changes	**Toxic granulation, Döhle bodies**, vacuoles	Absent
Basophils, eosinophils	Not increased	**Increased**
LAP score	**High**	**Low**
Spleen	Normal or slightly enlarged	Massive

Feature	Leukaemoid reaction	CML
Ph chromosome / BCR–ABL	Absent	**Present**
Course	Settles when the cause is treated	Progressive; needs TKI

⚠ EXAM TRAP

LAP or BCR–ABL? Many teachers still expect the **leukocyte alkaline phosphatase (LAP) score** (low in CML, high in a leukaemoid reaction). Write it, but add that the **definitive test is BCR–ABL** by karyotype (Ph chromosome), FISH or PCR; PCR also monitors response to imatinib.

The JAK2 family: PV, ET and primary myelofibrosis

JAK2 is the tyrosine kinase behind the receptors for erythropoietin (EPO), thrombopoietin (TPO) and G-CSF. The **V617F** mutation leaves it permanently on, so precursors grow without their growth factors; *CALR* and *MPL* mutations act on the same pathway.

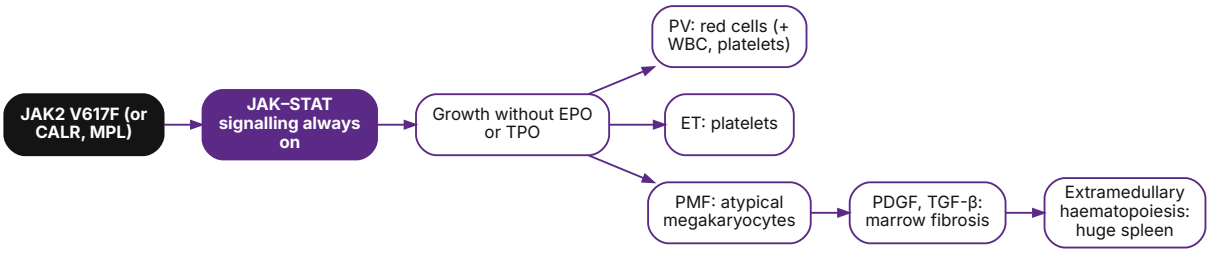


Figure 13.5 · One pathway, three diseases. Which lineage dominates decides the name.

MEMORY HOOK

One Philadelphia cousin, three JAK2 sisters. CML alone has **BCR–ABL**. **P**V makes **P**acked red cells, **E**T makes **E**xcess platelets, PM**F** makes **F**ibrosis. JAK2: nearly all PV, about half of ET and PMF.

POLYCYTHAEMIA VERA (PV)

An MPN with marked overproduction of **red cells**, usually with raised granulocytes and platelets too (panmyelosis). Precursors grow without EPO, so **serum EPO is low**: the key difference from secondary polycythaemia.

- Hb and haematocrit high (haematocrit may exceed 60%); WBC and platelets raised; basophilia
- Marrow hypercellular (all three lineages); large, clustered megakaryocytes
- Moderate splenomegaly (extramedullary haematopoiesis); congested organs
- Late **spent phase**: post-polycythaemic myelofibrosis; transformation to AML is uncommon
- Older adults: **plethora**, ruddy cyanosis, headache, dizziness, hypertension (hyperviscosity)
- **Itching after a warm bath** and peptic ulcer: histamine from basophils
- **Gout** from high cell turnover
- **Thrombosis** (stroke, MI, DVT, hepatic vein: Budd–Chiari) and, paradoxically, bleeding
- Phlebotomy with low-dose aspirin: survival of a decade or more

⚠ EXAM TRAP

A raised haematocrit has three explanations. Relative: plasma volume is low (dehydration, "stress" polycythaemia); red cell mass is normal. **Secondary**: EPO is **high**, from hypoxia (high altitude, COPD, cyanotic heart disease, heavy smoking) or an EPO-secreting tumour (renal cell carcinoma, hepatocellular carcinoma, cerebellar haemangioblastoma). **Primary (PV)**: EPO is **low**, JAK2 is mutated, and WBC, platelets and spleen are usually raised too.

ESSENTIAL THROMBOCYTHAEMIA (ET)

An MPN of megakaryocytes with a **persistent platelet count above 450,000/µL**. The marrow shows increased, **large, hyperlobated megakaryocytes** in clusters, with little fibrosis. Often asymptomatic; some have thrombosis, bleeding or **erythromelalgia** (burning hands and feet from platelet plugs in arterioles, relieved by aspirin). A **diagnosis of**

exclusion: first rule out reactive thrombocytosis (iron deficiency, infection, inflammation, cancer, splenectomy) and other MPNs. Indolent, with long survival.

PRIMARY MYELOFIBROSIS (PMF)

An MPN in which **obliterative marrow fibrosis** develops early and dominates. The fibroblasts are **not neoplastic**: they respond to **PDGF and TGF-β** released by neoplastic megakaryocytes. As the marrow scars, haematopoiesis moves to the spleen, liver and nodes.

PERIPHERAL SMEAR

- Normocytic anaemia
- **Leukoerythroblastosis**: nucleated red cells + immature granulocytes
- **Teardrop cells (dacrocytes)**: squeezed out of a scarred marrow
- Large abnormal platelets; basophilia

MARROW (TREPHINE BIOPSY)

- Aspiration gives a **dry tap**
- Diffuse **reticulin, then collagen, fibrosis**; marrow becomes hypocellular
- Clusters of **atypical megakaryocytes** with cloud-like nuclei; dilated sinusoids
- Late new bone formation (osteosclerosis)

Clinical: usually over 60; progressive anaemia, **massive splenomegaly** (early satiety, infarcts), weight loss, night sweats, gout. Median survival about 3–5 years; some transform to AML. Ruxolitinib (a JAK2 inhibitor) shrinks the spleen; only stem-cell transplantation cures.

13.5 Myelodysplastic syndromes (MDS)

DEFINITION

MDS is a group of clonal stem-cell disorders with **dysplastic, ineffective haematopoiesis**: the marrow is usually hypercellular, yet the blood shows cytopenias, because many precursors die in the marrow. Blasts are **below 20%**, and the risk of transformation to **AML** is high.

Primary MDS affects older adults (median about 70); **therapy-related** MDS follows alkylating drugs or radiation by 2–8 years and does worse. Mutations hit epigenetic regulators (*TET2*, *DNMT3A*), RNA-splicing genes (*SF3B1*: ring sideroblasts) and *TP53* (bad); typical karyotypes are monosomy 5 or 7, del(5q), del(7q), trisomy 8. Patients have weakness, infections and bleeding; about 10–40% progress to AML and median survival is short (a few years at most). The **5q− syndrome** of older women does better and responds to lenalidomide.

PERIPHERAL SMEAR

- Cytopenias; oval macrocytes
- **Pseudo-Pelger–Huët** neutrophils (bilobed), hypogranular neutrophils
- Giant, abnormal platelets

MARROW

- **Ring sideroblasts** (Prussian blue): iron-laden mitochondria ringing the nucleus
- Megaloblastoid erythroid precursors, nuclear budding
- **Micromegakaryocytes**; blasts < 20%

LAB SPOTTER

Leukaemia smears in 10 seconds. (1) Mostly **blasts**: acute leukaemia; **Auer rods** = AML, small agranular blasts = ALL. (2) **Every stage** of granulocyte + **basophils**: CML. (3) **Small mature lymphocytes** + **smudge cells** in an older adult: CLL. (4) **Teardrops** + nucleated red cells: marrow fibrosis or infiltration. (5) Left shift with **toxic granulation**: leukaemoid reaction.

PAKISTAN FOCUS

Leukaemia, mainly **ALL**, is the **commonest childhood cancer** in Pakistan, as it is worldwide. A child with fever, pallor, bone pain and bruising needs a blood count and smear before empirical treatment for typhoid or malaria. Cousin marriage is common, raising the risk of **autosomal recessive** marrow-failure syndromes such as **Fanconi anaemia**, which predispose to AML.

EXAM ZONE Past-Paper Pattern SEQs — Solved

PROF PATTERN **a)** Classify leukaemias. **b)** Differentiate between acute lymphoblastic and acute myeloid leukaemia. **2 + 3 marks**

MODEL ANSWER PLAN

a) By maturity (acute vs chronic) and lineage (myeloid vs lymphoid): ALL, AML, CLL, CML. WHO: myeloid neoplasms = AML, MDS, MPN (CML, PV, ET, PMF); lymphoid neoplasms = precursor B/T (ALL/LBL), mature B (CLL), T/NK. **b)** Table of 6–8 points: age; blast morphology (chromatin, nucleoli, granules); Auer rods; MPO/Sudan black (AML+); PAS block and TdT (ALL+); markers (CD10/CD19 vs CD13/CD33/CD117); CNS/testis vs gums/skin; DIC in APL; prognosis.

Score booster: name one translocation for each (t(12;21) in ALL; t(15;17) in AML-M3) to show depth.

MODULE PATTERN A 50-year-old man has fatigue and a dragging sensation in the left hypochondrium. The spleen is palpable 12 cm below the costal margin. WBC 180,000/µL with myelocytes, metamyelocytes and neutrophils; basophils 6%; blasts 3%. **a)** Most likely diagnosis? **b)** Name the cytogenetic abnormality and explain its molecular effect. **c)** Give four features that distinguish this from a leukaemoid reaction. **1 + 2 + 2 marks**

MODEL ANSWER PLAN

a) CML, chronic phase. **b)** t(9;22)(q34;q11): Philadelphia chromosome; *ABL* fuses with *BCR* → BCR–ABL, a constitutively active tyrosine kinase → growth-factor-independent proliferation of a pluripotent stem cell; maturation preserved; inhibited by imatinib. **c)** Basophilia/eosinophilia; myelocyte bulge; low LAP; no toxic granulation; massive spleen; Ph/BCR–ABL positive.

PROF PATTERN A 28-year-old woman has gum bleeding, bruises and fever. Hb 7.5 g/dL, WBC 3,000/µL, platelets 18,000/µL; PT and aPTT prolonged, fibrinogen low, D-dimer high. The marrow is replaced by abnormal granular promyelocytes with bundles of Auer rods. **a)** Diagnosis and genetic lesion? **b)** Explain her bleeding. **c)** State the principle of specific treatment. **1 + 2 + 2 marks**

MODEL ANSWER PLAN

a) Acute promyelocytic leukaemia (AML-M3), t(15;17) PML–RARA. **b)** Thrombocytopenia from marrow replacement plus **DIC**: granules release tissue factor-like procoagulants and fibrinolytic activators → consumption of platelets and factors, low fibrinogen, raised D-dimer. **c)** ATRA overcomes the differentiation block (cells mature and die); arsenic trioxide degrades PML–RARA; supportive platelets/cryoprecipitate.

Score booster: say that APL is a medical emergency because early haemorrhage is the main cause of death, yet with ATRA it is the most curable AML.

BCQs One-Best-Answer Practice

1 A 4-year-old boy has fever, pallor, bone pain and petechiae. The smear shows many small blasts with scant agranular cytoplasm and inconspicuous nucleoli. Which marker best confirms that these are precursor lymphoid cells?

A. Myeloperoxidase
B. CD117
C. TdT
D. Non-specific esterase
E. CD13

2 A 32-year-old woman has gum bleeding and ecchymoses. PT and aPTT are prolonged and fibrinogen is low. The marrow is full of heavily granular cells containing bundles of Auer rods. The genetic lesion is:

A. t(9;22)
B. t(8;21)
C. inv(16)
D. t(15;17)
E. t(12;21)

3 The drug that makes the leukaemic cells in the previous patient mature is:

A. Imatinib
B. All-trans retinoic acid
C. Rituximab
D. Hydroxyurea
E. Ruxolitinib

4 Which finding on a Romanowsky-stained smear establishes that blasts are myeloid?

A. Auer rods
B. Smudge cells
C. Döhle bodies
D. Teardrop cells
E. Rouleaux

5 A 68-year-old man has a lymphocyte count of 42,000/µL. The smear shows small mature lymphocytes and many smudge cells. Flow cytometry will most likely show:

A. CD10+, CD19+, TdT+
B. CD5+, CD23−, cyclin D1+
C. CD3+, CD4+, CD7+
D. CD13+, CD33+, CD117+
E. CD5+, CD23+, CD19+

6 The BCR–ABL protein of chronic myeloid leukaemia acts as a:

A. Repressor of retinoic acid target genes
B. Constitutively active tyrosine kinase
C. Mitochondrial anti-apoptotic protein
D. Cyclin driving G1 to S transition
E. Defective DNA mismatch-repair enzyme

7 A 40-year-old man with severe pneumonia has WBC 62,000/µL with many band forms and metamyelocytes. Which finding favours a leukaemoid reaction over CML?

A. Basophilia
B. Massive splenomegaly
C. High LAP score with toxic granulation
D. Myelocytes outnumbering metamyelocytes
E. Philadelphia chromosome

8 A 62-year-old man has itching after hot baths, a ruddy face, Hb 20.5 g/dL, platelets 640,000/µL, splenomegaly and a low serum erythropoietin. The most likely mutation is:

A. BCR–ABL
B. PML–RARA
C. KIT
D. JAK2 V617F
E. NPM1

9 The smear shows teardrop cells, nucleated red cells and myelocytes. Marrow aspiration yields a dry tap and the spleen is massive. The diagnosis is:

A. Primary myelofibrosis
B. Chronic lymphocytic leukaemia
C. Immune thrombocytopenic purpura
D. Iron deficiency anaemia
E. Acute promyelocytic leukaemia

10 Which is a favourable prognostic feature in childhood ALL?

A. Age under 1 year
B. WBC over 100,000/µL
C. t(9;22)
D. Hypodiploidy
E. t(12;21) ETV6–RUNX1

11 A 70-year-old has pancytopenia. The marrow is hypercellular with ring sideroblasts, pseudo-Pelger–Huët neutrophils, micromegakaryocytes and 6% blasts. The diagnosis is:

A. Acute myeloid leukaemia
B. Aplastic anaemia
C. Myelodysplastic syndrome
D. Vitamin B12 deficiency
E. CML in blast crisis

12 A 55-year-old with AML has swollen, bleeding gums and nodular skin infiltrates. The blasts are most likely:

A. Monoblasts, non-specific esterase positive
B. Lymphoblasts, TdT positive
C. Promyelocytes with t(15;17)
D. Megakaryoblasts with marrow fibrosis
E. Erythroblasts, PAS positive

ANSWER KEY WITH ONE-LINE REASONS

1. **C** TdT is expressed by pre-B and pre-T lymphoblasts in over 95% of ALL.
2. **D** APL: PML–RARA fusion from t(15;17), with DIC.
3. **B** Pharmacological doses of ATRA overcome the PML–RARA block.
4. **A** Auer rods are aggregates of azurophilic granules found only in neoplastic myeloid cells (blasts, abnormal promyelocytes).
5. **E** CLL: B cells that co-express CD5 and CD23.
6. **B** Permanent kinase activity drives growth-factor-independent proliferation.
7. **C** Reactive neutrophils show toxic changes and a high LAP score.
8. **D** Polycythaemia vera: JAK2 V617F in over 95%; EPO is suppressed.
9. **A** Marrow fibrosis gives leukoerythroblastosis, dacrocytes and a dry tap.
10. **E** t(12;21) and hyperdiploidy predict an excellent outcome.
11. **C** Dysplasia in several lineages with blasts below 20% = MDS.
12. **A** Monocytic AML (M4/M5) infiltrates gums and skin; NSE positive.

VIVA Rapid-Fire Viva Voce

Q Define leukaemia.	A clonal malignant proliferation of haematopoietic cells that replaces the marrow and usually floods the blood.
Q What blast count defines AML?	At least 20% myeloid blasts in marrow or blood.
Q What are Auer rods?	Red, needle-like aggregates of azurophilic granules in myeloblasts; MPO-positive and specific for the myeloid lineage.
Q What is a faggot cell?	An abnormal promyelocyte with bundles of Auer rods, typical of APL.
Q Why does APL cause DIC?	Its granules release tissue factor-like procoagulants and fibrinolytic activators.
Q What is the Philadelphia chromosome?	The shortened chromosome 22 produced by t(9;22), carrying the BCR–ABL fusion gene.
Q What are smudge cells?	Fragile CLL lymphocytes crushed during smear preparation.
Q What are proliferation centres?	Pale nodal foci of prolymphocytes and paraimmunoblasts; pathognomonic of CLL/SLL.
Q What is Richter transformation?	Transformation of CLL into diffuse large B-cell lymphoma, presenting as a rapidly enlarging mass.
Q Name the phases of CML.	Chronic, accelerated (blasts 10–19%) and blast crisis (blasts 20% or more).
Q What is a dry tap?	Failed marrow aspiration because of fibrosis (myelofibrosis) or packed marrow; a trephine biopsy is needed.
Q What is a pseudo-Pelger–Huët cell?	A bilobed, hypogranular neutrophil: a dysplastic feature of MDS.

OSPE Practice Stations

STATION 1 · PERIPHERAL SMEAR — 2 minutes

TASK

A 5-year-old: Hb 6.8 g/dL, WBC 58,000/µL, platelets 22,000/µL. Most cells are small blasts with scant agranular cytoplasm, condensed chromatin and inconspicuous nucleoli; no Auer rods. (a) Diagnosis? (b) Two tests to confirm the lineage. (c) One favourable genetic finding.

ANSWER

(a) Acute lymphoblastic leukaemia. (b) TdT positive; flow cytometry CD19/CD10 (B) or CD3/CD7 (T); MPO negative. (c) t(12;21) ETV6–RUNX1 or hyperdiploidy.

STATION 2 · SMEAR + COAGULATION DATA — 2 minutes

TASK

An adult with gum bleeding. Platelets 20,000/µL; PT 21 s (control 13); aPTT 52 s (control 32); fibrinogen 80 mg/dL; D-dimer markedly raised. Smear: large cells with dense purple granules, bilobed nuclei and bundles of rod-like inclusions. (a) Diagnosis? (b) What are the inclusions? (c) Interpret the coagulation profile.

ANSWER

(a) Acute promyelocytic leukaemia, t(15;17). (b) Auer rods (faggot cells). (c) DIC: consumption of platelets and factors (long PT/aPTT, low fibrinogen) with fibrinolysis (high D-dimer).

STATION 3 · PERIPHERAL SMEAR 2 minutes

TASK

A 45-year-old with a massive spleen; WBC 186,000/µL. Granulocytes at every stage, myelocytes and neutrophils most numerous; basophils and eosinophils increased; platelets 510,000/µL. (a) Diagnosis? (b) Confirmatory test. (c) One drug and its target.

ANSWER

(a) Chronic myeloid leukaemia, chronic phase. (b) Karyotype for the Ph chromosome, or FISH/PCR for BCR–ABL. (c) Imatinib, a tyrosine kinase inhibitor of BCR–ABL.

RAPID REVISION The Night-Before Page

15 ONE-LINERS

01 Acute = blasts that cannot mature; chronic = mature cells that cannot stop.

02 AML needs ≥ 20% myeloid blasts in marrow or blood.

03 ALL: children, TdT+, PAS block, MPO−, CNS and testis relapse.

04 T-ALL: adolescent boy with a mediastinal mass; NOTCH1.

05 Childhood ALL: t(12;21) and hyperdiploidy good; Ph+ and age under 2 bad.

06 AML: fine chromatin, 3–5 nucleoli, Auer rods, MPO+.

07 APL: t(15;17) PML–RARA, faggot cells, DIC; ATRA + arsenic.

08 Monocytic AML (M4/M5): gums and skin, NSE+.

09 t(8;21) and inv(16): good-prognosis AML.

10 CLL: elderly, smudge cells, CD5+ CD23+, proliferation centres.

11 CLL complications: infections, AIHA, Richter (DLBCL).

12 CML: t(9;22) BCR–ABL, myelocyte bulge, basophils, huge spleen.

13 CML vs leukaemoid: low vs high LAP; basophils; Ph chromosome.

14 PV: JAK2 V617F, low EPO, pruritus, thrombosis.

15 PMF: teardrops, leukoerythroblastosis, dry tap; MDS: ring sideroblasts.

MEMORY HOOKS

- **"3 bleeds, 4–5 chew, 7 has Down"**: M3 DIC, M4/M5 gums, M7 Down syndrome
- **One Philadelphia cousin, three JAK2 sisters**: CML vs PV, ET, PMF
- **P**V packed cells · **E**T excess platelets · PM**F** fibrosis
- **CLL accumulates**; it does not race (BCL2)

TEST YOURSELF: FILL THE GAPS

1. TdT is positive in _____ blasts.
2. APL fusion gene: _____.
3. CLL B cells co-express CD5 and _____.
4. LAP score in CML is _____
5. Blast crisis: blasts ≥ _____%.
6. Teardrop cells + dry tap = _____.

Answers: lymphoid (ALL) · PML–RARA · CD23 · low · 20 · primary myelofibrosis

DRAW FROM MEMORY

A myeloblast with an Auer rod beside a lymphoblast; label five differences.

DRAW FROM MEMORY

The CML pathway from t(9;22) to massive spleen, and the three phases (Figures 13.3–13.4).

SPACED REVISION ☐ Day 1 ☐ Day 3 ☐ Day 7 ☐ Day 14 ☐ Day 30

14

Lymphomas

An enlarged lymph node that is painless, rubbery and growing is asking a question only a biopsy can answer. Every lymphoma is a frozen snapshot of a normal stage in a lymphocyte's life: know the stage, and the tumour's marker, translocation and behaviour fall into place.

BY THE END OF THIS CHAPTER YOU CAN

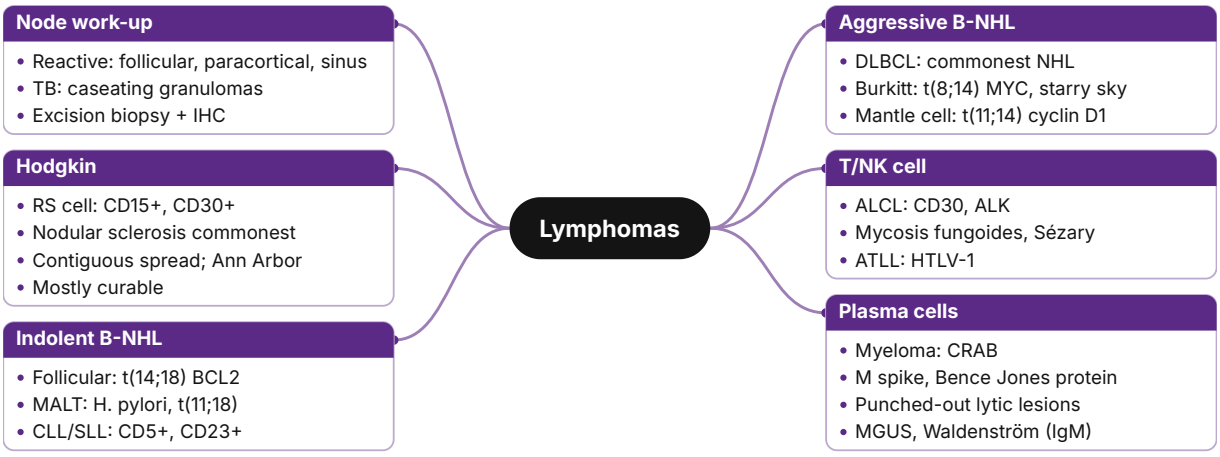
1. Outline an approach to lymphadenopathy and distinguish reactive hyperplasia from lymphoma
2. Describe the Reed–Sternberg cell, the subtypes of Hodgkin lymphoma and Ann Arbor staging
3. Classify lymphoid neoplasms (WHO) and differentiate Hodgkin from non-Hodgkin lymphoma
4. Link follicular, Burkitt, mantle cell and MALT lymphomas with their translocations and markers
5. Describe the pathogenesis, morphology and laboratory diagnosis of multiple myeloma
6. Recognise MGUS, Waldenström macroglobulinaemia and the main T-cell lymphomas

EXAM RADAR

Theory SEQs	★★★
BCQs / MCQs	★★★
Viva voce	★★★
OSPE / slides	★★☆

Hodgkin lymphoma (RS cell, subtypes, staging), HL vs NHL and multiple myeloma are perennial long questions; translocations and CD markers are favourite BCQs.

ROBBINS LINK Robbins Basic Pathology — *Hematopoietic and Lymphoid Systems* (PBD: *Diseases of White Blood Cells, Lymph Nodes, Spleen, and Thymus*): reactive lymphadenitis, lymphoid neoplasms (precursor and mature B-, T- and NK-cell), plasma cell neoplasms, Hodgkin lymphoma.



Chapter at a glance. Six groups cover every lymphoma question.

14.1 Lymphadenopathy: reactive or neoplastic?

Lymph nodes enlarge for two reasons: lymphocytes and macrophages **react** to antigen, or a clone **takes over**. Reactive nodes keep their architecture and contain a mixture of cells; a lymphoma **effaces** the architecture with a monotonous, clonal population.

PATTERNS OF REACTIVE LYMPHADENITIS

Pattern	What expands	Typical causes
Acute nonspecific	Neutrophils around follicles and in sinuses; may suppurate	Bacterial infection in the drainage area; nodes **tender**
Follicular hyperplasia	B-cell follicles with large germinal centres	Rheumatoid arthritis, toxoplasmosis, early HIV, syphilis
Paracortical hyperplasia	T-cell zone, with immunoblasts	Viral infections (EBV: infectious mononucleosis), vaccination, phenytoin
Sinus histiocytosis	Sinuses packed with histiocytes	Nodes draining cancers (e.g. axillary nodes in breast cancer)

Pattern	What expands	Typical causes
Granulomatous	Epithelioid granulomas	**Tuberculosis** (caseation), sarcoidosis, cat-scratch disease (stellate necrotising), fungi

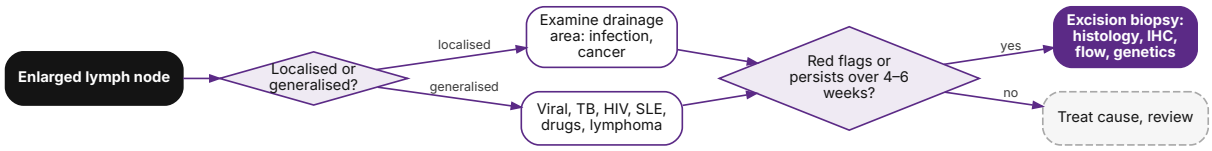


Figure 14.1 · Approach to an enlarged lymph node. Red flags and persistence, not size alone, decide the biopsy.

RED FLAGS IN A NODE

- Size over 2 cm; **hard** (carcinoma) or **rubbery** (lymphoma); fixed or matted
- **Supraclavicular** node (always abnormal); painless and progressive growth
- **B symptoms**: fever, drenching night sweats, weight loss over 10%
- Hepatosplenomegaly, cytopenias, older age

FNA is useful for metastatic carcinoma and tuberculosis, but a **lymphoma is diagnosed on an excised whole node**: architecture, IHC and clonality are needed.

PAKISTAN FOCUS

Pakistan is among the five highest **tuberculosis**-burden countries (incidence about 266 per 100,000). Tuberculous lymphadenitis is therefore a very common cause of neck swelling and the chief mimic of lymphoma. **Matted nodes with caseating granulomas** mean TB; a **monotonous infiltrate effacing the node** means lymphoma. A node that fails to respond to anti-TB therapy, or shows no granulomas, needs excision with IHC and GeneXpert.

14.2 Lymphoid neoplasms: the principles

About **85–90%** of lymphoid neoplasms are of **B-cell** origin; the rest are T or NK cell. Each is a **monoclonal** expansion (one light chain, κ or λ, in B-cell tumours; one T-cell receptor rearrangement in T-cell tumours) of a cell frozen at a normal stage of development. That stage decides its markers and its behaviour.

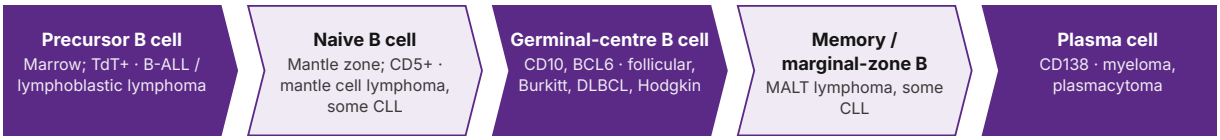


Figure 14.2 · Every lymphoma is a frozen stage of B-cell life. Translocations often strike in the germinal centre, where B cells deliberately cut and rejoin their immunoglobulin genes.

WHO CLASSIFICATION OF LYMPHOID NEOPLASMS (SIMPLIFIED)

Group	B-cell	T/NK-cell
Precursor (lymphoblastic)	B-ALL / lymphoblastic lymphoma	T-ALL / lymphoblastic lymphoma
Mature: indolent	CLL/SLL, **follicular**, marginal zone (**MALT**), lymphoplasmacytic, hairy cell leukaemia	Mycosis fungoides, T-cell large granular lymphocytic leukaemia
Mature: aggressive	**DLBCL**, **Burkitt**, **mantle cell**	Peripheral T-cell lymphoma NOS, **ALCL**, **ATLL**, extranodal NK/T-cell lymphoma
Plasma cell neoplasms	Myeloma, plasmacytoma, MGUS	—
Hodgkin lymphoma	Classical (4 subtypes); nodular lymphocyte predominant	—

CONCEPT CLEAR

The lymphoma paradox. Indolent lymphomas (follicular, CLL/SLL, MALT) grow slowly, so patients live for years, but chemotherapy, which kills dividing cells, rarely cures them. **Aggressive** lymphomas (DLBCL, Burkitt) kill within months if untreated, yet because their cells divide fast, intensive chemotherapy cures many. Slow is not always good; fast is not always fatal.

MOLECULAR KEY

Markers you must know: CD45 = all leukocytes · **CD19, CD20, PAX5** = B cells · **CD3** = T cells · **TdT** = lymphoblasts · **CD10, BCL6** = germinal centre · **CD5 on a B cell** = CLL or mantle cell · **CD23** = CLL · **cyclin D1** = mantle cell · **BCL2 in follicles** = follicular lymphoma · **CD15 + CD30** = Reed–Sternberg cell · **CD30 + ALK** = ALCL · **CD138** = plasma cells.

14.3 Hodgkin lymphoma (HL)

DEFINITION

Hodgkin lymphoma is a lymphoid neoplasm, usually arising in a single node or chain of nodes, in which a few large neoplastic **Reed–Sternberg (RS) cells** (germinal-centre B-cell origin) lie in an abundant **reactive** background of lymphocytes, eosinophils, plasma cells, histiocytes and fibrosis. It spreads in an orderly fashion to **anatomically contiguous** nodes.

One of the commonest cancers of **young adults**, with a second peak in older age. The RS cells are a small minority of the tumour mass; the rest is an army they have recruited.

THE REED–STERNBERG CELL

- **Large** (about 45 µm), with **two mirror-image nuclei** or a bilobed nucleus
- Each lobe has a **large, eosinophilic, inclusion-like nucleolus** the size of a small lymphocyte, with a clear halo: the **"owl-eye"** appearance
- Abundant cytoplasm
- **Classical HL:** CD15+, CD30+, PAX5 weak; **CD45−**, usually CD20−

VARIANTS

- **Hodgkin cell**: mononuclear variant
- **Lacunar cell**: folded nucleus in a clear space (cytoplasm retracts in formalin); nodular sclerosis
- **LP ("popcorn") cell**: multilobed nucleus; NLPHL; CD20+, CD15−, CD30−
- **Mummified cell**: shrunken, dark, apoptotic

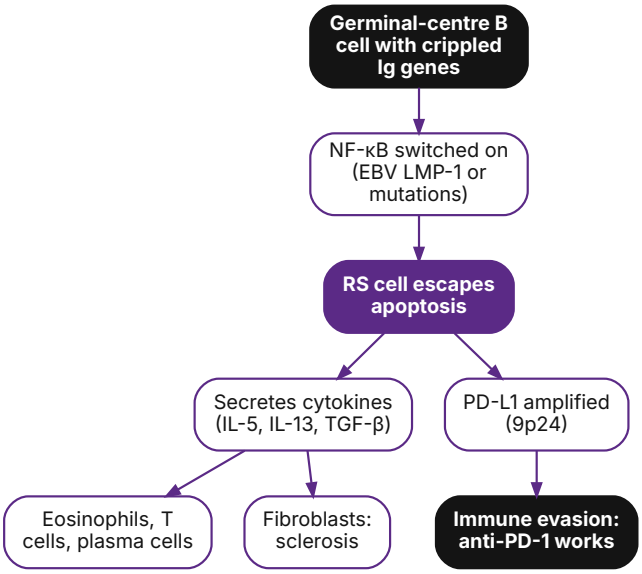


Figure 14.3 · How the RS cell builds its tumour.

SUBTYPES OF HODGKIN LYMPHOMA (WHO)

Subtype	Who / where	Morphology	Phenotype, EBV	Outlook
Nodular sclerosis	**Commonest (65–70%)**; adolescents, young adults; women as often as men; **mediastinal**, lower cervical, supraclavicular nodes	**Collagen bands** divide the node into nodules; **lacunar cells**	CD15+, CD30+; EBV rare	Excellent
Mixed cellularity	About 25%; men, older patients; B symptoms, advanced stage	Many classic RS cells in a mixed infiltrate (lymphocytes, eosinophils, plasma cells, histiocytes)	CD15+, CD30+; **EBV in about 70%**	Good
Lymphocyte-rich	Uncommon; older adults	Reactive lymphocytes predominate; few RS cells	CD15+, CD30+; EBV in about 40%	Very good

Subtype	Who / where	Morphology	Phenotype, EBV	Outlook
Lymphocyte-depleted	Rare; elderly, HIV, developing countries	Few lymphocytes; many RS cells, often bizarre	CD15+, CD30+; EBV in over 90%	Least favourable
Nodular lymphocyte predominant (NLPHL)	About 5%; young men; single cervical or axillary node	Nodules of small B cells with **LP (popcorn) cells**	**CD20+**, BCL6+; CD15−, CD30−; EBV−	Excellent; may become DLBCL

ANN ARBOR STAGING (HL AND NHL)

Stage	Extent
I	One lymph node region (I) or one extralymphatic organ or site (IE)
II	Two or more node regions on the **same side** of the diaphragm, ± limited contiguous extralymphatic spread (IIE)
III	Node regions on **both sides** of the diaphragm; ± spleen (IIIS) or limited contiguous extralymphatic spread (IIIE)
IV	**Diffuse or disseminated** involvement of extralymphatic organs (liver, marrow, lung) ± nodes

A = no systemic symptoms. **B** = fever, drenching night sweats, or unexplained loss of over 10% of body weight in 6 months.

CLINICAL FEATURES AND OUTCOME

- **Painless**, rubbery enlargement of cervical, supraclavicular, axillary or mediastinal nodes
- B symptoms, **pruritus**; classic but uncommon: cyclical **Pel–Ebstein fever** and pain in nodes after alcohol
- Defective cell-mediated immunity; anergy
- Spreads **nodes → spleen → liver → marrow**; staging guides therapy
- Stage I–IIA: cure in about 90%; even advanced disease is often curable
- **Late effects** of therapy: second cancers (AML/MDS, lung, breast after mediastinal radiation), heart and lung damage, infertility

HODGKIN VS NON-HODGKIN LYMPHOMA

Feature	Hodgkin lymphoma	Non-Hodgkin lymphoma
Neoplastic cell	**RS cell**, a small minority in a reactive background	Neoplastic lymphocytes form the bulk
Lineage	Germinal-centre B cell	85–90% B cell; rest T/NK
Nodal involvement	Often a **single axial group** (cervical, mediastinal, para-aortic)	Often **multiple peripheral** nodes
Spread	**Orderly, contiguous**	Non-contiguous, unpredictable
Mesenteric nodes, Waldeyer ring	Rarely involved	Commonly involved
Extranodal disease	Uncommon	Common (gut, skin, CNS, marrow)
Leukaemic phase	Absent	In some (CLL, mantle cell)
Markers	CD15+, CD30+, CD45−	CD45+; CD20 (B) or CD3 (T)
Outcome	Mostly curable	Variable by type

14.4 B-cell non-Hodgkin lymphomas

Follicular lymphoma

The commonest **indolent** NHL; middle-aged adults, both sexes. A neoplasm of **germinal-centre B cells** driven by **t(14;18)(q32;q21)**, which places *BCL2* next to the *IGH* locus. BCL2 blocks apoptosis, so cells that should die in the germinal centre survive. Mutations of the histone methyltransferase *KMT2D* are also common.

GROSS

- Painless, often generalised lymphadenopathy that may wax and wane
- Cut surface may show vague nodularity
- Marrow involved in most patients (**paratrabecular** aggregates)

MICROSCOPY

- **Nodular (follicular)** growth effacing the node
- **Centrocytes**: small cells with cleaved, angular nuclei (most)
- **Centroblasts**: larger cells, open chromatin, several nucleoli (few)
- CD19, CD20, **CD10**, BCL6 and **BCL2 positive**

Course: patients live for many years, but the disease is rarely cured. In **30–50%** it transforms to DLBCL, after which survival is short.

REACTIVE FOLLICULAR HYPERPLASIA VS FOLLICULAR LYMPHOMA

Feature	Reactive follicular hyperplasia	Follicular lymphoma
Architecture	Preserved; follicles mainly in the cortex	**Effaced**; crowded follicles throughout
Follicles	Vary in size and shape	**Uniform**, back-to-back
Mantle zone	Well formed	Thin or absent
Germinal centre	Polarised; mixed cells; many mitoses	Monotonous centrocytes; few mitoses
Tingible-body macrophages	**Many**	Few or none
BCL2 in follicles	Negative	**Positive**
Clonality	Polyclonal (κ and λ)	**Monoclonal**

Diffuse large B-cell lymphoma (DLBCL)

The **commonest NHL** of adults (about a third). Median age about 60, but it occurs at any age. It arises de novo or by transformation of an indolent lymphoma. About a third have rearrangement of **BCL6** (3q27); others carry t(14;18) or MYC rearrangements. Typical presentation: a **rapidly enlarging mass** at one nodal or extranodal site (Waldeyer ring, gut, skin, bone, brain).

GROSS

- Fleshy, white, "fish-flesh" mass
- Often extranodal; may show necrosis

MICROSCOPY

- **Diffuse sheets of large B cells** (nuclei several times a small lymphocyte)
- Vesicular nuclei, prominent nucleoli (centroblast- or immunoblast-like)
- CD19, CD20; variable CD10, BCL6

Special settings: EBV-driven DLBCL in AIDS and after transplantation; **primary effusion lymphoma** (HHV-8/KSHV) in body cavities; primary mediastinal large B-cell lymphoma in young women. Aggressive and fatal if untreated, but rituximab plus combination chemotherapy (R-CHOP) brings remission in most and cures about half.

Burkitt lymphoma

A highly aggressive germinal-centre B-cell lymphoma, the **fastest-growing human tumour**, defined by **translocation of MYC** on chromosome 8, usually **t(8;14)** with *IGH* (or t(2;8), t(8;22) with the light-chain genes). MYC drives growth and the Warburg (aerobic glycolytic) metabolism needed to build new cells.

- **Endemic (African):** children; **jaw (mandible/maxilla)** mass; **EBV in nearly all**
- **Sporadic:** children and young adults; **ileocaecal** or peritoneal mass, ovaries, kidneys; EBV in a minority
- **Immunodeficiency-associated:** HIV
- Responds to intensive chemotherapy: most children and young adults are cured; risk of **tumour lysis syndrome**

MICROSCOPY

- Diffuse sheets of **monotonous intermediate-sized** cells, round nuclei, 2–5 small nucleoli
- Deeply basophilic cytoplasm with **lipid vacuoles** (smear/touch prep)
- Very high mitotic and apoptotic rate (Ki-67 near 100%)
- **Tingible-body macrophages** with nuclear debris: clear spaces in a blue sea = **"starry sky"**
- CD19, CD20, CD10, BCL6+; **BCL2 negative**

Mantle cell lymphoma

A neoplasm of **naive B cells** of the mantle zone; usually **older men**. **t(11;14)(q13;q32)** joins the *CCND1* gene to *IGH*, causing **cyclin D1** overexpression, which pushes cells from G1 into S phase. Most patients present with generalised lymphadenopathy, splenomegaly and **marrow involvement**; the gut may show multiple polyps (**lymphomatoid polyposis**). Small-to-intermediate cells with irregular nuclei, diffuse or vaguely nodular, with **no proliferation centres**. Phenotype: CD19, CD20, **CD5+**, **CD23−**, **cyclin D1+**. Moderately aggressive and incurable with standard chemotherapy.

Extranodal marginal zone (MALT) lymphoma

Indolent B-cell lymphomas that arise in **mucosa or glands affected by chronic inflammation**: the **stomach** in *H. pylori* gastritis, **salivary glands** in Sjögren syndrome, the **thyroid** in Hashimoto thyroiditis. They show how chronic antigen stimulation becomes a cancer.

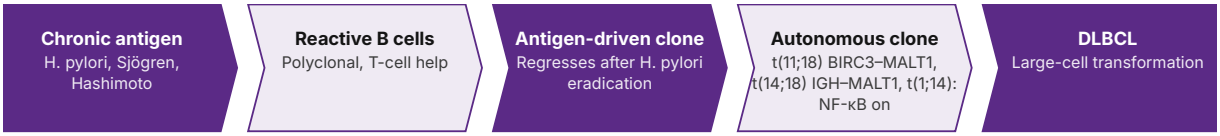


Figure 14.4 · From gastritis to lymphoma. While the clone still needs antigen, killing H. pylori can cure it.

Microscopy: marginal-zone cells (small, with abundant pale cytoplasm; "monocytoid"), plasma-cell differentiation and **lymphoepithelial lesions** (tumour cells invading glandular epithelium). CD20+, CD5−, CD10−, CD23−. Usually localised for long periods.

CLL / small lymphocytic lymphoma

The same neoplasm as CLL (Chapter 13), called SLL when the nodes rather than the blood dominate. Nodes are effaced by small round lymphocytes with pale **proliferation centres**; the cells are **CD5+ and CD23+**.

TRANSLOCATIONS IN LYMPHOID NEOPLASMS: THE HIGH-YIELD TABLE

Neoplasm	Translocation	Genes	Effect
Follicular lymphoma	**t(14;18)**(q32;q21)	*IGH–BCL2*	Blocks apoptosis
Burkitt lymphoma	**t(8;14)**(q24;q32)	*MYC–IGH*	Drives growth and metabolism
Mantle cell lymphoma	**t(11;14)**(q13;q32)	*CCND1–IGH*	Cyclin D1: G1 → S
MALT lymphoma	**t(11;18)**(q21;q21)	*BIRC3–MALT1*	NF-κB on; antigen-independent
DLBCL	3q27 rearrangement	*BCL6*	Blocks germinal-centre exit
Anaplastic large cell lymphoma	**t(2;5)**(p23;q35)	*NPM1–ALK*	Active tyrosine kinase
Multiple myeloma	14q32 translocations, e.g. t(11;14), t(4;14)	*IGH* with *CCND1*, *FGFR3*	Cell-cycle and growth signals

MEMORY HOOK

IGH lives on 14

Its partner names the tumour: **8** = MYC, Burkitt (races) · **11** = cyclin D1, Mantle (cycles) · **18** = BCL2, Follicular (refuses to die).

EXAM TRAP

Same number, different gene. t(14;18) means *BCL2* in follicular lymphoma but *IGH–MALT1* in some MALT lymphomas (both genes sit on 18q21). And t(11;14) is cyclin D1 in mantle cell lymphoma **and** in some myelomas.

14.5 T-cell and NK-cell lymphomas (brief)

Neoplasm	Key points
Peripheral T-cell lymphoma, NOS	Heterogeneous mature T-cell tumours; generalised nodes, eosinophilia, pruritus; poor prognosis
Anaplastic large cell lymphoma (ALCL)	Large cells with horseshoe-shaped nuclei ("hallmark cells"); **CD30+**; **ALK** rearrangement [t(2;5)] in children and young adults carries a good prognosis
Adult T-cell leukaemia/lymphoma	**HTLV-1** (Japan, Caribbean, West Africa); skin lesions, nodes, hepatosplenomegaly, **hypercalcaemia**; CD4+ cells with multilobed ("flower") nuclei; poor prognosis
Mycosis fungoides / Sézary syndrome	CD4+ skin-homing T cells with **cerebriform** nuclei; patch → plaque → tumour; epidermal **Pautrier microabscesses**; Sézary = erythroderma + circulating tumour cells
Extranodal NK/T-cell lymphoma	**EBV**-associated; commoner in Asia; destructive nasal and midline facial masses with angiocentric necrosis

T-ALL/lymphoblastic lymphoma (mediastinal mass in adolescent boys): Chapter 13.

14.6 Plasma cell neoplasms

Plasma cell neoplasms are clonal expansions of immunoglobulin-secreting cells. They usually make a complete immunoglobulin or a free light chain, detected as a **monoclonal (M) protein**. The M protein, the bone damage and the marrow takeover explain almost every feature.

Multiple myeloma

DEFINITION

Multiple myeloma is a neoplasm of plasma cells that grows mainly in the bone marrow, produces **multifocal lytic bone lesions** and usually secretes an **M protein** into serum and/or urine.

Older adults (median about 70); men and people of African descent more often. Early lesions are translocations joining *IGH* (14q32) to genes such as *CCND1* or *FGFR3*; *MYC* rearrangement and *TP53* loss come later. Myeloma cells grow on **IL-6** from marrow stromal cells.

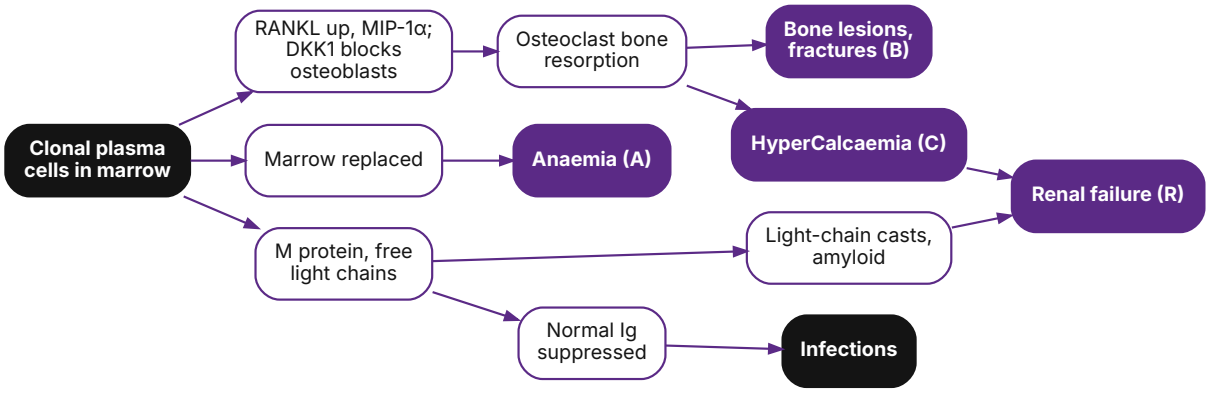


Figure 14.5 · Pathogenesis of myeloma: how one clone produces CRAB.

SKELETON AND KIDNEY

- Multiple **punched-out lytic lesions** 1–4 cm, filled with soft red gelatinous tissue
- Vertebrae (commonest), ribs, **skull**, pelvis, femur, clavicle
- Pathological fractures; vertebral collapse
- X-ray skull: multiple "punched-out" holes; bone scan often cold

MARROW AND KIDNEY

- Marrow: **sheets of plasma cells**, often over 30% of cellularity; clock-face chromatin, perinuclear halo, binucleate and immature forms
- Inclusions: **Russell bodies** (cytoplasmic Ig), **Dutcher bodies** (intranuclear), Mott (grape) and flame cells
- Blood: **rouleaux** (stacked red cells), very high ESR
- **Myeloma kidney**: pink-blue **light-chain (Bence Jones) casts** in distal tubules and collecting ducts, ringed by **multinucleate giant cells**; metastatic calcification; AL amyloid

LAB VALUES

Serum electrophoresis narrow **M spike**

(γ region); IgG commonest (about 60%), then IgA (20–25%); light chains only in 15–20%

Urine **Bence Jones protein**

= free κ or λ light chains (missed by dipstick)

Marrow clonal plasma cells ≥ 10% (diagnostic threshold)

Blood high Ca^{2+}, creatinine, ESR, β2-microglobulin; **normal ALP**

CLINICAL FEATURES

- **Bone pain** (back), pathological fractures, cord compression
- **Hypercalcaemia**: confusion, weakness, constipation, polyuria
- **Recurrent bacterial infections** (pneumococcus, *S. aureus*, *E. coli*): the leading cause of death, with renal failure second
- Normocytic anaemia; hyperviscosity in some (especially IgA)
- Bortezomib, lenalidomide and stem-cell transplantation prolong survival, but cure is rare

CONCEPT CLEAR

Why is ALP normal and the bone scan cold in myeloma? Myeloma lesions are purely **lytic**: osteoclasts are switched on while osteoblasts are switched off (DKK1). ALP and isotope uptake come from osteoblast activity, so they stay normal. Plain X-rays (or MRI/CT), not a bone scan, show myeloma; a raised ALP points instead to fracture healing, liver disease or osteoblastic metastases.

Other plasma cell neoplasms

Disorder	Key points
MGUS (monoclonal gammopathy of undetermined significance)	Commonest monoclonal gammopathy; about 3% of people over 50. M protein < 3 g/dL, marrow plasma cells < 10%, **no CRAB**. Progresses to myeloma at about **1% per year**, so it needs follow-up
Solitary plasmacytoma	A single plasma-cell mass in bone or soft tissue (often the upper respiratory tract). Bone lesions often progress to myeloma; soft-tissue lesions are often cured by local treatment
Lymphoplasmacytic lymphoma (Waldenström macroglobulinaemia)	Older adults; mixture of B cells, plasmacytoid lymphocytes and plasma cells in marrow, nodes, spleen; **IgM** M protein → **hyperviscosity** (visual blurring, headache, bleeding), cryoglobulins, Raynaud phenomenon; *MYD88* L265P mutation in most; **no lytic bone lesions**

LAB SPOTTER

Node or marrow slide in 10 seconds. (1) Architecture **preserved** with varied follicles and tingible-body macrophages: reactive. (2) Node **effaced** by crowded uniform follicles: follicular lymphoma. (3) **Starry sky** of pale macrophages in blue sheets: Burkitt. (4) Bands of **collagen** and cells in clear spaces in a young adult: nodular sclerosis HL. (5) Owl-eye cell among eosinophils: classical HL. (6) Marrow sheets of **eccentric-nucleus cells with a perinuclear halo**: myeloma.

EXAM ZONE Past-Paper Pattern SEQs — Solved

PROF PATTERN **a)** Classify Hodgkin lymphoma. **b)** Describe the Reed–Sternberg cell and its variants, with their immunophenotype. **2 + 3 marks**

MODEL ANSWER PLAN

a) Classical HL: nodular sclerosis, mixed cellularity, lymphocyte-rich, lymphocyte-depleted; nodular lymphocyte predominant HL. One line on frequency/age for each (NS commonest, young adults, mediastinum). **b)** Large binucleate/bilobed cell, owl-eye eosinophilic nucleoli, abundant cytoplasm; variants: mononuclear Hodgkin cell, lacunar cell (NS), LP/popcorn cell (NLPHL), mummified cell. Classical: CD15+, CD30+, CD45−; LP cells CD20+, CD15−, CD30−.

Score booster: a labelled sketch of the owl-eye RS cell earns the description marks quickly.

MODULE PATTERN A 66-year-old man has back pain and tiredness. Hb 8.9 g/dL, ESR 110 mm/h, calcium 12.1 mg/dL, creatinine 2.6 mg/dL. Skull X-ray shows multiple punched-out lesions. **a)** Most likely diagnosis? **b)** Explain the bone lesions and hypercalcaemia. **c)** List the investigations that confirm it and the expected results. **1 + 2 + 2 marks**

MODEL ANSWER PLAN

a) Multiple myeloma. **b)** Myeloma cells raise RANKL and MIP-1α → osteoclast activation; DKK1 inhibits osteoblasts → purely lytic lesions; calcium released from bone. **c)** SPEP/immunofixation: M spike (IgG or IgA); urine: Bence Jones protein; marrow: clonal plasma cells ≥ 10% in sheets; smear: rouleaux; normal ALP; raised β2-microglobulin.

Score booster: write CRAB as a mini-flowchart (Figure 14.5) and add "infection is the leading cause of death".

PROF PATTERN **a)** Give the Ann Arbor staging of lymphomas. **b)** Differentiate between Hodgkin and non-Hodgkin lymphoma. **2 + 3 marks**

MODEL ANSWER PLAN

a) I one node region/IE; II ≥ 2 regions same side of diaphragm; III both sides (± S, E); IV disseminated extralymphatic (liver, marrow, lung); suffix A/B with the three B symptoms. **b)** 5–6 points: RS cell minority vs bulk neoplastic cells; single axial group vs multiple peripheral nodes; contiguous vs non-contiguous spread; mesenteric/Waldeyer rarely vs often; extranodal uncommon vs common; CD15/CD30 vs CD20/CD3.

CLASSIC Name three lymphomas with characteristic translocations and the genes involved. Describe the morphology of Burkitt lymphoma. **5 marks**

MODEL ANSWER PLAN

Follicular t(14;18) *IGH–BCL2*; Burkitt t(8;14) *MYC–IGH*; mantle cell t(11;14) *CCND1–IGH* (MALT t(11;18) as a bonus). Burkitt: diffuse sheets of monotonous intermediate cells, round nuclei, 2–5 nucleoli, basophilic vacuolated cytoplasm, very high mitotic/apoptotic rate, tingible-body macrophages → starry sky; CD10+, BCL2−, Ki-67 ~100%; endemic jaw (EBV) vs sporadic ileocaecal forms.

BCQs One-Best-Answer Practice

1 A 24-year-old woman has a mediastinal mass. Node biopsy shows nodules separated by collagen bands and large cells lying in clear spaces. These cells are most likely:

A. CD20+, CD15−, CD30−
B. CD3+, CD4+
C. CD15+, CD30+, CD45−
D. CD5+, CD23+
E. CD138+, CD38+

2 The 'popcorn' cells of nodular lymphocyte predominant Hodgkin lymphoma characteristically express:

A. CD20 and BCL6
B. CD15 and CD30
C. Cyclin D1
D. TdT
E. ALK

3 Of the following Hodgkin lymphoma subtypes, EBV is most often found in the Reed–Sternberg cells of:

A. Nodular sclerosis
B. Mixed cellularity
C. Nodular lymphocyte predominant
D. Lymphocyte-rich
E. All equally

4 A patient has involved nodes in the neck and the para-aortic region, and an enlarged involved spleen; marrow is clear. The Ann Arbor stage is:

A. IE
B. IIS
C. IIE
D. IIIS
E. IV

5 An abdominal mass in a 9-year-old shows sheets of intermediate cells with numerous mitoses and scattered pale macrophages. Ki-67 is near 100%. The key gene is:

A. BCL2
B. CCND1
C. ALK
D. BCL6
E. MYC

6 In follicular lymphoma, t(14;18) causes overexpression of a protein that:

A. Inhibits apoptosis
B. Drives aerobic glycolysis
C. Activates NF-κB
D. Pushes cells from G1 to S
E. Is a constitutive tyrosine kinase

7 A 65-year-old man has generalised lymphadenopathy and multiple colonic polyps. The lymphoma cells are CD5+, CD23− and cyclin D1+. The diagnosis is:

A. CLL/SLL
B. Follicular lymphoma
C. Mantle cell lymphoma
D. MALT lymphoma
E. Burkitt lymphoma

8 An early, localised gastric MALT lymphoma without t(11;18) is best treated first with:

A. Total gastrectomy
B. Eradication of Helicobacter pylori
C. High-dose chemotherapy
D. Radiotherapy alone
E. Imatinib

9 Bence Jones protein in the urine of a patient with myeloma consists of:

A. Intact IgG
B. IgM pentamers
C. Free immunoglobulin light chains
D. β2-microglobulin
E. Albumin

10 A 70-year-old has an IgM M protein, blurred vision with engorged retinal veins and no lytic bone lesions. The diagnosis is:

A. Multiple myeloma
B. MGUS
C. Solitary plasmacytoma
D. Heavy-chain disease
E. Waldenström macroglobulinaemia

11 A 60-year-old has an M protein of 1.5 g/dL, 6% marrow plasma cells, normal calcium, renal function and haemoglobin, and no bone lesions. The best label is:

A. Smouldering myeloma
B. Plasma cell leukaemia
C. Multiple myeloma
D. MGUS
E. Waldenström macroglobulinaemia

12 A Japanese man has skin lesions, lymphadenopathy, hypercalcaemia and circulating CD4+ cells with flower-shaped nuclei. The causative agent is:

A. EBV
B. HHV-8
C. HIV
D. HTLV-1
E. Hepatitis C virus

ANSWER KEY WITH ONE-LINE REASONS

1. **C** Lacunar cells of nodular sclerosis HL have the classical RS phenotype.
2. **A** LP cells keep the B-cell programme: CD20+, BCL6+, CD15−, CD30−.
3. **B** EBV is present in about 70% of mixed cellularity HL (the rare lymphocyte-depleted type is higher still).
4. **D** Nodes on both sides of the diaphragm with splenic involvement = IIIS.
5. **E** Starry-sky Burkitt lymphoma: MYC translocation, usually t(8;14).
6. **A** BCL2 blocks the mitochondrial pathway of apoptosis.
7. **C** CD5+, CD23−, cyclin D1+ with lymphomatoid polyposis = mantle cell lymphoma.
8. **B** Antigen-dependent clones often regress when H. pylori is killed.
9. **C** Free κ or λ light chains filtered into urine.
10. **E** IgM → hyperviscosity; lymphoplasmacytic lymphoma lacks lytic lesions.
11. **D** M protein < 3 g/dL, plasma cells < 10%, no CRAB = MGUS.
12. **D** Adult T-cell leukaemia/lymphoma is caused by HTLV-1.

VIVA Rapid-Fire Viva Voce

Q What is a Reed–Sternberg cell?	A large binucleate or bilobed cell with owl-eye eosinophilic nucleoli; CD15+, CD30+, CD45−.
Q What is a lacunar cell?	An RS variant in a clear space from cytoplasmic retraction; nodular sclerosis HL.
Q Commonest subtype of Hodgkin lymphoma?	Nodular sclerosis.
Q What are B symptoms?	Fever, drenching night sweats and loss of over 10% body weight in 6 months.
Q How does Hodgkin lymphoma spread?	Predictably, to anatomically contiguous node groups, then spleen, liver and marrow.
Q Commonest non-Hodgkin lymphoma?	Diffuse large B-cell lymphoma.
Q What is the starry-sky pattern?	Tingible-body macrophages scattered among sheets of tumour cells in Burkitt lymphoma.
Q Which marker separates follicular lymphoma from reactive follicles?	BCL2: positive in neoplastic follicles, negative in reactive germinal centres.
Q Name a lymphoma that can regress with antibiotics.	Gastric MALT lymphoma, after eradication of H. pylori.
Q What does CRAB stand for?	HyperCalcaemia, Renal failure, Anaemia, Bone lesions: end-organ damage in myeloma.
Q What are Russell and Dutcher bodies?	Immunoglobulin inclusions in plasma cells: cytoplasmic (Russell) and intranuclear (Dutcher).
Q What are rouleaux?	Red cells stacked like coins because M protein reduces their surface charge.
Q Which IgM disorder causes hyperviscosity?	Waldenström macroglobulinaemia (lymphoplasmacytic lymphoma).

OSPE Practice Stations

STATION 1 · SLIDE — 2 minutes

TASK

Cervical lymph node from a 22-year-old: architecture effaced; broad bands of collagen divide the node into nodules; scattered large cells with folded nuclei sit in clear spaces, among lymphocytes, eosinophils and histiocytes. (a) Identify. (b) Name the large cells and two markers they express. (c) Which mediastinal finding often accompanies it?

ANSWER

(a) Classical Hodgkin lymphoma, nodular sclerosis type. (b) Lacunar cells (RS variant): CD15+, CD30+ (CD45−). (c) A mediastinal mass.

STATION 2 · SLIDE — 2 minutes

TASK

Ileocaecal mass in a 10-year-old: diffuse sheets of uniform medium-sized cells with round nuclei and many mitoses; scattered pale macrophages containing nuclear debris give a speckled look. (a) Diagnosis? (b) Name the pattern. (c) Translocation and gene.

ANSWER

(a) Burkitt lymphoma (sporadic). (b) Starry-sky pattern (tingible-body macrophages). (c) t(8;14), MYC.

STATION 3 · LAB DATA 2 minutes

TASK

A 68-year-old: Hb 8.4 g/dL, ESR 120 mm/h, total protein 10.2 g/dL with albumin 3.1 g/dL, calcium 11.8 mg/dL, creatinine 2.4 mg/dL, ALP normal. Smear: red cells in stacks. Serum electrophoresis: narrow band in the γ region. (a) Diagnosis? (b) Name the smear finding and the band. (c) Two further tests.

ANSWER

(a) Multiple myeloma. (b) Rouleaux; M spike (monoclonal Ig). (c) Urine Bence Jones protein / serum free light chains; bone marrow for clonal plasma cells; skeletal survey.

RAPID REVISION The Night-Before Page

15 ONE-LINERS

01 Reactive nodes keep architecture; lymphoma effaces it with a clone.

02 Paracortical hyperplasia = viral (EBV); sinus histiocytosis = draining cancer.

03 Lymphoma needs an excision biopsy, not just FNA.

04 RS cell: owl-eye nucleoli, CD15+, CD30+, CD45−.

05 Nodular sclerosis: commonest HL, young women and men, mediastinum, lacunar cells.

06 Mixed cellularity: older men, EBV, many RS cells.

07 NLPHL: popcorn cells, CD20+, CD15/30 negative.

08 HL spreads contiguously; NHL jumps and goes extranodal.

09 Ann Arbor: II one side, III both sides, IV extralymphatic organs.

10 Follicular: t(14;18) BCL2, centrocytes, BCL2+ follicles, indolent.

11 DLBCL: commonest NHL; BCL6; R-CHOP cures about half.

12 Burkitt: t(8;14) MYC, starry sky, jaw (endemic) or ileocaecum.

13 Mantle cell: t(11;14) cyclin D1, CD5+ CD23−, gut polyps.

14 MALT: H. pylori, Sjögren, Hashimoto; lymphoepithelial lesions.

15 Myeloma: CRAB, M spike, Bence Jones, rouleaux, normal ALP.

MEMORY HOOKS

- **IGH lives on 14**: 8 MYC (Burkitt), 11 cyclin D1 (Mantle), 18 BCL2 (Follicular)
- **CRAB**: Calcium, Renal, Anaemia, Bone
- **Slow is not always good**: indolent lymphomas are incurable; aggressive ones often curable

TEST YOURSELF: FILL THE GAPS

1. RS cells are CD15+ and CD_____+.
2. Commonest HL subtype: _____.
3. Burkitt gene: _____.
4. Mantle cell lymphoma overexpresses _____.
5. Waldenström M protein class: _____.
6. MGUS → myeloma at about _____% per year.

Answers: 30 · nodular sclerosis · MYC · cyclin D1 · IgM · 1

DRAW FROM MEMORY

A classic Reed–Sternberg cell and a lacunar cell, with their markers.

DRAW FROM MEMORY

The myeloma CRAB flowchart (Figure 14.5).

SPACED REVISION ☐ Day 1 ☐ Day 3 ☐ Day 7 ☐ Day 14 ☐ Day 30

15

Bleeding Disorders

Ask two questions of any patient who bleeds: where does the blood come out, and what do four simple tests show? Oozing into skin and mucosa points to vessels or platelets; bleeding into joints and muscles points to clotting factors. The platelet count, bleeding time, PT and aPTT then name the culprit.

BY THE END OF THIS CHAPTER YOU CAN

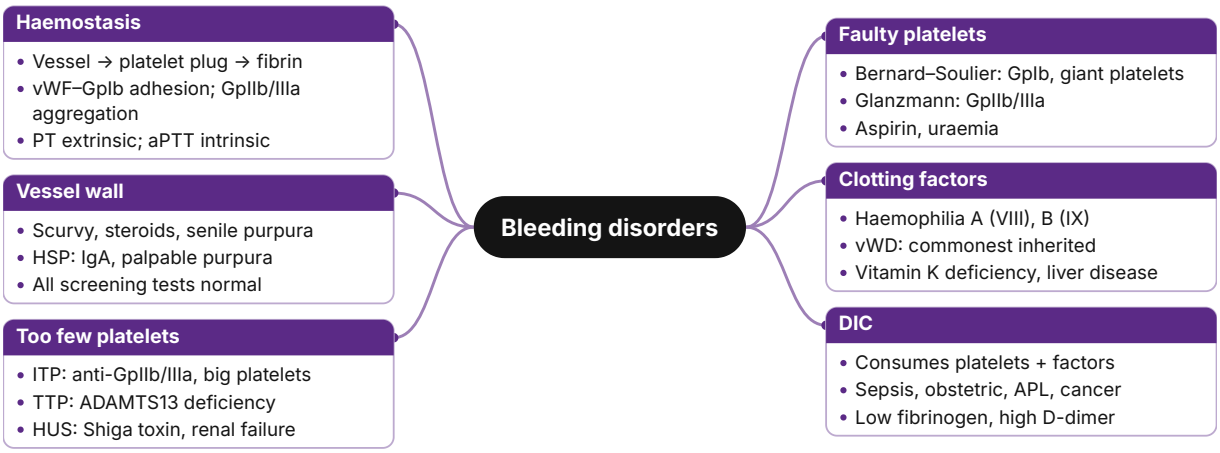
1. Recap primary and secondary haemostasis and link each step to a laboratory test
2. Interpret the platelet count, bleeding time, PT, aPTT and thrombin time in a bleeding patient
3. Describe the vascular purpuras and the causes of thrombocytopenia, including ITP, TTP and HUS
4. Differentiate Bernard–Soulier syndrome, Glanzmann thrombasthenia and von Willebrand disease
5. Describe haemophilia A and B, vitamin K deficiency and the coagulopathy of liver disease
6. Explain the causes, pathogenesis, morphology and laboratory diagnosis of DIC

EXAM RADAR

Theory SEQs	★★★
BCQs / MCQs	★★★
Viva voce	★★★
OSPE / slides	★★★

DIC, ITP and haemophilia are repeated long questions; interpreting a coagulation profile is a standard OSPE data station.

ROBBINS LINK Robbins Basic Pathology — *Hematopoietic and Lymphoid Systems*; PBD — *Hemodynamic Disorders, Thromboembolic Disease, and Shock* (normal haemostasis) and *Red Blood Cell and Bleeding Disorders*: vessel-wall, platelet and coagulation-factor disorders, thrombotic microangiopathies, DIC.



Chapter at a glance. Every bleeding disorder fails at one of three levels: vessel, platelet or clotting factor.

15.1 Normal haemostasis: a recap

Haemostasis keeps blood fluid inside normal vessels yet seals a breach within minutes. Three players act in sequence: the **vessel wall**, the **platelets** (primary haemostasis: the plug) and the **coagulation cascade** (secondary haemostasis: fibrin). Every bleeding disorder fails at one of these steps, and every screening test examines one of them.



Figure 15.1 · The sequence of haemostasis. Steps 2–4 are primary haemostasis; step 5 is secondary haemostasis.

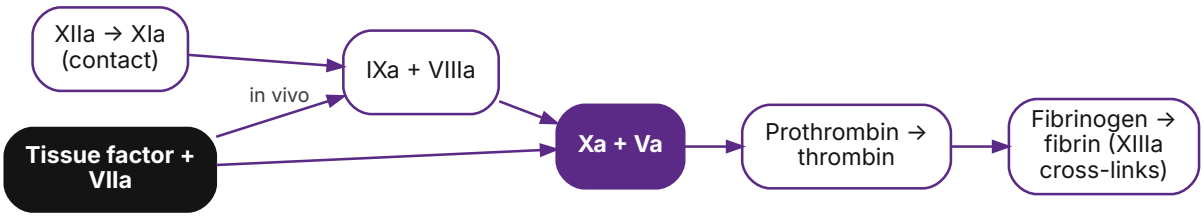


Figure 15.2 · The coagulation cascade as the tests see it. aPTT reads the intrinsic limb (XII, XI, IX, VIII), PT the tissue-factor limb (VII); both read the common pathway (X, V, II, fibrinogen).

WHAT EACH TEST MEASURES (NORMAL VALUES VARY BY LABORATORY: COMPARE WITH THE CONTROL)

- **Platelet count** (150,000–450,000/μL): spontaneous bleeding below about 20,000; post-traumatic bleeding at 20,000–50,000
- **Bleeding time** (2–9 min): the platelet plug: platelet number and function, vWF, vessel wall
- **PT** (about 11–14 s; INR about 1): **VII** + common pathway (X, V, II, fibrinogen); monitors warfarin
- **aPTT** (about 25–40 s): XII, XI, **IX, VIII** + common pathway; monitors heparin
- **Thrombin time:** fibrinogen → fibrin only; long with low fibrinogen (normal 200–400 mg/dL), heparin, FDPs
- **Mixing study:** corrects with normal plasma → factor deficiency; fails → inhibitor. **D-dimer** rises in DIC and thrombosis

MEMORY HOOK

Play Tennis outside, Play Table Tennis inside: PT = extrinsic, PTT = intrinsic.

"1972": vitamin K-dependent factors **X, IX, VII, II** (plus proteins C and S).

CONCEPT CLEAR

Why do haemophiliacs bleed if TF–VIIa starts clotting? In the body TF–VIIa makes only a small "spark" of thrombin before it is shut off. The fire is kept alight by TF–VIIa activating **factor IX** and by the **IXa–VIIIa** complex making Xa on platelets. Without VIII or IX the spark never becomes a fire, though the test-tube PT stays normal.

15.2 Approach to a bleeding patient

THE BLEEDING PATTERN POINTS TO THE LEVEL OF THE DEFECT

Feature	Platelet or vessel-wall defect	Coagulation factor defect
Site	**Skin and mucosa**: epistaxis, gums, menorrhagia, GI tract	**Deep tissues**: joints (haemarthrosis), muscles
Skin lesions	**Petechiae**, purpura, small bruises	Large spreading ecchymoses; petechiae rare
After a cut or extraction	Bleeds **at once**, persistently	Stops, then bleeds again **hours later**
Examples	ITP, vWD, aspirin, scurvy	Haemophilia A and B, vitamin K deficiency

INTERPRETING THE SCREENING TESTS: LEARN THIS TABLE AND EVERY DATA STATION IS EASY

Disorder	Platelets	BT	PT	aPTT	Clincher
Vascular purpura (scurvy, HSP)	N	N	N	N	All normal: diagnosis is clinical
ITP	↓	↑	N	N	Large platelets; marrow megakaryocytes ↑
TTP / HUS	↓	↑	N	N	**Schistocytes**, high LDH
Bernard–Soulier	Mild ↓	↑	N	N	**Giant platelets**; no ristocetin response
Glanzmann; aspirin	N	↑	N	N	Poor aggregation with ADP, collagen
von Willebrand disease	N	↑	N	N or ↑	Low vWF and ristocetin cofactor
Haemophilia A or B	N	N	N	↑	Low VIII or IX; mixing corrects
Factor VII deficiency; early warfarin	N	N	↑	N	VII has the shortest half-life
Vitamin K deficiency; warfarin	N	N	↑↑	↑	Corrects with vitamin K
Liver disease	N or ↓	N or ↑	↑↑	↑	No correction with vitamin K; VIII normal

Disorder	Platelets	BT	PT	aPTT	Clincher
Heparin	N	N	N	↑↑	Thrombin time prolonged
DIC	↓	↑	↑	↑	Low fibrinogen, **high D-dimer**, schistocytes

15.3 Vascular (non-thrombocytopenic) purpuras

Common and usually not serious. The vessel wall or its supporting tissue is weak or inflamed, so blood leaks into skin and mucosa as **petechiae and ecchymoses**, yet the **platelet count and clotting tests are normal**.

- **Infections** (meningococcaemia, infective endocarditis, rickettsiae) and **drug reactions**: vasculitis
- **Scurvy:** faulty collagen hydroxylation → fragile capillaries; **perifollicular haemorrhages**, bleeding gums
- **Ehlers–Danlos syndrome** (defective collagen); **Cushing syndrome**, steroids and **senile purpura** (lost perivascular support)
- **Henoch–Schönlein purpura** (IgA vasculitis): **palpable purpura on buttocks and extensor legs**, abdominal pain, arthralgia, IgA nephritis (Chapters 16, 25)
- **Hereditary haemorrhagic telangiectasia**: autosomal dominant (*ENG*, *ACVRL1*); thin dilated vessels → epistaxis, GI bleeding
- **Amyloidosis:** perivascular amyloid → fragile vessels (periorbital purpura)

> ⚠ EXAM TRAP
>
> **Palpable vs flat purpura. Palpable** = an inflamed vessel (vasculitis: HSP, drug reaction). **Flat** with normal tests = a fragile vessel (scurvy, steroids, old age). Flat petechiae with a **low platelet count** = thrombocytopenia.

15.4 Thrombocytopenia

A platelet count below **150,000/µL**, with bleeding from small vessels: petechiae, epistaxis, gum and GI bleeding and, most feared, **intracranial haemorrhage**. Causes: **decreased production** (aplastic anaemia, leukaemia, marrow metastases, drugs, alcohol, megaloblastic anaemia, HIV); **immune destruction** (ITP, drugs such as quinine, sulfonamides and heparin, SLE, CLL, post-transfusion); **non-immune consumption** (DIC, TTP/HUS, prosthetic valves); **sequestration** (hypersplenism); and **dilution** (massive transfusion).

Immune thrombocytopenic purpura (ITP)

> DEFINITION
>
> **ITP** is thrombocytopenia caused by **autoantibodies against platelet membrane glycoproteins**, most often **GpIIb/IIIa** or GpIb–IX; splenic macrophages remove the coated platelets. It is primary, or secondary to SLE, HIV or B-cell neoplasms such as CLL.

- The **spleen** both makes the antibodies and destroys the platelets, so **splenectomy** helps most patients with chronic ITP
- **Spleen:** normal size; congested sinusoids, prominent germinal centres
- **Marrow: increased megakaryocytes**, many large and immature
- **Smear:** few but **large platelets**; red and white cells normal
- **Splenomegaly or lymphadenopathy** are not features: look for another cause

Feature	Acute ITP	Chronic ITP
Who	**Children**, both sexes	**Adult women** under 40 (F:M about 3:1)
Onset	Abrupt, about 2 weeks after a **viral illness**	Insidious
Course	**Self-limited** within 6 months; about a fifth become chronic	Relapsing, persistent
Treatment	Often none; steroids if severe	Steroids, splenectomy, rituximab, TPO mimetics

> ⚠ EXAM TRAP
> **Heparin-induced thrombocytopenia clots, it does not bleed.** About 1–2 weeks after starting heparin, IgG antibodies bind **platelet factor 4–heparin complexes** and activate platelets: the count falls, but the danger is **venous and arterial thrombosis**. Stop all heparin and use a non-heparin anticoagulant.

Thrombotic microangiopathies: TTP and HUS

Both give **thrombocytopenia + microangiopathic haemolytic anaemia (MAHA)** from **platelet-rich thrombi** in small vessels, which slice red cells into **schistocytes**. The clotting cascade is **not** activated, so **PT and aPTT are normal**: the key difference from DIC.

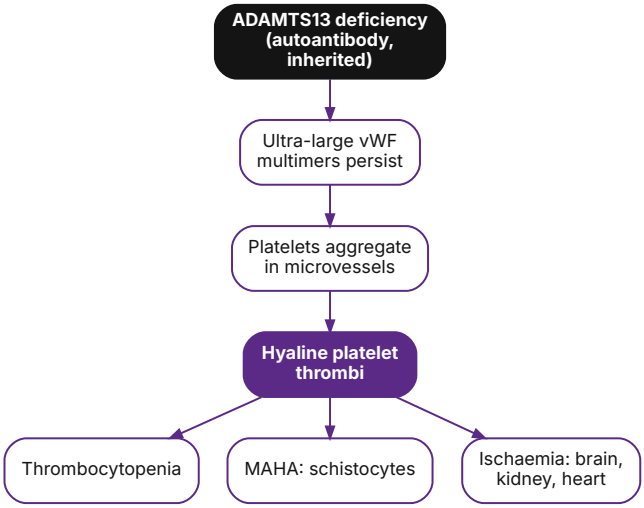


Figure 15.3 · Pathogenesis of TTP.

TTP: the classic pentad of **fever, thrombocytopenia, MAHA, transient neurological deficits and renal failure** (all five are uncommon together). Usually adult women with an **acquired autoantibody** against ADAMTS13, the protease that trims vWF multimers; rarely inherited. Fatal if untreated; **plasma exchange** (removes antibody, supplies ADAMTS13) saves most patients.

HUS: acute renal failure dominates. **Typical**: children after **bloody diarrhoea** from *E. coli* O157:H7, whose Shiga-like toxin injures endothelium. **Atypical**: inherited defects of complement regulators (factor H, CD46, factor I), drugs, radiation. ADAMTS13 is normal.

15.5 Defects of platelet function

The count is normal or near normal, but the platelets fail, so the **bleeding time is prolonged**. Inherited defects hit **adhesion** (Bernard–Soulier), **aggregation** (Glanzmann) or **secretion** (storage-pool disorders). Acquired defects are far commoner: **aspirin** irreversibly blocks cyclo-oxygenase, so no TxA2 is made for the platelet's whole life (7–10 days); **uraemia** impairs adhesion, secretion and aggregation.

BERNARD–SOULIER VS GLANZMANN VS VON WILLEBRAND DISEASE

Feature	Bernard–Soulier syndrome	Glanzmann thrombasthenia	von Willebrand disease
Inheritance	Autosomal recessive	Autosomal recessive	Mostly autosomal dominant
Missing molecule	**GpIb–IX** (vWF receptor)	**GpIIb/IIIa** (fibrinogen receptor)	**vWF** (the ligand)
Step that fails	Adhesion	Aggregation	Adhesion (+ low VIII)
Platelet count, size	Mildly low; **giant platelets**	Normal	Normal
Aggregation with ADP, collagen	Normal	**Absent**	Normal
Ristocetin agglutination	**Absent**; not corrected by normal plasma	Normal	**Reduced**; corrected by normal plasma

PAKISTAN FOCUS

Cousin marriage is common in Pakistan, which raises the frequency of **autosomal recessive** disease. Rare recessive bleeding disorders (**Glanzmann thrombasthenia**, **Bernard–Soulier syndrome**, **type 3 vWD**, recessive deficiencies of factors VII, X and XIII) are therefore seen more often than where such marriages are rare. Ask about consanguinity and affected siblings in every child who bleeds.

15.6 Coagulation factor disorders

Haemophilia A and B

Haemophilia A (factor VIII deficiency) is the commonest hereditary cause of **serious bleeding**. It is **X-linked recessive**, so males are affected and carrier females are usually well; about 30% have no family history (new mutations), and an **intron 22 inversion** causes many severe cases. Severity follows the factor level: **below 1% severe**, 2–5% moderate, 6–50% mild. **Haemophilia B** (Christmas disease, factor IX) is also X-linked and clinically identical; only a factor assay separates them.

- Easy bruising; **massive bleeding after trauma or surgery** (circumcision, tooth extraction)
- **Spontaneous haemarthroses** (knees, ankles, elbows) → crippling joint deformity; muscle haematomas
- **Petechiae are characteristically absent**
- **aPTT prolonged**; PT, BT and platelets normal; low factor assay; mixing corrects
- Factor VIII concentrate; about 15% of severe cases develop **inhibitory antibodies**

von Willebrand disease (vWD)

The **commonest inherited bleeding disorder** (about 1% in some surveys, mostly mild). von Willebrand factor, a large multimer made by endothelium and megakaryocytes, **glues platelet GpIb to subendothelial collagen** and **carries and stabilises factor VIII**. So vWD gives **mucosal bleeding** (epistaxis, menorrhagia, bleeding after dental work) with a long BT, plus a long aPTT when VIII falls. **Type 1** (about 70%): dominant, partial deficiency, mild. **Type 2** (about 25%): qualitative (2A loses large multimers). **Type 3**: recessive, near-absent vWF, haemophilia-like. Labs: low vWF antigen and ristocetin cofactor activity. **Desmopressin** releases stored vWF from endothelial Weibel–Palade bodies (type 1).

HAEMOPHILIA A VS VON WILLEBRAND DISEASE

Feature	Haemophilia A	von Willebrand disease
Inheritance	X-linked recessive; males	Autosomal dominant (most); both sexes
Bleeding pattern	**Deep**: haemarthroses, muscle	**Mucocutaneous**: epistaxis, menorrhagia
Bleeding time	Normal	**Prolonged**
aPTT	**Prolonged**	Normal or prolonged
Factor VIII; vWF	VIII low; vWF normal	VIII normal or low; **vWF and ristocetin cofactor low**

Vitamin K deficiency and liver disease

Vitamin K is the cofactor for **γ-carboxylation** of factors **II, VII, IX and X** (and proteins C and S), which lets them bind calcium and phospholipid; warfarin blocks its recycling. **PT rises first** (factor VII has the shortest half-life), then aPTT; platelets are normal; injected vitamin K corrects it. Causes: **newborns** (poor placental transfer, sterile gut, little in breast milk → vitamin K deficiency bleeding, prevented by vitamin K at birth); **fat malabsorption** (obstructive jaundice, coeliac disease); **broad-spectrum antibiotics** with poor intake; **warfarin** or rat-poison overdose.

In **liver disease** almost every factor falls (VIII, partly endothelial, does not), so **PT/INR and aPTT are prolonged**; the INR is a sensitive marker of synthetic function. Platelets fall from **hypersplenism** and alcohol, and activated factors are cleared poorly, so a DIC-like picture can develop.

CONCEPT CLEAR

The vitamin K test separates two jaundiced patients. In **obstructive jaundice** the liver is healthy but cannot absorb vitamin K: inject it and the PT corrects within a day. In **cirrhosis** the liver cannot make the factors: the PT stays long. **Factor VIII** then separates liver disease (normal or high) from DIC (low).

15.7 Disseminated intravascular coagulation (DIC)

DEFINITION

DIC is an acquired thrombohaemorrhagic disorder, always **secondary** to another disease, in which widespread activation of coagulation forms **microthrombi** throughout the microcirculation. Platelets and clotting factors are **consumed** and fibrinolysis is activated, so the patient both **clots and bleeds** ("consumption coagulopathy").

TWO TRIGGERS

- **Tissue factor enters the blood:** obstetric accidents (**abruptio placentae**, retained dead fetus, septic abortion, **amniotic fluid embolism**, eclampsia); **acute promyelocytic leukaemia**; mucin-secreting adenocarcinoma (pancreas, lung, stomach, prostate); trauma, burns, surgery; snake venom
- **Widespread endothelial injury: sepsis** (gram-negative endotoxin, **meningococcaemia**), immune-complex vasculitis, heat stroke, shock

MEMORY HOOK

STOMP out clotting: Sepsis · **T**rauma, burns · **O**bstetric accidents · **M**alignancy (APL, adenocarcinoma) · **P**oisons (snake venom)

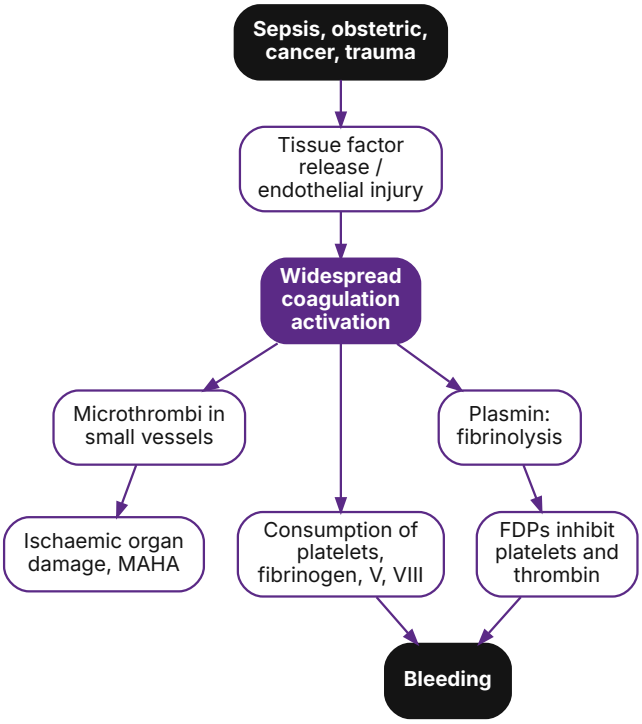


Figure 15.4 · Pathogenesis of DIC. One process, two opposite results.

GROSS

- **Petechiae, ecchymoses**; oozing from venepuncture and surgical sites
- **Kidneys:** bilateral renal cortical necrosis
- **Adrenals:** massive haemorrhage in meningococcaemia (**Waterhouse–Friderichsen**)
- **Pituitary:** postpartum necrosis (Sheehan syndrome)

MICROSCOPY

- **Fibrin microthrombi** in arterioles and capillaries: glomeruli, brain, lungs, heart, adrenals, placenta
- Microinfarcts and haemorrhages around them
- Smear: **schistocytes**, few platelets

Clinical: acute DIC (obstetric, sepsis, trauma) is dominated by **bleeding**, with shock, oliguria, dyspnoea, coma; **chronic DIC** (cancer) by **thrombosis**. **Labs:** platelets ↓; PT, aPTT, TT ↑; **fibrinogen ↓**; **D-dimer ↑↑**; schistocytes. **Treatment:** remove the trigger (deliver the placenta, antibiotics, ATRA in APL); give platelets, fresh frozen plasma, cryoprecipitate.

DIC VS TTP/HUS VS LIVER DISEASE: THREE CAUSES OF A LOW PLATELET COUNT WITH ABNORMAL TESTS

Feature	DIC	TTP / HUS	Liver disease
Basic defect	Coagulation activated and consumed	Platelet thrombi (ADAMTS13 loss, endothelial injury)	Factors not made; spleen pools platelets
PT, aPTT	**Prolonged**	**Normal**	Prolonged

Feature	DIC	TTP / HUS	Liver disease
Fibrinogen; D-dimer	**Low; very high**	Normal; normal or mildly raised	Normal or low; mildly raised
Factor VIII	Low	Normal	**Normal or high**
Schistocytes	Present	**Prominent**	Absent

EXAM ZONE Past-Paper Pattern SEQs — Solved

PROF PATTERN **a)** Define disseminated intravascular coagulation and list four causes. **b)** Describe its pathogenesis and laboratory findings. **2 + 3 marks**

MODEL ANSWER PLAN

a) Secondary thrombohaemorrhagic disorder: widespread microthrombi, consumption of platelets and factors, secondary fibrinolysis. Causes: abruptio placentae/amniotic fluid embolism, gram-negative sepsis/meningococcaemia, APL and mucinous adenocarcinoma, trauma/burns. **b)** Two triggers (tissue factor release; endothelial injury) → thrombin generation → microthrombi (ischaemia, MAHA) + consumption and plasmin/FDPs (bleeding). Labs: low platelets, long PT/aPTT/TT, low fibrinogen, high D-dimer, schistocytes.

Score booster: draw Figure 15.4 and mention one organ lesion (renal cortical necrosis or Waterhouse–Friderichsen).

MODULE PATTERN A 6-year-old girl develops petechiae and nosebleeds two weeks after a viral fever. She is otherwise well; there is no splenomegaly. Hb 12 g/dL, WBC 7,500/µL, platelets 14,000/µL; PT and aPTT normal. **a)** Most likely diagnosis? **b)** Explain the pathogenesis and the expected marrow findings. **c)** Differentiate it from the chronic form. **1 + 2 + 2 marks**

MODEL ANSWER PLAN

a) Acute immune thrombocytopenic purpura. **b)** Antibodies (often after viral infection) coat platelet GpIIb/IIIa or GpIb–IX → splenic macrophages remove them; marrow shows increased, large megakaryocytes; smear shows large platelets. **c)** Acute: children, abrupt, post-viral, self-limited within 6 months; chronic: adult women, insidious, relapsing, needs steroids/splenectomy; secondary causes (SLE, HIV, CLL).

PROF PATTERN **a)** Differentiate between haemophilia A and von Willebrand disease. **b)** A 9-year-old boy has a swollen, painful knee after a minor fall; his maternal uncle had a similar problem. Give the expected screening tests and explain why his joint bleeds. **2 + 3 marks**

MODEL ANSWER PLAN

a) Table: inheritance, sex, bleeding pattern, BT, aPTT, VIII, vWF/ristocetin. **b)** Haemophilia A: aPTT prolonged; PT, BT, platelets normal; factor VIII low; mixing study corrects. Platelet plug forms, but without IXa–VIIIa there is no sustained thrombin and fibrin, so the plug is unstable and deep bleeding (joints, muscles) recurs hours after trauma.

BCQs One-Best-Answer Practice

1 Adhesion of platelets to exposed subendothelial collagen depends mainly on von Willebrand factor binding to:

A. GpIIb/IIIa
B. GpIb–IX
C. The ADP (P2Y12) receptor
D. Fibrinogen
E. The thromboxane receptor

2 A 7-year-old has petechiae and epistaxis two weeks after a viral illness. Platelets are 12,000/µL; haemoglobin and WBC are normal; there is no splenomegaly. The marrow would most likely show:

A. Reduced megakaryocytes
B. Increased megakaryocytes
C. Reticulin fibrosis
D. Sheets of blasts
E. Ring sideroblasts

3 A 32-year-old woman has fever, confusion, schistocytes, platelets of 18,000/µL and a raised creatinine. PT and aPTT are normal. The deficient enzyme is:

A. ADAMTS13
B. Factor VIII
C. Antithrombin
D. Protein C
E. Plasminogen

4 A 10-year-old boy has recurrent knee haemarthroses; his maternal uncle was similarly affected. aPTT is prolonged; PT, bleeding time and platelet count are normal. The diagnosis is:

A. von Willebrand disease type 1
B. Glanzmann thrombasthenia
C. Vitamin K deficiency
D. Haemophilia A
E. Immune thrombocytopenia

5 A child's platelets are normal in number and size and fail to aggregate with ADP, collagen and adrenaline, but agglutinate normally with ristocetin. The missing molecule is:

A. von Willebrand factor
B. GpIb–IX
C. GpIIb/IIIa
D. Factor VIII
E. Cyclo-oxygenase

6 A boy has giant platelets, a mildly low count and absent ristocetin-induced agglutination that does not correct with normal plasma. The diagnosis is:

A. von Willebrand disease
B. Glanzmann thrombasthenia
C. Aspirin effect
D. Immune thrombocytopenia
E. Bernard–Soulier syndrome

7 The commonest inherited bleeding disorder is:

A. Haemophilia A
B. Haemophilia B
C. von Willebrand disease
D. Glanzmann thrombasthenia
E. Factor VII deficiency

8 Desmopressin helps in type 1 von Willebrand disease and mild haemophilia A because it:

A. Releases vWF (and factor VIII) from endothelial stores
B. Inhibits fibrinolysis
C. Supplies vitamin K
D. Stimulates megakaryocytes
E. Neutralises factor VIII inhibitors

9 An ICU patient on broad-spectrum antibiotics and nil by mouth starts bleeding. PT is markedly prolonged, aPTT mildly prolonged, platelets normal. The factor that falls first is:

A. Factor VIII
B. Fibrinogen
C. Factor XII
D. Factor VII
E. von Willebrand factor

10 Which finding best separates DIC from the coagulopathy of liver disease?

A. Prolonged PT
B. Low platelet count
C. Low factor VIII
D. Prolonged aPTT
E. Raised bilirubin

11 After abruptio placentae a woman oozes from venepuncture sites. Platelets 40,000/µL, PT and aPTT prolonged, fibrinogen low, D-dimer very high. The initiating mechanism is:

A. Antibodies to GpIIb/IIIa
B. Release of tissue factor into the circulation
C. Vitamin K deficiency
D. ADAMTS13 deficiency
E. An acquired factor VIII inhibitor

12 On day 8 of unfractionated heparin, a patient's platelets fall from 250,000 to 90,000/µL and a new leg DVT appears. The mechanism is:

A. Direct toxicity of heparin to platelets
B. Splenic sequestration
C. Marrow suppression
D. Vitamin K antagonism
E. Antibodies to platelet factor 4–heparin complexes

ANSWER KEY WITH ONE-LINE REASONS

1. **B** vWF bridges collagen to platelet GpIb–IX; GpIIb/IIIa mediates aggregation.
2. **B** In ITP, peripheral destruction drives megakaryocytic hyperplasia.
3. **A** TTP: loss of the vWF-cleaving protease ADAMTS13, usually from an autoantibody.
4. **D** X-linked deep bleeding with isolated long aPTT = factor VIII (or IX) deficiency.
5. **C** Glanzmann thrombasthenia: no fibrinogen receptor, so no aggregation.
6. **E** Bernard–Soulier: the GpIb–IX receptor itself is missing.
7. **C** vWD affects about 1% in some surveys, mostly mildly.
8. **A** It releases vWF from Weibel–Palade bodies; vWF also stabilises factor VIII.
9. **D** Vitamin K deficiency: factor VII has the shortest half-life, so PT rises first.
10. **C** VIII is consumed in DIC but normal or high in liver disease.
11. **B** Placental tissue factor triggers DIC.
12. **E** HIT: activating antibodies cause thrombosis despite thrombocytopenia.

VIVA Rapid-Fire Viva Voce

Q What does the bleeding time test?	The platelet plug: platelet number and function, vWF and the vessel wall.
Q What does the PT measure?	Extrinsic and common pathways: factors VII, X, V, II and fibrinogen.
Q What does the aPTT measure?	Intrinsic and common pathways: XII, XI, IX, VIII, X, V, II and fibrinogen.
Q Name the vitamin K-dependent factors.	II, VII, IX and X, plus proteins C and S.
Q Target antigens in chronic ITP?	Platelet GpIIb/IIIa or GpIb–IX.
Q Pentad of TTP?	Fever, thrombocytopenia, MAHA, neurological deficits and renal failure.
Q Cause of typical HUS?	Shiga-like toxin of E. coli O157:H7 after bloody diarrhoea, mainly in children.
Q Defect in Glanzmann thrombasthenia?	Absent GpIIb/IIIa, so platelets cannot aggregate.
Q Two functions of von Willebrand factor?	Links platelet GpIb to collagen; carries and stabilises factor VIII.
Q Grading of haemophilia A?	Factor VIII below 1% severe, 2–5% moderate, 6–50% mild.
Q Laboratory picture of DIC?	Low platelets and fibrinogen; long PT, aPTT and TT; high D-dimer; schistocytes.
Q What is Waterhouse–Friderichsen syndrome?	Massive bilateral adrenal haemorrhage in meningococcal septicaemia with DIC.

OSPE Practice Stations

STATION 1 · COAGULATION DATA — 2 minutes

TASK

A woman bleeds heavily after delivery of a macerated stillborn fetus. Platelets 45,000/µL; PT 25 s (control 13); aPTT 58 s (control 32); fibrinogen 70 mg/dL; D-dimer markedly raised. Smear: fragmented red cells. (a) Diagnosis? (b) Name the red cells. (c) Explain the low fibrinogen.

ANSWER

(a) DIC due to retained dead fetus/obstetric accident. (b) Schistocytes (microangiopathic haemolysis). (c) Fibrinogen is consumed in microthrombi and degraded by plasmin (hence high D-dimer).

STATION 2 · COAGULATION DATA — 2 minutes

TASK

A 3-year-old boy bled for two days after circumcision. Platelets 280,000/µL; BT 4 min; PT 13 s (control 13); aPTT 78 s (control 32); aPTT corrects on mixing with normal plasma; factor VIII 2%. (a) Diagnosis and severity? (b) Inheritance? (c) Meaning of the mixing result.

ANSWER

(a) Haemophilia A, moderate (2–5%). (b) X-linked recessive. (c) Correction means a factor deficiency, not an inhibitor.

STATION 3 · BLOOD COUNT AND SMEAR 1 minute

TASK

A 30-year-old woman has petechiae and menorrhagia; no splenomegaly. Hb 11.8 g/dL, WBC 6,800/µL, platelets 22,000/µL; PT and aPTT normal. The smear shows few, large platelets and normal red and white cells. (a) Likely diagnosis? (b) Expected marrow finding. (c) Two secondary causes to exclude.

ANSWER

(a) Chronic immune thrombocytopenic purpura. (b) Increased (large, immature) megakaryocytes. (c) SLE, HIV (also CLL, drugs).

RAPID REVISION The Night-Before Page

15 ONE-LINERS

01 Mucosal bleeding and petechiae = platelets/vessels; joints and muscles = factors.

02 vWF–GpIb = adhesion; fibrinogen–GpIIb/IIIa = aggregation.

03 PT = VII + common; aPTT = XII, XI, IX, VIII + common.

04 Vascular purpura: all screening tests normal.

05 HSP: IgA, palpable purpura on buttocks, abdominal pain, nephritis.

06 ITP: anti-GpIIb/IIIa, large platelets, increased megakaryocytes, no splenomegaly.

07 Acute ITP: post-viral child, self-limited; chronic ITP: young women.

08 HIT: low platelets but thrombosis.

09 TTP: ADAMTS13 loss, pentad, normal PT/aPTT, plasma exchange.

10 HUS: E. coli O157:H7 toxin, child, renal failure.

11 Bernard–Soulier: GpIb, giant platelets; Glanzmann: GpIIb/IIIa.

12 Haemophilia A: X-linked, VIII, isolated long aPTT, haemarthroses.

13 vWD: commonest inherited; long BT ± aPTT; desmopressin.

14 Vitamin K deficiency: PT first; corrects with vitamin K; liver disease does not.

15 DIC: low platelets and fibrinogen, long PT/aPTT, high D-dimer.

MEMORY HOOKS

- **Play Tennis outside (PT, extrinsic), Table Tennis inside (PTT, intrinsic)**
- **"1972"**: vitamin K factors X, IX, VII, II
- **STOMP**: DIC triggers (Sepsis, Trauma, Obstetric, Malignancy, Poisons)
- **Big platelets, no adhesion**: Bernard–Soulier

TEST YOURSELF: FILL THE GAPS

1. TTP is caused by deficiency of _____.
2. Glanzmann lacks Gp_____.
3. Severe haemophilia: factor VIII below _____%.
4. Factor with the shortest half-life: _____.
5. Factor VIII in liver disease is _____
6. Meningococcal DIC bleeds into the _____.

Answers: ADAMTS13 · IIb/IIIa · 1 · VII · normal or high · adrenals

DRAW FROM MEMORY

The coagulation cascade, marking what PT and aPTT measure (Figure 15.2).

DRAW FROM MEMORY

The DIC flowchart from trigger to clotting and bleeding (Figure 15.4).

SPACED REVISION ☐ Day 1 ☐ Day 3 ☐ Day 7 ☐ Day 14 ☐ Day 30

16

Glomerular Diseases: Nephritic & Nephrotic Syndromes

The glomerulus tells you what is wrong with it through the urine. Red cells and casts mean it is inflamed; a flood of protein means the filter is leaking. Learn where the injury sits in the capillary wall and the syndrome follows.

BY THE END OF THIS CHAPTER YOU CAN

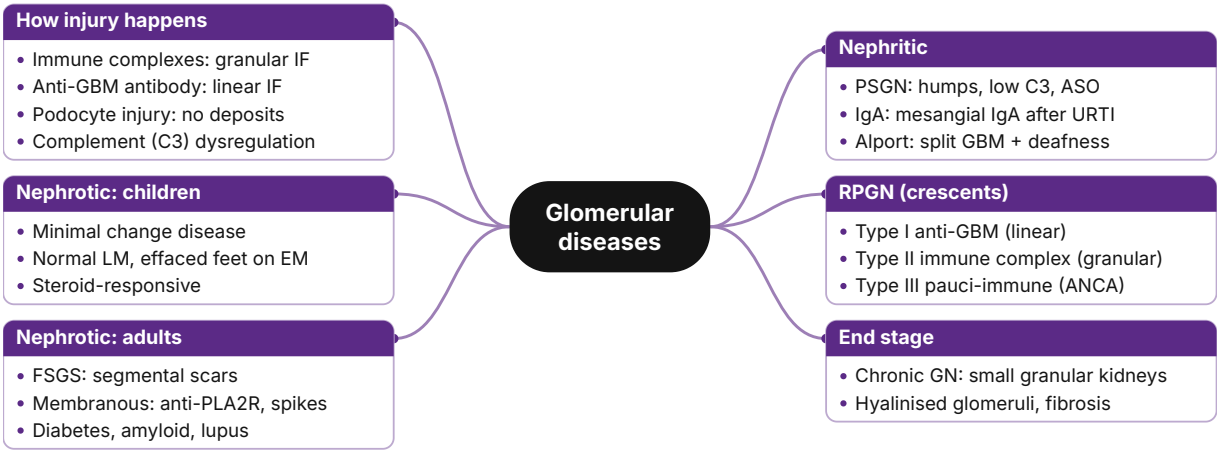
1. Explain how immune complexes, anti-GBM antibodies and podocyte injury damage the glomerulus
2. Differentiate nephritic, nephrotic and rapidly progressive syndromes at the bedside and in the lab
3. Describe the pathogenesis and morphology (LM, IF, EM) of minimal change disease, FSGS and membranous nephropathy
4. Describe post-streptococcal GN and IgA nephropathy as the classic nephritic diseases
5. Classify crescentic (RPGN) glomerulonephritis by immunofluorescence pattern
6. Recognise the end-stage kidney of chronic glomerulonephritis

EXAM RADAR

Theory SEQs	★★★
BCQs / MCQs	★★★
Viva voce	★★★
OSPE / slides	★★☆

Nephritic vs nephrotic, PSGN and minimal change disease are classic long questions; IF/EM patterns are favourite BCQs.

ROBBINS LINK Robbins Basic Pathology — *The Kidney and Its Collecting System*: glomerular diseases, nephrotic and nephritic syndromes, RPGN, chronic GN.



Chapter at a glance. Six ideas carry the whole chapter.

16.1 How glomeruli get injured

The glomerular filter has three layers: a **fenestrated endothelium**, the **glomerular basement membrane** (GBM: type IV collagen, laminin and negatively charged heparan sulfate) and the **podocytes**, whose interdigitating foot processes are bridged by the slit diaphragm (nephrin, podocin). The **mesangium** holds the capillary tuft together. Damage any layer and the filter fails, but **how** it fails depends on **which layer** is hit.

ANTIBODY-MEDIATED INJURY (MOST GN)

- **In situ immune complexes**: antibodies bind antigens already in the glomerulus. Intrinsic antigen: **PLA2R** on podocytes (membranous). Planted antigen: bacterial, DNA, drugs.
- **Circulating immune complexes** trap in the glomerulus: lupus, PSGN, IgA nephropathy, HBV/HCV.

NON-ANTIBODY INJURY

- **Podocyte injury** (toxins, cytokines, genetic: nephrin, podocin) → foot-process effacement → proteinuria. MCD and FSGS.
- **Complement dysregulation** of the alternative pathway → C3 glomerulopathy.
- **Nephron loss** → remaining glomeruli hyperfilter → secondary FSGS → progression of any chronic

- **Anti-GBM antibodies** against the α3 chain of type IV collagen: Goodpasture. disease.

> **CONCEPT CLEAR**
>
> **Where the deposit sits decides the syndrome.** Deposits on the **blood side** (subendothelial, mesangial) are in contact with circulating leukocytes. They draw in neutrophils and monocytes, the tuft becomes inflamed and hypercellular, and red cells leak: a **nephritic** picture. Deposits on the **urinary side** (subepithelial) are walled off from leukocytes by the GBM. Complement (C5b–9) attacks the podocytes quietly and the filter leaks protein: a **nephrotic** picture. PSGN is the exception that proves the rule: it has subepithelial humps but also intense complement activation and subendothelial and mesangial deposits, so it is nephritic.

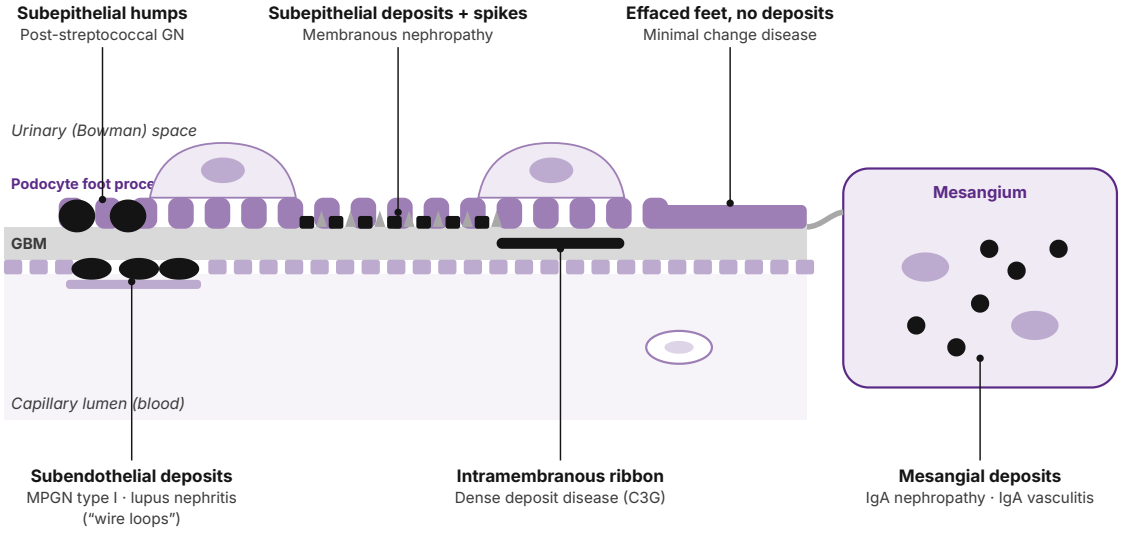


Figure 16.1 · Deposit map of the glomerular capillary wall. Learn this one picture and most EM questions answer themselves.

IMMUNOFLUORESCENCE (IF) PATTERNS: THE FIRST THING TO READ ON A RENAL BIOPSY

Pattern	What it means	Examples
Granular ("lumpy-bumpy")	Discrete immune-complex deposits	PSGN, membranous, lupus, MPGN type I, IgA (mesangial)
Linear	Antibody coating the whole GBM evenly	Anti-GBM disease / Goodpasture syndrome
Negative (pauci-immune)	Little or no immunoglobulin	MCD, ANCA-associated RPGN (GPA, MPA)
C3 only, no Ig	Alternative-pathway complement activation	C3 glomerulopathy (dense deposit disease, C3GN)

16.2 The five glomerular syndromes

Clinicians name a syndrome; pathologists name the disease behind it. The urine findings sort almost every patient into one of five boxes.

NEPHRITIC VS NEPHROTIC SYNDROME: THE MOST-ASKED COMPARISON IN RENAL PATHOLOGY

Feature	Nephritic syndrome	Nephrotic syndrome
Core lesion	Inflammation, proliferation, GBM rupture	Podocyte / GBM permeability defect
Proteinuria	Mild–moderate (< 3.5 g/day)	**> 3.5 g/day** (adult); > 40 mg/m²/h (child)
Haematuria	**Yes**: smoky/cola urine, **RBC casts**	Usually absent or mild
Serum albumin	Normal or slightly low	**< 3 g/dL**
Oedema	Mild, periorbital (salt and water retention)	**Generalised** (anasarca), pitting
Blood pressure	**Raised**	Usually normal

Feature	Nephritic syndrome	Nephrotic syndrome
Renal function	Oliguria, azotaemia	Usually preserved early
Lipids	Normal	Hyperlipidaemia, **lipiduria** (oval fat bodies, Maltese crosses)
Prototype	Post-streptococcal GN	Minimal change disease (child), membranous / FSGS (adult)

- **Rapidly progressive GN (RPGN):** nephritic syndrome with loss of renal function over **days to weeks**; crescents on biopsy.
- **Asymptomatic haematuria / proteinuria:** IgA nephropathy, thin basement membrane disease, early Alport.
- **Chronic kidney disease:** slowly rising creatinine, hypertension, small kidneys: the end point of most progressive GN.
- **Mixed nephritic–nephrotic:** MPGN, lupus nephritis (class IV).

LAB VALUES

Nephrotic range
proteinuria > 3.5 g/24 h (urine PCR > 300 mg/mmol)

Hypoalbuminaemia serum albumin < 3 g/dL

Selective proteinuria: mainly albumin (MCD)

Non-selective albumin + globulins (FSGS, membranous)

16.3 Nephrotic syndrome

All the features of nephrotic syndrome flow from one event: the filter lets **large amounts of plasma protein** through. Draw this chain in every answer on nephrotic syndrome; it earns the "pathogenesis" marks.

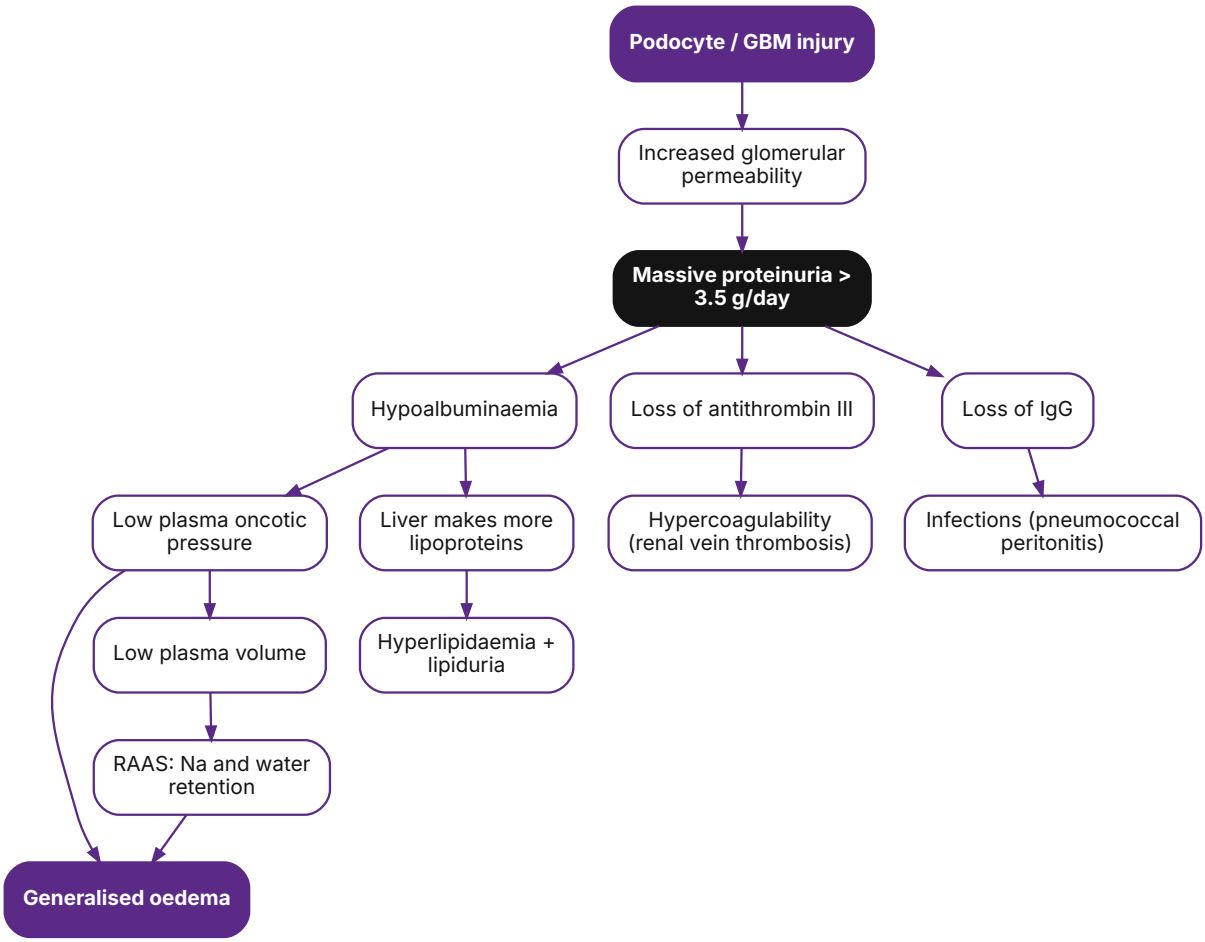


Figure 16.2 · Pathophysiology of nephrotic syndrome. One leak, five consequences.

CAUSES IN CHILDREN

- **Minimal change disease** (about two-thirds)
- FSGS
- MPGN, membranous (uncommon)

CAUSES IN ADULTS

- Primary: **FSGS** and **membranous** (each about one-third)
- Secondary: **diabetes mellitus** (commonest systemic cause), **amyloidosis**, **SLE**

Minimal change disease (lipoid nephrosis)

The commonest cause of nephrotic syndrome in **children (peak 2–6 years)**. Often follows a respiratory infection or immunisation. A circulating T-cell factor is thought to injure podocytes. The lesion is invisible on light microscopy, which gives the disease its name.

GROSS

- Kidneys normal or slightly swollen
- Cortex may look pale-yellow from lipid in tubules

MICROSCOPY

- **LM:** glomeruli **normal**
- PCT cells packed with **lipid droplets** (hence "lipoid nephrosis")
- **IF:** negative
- **EM:** diffuse **effacement of foot processes**; no deposits

Clinical: selective proteinuria, no hypertension, normal renal function. **> 90%** of children respond to corticosteroids; relapses are common but chronic renal failure is rare. In adults it can follow NSAIDs or accompany Hodgkin lymphoma.

Focal segmental glomerulosclerosis (FSGS)

Focal = some glomeruli; **segmental** = part of each affected tuft. Podocyte injury causes scarring that starts in juxtamedullary glomeruli, so a superficial biopsy can miss it and be mislabelled MCD.

TYPES

- **Primary** (idiopathic): circulating permeability factor; recurs after transplant
- **Secondary:** HIV (**collapsing variant**), heroin, sickle cell disease, obesity, reflux nephropathy, nephron loss
- **Genetic:** NPHS1 (nephrin), NPHS2 (podocin), APOL1 variants

VERSUS MCD

- Haematuria and **hypertension** common
- **Non-selective** proteinuria
- **Poor steroid response**
- About half progress to end-stage renal disease within 10 years

GROSS

- Normal early; small, scarred kidneys late

MICROSCOPY

- **LM:** segmental sclerosis with **hyalinosis** and lipid foam cells in some glomeruli
- **IF:** IgM and C3 trapped in scars (non-specific)
- **EM:** effacement of foot processes
- Collapsing variant: tuft collapses, podocytes proliferate; worst prognosis

Membranous nephropathy

A leading cause of nephrotic syndrome in **adults (30–60 years)**. It is a chronic **immune-complex** disease in which IgG and complement accumulate under the podocytes. The GBM lays down new matrix between the deposits, producing the famous spikes.

PRIMARY (ABOUT 75%)

- Autoantibodies to PLA2R (phospholipase A2 receptor) on podocytes in most cases; THSD7A in a few
- Serum anti-PLA2R is used for diagnosis and follow-up

SECONDARY: "DIMS"

- **D**rugs: penicillamine, captopril, gold, NSAIDs
- **I**nfections: HBV, HCV, syphilis, malaria
- **M**alignancy: carcinoma of lung/colon, melanoma
- **S**LE (class V), autoimmune thyroiditis

GROSS

- Enlarged, pale kidneys early
- Look for **renal vein thrombosis**, which is classically linked to membranous nephropathy

MICROSCOPY

- **LM:** diffuse uniform **thickening of capillary walls**, no hypercellularity
- **Silver stain: "spikes"** of GBM between deposits
- **IF:** granular IgG and C3 along GBM
- **EM: subepithelial deposits** ("spike and dome")

Course: unpredictable. About a third remit, a third persist, and roughly 40% progress to renal failure over years. Poorly steroid-responsive; immunosuppression is used for high-risk patients.

Membranoproliferative GN (MPGN) and C3 glomerulopathy

MPGN gives a **mixed nephritic–nephrotic** picture. The GBM is thickened (membrano-) and the tuft is hypercellular (proliferative). Mesangial cells grow into the capillary wall and lay down new basement membrane, so silver stain shows a **double contour ("tram-track")**.

Type	Mechanism	EM / IF	Associations
MPGN (immune complex; old type I)	Circulating immune complexes, classical pathway	**Subendothelial** deposits; granular IgG + C3	**HCV** (cryoglobulinaemia), HBV, SLE, chronic infections
Dense deposit disease (old type II)	**C3 nephritic factor** (autoantibody that stabilises C3 convertase) → persistent alternative-pathway activation	Ribbon-like **intramembranous** dense deposits; C3 only	Low serum C3; partial lipodystrophy

PAKISTAN FOCUS

Hepatitis C is a major cause of **membranoproliferative GN** through mixed cryoglobulinaemia. Pakistan carries one of the largest HCV burdens in the world: the national survey of 2007–08 found anti-HCV in about **4.8%** and HBsAg in about **2.5%** of the general population. In any Pakistani adult with proteinuria, haematuria and low C3, test for HCV and cryoglobulins. Diabetic kidney disease is another leading cause of end-stage renal disease in the country.

Source: PMRC national survey of hepatitis B and C, 2007–08.

Secondary nephrotic syndrome: three you must recognise

Disease	Key glomerular lesion
Diabetic nephropathy	GBM thickening, diffuse mesangial sclerosis, **nodular glomerulosclerosis (Kimmelstiel–Wilson nodules)**, hyaline arteriolosclerosis of **both afferent and efferent** arterioles (see Chapter 24)
Amyloidosis	Amorphous pink deposits in mesangium and capillary walls; **Congo red** gives **apple-green birefringence** under polarised light
Lupus nephritis	Six ISN/RPS classes; class IV (diffuse proliferative) is the commonest and most severe: subendothelial deposits make thick "**wire-loop**" capillaries; class V is membranous

16.4 Nephritic syndrome

Acute post-streptococcal glomerulonephritis (PSGN)

DEFINITION

PSGN is an immune-complex glomerulonephritis that follows infection with nephritogenic strains of **group A β-haemolytic streptococci**, typically **1–4 weeks** after pharyngitis or skin infection (impetigo), and presents as acute nephritic syndrome.

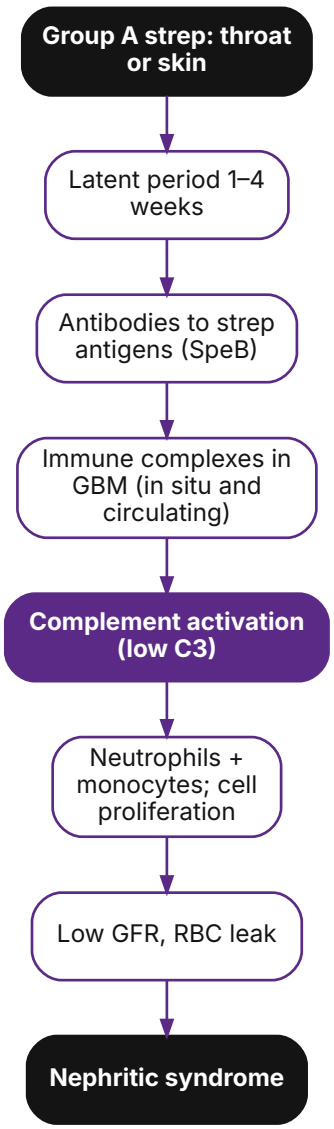


Figure 16.3 · Pathogenesis of PSGN.

CLINICAL PICTURE

- Child of **6–10 years**; sudden onset after sore throat or impetigo
- **Smoky / cola-coloured urine**, RBC casts
- **Periorbital oedema**, oliguria, mild hypertension
- **Raised ASO titre** (throat) or anti-DNase B (skin); **low serum C3** that normalises in 6–8 weeks
- **> 95% of children recover** fully; adults do worse (RPGN or chronic disease in a minority)

GROSS

- Kidneys symmetrically enlarged and pale
- Petechial haemorrhages may dot the cortex

MICROSCOPY

- **LM: diffuse** enlargement and **hypercellularity** of glomeruli: proliferating endothelial and mesangial cells plus **neutrophils** and monocytes; occasional crescents
- **IF: granular** IgG, IgM and C3 ("starry sky")
- **EM: subepithelial humps**

MEMORY HOOK

PSGN = "4 Ps + 1 C"

Pharynx/skin infection first · **P**eriorbital puffiness · **P**ee is cola-coloured · **P**ressure (BP) up · **C**3 down

IgA nephropathy (Berger disease)

The **commonest glomerulonephritis worldwide** on biopsy. Children and young adults get **visible haematuria within 1–2 days** of a respiratory or GI infection (*synpharyngitic*), recurring every few months. A defect in IgA1 glycosylation (**galactose-deficient IgA1**) makes it immunogenic. Autoantibodies form complexes that lodge in the **mesangium** and activate the alternative and lectin complement pathways.

MICROSCOPY

- **LM:** normal to **mesangial proliferation**
- **IF: mesangial IgA** (often with C3): diagnostic
- **EM:** mesangial deposits

- **Secondary IgA nephropathy:** coeliac disease, chronic liver disease/cirrhosis (poor hepatic clearance of IgA)
- **IgA vasculitis (Henoch–Schönlein purpura):** the systemic form, with purpura on buttocks and legs, arthritis and abdominal pain
- Serum C3 normal; slow progression to chronic renal failure in about a quarter to a half over 20 years

EXAM TRAP

PSGN vs IgA nephropathy. Both follow a sore throat. The **interval** separates them: PSGN appears **1–4 weeks later** with **low C3**; IgA haematuria appears **within 1–2 days** with **normal C3** and recurs.

Hereditary nephritis: Alport syndrome

Mutation of type IV collagen, usually **X-linked (COL4A5)**, so boys are worst affected. The triad is **haematuria progressing to renal failure**, **sensorineural deafness** and **eye lesions** (anterior lenticonus, lens dislocation, cataract). EM shows a thin, then irregularly thickened and **split, laminated ("basket-weave") GBM**. **Thin basement membrane disease** (heterozygous COL4A3/4) causes benign familial haematuria.

16.5 Rapidly progressive (crescentic) GN

RPGN is a clinical syndrome: a **nephritic** picture with severe oliguria and a fall of GFR over **days to weeks**. The common microscopic lesion is the crescent: severe injury ruptures the capillary wall and fibrin leaks into Bowman's space. Parietal epithelial cells proliferate and monocytes/macrophages migrate in, forming a crescent that compresses the tuft and later scars.

CLASSIFICATION OF RPGN BY IMMUNOFLUORESCENCE

Type	IF pattern	Mechanism	Causes
Type I: anti-GBM	**Linear** IgG + C3	Antibodies to the α3 chain (NC1 domain) of type IV collagen	Kidney only, or **Goodpasture syndrome** (lung + kidney)
Type II: immune complex	**Granular**	Any severe immune-complex GN	PSGN, lupus nephritis, IgA nephropathy / IgA vasculitis
Type III: pauci-immune	**None / scanty**	ANCA-activated neutrophils damage capillaries	**GPA** (PR3-ANCA, c-ANCA), **microscopic polyangiitis** (MPO-ANCA, p-ANCA), EGPA

CLINICAL CORRELATION

Goodpasture syndrome: the same antibody attacks alveolar and glomerular basement membranes, giving **haemoptysis + RPGN**, usually in young men. Smoking and lung infections expose the hidden antigen. Treatment: **plasmapheresis** plus immunosuppression.

MOLECULAR KEY

ANCA targets: PR3 (cytoplasmic pattern, c-ANCA) → granulomatosis with polyangiitis. **MPO** (perinuclear pattern, p-ANCA) → microscopic polyangiitis, EGPA. Anti-GBM: α3(IV) NC1 domain; linked to **HLA-DRB1**.

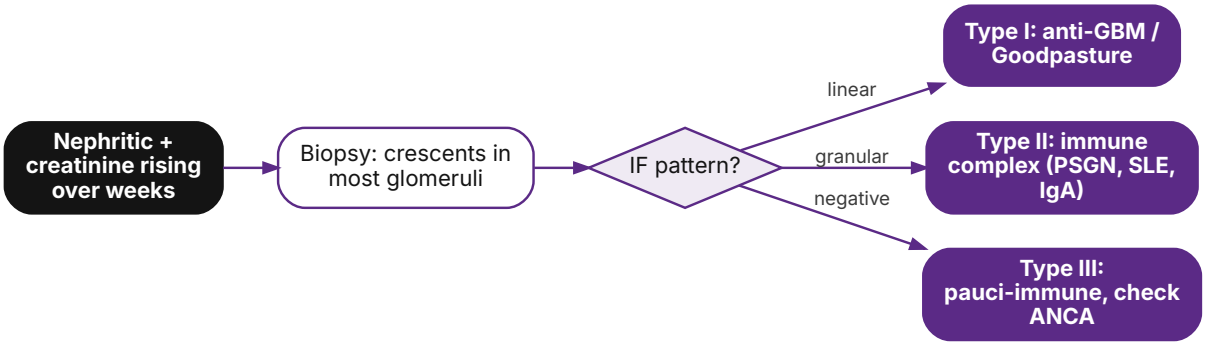


Figure 16.4 · Approach to RPGN from the biopsy. Immunofluorescence is the branch point.

16.6 Chronic glomerulonephritis

The common end point of many glomerular diseases: **RPGN** (most cases), **FSGS**, **MPGN**, membranous and IgA nephropathy. PSGN in children rarely ends here. Patients present with hypertension, anaemia and uraemia, often without any history of renal disease.

GROSS

- Kidneys **symmetrically contracted**
- **Diffusely granular** cortical surface; thin cortex
- Increased peripelvic fat

MICROSCOPY

- **Obliterated, hyalinised glomeruli** ("glomerular ghosts")
- Tubular atrophy, **interstitial fibrosis**, lymphocytic infiltrate
- Thick-walled arteries and arterioles from hypertension

Uraemic complications: fibrinous pericarditis, uraemic gastroenteritis, secondary hyperparathyroidism with renal osteodystrophy, normocytic anaemia (low erythropoietin), uraemic pneumonitis.

LAB SPOTTER

Glomerulus under the microscope in 10 seconds: (1) **Count cells** in the tuft. More than 3 nuclei per mesangial area, or neutrophils, means proliferative GN (PSGN). (2) **Look at the capillary walls.** Thick, rigid walls point to membranous; tram-tracks point to MPGN; wire loops point to lupus. (3) **Look in Bowman's space** for a crescent (RPGN). (4) A tuft that looks entirely normal in a nephrotic child is MCD.

CLINICAL CASE

A 58-year-old man has anasarca for 3 months. Urine protein 6 g/day, albumin 2.1 g/dL, cholesterol high, no RBC casts, BP normal. Biopsy: thick capillary walls, silver spikes, granular IgG along GBM.

Reasoning: nephrotic range + adult + thick walls without proliferation + spikes → **membranous nephropathy**. Next steps: anti-PLA2R antibody, and screen for the secondary causes in "DIMS". At his age, look carefully for a **hidden carcinoma**.

EXAM ZONE Past-Paper Pattern SEQs — Solved

PROF PATTERN **a)** Differentiate between nephritic and nephrotic syndrome. **b)** Describe the pathogenesis and morphology of acute post-streptococcal glomerulonephritis. **2 + 3 marks**

MODEL ANSWER PLAN

a) Table of 5 points: haematuria with RBC casts vs none; proteinuria < 3.5 vs > 3.5 g/day; hypertension vs normal BP; mild periorbital vs generalised oedema; hyperlipidaemia/lipiduria only in nephrotic. One prototype each (PSGN; MCD).

b) Group A strep (pharynx/skin) → 1–4 week latency → antibodies to strep antigens (SpeB) → immune complexes in GBM → complement activation (low C3) → neutrophil influx and proliferation → nephritic syndrome. Morphology: enlarged pale kidneys; LM diffuse hypercellular glomeruli with neutrophils; IF granular IgG/C3; EM subepithelial humps.

Score booster: draw the pathogenesis flowchart (Fig 16.3); it takes 1 minute and shows the examiner the whole sequence.

MODULE PATTERN A 5-year-old boy develops puffiness of the eyes that spreads to the legs and abdomen two weeks after a cold. Urine: protein 4+, no RBCs. BP is normal. **a)** What is the most likely diagnosis? **b)** Explain the mechanism of his oedema. **c)** Describe the biopsy findings. **1 + 2 + 2 marks**

MODEL ANSWER PLAN

a) Nephrotic syndrome due to **minimal change disease**. **b)** Podocyte injury → massive selective proteinuria → hypoalbuminaemia → low oncotic pressure → fluid moves to interstitium; low plasma volume activates RAAS → Na/water retention. **c)** LM normal glomeruli, lipid in PCT cells; IF negative; EM diffuse foot-process effacement, no deposits.

Score booster: add "excellent steroid response (> 90%)"; examiners often hold back a mark for prognosis.

PROF PATTERN **a)** Classify rapidly progressive glomerulonephritis. **b)** Write a note on Goodpasture syndrome. **2 + 3 marks**

MODEL ANSWER PLAN

a) Three types by IF: I anti-GBM (linear), II immune complex (granular; PSGN, SLE, IgA), III pauci-immune (ANCA: GPA, MPA). Mention crescents as the common lesion. **b)** Antibodies to α3(IV) collagen NC1 domain → pulmonary haemorrhage + RPGN; young men; linear IgG/C3 on IF; crescents; treat with plasmapheresis + steroids/cyclophosphamide.

PROF PATTERN Enumerate the causes of membranous nephropathy and describe its morphology. **5 marks**

MODEL ANSWER PLAN

Primary (anti-PLA2R) about 75%; secondary "DIMS": drugs, infections (HBV, HCV, syphilis, malaria), malignancy, SLE. Morphology: LM uniform capillary wall thickening; silver spikes; IF granular IgG/C3; EM subepithelial deposits. Clinical: adult nephrotic syndrome, renal vein thrombosis, variable course.

BCQs One-Best-Answer Practice

1 A 6-year-old has generalised oedema, 4+ proteinuria and no haematuria. Light microscopy of the biopsy is normal. Which electron-microscopic finding is expected?

A. Subepithelial humps
B. Diffuse effacement of podocyte foot processes
C. Split, laminated GBM
D. Subendothelial deposits
E. Intramembranous ribbon-like deposits

2 Which immunofluorescence pattern is characteristic of anti-GBM disease?

A. Granular IgG and C3
B. Mesangial IgA
C. Linear IgG along the GBM
D. C3 only, no immunoglobulin
E. Negative staining

3 Silver stain of a renal biopsy from a 50-year-old with nephrotic syndrome shows spikes projecting from the GBM. The most likely target antigen is:

A. α3 chain of type IV collagen
B. Myeloperoxidase
C. Proteinase 3
D. Nephrin
E. Phospholipase A2 receptor

4 A 9-year-old has cola-coloured urine and periorbital oedema 2 weeks after impetigo. Which serum finding is most likely?

A. Low C3
B. Raised IgA
C. Positive c-ANCA
D. Anti-GBM antibody
E. Normal complement with high IgE

5 A 20-year-old man has visible haematuria one day after each upper respiratory infection. Serum C3 is normal. The biopsy is most likely to show:

A. Crescents with linear IgG
B. Subepithelial humps
C. Mesangial IgA deposits
D. Kimmelstiel–Wilson nodules
E. Tram-track GBM with low C3

6 The collapsing variant of FSGS is most strongly associated with:

A. Hepatitis B
B. HIV infection
C. Penicillamine
D. Hodgkin lymphoma
E. Streptococcal pharyngitis

7 Which of the following is NOT a feature of nephrotic syndrome?

A. Hypoalbuminaemia
B. Hyperlipidaemia
C. Lipiduria
D. RBC casts
E. Hypercoagulability

8 C3 nephritic factor is an autoantibody that:

A. Binds PLA2R on podocytes
B. Destroys C1 inhibitor
C. Targets type IV collagen
D. Activates neutrophils directly
E. Stabilises the alternative-pathway C3 convertase

9 A 28-year-old man has haemoptysis and rapidly rising creatinine. Biopsy shows crescents in 80% of glomeruli with linear IgG along the GBM. The first-line specific treatment is:

A. Plasmapheresis
B. Oral penicillin
C. Dietary protein restriction
D. Gluten-free diet
E. ACE inhibitor alone

10 Pauci-immune crescentic GN with c-ANCA (anti-PR3) positivity suggests:

A. Microscopic polyangiitis
B. Lupus nephritis
C. Granulomatosis with polyangiitis
D. IgA vasculitis
E. Alport syndrome

11 A boy has haematuria, sensorineural deafness and anterior lenticonus. The defect lies in:

A. Nephrin
B. Podocin
C. Fibrillin
D. Type IV collagen
E. Type I collagen

12 Crescents in RPGN are formed mainly by:

A. Mesangial cells and IgA
B. Endothelial cells and platelets
C. Podocytes and amyloid
D. Tubular cells and casts
E. Parietal epithelial cells and macrophages

13 Hepatitis C infection is most typically associated with which glomerular lesion?

A. Membranoproliferative GN with cryoglobulins
B. Minimal change disease
C. Alport syndrome
D. Anti-GBM disease
E. Thin basement membrane disease

14 End-stage kidneys of chronic glomerulonephritis are typically:

A. Enlarged with multiple cysts
B. Small with coarse irregular U-shaped scars
C. Symmetrically small with a diffusely granular surface
D. Enlarged, pale and waxy
E. Normal-sized with papillary necrosis

15 Hypercoagulability in nephrotic syndrome is mainly due to urinary loss of:

A. Fibrinogen
B. Platelets
C. Factor VIII
D. Antithrombin III
E. Von Willebrand factor

ANSWER KEY WITH ONE-LINE REASONS

1. **B** Minimal change disease: the only lesion is foot-process effacement.
2. **C** Antibody coats the GBM evenly, giving linear staining.
3. **E** Primary membranous nephropathy: anti-PLA2R in most cases.
4. **A** PSGN consumes complement; C3 is low and recovers in 6–8 weeks.
5. **C** Synpharyngitic haematuria with normal C3 = IgA nephropathy.
6. **B** HIV-associated nephropathy is the classic collapsing glomerulopathy.
7. **D** RBC casts indicate glomerular inflammation: a nephritic feature.
8. **E** Persistent C3 activation causes dense deposit disease with low C3.
9. **A** Goodpasture: remove circulating anti-GBM antibody plus immunosuppression.
10. **C** PR3/c-ANCA is the marker of GPA (Wegener).
11. **D** Alport syndrome: usually X-linked COL4A5 mutation.
12. **E** Fibrin in Bowman's space triggers parietal cell proliferation and monocyte influx.
13. **A** HCV → mixed cryoglobulinaemia → immune-complex MPGN.
14. **C** Symmetric contraction and fine granularity; coarse scars suggest chronic pyelonephritis.
15. **D** Loss of antithrombin III favours renal vein and deep vein thrombosis.

VIVA Rapid-Fire Viva Voce

Q Define nephrotic syndrome.	Proteinuria > 3.5 g/day, serum albumin < 3 g/dL, generalised oedema, hyperlipidaemia and lipiduria.
Q Commonest cause of nephrotic syndrome in children?	Minimal change disease.
Q Why is MCD called lipoid nephrosis?	Proximal tubular cells are loaded with lipid reabsorbed from filtered lipoproteins.
Q What is selective proteinuria?	Loss mainly of small proteins like albumin, not globulins; typical of MCD.
Q What do you see in Bowman's space in RPGN?	Crescents: proliferating parietal epithelial cells with macrophages and fibrin.
Q Name the three immunological types of RPGN.	Anti-GBM (linear), immune complex (granular), pauci-immune (ANCA, no deposits).
Q What are 'humps'?	Large subepithelial immune deposits seen on EM in PSGN.

Q What are 'spikes'?	GBM projections between subepithelial deposits on silver stain in membranous nephropathy.
Q What is a tram-track appearance?	Duplicated GBM from mesangial interposition in MPGN.
Q What are wire loops?	Thickened capillary walls from subendothelial deposits in lupus nephritis class IV.
Q Which GNs have low serum C3?	PSGN, MPGN/C3 glomerulopathy, lupus nephritis, cryoglobulinaemic GN.
Q Most common GN worldwide?	IgA nephropathy (Berger disease).
Q Antigen in Goodpasture syndrome?	NC1 domain of the α3 chain of type IV collagen.
Q Which nephrotic disease is linked to renal vein thrombosis?	Membranous nephropathy.

OSPE Practice Stations

STATION 1 · SLIDE — 2 minutes

TASK

Section of kidney: glomeruli are enlarged and fill Bowman's space. The tufts are packed with nuclei, including many neutrophils; capillary lumens are obliterated. (a) Identify the lesion. (b) Name the expected EM and serum findings.

ANSWER

(a) Acute proliferative (post-streptococcal) glomerulonephritis. (b) EM: subepithelial humps. Serum: low C3, raised ASO titre.

STATION 2 · LAB DATA — 2 minutes

TASK

Adult: urine protein 5.2 g/24 h, serum albumin 2.3 g/dL, cholesterol 310 mg/dL, urine shows Maltese crosses under polarised light, no RBC casts. (a) Name the syndrome. (b) What are the Maltese crosses? (c) Give two likely primary causes.

ANSWER

(a) Nephrotic syndrome. (b) Oval fat bodies: lipid-laden tubular cells/droplets showing cholesterol birefringence. (c) FSGS, membranous nephropathy.

STATION 3 · GROSS SPECIMEN — 1 minute

TASK

Two small kidneys of equal size with a finely granular surface and thin cortex. (a) Diagnosis? (b) Two microscopic features. (c) One complication.

ANSWER

(a) End-stage kidney of chronic glomerulonephritis. (b) Hyalinised glomeruli; interstitial fibrosis with tubular atrophy. (c) Uraemia: pericarditis, renal osteodystrophy or anaemia.

RAPID REVISION

The Night-Before Page

15 ONE-LINERS

01 Subepithelial deposits → nephrotic; subendothelial/mesangial → nephritic.

02 Granular IF = immune complex; linear = anti-GBM; negative = MCD or ANCA.

03 Nephrotic: protein > 3.5 g/day, albumin < 3 g/dL, oedema, lipids.

04 Nephritic: haematuria with RBC casts, hypertension, oliguria.

05 MCD: normal LM, effaced feet on EM, steroid-responsive child.

06 FSGS: segmental scars, hypertension, poor steroid response; HIV → collapsing.

07 Membranous: anti-PLA2R, silver spikes, subepithelial deposits.

08 Membranous is the nephrotic disease of renal vein thrombosis.

09 MPGN: tram-tracks; HCV with cryoglobulins; low C3.

10 Dense deposit disease: C3 nephritic factor, ribbon deposits.

11 PSGN: 1–4 weeks after strep, humps, low C3, ASO up.

12 IgA: haematuria 1–2 days after URTI, mesangial IgA, normal C3.

13 Alport: COL4A5, deafness, lenticonus, split GBM.

14 RPGN: crescents; types I linear, II granular, III pauci-immune.

15 Chronic GN: small symmetric granular kidneys, hyalinised glomeruli.

MEMORY HOOKS

- **DIMS**: secondary membranous (Drugs, Infections, Malignancy, SLE)
- **4 Ps + 1 C**: PSGN (pharynx, periorbital, pee cola, pressure up, C3 down)
- **Low C3 club**: PSGN, MPGN, lupus, cryoglobulins
- **"Linear = Lung + Loin"**: anti-GBM hits lung and kidney

TEST YOURSELF: FILL THE GAPS

1. MCD: EM shows _____ of foot processes.
2. Anti-GBM antigen: _____ chain of type IV collagen.
3. c-ANCA targets _____.
4. Silver-stain spikes = _____ nephropathy.
5. Tram-tracks are seen in _____.
6. IgA haematuria starts _____ after infection.

Answers: effacement · α3 · PR3 · membranous · MPGN · 1–2 days

DRAW FROM MEMORY

The capillary wall with the six deposit sites (Figure 16.1).

DRAW FROM MEMORY

The nephrotic pathophysiology chain from proteinuria to oedema, lipids, clots and infection.

SPACED REVISION ☐ Day 1 ☐ Day 3 ☐ Day 7 ☐ Day 14 ☐ Day 30

17

Tubulointerstitial Disease, Cystic Kidney & Renal Function

Most acute kidney injury is not glomerular at all: it is tubules dying of shock, poison or infection. Learn to read a cast, a creatinine and a blood gas, add the two great cystic kidneys, and half the renal OSPE is answered before you look down the microscope.

BY THE END OF THIS CHAPTER YOU CAN

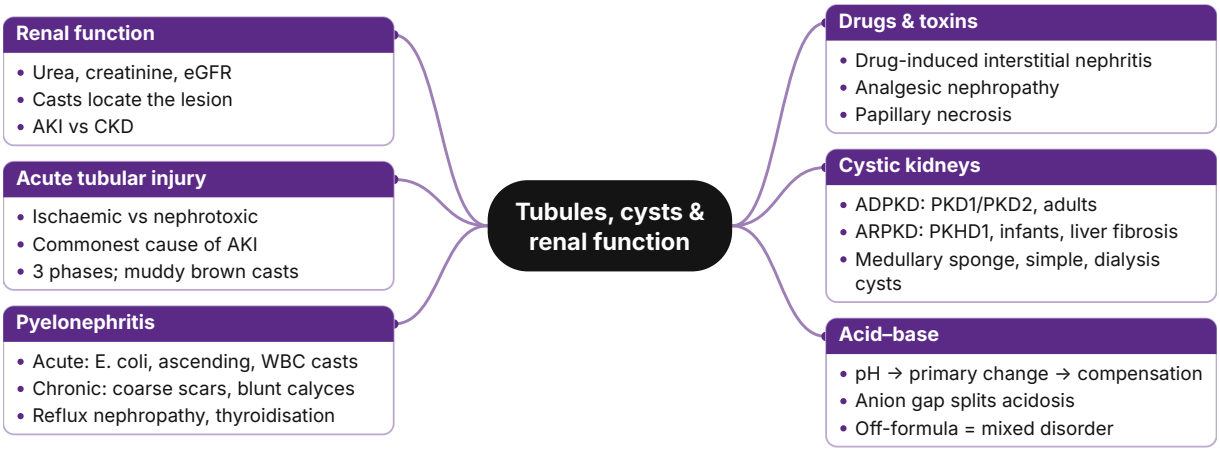
1. Interpret urea, creatinine, eGFR and urinalysis, and use urinary casts to localise renal disease
2. Differentiate prerenal azotaemia, acute tubular injury and chronic kidney disease from laboratory data
3. Describe the causes, pathogenesis, morphology and clinical phases of acute tubular injury (ATN)
4. Compare acute and chronic pyelonephritis, including reflux nephropathy and papillary necrosis
5. Describe drug-induced acute interstitial nephritis and analgesic nephropathy
6. Differentiate ADPKD from ARPKD and recognise the other cystic diseases of the kidney
7. Interpret an arterial blood gas step by step using the anion gap and compensation formulae

EXAM RADAR

Theory SEQs	★★★
BCQs / MCQs	★★★
Viva voce	★★★
OSPE / slides	★★★

ATN, acute vs chronic pyelonephritis and ADPKD are long-question favourites; casts, RFT and ABG slips are standard OSPE data stations.

ROBBINS LINK Robbins Basic Pathology — *The Kidney and Its Collecting System*: clinical manifestations of renal disease, acute tubular injury, tubulointerstitial nephritis (pyelonephritis, drug-induced), cystic diseases. Acid–base rules follow standard chemical pathology.



Chapter at a glance. Tubules, interstitium, cysts and the numbers that tell you how the kidney is coping.

17.1 Renal function and the renal syndromes

Glomeruli are injured mainly by immune mechanisms (Chapter 16). Tubules and interstitium are injured by **ischaemia, toxins and infection**, and betray themselves not by heavy proteinuria or RBC casts but by **failure of tubular work**: poor concentration (polyuria, nocturia), salt wasting and metabolic acidosis.

DEFINITION

Azotaemia is a rise in blood urea and creatinine caused by a fall in GFR. It is **prerenal** when perfusion is poor, **postrenal** when urine flow is obstructed. **Uraemia** is azotaemia plus the clinical and biochemical failure of many organs (pericarditis, gastritis, neuropathy, bleeding, acidosis).

DEFINITION

Acute kidney injury (AKI) is a rapid fall in GFR over hours to days with rising creatinine, often with **oliguria** (< 400 mL/day). **Chronic kidney disease (CKD)** is GFR < 60 mL/min/1.73 m², or evidence of kidney damage such as albuminuria, lasting **more than 3 months**.

LAB VALUES · RENAL FUNCTION TESTS

Serum urea 15–40 mg/dL (BUN 7–20 mg/dL)

GFR

about 90–120 mL/min/1.73 m²; **BUN : creatinine > 20** = prerenal

Specific gravity 1.003–1.030; fixed at 1.010 in CKD

CKD (eGFR)

G1 ≥ 90 · G2 60–89 · G3 30–59 · G4 15–29 · G5 < 15

Serum creatinine

about 0.6–1.2 mg/dL; rises only after about half the GFR is lost (use **eGFR**)

Urine volume oliguria < 400, anuria < 100 mL/day

Proteinuria

normal < 150 mg/day; ACR 30–300 mg/g = early diabetic

Urinalysis: the casts tell you where the injury is

A cast is a **mould of a tubule**: Tamm–Horsfall protein (uromodulin) from the thick ascending limb gels in the distal tubule and traps passing cells, proving that they came **from the kidney**, not the bladder.

URINARY CASTS AND WHAT THEY MEAN

Cast	Made of	Points to
Hyaline	Pure Tamm–Horsfall protein	Few are normal; concentrated urine, exercise, fever, **prerenal** state
RBC cast	Red cells in protein matrix	**Glomerulonephritis** (nephritic), vasculitis
WBC cast	Neutrophils	**Acute pyelonephritis**, acute interstitial nephritis
Muddy brown granular; epithelial	Shed, degenerated tubular cells ± pigment	**Acute tubular injury** (ischaemic, toxic, myoglobin)
Fatty cast, oval fat bodies	Lipid; Maltese crosses in polarised light	**Nephrotic syndrome**
Waxy and broad casts	Old casts formed in dilated, atrophic tubules	**Advanced CKD** ("renal failure casts")

Urinary **eosinophils** suggest drug-induced interstitial nephritis; **dysmorphic RBCs** mean glomerular bleeding. Crystals: Chapter 18.

AKI has three doors. Prerenal: a healthy kidney underperfused (haemorrhage, diarrhoea, burns, heart failure, sepsis). **Intrinsic renal**: **ATN** (commonest), RPGN, acute interstitial nephritis, HUS, malignant hypertension. **Postrenal**: obstruction of both outflows (BPH, stones, pelvic tumours).

PRERENAL AZOTAEMIA VS ACUTE TUBULAR INJURY: THE COMMONEST AKI DATA QUESTION

Index	Prerenal azotaemia	Acute tubular injury (ATN)
Urine Na	**< 20 mEq/L**	**> 40 mEq/L**
Fractional excretion of Na (FE_{Na})	**< 1%**	**> 2%**
Urine osmolality / specific gravity	> 500 mOsm/kg; SG > 1.020	< 350 mOsm/kg; SG about 1.010

Index	Prerenal azotaemia	Acute tubular injury (ATN)
BUN : creatinine	**> 20 : 1**	10–15 : 1
Sediment	Bland; hyaline casts	**Muddy brown granular** and epithelial casts
Fluid challenge	Prompt improvement	No improvement

AKI VS CKD

Feature	Acute kidney injury	Chronic kidney disease
Kidney size (ultrasound)	Normal or swollen	**Small** (not diabetes, ADPKD, amyloid)
Anaemia	Usually absent	**Normocytic** (low erythropoietin)
Calcium, phosphate, PTH	Usually normal	Low Ca, high PO_4, high PTH: **osteodystrophy**
Outcome	Often reversible	Progressive and irreversible

PAKISTAN FOCUS

Diabetes and hypertension are the leading causes of CKD in Pakistan, where about one adult in four has diabetes. Common **consanguineous marriage** raises the frequency of autosomal recessive kidney diseases (ARPKD, cystinuria): ask about it when a child has unexplained renal failure.

Source: IDF Diabetes Atlas 2021.

17.2 Acute tubular injury (acute tubular necrosis)

DEFINITION

Acute tubular injury / necrosis (ATN) is a clinicopathological entity in which **damage to tubular epithelial cells** causes an acute fall in renal function, usually with granular casts and tubular cells in the urine. It is the **commonest cause of AKI**. Robbins prefers "injury" because frank necrosis is often scanty.

ISCHAEMIC VS NEPHROTOXIC ATN

Feature	Ischaemic ATN	Nephrotoxic ATN
Causes	Shock of any kind: haemorrhage, **sepsis**, burns, severe dehydration, pancreatitis, major surgery	**Aminoglycosides** (gentamicin), radiocontrast, cisplatin, heavy metals (**mercury**), organic solvents (CCl_4), **ethylene glycol** (ballooned PCT cells, **oxalate crystals**), **myoglobin** (crush injury), **haemoglobin** (mismatched transfusion)
Segments hit	**Straight part of PCT** and **thick ascending limb**	Mainly **PCT** along its length
Pattern; basement membrane	Patchy skip lesions; BM often ruptured (**tubulorrhexis**)	Extensive, continuous; BM usually intact
Prognosis	Death rate over 50% with sepsis or burns	About 95% recover once toxin removed

CONCEPT CLEAR

Why the PCT and thick ascending limb? They pump most of the filtered sodium and burn the most oxygen. The thick limb lies in the oxygen-poor **outer medulla**, so it suffocates first in shock; the PCT **reabsorbs and concentrates toxins**.

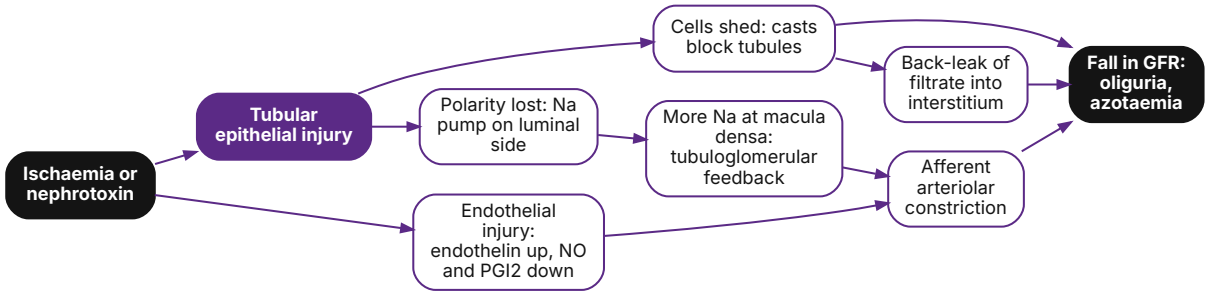


Figure 17.1 · Pathogenesis of acute tubular injury. Tubular and vascular events both end in a falling GFR.

GROSS

- Kidneys **swollen and heavy**
- **Pale cortex**, dark congested medulla

MICROSCOPY

- Loss of brush border, flattening and **vacuolation** of PCT cells; scattered necrotic cells
- Cells **sloughed into lumens**; eosinophilic hyaline and **pigmented granular casts** in distal tubules and collecting ducts
- Interstitial **oedema**; leukocytes in dilated vasa recta
- Recovery: flattened **regenerating cells** with mitoses; **glomeruli normal**

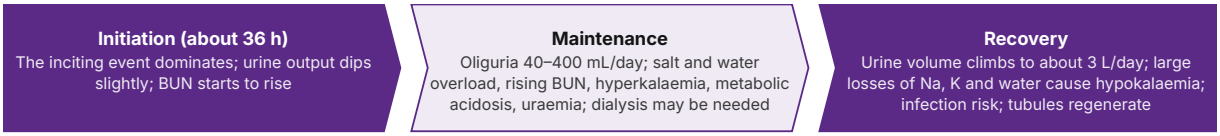


Figure 17.2 · Clinical course of ATN. Hyperkalaemia kills in the middle phase; hypokalaemia and infection in the last.

Exam trap: about half of patients (especially with nephrotoxins) keep a normal urine volume: **non-oliguric ATN**, with a milder course.

17.3 Pyelonephritis

Tubulointerstitial nephritis (TIN) is inflammation falling mainly on tubules and interstitium, sparing glomeruli until late; **acute** (oedema, leukocytes) or **chronic** (mononuclear cells, fibrosis, atrophy). Causes: **infection**, **drugs**, urate, hypercalcaemia, obstruction, myeloma casts, immune disease.

Acute pyelonephritis

A **suppurative bacterial inflammation of the kidney and renal pelvis**. Gut gram-negative bacilli cause most cases: **E. coli** by far the commonest, then *Proteus*, *Klebsiella*, *Enterobacter*; *Pseudomonas* and enterococci follow instrumentation. Infection usually **ascends**; the **haematogenous** route (septicaemia, endocarditis, staphylococci, TB, fungi) is less common and seeds abscesses at random.

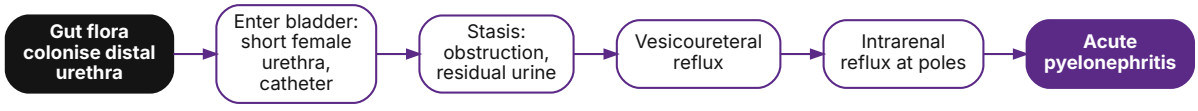


Figure 17.3 · Ascending infection: the five steps. Remove any step (catheter, obstruction, reflux) and infection is much less likely.

PREDISPOSING FACTORS

- **Obstruction** (BPH, stones, gravid uterus); **catheters**
- **Vesicoureteral reflux** (congenital, neurogenic bladder)
- **Pregnancy**, female sex, **diabetes**, immunosuppression

CONCEPT CLEAR

Why reflux matters. The ureter tunnels obliquely through the bladder wall, so voiding squeezes it shut. A **short tunnel** or atonic bladder lets urine shoot up and fall back as infected **residual urine**. **Compound papillae at the poles** have open ducts that admit it, so scars cluster at the poles.

GROSS

- One or both kidneys; patchy or diffuse
- Discrete **yellowish raised abscesses** on the cortical surface
- Yellow streaks of pus in the medulla

MICROSCOPY

- Patchy **suppurative interstitial inflammation**
- **Neutrophil aggregates within tubules** (the source of WBC casts) and tubular necrosis
- **Glomeruli are spared**: they resist infection

Complications: papillary necrosis (diabetes, obstruction); **pyonephrosis** (pus filling an obstructed pelvicalyceal system); **perinephric abscess** (pus breaks through the capsule). Healing leaves fibrous cortical scars over deformed calyces.

Clinical: sudden **fever with chills**, **loin pain and tenderness**, dysuria and frequency; pyuria, bacteriuria ($\geq 10^5$/mL) and **WBC casts**, which prove renal involvement. It settles with antibiotics; recurrence signals obstruction or reflux.

PAPILLARY NECROSIS

Ischaemic (coagulative) necrosis of the renal papillae. The tips or distal two-thirds of the pyramids turn **grey-white to yellow**; microscopy shows **ghost outlines of tubules** with a band of neutrophils at the junction with living tissue. A sloughed papilla can cause **colic, haematuria or obstruction**.

MEMORY HOOK

Papillary necrosis = "DOSA"

Diabetes mellitus · **O**bstruction with infection · **S**ickle cell disease (or trait) · **A**nalgesic abuse. All four starve the papilla, the end of a long, thin blood supply.

Chronic pyelonephritis and reflux nephropathy

DEFINITION

Chronic pyelonephritis is chronic tubulointerstitial inflammation and scarring that **involves the calyces and pelvis**, the feature that separates it from every other scarred kidney. Two forms: **reflux nephropathy** (commoner), infection on **congenital vesicoureteral and intrarenal reflux** in early childhood, often silent until hypertension appears; and **chronic obstructive pyelonephritis**, recurrent infection behind an obstruction (bilateral with posterior urethral valves, unilateral with a stone).

GROSS

- One or both kidneys; if both, **asymmetrically** shrunken
- **Coarse, discrete, U-shaped corticomedullary scars**, mostly at the **poles**
- Underlying calyces **dilated, blunted or deformed**; papillae flattened

MICROSCOPY

- Uneven **interstitial fibrosis** with lymphocytes, plasma cells (± neutrophils)
- Atrophic tubules; dilated tubules filled with pink, glassy colloid casts: **"thyroidisation"**
- Chronic inflammation and fibrosis of the **calyceal mucosa**
- Glomeruli normal or periglomerular fibrosis; later **secondary FSGS**

ACUTE VS CHRONIC PYELONEPHRITIS

Feature	Acute pyelonephritis	Chronic pyelonephritis
Basic lesion	Suppurative bacterial inflammation	Chronic inflammation and **scarring** with calyceal deformity
Setting	Ascending UTI, catheter, pregnancy, diabetes	**Vesicoureteral reflux** in childhood, chronic obstruction
Onset	Sudden: fever, chills, loin pain	Insidious; recurrent UTI, hypertension
Gross	Swollen; yellow cortical abscesses	Small, **asymmetric**, coarse polar scars, **blunted calyces**
Microscopy	Neutrophils in interstitium and tubules	Lymphocytes, plasma cells, fibrosis, **thyroidisation**
Urine	Pyuria, **WBC casts**, bacteriuria	Pyuria ± bacteriuria; polyuria, **nocturia**; proteinuria late (FSGS)
Outcome	Recovers with antibiotics	Hypertension, **CKD**, end-stage renal disease

⚠ EXAM TRAP

Three scarred kidneys. Chronic pyelonephritis: coarse, asymmetric scars over **deformed calyces**. **Chronic GN** (Chapter 16) and **benign nephrosclerosis**: symmetric, finely granular, calyces normal.

17.4 Drug-induced interstitial nephritis and analgesic nephropathy

Acute drug-induced (hypersensitivity) interstitial nephritis

An **immune reaction**, not direct toxicity, so it is unrelated to dose. The drug acts as a **hapten**: it binds tubular cell or matrix proteins and makes them immunogenic. Raised IgE and eosinophils point to a **type I** reaction; granulomas and T-cell infiltrates to a **type IV** reaction.

COMMON CULPRITS

- **Antibiotics**: penicillins (methicillin), rifampicin, sulfonamides
- **NSAIDs**, **proton pump inhibitors**, **thiazides**, allopurinol

CLINICAL PICTURE

- About **2 weeks** (2–40 days) after starting the drug
- **Fever, eosinophilia, rash**; haematuria, mild proteinuria, **urinary eosinophils**
- Rising creatinine, AKI in about half; recovers on **stopping the drug**

MICROSCOPY

- Interstitial **oedema** with infiltrates of **lymphocytes and macrophages**
- **Eosinophils** (and neutrophils) may be numerous
- **Tubulitis**: lymphocytes crossing the tubular basement membrane
- Non-necrotising **granulomas with giant cells** (methicillin, thiazides, rifampicin)
- **Glomeruli normal**, except NSAIDs, which can add minimal change disease (nephrotic syndrome)

Analgesic nephropathy

Chronic renal disease from **years of heavy use of analgesic mixtures** (classically phenacetin + aspirin + caffeine). The first lesion is **papillary necrosis** from two hits: **paracetamol metabolites** cause oxidative injury and **aspirin** blocks vasodilator prostaglandins, making the medulla ischaemic. Cortical tubulointerstitial nephritis follows.

Morphology: papillae **brown, shrunken** and necrotic (coagulative necrosis, ghost tubules, calcification) or sloughed; the cortex over them is **depressed** with tubular atrophy and fibrosis, while spared columns of Bertin bulge. **Clinical:** polyuria, nocturia, hypertension, anaemia, UTI, colic from a sloughed papilla; risk of **urothelial carcinoma of the renal pelvis** (Chapter 18).

CLINICAL CORRELATION

NSAIDs and the kidney: four lesions. (1) **Haemodynamic AKI** (loss of vasodilator prostaglandins in dehydration or heart failure). (2) **Hypersensitivity interstitial nephritis**. (3) **Minimal change** or membranous nephropathy. (4) **Papillary necrosis** with long-term use.

17.5 Cystic diseases of the kidney

Renal cysts are dilated nephron segments. Hereditary forms cause **renal failure** and carry **extrarenal** lesions; a simple cyst must not be mistaken for a **tumour**.

Autosomal dominant (adult) polycystic kidney disease (ADPKD)

A hereditary disease in which **multiple expanding cysts of both kidneys** gradually destroy the parenchyma. It affects about 1 in 400–1000 people and accounts for roughly 5–10% of end-stage renal disease. Penetrance is high, but symptoms usually wait until the **fourth decade**.

MOLECULAR KEY

PKD1 (**16p13.3**) encodes **polycystin-1**, a receptor-like membrane protein: about **85%** of cases, more severe. **PKD2** (**4q21**) encodes **polycystin-2**, a calcium channel: about 15%, slower course. Together they form a **flow sensor in the primary cilium**. Loss lowers cell calcium and **raises cAMP**, driving epithelial **proliferation and fluid secretion**. Cysts form only where a **second somatic hit** removes the normal allele; the V2 blocker **tolvaptan** lowers cAMP and slows cyst growth.

GROSS

- Both kidneys **massively enlarged** (up to 4 kg each)
- Surface made entirely of **cysts up to 3–4 cm**, no intervening parenchyma visible
- Cysts hold clear serous, turbid or **red-brown haemorrhagic** fluid

MICROSCOPY

- Cysts arise from **any part of the nephron**, even Bowman's capsule
- Lined by flattened or cuboidal, sometimes papillary epithelium
- Compressed functioning nephrons, interstitial fibrosis between cysts

Clinical: flank pain and **dragging heaviness**, bilateral masses, **haematuria**, hypertension, UTI, stones; **end-stage renal disease around 50 years** (later with PKD2). **Extrarenal: liver cysts** (about 40%), **berry aneurysms** (subarachnoid haemorrhage), **mitral valve prolapse** (about a quarter), colonic diverticula.

Autosomal recessive (childhood) polycystic kidney disease (ARPKD)

Caused by mutations of **PKHD1** (6p), encoding the ciliary protein **fibrocystin**; perinatal, neonatal, infantile and juvenile forms (severe perinatal cases: oligohydramnios, **Potter sequence**). Kidneys are enlarged with a **smooth surface**; the cut surface is **sponge-like**, with cuboidal-lined dilated **collecting ducts** running **at right angles to the capsule**. The liver always shows **congenital hepatic fibrosis** (portal fibrosis, bile-duct proliferation), bringing portal hypertension in survivors.

ADPKD VS ARPKD

Feature	ADPKD (adult)	ARPKD (childhood)
Inheritance; frequency	Autosomal dominant; common (1 in 400–1000)	Autosomal recessive; rare (1 in 20,000)
Gene; protein	**PKD1** (16p) polycystin-1; **PKD2** (4q) polycystin-2	**PKHD1** (6p) fibrocystin
Age at presentation	Adults, 30–50 years	Perinatal period, infancy, childhood
Kidney surface	Bosselated: large cysts bulge out	**Smooth**
Cysts	Large, few to many, from **all nephron segments**	Small, uniform, radial, from **collecting ducts**
Liver	Liver cysts (about 40%)	**Congenital hepatic fibrosis** (constant)
Other	Berry aneurysm, mitral valve prolapse	Potter sequence, portal hypertension

OTHER CYSTIC LESIONS YOU MUST RECOGNISE

Lesion	Key features
Simple cysts	Common, incidental **cortical** cysts 1–5 cm; thin translucent wall, **clear fluid**, flat lining; smooth and avascular on imaging, unlike tumours.
Acquired (dialysis-associated) cystic disease	Multiple cysts in end-stage kidneys on **long-term dialysis**; oxalate crystals, bleeding; risk of **renal cell carcinoma** (about 7% over 10 years).
Medullary sponge kidney	Adults; small cystic dilatations of **medullary collecting ducts**; normal function; **stones, nephrocalcinosis**, haematuria, infection.

17.6 Acid–base balance and the ABG

Blood pH depends on the **ratio** HCO_3^- / $PaCO_2$. The lungs adjust the denominator within minutes; the kidneys adjust the numerator over **3–5 days** (reclaiming bicarbonate, excreting H^+ as ammonium and titratable acid). So metabolic disorders are compensated fast, respiratory ones slowly, which is why respiratory disorders are called **acute** or **chronic**.

LAB VALUES · ABG AND COMPENSATION

Normal

pH 7.35–7.45 · $PaCO_2$ 35–45 mmHg · HCO_3 22–26 mEq/L

Anion gap

Na − (Cl + HCO_3) = 8–12; falls 2.5 per 1 g/dL fall in albumin

Metabolic acidosis

expected $PaCO_2$ = 1.5 × HCO_3 + 8 (± 2) (Winter)

Metabolic alkalosis

$PaCO_2$ rises about 0.7 per 1 mEq/L rise in HCO_3

Respiratory acidosis

HCO_3 rises 1 (acute) or 3.5 (chronic) per 10 mmHg rise in $PaCO_2$

Respiratory alkalosis

HCO_3 falls 2 (acute) or 4–5 (chronic) per 10 mmHg fall in $PaCO_2$

Rule

compensation never overshoots; off target = **mixed disorder**

Delta ratio

ΔAG / ΔHCO_3: < 1 extra normal-AG acidosis; > 2 extra alkalosis

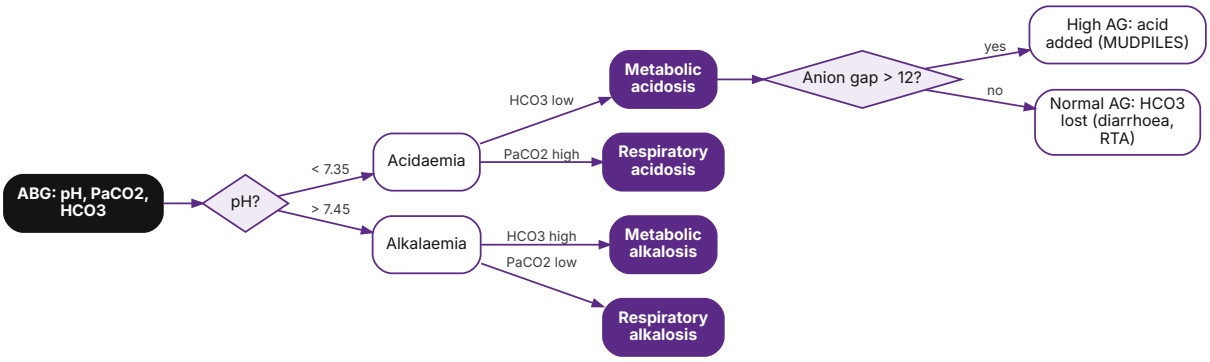


Figure 17.4 · Reading an ABG. pH first, then the culprit, then the compensation; in acidosis always add the anion gap.

CONCEPT CLEAR

What the anion gap really measures. Blood is electrically neutral, but routine tests measure only Na^+, Cl^- and HCO_3^-. The "gap" is the unmeasured anions, mostly **albumin**. When an acid (HA) is added, H^+ uses up bicarbonate and the $\mathbf{A^-}$ **stays behind** as a new unmeasured anion: the gap widens (ketoacids, lactate, formate). When bicarbonate itself is lost (diarrhoea, RTA), the kidney holds on to **chloride** instead: the gap stays normal and the acidosis is **hyperchloraemic**.

THE FOUR PRIMARY DISORDERS

Disorder	Primary change	Compensation	Common causes
Metabolic acidosis	HCO_3 low	Hyperventilation (Kussmaul): $PaCO_2$ falls	**High AG**: DKA, lactic acidosis, **renal failure**, toxic alcohols, salicylate. **Normal AG**: diarrhoea, RTA, acetazolamide
Metabolic alkalosis	HCO_3 high	Hypoventilation: $PaCO_2$ rises	**Vomiting**, NG suction, diuretics, hyperaldosteronism, hypokalaemia
Respiratory acidosis	$PaCO_2$ high	Kidneys retain HCO_3 (3–5 days)	**COPD**, opioid/sedative overdose, neuromuscular weakness, chest-wall disease
Respiratory alkalosis	$PaCO_2$ low	Kidneys excrete HCO_3	Anxiety, **hypoxaemia** (pneumonia, embolism, altitude), early salicylate, sepsis, pregnancy

MEMORY HOOK

ROME · MUDPILES

Respiratory **O**pposite (pH and $PaCO_2$ move in opposite directions), **M**etabolic **E**qual (pH and HCO_3 move together). High-AG acidosis: **M**ethanol · **U**raemia · **D**KA · **P**ropylene glycol · **I**soniazid, iron · **L**actic acidosis · **E**thylene glycol · **S**alicylates.

CLINICAL CASE · MIXED DISORDER

A 30-year-old woman swallowed a bottle of aspirin. pH 7.44, $PaCO_2$ 22, HCO_3 15, Na 140, Cl 103. **Reasoning:** pH "normal", but $PaCO_2$ and HCO_3 are both far off. AG = 140 − 118 = **22** → high-AG metabolic acidosis; expected $PaCO_2$ = 1.5 × 15 + 8 = 30.5 (± 2), but measured 22 is **lower** → an added **respiratory alkalosis** (salicylate drives the respiratory centre). **Diagnosis:** mixed high-AG metabolic acidosis + respiratory alkalosis, classic of **salicylate poisoning**. A normal pH never excludes an acid–base disorder.

EXAM ZONE Past-Paper Pattern SEQs — Solved

PROF PATTERN **a)** Define acute tubular injury (necrosis) and enumerate its causes. **b)** Describe its pathogenesis and clinical course. **2 + 3 marks**

MODEL ANSWER PLAN

a) Damage to tubular epithelial cells with acute fall in GFR; commonest cause of AKI. Ischaemic: shock, sepsis, haemorrhage, burns. Nephrotoxic: aminoglycosides, radiocontrast, mercury, ethylene glycol, myoglobin, haemoglobin.

b) Loss of polarity → tubuloglomerular feedback; endothelin up, NO down → afferent vasoconstriction; casts → obstruction, back-leak → fall in GFR. Course: initiation (~36 h), maintenance (oliguria, high K, acidosis), recovery (diuresis, low K, infection).

Score booster: draw Figure 17.1 and the three phases; add "non-oliguric in about half".

CLASSIC **a)** Differentiate between acute and chronic pyelonephritis. **b)** Describe the morphology of chronic pyelonephritis. **2 + 3 marks**

MODEL ANSWER PLAN

a) Five rows of the table: suppurative vs scarring; UTI vs reflux/obstruction; acute fever vs insidious hypertension, nocturia; abscesses vs coarse asymmetric scars; neutrophils, WBC casts vs lymphocytes, thyroidisation.

b) Gross: asymmetric, coarse U-shaped polar scars over blunted, deformed calyces. Micro: patchy interstitial fibrosis, lymphocytes and plasma cells, tubular atrophy, thyroidisation, calyceal mucosal fibrosis, periglomerular fibrosis and later secondary FSGS.

MODULE PATTERN A 42-year-old hypertensive man has haematuria and bilateral flank masses; his father died of a "brain bleed". **a)** Diagnosis? **b)** Genetic basis. **c)** Gross kidney and two extrarenal lesions. **1 + 2 + 2 marks**

MODEL ANSWER PLAN

a) ADPKD. **b)** PKD1 (16p, polycystin-1, ~85%) or PKD2 (4q, polycystin-2); ciliary defect → high cAMP → proliferation, secretion; second hit. **c)** Huge kidneys of cysts up to 3–4 cm, clear or haemorrhagic fluid; liver cysts, berry aneurysm, mitral valve prolapse.

MODULE PATTERN A 22-year-old diabetic is vomiting and breathing deeply. pH 7.18, $PaCO_2$ 22 mmHg, HCO_3 8 mEq/L, Na 135, Cl 97, glucose 450 mg/dL. **a)** Name the acid–base disorder. **b)** Calculate the anion gap and comment on compensation. **c)** List four other causes of this type of disorder. **1 + 2 + 2 marks**

MODEL ANSWER PLAN

a) High-AG metabolic acidosis (DKA). **b)** AG = 135 − (97 + 8) = 30. Expected $PaCO_2$ = 1.5 × 8 + 8 = 20 ± 2; measured 22 → appropriate compensation. **c)** Lactic acidosis, renal failure, methanol, ethylene glycol, salicylates.

Score booster: show each formula with numbers substituted; marks go to the working.

BCQs One-Best-Answer Practice

1 After 10 hours trapped under rubble with crushed legs, a man passes scanty dark urine with pigmented granular casts. The renal lesion is most likely:

A. Acute pyelonephritis
B. Acute tubular injury
C. Post-streptococcal glomerulonephritis
D. Drug-induced interstitial nephritis
E. Renal papillary necrosis

2 In ischaemic acute tubular injury, the nephron segments most vulnerable are:

A. Glomerulus and Bowman capsule
B. Distal convoluted tubule and collecting duct
C. Straight proximal tubule and thick ascending limb
D. Thin descending limb only
E. Juxtaglomerular apparatus

3 Recovering from ATN, a patient passes 3.5 L of urine a day. The main electrolyte risk is:

A. Hyperkalaemia
B. Hypercalcaemia
C. Hypermagnesaemia
D. Hypokalaemia
E. Hypernatraemia

4 After 3 days of diarrhoea, an elderly woman has a raised creatinine. Which finding favours prerenal azotaemia over ATN?

A. Fractional excretion of sodium below 1%
B. Urine sodium above 40 mEq/L
C. Muddy brown granular casts
D. Urine osmolality below 350 mOsm/kg
E. Specific gravity fixed at 1.010

5 A pregnant woman has fever, loin pain and dysuria. Which urinary finding proves the kidney itself is infected?

A. Nitrites
B. Leukocyte esterase
C. Bacteria on Gram stain
D. Uniform red cells
E. White cell casts

6 A small scarred kidney shows coarse polar scars over blunted calyces; microscopy shows dilated tubules filled with eosinophilic colloid. The diagnosis is:

A. Chronic glomerulonephritis
B. Chronic pyelonephritis
C. Benign nephrosclerosis
D. ADPKD
E. Acute tubular injury

7 Two weeks after starting a penicillin, a man develops fever, rash, eosinophilia and a rising creatinine. Biopsy is most likely to show:

A. Interstitial oedema with eosinophils and granulomas
B. Crescents in Bowman space
C. Subepithelial humps
D. Massive cortical necrosis
E. Glomerular wire loops

8 Renal papillary necrosis is most typically associated with:

A. Minimal change disease
B. Medullary sponge kidney
C. Diabetes mellitus with pyelonephritis
D. IgA nephropathy
E. Simple renal cysts

9 The gene most often mutated in autosomal dominant polycystic kidney disease encodes:

A. Fibrocystin
B. Nephrin
C. Uromodulin
D. Polycystin-2
E. Polycystin-1

10 A neonate with enlarged smooth kidneys and radially arranged collecting-duct cysts will also have:

A. Berry aneurysms
B. Mitral valve prolapse
C. Colonic diverticula
D. Congenital hepatic fibrosis
E. Pancreatic cysts only

11 A man on dialysis for 12 years has multiple bilateral renal cysts. The most serious complication is:

A. Berry aneurysm rupture
B. Congenital hepatic fibrosis
C. Renal cell carcinoma
D. Papillary necrosis
E. Medullary nephrocalcinosis

12 ABG: pH 7.25, PaCO2 26 mmHg, HCO3 12 mEq/L; Na 140, Cl 104. The best interpretation is:

A. Respiratory acidosis with renal compensation
B. High anion gap metabolic acidosis with appropriate compensation
C. Normal anion gap metabolic acidosis
D. Metabolic alkalosis
E. Mixed respiratory and metabolic acidosis

ANSWER KEY WITH ONE-LINE REASONS

1. **B** Myoglobin from crushed muscle causes pigment ATN.
2. **C** Highest oxygen demand; outer medulla is hypoxic.
3. **D** The diuretic phase wastes potassium.
4. **A** Healthy tubules conserve sodium: FENa < 1%.
5. **E** Casts form only in tubules: renal origin.
6. **B** Thyroidisation + calyceal deformity.
7. **A** Hypersensitivity AIN: eosinophils, tubulitis, granulomas.
8. **C** DOSA: diabetes, obstruction, sickle cell, analgesics.
9. **E** PKD1 (16p13.3) causes about 85%.
10. **D** ARPKD almost always has congenital hepatic fibrosis.
11. **C** Acquired cystic disease predisposes to RCC.
12. **B** AG 24; expected PaCO2 1.5 × 12 + 8 = 26 ± 2.

VIVA Rapid-Fire Viva Voce

Q Commonest cause of AKI?	Acute tubular injury (necrosis).
Q Why is FENa low in prerenal azotaemia?	Healthy tubules avidly reabsorb sodium.
Q Which casts suggest ATN?	Muddy brown granular and epithelial cell casts.
Q What is tubulorrhexis?	Rupture of tubular basement membrane (ischaemic ATN).
Q Three nephrotoxins causing ATN?	Gentamicin, radiocontrast, mercury (also ethylene glycol).
Q Commonest organism of acute pyelonephritis?	Escherichia coli, ascending infection.
Q What is thyroidisation of the kidney?	Dilated tubules filled with colloid casts: chronic pyelonephritis.
Q What is reflux nephropathy?	Chronic pyelonephritis from childhood reflux plus infection.
Q Causes of papillary necrosis?	Diabetes, obstruction, sickle cell disease, analgesics.
Q Extrarenal lesions of ADPKD?	Liver cysts, berry aneurysm, mitral valve prolapse.
Q Normal anion gap?	About 8–12 mEq/L, mostly albumin.
Q Two causes of normal-AG acidosis?	Diarrhoea and renal tubular acidosis.

OSPE Practice Stations

STATION 1 · LAB DATA (RFT) — 2 minutes

TASK

Man, 70, after 4 days of diarrhoea: BUN 56 mg/dL, creatinine 2.0 mg/dL, urine Na 12 mEq/L, FE_{Na} 0.5%, SG 1.028, hyaline casts only. (a) Type of AKI? (b) Two supporting values. (c) Urine findings if ATN developed?

ANSWER

(a) Prerenal azotaemia. (b) BUN:creatinine 28 (> 20), FE_{Na} < 1%, urine Na < 20, high SG. (c) Urine Na > 40, FE_{Na} > 2%, SG about 1.010, muddy brown granular casts.

STATION 2 · SLIDE 2 minutes

TASK

Kidney: patchy interstitial fibrosis with lymphocytes and plasma cells; many tubules are atrophic, others dilated and filled with homogeneous pink colloid; the pelvic mucosa is fibrosed; glomeruli show periglomerular fibrosis. (a) Diagnosis? (b) Name the colloid-tubule change. (c) Commonest underlying cause.

ANSWER

(a) Chronic pyelonephritis. (b) Thyroidisation. (c) Vesicoureteral reflux with recurrent infection (reflux nephropathy).

RAPID REVISION The Night-Before Page

15 ONE-LINERS

01 Tubulointerstitial disease = polyuria, nocturia, salt wasting, acidosis.

02 Creatinine rises late; use eGFR. BUN:Cr > 20 = prerenal.

03 Prerenal: FE_{Na} < 1%, urine Na < 20; ATN: FE_{Na} > 2%, Na > 40.

04 RBC casts = GN; WBC casts = pyelonephritis/AIN; waxy = CKD.

05 ATN is the commonest cause of AKI.

06 Ischaemic ATN: patchy, straight PCT + TAL, tubulorrhexis.

07 ATN phases: initiation, maintenance (high K), recovery (low K).

08 Acute pyelonephritis: E. coli, ascending, abscesses, glomeruli spared.

09 Chronic pyelonephritis: coarse polar scars, blunt calyces, thyroidisation.

10 Papillary necrosis: diabetes, obstruction, sickle cell, analgesics.

11 Drug AIN: 2 weeks, fever, rash, eosinophils, granulomas.

12 ADPKD: PKD1 16p polycystin-1; huge kidneys, berry aneurysms.

13 ARPKD: PKHD1 fibrocystin; smooth kidneys, hepatic fibrosis.

14 Dialysis cysts → renal cell carcinoma.

15 Metabolic acidosis: Winter $PaCO_2 = 1.5 \times HCO_3 + 8 \pm 2$; check AG.

MEMORY HOOKS

- **DOSA**: papillary necrosis (Diabetes, Obstruction, Sickle, Analgesics)
- **ROME**: Respiratory Opposite, Metabolic Equal
- **MUDPILES**: high anion gap acidosis
- **"Healthy but thirsty"** prerenal kidney vs **"sick and leaky"** ATN kidney

TEST YOURSELF: FILL THE GAPS

1. Casts are moulds made of _____ protein.
2. ATN casts are described as _____.
3. Commonest form of chronic pyelonephritis: _____ nephropathy.
4. ARPKD gene: _____.
5. Normal anion gap: _____ mEq/L.
6. Recovery phase of ATN risks _____.

Answers: Tamm–Horsfall · muddy brown granular · reflux · PKHD1 · 8–12 · hypokalaemia

DRAW FROM MEMORY

The pathogenesis of ATN: tubular and vascular arms meeting at a falling GFR (Figure 17.1).

DRAW FROM MEMORY

The five-step ABG flowchart with the four primary disorders and the anion gap branch.

SPACED REVISION ☐ Day 1 ☐ Day 3 ☐ Day 7 ☐ Day 14 ☐ Day 30

18

Renal Tumours, Urothelial Carcinoma & Urolithiasis

Painless haematuria in a middle-aged smoker is cancer until proved otherwise: a golden-yellow clear cell carcinoma creeping up the renal vein, or a papillary bladder tumour that keeps coming back. Painful haematuria in a young man in the Pakistani summer is usually a stone.

BY THE END OF THIS CHAPTER YOU CAN

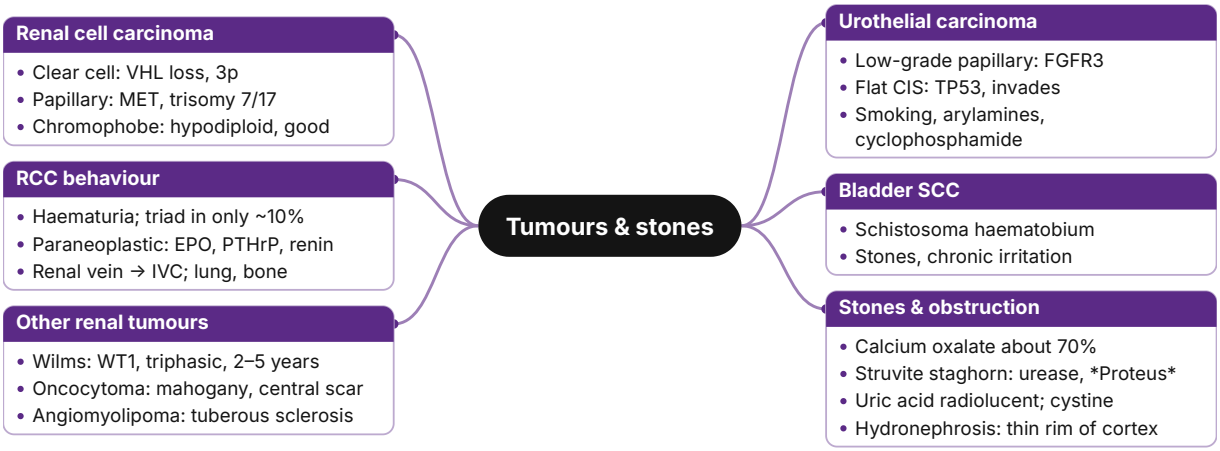
1. Classify renal cell carcinoma and link each subtype to its gene (VHL, MET) and morphology
2. Describe the clinical features, paraneoplastic syndromes and spread of renal cell carcinoma
3. Describe Wilms tumour, its syndromes (WT1) and triphasic histology, and distinguish it from neuroblastoma
4. List the risk factors for bladder cancer and explain the papillary and flat (CIS) pathways of urothelial carcinoma
5. Grade and stage urothelial tumours and recognise urothelial carcinoma on a slide
6. Classify urinary stones, explain how they form and describe hydronephrosis

EXAM RADAR

Theory SEQs	★★★
BCQs / MCQs	★★★
Viva voce	★★☆
OSPE / slides	★★★

RCC (types, clear cell morphology, paraneoplastic syndromes) and bladder cancer risk factors are classic SEQs; RCC and TCC are lab slides; stone types are favourite BCQs.

ROBBINS LINK Robbins Basic Pathology — *The Kidney and Its Collecting System*: renal tumours, urolithiasis, obstructive uropathy; *The Lower Urinary Tract and Male Genital System*: urothelial tumours of the bladder.



Chapter at a glance. Three tumours, two bladder cancers and one stone table.

18.1 Renal cell carcinoma and benign renal tumours

DEFINITION

Renal cell carcinoma (RCC) is a malignant tumour of **renal tubular epithelium**, arising in the cortex. It forms about 3% of adult cancers, peaks in the **sixth and seventh decades** and is about **twice as common in men**.

RISK FACTORS

- **Tobacco** (the most important), **obesity**, **hypertension**
- Asbestos, petroleum products, heavy metals; unopposed oestrogen
- **End-stage kidneys with acquired cystic disease** (Chapter 17), tuberous sclerosis

FAMILIAL RCC (A FEW PER CENT)

- **von Hippel–Lindau**: bilateral clear cell RCC, cerebellar and retinal haemangioblastomas
- Hereditary papillary RCC (**MET**); Birt–Hogg–Dubé (FLCN)

CLASSIFICATION OF RENAL CELL CARCINOMA (RARER TYPES: COLLECTING DUCT, XP11 TRANSLOCATION)

Type	Share; origin	Genetics	Morphology	Behaviour
Clear cell	Commonest (about 70%); proximal tubule	**VHL** loss with **3p deletion**; also PBRM1, SETD2, BAP1	Golden-yellow, variegated; clear cells, delicate vessels	Renal vein invasion; most metastases
Papillary	10–15%; proximal tubule	**MET** activation; **trisomy 7, 16, 17**, loss of Y	Multifocal, bilateral; papillae with foamy macrophages, psammoma bodies	Seen in dialysis cystic disease
Chromophobe	About 5%; intercalated cells of collecting duct	Extreme **hypodiploidy** (loss of whole chromosomes)	Tan-brown; pale cells with **crisp membranes** and **perinuclear halos**	**Excellent** prognosis

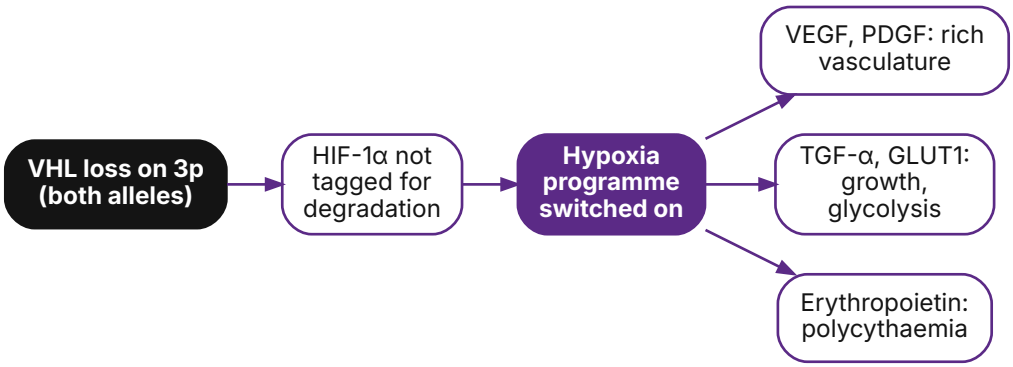


Figure 18.1 · Why VHL loss makes a clear, vascular tumour. HIF is meant to be destroyed when oxygen is plentiful.

CONCEPT CLEAR

One gene explains the whole picture. The VHL protein is part of a ubiquitin ligase that destroys **hypoxia-inducible factor (HIF)**. Without it, the tumour cell behaves as if permanently starved of oxygen: it builds **new blood vessels** (hence haemorrhage and early spread into veins), switches to glycolysis and stores **glycogen and lipid** (dissolved in processing, leaving **clear** cytoplasm and a **yellow** cut surface), and may secrete **erythropoietin**. Anti-VEGF drugs and mTOR inhibitors are used against advanced disease.

Clear cell carcinoma: morphology

GROSS

- Solitary, spherical, **3–15 cm**, anywhere in the cortex; well demarcated, bulging
- **Variegated** cut surface: **golden-yellow** (lipid) with grey-white, red-brown **haemorrhagic**, necrotic and cystic areas
- May grow as a **solid column up the renal vein into the IVC**, even the right heart

MICROSCOPY

- Sheets, nests or tubules of round to polygonal cells with abundant **clear cytoplasm** (glycogen, lipid) and distinct membranes
- Small round nuclei; nucleolar prominence sets the **WHO/ISUP grade (1–4)**
- Rich, **delicate branching vascular network** between nests
- Sarcomatoid (spindle) change = high grade, poor prognosis

CLINICAL FEATURES AND SPREAD

- Classic triad of **haematuria, flank pain, palpable mass** in only about 10%; **haematuria** is the most reliable sign
- "The internist's tumour": fever, weight loss, paraneoplastic syndromes
- About a quarter have **metastases at presentation**: **lungs** (cannon-ball deposits, over half), **bones** (lytic, about a third), nodes, liver, adrenals, brain

PARANEOPLASTIC SYNDROMES

- **Polycythaemia**: erythropoietin
- **Hypercalcaemia**: PTH-related peptide
- **Hypertension**: renin
- **Cushing syndrome**: ACTH-like hormone
- Feminisation or masculinisation: gonadotrophins
- Eosinophilia, leukaemoid reaction, amyloidosis

- Left tumour in the renal vein may cause a **left varicocele**
- 5-year survival about 70% (up to 95% without metastases)

MEMORY HOOK

RCC secretes a "PEAR"

PTHrP → hypercalcaemia · **E**rythropoietin → polycythaemia · **A**CTH-like hormone → Cushing syndrome · **R**enin → hypertension

LAB SPOTTER

RCC slide in 10 seconds: nests and sheets of cells with **water-clear cytoplasm** and sharp cell borders, each nest wrapped by **thin-walled capillaries**, with fresh haemorrhage around. Small round dark nuclei. Look-alikes: adrenal cortex (finely vacuolated, not empty), clear cell areas of other tumours. Ask about **VHL/3p** and **polycythaemia**.

BENIGN RENAL TUMOURS

Tumour	Key features
Papillary adenoma	Tiny (< 1.5 cm) cortical papillary tumour, common at autopsy
Oncocytoma	5–15% of renal tumours; from intercalated cells. **Mahogany-brown**, encapsulated, **central stellate scar**. Large **eosinophilic granular cells packed with mitochondria**. Benign; chief mimic is chromophobe RCC
Angiomyolipoma	**Vessels + smooth muscle + fat** (a PEComa). In 25–50% of patients with **tuberous sclerosis** (TSC1/TSC2). Main danger: spontaneous **retroperitoneal haemorrhage**

18.2 Wilms tumour (nephroblastoma)

The commonest primary renal tumour of **childhood**, peaking at **2–5 years**; about 95% occur before 10 years and 5–10% are bilateral. It arises from persistent embryonic kidney (metanephric blastema); **nephrogenic rests** in the adjacent kidney are its precursor lesions.

SYNDROMES WITH A HIGH RISK OF WILMS TUMOUR

Syndrome	Features	Genetics
WAGR	**W**ilms, **A**niridia, **G**enital anomalies, intellectual disability (**R**etardation); about a third develop Wilms	Deletion of **11p13** (**WT1** + PAX6)
Denys–Drash	Gonadal dysgenesis, early nephropathy (diffuse mesangial sclerosis); about 90% develop Wilms	Dominant-negative **WT1** point mutation
Beckwith–Wiedemann	Organomegaly, macroglossia, hemihypertrophy, omphalocele; also hepatoblastoma, adrenal tumours	**11p15.5 (WT2)**: loss of imprinting of **IGF2**

Sporadic tumours: WT1 or **β-catenin (CTNNB1)** mutations in about 10% each; microRNA-processing genes in others.

GROSS

- **Large, solitary, well-circumscribed** mass; 5–10% bilateral or multicentric
- **Soft, homogeneous, tan-grey** ("fish-flesh") cut surface with haemorrhage, cysts and necrosis

MICROSCOPY

- **Triphasic**: (1) **blastema**: sheets of small blue cells; (2) **epithelial**: abortive tubules and glomeruli; (3) **stroma**: fibrocytic or myxoid, sometimes skeletal muscle
- **Anaplasia** (about 5%): huge hyperchromatic nuclei, abnormal mitoses; **TP53** mutation, chemoresistance

Clinical: a large **abdominal mass** found by the mother while bathing the child, sometimes with haematuria, pain or hypertension; spreads to the **lungs**. With surgery, chemotherapy and radiotherapy, long-term survival is about **90%**; anaplasia is the main adverse histological feature.

WILMS TUMOUR VS NEUROBLASTOMA: THE TWO BIG ABDOMINAL MASSES OF EARLY CHILDHOOD

Feature	Wilms tumour	Neuroblastoma
Origin; age	Kidney; **2–5 years**	Adrenal medulla / sympathetic chain; median about **18 months**
Mass	Smooth, flank; rarely crosses midline	Irregular, often **crosses the midline**; **calcification** common
Urine catecholamines	Normal	**Raised VMA and HVA**
Histology; gene	Triphasic; **WT1**, WT2	Small round cells, **Homer Wright rosettes**; **MYCN** amplification

18.3 Urothelial tumours of the bladder

About **90%** of bladder cancers are **urothelial (transitional cell) carcinomas**; squamous cell carcinoma (3–7% in the West, far more where schistosomiasis is endemic) and adenocarcinoma make up the rest. Men are affected about three times as often, mostly at **50–80 years**. The same tumour arises, less often, in the renal pelvis and ureter.

RISK FACTORS

- **Cigarette smoking**: the most important; raises risk 3- to 7-fold
- **Arylamines** (2-naphthylamine, benzidine): **aniline dye**, rubber, leather and textile workers; latency 15–40 years
- Long-term **cyclophosphamide** (acrolein; also haemorrhagic cystitis)
- Pelvic **radiation**; arsenic; analgesic (phenacetin) abuse for renal pelvis tumours
- **Schistosoma haematobium** → mainly **squamous** carcinoma

PAKISTAN FOCUS

Tobacco is the bladder's carcinogen. In the Global Adult Tobacco Survey (2014), about **19%** of Pakistani adults used tobacco in some form (men about 32%, women about 6%). Aromatic amines from smoke are excreted and concentrated in urine, which then bathes the urothelium for hours, so stopping smoking is the single most effective prevention.

Source: GATS Pakistan 2014.

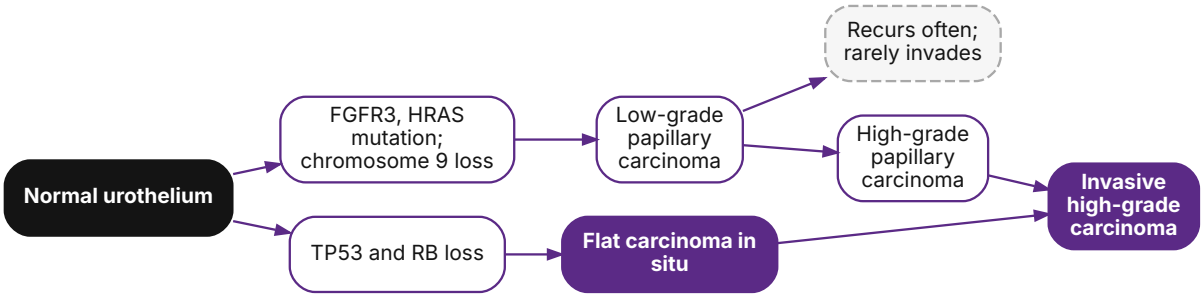


Figure 18.2 · Two roads to urothelial carcinoma. The papillary road recurs; the flat road invades.

WHO/ISUP GRADING OF PAPILLARY UROTHELIAL NEOPLASMS

Lesion	Microscopy	Behaviour
Urothelial papilloma	Delicate fronds lined by normal-looking urothelium	Benign; rarely recurs
PUNLMP (papillary urothelial neoplasm of low malignant potential)	Thicker urothelium, slight nuclear enlargement, orderly	Recurs; almost never invades
Low-grade papillary carcinoma	Orderly, cohesive; mild variation in nuclear size; few mitoses	Recurs; invasion uncommon
High-grade papillary carcinoma	Discohesive cells, **large hyperchromatic pleomorphic nuclei**, loss of polarity, many mitoses	**Invasion and metastasis** common
Carcinoma in situ (CIS)	**Flat** urothelium with cytologically malignant cells; cells fall apart and **shed into urine**	Often multifocal; untreated, 50–75% become **muscle-invasive**

GROSS

- **Papillary**: soft, frond-like, cauliflower growths, mostly on the **lateral and posterior walls near the trigone**; often **multiple**
- CIS: flat, velvety red patches or invisible
- Invasive: nodular, ulcerated, firm, thickening the wall

MICROSCOPY

- Branching **papillae with fibrovascular cores** lined by thickened **multilayered urothelium**
- Atypia grades the tumour (table above)
- Invasion: nests of atypical cells in lamina propria or **muscularis propria** (detrusor)

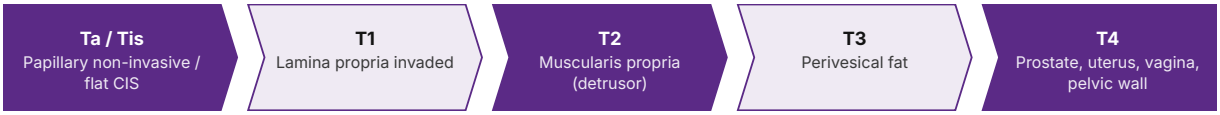


Figure 18.3 · Staging (pT) of bladder carcinoma. Invasion of the muscularis propria is the most important prognostic line.

Clinical: painless haematuria, often the only symptom; CIS causes frequency, urgency and dysuria. A tumour over a ureteric orifice causes **hydronephrosis**. The whole urothelium is exposed to the same urinary carcinogens (**field effect**), so tumours are **multifocal and recur** at new sites. Diagnosis: cystoscopy and biopsy; **urine cytology** is sensitive for high-grade tumours and CIS. Treatment: transurethral resection; **intravesical BCG** for CIS and high-risk superficial tumours; radical cystectomy for muscle invasion.

EXAM TRAP

Squamous vs urothelial. Schistosoma haematobium eggs in the bladder wall cause chronic inflammation → **squamous metaplasia → dysplasia → keratinising SCC**, usually invasive at diagnosis (Egypt, Sudan). Long-standing stones and catheters do the same. **Adenocarcinoma** arises at the dome from **urachal remnants** or in bladder exstrophy.

LAB SPOTTER

Transitional cell carcinoma slide in 10 seconds: finger-like **papillae with thin fibrovascular cores** covered by **thick, many-layered urothelium**. Then grade: orderly cells with uniform nuclei = low grade; crowded, discohesive cells with large dark pleomorphic nuclei and mitoses at all levels = high grade. Always check the base for **invasion of muscle**.

18.4 Urolithiasis

DEFINITION

Urolithiasis is the formation of calculi anywhere in the urinary collecting system, most often in the **kidney**. It is common, commoner in **men**, peaks at **20–30 years** and runs in families.

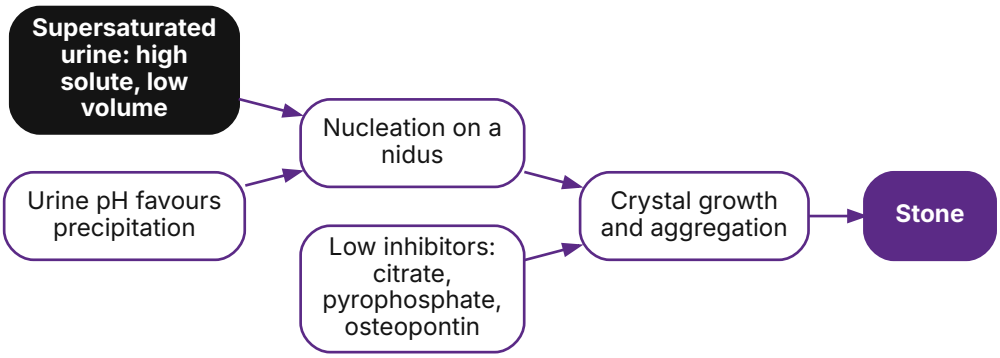


Figure 18.4 · How a stone forms. Supersaturation is the engine; a nidus and missing inhibitors let it run.

TYPES OF URINARY STONES

Stone (share)	Causes	Urine pH; X-ray	Points to remember
Calcium oxalate ± phosphate (about 70%)	**Idiopathic hypercalciuria** (commonest); hypercalcaemia (hyperparathyroidism, vitamin D, sarcoidosis, bone disease); **hyperoxaluria** (enteric: Crohn disease, gut resection; primary); hyperuricosuria; hypocitraturia	Phosphate in alkaline urine; **radio-opaque**	Envelope-shaped (oxalate) crystals; often small, jagged

Stone (share)	Causes	Urine pH; X-ray	Points to remember
Struvite (Mg NH_4 PO_4, "triple phosphate") (about 15%)	**Urease-producing bacteria** (*Proteus*, some staphylococci) split urea to ammonia	**Alkaline**; radio-opaque	Largest stones: **staghorn calculi**; coffin-lid crystals; women (UTI)
Uric acid (5–10%)	Gout, leukaemia and chemotherapy (high cell turnover); over half have neither, only **persistently acid urine (pH < 5.5)**	Acid; **radiolucent**	Dissolve with urine alkalinisation; rhomboid crystals
Cystine (1–2%)	**Cystinuria** (AR): poor tubular reabsorption of **COLA**: Cystine, Ornithine, Lysine, Arginine	Acid; faintly opaque	**Hexagonal** crystals; may form staghorns

Morphology: unilateral in about 80%; usually in the **renal pelvis, calyces and bladder**. Most are small (2–3 mm), smooth or jagged; a **staghorn calculus** is a branching cast of the pelvicalyceal system.

Clinical: silent, or **renal colic** (severe loin pain radiating to the groin) as a stone passes down the ureter, with **haematuria**. Complications: **obstruction and hydronephrosis**, **infection** (pyelonephritis, pyonephrosis), ulceration and, in the bladder, squamous metaplasia.

CONCEPT CLEAR

Small stones hurt; big stones hide. A small stone can enter the ureter, where it obstructs and triggers violent peristalsis: colic. A large staghorn cannot leave the pelvis, so it may stay silent for years, announcing itself only by **haematuria, infection** or slow destruction of the kidney.

PAKISTAN FOCUS

Pakistan lies in the Afro-Asian **"stone belt"**: hot summers, outdoor labour and low fluid intake keep urine concentrated, so urolithiasis is very common. **Calcium oxalate** stones predominate, and **bladder stones in children** are still seen in poor rural areas. The cheapest prevention is drinking enough water to pass at least 2 litres of urine a day.

18.5 Hydronephrosis

DEFINITION

Hydronephrosis is dilatation of the renal pelvis and calyces, with progressive **atrophy of the parenchyma**, caused by obstruction to the outflow of urine.

CAUSES

- **Congenital**: pelviureteric junction obstruction, **posterior urethral valves**, urethral stricture, severe reflux
- **Stones**, sloughed papillae, blood clot
- **Tumours**: BPH, carcinoma of prostate, bladder or cervix; retroperitoneal lymphoma
- Inflammation, retroperitoneal fibrosis; **neurogenic bladder**; pregnancy (mild)
- Obstruction **below the bladder neck → bilateral**; above it → unilateral

PATHOGENESIS

Filtration continues after obstruction; filtrate leaks back into the interstitium and the **raised pelvic pressure** is transmitted to the collecting ducts. This compresses the medullary vessels, causing venous stasis and **ischaemic atrophy**, first of the papillae and then of the cortex. **Tubular function (concentration) fails first**; GFR falls later.

GROSS

- Pelvis and calyces **dilated**; papillae flattened, then **cupped**
- Parenchyma compressed to a **thin rim**; late, a large thin-walled cystic sac

MICROSCOPY

- Dilated, then **atrophic tubules**; interstitial fibrosis and inflammation
- **Glomeruli preserved** until late

Clinical: acute obstruction is painful; unilateral obstruction may be silent while the other kidney copes. Partial bilateral obstruction causes **polyuria** (concentrating defect), distal tubular acidosis, salt wasting and hypertension; complete bilateral obstruction causes **anuria**, and relief is followed by a **post-obstructive diuresis**.

EXAM ZONE Past-Paper Pattern SEQs — Solved

PROF PATTERN **a)** Classify renal cell carcinoma with the genetic basis of each type. **b)** Describe the morphology of clear cell carcinoma and list its paraneoplastic syndromes. **2 + 3 marks**

MODEL ANSWER PLAN

a) Clear cell (~70%, VHL/3p), papillary (10–15%, MET, trisomy 7/17), chromophobe (~5%, hypodiploid, good prognosis), collecting duct and Xp11 translocation (rare). **b)** Gross: solitary 3–15 cm, golden-yellow variegated with haemorrhage, necrosis, cysts; renal vein invasion. Micro: clear cells in nests with delicate vessels; WHO/ISUP grade. Paraneoplastic: polycythaemia, hypercalcaemia, hypertension, Cushing, feminisation, amyloidosis.

Score booster: draw the VHL → HIF → VEGF/EPO chain (Figure 18.1); it explains clear cells, vascularity and polycythaemia in one line.

MODULE PATTERN A 3-year-old boy has a smooth flank mass noticed by his mother; he has no irises. **a)** Most likely diagnosis? **b)** Name the syndrome and its genetic basis. **c)** Describe the microscopy. **1 + 2 + 2 marks**

MODEL ANSWER PLAN

a) Wilms tumour. **b)** WAGR syndrome: deletion of 11p13 including WT1 and PAX6 (aniridia); mention Denys–Drash (WT1) and Beckwith–Wiedemann (WT2, IGF2). **c)** Triphasic: blastema, abortive tubules/glomeruli, stroma; anaplasia predicts a poor response.

PROF PATTERN **a)** Enumerate the risk factors of carcinoma of the urinary bladder. **b)** Describe the morphology and grading of urothelial carcinoma. **2 + 3 marks**

MODEL ANSWER PLAN

a) Smoking, arylamines (aniline dyes, rubber), cyclophosphamide, radiation, analgesic abuse, Schistosoma haematobium (SCC), chronic irritation. **b)** Gross papillary, flat or invasive, lateral walls/trigone, multifocal. Micro: papillae with fibrovascular cores, multilayered urothelium; WHO/ISUP: papilloma, PUNLMP, low-grade, high-grade, CIS. Two pathways: FGFR3 papillary vs TP53 flat CIS. Stage by depth: muscularis propria is the key line.

Score booster: add the two-pathway diagram and the pT chevron; both are quick to draw.

CLASSIC **a)** Name the types of renal stones and the cause of each. **b)** Describe the pathogenesis and complications of urolithiasis. **2 + 3 marks**

MODEL ANSWER PLAN

a) Calcium oxalate/phosphate (~70%: hypercalciuria, hyperoxaluria); struvite (~15%: urease, Proteus, staghorn); uric acid (5–10%: gout, leukaemia, acid urine); cystine (1–2%: cystinuria). **b)** Supersaturation, pH, low inhibitors, nidus → nucleation → growth. Complications: colic, haematuria, hydronephrosis, infection, squamous metaplasia.

BCQs One-Best-Answer Practice

1 A 30-year-old man with cerebellar haemangioblastoma develops bilateral renal tumours. The tumours most likely show:

A. Papillae with foamy macrophages
B. Clear cells with delicate vasculature
C. Triphasic blastemal histology
D. Cells with perinuclear halos
E. Vessels, smooth muscle and fat

2 A 60-year-old man with a renal mass has a haemoglobin of 19 g/dL. The cause is tumour secretion of:

A. Renin
B. PTH-related peptide
C. ACTH
D. Erythropoietin
E. Gonadotrophin

3 Papillary renal cell carcinoma is linked to activation of:

A. MET
B. VHL
C. WT1
D. TSC2
E. FGFR3

4 A well-circumscribed mahogany-brown renal tumour with a central stellate scar, made of large eosinophilic cells full of mitochondria, is:

A. Chromophobe RCC
B. Clear cell RCC
C. Angiomyolipoma
D. Wilms tumour
E. Oncocytoma

5 A young woman with seizures, facial angiofibromas and a fatty renal mass is at risk of which complication of the mass?

A. Polycythaemia
B. Hypercalcaemia
C. Spontaneous retroperitoneal haemorrhage
D. Lung metastases
E. Renal vein thrombosis

6 A 4-year-old with aniridia and genital anomalies is most at risk of a tumour caused by deletion at:

A. 11p15.5
B. 3p25
C. 17p13
D. 11p13
E. 13q14

7 An 18-month-old has an irregular abdominal mass crossing the midline with stippled calcification and raised urinary VMA. The diagnosis is:

A. Wilms tumour
B. Neuroblastoma
C. Mesoblastic nephroma
D. Hepatoblastoma
E. ARPKD

8 The single most important risk factor for urothelial carcinoma of the bladder is:

A. Cigarette smoking
B. Cyclophosphamide
C. Schistosomiasis
D. Radiation
E. Chronic catheterisation

9 A farmer from the Nile valley with long-standing haematuria has an invasive bladder tumour. Histology most likely shows:

A. Low-grade papillary urothelial carcinoma
B. Adenocarcinoma
C. Small cell carcinoma
D. Keratinising squamous cell carcinoma
E. Urothelial papilloma

10 Flat urothelial carcinoma in situ differs from low-grade papillary carcinoma in that CIS:

A. Has FGFR3 mutations
B. Never sheds cells into urine
C. Is always solitary
D. Rarely progresses
E. Is high grade with TP53 changes and often invades

11 A woman with recurrent Proteus UTIs has a large branching stone filling the pelvis and calyces. It is composed of:

A. Calcium oxalate
B. Uric acid
C. Magnesium ammonium phosphate
D. Cystine
E. Xanthine

12 A man with gout has renal colic; the stone is invisible on plain X-ray. Its most likely composition is:

A. Calcium oxalate
B. Struvite
C. Calcium phosphate
D. Cystine
E. Uric acid

ANSWER KEY WITH ONE-LINE REASONS

1. **B** VHL syndrome: bilateral, multiple clear cell RCC.
2. **D** EPO from RCC causes polycythaemia.
3. **A** MET proto-oncogene; trisomy 7, 16, 17.
4. **E** Benign oncocytoma from intercalated cells.
5. **C** Angiomyolipoma of tuberous sclerosis bleeds.
6. **D** WAGR: 11p13 deletion of WT1 and PAX6.
7. **B** Neuroblastoma: adrenal, crosses midline, calcified, catecholamines.
8. **A** Smoking raises risk 3- to 7-fold.
9. **D** Schistosoma haematobium → squamous metaplasia → SCC.

10. **E** CIS: TP53/RB pathway; 50–75% become invasive untreated.
11. **C** Urease-split urea makes alkaline urine: struvite staghorn.
12. **E** Uric acid stones are radiolucent and form in acid urine.

VIVA Rapid-Fire Viva Voce

Q Commonest type of RCC and its gene?	Clear cell carcinoma; VHL on 3p.
Q What does VHL protein normally do?	Targets HIF for destruction in normoxia.
Q Classic triad of RCC?	Haematuria, flank pain, palpable mass (only ~10%).
Q Why is RCC called the internist's tumour?	Fever and many paraneoplastic syndromes.
Q Commonest sites of RCC metastasis?	Lungs and bones.
Q Benign renal tumour with central scar?	Oncocytoma.
Q Name the three components of Wilms tumour.	Blastema, epithelium (tubules), stroma.
Q Adverse histology in Wilms tumour?	Anaplasia (TP53 mutation).
Q Two molecular pathways of urothelial carcinoma?	FGFR3 papillary (low grade); TP53/RB flat CIS (invasive).
Q Most important prognostic factor in bladder cancer?	Depth of invasion (muscularis propria) and stage.
Q Which stone is radiolucent?	Uric acid.
Q Organism behind staghorn calculi?	Proteus (urease producers).

OSPE Practice Stations

STATION 1 · SLIDE — 2 minutes

TASK

Kidney tumour: nests of polygonal cells with empty-looking cytoplasm and distinct borders, separated by a fine network of thin-walled vessels; areas of haemorrhage. (a) Identify. (b) Why is the cytoplasm clear? (c) Name the gene involved.

ANSWER

(a) Clear cell renal cell carcinoma. (b) Glycogen and lipid dissolved during processing. (c) VHL (chromosome 3p).

STATION 2 · SLIDE — 2 minutes

TASK

Bladder biopsy: branching papillae with thin fibrovascular cores covered by thick, many-layered urothelium; cells are crowded with large, hyperchromatic, pleomorphic nuclei and mitoses. (a) Diagnosis and grade. (b) Two risk factors. (c) What must the pathologist look for at the base?

ANSWER

(a) High-grade papillary urothelial (transitional cell) carcinoma. (b) Smoking, aniline dyes, cyclophosphamide. (c) Invasion of the muscularis propria.

STATION 3 · GROSS SPECIMEN — 1 minute

TASK

Bisected kidney: a large branching stone fills a dilated pelvicalyceal system; the parenchyma is reduced to a thin rim. (a) Name the stone and the change in the kidney. (b) Its composition and cause. (c) One complication.

ANSWER

(a) Staghorn calculus with hydronephrosis. (b) Struvite ($Mg\ NH_4\ PO_4$) from urease-producing Proteus infection. (c) Pyonephrosis, pyelonephritis or renal failure.

RAPID REVISION The Night-Before Page

15 ONE-LINERS

01 RCC: tubular epithelium; smoking, obesity, hypertension, dialysis cysts.

02 Clear cell RCC (~70%): VHL/3p → HIF → VEGF, EPO.

03 Papillary RCC: MET, trisomy 7/17, multifocal.

04 Chromophobe: hypodiploid, halos, best prognosis.

05 Clear cell gross: golden-yellow, variegated, renal vein invasion.

06 RCC triad in ~10%; haematuria is commonest sign.

07 Paraneoplastic: EPO, PTHrP, renin, ACTH.

08 Oncocytoma: mahogany, central scar, mitochondria.

09 Angiomyolipoma: fat + muscle + vessels; tuberous sclerosis.

10 Wilms: 2–5 years, triphasic, WT1 (11p13), WT2 (11p15).

11 Urothelial carcinoma: smoking, arylamines, cyclophosphamide.

12 Papillary pathway FGFR3 recurs; flat CIS TP53 invades.

13 Schistosoma haematobium → bladder SCC.

14 Stones: Ca oxalate 70, struvite 15, urate 5–10, cystine 1–2%.

15 Hydronephrosis: concentrating defect first; thin cortical rim.

MEMORY HOOKS

- **VHL → HIF**: the tumour thinks it is suffocating (vessels, EPO)
- **WAGR**: Wilms, Aniridia, Genital, Retardation
- **PEAR**: RCC paraneoplastic hormones
- **COLA**: cystinuria amino acids
- **"Small stones hurt, big stones hide"**

TEST YOURSELF: FILL THE GAPS

1. Clear cell RCC loses chromosome _____
2. Papillary RCC gene: _____.
3. Wilms: commonest age _____ years.
4. Low-grade papillary bladder cancer: _____ mutation.
5. Staghorn stones are made of _____.
6. Radiolucent stone: _____.

Answers: 3p · MET · 2–5 · FGFR3 · struvite · uric acid

DRAW FROM MEMORY

The VHL → HIF → VEGF/EPO chain and how it explains a clear, vascular tumour.

DRAW FROM MEMORY

The two pathways of urothelial carcinoma and the pT staging ladder.

SPACED REVISION ☐ Day 1 ☐ Day 3 ☐ Day 7 ☐ Day 14 ☐ Day 30

19

Prostate & Testis

Almost every old man has a big prostate, and many have a small cancer in it; the trick is knowing where each one starts, what symptoms that zone gives and what PSA can and cannot tell you. In the young man the story flips: a painless lump in the testis is a germ cell tumour until proved otherwise, and one of the most curable cancers in medicine.

BY THE END OF THIS CHAPTER YOU CAN

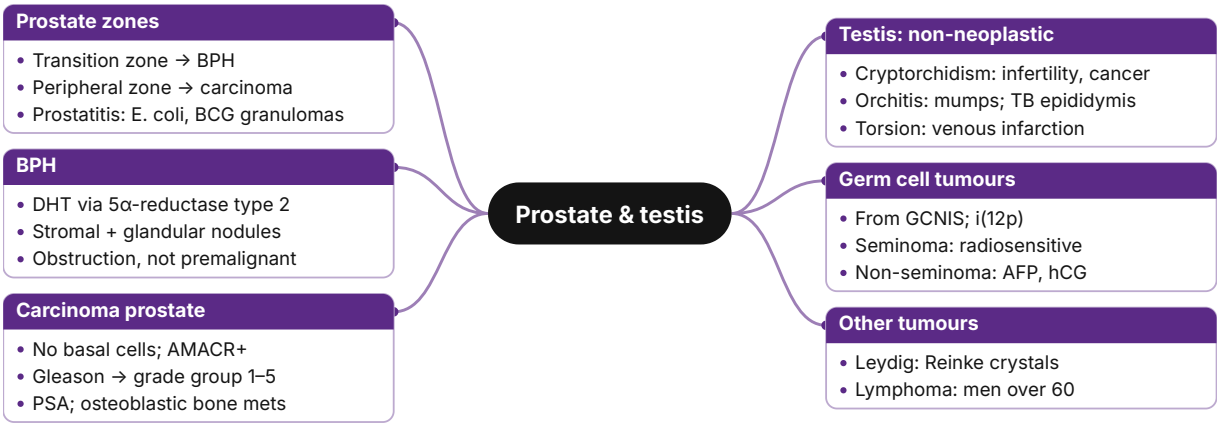
1. Describe the zones of the prostate and classify prostatitis
2. Explain the pathogenesis and morphology of benign prostatic hyperplasia and its complications
3. Describe carcinoma of the prostate: risk factors, molecular changes, morphology, Gleason grading, PSA and spread
4. Differentiate BPH from carcinoma of the prostate
5. Describe cryptorchidism, testicular atrophy, orchitis and torsion
6. Classify testicular tumours and differentiate seminoma from non-seminomatous germ cell tumours, using AFP, hCG and LDH

EXAM RADAR

Theory SEQs	★★★
BCQs / MCQs	★★★
Viva voce	★★★
OSPE / slides	★★★

BPH vs carcinoma prostate and classification of testicular tumours with seminoma are perennial long questions; BPH and seminoma are lab slides; tumour markers are favourite BCQs.

ROBBINS LINK Robbins Basic Pathology — *The Lower Urinary Tract and Male Genital System*: prostate (prostatitis, BPH, carcinoma), testis and epididymis (cryptorchidism, atrophy, inflammation, torsion, testicular neoplasms).



Chapter at a glance. The prostate for the old, the testis for the young.

19.1 Prostate: zones and prostatitis

The adult prostate (about 20 g) has a **peripheral zone** (about 70% of glands, posterior, against the rectum), a **central zone**, a periurethral **transition zone** and anterior fibromuscular stroma. Glands have **two cell layers**: a flat **basal layer** and tall secretory cells. Hence: **hyperplasia starts in the transition zone** and squeezes the urethra early; **carcinoma starts in the peripheral zone**, is felt on rectal examination and obstructs late.

PROSTATITIS

Type	Cause and features
Acute bacterial	Urinary pathogens (**E. coli**, other gram-negative rods, enterococci, staphylococci) reach the gland by **intraprostatic reflux of urine** or after catheterisation, cystoscopy or TURP. Fever, chills, dysuria; **tender, boggy prostate** (do not massage or biopsy: bacteraemia). Abscesses, oedema, neutrophils
Chronic bacterial	Recurrent UTI with the same organism; low back and perineal discomfort; leukocytes and bacteria in prostatic secretions
Chronic abacterial (commonest)	Chronic pelvic pain syndrome: leukocytes in secretions but **negative cultures**; no history of recurrent UTI
Granulomatous	Specific: **BCG instillation** for bladder cancer (commonest in the West), TB, fungi; non-specific reaction to leaked secretions; after TURP

19.2 Benign prostatic hyperplasia (nodular hyperplasia)

DEFINITION

Benign prostatic hyperplasia (BPH) is a **non-neoplastic** proliferation of **stromal and glandular** elements forming nodules in the **transition zone**, enlarging the gland and obstructing urine flow. It is present in about 20% of men at 40 years, 70% at 60 and **90% by 80**. It is **not premalignant**.

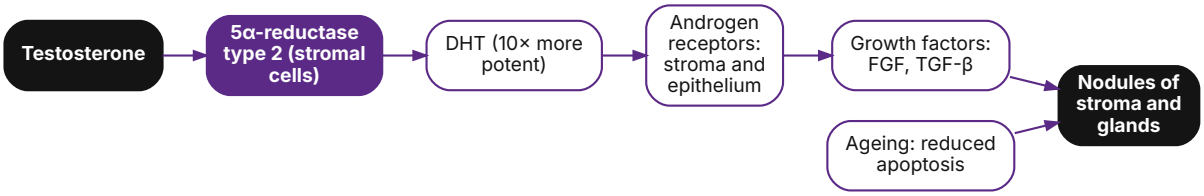


Figure 19.1 · Pathogenesis of BPH. Androgen-driven growth plus slower cell death.

CONCEPT CLEAR

Why the drugs work. BPH needs **dihydrotestosterone**, not testosterone itself; men castrated before puberty never develop it. **Finasteride** blocks 5α-reductase and shrinks the gland over months. Part of the obstruction is **smooth-muscle tone** in the stroma and capsule, so **α1-blockers** (tamsulosin) relieve symptoms within days.

GROSS

- Enlarged, **60–100 g** or more; firm, rubbery
- **Well-circumscribed nodules** bulging from the cut surface in the **transition (periurethral) zone**
- Glandular nodules: yellow-pink, soft, small **cysts with milky fluid**; stromal nodules: grey-white, firm
- Urethra compressed to a **slit**; a median lobe may project into the bladder

MICROSCOPY

- Nodules of **hyperplastic glands** and **fibromuscular stroma**
- Glands lined by tall columnar cells with a **preserved basal cell layer** (two layers)
- **Papillary infoldings** into dilated lumens
- Lumens contain **corpora amylacea** (laminated concretions)
- Foci of infarction with squamous metaplasia

Clinical: lower urinary tract symptoms (**hesitancy, poor stream, intermittency, dribbling, frequency, nocturia, urgency**), acute retention. **Complications:** bladder hypertrophy and **trabeculation**, diverticula, residual urine → **UTI** and stones, **bilateral hydronephrosis** and renal failure. PSA may be mildly raised.

LAB SPOTTER

BPH slide in 10 seconds: nodules of **glands of varying size** with **papillary infoldings**, lined by **two layers** of cells, many containing pink, laminated **corpora amylacea**, set in plenty of **fibromuscular stroma**. No crowding, no prominent nucleoli, no single-layered glands invading between others.

19.3 Carcinoma of the prostate

One of the commonest cancers of older men and a leading cause of male cancer death in the West; a disease of men **over 50**, with small latent cancers common at autopsy.

RISK FACTORS

- **Age**; **race** (highest in men of African ancestry)
- **Family history**; germline **BRCA2**, HOXB13
- **Androgens**: does not develop in men castrated before puberty
- Diet and environment: risk rises in Asian migrants to the West

MOLECULAR KEY

- **TMPRSS2–ETS (ERG) fusion**: an androgen-driven promoter switches on an ETS transcription factor (about half of cases)
- **PTEN** loss → PI3K/AKT activation
- **GSTP1** silenced by promoter hypermethylation (very common, early)
- **AR** amplification or mutation in castration-resistant disease
- Precursor: **high-grade prostatic intraepithelial neoplasia (HGPIN)**: atypical cells with prominent nucleoli in glands that keep a basal layer

GROSS

- About 70% arise in the **peripheral zone**, classically **posteriorly**: palpable as a **hard, irregular nodule** on rectal examination
- Firm, **gritty, grey-white to yellow**, poorly defined
- Local spread to **seminal vesicles**, bladder base (ureteric obstruction); rectum rarely (Denonvilliers fascia)

MICROSCOPY

- **Small, crowded, back-to-back glands** infiltrating stroma
- Glands lined by a **single layer** of cuboidal cells: **no basal cell layer**
- Enlarged nuclei with **prominent nucleoli**
- **Perineural invasion**; intraluminal crystalloids, blue mucin
- IHC: **AMACR positive**; basal markers (**p63**, high-molecular-weight keratin) **negative**; PSA/NKX3.1 confirm prostatic origin of metastases

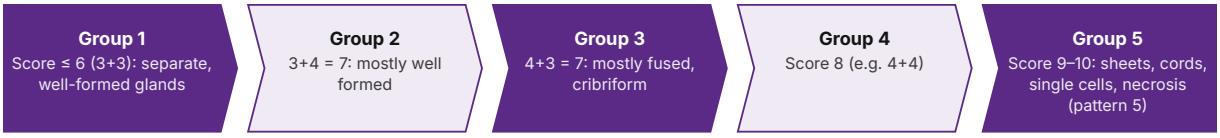


Figure 19.2 · Gleason score to ISUP grade group. Score = commonest + second commonest pattern; low-power architecture only.

Gleason grading uses **gland architecture at low power**, not nuclear atypia; patterns 1–2 are no longer diagnosed, so the lowest practical score is 6. Grade group, stage and PSA are the main prognostic factors.

SPREAD AND CLINICAL PICTURE

- Most are **silent** (peripheral, away from the urethra): found by PSA, rectal examination or metastases
- Lymphatics: **obturator** and pelvic, then para-aortic nodes
- Blood: **bone**, especially the axial skeleton (lumbar spine, pelvis, femur), via the vertebral venous plexus
- Bone metastases are typically **osteoblastic** (sclerotic): back pain, raised **alkaline phosphatase**
- Treatment: surgery, radiotherapy, **androgen deprivation**; active surveillance for small low-grade cancers

LAB VALUES · PSA

Prostate-specific antigen is a serine protease (kallikrein) of prostatic epithelium that liquefies semen. It is **organ-specific, not cancer-specific**.

- Traditional cut-off **4 ng/mL**; 20–40% of organ-confined cancers lie below it
- Also raised in **BPH, prostatitis**, infarction, after catheterisation or biopsy
- Refinements: PSA **density** (per gram of gland), **velocity**, age-specific ranges, **free-to-total ratio** (lower free fraction in cancer)
- Best use: **monitoring** recurrence after treatment

BPH VS CARCINOMA OF THE PROSTATE

Feature	Benign prostatic hyperplasia	Carcinoma prostate
Nature	Hyperplasia; not premalignant	Malignant neoplasm (adenocarcinoma)
Zone	**Transition** (periurethral)	**Peripheral** (posterior)

Feature	Benign prostatic hyperplasia	Carcinoma prostate
Rectal examination	Smooth, rubbery, symmetric enlargement	**Hard, irregular nodule**; fixed when advanced
Symptoms	**Early** obstructive symptoms	Usually silent; **late** obstruction; bone pain
Gross	Circumscribed nodules, cysts with milky fluid	Ill-defined, gritty, grey-white to yellow
Microscopy	Glands + stroma; **two cell layers**; papillary infolding; corpora amylacea	Small crowded glands; **single layer, no basal cells**; prominent nucleoli; **perineural invasion**
Markers	p63 / basal keratin positive	**AMACR** positive; basal markers negative
PSA	Normal or mildly raised	Often high and rising
Spread	None	Seminal vesicles, nodes, **osteoblastic bone metastases**

19.4 Testis: congenital and inflammatory disease

Cryptorchidism

DEFINITION

Cryptorchidism is failure of the testis to descend into the scrotum. It affects about **1% of boys at 1 year**, is usually **unilateral** (bilateral in about a quarter) and the testis most often lies in the **inguinal canal**. Descent has a transabdominal phase and an androgen-dependent inguinoscrotal phase.

MICROSCOPY

- Germ cell development arrests; damage visible **from about 2 years**
- Seminiferous tubules shrink with **thick, hyalinised basement membranes**, ending as hyaline cords
- **Leydig cells spared**, so they look prominent
- Testis small and firm; the descended one may also be abnormal

TESTICULAR ATROPHY: CAUSES

- Atherosclerosis (old age), end-stage orchitis, cryptorchidism
- Hypopituitarism, cachexia, irradiation, oestrogen/antiandrogen therapy, **cirrhosis**
- **Klinefelter syndrome** (47,XXY): primary failure

COMPLICATIONS

- **Infertility**, especially if bilateral
- **Germ cell tumour**: risk raised 3–5-fold, also in the descended testis
- Inguinal hernia, torsion, trauma against the pubis
- **Orchiopexy** early in infancy (ideally by 1–2 years) reduces atrophy and infertility, but not all of the cancer risk

CLINICAL CORRELATION

Torsion: twisting of the spermatic cord **blocks the thin-walled veins first** while arteries still pump blood in, so the testis becomes engorged and undergoes **haemorrhagic (venous) infarction**. **Neonatal** torsion occurs around birth (extravaginal). **Adult** torsion occurs in adolescents with a bilateral **"bell-clapper" deformity** (intravaginal): sudden severe pain, often after exertion or in sleep, absent cremasteric reflex. A **surgical emergency**: untwisted within about 6 hours, the testis usually survives; the other side is fixed too.

ORCHITIS AND EPIDIDYMITIS

Type	Key points
Non-specific epididymitis / orchitis	Children: gram-negative rods with urinary tract anomalies. Sexually active men under 35: **Chlamydia trachomatis**, **Neisseria gonorrhoeae**. Older men: **E. coli**, *Pseudomonas* from UTI. Starts in the epididymis, may spread to the testis
Mumps orchitis	Rare in children; affects a sizeable minority (roughly a quarter) of adult men with mumps, usually unilateral. Interstitial oedema with lymphocytes and plasma cells; patchy atrophy; sterility uncommon
Tuberculosis	**Begins in the epididymis**, then spreads to the testis; often with prostatic and seminal vesicle TB; **caseating granulomas**; may form a scrotal sinus

Type	Key points
Syphilis	**Testis first**: gumma or diffuse interstitial inflammation with plasma cells and **obliterative endarteritis**

PAKISTAN FOCUS

Pakistan is among the **five highest-burden countries for tuberculosis** (incidence about 266 per 100,000). Genital TB presents as a hard, craggy **epididymis**, a "beaded" vas or a scrotal sinus, and can mimic a tumour: send urine for AFB culture or GeneXpert before orchidectomy.

Source: WHO Global Tuberculosis Report (approximate figure).

Scrotal swellings: Hydrocele: serous fluid in the tunica vaginalis, commonest, **transilluminates**; congenital (patent processus vaginalis) or secondary. **Varicocele**: dilated pampiniform plexus ("bag of worms"), **left-sided** (left testicular vein joins the renal vein at a right angle), causes infertility; a new one in an older man suggests left **RCC** (Chapter 18). **Spermatocele**: epididymal cyst with sperm.

19.5 Testicular tumours

A firm, **painless testicular enlargement** is a tumour until proved otherwise. Testicular tumours peak at **15–34 years**, where they are the commonest cancer of men. About **95% are germ cell tumours (GCTs)**: in adults almost all are **malignant**. Sex cord–stromal tumours are usually benign.

CLASSIFICATION (WHO, SIMPLIFIED)

- **GCTs from germ cell neoplasia in situ (GCNIS)**: **seminoma**; non-seminomatous: **embryonal carcinoma**, **yolk sac tumour**, **choriocarcinoma**, **teratoma** (postpubertal); **mixed GCT**
- **GCTs not from GCNIS**: spermatocytic tumour (old men), prepubertal teratoma and yolk sac tumour (children)
- **Sex cord–stromal**: Leydig cell, Sertoli cell tumours
- **Lymphoma**: the commonest testicular tumour in men **over 60** (diffuse large B-cell)

RISK FACTORS

- **Cryptorchidism**; androgen insensitivity, gonadal dysgenesis
- Family history; tumour in the other testis

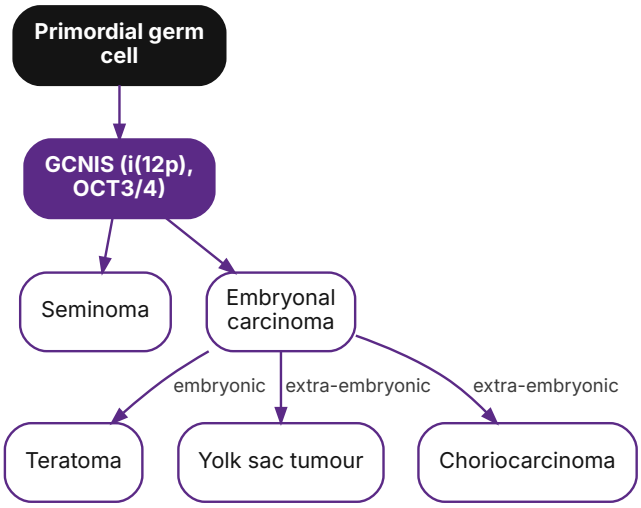


Figure 19.3 · Histogenesis of germ cell tumours.

MOLECULAR KEY

GCNIS and i(12p). Germ cell neoplasia in situ (formerly intratubular germ cell neoplasia) is a row of large atypical germ cells with **clear cytoplasm** and big nuclei along the basement membrane of seminiferous tubules; about half progress to invasive GCT within 5 years. GCNIS and its tumours share an **isochromosome 12p, i(12p)**, found in virtually all postpubertal GCTs, and express the pluripotency factors **OCT3/4 and NANOG**. Seminomas often carry **KIT** mutations.

Seminoma

The commonest GCT (about half of all), peaking in the **30s**; it never occurs in infants. It is the testicular counterpart of the ovarian dysgerminoma.

GROSS

- Bulky, **well-demarcated**, soft
- **Homogeneous grey-white to cream, lobulated** cut surface ("potato-like")
- **No haemorrhage or necrosis** (if present, suspect a non-seminoma component)
- Tunica albuginea usually intact

MICROSCOPY

- **Sheets of uniform large cells** divided into lobules by delicate **fibrous septa with lymphocytes**
- Round to polygonal cells with **clear, glycogen-rich cytoplasm**, distinct membranes, central nucleus with **prominent nucleoli** ("fried-egg" cells)
- **Granulomas** in some
- About 15% contain **syncytiotrophoblasts** (mild hCG rise)
- IHC: OCT3/4, **KIT (CD117)**, PLAP, D2-40 positive

LAB SPOTTER

Seminoma slide in 10 seconds: sheets of **large clear cells** with central nuclei and big nucleoli, like fried eggs, chopped into lobules by **thin fibrous septa packed with lymphocytes**. Two-component pattern (tumour cells + lymphocytes) = seminoma. Ask: **radiosensitive**; AFP **normal**; spreads to **para-aortic** nodes.

NON-SEMINOMATOUS GERM CELL TUMOURS

Tumour	Age; gross	Microscopy	Marker
Embryonal carcinoma	20–30 years; small, **variegated**, haemorrhage and necrosis, invades epididymis	Sheets, glands, papillae of **large anaplastic cells**, indistinct borders, prominent nucleoli; CD30, cytokeratin, OCT3/4 +	± AFP, hCG (mixed)
Yolk sac tumour	**Commonest testicular tumour under 3 years** (good prognosis); in adults, part of mixed GCT; yellow, mucoid	Lace-like (reticular) network; **Schiller–Duval bodies** (glomeruloid: vessel cuffed by tumour cells); **hyaline globules** of AFP and α1-antitrypsin	**AFP**
Choriocarcinoma	Pure form rare; primary often tiny, haemorrhagic; **early blood-borne metastases** (lung, liver, brain)	**Syncytiotrophoblasts** (large, multinucleated, hCG+) with **cytotrophoblasts**; **no villi**	**hCG** very high (gynaecomastia, hyperthyroidism)
Teratoma	Children: prepubertal type, benign. Adults: postpubertal type, **malignant even when mature**; solid and cystic	Tissues of more than one germ layer: neural, cartilage, muscle, squamous, glandular epithelium	Usually none

SEMINOMA VS NON-SEMINOMATOUS GERM CELL TUMOURS (NSGCT)

Feature	Seminoma	NSGCT
Peak age	30s	20s (yolk sac: infants)
Gross	Homogeneous, lobulated, grey-white; no necrosis	Variegated, haemorrhage, necrosis, cysts
Microscopy	Uniform clear cells, septa with lymphocytes	Anaplastic, glandular, trophoblastic or teratomatous
Markers	**AFP normal**; hCG normal or slightly raised	**AFP and/or hCG** often raised
Spread	Lymphatic (para-aortic) first, late haematogenous	Earlier haematogenous (lung, liver; brain in chorio)
Therapy; prognosis	**Very radiosensitive** and chemosensitive; best prognosis	Radioresistant; **cisplatin chemotherapy**; still highly curable

MEMORY HOOK

Yolk → AFP · Chorio → hCG · Load → LDH

AFP comes from yolk sac (fetal) elements; **hCG** from syncytiotrophoblast, sky-high in **c**horiocarcinoma; **LDH** tracks tumour **L**oad. A pure seminoma never raises AFP.

LAB VALUES · TUMOUR MARKERS

- **AFP**: yolk sac elements; **never raised in pure seminoma**, so a raised AFP means a non-seminoma component
- **hCG**: syncytiotrophoblast; very high in choriocarcinoma, mild in some seminomas
- **LDH**: tumour bulk; prognosis
- Uses: diagnosis, staging, response and **early detection of relapse**

Clinical rules: remove by **radical inguinal orchiectomy**; scrotal biopsy is avoided because it opens a lymphatic route through scrotal skin. Spread follows the testicular vessels to **para-aortic nodes** (not inguinal). Overall cure exceeds 90%.

Leydig cell tumour: golden-brown, homogeneous; polygonal cells with eosinophilic cytoplasm and rod-shaped **Reinke crystalloids**; androgens → **precocious puberty** in boys, oestrogens → **gynaecomastia** in men; about 10% malignant. **Sertoli cell tumour**: grey-white, tubule-forming, usually benign and hormonally silent.

EXAM ZONE Past-Paper Pattern SEQs — Solved

CLASSIC **a)** Discuss the pathogenesis of benign prostatic hyperplasia. **b)** Differentiate between BPH and carcinoma of the prostate. **2 + 3 marks**

MODEL ANSWER PLAN

a) Testosterone → DHT by 5α-reductase type 2 in stromal cells → androgen receptors → growth factors (FGF, TGF-β) → proliferation of stroma and glands; reduced apoptosis with age; smooth-muscle tone adds obstruction. **b)** Five rows: transition vs peripheral zone; smooth rubbery vs hard nodule; early vs late obstruction; two cell layers vs no basal layer, nucleoli, perineural invasion; no spread vs osteoblastic bone metastases.

Score booster: say "BPH is not premalignant; the two arise in different zones" in one line.

PROF PATTERN **a)** Describe the morphology of carcinoma of the prostate and its grading. **b)** Write a note on the role of PSA. **2 + 3 marks**

MODEL ANSWER PLAN

a) Peripheral zone, gritty grey-yellow, ill-defined; small crowded glands, single layer, no basal cells, prominent nucleoli, perineural invasion; AMACR+, p63−. Gleason patterns 1–5, score = primary + secondary; grade groups 1–5. **b)** Kallikrein protease; organ- not cancer-specific; cut-off 4 ng/mL with false negatives; raised in BPH, prostatitis; density, velocity, free/total ratio; main value in follow-up after treatment.

PROF PATTERN **a)** Classify testicular tumours. **b)** Describe the gross and microscopic features of seminoma. **2 + 3 marks**

MODEL ANSWER PLAN

a) GCTs (95%) from GCNIS (seminoma; embryonal, yolk sac, chorio, teratoma; mixed) or not (spermatocytic, prepubertal); sex cord–stromal (Leydig, Sertoli); lymphoma. **b)** Homogeneous grey-white lobulated, no necrosis; clear cells with nucleoli, lymphocytic septa, granulomas; KIT, OCT3/4.

Score booster: add i(12p) and "radiosensitive, para-aortic spread" for the extra mark.

MODULE PATTERN A 24-year-old man has a painless, heavy left testis. Serum AFP and hCG are raised; ultrasound shows a heterogeneous mass. **a)** Which category of tumour is likely? **b)** Explain what each marker tells you. **c)** How does it spread, and why is scrotal biopsy avoided? **1 + 2 + 2 marks**

MODEL ANSWER PLAN

a) Non-seminomatous (mixed) germ cell tumour. **b)** AFP: yolk sac elements (excludes pure seminoma); hCG: syncytiotrophoblast; both used for staging and to detect relapse; LDH reflects bulk. **c)** Para-aortic nodes, then haematogenous to lungs; inguinal orchiectomy because scrotal violation opens inguinal lymphatic spread.

BCQs One-Best-Answer Practice

1 A 68-year-old with nocturia and poor stream improves after months of finasteride. The drug acts by inhibiting:

A. Androgen receptor
B. α1-adrenergic receptor
C. 5α-reductase type 2
D. Aromatase
E. GnRH receptor

2 A prostate biopsy shows small crowded glands. Which finding most strongly supports carcinoma?

A. Absent basal cell layer with prominent nucleoli
B. Corpora amylacea
C. Papillary infolding
D. Two-layered epithelium
E. Stromal nodules

3 A 72-year-old man has back pain; X-ray shows sclerotic lesions in the lumbar spine and pelvis. The most likely primary is:

A. Kidney
B. Lung
C. Thyroid
D. Prostate
E. Stomach

4 A biopsy has Gleason pattern 4 as the commonest and pattern 3 as the second. The ISUP grade group is:

A. 1
B. 3
C. 2
D. 4
E. 5

5 Which statement about PSA is correct?

A. It is specific for cancer
B. It is never raised in BPH
C. Values below 4 ng/mL exclude cancer
D. It is produced by seminal vesicles
E. It is organ-specific and useful for follow-up after treatment

6 An undescended testis removed from a 10-year-old shows small tubules with thick hyaline basement membranes and prominent Leydig cells. This patient also has an increased risk of:

A. Leydig cell tumour
B. Testicular lymphoma
C. Germ cell tumour in either testis
D. Adenomatoid tumour
E. Mumps orchitis

7 A 15-year-old wakes with sudden severe scrotal pain; the cremasteric reflex is absent. The underlying mechanism of testicular damage is:

A. Venous obstruction causing haemorrhagic infarction
B. Arterial embolism
C. Bacterial suppuration
D. Immune-complex vasculitis
E. Granulomatous inflammation

8 A 35-year-old has a homogeneous grey-white lobulated testicular mass. Histology: sheets of clear cells with central nuclei, septa with lymphocytes. Which marker profile fits?

A. AFP very high
B. hCG very high with normal AFP
C. AFP and hCG both very high
D. Normal AFP, normal or mildly raised hCG
E. Raised PSA

9 A 2-year-old has a testicular mass and a raised AFP. Histology shows glomeruloid structures with a central vessel. The tumour is:

A. Seminoma
B. Embryonal carcinoma
C. Choriocarcinoma
D. Teratoma
E. Yolk sac tumour

10 A 25-year-old with haemoptysis and gynaecomastia has a tiny haemorrhagic testicular nodule. The most likely tumour is:

A. Seminoma
B. Choriocarcinoma
C. Leydig cell tumour
D. Yolk sac tumour
E. Teratoma

11 Genital tuberculosis in men most typically begins in the:

A. Testis
B. Tunica vaginalis
C. Epididymis
D. Penile urethra
E. Spermatic cord veins

12 The commonest testicular neoplasm in a 70-year-old man is:

A. Seminoma
B. Yolk sac tumour
C. Leydig cell tumour
D. Diffuse large B-cell lymphoma
E. Choriocarcinoma

ANSWER KEY WITH ONE-LINE REASONS

1. **C** Blocks testosterone → DHT conversion.
2. **A** Loss of basal cells (p63-negative) with nucleoli; AMACR positive.
3. **D** Prostate cancer gives osteoblastic bone metastases.
4. **B** 4 + 3 = 7 is grade group 3.
5. **E** Organ-, not cancer-specific; best in monitoring.
6. **C** Cryptorchidism raises GCT risk 3–5-fold, both sides.
7. **A** Torsion blocks veins first: haemorrhagic infarction.
8. **D** Pure seminoma never raises AFP.

9. **E** Schiller–Duval bodies; commonest testicular tumour in infants.
10. **B** Very high hCG; early haematogenous spread to lungs.
11. **C** TB starts in the epididymis; syphilis in the testis.
12. **D** Lymphoma is the commonest testicular tumour over 60.

VIVA Rapid-Fire Viva Voce

Q Zone of origin of BPH and of carcinoma?	BPH: transition zone. Carcinoma: peripheral zone.
Q Hormone essential for BPH?	Dihydrotestosterone, made by 5α-reductase type 2.
Q What are corpora amylacea?	Laminated eosinophilic concretions in prostatic gland lumens.
Q Precursor of prostate cancer?	High-grade prostatic intraepithelial neoplasia (HGPIN).
Q Commonest gene fusion in prostate cancer?	TMPRSS2–ERG (ETS family).
Q On what is Gleason grading based?	Glandular architecture at low power, not cytology.
Q Type of bone metastasis in prostate cancer?	Osteoblastic (sclerotic).
Q Why is varicocele usually left-sided?	Left testicular vein drains at a right angle into the left renal vein.
Q Precursor of most testicular GCTs?	Germ cell neoplasia in situ (GCNIS).
Q Chromosomal marker of GCTs?	Isochromosome 12p, i(12p).
Q What are Schiller–Duval bodies?	Glomeruloid structures of yolk sac tumour.
Q First lymph nodes involved by testicular tumours?	Retroperitoneal para-aortic nodes.

OSPE Practice Stations

STATION 1 · SLIDE — 2 minutes

TASK

Prostate: nodules of glands of varying size with papillary infoldings, lined by an inner columnar and an outer flat cell layer; some lumens contain pink laminated bodies; abundant fibromuscular stroma. (a) Diagnosis. (b) Name the pink bodies. (c) Which zone is affected and why does it cause early symptoms?

ANSWER

(a) Benign prostatic (nodular) hyperplasia. (b) Corpora amylacea. (c) Transition (periurethral) zone; nodules compress the prostatic urethra.

STATION 2 · SLIDE — 2 minutes

TASK

Testis: sheets of large uniform cells with clear cytoplasm, distinct borders and central nuclei with prominent nucleoli, divided by thin fibrous septa infiltrated by lymphocytes. (a) Diagnosis. (b) Two features of the gross specimen. (c) Which tumour marker should be normal?

ANSWER

(a) Seminoma. (b) Homogeneous grey-white lobulated cut surface; no haemorrhage or necrosis. (c) AFP.

STATION 3 · LAB DATA 2 minutes

TASK

A 70-year-old man: PSA 48 ng/mL (normal < 4), serum alkaline phosphatase raised, isotope bone scan shows multiple hot spots in the spine and pelvis. (a) Most likely diagnosis. (b) Type of bone lesion expected on X-ray. (c) Two other causes of a raised PSA.

ANSWER

(a) Metastatic carcinoma of the prostate. (b) Osteoblastic (sclerotic) metastases. (c) BPH, prostatitis (also infarction, instrumentation).

RAPID REVISION The Night-Before Page

15 ONE-LINERS

01 Transition zone → BPH; peripheral zone → carcinoma.

02 BPH: DHT via 5α-reductase type 2; stroma + glands; not premalignant.

03 BPH glands: two layers, papillary infolds, corpora amylacea.

04 BPH complications: retention, trabeculated bladder, UTI, hydronephrosis.

05 Carcinoma: no basal layer, nucleoli, perineural invasion, AMACR+.

06 TMPRSS2–ERG fusion, PTEN loss, GSTP1 methylation.

07 Gleason = two commonest patterns; 4+3 = grade group 3.

08 PSA: organ- not cancer-specific; cut-off 4 ng/mL.

09 Prostate metastases to bone are osteoblastic.

10 Cryptorchidism: infertility, 3–5× GCT risk, hyalinised tubules.

11 Torsion: veins block first → haemorrhagic infarction; 6-hour window.

12 TB starts in epididymis; syphilis in testis.

13 GCTs: 95%; GCNIS precursor; i(12p).

14 Seminoma: clear cells, lymphocytic septa, AFP normal, radiosensitive.

15 Yolk sac: Schiller–Duval, AFP; chorio: hCG; lymphoma over 60.

MEMORY HOOKS

- **"Transition traps the urethra; peripheral is palpable"**
- **Seminoma = fried eggs in a lymphocyte net**
- **Yolk → AFP, Chorio → hCG, Load → LDH**
- **TB → Tail (epididymis) first; Syphilis → Seminiferous testis first**

TEST YOURSELF: FILL THE GAPS

1. BPH drug that blocks DHT: _____.
2. Carcinoma marker positive on IHC: _____.
3. Gleason 3 + 4 = grade group _____.
4. Commonest testicular tumour under 3 years: _____.
5. Reinke crystalloids: _____ cell tumour.
6. Testicular tumours spread first to _____ nodes.

Answers: finasteride · AMACR · 2 · yolk sac tumour · Leydig · para-aortic

DRAW FROM MEMORY

The prostate zones, marking where BPH and carcinoma start (and why one obstructs early).

DRAW FROM MEMORY

The germ cell tumour family tree from GCNIS (Figure 19.3) with each tumour's marker.

SPACED REVISION ☐ Day 1 ☐ Day 3 ☐ Day 7 ☐ Day 14 ☐ Day 30

20

Vulva, Cervix & Endometrium

One virus explains almost every cervical cancer, and one hormone left unopposed explains most endometrial cancers. Follow HPV through the transformation zone and oestrogen through the endometrium, and the lower genital tract becomes a single, logical story.

BY THE END OF THIS CHAPTER YOU CAN

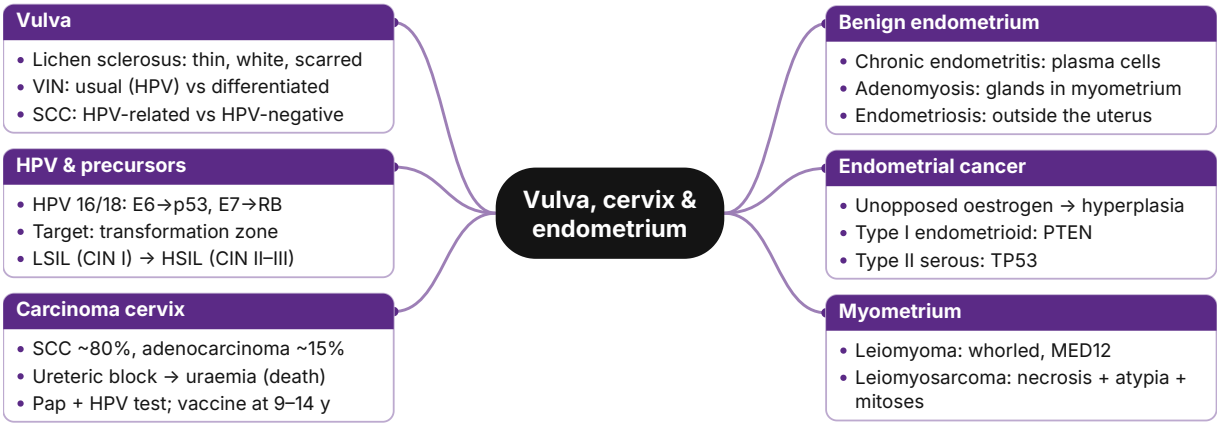
1. Differentiate lichen sclerosus, lichen simplex chronicus and the two pathways of vulvar squamous carcinoma
2. Explain how high-risk HPV E6 and E7 proteins transform cervical epithelium
3. Grade cervical precursor lesions (LSIL/HSIL, CIN I–III) and describe Pap smear screening and HPV vaccination
4. Describe the morphology, spread and staging of carcinoma of the cervix
5. Compare adenomyosis with endometriosis and explain the theories of endometriosis
6. Classify endometrial hyperplasia and contrast type I and type II endometrial carcinoma
7. Differentiate leiomyoma from leiomyosarcoma

EXAM RADAR

Theory SEQs	★★★
BCQs / MCQs	★★★
Viva voce	★★★
OSPE / slides	★★★

HPV and CIN, carcinoma cervix, endometriosis and endometrial carcinoma are favourite long questions; adenomyosis and endometrial carcinoma are lab slides.

ROBBINS LINK Robbins Basic Pathology — *Female Genital System and Breast*: vulva, cervix (HPV, SIL, carcinoma), body of uterus and endometrium (endometritis, adenomyosis, endometriosis, hyperplasia, carcinoma), myometrial tumours.



Chapter at a glance. Two drivers (HPV and oestrogen) and two smooth-muscle tumours.

20.1 Vulva

The vulva is covered by skin, so most of its diseases behave like skin diseases: itchy white patches, warts and squamous neoplasia. A white plaque (**leukoplakia**) is a **clinical description, not a diagnosis**: it may be a dermatosis, VIN or invasive carcinoma, so it must be biopsied.

THE TWO CLASSIC NON-NEOPLASTIC WHITE LESIONS OF THE VULVA

Feature	Lichen sclerosus	Lichen simplex chronicus (squamous hyperplasia)
Cause	Inflammatory dermatosis of postmenopausal women; autoimmune T-cell reaction suspected	Chronic rubbing and scratching (itch–scratch cycle)
Gross	Smooth white "parchment" plaques, labial atrophy, narrowed introitus	Thick, leathery white skin
Epidermis	**Thinned**, loss of rete pegs, hydropic basal cells	**Thickened** (acanthosis) with hyperkeratosis
Dermis	**Homogeneous, acellular hyalinised collagen band** with a band of lymphocytes beneath	Mild chronic inflammation

Feature	Lichen sclerosus	Lichen simplex chronicus (squamous hyperplasia)
Cancer risk	Slightly raised (HPV-negative SCC)	Not premalignant, but often seen at the edge of a carcinoma

Condyloma acuminatum is a sexually transmitted wart caused by **low-risk HPV 6 and 11**: papillary, exophytic lesions whose surface cells show **koilocytosis**. It is not a precursor of cancer.

Vulvar intraepithelial neoplasia (VIN) and squamous cell carcinoma

Squamous cell carcinoma makes up about **90%** of vulvar cancers, mostly in women over 60. It arises along **two separate pathways** that differ in age, cause and precursor. Learn them as a pair.

Feature	HPV-associated pathway	HPV-independent pathway
Frequency	Less common	**More common**
Age	Younger and middle-aged women, often **smokers**	**Older** women
Precursor	**Usual (classic) VIN** = vulvar HSIL; often multifocal, may coexist with cervical or vaginal HSIL	**Differentiated VIN**: atypia confined to the basal layer, arising in **lichen sclerosus** or squamous hyperplasia
Molecular driver	High-risk HPV (mainly **HPV 16**)	**TP53** mutation
Tumour type	Basaloid or warty carcinoma, poorly differentiated	**Keratinising**, well-differentiated SCC

Spread is first to **inguinal** and then pelvic lymph nodes. Prognosis depends on tumour size, depth of invasion and node status.

20.2 HPV and cervical precursor lesions

Almost every cervical carcinoma begins with a **persistent high-risk HPV infection**. The virus is very common in young sexually active women, but most infections are transient: about **90% are cleared within 2 years**. Cancer appears only when infection persists and the viral oncogenes take over the cell cycle.

> **CONCEPT CLEAR**
>
> **Why the transformation zone?** HPV can only infect **immature basal cells** that it reaches through tiny breaks in the epithelium; the virus then replicates in the cells as they mature (a productive infection, seen as koilocytes). The **transformation zone**, where columnar epithelium is being replaced by squamous metaplasia, is full of **immature metaplastic squamous cells**, so it is the most vulnerable target. That is why nearly all CIN and carcinoma arise there, and why the Pap smear must sample the squamocolumnar junction.

HPV TYPES

- **Low-risk** (6, 11): condylomas, some LSIL; virus stays **episomal**
- **High-risk** (16, 18, 31, 33, 45, 52, 58 and others): HSIL and carcinoma; **HPV 16** causes almost **60%** and **HPV 18** about **10%** of cervical cancers
- HPV 18 is relatively more frequent in **adenocarcinoma**

RISK FACTORS (HELP HPV PERSIST)

- Early first intercourse; multiple partners; male partner with many partners
- **Smoking**; **immunodeficiency** (HIV)
- High parity; long-term oral contraceptives; other STIs (Chlamydia)

> **MOLECULAR KEY**
>
> **E6 and E7, the oncoproteins. E6** binds **p53** and marks it for ubiquitin-mediated destruction; it also activates **TERT** (telomerase), so cells become immortal. **E7** binds hypophosphorylated **RB**, releasing **E2F** to drive the cell into S phase; it also inactivates the CDK inhibitors **p21** and **p27**. **Integration** of the viral genome breaks the viral **E2** gene, which normally restrains E6/E7, so their expression rises. With RB disabled, the cell over-produces **$p16^{INK4a}$**: strong, diffuse ("block") p16 staining is a surrogate marker of high-risk HPV-driven HSIL.

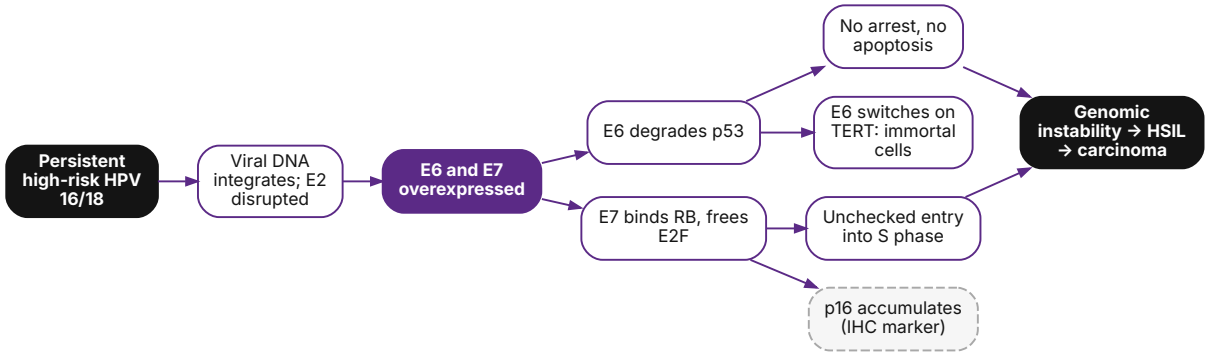


Figure 20.1 · How high-risk HPV transforms a cervical cell. Two viral proteins switch off the two great tumour suppressors.

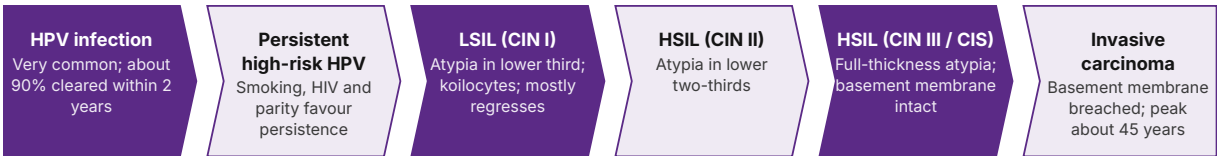


Figure 20.2 · From HPV infection to invasive carcinoma. The whole sequence usually takes years, which is what makes screening work.

Grading the precursors: SIL and CIN

The old three-tier CIN system has been folded into the two-tier **Bethesda (SIL)** system, because CIN II and III behave alike. **LSIL** is essentially a **productive viral infection**; **HSIL** is a **true neoplasm** in which deregulated E6/E7 have stopped the cells maturing.

Bethesda	CIN	Extent of atypia	Key microscopic features
LSIL	CIN I	Lower **one-third**	**Koilocytes** in upper layers: perinuclear halo, enlarged hyperchromatic "raisin-like" nuclei, binucleation; episomal HPV
HSIL	CIN II	Lower **two-thirds**	Immature basal-type cells expand upward; mitoses above the basal layer
	CIN III (includes carcinoma in situ)	**Full thickness**	No maturation, high N:C ratio, atypical mitoses at all levels; **basement membrane intact**; HPV often integrated

NATURAL HISTORY (APPROXIMATE, ROBBINS)

Lesion	Regress	Persist	Progress
LSIL	60%	30%	10% to HSIL; < 1% to carcinoma
HSIL	30%	60%	> 10% to carcinoma

PAP REPORT (BETHESDA)

NILM
negative for intraepithelial lesion or malignancy

ASC-US / ASC-H
atypical squamous cells, undetermined / cannot exclude HSIL

LSIL · HSIL
low- / high-grade lesion

AGC · AIS
atypical glandular cells · adenocarcinoma in situ

Screening and prevention

- **Pap smear** (exfoliative cytology): cells scraped from the transformation zone, Papanicolaou-stained (or liquid-based). Dysplastic cells show nuclear enlargement, hyperchromasia, irregular membranes and a high N:C ratio.
- **HPV DNA test** for high-risk types: more sensitive; used with cytology (co-testing) or as the primary screen. WHO now favours primary HPV testing from age 30.
- **Why screening works:** the precursor phase lasts years and the cervix is easy to sample. Robbins (US guidance): cytology from **21 years** every 3 years; from **30 to 65**, cytology plus HPV co-testing every 5 years.
- **HPV vaccines** are prophylactic virus-like particles of the L1 capsid protein: bivalent (16, 18), quadrivalent (6, 11, 16, 18) and nine-valent (adds 31, 33, 45, 52, 58). Best given **before sexual debut** (girls 9–14 years).

- **Colposcopy** after an abnormal result: acetic acid turns abnormal epithelium **acetowhite**; directed biopsy confirms the grade.
- HSIL is excised (LEEP, cone biopsy); LSIL is usually followed up.
- Vaccines prevent infection but do not treat it and do not cover every oncogenic type, so **vaccinated women still need screening**.

PAKISTAN FOCUS

Pakistan reports a lower incidence of cervical cancer than many countries, but **most women have never been screened**, so cases present late. Pakistan began **HPV vaccination of girls aged 9–14 years** in a national campaign in **September 2025**. For the next generation, vaccination plus periodic HPV testing could make this cancer rare.

20.3 Carcinoma of the cervix

Invasive carcinoma means the neoplastic cells have broken through the basement membrane. The peak age is about **45 years**, roughly 15 years after the peak of HSIL. All the main types are HPV-related.

- **Types: squamous cell carcinoma** about **80%**; **adenocarcinoma** about **15%** (precursor adenocarcinoma in situ; harder to detect on Pap smear); adenosquamous and **neuroendocrine (small cell)** carcinoma about 5%, the latter most aggressive.
- **Clinical:** early disease is silent (found by screening); then **postcoital** or intermenstrual bleeding, discharge, dyspareunia; late pelvic pain, fistulae and **uraemia**.

GROSS

- **Exophytic**: fungating, cauliflower-like, friable growth that bleeds on touch
- **Ulcerative**: crater replacing the cervix
- **Endophytic**: infiltrates the wall, giving a hard "barrel-shaped" cervix

MICROSCOPY

- Irregular **nests and tongues** of malignant squamous cells invading the stroma
- Keratinising type: **keratin pearls**, intercellular bridges; non-keratinising: sheets of polygonal cells
- Desmoplastic stroma with lymphocytes; often HSIL at the edge

SPREAD

- **Direct**: vagina, parametrium, pelvic wall; bladder in front, rectum behind
- Parametrial spread encircles the **ureters** → hydronephrosis → **renal failure, the commonest cause of death**
- **Lymphatic**: paracervical, obturator, internal and external iliac, then para-aortic nodes
- **Blood** (late): liver, lungs, bone

FIGO STAGING (SIMPLIFIED)

I	Confined to cervix. IA: microscopic, invasion ≤ 5 mm deep; IB: larger
II	Beyond uterus but not to pelvic wall or lower third of vagina (IIB: parametrium)
III	Pelvic wall, lower third of vagina, or **hydronephrosis**; IIIC: pelvic/para-aortic nodes
IV	IVA: bladder or rectal mucosa; IVB: distant metastasis

Prognosis follows **stage** and node status: microinvasive disease is almost always curable.

CLINICAL CASE

A 45-year-old mother of six, never screened, has postcoital bleeding for 4 months. A friable cauliflower growth replaces the cervix; the parametrium is thick up to the pelvic wall; creatinine 3.1 mg/dL with bilateral hydronephrosis. **Reasoning:** invasive cervical carcinoma (biopsy: keratinising SCC); pelvic wall plus hydronephrosis = **stage III**; the raised creatinine is the classic killer, **ureteric obstruction**.

20.4 Endometritis, adenomyosis and endometriosis

Endometrium grows, bleeds and sheds every month. Misplaced endometrium still obeys the cycle, and that misplaced bleeding explains the pain, scarring and cysts of endometriosis.

Endometritis

- **Acute endometritis**: bacterial infection after **delivery or abortion**, usually with retained products of conception; neutrophils in the stroma. Treated with curettage and antibiotics.
- **Chronic endometritis**: the diagnostic clue is **plasma cells in the endometrial stroma** (normal endometrium has lymphocytes but no plasma cells). Causes: pelvic inflammatory disease (gonococcus, chlamydia), retained products, **intrauterine device**, and **tuberculosis**.
- **Tuberculous endometritis** usually spreads from tuberculous salpingitis and is an important cause of **infertility** where TB is common, as in Pakistan. Granulomas are small and seldom caseate, because the endometrium is shed each month; they are best seen in a premenstrual biopsy.

Adenomyosis

DEFINITION

Adenomyosis is the presence of endometrial glands and stroma **deep within the myometrium**, between the muscle bundles, conventionally more than about one low-power field (2–3 mm) below the endometrial–myometrial junction.

It results from downgrowth of the **basal layer** of the endometrium into the myometrium, and is found in up to a fifth of hysterectomy specimens. Because the misplaced tissue is basalis, it usually does **not** bleed cyclically. The surrounding smooth muscle reacts with hypertrophy, so the wall thickens.

GROSS

- Uterus **diffusely enlarged and globular**, wall thickened
- Cut surface coarsely trabeculated with small **haemorrhagic or cystic spaces**
- **No capsule**, no sharp edge (unlike a fibroid); a localised nodular form is an **adenomyoma**

MICROSCOPY

- **Islands of endometrial glands with surrounding endometrial stroma** deep in the myometrium
- Glands usually inactive (basalis type)
- **Hypertrophied smooth muscle** around the islands

Clinical: parous women in their 40s with **menorrhagia**, colicky **dysmenorrhoea**, dyspareunia and premenstrual pelvic pain; the uterus is bulky and tender.

Endometriosis

DEFINITION

Endometriosis is the presence of endometrial glands and stroma **outside the uterus**. It affects up to about **10%** of women of reproductive age and nearly **half of infertile women**.

Sites, commonest first: ovaries (often bilateral) → **uterine ligaments**, rectovaginal septum, **pouch of Douglas** → pelvic peritoneum, bowel, appendix, bladder → cervix, vagina, tubes, laparotomy scars → rarely umbilicus, lymph nodes, lungs. **Theories: regurgitation** (Sampson; retrograde menstruation, the leading theory for pelvic disease), **benign metastasis** by lymphatics or veins (explains nodes and lung), **metaplasia** of coelomic epithelium, and **stem/progenitor cells** from bone marrow or basalis.

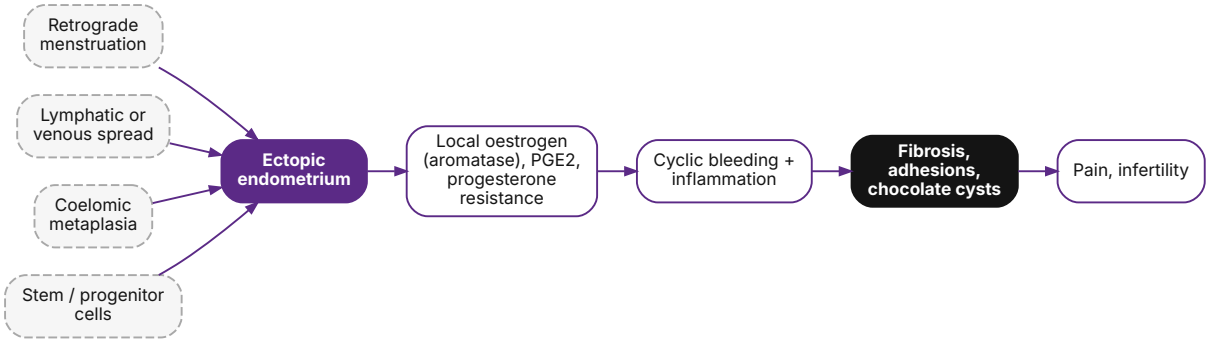


Figure 20.3 · Pathogenesis of endometriosis. Four routes in, one hormone-driven cycle of bleeding and scarring.

GROSS

- Red-blue to yellow-brown **"powder-burn" nodules** on serosal surfaces
- Ovary: cysts full of thick, old brown blood: **chocolate cysts (endometriomas)**
- Dense **fibrous adhesions**: tubo-ovarian masses, obliterated pouch of Douglas, "frozen pelvis"

MICROSCOPY

- Diagnosis needs **two of three**: **endometrial glands**, **endometrial stroma**, **haemosiderin-laden macrophages**
- In old lesions the glands may disappear, leaving fibrosis and pigment

Clinical: severe **dysmenorrhoea**, chronic pelvic pain, **dyspareunia**, painful defecation (rectal wall), dysuria (bladder), **infertility** (tubal distortion, adhesions). Ovarian endometriosis carries a small risk of **endometrioid and clear cell carcinoma** (ARID1A mutations; see Chapter 21).

ADENOMYOSIS VS ENDOMETRIOSIS: A PERENNIAL "DIFFERENTIATE" QUESTION

Feature	Adenomyosis	Endometriosis
Where is the endometrium?	**Inside the myometrium**	**Outside the uterus**
Origin	Downgrowth of basalis	Retrograde menstruation (mainly), metastasis, metaplasia, stem cells
Typical patient	Parous woman in her 40s	Woman of 20–40, often nulliparous or infertile
Cyclical bleeding	Usually absent (basalis)	**Present** → haemosiderin, chocolate cysts
Uterus	Diffusely enlarged, globular, tender	Usually normal size; may be fixed by adhesions
Microscopy	Glands + stroma between muscle bundles	2 of 3: glands, stroma, haemosiderin
Main symptoms	Menorrhagia, dysmenorrhoea	Dysmenorrhoea, pelvic pain, dyspareunia, infertility
Malignant link	Not significant	Endometrioid and clear cell ovarian carcinoma

MEMORY HOOK

Adeno-MYO-sis stays in the MYOmetrium

Endometri-**OSIS** lives **O**utside, **S**eeds the **I**ntraperitoneal **S**urfaces. The one that lives outside bleeds every month, so it is the one with chocolate cysts and adhesions.

20.5 Endometrial hyperplasia and carcinoma

CONCEPT CLEAR

Oestrogen grows the endometrium; progesterone matures it and lets it shed. In every ovulatory cycle the progesterone of the corpus luteum switches the glands to secretion and then withdrawal causes menstruation. When ovulation fails, or oestrogen comes from elsewhere, the glands keep proliferating with no brake and no shedding. Years of this **unopposed oestrogen** produce hyperplasia, and once mutations such as **PTEN** loss are added, atypical hyperplasia and endometrioid carcinoma follow.

SOURCES OF UNOPPOSED OESTROGEN

- **Anovulatory cycles** (perimenopause), **PCOS**
- **Obesity**: adipose aromatase converts androstenedione to oestrone
- Oestrogen-only hormone replacement
- Oestrogen-secreting ovarian tumours (**granulosa cell tumour**)
- **Tamoxifen** (weak oestrogen agonist on endometrium)

ENDOMETRIAL HYPERPLASIA (WHO TWO-TIER)

- **Hyperplasia without atypia**: increased gland-to-stroma ratio; glands of irregular size and shape, some cystically dilated; normal cytology. Low risk of carcinoma (about 1–3%).
- **Atypical hyperplasia / endometrioid intraepithelial neoplasia (EIN)**: crowded, back-to-back complex glands with **nuclear atypia** (rounded vesicular nuclei, nucleoli, loss of polarity). **High risk**: many already

harbour carcinoma, so treatment is hysterectomy (progestins if fertility is wanted).

Endometrial carcinoma

The commonest invasive cancer of the female genital tract in high-income countries. The classic presentation is **postmenopausal bleeding**; every episode must be investigated (transvaginal ultrasound of endometrial thickness, then endometrial biopsy). A Pap smear is not a reliable test for it. Robbins describes two clinicopathological types.

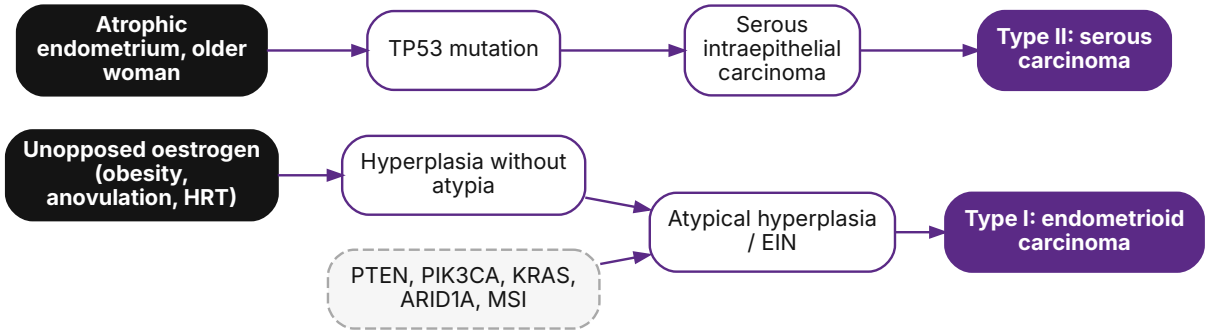


Figure 20.4 · Two roads to endometrial carcinoma. Oestrogen-driven type I and oestrogen-independent type II.

TYPE I VS TYPE II ENDOMETRIAL CARCINOMA

Feature	Type I (endometrioid)	Type II (serous)
Frequency	**About 80%**	About 15%
Age	55–65 years (peri- and early postmenopause)	**65–75 years**
Oestrogen link	**Yes**: unopposed oestrogen	No
Associations	Obesity, diabetes, hypertension, nulliparity, PCOS; **Lynch syndrome**, Cowden	Endometrial atrophy; thin women
Precursor	**Atypical hyperplasia / EIN**	**Serous endometrial intraepithelial carcinoma**
Key genes	**PTEN** (commonest), PIK3CA, KRAS, ARID1A, β-catenin, mismatch repair (MSI)	**TP53** (nearly all)
Microscopy	Gland-forming, resembles normal endometrium; often squamous foci; grade 1–3	Complex **papillae**, marked nuclear atypia, **psammoma bodies**; always high grade
Spread / behaviour	Myometrial invasion, then lymphatics; indolent	Spreads early via tubes to **peritoneum**, even with little invasion; aggressive
Prognosis	**Good**	**Poor**

GROSS

- **Polypoid** localised mass or **diffuse** tumour lining the cavity
- Tan-white, soft, friable; may fill an enlarged uterus
- Depth of myometrial invasion and cervical extension seen on cut surface

MICROSCOPY

- **Endometrioid:** crowded **back-to-back glands without intervening stroma**, cribriform areas, stratified atypical columnar cells, invasion of myometrium
- Graded by solid (non-glandular) growth: G1 ≤ 5%, G2 6–50%, G3 > 50%
- **Serous:** papillary tufts, high-grade nuclei, psammoma bodies

Staging (FIGO, simplified): I corpus only (IA < half, IB ≥ half of myometrium) · II cervical stroma · III serosa, adnexa, vagina, parametrium or pelvic/para-aortic nodes · IV bladder/bowel mucosa or distant. **Prognosis** depends on stage, grade and histological type.

MOLECULAR KEY

Lynch syndrome (germline mismatch repair defect: MLH1, MSH2, MSH6, PMS2): endometrial carcinoma is its **commonest extracolonic cancer**. Sporadic MSI arises from **MLH1 promoter methylation**. Genomic studies now sort tumours into four groups: POLE-ultramutated (best outlook), MSI, copy-number-low (most endometrioid) and copy-number-high/TP53 (serous-like, worst). **Carcinosarcoma** (malignant mixed Müllerian tumour), with malignant glands plus malignant stroma, arises from one epithelial clone and behaves like a high-grade type II cancer.

20.6 Tumours of the myometrium

Leiomyoma (fibroid)

The **commonest benign tumour in women**, present in about 30–50% of women of reproductive age. It is a benign smooth-muscle tumour that depends on **oestrogen**: it grows in pregnancy and on oestrogen therapy and shrinks after menopause. About **70%** carry **MED12** mutations; others have **HMGA2** rearrangements; germline **fumarate hydratase (FH)** mutation causes hereditary leiomyomatosis with renal cell carcinoma.

GROSS

- **Sharply circumscribed**, round, firm, grey-white nodules; usually **multiple**
- Cut surface bulges and shows a **whorled** pattern
- Sites: **intramural** (commonest), **submucosal** (bleeding), **subserosal** (may be pedunculated)

MICROSCOPY

- Interlacing **whorled fascicles** of uniform smooth-muscle cells
- Elongated, blunt-ended **"cigar-shaped" nuclei**, eosinophilic cytoplasm
- Few mitoses, **no tumour necrosis**, no atypia

Secondary (degenerative) changes: hyaline (commonest), cystic, myxoid, calcification (postmenopausal), and **red (carneous) degeneration** in pregnancy: haemorrhagic infarction with acute pain. Malignant change is **extremely rare**. **Clinical:** often silent; menorrhagia with iron-deficiency anaemia, pelvic pressure, urinary frequency, infertility, miscarriage.

Leiomyosarcoma

A malignant smooth-muscle tumour that arises **de novo**, not from a leiomyoma, usually in women aged 40–60. It recurs locally and metastasises by blood, mainly to the **lungs**; five-year survival is about **40%**.

LEIOMYOMA VS LEIOMYOSARCOMA

Feature	Leiomyoma	Leiomyosarcoma
Number	Usually multiple	**Solitary**
Edge	Sharply circumscribed, shells out	**Infiltrative**, poorly defined
Cut surface	Firm, grey-white, whorled	**Soft, fleshy**, with **haemorrhage and necrosis**
Atypia	Absent	Moderate to severe, diffuse
Mitoses	Rare	**Numerous** (often ≥ 10 per 10 HPF)
Tumour cell necrosis	Absent (infarct-type only)	**Coagulative tumour cell necrosis**
Behaviour	Benign	Recurs; lung metastases

EXAM TRAP

Diagnose leiomyosarcoma on a combination (usually any two of: tumour cell necrosis, diffuse atypia, brisk mitoses). Bizarre nuclei alone, or mitoses alone in a bland tumour, are still benign. A "fibroid" that grows after menopause is suspicious.

LAB SPOTTER

Lab slides in 10 seconds. Adenomyosis: a section that is mostly **smooth muscle**, with islands of **glands plus cuffs of blue endometrial stroma** far from the surface. **Endometrial carcinoma:** glands packed **back to back** with no stroma between them, lined by stratified, atypical columnar cells and pushing into muscle.

EXAM ZONE Past-Paper Pattern SEQs — Solved

PROF PATTERN **a)** Discuss the role of human papillomavirus in the pathogenesis of carcinoma of the cervix. **b)** Classify cervical intraepithelial lesions and outline how they are detected. **2 + 3 marks**

MODEL ANSWER PLAN

a) High-risk HPV 16/18 infects immature metaplastic cells of the transformation zone; persistence (smoking, HIV, parity) is key. Integration disrupts E2 → E6/E7 overexpressed: E6 degrades p53 and activates TERT; E7 binds RB (E2F released) and blocks p21/p27 → genomic instability → HSIL → invasion.

b) LSIL = CIN I (lower third, koilocytes); HSIL = CIN II (lower two-thirds) and CIN III/CIS (full thickness, BM intact). Detection: Pap smear from the transformation zone, HPV DNA test, colposcopy (acetowhite areas) with biopsy; p16 block staining in HSIL.

Score booster: draw Figure 20.1 (E6/p53, E7/RB) in one line each; examiners give the mechanism marks only if both proteins and both targets are named.

MODULE PATTERN A 60-year-old obese, diabetic, nulliparous woman presents with postmenopausal bleeding. Ultrasound shows a thickened endometrium. **a)** What is the most likely diagnosis? **b)** Explain the role of oestrogen in its pathogenesis. **c)** Tabulate the differences between its two types. **1 + 2 + 2 marks**

MODEL ANSWER PLAN

a) Endometrial carcinoma, type I (endometrioid). **b)** Adipose aromatase converts androstenedione to oestrone → unopposed oestrogen (no progesterone) → persistent proliferation → hyperplasia → atypical hyperplasia/EIN with PTEN loss → carcinoma. **c)** Type I vs II: age, oestrogen link, precursor (EIN vs SEIC), genes (PTEN vs TP53), histology (endometrioid vs serous with psammoma bodies), prognosis.

CLASSIC **a)** Define endometriosis. **b)** Give the theories of its pathogenesis. **c)** Describe its morphology and differentiate it from adenomyosis. **1 + 2 + 2 marks**

MODEL ANSWER PLAN

a) Endometrial glands and stroma outside the uterus. **b)** Regurgitation (retrograde menstruation), benign metastasis, coelomic metaplasia, stem cell theory; ectopic tissue makes local oestrogen and PGE2 and is progesterone-resistant. **c)** Powder-burn nodules, chocolate cysts of ovary, adhesions; micro 2 of 3: glands, stroma, haemosiderin. Table with adenomyosis: site, origin, cyclic bleeding, uterus size, symptoms.

PROF PATTERN **a)** Describe the gross and microscopic features and secondary changes of uterine leiomyoma. **b)** How will you differentiate it from leiomyosarcoma? **3 + 2 marks**

MODEL ANSWER PLAN

a) Multiple, circumscribed, firm, grey-white whorled nodules (intramural, submucosal, subserosal); micro: whorled fascicles of uniform spindle cells, cigar-shaped nuclei, rare mitoses. Secondary changes: hyaline, cystic, myxoid, calcific, red degeneration in pregnancy. **b)** Leiomyosarcoma: solitary, infiltrative, soft, haemorrhagic and necrotic; micro: coagulative tumour cell necrosis, diffuse atypia, ≥ 10 mitoses/10 HPF; arises de novo; lung metastases.

BCQs One-Best-Answer Practice

1 The HPV E7 oncoprotein promotes cell proliferation chiefly by binding and inactivating:

A. p53
B. RB
C. BCL2
D. TERT
E. E-cadherin

2 A Pap smear shows superficial squamous cells with enlarged, wrinkled hyperchromatic nuclei surrounded by a clear perinuclear halo. These cells indicate:

A. Candida infection
B. Atrophic vaginitis
C. Squamous metaplasia
D. Cytopathic effect of HPV
E. Invasive carcinoma

3 Nearly all cervical intraepithelial neoplasia arises in the:

A. Transformation zone
B. Ectocervix far from the os
C. Endometrial cavity
D. Vaginal fornix
E. Endocervical crypt base

4 The commonest cause of death in advanced carcinoma of the cervix is:

A. Pulmonary metastasis
B. Haemorrhage
C. Brain metastasis
D. Sepsis from fistula
E. Renal failure from ureteric obstruction

5 An endometrial biopsy from a woman with an intrauterine device and pelvic pain shows numerous plasma cells in the stroma. The diagnosis is:

A. Normal secretory endometrium
B. Atypical hyperplasia
C. Chronic endometritis
D. Adenomyosis
E. Endometrial polyp

6 The commonest site of endometriosis is the:

A. Ovary
B. Umbilicus
C. Lung
D. Appendix
E. Cervix

7 Histological diagnosis of endometriosis requires two of three features. Which set is correct?

A. Glands, smooth muscle, plasma cells
B. Stroma, granulomas, neutrophils
C. Glands, psammoma bodies, mucin
D. Endometrial glands, endometrial stroma, haemosiderin
E. Stroma, keratin, giant cells

8 A 70-year-old thin woman has postmenopausal bleeding. Hysterectomy shows a papillary tumour with marked nuclear atypia, psammoma bodies and an atrophic background endometrium. The gene most likely mutated is:

A. PTEN
B. TP53
C. KRAS
D. MLH1
E. CTNNB1

9 The most frequently altered gene in type I (endometrioid) endometrial carcinoma and its precursor is:

A. TP53
B. BRCA1
C. HER2
D. RB
E. PTEN

10 A 30-year-old woman develops sudden severe pain over a known fibroid in the second trimester of pregnancy. The fibroid most likely shows:

A. Sarcomatous change
B. Calcification
C. Red (carneous) degeneration
D. Hyaline change
E. Cystic change

11 Which feature best supports leiomyosarcoma over leiomyoma?

A. Coagulative tumour cell necrosis with atypia and brisk mitoses
B. Multiple well-circumscribed nodules
C. Whorled grey-white cut surface
D. Hyaline degeneration
E. MED12 mutation

12 A 75-year-old woman with long-standing lichen sclerosus develops an ulcerated vulvar mass. The tumour is most likely:

A. HPV 16-positive basaloid carcinoma
B. Melanoma
C. Extramammary Paget disease
D. Keratinising SCC with TP53 mutation, from differentiated VIN
E. Condyloma acuminatum

ANSWER KEY WITH ONE-LINE REASONS

1. **B** E7 binds hypophosphorylated RB and releases E2F; E6 targets p53.
2. **D** Koilocytes are the hallmark of productive HPV infection (LSIL).
3. **A** Immature metaplastic squamous cells of the transformation zone are most susceptible to HPV.
4. **E** Parametrial spread compresses the ureters, causing hydronephrosis and uraemia.
5. **C** Plasma cells are not present in normal endometrium; they define chronic endometritis.
6. **A** Ovaries (often bilateral, with chocolate cysts), then uterine ligaments and pouch of Douglas.
7. **D** Glands, stroma and haemosiderin-laden macrophages: any two suffice.
8. **B** Serous (type II) carcinoma: TP53 in nearly all cases; no hyperplasia background.
9. **E** PTEN loss is the commonest change in endometrioid carcinoma and atypical hyperplasia.
10. **C** Red degeneration is haemorrhagic infarction of a fibroid, typically in pregnancy.
11. **A** Necrosis, atypia and mitoses together define malignancy; the others favour leiomyoma.
12. **D** Older women with lichen sclerosus get HPV-negative keratinising SCC via differentiated VIN.

VIVA Rapid-Fire Viva Voce

Q Which HPV types cause cervical cancer most often?	HPV 16 (almost 60%) and HPV 18 (about 10%).
Q What do E6 and E7 do?	E6 degrades p53 and activates telomerase; E7 binds RB, releasing E2F.
Q What is a koilocyte?	HPV-infected squamous cell with perinuclear halo and enlarged, wrinkled hyperchromatic nucleus.
Q LSIL and HSIL correspond to which CIN grades?	LSIL = CIN I; HSIL = CIN II and CIN III (including carcinoma in situ).
Q Which IHC marker supports HSIL?	Diffuse block-type p16 staining.
Q Commonest histological type of cervical cancer?	Squamous cell carcinoma, about 80%.
Q Can a vaccinated girl skip screening?	No. Vaccines do not cover all oncogenic types and do not treat existing infection.
Q Hallmark of chronic endometritis?	Plasma cells in the endometrial stroma.
Q What is a chocolate cyst?	Ovarian endometriotic cyst filled with altered brown blood.
Q Why does adenomyosis rarely bleed cyclically?	The misplaced tissue comes from the basalis, which does not respond fully to hormones.
Q Precursor of serous endometrial carcinoma?	Serous endometrial intraepithelial carcinoma, on an atrophic endometrium.
Q Commonest secondary change in a fibroid?	Hyaline degeneration; red degeneration is typical in pregnancy.

OSPE Practice Stations

STATION 1 · SLIDE — 2 minutes

TASK

Section of uterine wall: bundles of smooth muscle with scattered islands of endometrial glands, each surrounded by cellular endometrial stroma, lying deep within the muscle far from the endometrial surface. (a) Identify. (b) Give two features. (c) Name two symptoms.

ANSWER

(a) Adenomyosis. (b) Glands with endometrial stroma between muscle bundles; hypertrophied smooth muscle around them. (c) Menorrhagia and dysmenorrhoea (bulky tender uterus).

STATION 2 · SLIDE — 2 minutes

TASK

Endometrial curettings: closely packed, back-to-back glands with no stroma in between, lined by stratified columnar cells with enlarged vesicular nuclei and mitoses; a few cribriform areas. (a) Diagnosis. (b) Two risk factors. (c) Which gene is most often lost?

ANSWER

(a) Endometrioid (type I) endometrial adenocarcinoma. (b) Obesity, unopposed oestrogen (anovulation/PCOS/HRT), diabetes, nulliparity, Lynch syndrome. (c) PTEN.

STATION 3 · CYTOLOGY 2 minutes

TASK

Pap smear of a 32-year-old: many small cells with scanty cytoplasm, high N:C ratio, dark irregular nuclei, singly and in clusters; mature superficial cells are few. (a) Report this smear in Bethesda terms. (b) What is the next step? (c) Which virus is responsible?

ANSWER

(a) HSIL (CIN II–III). (b) Colposcopy with directed biopsy, then excision (LEEP/cone). (c) High-risk HPV, most often type 16.

RAPID REVISION The Night-Before Page

15 ONE-LINERS

01 Leukoplakia is a clinical sign; biopsy every white vulvar patch.

02 Lichen sclerosus: thin epidermis, hyalinised dermis, band of lymphocytes.

03 Vulvar SCC: HPV pathway (younger, usual VIN) vs TP53 pathway (older, dVIN).

04 HPV 16 ≈ 60%, HPV 18 ≈ 10% of cervical cancers; 6 and 11 cause warts.

05 E6 → p53 destroyed; E7 → RB inactivated; integration removes E2 brake.

06 Transformation zone: immature metaplastic cells are HPV's target.

07 LSIL = CIN I; HSIL = CIN II + III; p16 block-positive in HSIL.

08 Cervical SCC ~80%; peak ~45 years; death from ureteric obstruction.

09 Vaccines are L1 virus-like particles; vaccinated women still need screening.

10 Chronic endometritis = plasma cells; TB causes infertility.

11 Adenomyosis: basalis in myometrium; bulky globular uterus; menorrhagia.

12 Endometriosis: ovary commonest; 2 of 3 (glands, stroma, haemosiderin).

13 Unopposed oestrogen → hyperplasia → EIN → endometrioid carcinoma (PTEN).

14 Serous carcinoma: old, atrophic endometrium, TP53, psammoma, peritoneal spread.

15 Fibroid: multiple, whorled, MED12; leiomyosarcoma: necrosis, atypia, mitoses.

MEMORY HOOKS

- **Lower number, earlier letter**: E6 → p53, E7 → RB (6 before 7, p before R)
- **Adeno-MYO-sis** stays in the MYOmetrium; endometri-OSIS lives Outside
- **Type I = "I"nsulin-resistant, obese, oestrogen, PTEN**; Type II = "Too old, TP53, Terrible"
- **LMS = Lots of Mitoses, Spots of necrosis**, Solitary

TEST YOURSELF: FILL THE GAPS

1. E6 binds and degrades _____.
2. CIN III equals _____ in Bethesda terms.
3. Commonest cause of death in Ca cervix: _____.
4. Diagnostic cell of chronic endometritis: _____.
5. Ovarian endometrioma is called a _____ cyst.
6. Commonest gene in leiomyoma: _____.

Answers: p53 · HSIL · uraemia (ureteric obstruction) · plasma cell · chocolate · MED12

DRAW FROM MEMORY

HPV E6/E7 acting on p53 and RB, ending in HSIL and carcinoma (Figure 20.1).

DRAW FROM MEMORY

The two pathways of endometrial carcinoma side by side (Figure 20.4).

SPACED REVISION ☐ Day 1 ☐ Day 3 ☐ Day 7 ☐ Day 14 ☐ Day 30

21

Ovarian Tumours & Gestational Trophoblastic Disease

The ovary holds three kinds of tissue: a surface lining, germ cells and hormone-making stroma. Each gives its own family of tumours. The placenta adds a fourth story, in which a pregnancy with too many paternal genes grows into a bunch of grapes.

BY THE END OF THIS CHAPTER YOU CAN

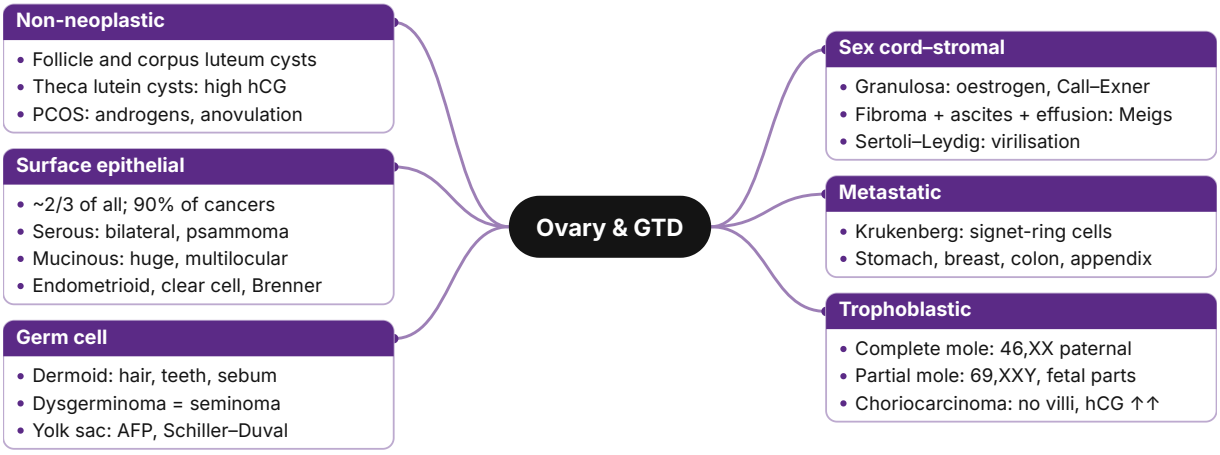
1. Describe functional ovarian cysts and the pathogenesis and morphology of polycystic ovary syndrome
2. Classify ovarian tumours by cell of origin and list their tumour markers
3. Differentiate serous from mucinous tumours and describe borderline and malignant forms
4. Describe mature cystic teratoma, dysgerminoma and yolk sac tumour
5. Describe granulosa cell tumour, fibroma with Meigs syndrome and Krukenberg tumour
6. Differentiate complete from partial hydatidiform mole, and invasive mole from choriocarcinoma

EXAM RADAR

Theory SEQs	★★★
BCQs / MCQs	★★★
Viva voce	★★★
OSPE / slides	★★★

Classification of ovarian tumours, teratoma and complete vs partial mole are classic SEQs; markers (AFP, hCG, inhibin, CA-125) are BCQ favourites; mucinous cystadenoma and mole are lab slides.

ROBBINS LINK Robbins Basic Pathology — *Female Genital System and Breast*: ovary (non-neoplastic cysts, polycystic ovary syndrome, ovarian tumours) and gestational trophoblastic disease.



Chapter at a glance. Know the cell of origin and the marker, and most ovarian questions answer themselves.

21.1 Non-neoplastic cysts and polycystic ovary syndrome

Most ovarian "lumps" in young women are **functional cysts**: follicles or corpora lutea that failed to involute. They usually regress; their danger is **rupture or torsion**. (Endometriotic "chocolate" cysts: Chapter 20.)

- **Follicle cyst** (commonest): unruptured Graafian follicle lined by granulosa/theca cells; thin wall, clear fluid, usually 1–1.5 cm; may make oestrogen.
- **Corpus luteum cyst**: yellow luteinised lining, haemorrhagic contents; rupture causes **haemoperitoneum** that mimics ectopic pregnancy.
- **Theca lutein cysts**: **bilateral**, multiple; luteinised theca cells driven by **high hCG** (mole, choriocarcinoma, multiple pregnancy, ovulation induction).

Polycystic ovary syndrome (Stein–Leventhal syndrome)

A common endocrine disorder of young women (about 6–10% of women of reproductive age) defined by **hyperandrogenism**, **chronic anovulation** (oligomenorrhoea, infertility) and **polycystic ovaries**; two of the three are enough (Rotterdam criteria). Many patients are obese and **insulin-resistant**, and they carry a higher risk of type 2 diabetes and endometrial carcinoma.

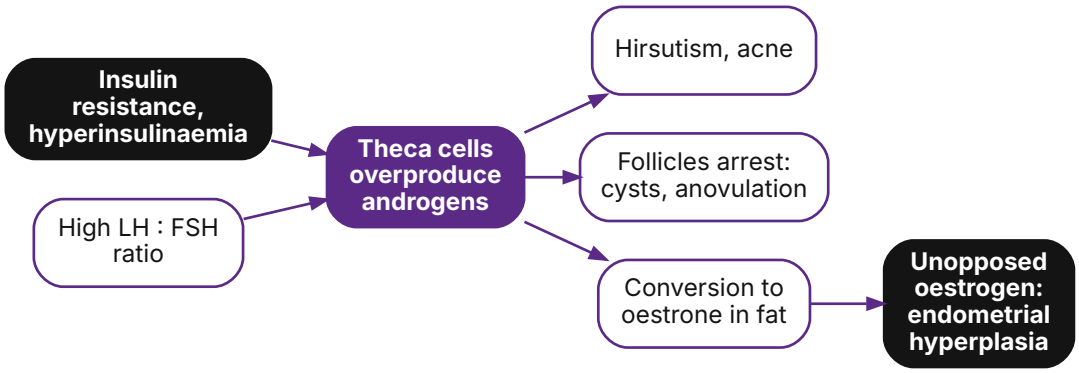


Figure 21.1 · Pathogenesis of PCOS. Insulin and LH drive theca cells to make androgens (low SHBG raises the free fraction); follicles stall and ovulation stops.

Morphology: ovaries **twice normal size**, grey-white, with a smooth **thickened fibrotic cortex** over many **subcortical cysts** 0.5–1.5 cm ("string of pearls" on ultrasound). The cysts are lined by granulosa cells over a **hyperplastic, luteinised theca interna**; **corpora lutea are absent**.

21.2 Classification of ovarian tumours

Ovarian tumours are classified by the **tissue they resemble**: the surface (Müllerian-type) epithelium, the germ cells, or the sex cord–stroma that makes hormones. A fourth group comes from elsewhere. Age is a strong clue: germ cell tumours dominate in children and young women, epithelial tumours after 40.

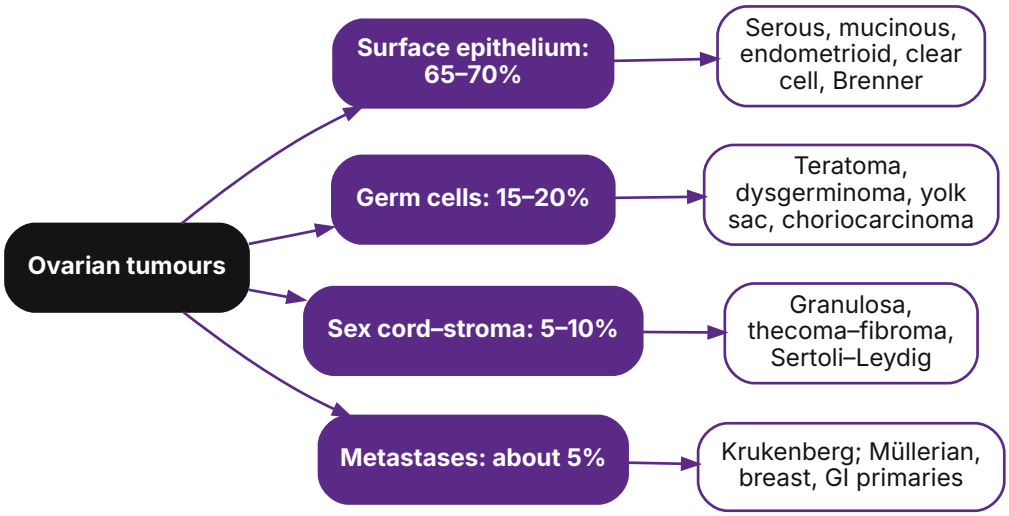


Figure 21.2 · Ovarian tumours by cell of origin (share of all ovarian tumours). Of *malignant* ovarian tumours, about 90% are surface epithelial, 3–5% germ cell, 2–3% sex cord–stromal and about 5% metastatic.

RISK FACTORS FOR OVARIAN CARCINOMA

- **Nulliparity** and **family history** (the two most important)
- Germline **BRCA1/BRCA2** (high-grade serous); Lynch syndrome
- **Endometriosis** (endometrioid, clear cell)

TUMOUR MARKERS: MATCH THE MARKER TO THE TUMOUR

Marker	Tumour
CA-125	Serous and endometrioid carcinoma (> 80%, but only half of stage I); also raised in endometriosis, PID, pregnancy; used to **monitor**, not screen
CEA, CA 19-9	Mucinous tumours

- **Protective:** long-term oral contraceptives, multiparity (fewer ovulations)

Marker	Tumour
AFP	Yolk sac tumour (also immature teratoma, embryonal)
hCG	Choriocarcinoma (also embryonal)
LDH, PLAP	Dysgerminoma
Inhibin	Granulosa cell tumour
Oestrogen / androgen	Granulosa, thecoma / Sertoli–Leydig

Why ovarian cancer is found late: a free-floating ovary causes only vague bloating, while malignant cells shed onto the peritoneum (omental cake, **ascites**). Most cases present in **stage III–IV**, making it the most lethal gynaecological cancer.

21.3 Surface epithelial tumours

Each epithelial type comes in three grades of behaviour: **benign** (cystadenoma), **borderline** (low malignant potential: epithelial proliferation and atypia but **no stromal invasion**) and **malignant** (cystadenocarcinoma, with stromal invasion). The more solid tissue and papillae in a cyst, the more likely it is malignant.

Serous tumours

The commonest ovarian epithelial tumours, with cells resembling **fallopian tube epithelium** (ciliated columnar). Most are benign, but serous carcinoma is the **commonest malignant ovarian tumour**. It follows two pathways. **Low-grade** serous carcinoma develops stepwise from cystadenoma and **borderline tumour** (**KRAS, BRAF** mutations; indolent). **High-grade** serous carcinoma, far commoner, begins as **serous tubal intraepithelial carcinoma (STIC)** in the fimbria, carries **TP53** mutations (plus **BRCA1/2** in hereditary cases) and presents with peritoneal spread.

Mucinous tumours

Lined by **tall columnar mucin-secreting cells without cilia**, resembling endocervical or intestinal epithelium. Most are benign; **KRAS** mutations are common even in benign ones. They may grow to enormous size.

GROSS

- **Large, multilocular** cystic mass, often weighing kilograms
- Smooth outer surface; loculi filled with **sticky, gelatinous mucin**
- Usually **unilateral** (only about 5% bilateral)

MICROSCOPY

- Single layer of **tall columnar cells** with **basal nuclei and apical mucin** (no cilia)
- Borderline: stratification, atypia, no invasion
- Carcinoma: crowded atypical glands invading stroma

SEROUS VS MUCINOUS TUMOURS: THE CLASSIC "DIFFERENTIATE" PAIR

Feature	Serous	Mucinous
Frequency	**Commonest** epithelial tumour; commonest ovarian cancer	Less common; mostly benign
Laterality	**Often bilateral**	**Usually unilateral**; bilateral → suspect metastasis
Size and loculi	Smaller; usually **unilocular**	Often **huge**, **multilocular**
Contents	Clear watery fluid	**Thick, sticky mucin**
Papillae, solid areas	Common (more in borderline and malignant)	Uncommon; smooth surface
Lining	Tubal-type **ciliated** columnar cells	**Non-ciliated mucin-secreting** columnar cells
Psammoma bodies	**Common**: concentric laminated calcifications	Rare
Carcinoma	Complex branching papillae, atypia, stromal invasion; bilateral in about two-thirds	Crowded atypical glands invading stroma

Feature	Serous	Mucinous
Key genes	Low-grade KRAS/BRAF; high-grade TP53, BRCA	KRAS
Marker	CA-125	CEA, CA 19-9

⚠ EXAM TRAP

Pseudomyxoma peritonei (abdomen filled with jelly-like mucin) is usually spread from a mucinous tumour of the **appendix**, not a primary ovarian tumour. Likewise, **bilateral** mucinous ovarian tumours should make you look for a primary in the GI tract (stomach, colon, appendix, pancreas).

Other epithelial tumours. Endometrioid carcinoma: endometrium-like tubular glands; arises in **endometriosis**; PTEN, ARID1A, β-catenin mutations; 15–30% coexist with endometrial carcinoma. **Clear cell carcinoma**: clear glycogen-rich and **hobnail** cells; endometriosis, **ARID1A** loss. **Brenner tumour**: nests of **transitional-type cells** with grooved "coffee-bean" nuclei in dense fibrous stroma; almost always benign.

21.4 Germ cell tumours

These mirror the testicular germ cell tumours, but in the ovary the overwhelming majority are **benign mature cystic teratomas**. Malignant germ cell tumours occur in **girls and young women** and announce themselves with markers (AFP, hCG, LDH).

Mature (benign) cystic teratoma: dermoid cyst

The commonest ovarian germ cell tumour, usually in women of reproductive age. It contains mature tissues from all three germ layers, with **ectoderm dominating**, hence "dermoid". It arises from an ovum after the first meiotic division, so it is **46,XX** (genetically female). About 10–15% are bilateral.

GROSS

- **Unilocular cyst** with a wall like skin
- Filled with **sebaceous (cheesy) material and matted hair**
- **Rokitansky protuberance** (dermoid plug): a nodule in the wall containing **teeth**, bone, cartilage

MICROSCOPY

- **Keratinised squamous epithelium** with hair follicles, sebaceous and sweat glands
- Mature neural (glial) tissue, cartilage, bone, respiratory and GI epithelium, thyroid
- All tissues mature: **no immature neuroepithelium**

Complications: torsion (acute abdomen), rupture with chemical (granulomatous) peritonitis, infection, **malignant transformation** in about 1% (usually **squamous cell carcinoma** in older women), and paraneoplastic **anti-NMDA receptor encephalitis** from its neural tissue.

- **Immature (malignant) teratoma**: adolescents (mean about 18 years); bulky, **solid**; graded by the amount of **immature neuroepithelium** (primitive neural tubules, rosettes).
- **Monodermal teratomas**: **struma ovarii** (thyroid tissue; may cause hyperthyroidism) and **ovarian carcinoid** (carcinoid syndrome without liver metastases, as its venous blood bypasses the liver).

Dysgerminoma

The ovarian **counterpart of the testicular seminoma**. Rare overall (about 2% of ovarian cancers) but about half of malignant germ cell tumours; occurs in the second and third decades, sometimes with **gonadal dysgenesis**.

GROSS	MICROSCOPY
• Usually **unilateral**, **solid**, often large • **Soft, fleshy**, grey-white to tan cut surface	• Sheets and cords of **large uniform cells** with **clear, glycogen-rich cytoplasm**, central nuclei, prominent nucleoli • Thin **fibrous septa infiltrated by lymphocytes**; occasional granulomas • IHC: **OCT3/4**, NANOG, **KIT (CD117)**, PLAP

Serum **LDH** raised (modest hCG if syncytiotrophoblast present). Malignant but **highly radio- and chemosensitive**: excellent outlook.

Yolk sac tumour (endodermal sinus tumour)

- Second commonest malignant germ cell tumour (children, young women); secretes **α-fetoprotein (AFP)** and α1-antitrypsin; grows fast but is chemosensitive.
- **Micro:** lacy (reticular) network of flattened to cuboidal cells; **Schiller–Duval bodies** (a central capillary wrapped by tumour cells, inside a space lined by tumour cells: a primitive glomerulus); **PAS-positive hyaline droplets** containing AFP.
- **Choriocarcinoma** (hCG) and **embryonal carcinoma** occur mostly within mixed germ cell tumours.

21.5 Sex cord–stromal tumours and metastases

These copy the hormone-making cells around the oocyte (granulosa, theca, Sertoli, Leydig) or the stroma, so the clue is often **hormonal**: feminisation or virilisation.

Granulosa cell tumour

Most occur in **postmenopausal women** (adult type, with a **FOXL2** mutation); a juvenile type affects girls. Many secrete **oestrogen**: precocious pseudopuberty in girls, and in older women **endometrial hyperplasia or carcinoma** and breast changes. All are **potentially malignant** and may recur many years later.

GROSS	MICROSCOPY
• Unilateral, solid and cystic • Cut surface **yellow** (lipid) with haemorrhage	• Small cuboidal to polygonal cells in cords, sheets, trabeculae • **Call–Exner bodies**: tiny follicle-like rosettes around eosinophilic fluid • Pale grooved **"coffee-bean" nuclei**; IHC and serum **inhibin**

FIBROMA, THECOMA AND MEIGS SYNDROME

- **Fibroma**: commonest sex cord–stromal tumour; **solid, hard, white**; bland fibroblasts and collagen; hormonally inactive.
- **Thecoma**: plump, lipid-laden spindle cells; **yellow** cut surface; may secrete oestrogen.
- **Meigs syndrome**: ovarian **fibroma** + **ascites** + **hydrothorax** (usually right), which **resolves after removal**. Fibromas also occur in Gorlin (basal cell naevus) syndrome.

SERTOLI–LEYDIG CELL TUMOUR

- Young women; secretes **androgens** → defeminisation then virilisation (amenorrhoea, breast atrophy, hirsutism, clitoromegaly)
- Tubules of Sertoli cells with Leydig cells in the stroma; **DICER1** mutations in many

METASTASES AND KRUKENBERG TUMOUR

- Commonest metastases come from **Müllerian** sites (uterus, tube, peritoneum); non-Müllerian from **breast** and **GI tract**
- **Krukenberg tumour**: **bilateral** ovarian metastases of **mucin-filled signet-ring cells** in a cellular stroma, classically from **gastric carcinoma**

21.6 Gestational trophoblastic disease

GTD is a spectrum of abnormal trophoblast proliferation: **hydatidiform mole** (benign) → **invasive mole** → **choriocarcinoma** (malignant), plus the rare placental site trophoblastic tumour. All make **hCG**, which is used to diagnose and follow them. Moles are commoner at the **extremes of reproductive age** and after a previous mole.

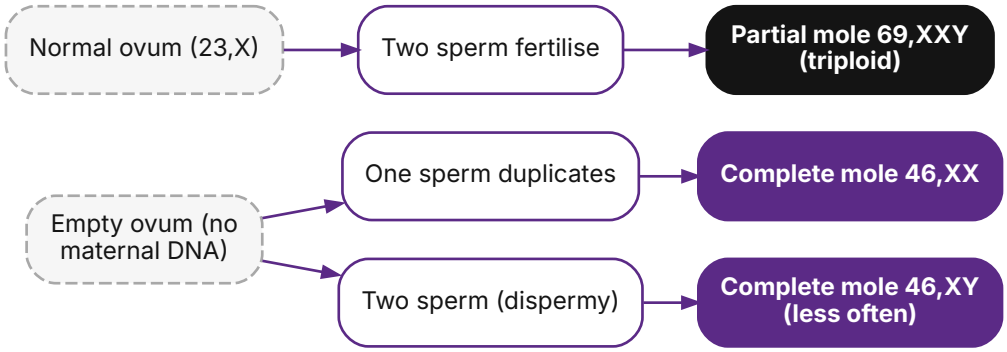


Figure 21.3 · Too many paternal genes make a mole. Paternal genes drive trophoblast, maternal genes the embryo. A complete mole is entirely paternal, so there is no embryo and every villus is molar; a partial mole has one maternal set, so some fetal tissue forms.

COMPLETE VS PARTIAL HYDATIDIFORM MOLE

Feature	Complete mole	Partial mole
Karyotype	**46,XX** (rarely 46,XY), all paternal	**Triploid** (69,XXY), extra paternal set
Fetal parts	**Absent**	**Present** (fetal tissue, nucleated red cells in villous vessels)
Villous oedema	**All villi**, with central cisterns	**Some villi**; others normal; scalloped outlines
Trophoblast proliferation	**Diffuse, circumferential**, often atypical	**Focal, slight**; no atypia
p57 immunostain	**Negative** (no maternal genome)	**Positive**
Serum hCG	**Markedly raised**	Less raised
Behaviour	About 10% invasive mole, **2–3% choriocarcinoma**	Small risk of persistent disease; **choriocarcinoma rare**

GROSS (COMPLETE MOLE)

- Uterine cavity filled with a delicate, friable mass of **thin-walled, translucent grape-like vesicles** ("bunch of grapes")
- No embryo, no normal placenta

MICROSCOPY

- Large, round **hydropic villi** with acellular, fluid-filled **central cisterns**
- Villous blood vessels scanty or absent
- **Circumferential proliferation** of cytotrophoblast and syncytiotrophoblast, with atypia

Clinical: bleeding in early pregnancy, passage of vesicles, uterus large for dates, no fetal heart, **"snowstorm"** ultrasound; very high hCG causes **hyperemesis**, early **pre-eclampsia** (< 20 weeks), **hyperthyroidism** and **theca lutein cysts**. **Management:** suction evacuation, then **serial hCG** to zero with contraception; a plateau or rise means persistent GTD.

Invasive mole and choriocarcinoma

Feature	Invasive mole	Gestational choriocarcinoma
Definition	Molar villi that **penetrate the myometrium**; hCG stays raised	**Malignant** tumour of cyto- and syncytiotrophoblast; hCG **very high**
Preceded by	Complete mole	Complete mole (about 50%), abortion (about 25%), normal pregnancy (about 22%), ectopic (rest)
Chorionic villi	**Present**	**Absent** (key diagnostic point)

Feature	Invasive mole	Gestational choriocarcinoma
Gross	Haemorrhagic, invasive molar tissue; may perforate the uterus	**Soft, fleshy, yellow-white mass with extensive haemorrhage and necrosis**
Spread	Villi may embolise to lung or brain but do not grow as true metastases	**Early blood-borne**: **lungs** (about half), vagina, brain, liver, bone, kidney
Treatment	Chemotherapy (e.g. methotrexate-based): invasive moles and **most gestational choriocarcinomas are cured**, even with metastases	

Placental site trophoblastic tumour is a rare tumour of intermediate (implantation-site) trophoblast: hPL-positive, only modest hCG, and relatively **resistant to chemotherapy**, so it is treated by hysterectomy.

⚠ EXAM TRAP

Villi = mole (complete, partial or invasive); **no villi = choriocarcinoma**. Gestational choriocarcinoma is highly chemosensitive; germ cell (non-gestational) choriocarcinoma is not.

LAB SPOTTER

Lab slides in 10 seconds. Mucinous cystadenoma: thin fibrous septa lined by a **single "picket fence" of tall columnar cells, basal nuclei, pale mucin**; no cilia. **Complete mole:** huge **round villi with empty central cisterns** and a **rim of proliferating trophoblast** all round.

EXAM ZONE Past-Paper Pattern SEQs — Solved

PROF PATTERN **a)** Classify ovarian tumours. **b)** Describe the morphology and complications of mature cystic teratoma. **2 + 3 marks**

MODEL ANSWER PLAN

a) By cell of origin: surface epithelial (serous, mucinous, endometrioid, clear cell, Brenner), germ cell (teratoma, dysgerminoma, yolk sac, choriocarcinoma), sex cord–stromal (granulosa, thecoma–fibroma, Sertoli–Leydig), metastatic (Krukenberg), with frequencies. **b)** Unilocular cyst with hair, sebum, Rokitansky protuberance with teeth; micro: skin with adnexa plus mature tissues of all three germ layers; 46,XX. Complications: torsion, rupture with chemical peritonitis, infection, malignant change (SCC, ~1%), anti-NMDA receptor encephalitis.

Score booster: a neat four-branch diagram (Figure 21.2) earns the classification marks faster than a list.

MODULE PATTERN **a)** Differentiate between serous and mucinous tumours of the ovary. **b)** Name the tumour markers useful in ovarian neoplasms. **3 + 2 marks**

MODEL ANSWER PLAN

a) Table: frequency, bilaterality, loculi, fluid, ciliated vs mucin-secreting cells, psammoma bodies, genes, CA-125 vs CEA. **b)** CA-125 (serous, endometrioid), CEA/CA 19-9 (mucinous), AFP (yolk sac), hCG (choriocarcinoma), LDH/PLAP (dysgerminoma), inhibin (granulosa).

PROF PATTERN A 19-year-old primigravida at 14 weeks has vaginal bleeding and passes grape-like tissue. The uterus is large for dates, serum hCG is very high and ultrasound shows a "snowstorm" pattern with no fetus. **a)** Diagnosis? **b)** Explain its genetic basis. **c)** Tabulate how it differs from a partial mole. **1 + 2 + 2 marks**

MODEL ANSWER PLAN

a) Complete hydatidiform mole. **b)** Empty ovum fertilised by one sperm that duplicates (46,XX) or by two sperm (46,XY): all chromosomes paternal → trophoblast overgrowth, no embryo. **c)** Karyotype, fetal parts, extent of villous oedema, trophoblast proliferation, p57, hCG level, risk of choriocarcinoma.

Score booster: add follow-up with serial hCG and the 2–3% risk of choriocarcinoma; examiners like the management link.

BCQs One-Best-Answer Practice

1 Which group accounts for most malignant ovarian tumours?

A. Germ cell tumours
B. Sex cord–stromal tumours
C. Surface epithelial tumours
D. Metastatic tumours
E. Lymphomas

2 A 55-year-old woman has bilateral ovarian masses with papillary excrescences, ascites and raised CA-125. Microscopy shows complex papillae with concentric laminated calcifications. The diagnosis is:

A. Serous cystadenocarcinoma
B. Mucinous cystadenoma
C. Brenner tumour
D. Dysgerminoma
E. Fibroma

3 A 40-year-old has a 25 cm unilateral multilocular ovarian cyst filled with sticky gelatinous fluid. The lining is a single layer of tall non-ciliated columnar cells with basal nuclei. The diagnosis is:

A. Serous cystadenoma
B. Follicular cyst
C. Endometriotic cyst
D. Mature cystic teratoma
E. Mucinous cystadenoma

4 The karyotype of a mature cystic teratoma of the ovary is usually:

A. 69,XXY
B. 46,XX
C. 46,XY
D. 45,X
E. 47,XXY

5 An 18-year-old has a rapidly growing ovarian mass and a very high serum AFP. Microscopy shows glomerulus-like structures with a central vessel. The tumour is:

A. Dysgerminoma
B. Granulosa cell tumour
C. Choriocarcinoma
D. Yolk sac tumour
E. Mature teratoma

6 Call–Exner bodies and coffee-bean nuclei are features of:

A. Granulosa cell tumour
B. Brenner tumour
C. Thecoma
D. Sertoli–Leydig cell tumour
E. Krukenberg tumour

7 A woman has a solid, hard, white ovarian tumour, ascites and a right pleural effusion that all resolve after oophorectomy. This is:

A. Krukenberg tumour
B. Pseudomyxoma peritonei
C. Meigs syndrome
D. Stein–Leventhal syndrome
E. Struma ovarii

8 Bilateral solid ovarian masses show signet-ring cells in a cellular stroma. The commonest primary site is:

A. Breast
B. Endometrium
C. Thyroid
D. Kidney
E. Stomach

9 A complete hydatidiform mole typically has which karyotype?

A. 69,XXY with maternal and paternal sets
B. 46,XX, entirely paternal
C. 46,XX, entirely maternal
D. 92,XXXY
E. 45,X

10 Evacuated products show some swollen villi with scalloped outlines, focal mild trophoblastic proliferation and fetal red cells in villous vessels. p57 staining is positive. The diagnosis is:

A. Complete mole
B. Choriocarcinoma
C. Invasive mole
D. Partial mole
E. Placental site trophoblastic tumour

11 The single most useful histological feature separating choriocarcinoma from invasive mole is:

A. Haemorrhage
B. Myometrial invasion
C. Absence of chorionic villi
D. Raised hCG
E. Lung metastases

12 Serum CA-125 is most useful for:

A. Monitoring treatment response and recurrence of epithelial ovarian cancer
B. Screening all women over 40
C. Diagnosing dysgerminoma
D. Distinguishing benign from malignant teratoma
E. Diagnosing hydatidiform mole

ANSWER KEY WITH ONE-LINE REASONS

1. **C** Surface epithelial tumours make up about 90% of ovarian cancers.
2. **A** Bilateral papillary tumour with psammoma bodies and CA-125 = serous carcinoma.
3. **E** Huge multilocular cyst with mucin-secreting columnar lining.

4. **B** It arises from an ovum after the first meiotic division, so it is 46,XX.
5. **D** Schiller–Duval bodies and AFP define yolk sac tumour.
6. **A** Call–Exner bodies are follicle-like rosettes of granulosa cells.
7. **C** Fibroma + ascites + hydrothorax = Meigs syndrome.
8. **E** Krukenberg tumour: classically metastatic gastric signet-ring carcinoma.
9. **B** An empty ovum fertilised by a sperm that duplicates: androgenetic diploidy.
10. **D** Fetal tissue, patchy oedema and p57 positivity (maternal genome present) = partial mole.
11. **C** Invasive moles contain villi; choriocarcinoma has none.
12. **A** CA-125 is non-specific and insensitive for early disease; it is best for follow-up.

VIVA Rapid-Fire Viva Voce

Q What causes theca lutein cysts?	High hCG: mole, choriocarcinoma, multiple pregnancy.
Q Rotterdam criteria for PCOS?	Two of: hyperandrogenism, anovulation, polycystic ovaries.
Q What is a borderline ovarian tumour?	Epithelial proliferation with atypia but no stromal invasion.
Q Where does high-grade serous carcinoma begin?	In serous tubal intraepithelial carcinoma of the fimbria; TP53 mutated.
Q What are psammoma bodies?	Concentric laminated calcifications, e.g. in serous carcinoma.
Q What is a Rokitansky protuberance?	Nodule in a dermoid cyst wall where teeth, bone and hair concentrate.
Q Ovarian counterpart of seminoma?	Dysgerminoma.
Q Marker of yolk sac tumour?	Alpha-fetoprotein (AFP).
Q What is struma ovarii?	Teratoma of thyroid tissue; may cause hyperthyroidism.
Q Define Krukenberg tumour.	Bilateral ovarian metastases of signet-ring cells, usually gastric.
Q Why is p57 negative in a complete mole?	p57 comes from the maternal allele; there is no maternal genome.
Q Commonest site of metastasis of choriocarcinoma?	Lungs (about half), then vagina.

OSPE Practice Stations

STATION 1 · SLIDE — 2 minutes

TASK

Section of an ovarian cyst wall: thin fibrous septa lined by a single layer of tall columnar cells with small basal nuclei and abundant pale apical mucin; no cilia, no papillae, no stromal invasion. (a) Identify. (b) Give two gross features. (c) Name the commonest genetic change.

ANSWER

(a) Mucinous cystadenoma of the ovary. (b) Large, unilateral, multilocular cyst filled with sticky gelatinous mucin; smooth outer surface. (c) KRAS mutation.

STATION 2 · SLIDE — 2 minutes

TASK

Uterine evacuation: large rounded avascular villi with central fluid-filled spaces, surrounded on all sides by sheets of proliferating cytotrophoblast and syncytiotrophoblast with atypia. No fetal tissue. (a) Diagnosis. (b) Expected karyotype. (c) How is the patient followed up?

ANSWER

(a) Complete hydatidiform mole. (b) 46,XX (all paternal). (c) Serial serum hCG until undetectable, with contraception; plateau or rise indicates persistent GTD.

STATION 3 · LAB DATA 2 minutes

TASK

Six weeks after evacuation of a complete mole, a 25-year-old's serum β-hCG readings (IU/L) are 1,200 → 1,250 → 1,900 over three weeks. Chest X-ray shows two round shadows. (a) Interpret. (b) Which histological feature would separate the two possible diagnoses? (c) Treatment?

ANSWER

(a) Rising hCG after a mole = persistent GTD with lung metastases (invasive mole or choriocarcinoma). (b) Chorionic villi present in invasive mole, absent in choriocarcinoma. (c) Chemotherapy (e.g. methotrexate-based); excellent cure rates.

RAPID REVISION The Night-Before Page

15 ONE-LINERS

01 Theca lutein cysts are bilateral and mean high hCG.

02 PCOS: androgens + anovulation + polycystic ovaries; insulin resistance; no corpora lutea.

03 Surface epithelial tumours: two-thirds of all, 90% of malignant ovarian tumours.

04 Borderline tumour: atypia without stromal invasion.

05 Serous: bilateral, unilocular, ciliated cells, psammoma bodies, CA-125.

06 High-grade serous: TP53, BRCA, starts in the fimbria (STIC).

07 Mucinous: huge, multilocular, unilateral, KRAS; bilateral → think metastasis.

08 Endometrioid and clear cell carcinoma arise in endometriosis (ARID1A).

09 Dermoid: 46,XX, hair, sebum, teeth, Rokitansky; SCC in ~1%.

10 Dysgerminoma: seminoma-like clear cells, lymphocytes, LDH, radiosensitive.

11 Yolk sac tumour: AFP, Schiller–Duval bodies, hyaline droplets.

12 Granulosa: oestrogen, Call–Exner, FOXL2, inhibin.

13 Meigs: fibroma + ascites + right hydrothorax; Krukenberg: gastric signet rings.

14 Complete mole 46,XX paternal, no fetus, p57 negative; partial 69,XXY with fetus.

15 Villi = mole; no villi = choriocarcinoma; lungs are the commonest metastatic site.

MEMORY HOOKS

- **A tumour secretes what its normal twin secretes**: yolk sac → AFP, chorion → hCG, granulosa → inhibin and oestrogen
- **Complete mole = Completely paternal**; Partial = Paternal ×2 + maternal ×1
- **Serous Spreads both Sides** (bilateral); Mucinous is Massive and Multilocular
- **Meigs = Mass + Moisture in belly + Moisture in chest**

TEST YOURSELF: FILL THE GAPS

1. Concentric calcifications in serous carcinoma: _____ bodies.
2. Germ cell tumour marker in yolk sac tumour: _____.
3. Rosettes in granulosa cell tumour: _____ bodies.
4. Karyotype of partial mole: _____.
5. Krukenberg primary is usually the _____.
6. Immunostain negative in complete mole: _____.

Answers: psammoma · AFP · Call–Exner · 69,XXY · stomach · p57

DRAW FROM MEMORY

The four-branch classification of ovarian tumours with two examples and one marker each.

DRAW FROM MEMORY

How complete and partial moles arise (empty vs normal ovum, one vs two sperm).

SPACED REVISION ☐ Day 1 ☐ Day 3 ☐ Day 7 ☐ Day 14 ☐ Day 30

22

Breast

A breast lump brings fear into the clinic, yet most lumps in young women are harmless fibroadenomas or cysts. The skill is to know which features are reassuring and which are not, and to read a carcinoma's receptors as carefully as its morphology, because they decide its treatment.

BY THE END OF THIS CHAPTER YOU CAN

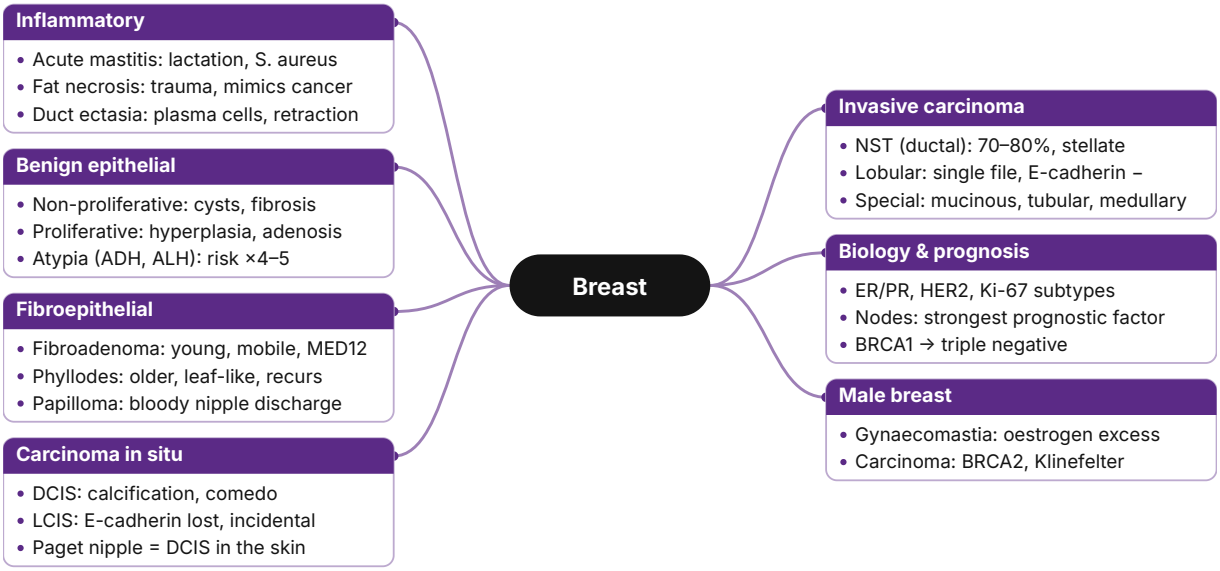
1. Describe acute mastitis, periductal mastitis, duct ectasia and fat necrosis, and why some mimic carcinoma
2. Classify benign epithelial lesions by their risk of later carcinoma
3. Differentiate fibroadenoma, phyllodes tumour and carcinoma as causes of a breast lump
4. List the risk factors for breast carcinoma and explain the role of BRCA1 and BRCA2
5. Differentiate DCIS from LCIS and invasive carcinoma of no special type from lobular carcinoma
6. Classify breast carcinoma into molecular subtypes and list prognostic and predictive factors
7. Describe gynaecomastia and carcinoma of the male breast

EXAM RADAR

Theory SEQs	★★★
BCQs / MCQs	★★★
Viva voce	★★★
OSPE / slides	★★★

Carcinoma breast (risk factors, morphology, prognostic factors) is a near-certain long question; fibroadenoma and carcinoma are lab slides; DCIS vs LCIS and receptors are BCQ favourites.

ROBBINS LINK Robbins Basic Pathology — *Female Genital System and Breast*: inflammatory disorders, benign epithelial lesions, fibroepithelial tumours, carcinoma of the breast, male breast.



Chapter at a glance. Seven ideas cover every breast question.

22.1 Where breast lesions arise; inflammatory disorders

The breast is a branching duct system that ends in the terminal duct lobular unit (TDLU). Each duct and lobule has two cell layers: inner **luminal epithelial cells** and outer **myoepithelial cells** on a basement membrane. **Rule of thumb:** benign and in situ lesions keep the myoepithelial layer; **invasive carcinoma loses it**. Knowing **where** along the tree a lesion arises predicts what it looks like and how it presents.

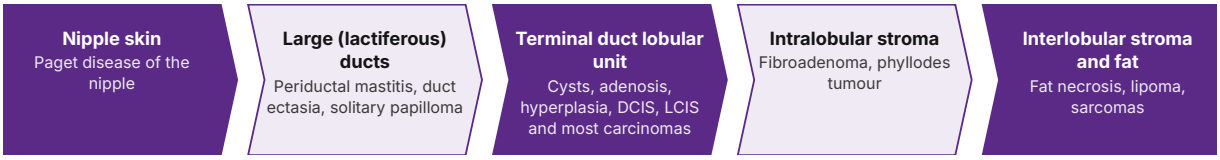


Figure 22.1 · Breast lesions mapped along the ductal tree, from the nipple to the lobule and its stroma.

CLINICAL CORRELATION

Triple assessment of every breast lump: (1) **clinical examination**, (2) **imaging** (ultrasound in young women, mammography in older women) and (3) **tissue sampling** (FNAC or core needle biopsy). A lump is labelled benign only when all three agree.

INFLAMMATORY LESIONS: TWO INFECTIONS AND TWO MIMICS OF CARCINOMA

Lesion	Who and why	Pathology	Clinical point
Acute mastitis	First month of **breastfeeding**; cracked nipple lets in bacteria	**S. aureus**: localised abscess; streptococci: diffuse spreading cellulitis	Red, hot, painful breast, fever; antibiotics, keep emptying milk, drain abscess
Periductal mastitis (squamous metaplasia of lactiferous ducts)	Middle-aged women, **smokers** in the great majority	Keratin plugs a metaplastic duct → duct ruptures → granulomatous reaction to keratin	Painful **subareolar abscess**, nipple inversion, recurrent **fistula** to the areola
Mammary duct ectasia	Older, usually multiparous women	Dilated ducts with inspissated secretions, **foamy macrophages**, periductal fibrosis and chronic inflammation rich in **plasma cells** ("plasma cell mastitis")	Thick nipple discharge, ill-defined periareolar mass, **nipple retraction**: **mimics carcinoma**
Fat necrosis	Trauma (often forgotten), surgery, radiotherapy	Early: haemorrhage, necrotic anucleate fat cells, neutrophils; then **lipid-laden macrophages**, giant cells, fibrosis, **calcification**	Painless **hard mass**, skin retraction, mammographic calcification: **mimics carcinoma**

EXAM TRAP

Three benign lesions that look malignant: fat necrosis (hard mass + calcification), **duct ectasia** (nipple retraction) and **sclerosing adenosis / radial scar** (distorted glands on mammogram and slide). Every hard lump with skin or nipple retraction needs tissue diagnosis.

22.2 Benign epithelial lesions (fibrocystic change)

These are very common changes in women of reproductive age, often felt as premenstrual lumpiness and pain. Robbins groups them by the **risk of later invasive carcinoma**, which is the only thing that matters clinically. The risk applies to **both breasts**: these lesions are markers of risk, not direct precursors.



Figure 22.2 · The risk ladder of benign breast lesions (relative risk of invasive carcinoma, Robbins).

NON-PROLIFERATIVE CHANGES

- **Cysts**: dilated lobules; large ones are bluish ("**blue-dome cysts**") with turbid fluid; lined by flattened or **apocrine metaplastic** cells (eosinophilic, granular cytoplasm); calcification common
- **Fibrosis**: ruptured cysts provoke inflammation and scarring
- **Adenosis**: more acini per lobule, may calcify

PROLIFERATIVE CHANGES

- **Epithelial hyperplasia**: more than two cell layers filling ducts, with irregular peripheral slit-like spaces and mixed, streaming cells
- **Sclerosing adenosis**: crowded acini compressed and distorted by stroma; myoepithelium kept; **mimics invasive carcinoma**
- **Radial scar (complex sclerosing lesion)**: stellate lesion with entrapped glands in a central scar
- **Atypical ductal hyperplasia**: monomorphic cells partly filling ducts, resembling but not amounting to DCIS. **Atypical lobular hyperplasia**: LCIS-like cells filling less than half of a lobule

22.3 Fibroepithelial tumours and intraductal papilloma

Fibroadenoma and phyllodes tumour are **biphasic**: both stroma and epithelium proliferate, but the **stroma is the neoplastic part**. Both often carry **MED12** mutations in their stromal cells. What separates them is how cellular and how aggressive that stroma is.

Fibroadenoma

The **commonest benign tumour of the female breast**, typically in women aged 20–35. It is hormone-responsive: it may enlarge in pregnancy and lactation, and regresses and calcifies after menopause. Multiple fibroadenomas occur in transplant patients on **cyclosporine**.

GROSS

- Well-circumscribed, **freely mobile** ("breast mouse"), rubbery nodule, usually a few cm
- Grey-white cut surface that **bulges**, with slit-like spaces

MICROSCOPY

- **Loose, delicate, cellular stroma** enclosing glands lined by **epithelial + myoepithelial** cells
- **Intracanalicular** pattern: stroma compresses ducts into curved slits; **pericanalicular**: ducts keep round lumens
- Sharp border; stroma bland, few mitoses

Not premalignant. A simple fibroadenoma adds no risk; a "complex" one (cysts, sclerosing adenosis, calcification) carries the small risk of proliferative disease.

Phyllodes tumour

Also arises from **intralobular stroma**, but is much less common, occurs about **10–20 years later** (peak in the sixth decade), grows larger and faster, and has **more cellular stroma** that overgrows the epithelium and pushes it into **leaf-like** (Greek *phyllon*) projections. Graded **benign, borderline or malignant** by stromal cellularity, atypia, mitoses, overgrowth and an infiltrative border. Most behave benignly but **recur locally** unless widely excised; malignant ones metastasise by blood (lymph-node spread is rare).

THE BREAST LUMP: FIBROADENOMA VS PHYLLODES TUMOUR VS CARCINOMA

Feature	Fibroadenoma	Phyllodes tumour	Carcinoma (NST)
Age	**20–35**	**About 50–60**	Rises with age; mostly over 50
Growth	Slow; stops at a few cm	Often **rapid**, may be large	Progressive
Feel	Rubbery, smooth, **very mobile**	Firm, bosselated, mobile	**Hard, irregular, fixed**; skin/nipple retraction
Cut surface	Bulging, grey-white, slit-like spaces	Fleshy with **leaf-like clefts** and cysts	**Stellate**, gritty, chalky streaks
Stroma	Bland, loose, low cellularity	**Hypercellular**, overgrowth, mitoses ± atypia	Desmoplastic, reactive
Myoepithelial layer	Present	Present	**Absent** around invasive nests
Behaviour	Benign	Local recurrence; malignant ones metastasise by blood	Lymph node and distant metastasis

Intraductal papilloma

A benign papillary growth in a duct. The common **solitary large-duct papilloma** sits in a lactiferous sinus beneath the nipple and presents with **serous or bloody nipple discharge**, sometimes a small subareolar lump. Each papilla has a **fibrovascular core** covered by **two cell layers**: luminal epithelium and **myoepithelium**. Papillary carcinoma lacks the myoepithelial layer. Multiple small-duct papillomas lie deeper; a papilloma with atypia raises the risk.

22.4 Carcinoma of the breast: risk and pathogenesis

The **commonest cancer of women worldwide** (excluding skin) and a leading cause of cancer death in women. In the USA the lifetime risk is about **1 in 8**. Almost all are **adenocarcinomas** arising in the TDLU.

PAKISTAN FOCUS

Breast cancer is the **commonest cancer in Pakistani women**, and Pakistan has one of the **highest breast cancer rates in Asia**; the lifetime risk is often quoted as about **1 in 9 women**. Many women present late, in **stage III–IV**, because of stigma, shyness and the absence of screening. Awareness campaigns such as Pink Ribbon Pakistan teach breast self-awareness and early consultation for any lump.

RISK FACTORS (ROBBINS)

- **Female sex** and **age** (rare before 25)
- **Family history** in first-degree relatives; **germline mutations**
- **Long oestrogen exposure**: early menarche, late menopause, nulliparity, **late first full-term pregnancy**
- Postmenopausal **hormone therapy** (oestrogen + progestin); **obesity** after menopause; alcohol
- **Atypical hyperplasia**; previous breast or endometrial cancer
- **Chest radiation** when young (e.g. for Hodgkin lymphoma); dense breasts
- **Protective: breastfeeding**, early first pregnancy, exercise

HEREDITARY BREAST CANCER

A minority (roughly one in ten) arise from an inherited high-penetrance mutation; suspect it with early onset, bilateral disease, male breast cancer, or ovarian cancer in the family.

BRCA1 (17q21)	DNA repair (homologous recombination); **triple-negative, basal-like**, high grade; ovarian cancer
BRCA2 (13q12)	DNA repair; usually **ER-positive**; **male breast**, ovary, prostate, pancreas
TP53 · PTEN	Li–Fraumeni · Cowden syndrome
CHEK2, ATM, PALB2	Moderate-risk DNA-repair genes
CDH1	Lobular carcinoma + diffuse gastric cancer

CONCEPT CLEAR

One morphology, three diseases. Breast carcinomas that look alike under the microscope follow three different molecular roads, and the road decides behaviour and treatment. Robbins groups them by two questions: **does the tumour express oestrogen receptor (ER)?** and **is HER2 (ERBB2) amplified?** Oestrogen drives the growth of ER-positive cells, which is why hormone exposure is a risk factor and why blocking oestrogen treats them.

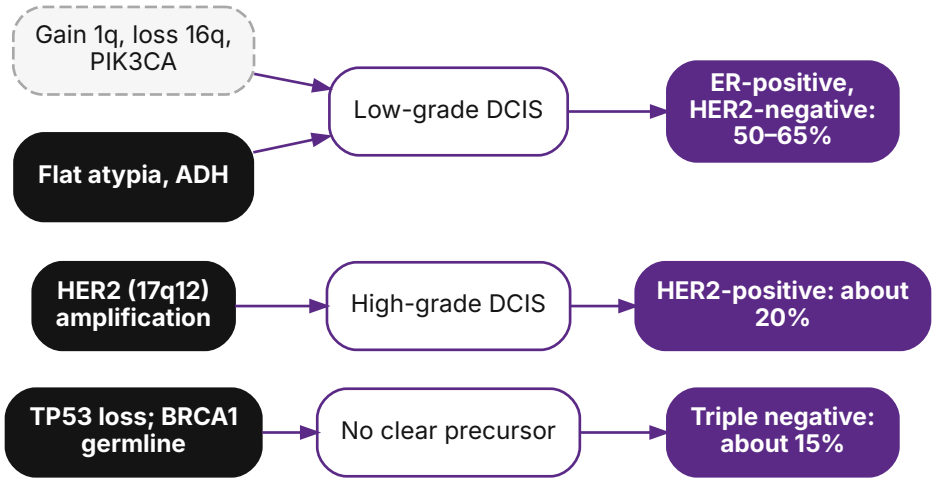


Figure 22.3 · Three pathways of breast carcinogenesis (approximate share of all carcinomas).

22.5 Carcinoma in situ and invasive carcinoma

In situ carcinoma is confined by the basement membrane, with the myoepithelial layer still present, so it cannot metastasise. The names "ductal" and "lobular" describe the **appearance of the cells**, not the site: both arise in the TDLU.

DCIS patterns. Comedo DCIS: high-grade cells around **central necrosis** that calcifies; cheesy necrotic plugs squeeze out of cut ducts like comedones. **Non-comedo DCIS**: **cribriform** (evenly spaced, round "cookie-cutter" spaces), solid, papillary or micropapillary; usually lower grade.

DCIS VS LCIS: A FAVOURITE "DIFFERENTIATE" QUESTION

Feature	Ductal carcinoma in situ (DCIS)	Lobular carcinoma in situ (LCIS)
How found	**Mammographic calcifications** (most); rarely a mass or nipple discharge	**Incidental** in a biopsy done for something else: no mass, no calcification
Cells	**Cohesive**; low to high nuclear grade	Small, **uniform, discohesive** round cells; **signet-ring** cells with mucin
Architecture	Fills and distends ducts: **comedo**, cribriform, solid, papillary, micropapillary	Fills and expands acini of lobules without distorting them
Necrosis, calcification	Common (central necrosis in comedo type)	Absent
E-cadherin	Present	**Lost** (CDH1 inactivated)
Bilateral	Less often	**More often** (bilateral and multicentric)
Later invasive cancer	Usually **same breast, same quadrant**, ductal (NST) type	**Either breast**; lobular or NST type
Meaning	**Direct precursor**	Marker of risk and non-obligate precursor
Management	Excision ± radiotherapy ± tamoxifen (or mastectomy)	Close follow-up, chemoprevention (e.g. tamoxifen)

CLINICAL CORRELATION

Paget disease of the nipple: **DCIS** (usually high grade, often HER2-positive) spreads up the lactiferous ducts into the **nipple epidermis** without crossing the basement membrane. The nipple shows a unilateral, red, scaly, crusted eruption that is easily mistaken for eczema. **Paget cells** are large, with pale abundant cytoplasm, lying singly or in groups in the epidermis. About half the patients have an underlying invasive carcinoma, usually with a palpable mass.

Invasive carcinoma of no special type (NST; invasive ductal carcinoma)

The commonest type, about **70–80%**. About half arise in the **upper outer quadrant**. It presents as a hard, painless lump; tethering to Cooper ligaments causes **skin dimpling**, involvement of ducts causes **nipple retraction**, and blocked dermal lymphatics cause **peau d'orange** (orange-peel skin).

GROSS

- **Hard** (scirrhous), **irregular, stellate** mass with infiltrating "crab-like" edges
- Grey-white, **gritty** cut surface with **chalky yellow-white streaks** (elastosis); grates like an unripe pear
- May tether skin, retract nipple, fix to pectoral fascia

MICROSCOPY

- Cords, nests, glands and sheets of **malignant ductal cells** infiltrating a **dense desmoplastic stroma**
- **No myoepithelial cells** around the invasive nests
- Lymphovascular invasion; accompanying DCIS
- **Nottingham grade** 1–3: tubule formation + nuclear pleomorphism + mitotic count

Invasive lobular carcinoma

- The second commonest type (roughly 10–15%); more often **bilateral and multicentric**
- Discohesive cells infiltrating **in single file** ("Indian file") or in **targetoid** rings around ducts; **signet-ring** cells; little desmoplasia
- Hallmark: **loss of E-cadherin** (CDH1); almost all **ER-positive**
- May form only a vague thickening, hard to feel or see on mammography
- **Unusual metastatic sites**: peritoneum and retroperitoneum, **leptomeninges**, GI tract, ovaries and uterus, bone marrow

SPECIAL TYPES AND A CLINICAL VARIANT

Type	Key features	Outlook
Mucinous (colloid)	Older women; soft, **gelatinous**, well-circumscribed; tumour clusters floating in **pools of extracellular mucin**; ER-positive	Good
Tubular	Small, mammographically detected; well-formed **tubules of single-layered cells**, no myoepithelium; ER-positive	Excellent

Type	Key features	Outlook
Carcinoma with medullary features	Soft, well-circumscribed; **syncytial sheets** of large pleomorphic cells, **pushing border**, dense **lymphoplasmacytic infiltrate**; triple negative; **BRCA1** carriers	Better than other triple-negative cancers
Inflammatory carcinoma	A **clinical** term: swollen, red, warm breast with peau d'orange, often no discrete mass; caused by **tumour emboli blocking dermal lymphatics**, not by inflammation	**Poor**

22.6 Molecular subtypes, prognosis and spread

Every invasive carcinoma is tested by immunohistochemistry for **ER, PR, HER2** and the proliferation marker **Ki-67**. These sort tumours into subtypes that mirror gene-expression profiles and predict treatment.

MOLECULAR SUBTYPES OF BREAST CARCINOMA

Subtype	ER / PR	HER2	Ki-67	Typical features	Treatment and outlook
Luminal A	+ / +	–	Low	Commonest; older women; low grade	Endocrine therapy; **best prognosis**
Luminal B	+ / ± (PR often low)	– or +	High	Higher grade	Endocrine + chemotherapy (± anti-HER2)
HER2-enriched	– / –	**Amplified**	High	High grade; younger women	**Trastuzumab** + chemotherapy; good response
Basal-like / triple negative	– / –	–	High	Young women, **BRCA1**; high grade, pushing border; brain and lung metastases	Chemotherapy only (PARP inhibitors in BRCA); **worst early outcome**

LAB VALUES · READING THE RECEPTOR REPORT

ER / PR positive nuclear staining in ≥ 1% of tumour cells

HER2 IHC

0 or 1+ negative · 2+ equivocal → in situ hybridisation (FISH) · 3+ positive

HER2 FISH

gene amplification = positive (eligible for trastuzumab)

Ki-67

% of tumour nuclei staining; high values mean a fast-growing tumour

PROGNOSTIC FACTORS

- **Distant metastasis** (stage IV): cure unlikely
- **Axillary lymph node status**: the **most important** prognostic factor in the absence of distant spread; assessed by **sentinel lymph node** biopsy
- **Tumour size**; locally advanced disease (skin, chest wall); inflammatory carcinoma
- **Histological grade** and type (special types do better)
- Proliferation (mitoses, Ki-67), lymphovascular invasion, molecular subtype

PREDICTIVE FACTORS (GUIDE THERAPY)

- **ER/PR**: predict response to **tamoxifen** or **aromatase inhibitors**
- **HER2** amplification: predicts response to **trastuzumab**; untreated it signals poor prognosis
- Response to neoadjuvant chemotherapy

SPREAD

- **Lymphatic**: axillary nodes (outer tumours), **internal mammary** (inner tumours), supraclavicular
- **Blood**: **bone** (mostly lytic), lungs, liver, brain, adrenals; ER-positive cancers can recur decades later

TNM STAGING (SIMPLIFIED)

T	Tis in situ · T1 ≤ 2 cm · T2 > 2–5 cm · T3 > 5 cm · T4 any size with chest-wall or skin involvement (ulcer, nodules, peau d'orange) or inflammatory carcinoma
N	N1 movable ipsilateral axillary nodes (1–3 on histology) · N2 fixed/matted axillary or internal mammary (4–9) · N3 infra- or supraclavicular nodes (≥ 10)
M	M1 distant metastasis

Stage	0 in situ · I small, node-negative · II larger or limited nodes · III locally advanced · IV distant spread

CLINICAL CASE

A 48-year-old woman has a 4 cm hard, painless lump in the upper outer quadrant with dimpled skin and two firm axillary nodes. Core biopsy: invasive carcinoma NST, grade 3. Immunohistochemistry: ER 0%, PR 0%, HER2 3+, Ki-67 60%.

Reasoning: T2 N1 → stage II. ER/PR negative with HER2 overexpression = **HER2-enriched** subtype. She needs chemotherapy with **trastuzumab**; endocrine therapy would be useless. Her nodes, not the tumour size, carry most prognostic weight.

22.7 The male breast

GYNAECOMASTIA

Enlargement of the male breast from an imbalance of **oestrogen (stimulates) over androgen (inhibits)**. Causes: **puberty** and **old age** (physiological), **Klinefelter syndrome**, **cirrhosis** (the liver fails to clear oestrogen), drugs (**spironolactone**, digoxin, cimetidine, anabolic steroids, alcohol, marijuana), and hormone-secreting tumours (testicular Leydig cell tumour, hCG-secreting tumours).

Morphology: button-like **subareolar** enlargement; dense collagenous stroma around ducts with **micropapillary epithelial hyperplasia**; lobules are rarely formed.

CARCINOMA OF THE MALE BREAST

- Rare: less than 1% of breast cancers; older men
- Risk: **BRCA2**, first-degree relatives, **Klinefelter**, oestrogen excess, testicular failure, chest radiation
- Subareolar mass or nipple discharge; because there is little breast tissue it **invades skin and chest wall early**
- Mostly **ER-positive** NST carcinoma; stage for stage, outlook is similar to women, but many present late

LAB SPOTTER

Lab slides in 10 seconds. Fibroadenoma: a **sharply bordered** nodule of loose pale stroma with **slit-like (or round) ducts** lined by two cell layers: "ducts squeezed by stroma". **Carcinoma (NST):** no border; **irregular nests and cords of pleomorphic cells** scattered through **dense pink fibrous stroma**, single cells invading fat, and no myoepithelial rim. **Lobular carcinoma:** small bland cells marching in **single file**.

EXAM ZONE Past-Paper Pattern SEQs — Solved

PROF PATTERN **a)** Enumerate the risk factors for carcinoma of the breast. **b)** Describe the gross and microscopic features of invasive carcinoma of no special type. **2 + 3 marks**

MODEL ANSWER PLAN

a) Female sex, age, family history, BRCA1/2 (and TP53, PTEN), early menarche, late menopause, nulliparity, late first pregnancy, HRT, postmenopausal obesity, alcohol, atypical hyperplasia, chest radiation, dense breasts; breastfeeding protective. **b)** Gross: hard, stellate, gritty grey-white mass with chalky streaks, skin tethering, nipple retraction. Micro: cords, nests, glands of malignant cells in desmoplastic stroma, no myoepithelium, lymphovascular invasion; Nottingham grade.

Score booster: close part (b) with "upper outer quadrant in about half; spreads to axillary nodes first".

MODULE PATTERN A 23-year-old woman has a 2 cm painless, rubbery, freely mobile lump in the right breast. **a)** What is the most likely diagnosis? **b)** Describe its morphology. **c)** How does it differ from phyllodes tumour? **1 + 2 + 2 marks**

MODEL ANSWER PLAN

a) Fibroadenoma. **b)** Circumscribed, bulging grey-white nodule with slit-like spaces; micro: loose cellular stroma around ducts with epithelial and myoepithelial layers; intracanalicular and pericanalicular patterns. **c)** Phyllodes: older (50s), larger, faster growth, hypercellular stroma with leaf-like clefts, mitoses; graded benign to malignant; recurs locally.

PROF PATTERN **a)** Give the molecular classification of breast carcinoma. **b)** Discuss the prognostic and predictive factors. **2 + 3 marks**

MODEL ANSWER PLAN

a) Luminal A (ER+/PR+, HER2−, low Ki-67), luminal B (ER+, high Ki-67 ± HER2), HER2-enriched (ER−, HER2 amplified), basal-like/triple negative (BRCA1). **b)** Prognostic: distant metastasis, axillary nodes (most important), tumour size, locally advanced/inflammatory, grade, histological type, proliferation, LVI. Predictive: ER/PR (endocrine), HER2 (trastuzumab).

Score booster: say clearly which factors are *predictive* (ER, PR, HER2); examiners often reserve a mark for this distinction.

CLASSIC **a)** Differentiate between ductal and lobular carcinoma in situ. **b)** What is Paget disease of the nipple? **3 + 2 marks**

MODEL ANSWER PLAN

a) Table: detection (calcification vs incidental), cells (cohesive vs discohesive, signet-ring), E-cadherin, necrosis/calcification, bilaterality, risk site (same quadrant vs either breast), significance, treatment. **b)** DCIS extending into nipple epidermis; red scaly eczema-like nipple; large pale Paget cells; often HER2-positive; underlying invasive carcinoma in about half.

BCQs One-Best-Answer Practice

1 Three weeks after delivery, a breastfeeding mother has a fluctuant, red, painful area in one breast and fever. The most likely organism is:

A. Streptococcus pyogenes
B. Escherichia coli
C. Mycobacterium tuberculosis
D. Staphylococcus aureus
E. Candida albicans

2 A woman who was hit by a car seat belt months ago has a hard breast lump with skin retraction. Biopsy shows foamy lipid-laden macrophages, giant cells and calcification. The diagnosis is:

A. Fat necrosis
B. Duct ectasia
C. Invasive lobular carcinoma
D. Fibroadenoma
E. Phyllodes tumour

3 Which benign breast lesion carries the highest relative risk of later invasive carcinoma?

A. Apocrine cyst
B. Sclerosing adenosis
C. Atypical ductal hyperplasia
D. Fibrosis
E. Simple fibroadenoma

4 A 22-year-old has a 2 cm mobile rubbery breast lump. Microscopy shows loose stroma compressing ducts into curved slits lined by epithelial and myoepithelial cells. The diagnosis is:

A. Phyllodes tumour
B. Intraductal papilloma
C. Medullary carcinoma
D. Tubular carcinoma
E. Fibroadenoma

5 A 40-year-old has bloody discharge from one nipple duct. A small subareolar lesion shows branching fibrovascular cores covered by luminal and myoepithelial cells. The diagnosis is:

A. Paget disease
B. Intraductal papilloma
C. Duct ectasia
D. Papillary carcinoma
E. Comedo DCIS

6 Which feature most favours phyllodes tumour over fibroadenoma?

A. Hypercellular stroma with leaf-like projections and mitoses
B. Age 20–30 years
C. Presence of myoepithelial cells
D. MED12 mutation
E. Well-circumscribed border

7 A mastectomy shows small uniform discohesive cells infiltrating in single file around ducts. Immunohistochemistry will most likely show loss of:

A. ER
B. HER2
C. Cytokeratin
D. E-cadherin
E. Ki-67

8 Mammography shows clustered linear branching microcalcifications. Biopsy shows ducts filled by high-grade cells around central necrosis, with intact myoepithelium. The diagnosis is:

A. LCIS
B. Atypical lobular hyperplasia
C. Comedo DCIS
D. Invasive carcinoma NST
E. Fat necrosis

9 A 35-year-old BRCA1 carrier develops breast cancer. The most likely phenotype is:

A. ER+/PR+/HER2− luminal A
B. Mucinous carcinoma
C. Tubular carcinoma
D. Invasive lobular carcinoma
E. ER−/PR−/HER2− basal-like

10 In a woman without distant metastases, the single most important prognostic factor in breast carcinoma is:

A. Tumour size
B. Axillary lymph node status
C. Patient age
D. Histological type
E. Ki-67 index

11 Trastuzumab is effective only in breast cancers that show:

A. ER positivity
B. PR positivity
C. HER2 amplification
D. BRCA1 mutation
E. E-cadherin loss

12 A 70-year-old man has a hard subareolar mass fixed to the skin. The germline mutation most associated with this tumour is:

A. BRCA2
B. APC
C. RET
D. VHL
E. NF1

ANSWER KEY WITH ONE-LINE REASONS

1. **D** Acute lactational mastitis with localised abscess is usually staphylococcal.
2. **A** Traumatic fat necrosis is a classic mimic of carcinoma.
3. **C** Proliferative disease with atypia: relative risk about 4–5.
4. **E** Intracanalicular pattern of fibroadenoma.
5. **B** Two cell layers on fibrovascular cores = papilloma; carcinoma lacks myoepithelium.
6. **A** Stromal hypercellularity and leaf-like architecture define phyllodes.
7. **D** Invasive lobular carcinoma: CDH1 (E-cadherin) inactivation.
8. **C** Comedo DCIS: high-grade, central necrosis that calcifies; in situ.
9. **E** BRCA1 tumours are usually triple-negative, basal-like, high grade.
10. **B** Nodal status (sentinel node) carries the greatest prognostic weight.
11. **C** Trastuzumab is a monoclonal antibody against HER2 (ERBB2).
12. **A** Male breast carcinoma is linked to BRCA2 (and Klinefelter syndrome).

VIVA Rapid-Fire Viva Voce

Q Commonest organism in lactational breast abscess?	Staphylococcus aureus.
Q Which inflammatory breast lesion is linked to smoking?	Periductal mastitis (squamous metaplasia of lactiferous ducts).
Q Name two benign lesions that mimic carcinoma.	Fat necrosis and sclerosing adenosis (also duct ectasia, radial scar).
Q Relative risk with atypical hyperplasia?	About 4–5 times, in either breast.
Q Commonest benign tumour of the female breast?	Fibroadenoma.
Q What does 'phyllodes' mean?	Leaf-like: stroma pushes epithelium into leaf-shaped projections.
Q Commonest cause of bloody nipple discharge in a young woman?	Intraductal (large-duct) papilloma.
Q What is the hallmark of lobular neoplasia?	Loss of E-cadherin, making cells discohesive.
Q What is Paget disease of the nipple?	DCIS cells spreading into the nipple epidermis.

Q What causes peau d'orange?	Tumour blocking dermal lymphatics, so skin swells around hair follicles.
Q Which quadrant is most often involved?	Upper outer quadrant, about half of cases.
Q Name two causes of gynaecomastia.	Cirrhosis and spironolactone (also puberty, Klinefelter, Leydig cell tumour).

OSPE Practice Stations

STATION 1 · SLIDE — 2 minutes

TASK

Breast lump from a 24-year-old: a well-circumscribed lesion of loose fibroblastic stroma with many ducts, some compressed into curved slit-like spaces; ducts lined by two cell layers; no atypia. (a) Identify. (b) Give two features. (c) Name its malignant-potential look-alike.

ANSWER

(a) Fibroadenoma. (b) Circumscribed border; intracanalicular/pericanalicular ducts with epithelial and myoepithelial cells in bland stroma. (c) Phyllodes tumour.

STATION 2 · SLIDE — 2 minutes

TASK

Breast tissue: irregular nests, cords and gland-like clusters of pleomorphic cells with hyperchromatic nuclei and mitoses infiltrating dense collagenous stroma and fat; no myoepithelial rim. (a) Diagnosis. (b) Two features. (c) Which three IHC markers are requested next?

ANSWER

(a) Invasive carcinoma of no special type (ductal). (b) Infiltrating nests and cords; desmoplastic stroma; loss of myoepithelium. (c) ER, PR and HER2 (plus Ki-67).

STATION 3 · GROSS SPECIMEN — 1 minute

TASK

Mastectomy specimen: a 3 cm hard, grey-white, stellate mass with chalky streaks beneath a retracted nipple; the overlying skin is puckered. (a) Identify. (b) Give two gross features. (c) Where should the pathologist look for spread?

ANSWER

(a) Carcinoma breast (invasive carcinoma NST). (b) Stellate infiltrative edge; gritty cut surface with chalky streaks; nipple retraction. (c) Axillary lymph nodes (sentinel node first).

STATION 4 · LAB DATA — 2 minutes

TASK

IHC report of an invasive breast carcinoma: ER 90% strong, PR 70%, HER2 1+ (negative), Ki-67 8%. (a) Assign the molecular subtype. (b) Which treatment does this predict? (c) Comment on prognosis.

ANSWER

(a) Luminal A. (b) Endocrine therapy (tamoxifen or an aromatase inhibitor); no anti-HER2 therapy. (c) Best prognosis of the subtypes, though late recurrences can occur.

RAPID REVISION The Night-Before Page

15 ONE-LINERS

01 Benign and in situ lesions keep myoepithelium; invasive carcinoma loses it.

02 Lactational mastitis: S. aureus abscess; periductal mastitis: smokers, fistula.

03 Fat necrosis and duct ectasia mimic carcinoma clinically.

04 Risk ladder: non-proliferative ×1, proliferative ×1.5–2, atypia ×4–5, CIS ×8–10.

05 Sclerosing adenosis and radial scar mimic carcinoma on slide and mammogram.

06 Fibroadenoma: 20–35 years, mobile, intracanalicular, MED12, not premalignant.

07 Phyllodes: older, cellular leaf-like stroma, local recurrence, blood-borne spread.

08 Papilloma: bloody discharge, fibrovascular cores with two cell layers.

09 BRCA1 → triple negative; BRCA2 → ER-positive and male breast cancer.

10 DCIS: calcification, comedo necrosis, same-quadrant risk; Paget = DCIS in nipple.

11 LCIS: incidental, discohesive, E-cadherin lost, risk in either breast.

12 NST: hard stellate gritty mass, desmoplasia, upper outer quadrant.

13 Lobular: single file, ER+, metastasises to peritoneum, meninges, ovary.

14 Nodes are the strongest prognostic factor; ER/PR/HER2 are predictive.

15 Gynaecomastia: cirrhosis, spironolactone, Klinefelter; male cancer: BRCA2.

MEMORY HOOKS

- **"Fibroadenoma is Free; Carcinoma is Caught"**: mobile vs fixed lump
- **LCIS = Loose cells, Lost E-cadherin, Left-or-right breast at risk**
- **BRCA1 = 1 negative triple; BRCA2 = 2 sexes** (men too)
- **Luminal A = "A"-grade prognosis**; HER2 → trastuzumab

TEST YOURSELF: FILL THE GAPS

1. Gene mutated in stromal cells of fibroadenoma: _____.
2. Protein lost in lobular carcinoma: _____.
3. Relative risk with carcinoma in situ: _____.
4. Orange-peel skin is called _____.
5. Strongest prognostic factor: _____ status.
6. Gene linked to male breast cancer: _____.

Answers: MED12 · E-cadherin · 8–10 · peau d'orange · axillary node · BRCA2

DRAW FROM MEMORY

The ductal tree from nipple to lobule with the lesions that arise at each level (Figure 22.1).

DRAW FROM MEMORY

The three molecular pathways: ER-positive, HER2-positive and triple negative (Figure 22.3).

SPACED REVISION ☐ Day 1 ☐ Day 3 ☐ Day 7 ☐ Day 14 ☐ Day 30

23

Thyroid & Parathyroid

The thyroid is the one endocrine gland you can see and feel. A TSH level, a hand on the neck and a fine needle answer most of its questions; the parathyroids, hidden behind it, speak only through calcium.

BY THE END OF THIS CHAPTER YOU CAN

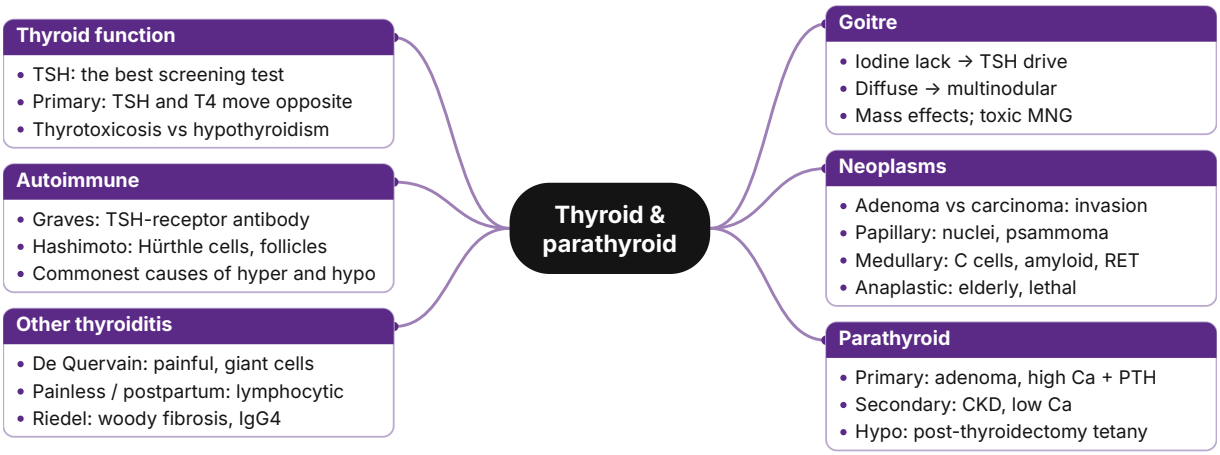
1. Interpret a thyroid profile and separate primary from central thyroid disease
2. Compare the pathogenesis and morphology of Graves disease and Hashimoto thyroiditis
3. Describe subacute granulomatous and Riedel thyroiditis and the evolution of a multinodular goitre
4. Differentiate follicular adenoma from follicular carcinoma
5. Compare papillary, follicular, medullary and anaplastic carcinomas: genes, morphology, spread and prognosis
6. Classify hyperparathyroidism and explain its effects on bone, kidney and mind
7. List the causes and features of hypoparathyroidism

EXAM RADAR

Theory SEQs	★★★
BCQs / MCQs	★★★
Viva voce	★★★
OSPE / slides	★★★

Thyroid carcinomas (papillary above all), Graves vs Hashimoto and adenoma vs carcinoma are perennial long questions; Hashimoto and papillary carcinoma are the lab slides.

ROBBINS LINK Robbins Basic Pathology — *The Endocrine System*: thyroid (hyperthyroidism, hypothyroidism, thyroiditis, Graves disease, goitre, neoplasms) and parathyroid glands.



Chapter at a glance. Six ideas carry the thyroid and parathyroid.

23.1 Thyroid function: reading the profile

Hypothalamic TRH drives pituitary **TSH**, and TSH drives follicular cells to iodinate thyroglobulin and release **T4** (mostly) and **T3** (the active hormone, largely made from T4 in the tissues). Free hormone feeds back on the pituitary, which detects even small changes; hence **serum TSH is the most sensitive single screening test**.

> **CONCEPT CLEAR**
>
> **Opposite directions = the gland; same direction = the pituitary.** In **primary** thyroid disease the pituitary is healthy and argues back: a failing gland (low T4) pushes TSH up, an overactive gland (high T4) shuts TSH down. When TSH and T4 move **together**, the fault lies upstream in the pituitary or hypothalamus (**central** disease).

THYROID FUNCTION TESTS: THE PATTERNS EXAMINERS GIVE

Pattern	TSH	Free T4 / T3	Think of
Primary hyperthyroidism	↓ (suppressed)	↑	Graves disease, toxic multinodular goitre, toxic adenoma
T3 toxicosis / subclinical	↓	T4 normal (T3 ↑ or normal)	Early Graves, autonomous nodule, excess thyroxine

Pattern	TSH	Free T4 / T3	Think of
Destructive thyroiditis (early)	↓	↑ but **radioiodine uptake low**	De Quervain, painless or postpartum thyroiditis
Primary hypothyroidism	↑	↓	Hashimoto, iodine deficiency, after surgery or radioiodine
Subclinical hypothyroidism	↑	Normal	Early Hashimoto
Central hypothyroidism	↓ or normal	↓	Pituitary or hypothalamic disease (tumour, Sheehan)
Central hyperthyroidism (rare)	↑ or normal	↑	TSH-secreting pituitary adenoma

Normal values (lab ranges vary): TSH about 0.4–4.0 mIU/L; free T4 about 0.8–1.8 ng/dL. **Radioiodine uptake**: diffusely high in Graves; a single hot nodule in toxic adenoma; low in thyroiditis or exogenous thyroxine.

DEFINITION

Thyrotoxicosis is the hypermetabolic state caused by raised circulating free T3 and T4. **Hyperthyroidism** is thyrotoxicosis caused by an overactive gland, its commonest cause. Leakage of stored hormone (thyroiditis) or swallowed thyroxine causes thyrotoxicosis without hyperthyroidism.

HYPERTHYROIDISM VS HYPOTHYROIDISM: THE CLINICAL MIRROR IMAGE

System	Hyperthyroidism (thyrotoxicosis)	Hypothyroidism
Metabolism, skin	Weight loss with good appetite, heat intolerance, warm moist skin	Weight gain, cold intolerance, dry coarse skin, **non-pitting myxoedema**
Heart	Tachycardia, **atrial fibrillation** (elderly), high-output failure	Bradycardia, pericardial effusion, high cholesterol
Nerves, muscle, gut	Tremor, anxiety, proximal myopathy; diarrhoea	Apathy, slow reflex relaxation; constipation
Eyes	Stare and lid lag in all; **ophthalmopathy only in Graves**	Periorbital puffiness
Emergencies	**Thyroid storm** (fever, tachyarrhythmia) after stress or surgery	Myxoedema coma; **cretinism** in infants

CAUSES OF THYROTOXICOSIS

- **With hyperthyroidism** (RAIU high): **Graves** (commonest), toxic multinodular goitre, toxic adenoma; rarely TSH-secreting adenoma
- **Without** (RAIU low): granulomatous and painless thyroiditis, struma ovarii, factitious thyroxine

CAUSES OF HYPOTHYROIDISM

- **Primary: iodine deficiency** (commonest worldwide), **Hashimoto** (commonest where iodine is adequate), thyroidectomy, radioiodine, lithium, amiodarone, congenital defects
- **Central:** pituitary tumour, Sheehan syndrome, hypothalamic lesions

Cretinism (hypothyroidism from infancy): impaired brain and skeletal growth, short stature, coarse face, protruding tongue, umbilical hernia; maternal iodine deficiency is the classic cause. **Myxoedema** (older children, adults): physical and mental slowing with mucopolysaccharide-rich oedema.

23.2 Autoimmune thyroid disease: Graves and Hashimoto

Graves disease

DEFINITION

Graves disease is an autoimmune disorder in which IgG autoantibodies **stimulate the TSH receptor**, producing a triad: **thyrotoxicosis with diffuse goitre**, **infiltrative ophthalmopathy** and, in a minority, **localised infiltrative dermopathy** (pretibial myxoedema). It is the commonest cause of endogenous hyperthyroidism; peak 20–40 years; women up to 10 times more often. Links: HLA-DR3, CTLA4, PTPN22.

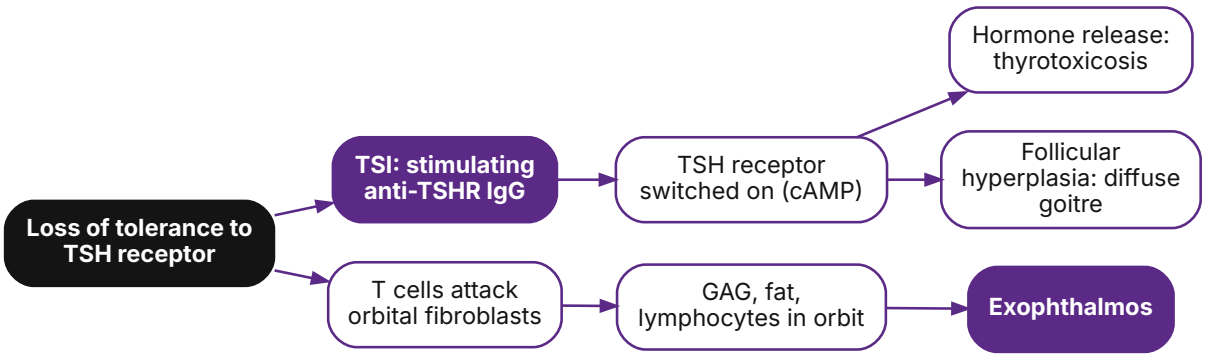


Figure 23.1 · Pathogenesis of Graves disease. One antibody explains the goitre and the hyperthyroidism; T cells in the orbit explain the eyes.

GROSS

- Symmetrically, **diffusely enlarged**; capsule intact
- Soft; cut surface **beefy, meaty red** (little colloid)

MICROSCOPY

- **Tall, crowded columnar cells** heaped into small papillae **without fibrovascular cores**
- **Pale colloid with scalloped margins**
- Lymphoid aggregates with germinal centres

Labs: low TSH, high free T4/T3, TSI positive, diffusely high radioiodine uptake. Pre-operative **iodine** makes the epithelium involute and colloid accumulate. Patients are prone to other autoimmune diseases (SLE, pernicious anaemia, type 1 diabetes).

Hashimoto thyroiditis (chronic lymphocytic thyroiditis)

The commonest cause of hypothyroidism where dietary iodine is sufficient; peak **45–65 years**, women 10–20 times more often. Tolerance to **thyroid peroxidase (TPO)** and **thyroglobulin** breaks down and the gland is slowly destroyed.

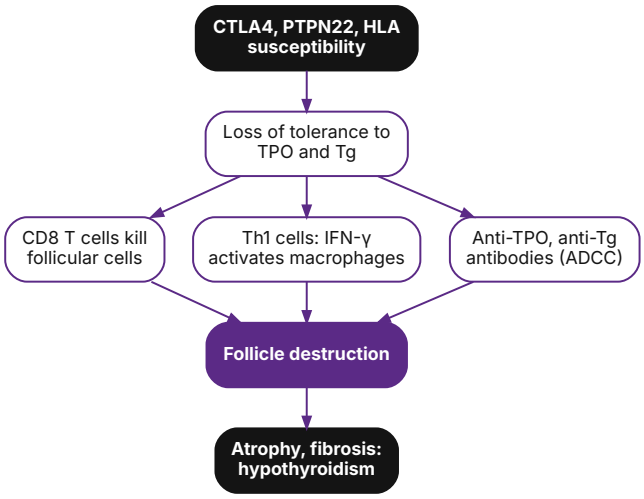


Figure 23.2 · Pathogenesis of Hashimoto thyroiditis. Three immune weapons, one result.

GROSS

- **Diffusely enlarged**, symmetrical; capsule intact
- Cut surface **pale, tan-grey, firm**, slightly friable

MICROSCOPY

- Dense lymphocytes and plasma cells forming **lymphoid follicles with germinal centres**
- Small, **atrophic follicles** lined by **Hürthle cells**: abundant **granular eosinophilic cytoplasm** (packed mitochondria)
- Interstitial fibrosis; fibrous and IgG4-related variants

Clinical: painless goitre with gradual hypothyroidism, sometimes preceded by transient thyrotoxicosis (**hashitoxicosis**) as follicles break open. Anti-TPO and anti-thyroglobulin antibodies are present. Other autoimmune diseases are common, and the risk of **extranodal marginal zone B-cell lymphoma** of the thyroid is increased.

GRAVES DISEASE VS HASHIMOTO THYROIDITIS

Feature	Graves disease	Hashimoto thyroiditis
Function	Hyperthyroidism	Hypothyroidism (hashitoxicosis early)
Typical patient	Woman, 20–40 years	Woman, 45–65 years
Key antibody	**Stimulating** anti-TSH-receptor IgG (TSI)	**Anti-TPO, anti-thyroglobulin**
Main effector	Antibody switches cells **on**	T cells **destroy** cells (plus ADCC)

Feature	Graves disease	Hashimoto thyroiditis
Gland	Soft, beefy red; tall hyperplastic cells, scalloped colloid	Firm, pale; atrophic follicles, **Hürthle cells**, **germinal centres**, fibrosis
Beyond the thyroid	Ophthalmopathy, pretibial myxoedema	Other autoimmune diseases; B-cell lymphoma
TSH / RAIU	Low TSH; uptake diffusely high	High TSH; uptake low

LAB SPOTTER

Hashimoto slide in 10 seconds: at low power the thyroid looks like a lymph node with **germinal centres**, but between them sit small colloid-filled follicles lined by **large pink granular Hürthle cells**. Normal thyroid has no lymphoid follicles; a lymph node has no colloid follicles (if it does, think metastatic papillary carcinoma).

Other thyroiditis

Type	Cause, patient	Clinical	Morphology
Subacute granulomatous (de Quervain)	After a viral illness (Coxsackie, mumps, adenovirus); HLA-B35; women 30–50 y	**Painful, tender** gland, fever, high ESR; transient thyrotoxicosis with **low RAIU**; resolves in 6–8 weeks	Follicle disruption, neutrophil microabscesses, then **granulomas with giant cells engulfing colloid**; heals by fibrosis
Subacute lymphocytic (painless, postpartum)	Autoimmune (anti-TPO); middle-aged women or after delivery	Painless goitre; brief hyper-, then hypothyroid phase; usually recovers	Lymphocytes, germinal centres; **no Hürthle cells or fibrosis**
Riedel thyroiditis	Rare; may be **IgG4-related disease**	**Rock-hard, fixed** mass mimicking carcinoma	Dense fibrosis spreading beyond the capsule into the neck

23.3 Goitre: diffuse and multinodular

A simple (non-toxic) goitre is the gland's answer to **impaired hormone synthesis**, most often **dietary iodine deficiency**: less hormone, more TSH, and TSH makes follicular cells grow. The patient usually stays **euthyroid**. **Endemic** goitre affects more than 10% of a population in iodine-poor mountains (Himalayas, Andes); **sporadic** goitre affects young women and follows **goitrogens** (cabbage, cauliflower, cassava) or inherited enzyme defects (**dyshormonogenetic goitre**).

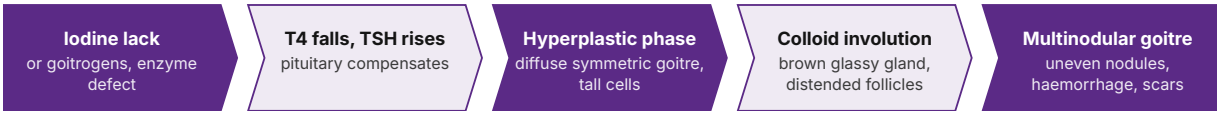


Figure 23.3 · From iodine lack to multinodular goitre. Repeated cycles of hyperplasia and involution make a smooth gland nodular.

GROSS: MULTINODULAR GOITRE

- **Multilobulated, asymmetrically enlarged**; the largest goitres (over 2 kg); may extend behind the sternum (**plunging goitre**)
- Nodules of **brown gelatinous colloid** with **haemorrhage, fibrosis, calcification, cysts**

MICROSCOPY

- **Colloid-rich follicles** lined by flat inactive epithelium, alternating with hyperplastic areas
- Haemorrhage, haemosiderin, scarring, calcification
- Nodules lack the **complete capsule** of an adenoma

Clinical: mass effects (**airway obstruction, dysphagia**, venous compression), cosmetic deformity; an autonomous nodule may cause thyrotoxicosis without eye signs (**toxic multinodular goitre, Plummer syndrome**). Malignancy is uncommon (under 5%), but a **dominant nodule** must be aspirated.

PAKISTAN FOCUS

Iodine-deficiency goitre was historically **endemic in the northern areas** of Pakistan (Gilgit-Baltistan, Khyber Pakhtunkhwa, Azad Kashmir). Universal salt iodisation has improved coverage, but large multinodular goitres remain a common surgical specimen.

23.4 Thyroid nodules and neoplasms

Most solitary nodules are **benign**: follicular adenomas or non-neoplastic lumps (dominant goitre nodule, cyst). **FNAC** is the first investigation. Clues to malignancy: **solitary** nodule, **young** or **male** patient, **past neck irradiation**, **cold** nodule on scan, hardness, fixity, hoarseness or a neck node. A **hot** nodule is almost always benign.

Follicular adenoma

A benign, usually solitary neoplasm of follicular epithelium. Toxic (hot) adenomas carry gain-of-function *TSHR* or *GNAS* mutations (TSH-independent cAMP signalling); non-functioning ones may carry *RAS*, *PIK3CA* or *PAX8–PPARG* changes.

GROSS

- **Solitary, spherical, encapsulated** nodule (about 3 cm) compressing the surrounding gland
- Grey-white to red-brown; haemorrhage, cystic change

MICROSCOPY

- **Uniform colloid-filled follicles** like normal thyroid; Hürthle cell variant
- Focal **endocrine atypia** is not malignancy
- **Intact capsule**, no capsular or vascular invasion

FOLLICULAR ADENOMA VS FOLLICULAR CARCINOMA

Feature	Follicular adenoma	Follicular carcinoma
Capsule / vessels	Intact; vessels not invaded	**Capsular invasion** (full thickness) and/or **vascular invasion**
Cytology	Follicular cells	Identical: FNAC reports only "follicular neoplasm"
Spread	None	Blood-borne: **lungs, bone, liver**
Treatment	Lobectomy	Total thyroidectomy + radioiodine

CONCEPT CLEAR

Look at the edge, not the centre. Benign and malignant follicular cells look the same, so a needle in the middle of the nodule cannot decide. Only the **capsule and its vessels** reveal invasion, so the whole nodule must be excised and its capsule sampled generously.

Thyroid carcinomas: the big four

Papillary **> 85%**, follicular **5–15%**, medullary about **5%**, anaplastic **< 5%**. Women predominate in early and middle adult life. **Ionising radiation** in the first two decades is the major risk factor (papillary); **iodine deficiency** favours follicular carcinoma.

MOLECULAR KEY

Papillary: MAPK pathway activated by **RET/PTC** rearrangements (radiation), *NTRK1* fusions or **BRAF V600E** (one-third to one-half; more aggressive). **Follicular: RAS** mutations, PI3K/AKT activation (*PTEN*, *PIK3CA*), **PAX8–PPARG** fusion t(2;3). **Anaplastic:** dedifferentiation with **TP53** loss. **Medullary:** activating **RET** mutation, germline in MEN2 and familial cases.

PAPILLARY CARCINOMA

Usually 25–50 years; the cancer that follows childhood neck irradiation. It may present first as an enlarged **cervical lymph node**.

GROSS

- Solitary or **multifocal**; circumscribed or infiltrative
- Granular, firm, **gritty** (calcification); sometimes cystic

MICROSCOPY

- **Branching papillae with fibrovascular cores**, lined by cuboidal cells
- **Nuclei make the diagnosis**, even without papillae: clear, empty **"Orphan Annie eye"** nuclei, **pseudo-inclusions**, **grooves**, overlapping
- **Psammoma bodies**: concentric laminated calcifications
- Lymphatic invasion common

Variants: follicular (encapsulated non-invasive forms are now NIFTP, low risk), tall cell (older, aggressive), diffuse sclerosing (young), microcarcinoma (< 1 cm). **Prognosis: 10-year survival over 95%**; nodal spread hardly alters it; worse with older age, extrathyroidal extension and lung metastases.

MEMORY HOOK

Papillary = the "P" cancer

Popular (commonest) · **P**ast radiation · **P**apillae with cores · **P**ale "Orphan Annie" nuclei · **P**sammoma bodies · **P**alpable neck nodes · **P**rognosis excellent

FOLLICULAR CARCINOMA

Women aged 40–60, commoner in iodine-deficient regions. A single nodule, **minimally invasive** (well circumscribed) or **widely invasive**; **small uniform follicles** with colloid; nuclei **lack** papillary features; diagnosis rests on **capsular or vascular invasion**. Spreads by **blood** to lungs and bone; metastases take up radioiodine and thyroglobulin is the follow-up marker.

MEDULLARY CARCINOMA

A neuroendocrine tumour of parafollicular **C cells** secreting **calcitonin** (marker), often CEA, sometimes VIP (diarrhoea) or ACTH. About **70% sporadic** (adults, solitary) and **30% familial** (MEN2A, MEN2B, familial MTC: younger, **bilateral, multicentric**); *RET* carriers have **prophylactic thyroidectomy** (MEN table, Chapter 24).

GROSS

- Solitary or bilateral nodules; firm, pale grey-tan, infiltrative

MICROSCOPY

- **Polygonal to spindle cells** in nests and trabeculae
- **Stromal amyloid** from altered calcitonin (Congo red positive)
- Background **C-cell hyperplasia** in familial cases

ANAPLASTIC CARCINOMA

Elderly patients with a **rapidly enlarging bulky neck mass** compressing trachea and oesophagus; about half have a past or coexisting differentiated carcinoma. **Pleomorphic giant cells**, **spindle cells** (sarcoma-like) or both; cytokeratin positive. Mortality is close to **100%**, usually within a year.

THE FOUR THYROID CARCINOMAS SIDE BY SIDE

Feature	Papillary	Follicular	Medullary	Anaplastic
Share	> 85%	5–15%	About 5%	< 5%
Cell of origin	Follicular	Follicular	**C cell**	Follicular (dedifferentiated)
Age	25–50 y	40–60 y	40–60 y; MEN2 younger	Elderly
Risk / link	Radiation	Iodine deficiency	RET (MEN2)	Prior differentiated tumour
Genes	RET/PTC, NTRK1, BRAF	RAS, PAX8–PPARG, PTEN	RET	TP53 (+ RAS/BRAF)
Hallmark	Clear grooved nuclei, psammoma bodies	Small follicles; capsular/vascular invasion	Nests in **amyloid** stroma	Giant and spindle cells

Feature	Papillary	Follicular	Medullary	Anaplastic
Spread	**Lymphatic** (neck nodes)	**Blood** (lung, bone)	Both	Local invasion
Marker	Thyroglobulin	Thyroglobulin	**Calcitonin**, CEA	Cytokeratin
Prognosis	Excellent	Good if minimally invasive	Intermediate	Dismal (< 1 year)

LAB SPOTTER

Papillary carcinoma slide in 10 seconds: finger-like **papillae with thin fibrovascular cores**; crowded, overlapping, **clear "empty" nuclei** (grooves and pink pseudo-inclusions at high power); **purple laminated psammoma bodies**. Graves papillae lack cores, and their nuclei are dark and round.

23.5 Parathyroid glands

Low free calcium, sensed by the **calcium-sensing receptor (CaSR)**, releases PTH. **PTH raises calcium and lowers phosphate**: more renal calcium reabsorption, more phosphate excretion, more vitamin D activation (1α-hydroxylation) and more osteoclast activity (indirectly, via **RANKL** on osteoblasts).

Primary hyperparathyroidism

Autonomous PTH overproduction: **adenoma 85–95%**, primary hyperplasia 5–10% (sporadic or MEN1/MEN2A), carcinoma about 1%. Usually women over 50. It is the **commonest cause of asymptomatic hypercalcaemia**; malignancy is the commonest cause of symptomatic hypercalcaemia. Genes: ***MEN1*** and **cyclin D1 (*CCND1*)** overexpression (sometimes by inversion next to the PTH gene promoter).

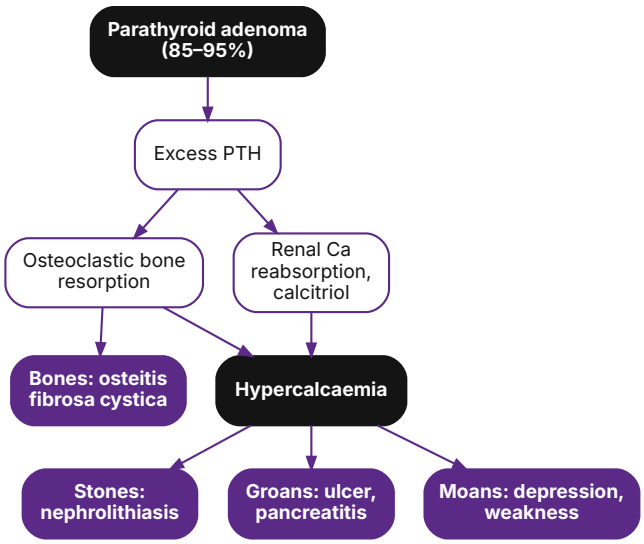


Figure 23.4 · Primary hyperparathyroidism. Bones, stones, groans and moans in one chain.

ADENOMA: GROSS

- **Solitary**, well-circumscribed, soft, **tan to red-brown** nodule, 0.5–5 g, thin capsule
- Other glands normal or **shrunken**

ADENOMA: MICROSCOPY

- Uniform polygonal **chief cells**, few oxyphil cells; **little or no fat**
- **Rim of compressed normal parathyroid** outside the capsule

Hyperplasia: all four glands enlarged. **Carcinoma:** diagnosed only by invasion or metastasis.

EFFECTS

- **Bone:** osteoporosis; **osteitis fibrosa cystica** (osteoclastic resorption, peritrabecular fibrosis, cysts); **brown tumours** (osteoclasts, giant cells, haemorrhage, haemosiderin) that mimic neoplasm
- **Kidney:** nephrolithiasis, **nephrocalcinosis**, polyuria
- **Metastatic calcification**: stomach, lungs, kidneys, vessels
- **Gut, mind:** constipation, **peptic ulcer, pancreatitis**, gallstones; depression, lethargy, weakness

MEMORY HOOK

"Painful bones, renal stones, abdominal groans, psychic moans"

The four faces of hypercalcaemia. Today most patients are found earlier, on a routine calcium.

Labs: calcium high, **PTH high or inappropriately normal**, phosphate low, urinary calcium high; alkaline phosphatase high with bone disease.

Secondary and tertiary hyperparathyroidism

PRIMARY, SECONDARY AND TERTIARY HYPERPARATHYROIDISM

Feature	Primary	Secondary	Tertiary
Cause	Adenoma, hyperplasia, carcinoma	**Chronic kidney disease** (commonest), vitamin D deficiency	Long-standing secondary turns autonomous (e.g. after transplant)
Glands	Usually one	All four hyperplastic	All four, autonomous
Calcium	**High**	Low or normal	**High**
Phosphate	Low	High in CKD; low in vitamin D deficiency	High or normal
PTH	High	High	Very high

CONCEPT CLEAR

Who is wrong: the gland or the body? In **primary** disease the gland is wrong: it makes PTH despite high calcium. In **secondary** disease the gland is right: failing kidneys retain phosphate and cannot activate vitamin D, calcium falls, and all four glands enlarge to correct it (renal osteodystrophy; vascular calcification may cause **calciphylaxis**). In **tertiary** disease the gland has been driven so long that it no longer switches off.

EXAM TRAP

Hypercalcaemia: check PTH first. High PTH = parathyroid cause. Suppressed PTH points to **malignancy** (PTH-related protein from squamous carcinomas, osteolytic metastases, myeloma), vitamin D excess or sarcoidosis. Medullary carcinoma secretes calcitonin but does **not** cause hypocalcaemia.

Hypoparathyroidism

CAUSES

- **Surgical**: glands removed or devascularised at thyroidectomy (commonest)
- **Congenital absence**: DiGeorge syndrome (22q11.2 deletion), with thymic aplasia and heart defects
- **Autoimmune**: APS-1 (*AIRE*) with mucocutaneous candidiasis and Addison disease
- **Autosomal dominant**: gain-of-function *CaSR* mutation

FEATURES OF HYPOCALCAEMIA

- **Tetany**: circumoral tingling, carpopedal spasm, laryngospasm, seizures
- **Chvostek sign** (tap facial nerve → twitch); **Trousseau sign** (BP cuff → carpal spasm)
- Psychosis, basal ganglia calcification, cataracts, **prolonged QT**, dental defects

Pseudohypoparathyroidism: end-organ resistance to PTH (*GNAS* defects), so low calcium and high phosphate coexist with **high PTH**; often short stature and short fourth and fifth metacarpals.

EXAM ZONE Past-Paper Pattern SEQs — Solved

PROF PATTERN **a)** Classify malignant tumours of the thyroid. **b)** Describe the morphology of papillary carcinoma of the thyroid. **2 + 3 marks**

MODEL ANSWER PLAN

a) Follicular-cell origin: papillary (> 85%), follicular (5–15%), anaplastic (< 5%); C-cell origin: medullary (about 5%). Others: lymphoma (on Hashimoto background), metastases.

b) Gross: solitary or multifocal, circumscribed or infiltrative, granular, gritty, may be cystic. Micro: branching papillae with fibrovascular cores; Orphan Annie (clear) nuclei, nuclear grooves, pseudo-inclusions; psammoma bodies; lymphatic invasion and cervical node metastasis.

Score booster: state that the nuclear features alone are diagnostic, even without papillae (follicular variant).

MODULE PATTERN A 30-year-old woman has weight loss despite a good appetite, palpitations, heat intolerance, protruding eyes and a diffuse, soft goitre with a bruit. **a)** Most likely diagnosis? **b)** Explain the pathogenesis. **c)** Expected thyroid profile and microscopy of the gland. **1 + 2 + 2 marks**

MODEL ANSWER PLAN

a) Graves disease. **b)** Loss of tolerance → TSI (stimulating IgG) binds TSH receptor → cAMP → hormone release (thyrotoxicosis) and hyperplasia (goitre); orbital fibroblasts with TSH receptor attacked by T cells → GAG, fat and lymphocytes → exophthalmos. HLA-DR3, CTLA4, PTPN22. **c)** TSH low, free T4/T3 high, RAIU diffusely high. Micro: tall crowded columnar cells, papillae without cores, pale scalloped colloid, lymphoid aggregates.

CLASSIC **a)** Describe the pathogenesis of Hashimoto thyroiditis. **b)** Describe its morphology and list two complications. **2 + 3 marks**

MODEL ANSWER PLAN

a) Genetic susceptibility (CTLA4, PTPN22) → loss of tolerance to TPO and thyroglobulin → CD8 T-cell killing, Th1/IFN-γ macrophage injury, antibody-dependent cytotoxicity → follicle destruction → hypothyroidism. **b)** Gross: diffuse, pale, firm goitre, intact capsule. Micro: lymphoplasmacytic infiltrate with germinal centres, atrophic follicles with Hürthle cells, fibrosis. Complications: hypothyroidism; B-cell (marginal zone) lymphoma; other autoimmune diseases.

Score booster: define Hürthle cell in one line (granular eosinophilic cytoplasm full of mitochondria).

PROF PATTERN **a)** How will you differentiate follicular adenoma from follicular carcinoma? **b)** Write a note on medullary carcinoma of the thyroid. **2 + 3 marks**

MODEL ANSWER PLAN

a) Capsular invasion and vascular invasion are the only reliable criteria; cytology is identical, so FNAC cannot separate them; carcinoma spreads by blood. **b)** C-cell origin; calcitonin and CEA markers; 70% sporadic, 30% familial (MEN2A/2B, RET); nests of polygonal/spindle cells in amyloid stroma (calcitonin-derived, Congo red positive); C-cell hyperplasia in familial cases; prophylactic thyroidectomy in RET carriers.

PROF PATTERN **a)** Enumerate the causes of primary hyperparathyroidism. **b)** Describe its effects on bones and kidneys. **2 + 3 marks**

MODEL ANSWER PLAN

a) Adenoma (85–95%), primary hyperplasia (5–10%, MEN1/MEN2A), carcinoma (about 1%). **b)** Bones: osteoporosis, osteitis fibrosa cystica, brown tumours, subperiosteal resorption. Kidneys: nephrolithiasis, nephrocalcinosis, polyuria, renal failure. Labs: high Ca, high PTH, low phosphate.

Score booster: finish with "bones, stones, groans and moans" and one line on secondary hyperparathyroidism in CKD.

BCQs One-Best-Answer Practice

1 A 48-year-old woman has a painless, firm, diffuse goitre, raised TSH and anti-TPO antibodies. Thyroid biopsy is most likely to show:

A. Granulomas with giant cells engulfing colloid
B. Lymphoid follicles with germinal centres and Hürthle cells
C. Tall crowded columnar cells with scalloped colloid
D. Dense fibrosis extending into strap muscles
E. Nests of cells in an amyloid stroma

2 The most sensitive single screening test for thyroid dysfunction is:

A. Free T4
B. Total T3
C. Radioactive iodine uptake
D. Serum TSH
E. Anti-TPO antibody

3 A 28-year-old woman has thyrotoxicosis, exophthalmos and thickened skin over both shins. The basic mechanism is:

A. Autoantibodies that stimulate the TSH receptor
B. Cytotoxic T-cell destruction of follicles
C. A TSH-secreting pituitary adenoma
D. Viral infection of follicular cells
E. Excess dietary iodine

4 After a flu-like illness, a woman has a painful tender thyroid, ESR 90 mm/h, high free T4 and very low radioiodine uptake. The diagnosis is:

A. Graves disease
B. Hashimoto thyroiditis
C. Riedel thyroiditis
D. Toxic multinodular goitre
E. Subacute granulomatous thyroiditis

5 The single most reliable criterion that separates follicular carcinoma from follicular adenoma is:

A. Nuclear pleomorphism
B. High mitotic count
C. Capsular and/or vascular invasion
D. Size over 4 cm
E. Hürthle cell change

6 A 26-year-old who received neck irradiation as a child has a thyroid nodule and an enlarged jugular node. FNAC shows grooved, clear nuclei with pseudo-inclusions. The genetic change most typical of radiation-associated tumours of this type is:

A. RET/PTC rearrangement
B. PAX8–PPARG fusion
C. TP53 mutation
D. Germline RET point mutation
E. MEN1 mutation

7 A thyroid tumour shows nests of polygonal cells in a stroma that stains with Congo red. Serum calcitonin is high. The tumour arises from:

A. Follicular cells
B. Hürthle cells
C. Intrathyroid lymphocytes
D. Parafollicular C cells
E. Sustentacular cells

8 A young man has medullary thyroid carcinoma, phaeochromocytoma, nodules on the lips and tongue and a marfanoid build. The syndrome is:

A. MEN1
B. MEN2B
C. MEN2A
D. Von Hippel–Lindau disease
E. Neurofibromatosis type 1

9 A 55-year-old woman with renal stones has calcium 11.8 mg/dL, low phosphate and raised PTH. The most likely lesion is:

A. Parathyroid carcinoma
B. Secondary parathyroid hyperplasia
C. Metastatic breast carcinoma
D. Vitamin D toxicity
E. Solitary parathyroid adenoma

10 A patient on long-term haemodialysis has low-normal calcium, high phosphate and markedly raised PTH. This is:

A. Primary hyperparathyroidism
B. Tertiary hyperparathyroidism
C. Secondary hyperparathyroidism
D. Hypoparathyroidism
E. Pseudohypoparathyroidism

11 Two days after total thyroidectomy a woman has perioral tingling and carpal spasm when a BP cuff is inflated. The cause is:

A. Hypocalcaemia from parathyroid removal or injury
B. Thyroid storm
C. Hyperkalaemia
D. Recurrent laryngeal nerve palsy
E. Hypermagnesaemia

12 A 70-year-old has a rapidly enlarging hard neck mass with stridor. Biopsy shows bizarre giant cells and spindle cells. The key genetic event is:

A. BRAF V600E
B. RET/PTC rearrangement
C. PAX8–PPARG fusion
D. TP53 inactivation
E. MEN1 mutation

ANSWER KEY WITH ONE-LINE REASONS

1. **B** Hashimoto: lymphoid follicles and Hürthle cells.
2. **D** The pituitary senses small changes in free hormone.
3. **A** Graves: TSI mimics TSH.
4. **E** De Quervain: hormone leaks out, so uptake is low.
5. **C** Cytology is identical; invasion decides.
6. **A** Radiation-linked papillary carcinoma: RET/PTC.
7. **D** Medullary carcinoma: calcitonin-derived amyloid.
8. **B** MEN2B: RET; neuromas, marfanoid, no HPT.
9. **E** Primary HPT; adenoma in 85–95%.
10. **C** CKD drives secondary hyperparathyroidism.
11. **A** Trousseau sign: post-surgical hypocalcaemia.
12. **D** Anaplastic carcinoma: TP53 loss.

VIVA Rapid-Fire Viva Voce

Q Thyrotoxicosis vs hyperthyroidism?	Any high free T3/T4 vs an overactive gland.
Q Triad of Graves disease?	Thyrotoxic goitre, ophthalmopathy, dermopathy.
Q Why do the eyes protrude in Graves?	T cells and TSI act on orbital fibroblasts: GAG, fat.
Q What are Hürthle cells?	Follicular cells stuffed with mitochondria.
Q Commonest cause of hypothyroidism?	Iodine lack worldwide; Hashimoto if iodine-replete.
Q What is Riedel thyroiditis?	IgG4-related fibrosis invading the neck; rock-hard.
Q What is an Orphan Annie eye nucleus?	Clear, empty nucleus of papillary carcinoma.
Q Name tumours with psammoma bodies.	Papillary thyroid, serous ovarian, meningioma.
Q Source of amyloid in medullary carcinoma?	Altered calcitonin from the tumour cells.
Q Which thyroid cancer spreads by blood?	Follicular carcinoma (lung, bone).
Q Commonest cause of primary hyperparathyroidism?	Solitary adenoma (85–95%).
Q What is a brown tumour?	Osteoclasts, haemorrhage, haemosiderin (HPT).

OSPE Practice Stations

STATION 1 · SLIDE — 2 minutes

TASK

Thyroid: sheets of lymphocytes and plasma cells form lymphoid follicles with germinal centres; scattered small follicles are lined by large cells with pink granular cytoplasm. (a) Identify. (b) Name the pink cells. (c) Which neoplasm is this patient at risk of?

ANSWER

(a) Hashimoto thyroiditis. (b) Hürthle (oxyphil) cells, rich in mitochondria. (c) Extranodal marginal zone B-cell lymphoma.

STATION 2 · SLIDE — 2 minutes

TASK

Thyroid nodule: branching papillae with fibrovascular cores lined by cells with overlapping, clear, grooved nuclei; laminated purple calcified bodies are present. (a) Diagnosis. (b) Name the calcified bodies. (c) Commonest route of spread.

ANSWER

(a) Papillary carcinoma of the thyroid. (b) Psammoma bodies. (c) Lymphatic: cervical lymph nodes.

STATION 3 · LAB DATA — 2 minutes

TASK

Patient A: TSH 26 mIU/L (0.4–4), free T4 low, anti-TPO strongly positive, firm goitre. **Patient B:** TSH < 0.01 mIU/L, free T4 high, painful thyroid, ESR 85 mm/h, radioiodine uptake very low. **Patient C:** calcium 11.6 mg/dL, phosphate low, PTH raised, renal stones. Interpret each and name the likely disease.

ANSWER

A: primary hypothyroidism; Hashimoto thyroiditis. **B:** thyrotoxicosis without hyperthyroidism (hormone leak); subacute granulomatous (de Quervain) thyroiditis. **C:** primary hyperparathyroidism, most likely a parathyroid adenoma.

RAPID REVISION

The Night-Before Page

15 ONE-LINERS

01 TSH is the best screening test; TSH and T4 move oppositely in primary disease.

02 Thyrotoxicosis with low radioiodine uptake = thyroiditis or exogenous hormone.

03 Graves: stimulating anti-TSH-receptor IgG; goitre, eyes, shins.

04 Graves gland: beefy red; tall cells, scalloped colloid, papillae without cores.

05 Hashimoto: anti-TPO, lymphoid follicles, Hürthle cells; hypothyroid.

06 Hashimoto raises the risk of marginal zone B-cell lymphoma.

07 De Quervain: painful, post-viral, giant cells engulf colloid, self-limited.

08 Riedel: woody fibrosis beyond the capsule; IgG4-related.

09 Goitre: iodine lack → TSH → hyperplasia → colloid → multinodular.

10 Hot nodule: almost always benign; cold, solitary, young, male, irradiated: suspect.

11 Follicular carcinoma = capsular or vascular invasion; spreads by blood.

12 Papillary: clear grooved nuclei, psammoma, nodes; RET/PTC, BRAF.

13 Medullary: C cells, calcitonin, amyloid stroma, RET, MEN2.

14 Anaplastic: elderly, giant and spindle cells, TP53, death within a year.

15 Primary HPT: adenoma 85–95%; high Ca, high PTH, low phosphate.

MEMORY HOOKS

- **Opposite = gland, same = pituitary**: reading TSH with T4
- **P cancer**: popular, past radiation, pale nuclei, psammoma, palpable nodes, prognosis good
- **Look at the edge**: follicular carcinoma is a diagnosis of the capsule
- **Bones, stones, groans, moans**: hypercalcaemia of primary HPT

TEST YOURSELF: FILL THE GAPS

1. Hürthle cells are packed with _____.
2. De Quervain thyroiditis follows a _____ infection.
3. Amyloid in medullary carcinoma comes from _____.
4. Follicular carcinoma metastasises via _____.
5. Commonest cause of secondary HPT: _____.
6. Tapping the facial nerve in hypocalcaemia: _____ sign.

Answers: mitochondria · viral · calcitonin · blood · chronic kidney disease · Chvostek

DRAW FROM MEMORY

Graves pathogenesis: TSI on the follicular cell and the orbital fibroblast (Figure 23.1).

DRAW FROM MEMORY

A papillary carcinoma papilla: fibrovascular core, clear grooved nuclei and a psammoma body.

SPACED REVISION ☐ Day 1 ☐ Day 3 ☐ Day 7 ☐ Day 14 ☐ Day 30

24

Pituitary, Adrenal & Diabetes Mellitus

Endocrine disease follows one rule: too much or too little of a hormone produces a syndrome you can predict from what the hormone normally does. Diabetes breaks the pattern: it is a slow sugar-coating of every blood vessel in the body.

BY THE END OF THIS CHAPTER YOU CAN

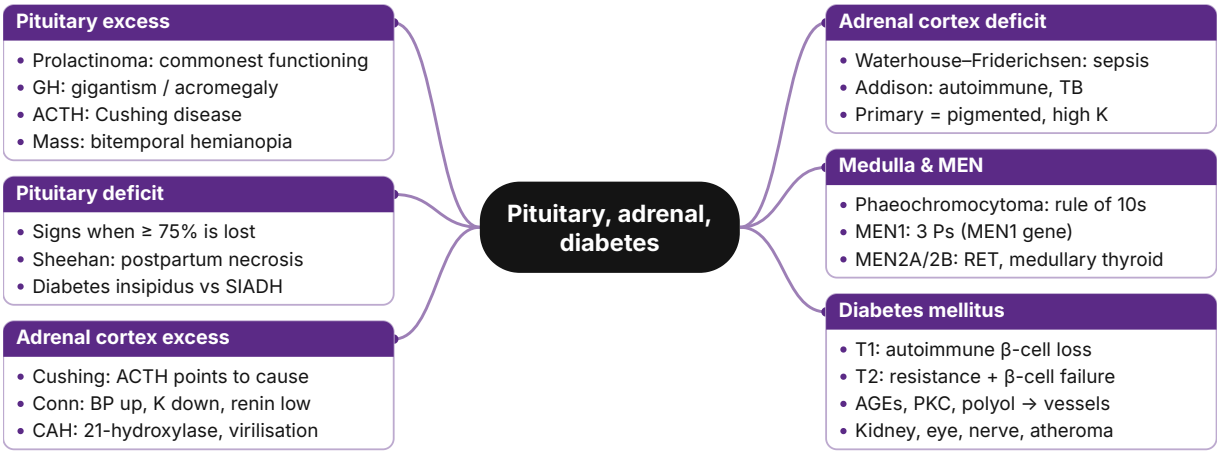
1. Classify pituitary adenomas and describe the effects of prolactin, GH and ACTH excess
2. Explain hypopituitarism (including Sheehan syndrome) and differentiate diabetes insipidus from SIADH
3. Work through the causes and laboratory localisation of Cushing syndrome
4. Describe hyperaldosteronism, congenital adrenal hyperplasia and acute and chronic adrenal insufficiency
5. Describe phaeochromocytoma and compare the MEN syndromes
6. Classify diabetes, state the diagnostic criteria and contrast type 1 with type 2 pathogenesis
7. Explain how hyperglycaemia injures vessels and describe the diabetic kidney, eye and nerve

EXAM RADAR

Theory SEQs	★★★
BCQs / MCQs	★★★
Viva voce	★★★
OSPE / slides	★★☆

Diabetes (classification, pathogenesis, complications) is a near-certain long question; Cushing, Addison, CAH, phaeochromocytoma and MEN are favourite short notes and BCQs.

ROBBINS LINK Robbins Basic Pathology — *The Endocrine System*: pituitary, adrenal cortex and medulla, multiple endocrine neoplasia, endocrine pancreas (diabetes mellitus).



Chapter at a glance. Excess, deficit, and the sugar that damages vessels.

24.1 Pituitary gland

The anterior pituitary has five cell types: **somatotrophs** (GH), **lactotrophs** (prolactin, held in check by hypothalamic **dopamine**), **corticotrophs** (ACTH, MSH), **thyrotrophs** (TSH) and **gonadotrophs** (FSH, LH). The posterior lobe stores hypothalamic **ADH** and oxytocin. Disease shows as **hormone excess** (usually an adenoma), **deficit**, or **mass effects**: bitemporal hemianopia (optic chiasm), raised intracranial pressure and **pituitary apoplexy** (sudden haemorrhage into an adenoma, an emergency).

Pituitary adenomas

Adenomas are classified by the hormone they make (immunohistochemistry and lineage factors PIT1, TPIT, SF1), as **functioning or non-functioning**, and by size: **microadenoma < 1 cm**, **macroadenoma > 1 cm**. Peak age 35–60 years. Genes: **GNAS** activation (constitutive Gsα → cAMP) in about 40% of somatotroph adenomas; germline *MEN1*, *AIP*, *CDKN1B*, *PRKAR1A* in familial cases.

GROSS

- Soft, well-circumscribed; small ones confined to the sella
- Large ones erode the sella and grow upwards through the diaphragma sellae onto the **optic chiasm**; some invade sinuses

MICROSCOPY

- Sheets or cords of **uniform, monomorphic polygonal cells**
- **Loss of the reticulin network** (the normal gland has mixed cells in acini)
- Pleomorphism does not predict behaviour; **carcinoma = metastasis**

FUNCTIONAL TYPES OF PITUITARY ADENOMA

Type	Hormone	Clinical picture	Key points
Lactotroph (prolactinoma)	Prolactin	Women: **galactorrhoea, amenorrhoea**, infertility; men and older women: loss of libido, late mass effects	**Commonest functioning adenoma** (about 30%); shrinks with dopamine agonists (bromocriptine, cabergoline)
Somatotroph	GH (→ IGF-1)	**Gigantism** before epiphyseal closure; **acromegaly** after	Second commonest; raised IGF-1, GH **not suppressed by oral glucose**
Corticotroph	ACTH	**Cushing disease**; Nelson syndrome after bilateral adrenalectomy	Usually microadenomas; basophilic, PAS positive
Gonadotroph / non-functioning	FSH, LH (inefficient) or none	Middle-aged; headache, visual loss, **hypopituitarism**	Present late, as macroadenomas
Thyrotroph	TSH	Central hyperthyroidism	Rare (about 1%)

ACROMEGALY AT THE BEDSIDE

Large hands and feet, **prognathism**, coarse face, broad nose, spaced teeth, big tongue; visceromegaly, arthritis, carpal tunnel syndrome; **diabetes**, hypertension, **cardiomyopathy and heart failure** (main cause of death), and a higher risk of colon cancer.

EXAM TRAP

Not every high prolactin is a prolactinoma. Pregnancy, dopamine-blocking drugs (antipsychotics, metoclopramide), primary hypothyroidism (TRH stimulates lactotrophs) and the **stalk effect** (any mass compressing the stalk cuts off dopamine) all raise prolactin modestly.

Hypopituitarism

Deficit appears when **about 75%** of the anterior pituitary is lost. Causes: **non-functioning adenomas** and other sellar masses (craniopharyngioma), head injury, surgery or radiation, **pituitary apoplexy**, **ischaemic necrosis** (Sheehan syndrome), **empty sella syndrome**, hypothalamic lesions and infiltrations (sarcoidosis, TB).

CONCEPT CLEAR: SHEEHAN SYNDROME

In pregnancy the anterior pituitary nearly **doubles in size** (lactotroph hyperplasia), but its **low-pressure portal venous supply** does not grow with it. A massive **postpartum haemorrhage** with shock drops perfusion below what the swollen gland needs, and the anterior lobe infarcts. The posterior lobe, fed directly by arteries, is spared. The first clue is **failure of lactation**, followed by amenorrhoea, loss of pubic and axillary hair, hypothyroidism and adrenal insufficiency. The infarct heals into a **shrunken fibrous nubbin**.

Effects: growth failure in children (GH); amenorrhoea, infertility, loss of body hair (gonadotrophins); hypothyroidism (TSH); hypoadrenalism **without pigmentation** (ACTH); failed lactation (prolactin).

Posterior pituitary: too little or too much ADH

DIABETES INSIPIDUS VS SIADH

Feature	Diabetes insipidus (DI)	SIADH
Defect	ADH deficiency (**central**) or renal unresponsiveness (**nephrogenic**)	ADH excess, not suppressed by low osmolality

Feature	Diabetes insipidus (DI)	SIADH
Causes	Central: head trauma, tumours, surgery. Nephrogenic: lithium, hypercalcaemia, inherited receptor defects	**Ectopic ADH from small cell carcinoma of lung**; CNS injury, meningitis; pneumonia; drugs (carbamazepine, SSRIs)
Urine	**Large volumes of dilute urine**, low specific gravity	Inappropriately concentrated
Serum	**Hypernatraemia**, high osmolality (if thirst is not met)	**Hyponatraemia**, low osmolality
Clinical	Polyuria, thirst, polydipsia, dehydration	Euvolaemic; confusion, seizures, cerebral oedema; **no peripheral oedema**

24.2 Adrenal cortex

Three zones, three hormone families, from outside in: **glomerulosa** (aldosterone, driven by renin–angiotensin), **fasciculata** (cortisol) and **reticularis** (androgens), both driven by ACTH. Remember **"GFR → salt, sugar, sex"**: the deeper you go, the sweeter it gets.

Cushing syndrome (hypercortisolism)

CAUSES OF CUSHING SYNDROME AND THE ADRENAL THEY LEAVE BEHIND

Cause	Share / notes	Adrenal morphology
Exogenous glucocorticoids (iatrogenic)	Commonest cause overall	Bilateral **cortical atrophy** (fasciculata and reticularis thin; glomerulosa normal)
Cushing disease: pituitary ACTH microadenoma	About **70% of endogenous** cases; young women	**Bilateral diffuse or nodular hyperplasia**: thick yellow cortex
Ectopic ACTH	About 10%; **small cell lung carcinoma**, carcinoid, medullary thyroid carcinoma	Bilateral hyperplasia
Primary adrenal (ACTH-independent)	About 10–20%: adenoma, carcinoma, macronodular hyperplasia, pigmented micronodular disease (Carney)	Unilateral tumour with **atrophy of adjacent and opposite cortex**

A cortisol-secreting **adenoma** is a yellow, encapsulated nodule (usually < 30 g); a **carcinoma** is large, with haemorrhage, necrosis and invasion. In the pituitary, high cortisol from any cause produces **Crooke hyaline change**: the granular cytoplasm of normal corticotrophs is replaced by homogeneous pale keratin filaments.

CLINICAL FEATURES

- **Central obesity**, moon face, buffalo hump; hypertension
- **Proximal myopathy**; osteoporosis; thin skin, bruising, **purple striae**
- Hyperglycaemia; infections; hirsutism, menstrual upset; depression, psychosis
- **Ectopic ACTH:** pigmentation, hypokalaemic alkalosis

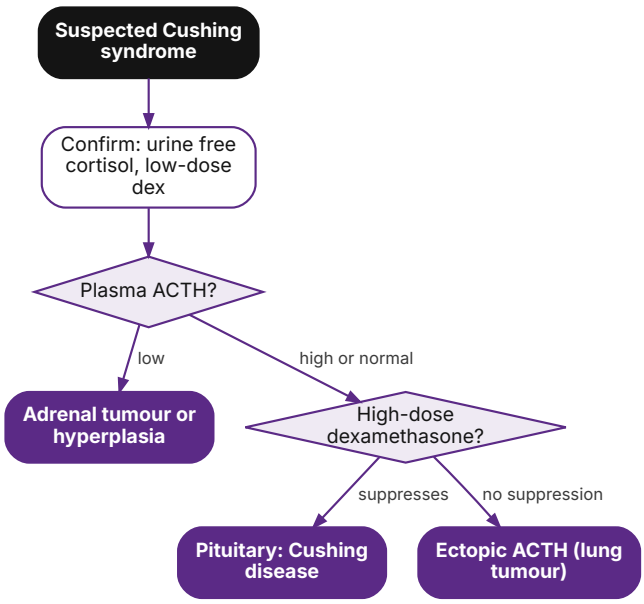


Figure 24.1 · Localising the cause of Cushing syndrome. Plasma ACTH is the branch point.

Hyperaldosteronism

Primary (low renin): autonomous aldosterone. Causes: **bilateral idiopathic hyperplasia** (about 60%), **aldosterone-producing adenoma = Conn syndrome** (about 35%; often *KCNJ5* potassium-channel mutations), familial glucocorticoid-remediable form, rarely carcinoma. It may be the commonest cause of secondary hypertension.

Secondary (high renin): renal artery stenosis, heart failure, cirrhosis, nephrotic syndrome, pregnancy.

CONN ADENOMA: GROSS

- **Solitary, small (< 2 cm)**, well circumscribed, **bright yellow**
- Adjacent cortex **not atrophic** (ACTH unaffected)

MICROSCOPY

- Lipid-laden cortical cells like fasciculata
- **Spironolactone bodies**: laminated eosinophilic inclusions after spironolactone

Clinical: hypertension, **hypokalaemia** (weakness, paraesthesias, arrhythmias), metabolic alkalosis; sodium is retained but oedema does not develop. Screening test: raised **aldosterone-to-renin ratio**.

Adrenogenital syndromes: congenital adrenal hyperplasia (CAH)

Androgen excess may come from a virilising cortical tumour or from **CAH**: autosomal recessive defects of cortisol synthesis. **21-hydroxylase (CYP21A2) deficiency** causes more than 90% of cases.

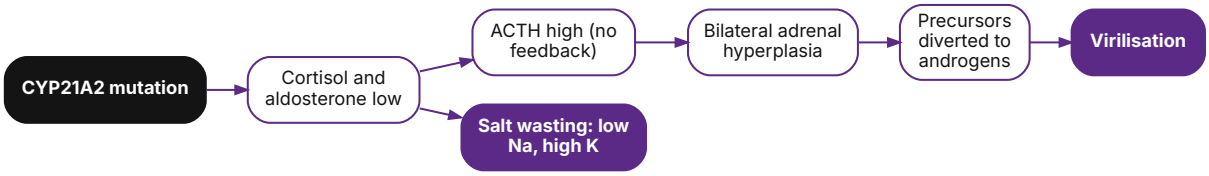


Figure 24.2 · Why 21-hydroxylase deficiency virilises and wastes salt. The block raises ACTH, and ACTH pushes precursors into the only open pathway.

- **Salt-wasting** (complete defect): neonate with vomiting, dehydration, **low Na, high K**, shock; ambiguous genitalia in girls.
- **Simple virilising** (partial; aldosterone adequate): genital ambiguity in girls, precocious puberty in boys.
- **Non-classic** (mild, late onset): hirsutism, acne, irregular menses; mimics polycystic ovary.

Morphology: both adrenals **hyperplastic** (up to 10–15 times normal), cortex thick and **brown** (lipid depleted). **Labs:** raised **17-hydroxyprogesterone**. Glucocorticoids suppress ACTH. **11β-hydroxylase** deficiency virilises but causes **hypertension** (deoxycorticosterone).

Adrenal insufficiency

ACUTE: WATERHOUSE–FRIDERICHSEN SYNDROME

Overwhelming bacterial sepsis, classically ***Neisseria meningitidis*** (also *Pseudomonas*, pneumococci, *H. influenzae*), mostly in children: endotoxin vascular injury and **DIC → shock**, purpura and **massive bilateral adrenal haemorrhage**; the glands become **sacs of clotted blood**. Other acute causes: stress in chronic insufficiency, **abrupt steroid withdrawal**.

CHRONIC PRIMARY: ADDISON DISEASE

Appears when about **90% of the cortex** is destroyed. Causes: **autoimmune adrenalitis** (60–70% in developed countries; alone or in APS-1 or APS-2), **tuberculosis** and fungi (histoplasmosis), AIDS (CMV, *M. avium*), metastatic carcinoma (lung, breast), amyloidosis. **Morphology:** autoimmune glands are **small and shrunken**, cortex replaced by lymphocytes, medulla spared; TB gives caseating granulomas.

PRIMARY VS SECONDARY ADRENAL INSUFFICIENCY

Feature	Primary (Addison)	Secondary (pituitary / hypothalamic)
ACTH	**High**	Low
Skin	**Hyperpigmentation** of skin and buccal mucosa (POMC → ACTH + MSH)	**No pigmentation** (pallor)
Aldosterone	Lost: **hyperkalaemia**, hyponatraemia, hypotension	Preserved (renin-driven): potassium normal

Feature	Primary (Addison)	Secondary (pituitary / hypothalamic)
Common features	Weakness, fatigue, anorexia, vomiting, weight loss, hypoglycaemia; stress may precipitate an **Addisonian crisis** (shock, coma)	

24.3 Adrenal medulla and MEN syndromes

Phaeochromocytoma

A tumour of **chromaffin cells** that secretes catecholamines: an uncommon but **surgically curable cause of hypertension**. Paroxysms bring **headache, palpitations and sweating** with tremor and anxiety; catecholamine excess may cause cardiomyopathy, arrhythmia, infarction and stroke. **Labs:** raised urinary and plasma **metanephrines** and urinary **VMA**. Remove after α-blockade.

GROSS

- Small and circumscribed to large and haemorrhagic
- Cut surface **yellow-tan** with haemorrhage, necrosis, cysts
- **Potassium dichromate turns it dark brown** (chromaffin reaction)

MICROSCOPY

- Polygonal to spindle chief cells in nests (**Zellballen**) wrapped by **sustentacular cells** and a rich capillary network
- Finely granular cytoplasm; **salt-and-pepper** chromatin; dense-core granules on EM
- Chromogranin, synaptophysin positive; S-100 in sustentacular cells
- **Malignancy is defined only by metastasis**

MEMORY HOOK

The traditional "rule of 10s"

10% extra-adrenal (paragangliomas, e.g. organ of Zuckerkandl, carotid body) · **10%** bilateral · **10%** malignant · **10%** without hypertension · **10%** familial. The last figure is outdated: about **25%** carry germline mutations (*RET*, *NF1*, *VHL*, *SDH* genes), and familial tumours are often bilateral.

Multiple endocrine neoplasia (MEN)

Autosomal dominant syndromes of endocrine tumours that are **younger**, **multifocal** and preceded by **hyperplasia**.

MEN SYNDROMES

Feature	MEN1 (Wermer)	MEN2A (Sipple)	MEN2B
Gene	*MEN1* (menin, tumour suppressor), 11q13	*RET* gain-of-function, 10q11	*RET* (a different codon, e.g. M918T)
Parathyroid	**Hyperplasia / adenomas** (80–95%; usually first)	Hyperplasia (10–20%)	**Not involved**
Pancreas, duodenum	Neuroendocrine tumours: **gastrinoma** (Zollinger–Ellison), insulinoma; leading cause of death	—	—
Pituitary	Adenoma, most often **prolactinoma**	—	—
Thyroid	—	**Medullary carcinoma** (virtually all)	**Medullary carcinoma** (earlier, more aggressive)
Adrenal medulla	—	**Phaeochromocytoma** (about 40–50%)	Phaeochromocytoma
Other	—	—	**Mucosal neuromas** (lips, tongue), **marfanoid habitus**, intestinal ganglioneuromas

MEMORY HOOK

MEN1 = 3 Ps · MEN2A = 2 Ps + M · MEN2B = 1 P + 3 Ms

MEN1: **P**ituitary, **P**arathyroid, **P**ancreas. MEN2A: **P**haeochromocytoma, **P**arathyroid, **M**edullary carcinoma. MEN2B: **P**haeochromocytoma, **M**edullary carcinoma, **M**arfanoid habitus, **M**ucosal neuromas. *RET* carriers have **prophylactic thyroidectomy**.

24.4 Diabetes mellitus

DEFINITION

Diabetes mellitus is a group of metabolic disorders sharing **hyperglycaemia** caused by defects in **insulin secretion, insulin action, or both**. Chronic hyperglycaemia damages blood vessels, kidneys, eyes and nerves; diabetes is a leading cause of end-stage renal disease, adult-onset blindness and non-traumatic leg amputation.

DIAGNOSTIC CRITERIA (ADA, AS USED BY ROBBINS)

Fasting plasma glucose ≥ 126 mg/dL (7.0 mmol/L)

Random glucose

≥ 200 mg/dL (11.1 mmol/L) with classic symptoms

2-hour OGTT (75 g) ≥ 200 mg/dL

HbA1c ≥ 6.5%

Prediabetes

FPG 100–125 (IFG); 2-h 140–199 (IGT); HbA1c 5.7–6.4%

Normal

FPG < 100 mg/dL. Without clear hyperglycaemia, repeat on another day

Classification: type 1 (5–10%; autoimmune β-cell destruction, absolute insulin deficiency); **type 2** (90–95%; insulin resistance with relative deficiency); **monogenic** (MODY: *GCK*, *HNF1A*; mitochondrial); **gestational**; and **secondary**: exocrine pancreatic disease (chronic pancreatitis, cystic fibrosis, haemochromatosis), endocrinopathies (Cushing, acromegaly, phaeochromocytoma, glucagonoma), drugs (glucocorticoids, thiazides), infections (congenital rubella, CMV) and genetic syndromes (Down, Turner, Klinefelter).

TYPE 1 VS TYPE 2 DIABETES

Feature	Type 1	Type 2
Onset	Usually childhood or adolescence	Usually adults (increasingly obese adolescents)
Body weight	Normal; weight loss at onset	**Obese** (central)
Basic defect	**Autoimmune β-cell destruction**; absolute deficiency	**Insulin resistance + β-cell dysfunction**; relative deficiency
Blood insulin	Very low	High early, falls later
Genetics	**HLA class II** (DR3, DR4); *CTLA4*, *PTPN22*, insulin gene	Polygenic (e.g. *TCF7L2*); **no HLA link**; strong family history
Islet antibodies	**Present** (insulin, GAD65, IA-2, ZnT8)	Absent
Islet pathology	**Insulitis**; small, β-cell-depleted islets	Mild β-cell loss; **amyloid deposition**
Acute crisis	**Diabetic ketoacidosis**	**Hyperosmolar hyperglycaemic state**

Pathogenesis: two routes to the same high glucose

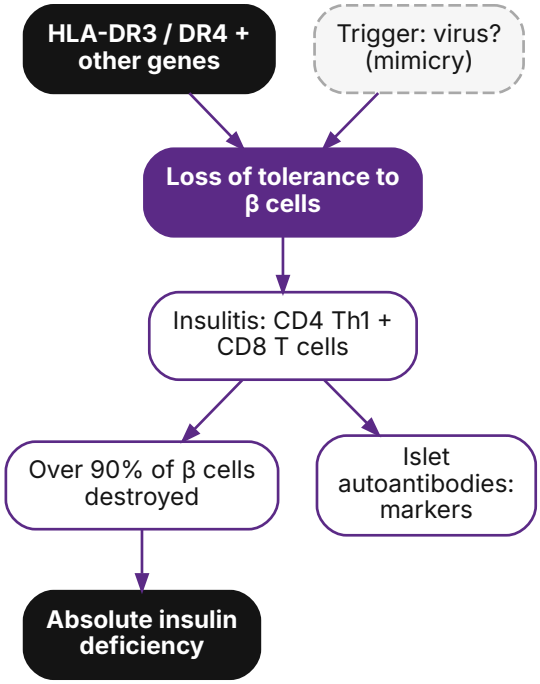


Figure 24.3 · Type 1 diabetes. The immune system deletes the β cells.

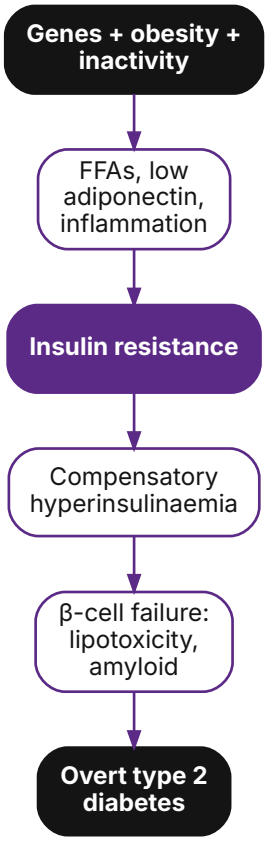


Figure 24.4 · Type 2 diabetes. Resistance first; diabetes only when β cells fail.

CONCEPT CLEAR

How fat blocks insulin. In **visceral obesity**, free fatty acids flood muscle and liver; their products (diacylglycerol, ceramide) activate kinases that put **serine phosphates on the insulin receptor substrate**, muting the signal. Fat also makes less **adiponectin** (an insulin-sensitiser), and FFAs activate the **inflammasome** (IL-1β). β cells compensate for years; diabetes appears when they fail through lipotoxicity, glucotoxicity, **islet amyloid (amylin)** and a weak incretin (GLP-1) effect.

Diabetic ketoacidosis (type 1): no insulin + excess glucagon → lipolysis → free fatty acids → hepatic **ketone bodies** (acetoacetate, β-hydroxybutyrate) → high anion gap **metabolic acidosis**. Glucose 250–600 mg/dL; osmotic diuresis, dehydration, vomiting, **Kussmaul breathing**, fruity breath; triggered by infection or missed insulin.

Hyperosmolar hyperglycaemic state (type 2): elderly patient who cannot drink enough; glucose often > 600 mg/dL, severe dehydration, confusion, but **no ketoacidosis** (residual insulin blocks lipolysis). **Hypoglycaemia** is the commonest acute complication of treatment.

Chronic complications: how sugar damages vessels

Nerves, lens, kidney and vessel walls take up glucose **without insulin**, so their cells fill with glucose whenever the blood does; three pathways turn it into **vascular damage**.

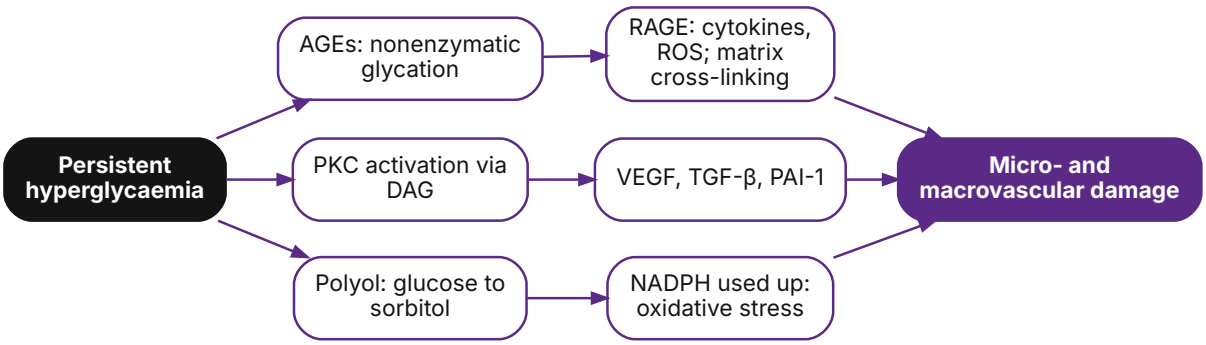


Figure 24.5 · Hyperglycaemia to complications. The microvascular injury (kidney, eye, nerve) and accelerated atherosclerosis share these pathways.

MOLECULAR KEY

AGEs form when glucose binds proteins without an enzyme (HbA1c is an early, measurable example reflecting the past 2–3 months). They act on **RAGE** (cytokines, procoagulant activity, smooth muscle growth) and **cross-link matrix**, trapping LDL in arteries and albumin in capillary basement membranes. **PKC** raises VEGF (retinal new vessels), TGF-β (matrix) and PAI-1. The **polyol pathway** (aldose reductase) uses up NADPH, so glutathione is not regenerated: oxidative stress in nerve and lens.

MORPHOLOGY OF CHRONIC COMPLICATIONS

Site	Lesion	Result
Large arteries	**Accelerated atherosclerosis** of aorta, coronary, cerebral and leg arteries	**Myocardial infarction (commonest cause of death)**, stroke, **gangrene of the legs**
Capillaries (microangiopathy)	**Diffuse basement membrane thickening** (retina, glomeruli, skin, muscle, nerves); vessels leak protein	Basis of nephropathy, retinopathy and part of neuropathy
Eye	**Non-proliferative**: microaneurysms, haemorrhages, exudates, oedema. **Proliferative**: new vessels, fibrosis; cataract, glaucoma	Vitreous haemorrhage, retinal detachment, **blindness**
Nerve	Axonal degeneration, segmental demyelination, vasa nervorum disease	**Distal symmetric** "glove-and-stocking" sensory loss; autonomic (bladder, bowel, erectile); mononeuropathy
Immunity	Impaired neutrophil function, poor blood supply	TB, pneumonia, pyelonephritis, skin infections, **mucormycosis**; foot ulcers

DIABETIC NEPHROPATHY

GROSS

- Kidneys normal to small with a **finely granular surface** (arteriolar nephrosclerosis)
- Look for **papillary necrosis** and pyelonephritic scars

MICROSCOPY

- **Glomerular capillary basement membrane thickening** (earliest, on EM)
- **Diffuse mesangial sclerosis**
- **Nodular glomerulosclerosis** (Kimmelstiel–Wilson): ovoid, often laminated, **PAS-positive** nodules of matrix at the periphery of the glomerulus
- **Hyaline arteriolosclerosis of both afferent and efferent arterioles**
- Pyelonephritis, **necrotising papillitis**

Course: microalbuminuria (30–300 mg/day) → overt, often nephrotic proteinuria → end-stage renal disease (see Chapter 16).

PAKISTAN FOCUS

About **one in four Pakistani adults** has diabetes: the IDF Diabetes Atlas 2021 estimated 26.7% (about 33 million), the third-largest number of people with diabetes in any country. With hypertension, diabetes is the **leading cause of chronic kidney disease** in Pakistan.

Source: IDF Diabetes Atlas 2021.

EXAM ZONE Past-Paper Pattern SEQs — Solved

PROF PATTERN **a)** Classify diabetes mellitus. **b)** Give the diagnostic criteria. **c)** Compare the pathogenesis of type 1 and type 2 diabetes. **1 + 1 + 3 marks**

MODEL ANSWER PLAN

a) Type 1, type 2, monogenic (MODY), gestational, secondary (pancreatic, endocrine, drugs, infections). **b)** FPG ≥ 126 mg/dL; random ≥ 200 with symptoms; 2-h OGTT ≥ 200; HbA1c ≥ 6.5%. **c)** T1: HLA-DR3/DR4 + trigger → loss of tolerance → insulitis (CD4 Th1, CD8) → β-cell destruction → absolute deficiency; autoantibodies. T2: genes + obesity → FFAs, adipokines, inflammation → insulin resistance → compensatory hyperinsulinaemia → β-cell failure (lipotoxicity, amyloid) → relative deficiency.

Score booster: draw Figures 24.3 and 24.4 side by side and end with a five-row T1 vs T2 table.

PROF PATTERN **a)** Describe the pathogenesis of the chronic complications of diabetes. **b)** Describe the morphology of diabetic nephropathy. **2 + 3 marks**

MODEL ANSWER PLAN

a) Insulin-independent tissues → intracellular hyperglycaemia → AGEs (RAGE, matrix cross-linking), PKC (VEGF, TGF-β, PAI-1), polyol pathway (sorbitol, NADPH loss, oxidative stress) → microangiopathy and atherosclerosis. **b)** BM thickening, diffuse mesangial sclerosis, Kimmelstiel–Wilson nodules (PAS+), afferent and efferent hyaline arteriolosclerosis, pyelonephritis and papillary necrosis.

MODULE PATTERN A 34-year-old woman has gained weight around the trunk, with a round face, purple abdominal striae, hypertension and raised blood glucose. **a)** Most likely diagnosis? **b)** List its causes. **c)** How will laboratory tests localise the cause? **1 + 2 + 2 marks**

MODEL ANSWER PLAN

a) Cushing syndrome. **b)** Exogenous steroids (commonest); Cushing disease (pituitary ACTH adenoma, about 70% of endogenous); ectopic ACTH (small cell lung carcinoma); adrenal adenoma, carcinoma or hyperplasia. **c)** Confirm hypercortisolism (24-h urine free cortisol, low-dose dexamethasone); ACTH low → adrenal; ACTH high + suppression with high-dose dexamethasone → pituitary; no suppression → ectopic.

CLASSIC **a)** Write a note on phaeochromocytoma. **b)** Tabulate the MEN syndromes. **3 + 2 marks**

MODEL ANSWER PLAN

a) Chromaffin cell tumour; rule of 10s (now about 25% familial); paroxysmal hypertension with headache, palpitations, sweating; gross yellow-tan, haemorrhagic, dichromate-brown; micro Zellballen with sustentacular cells, salt-and-pepper nuclei; malignancy only by metastasis; urinary metanephrines and VMA. **b)** MEN1 (MEN1: parathyroid, pancreas, pituitary); MEN2A (RET: medullary carcinoma, phaeochromocytoma, parathyroid); MEN2B (RET: medullary carcinoma, phaeochromocytoma, mucosal neuromas, marfanoid).

BCQs One-Best-Answer Practice

1 A 28-year-old woman has galactorrhoea and amenorrhoea. MRI shows an 8-mm sellar lesion. The most likely tumour and first-line treatment are:

A. Somatotroph adenoma; surgery
B. Corticotroph adenoma; ketoconazole
C. Lactotroph adenoma; dopamine agonist
D. Gonadotroph adenoma; radiotherapy
E. Thyrotroph adenoma; radioiodine

2 After a massive postpartum haemorrhage a woman cannot breastfeed and later has amenorrhoea and cold intolerance. The underlying lesion is:

A. Ischaemic necrosis of the enlarged anterior pituitary
B. Lymphocytic hypophysitis
C. Haemorrhage into a macroadenoma
D. Infarction of the posterior lobe
E. Metastatic carcinoma

3 A smoker with small cell carcinoma of the lung has serum sodium 121 mmol/L, concentrated urine and no oedema. The diagnosis is:

A. Central diabetes insipidus
B. Nephrogenic diabetes insipidus
C. Conn syndrome
D. Addison disease
E. SIADH

4 A cushingoid woman has high plasma ACTH, and cortisol falls after high-dose dexamethasone. The cause is:

A. Adrenal adenoma
B. Pituitary corticotroph adenoma
C. Ectopic ACTH from lung carcinoma
D. Exogenous glucocorticoids
E. Adrenal carcinoma

5 A hypertensive man has hypokalaemia, low plasma renin and a 1.5-cm bright yellow adrenal nodule; the rest of the cortex is not atrophic. The lesion is:

A. Phaeochromocytoma
B. Adrenocortical carcinoma
C. 21-hydroxylase deficiency
D. Aldosterone-producing adenoma
E. Myelolipoma

6 A 2-week-old girl with ambiguous genitalia has vomiting, sodium 124 and potassium 7.2 mmol/L. The deficient enzyme is:

A. 21-hydroxylase
B. 11β-hydroxylase
C. 17α-hydroxylase
D. Aromatase
E. 5α-reductase

7 A child dies of meningococcal septicaemia with purpura and shock. The adrenals at autopsy most likely show:

A. Shrunken glands with lymphocytes
B. Caseating granulomas
C. Massive bilateral haemorrhage
D. Bilateral nodular hyperplasia
E. Metastatic deposits

8 In a phaeochromocytoma, malignancy is established by:

A. Capsular invasion
B. Metastasis
C. Nuclear pleomorphism
D. Vascular invasion
E. Tumour weight over 100 g

9 A man has hypercalcaemia from parathyroid hyperplasia, recurrent peptic ulcers from a gastrinoma and a prolactinoma. The mutated gene is:

A. RET
B. VHL
C. NF1
D. MEN1 (menin)
E. SDHB

10 Which result, confirmed on repeat testing, establishes diabetes mellitus?

A. Fasting glucose 110 mg/dL
B. Random glucose 160 mg/dL without symptoms
C. HbA1c 6.0%
D. 2-hour OGTT glucose 170 mg/dL
E. Fasting glucose 130 mg/dL

11 A 12-year-old with weight loss and polyuria has antibodies to GAD65. Her islets would show:

A. Lymphocytic infiltrate (insulitis)
B. Amyloid deposition
C. Islet hyperplasia
D. Normal islets
E. Duct calcification with fibrosis

12 The Kimmelstiel–Wilson lesion of diabetic nephropathy is:

A. Hyaline thickening of the afferent arteriole only
B. Papillary necrosis
C. A cellular crescent
D. Subepithelial immune deposits
E. PAS-positive nodules of mesangial matrix

ANSWER KEY WITH ONE-LINE REASONS

1. **C** Prolactinoma: commonest functioning adenoma; cabergoline shrinks it.
2. **A** Sheehan syndrome; the posterior lobe is spared.
3. **E** Ectopic ADH: euvolaemic hyponatraemia.
4. **B** Cushing disease: pituitary ACTH still obeys high-dose feedback.
5. **D** Conn syndrome; ACTH is not suppressed, so no atrophy.
6. **A** Salt-wasting CAH: no cortisol or aldosterone, androgens in excess.
7. **C** Waterhouse–Friderichsen syndrome.
8. **B** Histology cannot predict behaviour; only metastasis proves malignancy.
9. **D** MEN1: parathyroid, pancreas, pituitary.

10. **E** FPG ≥ 126 mg/dL; the others fall in the prediabetic or normal range.
11. **A** Type 1 diabetes: T-cell insulitis destroys β cells.
12. **E** Nodular glomerulosclerosis at the periphery of the tuft.

VIVA Rapid-Fire Viva Voce

Q Commonest functioning pituitary adenoma?	Lactotroph adenoma (prolactinoma).
Q Why is the pituitary vulnerable in pregnancy?	It nearly doubles in size without more blood supply.
Q What is Crooke hyaline change?	Pale keratin in corticotrophs exposed to high cortisol.
Q Commonest cause of Cushing syndrome?	Exogenous steroids; endogenous: pituitary ACTH adenoma.
Q Commonest enzyme defect in CAH?	21-hydroxylase (CYP21A2): over 90%.
Q Why is Addison disease pigmented?	High POMC: more ACTH and MSH.
Q Define Waterhouse–Friderichsen syndrome.	Sepsis (meningococcal) with bilateral adrenal haemorrhage.
Q What are Zellballen?	Nests of chief cells ringed by sustentacular cells.
Q Best urine test for phaeochromocytoma?	Metanephrines (and VMA).
Q HbA1c cut-off for diabetes?	6.5% or more.
Q What does the polyol pathway do?	Glucose → sorbitol, using up NADPH: oxidative stress.
Q What is a Kimmelstiel–Wilson nodule?	PAS-positive mesangial nodule in diabetic kidney.

OSPE Practice Stations

STATION 1 · LAB DATA — 2 minutes

TASK

Classify each adult: **A** fasting glucose 118 mg/dL, HbA1c 6.1%. **B** fasting glucose 142 mg/dL on two days. **C** 2-h OGTT 168 mg/dL. **D** random glucose 340 mg/dL, pH 7.12, bicarbonate 9 mmol/L, urine ketones 3+.

ANSWER

A prediabetes (impaired fasting glucose). **B** diabetes. **C** impaired glucose tolerance. **D** diabetic ketoacidosis (high anion gap acidosis), typical of type 1.

STATION 2 · LAB DATA — 2 minutes

TASK

Three patients with confirmed hypercortisolism: **P** ACTH low. **Q** ACTH high, cortisol suppressed by high-dose dexamethasone. **R** ACTH very high, no suppression, hypokalaemia, heavy smoker. Localise each.

ANSWER

P adrenal (adenoma, carcinoma or hyperplasia). **Q** pituitary (Cushing disease). **R** ectopic ACTH, e.g. small cell lung carcinoma.

STATION 3 · SLIDE — 2 minutes

TASK

Kidney: glomeruli contain round, pink, acellular nodules at the periphery of the tuft (PAS positive); arterioles have thick, glassy pink walls. (a) Diagnosis. (b) Name the nodules. (c) Earliest clinical marker of this disease.

ANSWER

(a) Diabetic nephropathy (nodular glomerulosclerosis with hyaline arteriolosclerosis). (b) Kimmelstiel–Wilson nodules. (c) Microalbuminuria (30–300 mg/day).

STATION 4 · SLIDE 2 minutes

TASK

Adrenal mass from a 40-year-old with paroxysmal hypertension: nests of polygonal cells with finely granular cytoplasm and salt-and-pepper nuclei, surrounded by slender spindle cells and a rich capillary network. (a) Diagnosis. (b) Name the nested pattern. (c) Urine test that supports it.

ANSWER

(a) Phaeochromocytoma. (b) Zellballen (chief cells ringed by sustentacular cells). (c) Raised urinary metanephrines and VMA.

RAPID REVISION The Night-Before Page

15 ONE-LINERS

01 Prolactinoma: commonest functioning adenoma; dopamine agonists shrink it.

02 GH excess: gigantism before, acromegaly after epiphyseal closure; GNAS.

03 Hypopituitarism needs about 75% loss; Sheehan spares the posterior lobe.

04 DI: dilute polyuria, high Na. SIADH: low Na, euvolaemic; small cell lung.

05 Commonest Cushing: exogenous steroids; endogenous: pituitary (70%).

06 ACTH low = adrenal; high + suppression = pituitary; no suppression = ectopic.

07 Crooke hyaline change: keratin in corticotrophs from high cortisol.

08 Conn: hypertension, low K, low renin; yellow adenoma, no atrophy.

09 CAH: 21-hydroxylase; virilisation; salt wasting; raised 17-OH progesterone.

10 Waterhouse–Friderichsen: meningococcus, DIC, adrenal haemorrhage.

11 Addison: autoimmune or TB; pigmentation, high K, hypotension.

12 Phaeochromocytoma: Zellballen; malignant only if it metastasises.

13 MEN1 = menin; MEN2 = RET with medullary thyroid carcinoma.

14 DM: FPG ≥ 126, random/OGTT ≥ 200, HbA1c ≥ 6.5%.

15 Complications: AGEs, PKC, polyol → BM thickening, KW nodules, atheroma.

MEMORY HOOKS

- **GFR → salt, sugar, sex**: zones of the adrenal cortex
- **3 Ps / 2 Ps + M / 1 P + 3 Ms**: MEN1, MEN2A, MEN2B
- **Rule of 10s**: extra-adrenal, bilateral, malignant, normotensive (familial now 25%)
- **ACTH decides**: low = adrenal; high = pituitary or ectopic

TEST YOURSELF: FILL THE GAPS

1. GH not suppressed by an oral _____ load.
2. Commonest cause of endogenous Cushing: _____.
3. Spironolactone bodies are seen in a _____ adenoma.
4. Waterhouse–Friderichsen organism: _____.
5. T1DM is linked to HLA-_____ and DR4.
6. Polyol enzyme: _____ reductase.

Answers: glucose · pituitary ACTH adenoma · Conn (aldosterone) · Neisseria meningitidis · DR3 · aldose

DRAW FROM MEMORY

The Cushing localisation flowchart from screening to pituitary, adrenal or ectopic (Figure 24.1).

DRAW FROM MEMORY

Hyperglycaemia → AGEs, PKC, polyol → microangiopathy and atherosclerosis (Figure 24.5).

SPACED REVISION ☐ Day 1 ☐ Day 3 ☐ Day 7 ☐ Day 14 ☐ Day 30

25

Blood Vessels: Atherosclerosis, Aneurysms, Vasculitis & Vascular Tumours

Heart attacks, strokes and gangrene of the foot begin in the same place: the inner lining of an artery. Learn how a plaque grows and breaks, how a weak wall balloons or splits, and how inflamed vessels are sorted by size, and most of cardiovascular pathology falls into place.

BY THE END OF THIS CHAPTER YOU CAN

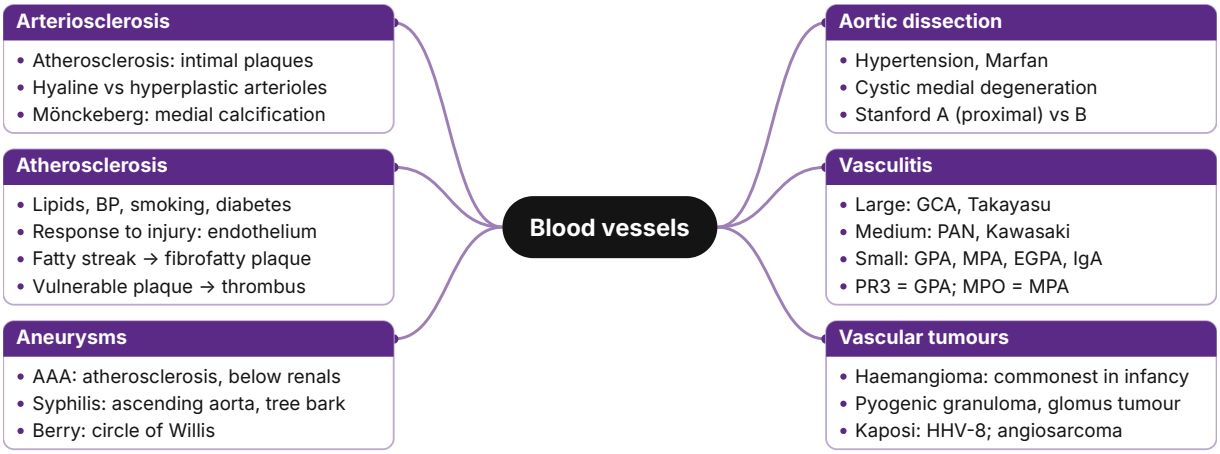
1. Classify arteriosclerosis and contrast hyaline with hyperplastic arteriolosclerosis
2. List the risk factors for atherosclerosis and explain the response-to-injury hypothesis
3. Describe fatty streaks and atheromatous plaques, and differentiate stable from vulnerable plaques
4. Describe abdominal aortic, syphilitic and berry aneurysms and classify aortic dissection
5. Classify vasculitis by vessel size and describe giant cell arteritis, Takayasu, PAN, Kawasaki disease, the ANCA vasculitides and Buerger disease
6. Describe haemangioma, pyogenic granuloma, glomus tumour, Kaposi sarcoma and angiosarcoma

EXAM RADAR

Theory SEQs	★★★
BCQs / MCQs	★★★
Viva voce	★★★
OSPE / slides	★★☆

Atherosclerosis (risk factors, pathogenesis, morphology, complications) is one of the most repeated long questions; vasculitis by vessel size, aortic dissection and Kaposi sarcoma are BCQ favourites.

ROBBINS LINK Robbins Basic Pathology — *Blood Vessels*: arteriosclerosis, atherosclerosis, hypertensive vascular disease, aneurysms and dissection, vasculitis, vascular tumours.



Chapter at a glance. Six ideas carry the vessel wall.

25.1 Arteriosclerosis and hypertensive vascular disease

Arteriosclerosis ("hardening of the arteries") means thickening and loss of elasticity of arterial walls. It has four patterns:

1. **Arteriolosclerosis** — small arteries and arterioles: **hyaline** or **hyperplastic**; narrows the lumen and causes downstream ischaemia (kidney).
2. **Mönckeberg medial sclerosis** — calcium in the **media** of muscular arteries, usually after 50 years; visible on X-ray and palpable as pipe-stem vessels, but the lumen is spared, so it is rarely of clinical importance.
3. **Fibromuscular intimal hyperplasia** — muscular arteries thickened by smooth muscle and matrix after injury: healed arteritis, transplant arteriopathy, **restenosis after angioplasty or stenting**.

4 **Atherosclerosis** — the commonest and most important (Section 25.2).

Hypertensive vascular disease

Hypertension is a sustained blood pressure above the guideline threshold: **≥ 130/80 mm Hg** in current American guidance (older and many local guidelines use ≥ 140/90). About **90–95%** is **essential** (polygenic, with salt, obesity, stress and inactivity acting on renal sodium handling and vascular resistance); the rest is **secondary**: renal (renal artery stenosis, glomerulonephritis), endocrine (Conn, Cushing, phaeochromocytoma, thyroid disease), coarctation of the aorta, raised intracranial pressure. A small fraction develops **malignant hypertension**: pressures above about 200/120 mm Hg with renal failure and retinal haemorrhages or papilloedema.

HYALINE VS HYPERPLASTIC ARTERIOLOSCLEROSIS

Feature	Hyaline arteriolosclerosis	Hyperplastic arteriolosclerosis
Setting	**Benign hypertension**, ageing, **diabetes**	**Malignant (severe) hypertension**
Mechanism	Plasma proteins leak across injured endothelium; smooth muscle lays down excess matrix	Smooth muscle proliferation and basement membrane reduplication in response to an acute, severe rise in pressure
Microscopy	**Homogeneous pink, glassy thickening** of the arteriolar wall; structure lost; lumen narrowed	**Concentric laminated "onion-skin" thickening**; lumen narrowed to obliterated; often **fibrinoid necrosis** (necrotising arteriolitis)
Kidney	**Benign nephrosclerosis**: small, symmetric, finely granular kidneys	**"Flea-bitten" kidney** with pinpoint cortical haemorrhages; acute renal failure

25.2 Atherosclerosis

DEFINITION

Atherosclerosis is a chronic inflammatory and healing response of the arterial wall to endothelial injury. It produces intimal lesions called **atheromas** (atheromatous or fibrofatty plaques) that bulge into the lumen, weaken the underlying media and can **rupture**, causing sudden thrombosis. It underlies most myocardial infarctions, strokes, abdominal aortic aneurysms and peripheral vascular disease.

RISK FACTORS FOR ATHEROSCLEROSIS

Non-modifiable	Modifiable (major)	Additional
Age (rises steadily after 40) **Male sex (women catch up after menopause)** **Family history, genetics (e.g. familial hypercholesterolaemia: LDL-receptor defect)**	**Hyperlipidaemia**: high LDL, low HDL **Hypertension** **Cigarette smoking** **Diabetes mellitus**	Inflammation (raised **CRP**), hyperhomocysteinaemia, metabolic syndrome, raised lipoprotein(a), prothrombotic states (PAI-1), obesity, inactivity, stress, clonal haematopoiesis

Risk factors **multiply** rather than add: two raise the risk about fourfold; hyperlipidaemia, hypertension and smoking together raise the rate of myocardial infarction about sevenfold.

Pathogenesis: the response-to-injury hypothesis

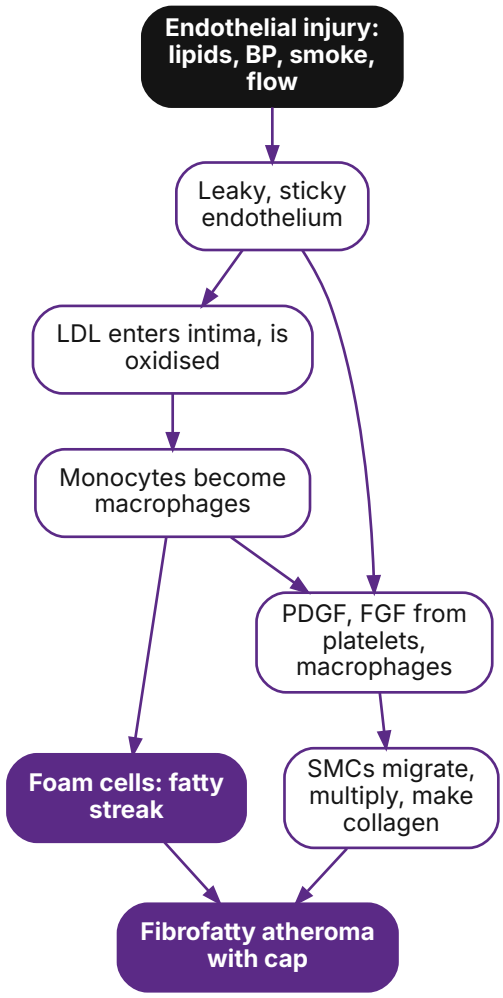


Figure 25.1 · Response to injury. An injured lining lets in lipid and inflammatory cells; growth factors build the cap.

1. **Endothelial dysfunction** (the first step): turbulent flow at branch points and ostia, hypercholesterolaemia, hypertension, smoking toxins and homocysteine make the endothelium **leaky**, **sticky** (VCAM-1) and prothrombotic, with less nitric oxide.
2. **Lipid accumulation**: LDL seeps into the intima and is **oxidised**; oxidised LDL is toxic and chemotactic.
3. **Monocytes** adhere, migrate in and become **macrophages** that swallow oxidised LDL through scavenger receptors: **foam cells**. T cells (Th1, IFN-γ) keep inflammation going; cholesterol crystals activate the **inflammasome** (IL-1).
4. **Platelets** adhere to the damaged lining.
5. **Growth factors** (PDGF, FGF, TGF-α) recruit **smooth muscle cells** from the media.
6. Smooth muscle cells multiply and secrete **collagen**, forming a **fibrous cap** over a lipid core: the mature atheroma.

CONCEPT CLEAR

Why LDL and inflammation are partners. Lipid alone does not make a plaque; the immune reaction to lipid does. Oxidised LDL is the fuel, macrophages are the fire and smooth muscle cells are the scar that tries to wall it off. Lowering LDL starves the process, which is why statins prevent infarction; CRP rises in the blood because the plaque is an inflamed lesion.

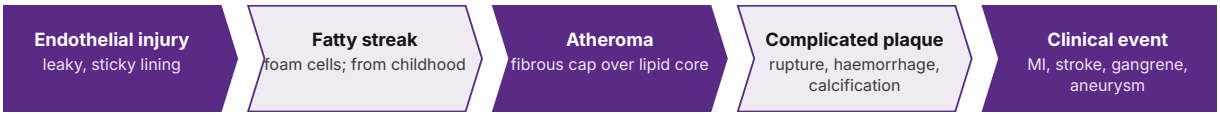


Figure 25.2 · Natural history of a plaque. Decades of silence, then an acute event when the cap breaks.

Morphology

Fatty streaks are the earliest lesion: multiple minute, flat, yellow spots that join into streaks 1 cm or more long. They are made of **lipid-filled foam cells**, are not raised and do not disturb flow. They appear in the aortas of some infants and in virtually all children over 10; not all progress, but coronary fatty streaks of adolescence form at the sites that later develop plaques.

GROSS: ATHEROMATOUS PLAQUE

- **Raised white-yellow plaques**, 0.3–1.5 cm, patchy and **eccentric**; may fuse into large masses
- Red-brown **thrombus** on ulcerated plaques; gritty calcification
- Order of involvement: **lower abdominal aorta** > coronary arteries > popliteal arteries > internal carotids > circle of Willis; worst around branch ostia

MICROSCOPY

- **Fibrous cap**: smooth muscle cells in dense collagen
- **Shoulder**: macrophages, smooth muscle cells, T cells
- **Necrotic core**: lipid, **cholesterol clefts**, foam cells, debris, fibrin, organised thrombus
- New vessels at the periphery; **thinned media** beneath; calcification

Three components: cells (smooth muscle cells, macrophages, T cells), extracellular matrix (collagen, elastic fibres, proteoglycans) and lipid (inside and outside cells). **Complicated plaques** show rupture or erosion with **thrombosis**, **haemorrhage into the plaque**, **atheroembolism** (cholesterol emboli), **aneurysmal dilatation** from medial atrophy, and dystrophic calcification.

STABLE VS VULNERABLE PLAQUE

Feature	Stable plaque	Vulnerable plaque
Fibrous cap	**Thick**, collagen-rich	**Thin**, weakened
Lipid core	Small	**Large**, soft
Inflammatory cells	Few	**Many macrophages** (metalloproteinases digest collagen) and T cells (IFN-γ stops collagen synthesis)
Smooth muscle cells	Many	Few
Typical outcome	**Fixed stenosis**: stable angina, claudication	**Rupture + thrombosis**: acute coronary syndrome, stroke, sudden death

CONSEQUENCES

- **Critical stenosis** (about 70% narrowing): demand ischaemia, stable angina, chronic ischaemic heart disease, bowel ischaemia, **intermittent claudication**
- **Acute plaque change** (rupture, erosion, haemorrhage) → **thrombosis** → MI, stroke
- **Embolism** of thrombus or cholesterol (atheroemboli)
- **Aneurysm**: weakened media, mainly abdominal aorta

EXAM TRAP

The dangerous plaque is often the small one. Most plaques that rupture and cause infarction were **not** critically stenotic beforehand. Stability depends on the cap and the inflammation, not on size. Adrenergic surges (emotional stress, the early morning hours) raise wall stress and can trigger rupture.

LAB SPOTTER

Atheroma in 10 seconds: gross aorta opened longitudinally with raised yellow-white plaques, some ulcerated with red thrombus, worst in the abdominal segment and around ostia. **Microscopy**: thickened intima with a pink collagen cap over a pale core of **empty needle-shaped cholesterol clefts** and foamy macrophages; the media beneath is thin.

PAKISTAN FOCUS

Ischaemic heart disease is a leading cause of death in Pakistan, and South Asians develop coronary disease at a younger age. The major risk factors are common: about one in three adults has hypertension, about one in four has diabetes, and about a fifth of adults use tobacco (31.8% of men; GATS 2014).

Source: Global Adult Tobacco Survey Pakistan 2014; IDF Diabetes Atlas 2021.

25.3 Aneurysms and aortic dissection

DEFINITION

Aneurysm: a localised abnormal dilatation of a blood vessel or the heart. A **true aneurysm** is bounded by all three layers of the (attenuated) wall: atherosclerotic, syphilitic and berry aneurysms, ventricular aneurysm after infarction. A **false aneurysm (pseudoaneurysm)** is a breach in the wall with an extravascular haematoma that communicates with the lumen: contained ventricular rupture, leaking graft anastomosis. Shapes: **saccular** (outpouching) or **fusiform** (circumferential).

CONCEPT CLEAR

Why walls weaken. (1) **Poor-quality connective tissue**: Marfan syndrome (defective fibrillin-1, excess TGF-β activity), Loeys–Dietz syndrome (TGF-β receptor mutations), Ehlers–Danlos (collagen), scurvy. (2) **Too much breakdown**: inflammatory cells release matrix metalloproteinases. (3) **Ischaemia of the media**: atherosclerotic intima starves the inner media; hypertension narrows the vasa vasorum and starves the outer media (**cystic medial degeneration**); syphilis obliterates the vasa vasorum. The two great predisposing conditions are **atherosclerosis** (abdominal aorta) and **hypertension** (ascending aorta).

ANEURYSMS YOU MUST KNOW

Type	Site, cause	Pathology	Consequences
Abdominal aortic (AAA)	Below the renal arteries, above the bifurcation; **atherosclerosis**; men over 50, smokers	Saccular or fusiform, up to 15 cm; severe atheroma, **laminated mural thrombus**, thinned destroyed media. Variants: inflammatory (IgG4-related), mycotic (e.g. *Salmonella*)	**Rupture** (massive retroperitoneal bleeding), branch occlusion, embolism, ureteric compression, vertebral erosion; pulsatile abdominal mass
Syphilitic (luetic)	Ascending aorta and arch; tertiary syphilis	**Obliterative endarteritis of the vasa vasorum** with plasma cells → medial scarring → wrinkled **"tree-bark" intima**	Aortic root dilatation → **aortic regurgitation**; compression of trachea, oesophagus, recurrent laryngeal nerve
Thoracic (non-syphilitic)	Hypertension, Marfan, Loeys–Dietz, bicuspid aortic valve	Cystic medial degeneration	Valve insufficiency, cough, dysphagia, dissection, rupture
Berry (saccular)	Branch points of the circle of Willis (anterior circulation); linked to adult polycystic kidney disease	Wall lacks **media and internal elastic lamina**	Rupture → **subarachnoid haemorrhage** (CNS chapter)

AAA rupture risk rises with diameter: rare below 4 cm, about 1% a year at 4–5 cm, about 11% a year at 5–6 cm and about 25% a year above 6 cm, so aneurysms of 5 cm or more are repaired. Most patients die of coronary disease, the companion of the same atherosclerosis.

Aortic dissection

DEFINITION

Aortic dissection: blood splits the laminar planes of the **media** to form a blood-filled channel within the aortic wall, which may rupture outwards with catastrophic haemorrhage. It is not an aneurysm in the strict sense, although the term "dissecting aneurysm" is still heard.

- **Who:** men aged 40–60 with **hypertension** (more than 90% of cases); younger patients with connective tissue disorders (**Marfan**); also cannulation injury, pregnancy, cocaine. Rare with atherosclerosis or syphilis, because medial scarring blocks propagation.
- **Lesion: cystic medial degeneration**: fragmented elastic tissue and pools of proteoglycan-rich matrix in the media, characteristically without inflammation.
- **Morphology:** a transverse **intimal tear** of 1–5 cm with jagged edges, usually in the **ascending aorta within 10 cm of the valve**; the haematoma runs between the **middle and outer thirds of the media**. It may rupture out (pericardium, pleura, peritoneum) or re-enter the lumen through a second tear, creating a **double-barrelled aorta**.

- **Clinical:** sudden, excruciating **tearing pain** in the anterior chest radiating to the back **between the scapulae** and moving downwards; may mimic MI; pulse and BP differences between arms.
- **Causes of death:** rupture into the **pericardium (tamponade)**, pleura or peritoneum; retrograde extension into the root (**aortic regurgitation**); compression of coronary (MI), carotid (stroke), renal, mesenteric or spinal arteries.

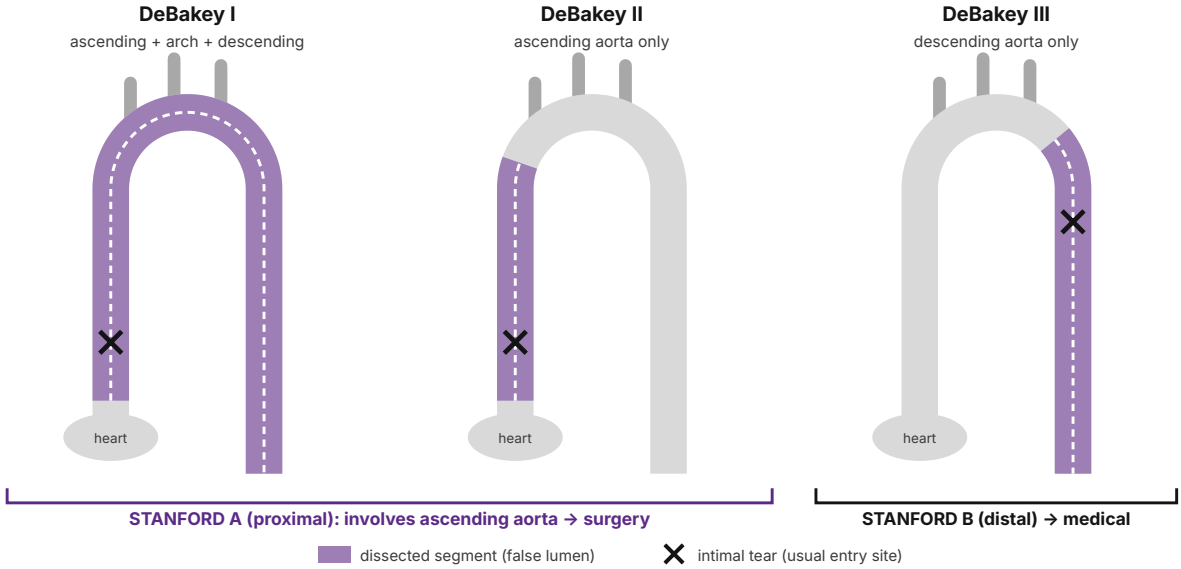


Figure 25.3 · Classification of aortic dissection. Stanford asks one question: is the ascending aorta involved? DeBakey describes the extent.

MEMORY HOOK

"A for Ascending, A for Acute surgery"

Stanford **type A** (DeBakey I and II) involves the ascending aorta, is commoner and deadlier, and needs urgent surgery. **Type B** (DeBakey III) starts beyond the left subclavian artery and is usually managed by lowering blood pressure.

25.4 Vasculitis

Vasculitis is inflammation of vessel walls. Patients have constitutional symptoms (fever, myalgia, arthralgia, malaise) plus the effects of ischaemia in whatever organ the vessels supply. Most vasculitis is **immune-mediated**; some is caused by **direct infection** (*Neisseria*, rickettsiae, *Pseudomonas*, *Aspergillus*, *Mucor*). The disorders are classified by the **size of vessel** they prefer.

- **Immune complexes**: SLE, cryoglobulinaemia (HCV), PAN with hepatitis B, IgA vasculitis, drug hypersensitivity
- **ANCA**: autoantibodies to neutrophil granule enzymes (pauci-immune: no deposits)
- **Anti-endothelial cell antibodies**: Kawasaki disease
- **Autoreactive T cells**: granulomatous GCA, Takayasu

MOLECULAR KEY

ANCA: PR3-ANCA (cytoplasmic, c-ANCA) → **granulomatosis with polyangiitis**; **MPO-ANCA** (perinuclear, p-ANCA) → **microscopic polyangiitis**, EGPA. Infection or drugs prime neutrophils, which move PR3 or MPO to their surface; ANCA binds and triggers degranulation and reactive oxygen species that kill endothelium. Titres roughly follow disease activity.

VASCULITIS BY VESSEL SIZE

Disease	Patient	Key pathology	Clinical clues, marker
Large vessels (aorta and major branches)			
Giant cell (temporal) arteritis	Over 50; commonest vasculitis in the elderly	**Granulomatous inflammation of the media** centred on the **internal elastic lamina**, giant cells, fragmented lamina; segmental (skip lesions)	Temporal headache, **jaw claudication**, **visual loss** (ophthalmic artery: emergency), polymyalgia rheumatica; **ESR high**; steroids

Disease	Patient	Key pathology	Clinical clues, marker
Takayasu arteritis	**Under 50**; young women, often Asian	Granulomatous aortitis of the arch and branches; adventitial fibrosis, intimal thickening → stenosis	**"Pulseless disease"**: weak arm pulses, BP difference between arms, visual and neurological symptoms
Medium vessels (main visceral arteries)			
Polyarteritis nodosa (PAN)	Young adults; about one-third have **chronic hepatitis B**	Segmental **transmural necrotising** arteritis with **fibrinoid necrosis**; **lesions of different ages** side by side; aneurysmal nodules; spares lungs and glomeruli	Renal (hypertension, renal failure), abdominal pain, melaena, **mononeuritis multiplex**; **ANCA negative**
Kawasaki disease	Children, 80% **under 4**	Transmural arteritis, less fibrinoid necrosis; **coronary arteries**	Fever, conjunctivitis, oral erythema, swollen red hands and feet with peeling, rash, cervical nodes; **coronary aneurysms**; IVIG + aspirin
Small vessels (arterioles, capillaries, venules)			
Granulomatosis with polyangiitis (GPA)	Men around 40	**Necrotising granulomas** of upper and lower airway + necrotising vasculitis + focal necrotising (crescentic) GN	Sinusitis, nasal ulcers, **saddle nose**, cavitating lung nodules, haematuria; **PR3-ANCA** (c-ANCA)
Microscopic polyangiitis (MPA)	Adults	Necrotising vasculitis of capillaries and venules; **all lesions of the same age**; **no granulomas**	**Necrotising GN**, pulmonary capillaritis (haemoptysis), palpable purpura; **MPO-ANCA** (p-ANCA)
Eosinophilic GPA (Churg–Strauss)	Adults with **asthma**	Necrotising vasculitis with **eosinophil-rich granulomas**	Asthma, eosinophilia, lung infiltrates; heart involvement is the main killer; MPO-ANCA in under half
IgA vasculitis (Henoch–Schönlein)	Children	Leukocytoclastic vasculitis with **IgA deposits**	**Palpable purpura** on buttocks and legs, arthralgia, abdominal pain, GN (Chapter 16)
Limb arteries with veins and nerves			
Thromboangiitis obliterans (Buerger)	Heavy **smokers**, usually men under 35	Segmental **thrombosing** acute and chronic vasculitis of tibial and radial arteries spreading to adjacent **veins and nerves**; microabscesses in the thrombus	Instep claudication, Raynaud-like cold sensitivity, rest pain, ulcers, **gangrene**; stopping smoking is the only effective treatment

GROSS: GIANT CELL ARTERITIS

- Affected temporal artery **nodular, thickened, tender**; lumen narrowed or thrombosed
- Involvement is **segmental**: biopsy at least 1 cm; a negative biopsy does not exclude it

MICROSCOPY

- **Granulomatous inflammation** of the inner media with T cells and macrophages
- **Multinucleate giant cells** along a **fragmented internal elastic lamina**
- Intimal thickening; healed lesions show medial scarring

EXAM TRAP

PAN vs microscopic polyangiitis. PAN hits **medium** arteries, is **ANCA negative**, is linked to **hepatitis B**, shows lesions of **different ages** and spares glomeruli and lungs. MPA hits **small** vessels, is **MPO-ANCA** positive, shows lesions of the **same age** and causes **glomerulonephritis and lung haemorrhage**. **GCA vs Takayasu:** the same granulomatous lesion; age (over or under 50) separates them.

MEMORY HOOK

Kawasaki: "CRASH and Burn"

Conjunctivitis · **R**ash · **A**denopathy (cervical) · **S**trawberry tongue and red lips · **H**ands and feet red, swollen, peeling · **Burn**: fever for five days or more. Watch the **coronaries**.

25.5 Vascular tumours

Benign vascular tumours form **well-formed channels** lined by a single layer of normal-looking endothelium and filled with blood; malignant ones are **solid, cellular and atypical** with poorly formed vessels. Endothelial markers: **CD31, CD34, ERG, von Willebrand factor**.

HAEMANGIOMA

The **commonest tumour of infancy and childhood**, mostly in the skin of the head and neck; the liver is the commonest internal site.

- **Capillary haemangioma**: red-blue, 1 to several cm; lobules of thin-walled **capillaries** with scant stroma. The juvenile "**strawberry**" type grows fast for months, fades by 1–3 years and regresses by about 7 years in most children.
- **Cavernous haemangioma**: large, **dilated, blood-filled channels**; poorly circumscribed, deep, does not regress; thrombosis and calcification; in the brain it may bleed.

PYOGENIC GRANULOMA AND GLOMUS TUMOUR

- **Pyogenic granuloma** (lobular capillary haemangioma): rapidly growing, **pedunculated red nodule** on skin, gingiva or oral mucosa that **bleeds easily** and ulcerates; often follows trauma. Lobules of proliferating capillaries with oedema and inflammation, like exuberant granulation tissue. **Pregnancy tumour** (granuloma gravidarum) of the gingiva regresses after delivery.
- **Glomus tumour**: benign, **exquisitely painful** tumour of the modified smooth muscle cells of the glomus body; typically **under a fingernail**; small, red-blue; branching vessels surrounded by nests of **uniform, round glomus cells**. Excision cures.

> ⚠ **EXAM TRAP**
>
> **A pyogenic granuloma is neither pyogenic nor a granuloma**: it is a benign capillary proliferation with no pus and no granulomas. **Bacillary angiomatosis** (*Bartonella* in immunosuppressed patients) mimics Kaposi sarcoma but has neutrophils and bacilli and is cured by antibiotics.

Kaposi sarcoma

An intermediate-grade (locally aggressive, occasionally metastasising) vascular neoplasm caused by **human herpesvirus 8 (HHV-8, KSHV)**. The virus infects endothelial cells; latent viral proteins drive proliferation and block death (**v-cyclin**, **LANA** inhibiting p53, **vFLIP** activating NF-κB), while **immune deficiency** lets infected cells escape and inflammatory cytokines feed the spindle cells.

Form	Who	Behaviour
Classic	Older men of Mediterranean, Middle Eastern or Eastern European (Ashkenazi) descent; HIV negative	Indolent red-purple plaques and nodules on the **lower legs**
Endemic African	HIV-negative young adults and children	Children: lymphadenopathic, aggressive
Transplant-associated	Organ recipients on immunosuppression	Nodes, mucosa, viscera; regresses when immunosuppression is reduced
AIDS-associated (epidemic)	HIV infection; an AIDS-defining illness	Skin, **oral mucosa**, lymph nodes, gut, lungs; less common with antiretroviral therapy

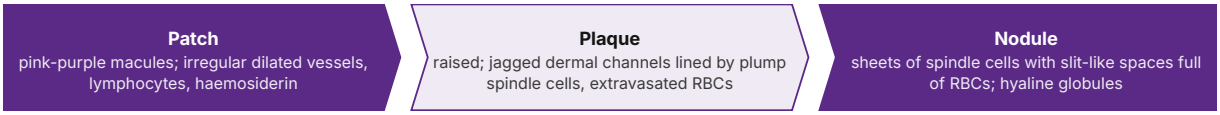


Figure 25.4 · Stages of Kaposi sarcoma. From a granulation-tissue look-alike to a frank spindle cell tumour.

Angiosarcoma

A malignant endothelial neoplasm of older adults: skin (scalp and face), soft tissue, breast and liver. Associations: **hepatic angiosarcoma** with **arsenic**, **Thorotrast** and **polyvinyl chloride**; **lymphoedema** after radical mastectomy (Stewart–Treves syndrome); radiation. **Gross:** small red nodules that grow into large, fleshy red-tan to grey-white masses with haemorrhage and necrosis. **Microscopy:** from plump atypical endothelial cells lining irregular channels to solid sheets of undifferentiated spindle cells; CD31 and ERG positive. Aggressive, with about 30% five-year survival.

EXAM ZONE Past-Paper Pattern SEQs — Solved

PROF PATTERN **a)** Enumerate the risk factors for atherosclerosis. **b)** Describe its pathogenesis. **2 + 3 marks**

MODEL ANSWER PLAN

a) Non-modifiable: age, male sex, family history/genetics (familial hypercholesterolaemia). Modifiable: hyperlipidaemia, hypertension, smoking, diabetes. Additional: CRP/inflammation, homocysteine, metabolic syndrome, lipoprotein(a), obesity, inactivity. **b)** Response to injury: endothelial dysfunction → LDL entry and oxidation → monocyte adhesion, macrophages, foam cells (fatty streak) → platelet adhesion → PDGF/FGF → SMC migration and proliferation → collagen cap over lipid core (atheroma).

Score booster: draw Figure 25.1 and add "risk factors multiply: three together raise MI risk about sevenfold".

PROF PATTERN **a)** Describe the morphology of an atheromatous plaque. **b)** What are its complications? Differentiate a stable from a vulnerable plaque. **2 + 3 marks**

MODEL ANSWER PLAN

a) Gross: raised white-yellow eccentric plaques 0.3–1.5 cm; sites in order (abdominal aorta, coronaries, popliteal, carotid, circle of Willis). Micro: fibrous cap, shoulder, necrotic core with cholesterol clefts, neovascularisation, thin media. **b)** Rupture/erosion with thrombosis, haemorrhage into plaque, atheroembolism, aneurysm, calcification; critical stenosis. Stable: thick cap, small core, few macrophages; vulnerable: thin cap, large core, many macrophages.

MODULE PATTERN A 55-year-old hypertensive man has sudden tearing chest pain radiating to his back. Radial pulses are unequal and the mediastinum is wide. **a)** Diagnosis? **b)** Describe the pathogenesis and morphology. **c)** Classify it and list the causes of death. **1 + 2 + 2 marks**

MODEL ANSWER PLAN

a) Aortic dissection. **b)** Hypertension (or Marfan) → cystic medial degeneration → intimal tear in ascending aorta within 10 cm of valve → haematoma between middle and outer thirds of media → outward rupture or re-entry (double-barrelled aorta). **c)** Stanford A (ascending; DeBakey I, II) and B (distal; DeBakey III). Death: haemopericardium with tamponade, rupture into pleura/peritoneum, aortic regurgitation, occlusion of coronary, carotid or visceral branches.

CLASSIC **a)** Classify vasculitis according to vessel size. **b)** Write a note on polyarteritis nodosa. **2 + 3 marks**

MODEL ANSWER PLAN

a) Large: GCA, Takayasu. Medium: PAN, Kawasaki. Small: ANCA-associated (GPA, MPA, EGPA), immune complex (IgA vasculitis, cryoglobulinaemic, anti-GBM). Buerger: limb arteries. **b)** Young adults; one-third HBV; medium visceral arteries, sparing lungs; segmental transmural fibrinoid necrosis; lesions of different ages; aneurysmal nodules, infarcts; hypertension, abdominal pain, mononeuritis multiplex; ANCA negative; responds to immunosuppression.

PROF PATTERN **a)** Differentiate hyaline from hyperplastic arteriolosclerosis. **b)** Write a note on Kaposi sarcoma. **2 + 3 marks**

MODEL ANSWER PLAN

a) Setting (benign hypertension, diabetes vs malignant hypertension); glassy pink wall vs onion-skin ± fibrinoid necrosis; benign nephrosclerosis vs flea-bitten kidney. **b)** HHV-8; four forms (classic, endemic, transplant, AIDS); patch → plaque → nodule; spindle cells with slit-like vascular spaces, extravasated RBCs, haemosiderin, hyaline globules.

BCQs One-Best-Answer Practice

1 The earliest visible lesion of atherosclerosis, found even in children, is the:

A. Fibrous plaque
B. Complicated plaque
C. Calcified plaque
D. Fatty streak
E. Aneurysm

2 Foam cells in an early atheroma are chiefly derived from:

A. Neutrophils
B. Monocyte-derived macrophages
C. Plasma cells
D. Mast cells
E. Endothelial cells

3 Which mediator, released by platelets and macrophages, recruits smooth muscle cells into the intima of a developing plaque?

A. PDGF
B. Erythropoietin
C. IL-2
D. Histamine
E. Bradykinin

4 Which feature makes a coronary plaque most likely to rupture?

A. Thick collagenous cap
B. Small lipid core
C. Abundant smooth muscle cells
D. Dense calcification
E. Thin cap with many macrophages

5 The arterial segment most severely affected by atherosclerosis is usually the:

A. Thoracic aorta
B. Renal artery
C. Lower abdominal aorta
D. Superior mesenteric artery
E. Subclavian artery

6 A kidney biopsy shows onion-skin arteriolar thickening with fibrinoid necrosis. The most likely setting is:

A. Benign hypertension
B. Malignant hypertension
C. Diabetes mellitus
D. Mönckeberg sclerosis
E. Amyloidosis

7 A 58-year-old hypertensive man dies of sudden tearing chest pain. The aortic media shows elastic fragmentation with mucoid pools and no inflammation. This is:

A. Granulomatous aortitis
B. Obliterative endarteritis of vasa vasorum
C. Atheroma with mural thrombus
D. Cystic medial degeneration
E. Fibrinoid necrosis

8 A dissection confined to the ascending aorta (DeBakey II) is classified as:

A. Stanford type A
B. Stanford type B
C. DeBakey III
D. False aneurysm
E. Berry aneurysm

9 A 72-year-old woman has headache, jaw pain on chewing, blurred vision and ESR 98 mm/h. Temporal artery biopsy would show:

A. Leukocytoclastic vasculitis
B. Necrotising airway granulomas
C. Fibrinoid necrosis of renal arteries
D. Eosinophil-rich granulomas
E. Medial granulomas with giant cells and a fragmented elastic lamina

10 A 40-year-old man has chronic sinusitis, nasal ulcers, cavitating lung nodules and haematuria. The antibody most likely present is:

A. Anti-GBM
B. MPO-ANCA (p-ANCA)
C. PR3-ANCA (c-ANCA)
D. Anti-dsDNA
E. Anti-centromere

11 Polyarteritis nodosa is classically associated with:

A. Hepatitis C cryoglobulinaemia
B. Bronchial asthma
C. Fever in a toddler
D. Chronic hepatitis B
E. Heavy smoking

12 An HIV-positive man has purple plaques on his legs. Biopsy shows spindle cells forming slit-like spaces filled with red cells. The causative agent is:

A. HHV-8
B. EBV
C. HPV
D. Bartonella
E. CMV

ANSWER KEY WITH ONE-LINE REASONS

1. **D** Flat yellow streaks of foam cells; not raised.
2. **B** Macrophages take up oxidised LDL by scavenger receptors.
3. **A** PDGF (with FGF, TGF-α) drives SMC migration and proliferation.
4. **E** Macrophage metalloproteinases thin the cap.
5. **C** Then coronaries, popliteal, internal carotid, circle of Willis.
6. **B** Hyperplastic arteriolosclerosis of malignant hypertension.
7. **D** The substrate of aortic dissection.
8. **A** Any involvement of the ascending aorta = type A.
9. **E** Giant cell arteritis.
10. **C** Granulomatosis with polyangiitis.
11. **D** About one-third of PAN patients carry HBV.

12. **A** Kaposi sarcoma herpesvirus (KSHV).

VIVA Rapid-Fire Viva Voce

Q Define atherosclerosis.	Intimal fibrofatty plaques that narrow, weaken and may rupture.
Q Four major modifiable risk factors?	Hyperlipidaemia, hypertension, smoking, diabetes.
Q What is a foam cell?	A macrophage (or SMC) full of oxidised LDL.
Q What makes a plaque vulnerable?	Thin cap, large lipid core, many macrophages.
Q What is Mönckeberg sclerosis?	Medial calcification of muscular arteries; lumen spared.
Q What is a flea-bitten kidney?	Petechial cortical bleeds of malignant hypertension.
Q What is a false aneurysm?	Wall breach with a haematoma connected to the lumen.
Q What is a tree-bark aorta?	Wrinkled intima over medial scars: syphilitic aortitis.
Q What is a double-barrelled aorta?	Dissection re-entering the lumen through a distal tear.
Q Main complication of Kawasaki disease?	Coronary artery aneurysm (thrombosis, MI).
Q What is Buerger disease?	Thrombosing vasculitis of limb vessels in young smokers.
Q What is a glomus tumour?	Painful benign glomus-cell tumour under a nail.

OSPE Practice Stations

STATION 1 · GROSS SPECIMEN — 1 minute

TASK

An opened aorta shows many raised yellow-white plaques, some ulcerated with adherent red-brown thrombus, most severe below the renal arteries and around branch openings. (a) Diagnosis. (b) Four modifiable risk factors. (c) Two complications.

ANSWER

(a) Complicated atherosclerosis of the aorta. (b) Hyperlipidaemia, hypertension, smoking, diabetes. (c) Abdominal aortic aneurysm; thrombosis or atheroembolism (renal, mesenteric, leg ischaemia).

STATION 2 · SLIDE — 2 minutes

TASK

Section of coronary artery: the intima is markedly thickened by a pink collagenous layer covering a pale area of needle-shaped empty clefts, foamy cells and debris; the media beneath is thin. (a) Identify. (b) Name the clefts. (c) Which feature would make it prone to rupture?

ANSWER

(a) Atheromatous (fibrofatty) plaque. (b) Cholesterol clefts. (c) A thin fibrous cap with many macrophages over a large lipid core.

STATION 3 · SLIDE — 2 minutes

TASK

Temporal artery biopsy from a 70-year-old woman: the media is infiltrated by lymphocytes and macrophages with multinucleate giant cells along a broken internal elastic lamina; the lumen is narrowed. (a) Diagnosis. (b) Most feared complication. (c) One blood test.

ANSWER

(a) Giant cell (temporal) arteritis. (b) Sudden blindness (ophthalmic artery). (c) ESR (markedly raised); CRP.

STATION 4 · SLIDE 2 minutes

TASK

Skin nodule from an HIV-positive man: sheets of plump spindle cells forming slit-like spaces containing red cells, with haemosiderin-laden macrophages and pink hyaline globules. (a) Diagnosis and stage. (b) Causative virus. (c) Name two other clinical forms.

ANSWER

(a) Kaposi sarcoma, nodular stage. (b) HHV-8 (KSHV). (c) Classic, endemic African, transplant-associated.

RAPID REVISION The Night-Before Page

15 ONE-LINERS

01 Hyaline arteriolosclerosis: benign hypertension, diabetes; glassy pink walls.

02 Hyperplastic: malignant hypertension; onion-skin, fibrinoid necrosis.

03 Mönckeberg: medial calcification, lumen spared, little significance.

04 Major risks: LDL, hypertension, smoking, diabetes; they multiply.

05 Response to injury: endothelium → LDL → macrophages → PDGF → SMC cap.

06 Fatty streak: earliest lesion, foam cells, seen in children.

07 Plaque: cap, shoulder, necrotic core with cholesterol clefts.

08 Worst site: lower abdominal aorta, then coronaries.

09 Vulnerable plaque: thin cap, big core, many macrophages; may be small.

10 AAA: below renals, atherosclerosis; repair at 5 cm or more.

11 Syphilitic aneurysm: ascending aorta, tree bark, aortic regurgitation.

12 Dissection: hypertension, cystic medial degeneration; Stanford A = ascending.

13 GCA over 50, Takayasu under 50; both granulomatous.

14 PAN: HBV, ANCA negative, mixed-age lesions; GPA PR3; MPA MPO.

15 Kaposi: HHV-8; patch → plaque → nodule; four clinical forms.

MEMORY HOOKS

- **Fuel, fire, scar**: LDL, macrophages, smooth muscle cap
- **A for Ascending, A for Acute surgery**: Stanford type A
- **CRASH and Burn**: Kawasaki disease
- **Size sorts vasculitis**: large GCA/Takayasu; medium PAN/Kawasaki; small ANCA and IgA

TEST YOURSELF: FILL THE GAPS

1. Foam cells take up oxidised LDL via _____ receptors.
2. Critical stenosis is about _____% narrowing.
3. Syphilis obliterates the _____.
4. Dissection tracks between the middle and _____ thirds of the media.
5. c-ANCA targets _____.
6. Stewart–Treves: angiosarcoma in _____.

Answers: scavenger · 70 · vasa vasorum · outer · PR3 · post-mastectomy lymphoedema

DRAW FROM MEMORY

An atheromatous plaque in cross-section: cap, shoulder, necrotic core, thin media.

DRAW FROM MEMORY

The aorta with DeBakey I, II, III and the Stanford A/B line (Figure 25.3).

SPACED REVISION ☐ Day 1 ☐ Day 3 ☐ Day 7 ☐ Day 14 ☐ Day 30

26

Heart: Ischaemic, Hypertensive & Valvular Disease, Cardiomyopathies

A coronary plaque that cracks at breakfast can kill by lunch, and the same heart, if it survives, writes a precise diary of the next eight weeks in its muscle. Learn to read that diary, and the valves and myocardium will tell you the rest of their story.

BY THE END OF THIS CHAPTER YOU CAN

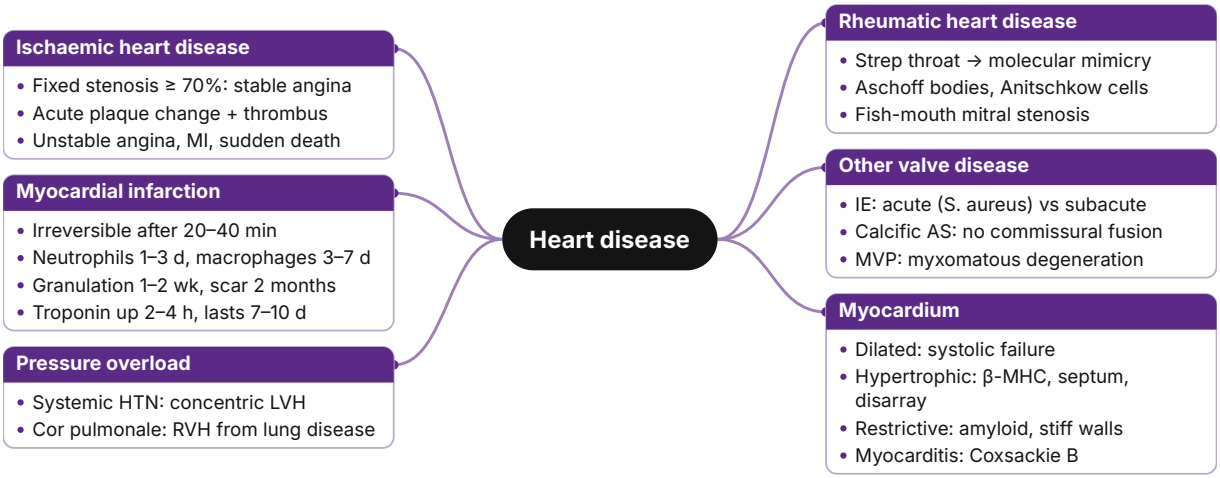
1. Explain how stable plaques, acute plaque change and thrombosis produce the spectrum of ischaemic heart disease
2. Describe the evolution of gross and microscopic changes in myocardial infarction and date an infarct
3. Interpret cardiac troponin and CK-MB curves and list the complications of MI by time
4. Describe hypertensive heart disease and cor pulmonale
5. Describe the pathogenesis, Jones criteria and morphology of rheumatic fever and chronic rheumatic heart disease
6. Differentiate acute from subacute infective endocarditis and compare the four types of vegetation
7. Compare dilated, hypertrophic and restrictive cardiomyopathies

EXAM RADAR

Theory SEQs	★★★
BCQs / MCQs	★★★
Viva voce	★★★
OSPE / slides	★★☆

MI morphology with time, complications of MI and rheumatic heart disease are perennial long questions; vegetations, markers and cardiomyopathy genes are favourite BCQs.

ROBBINS LINK Robbins Basic Pathology — *The Heart*: ischaemic heart disease, hypertensive heart disease, valvular heart disease, cardiomyopathies and myocarditis.



Chapter at a glance. Six areas carry the whole chapter.

26.1 Ischaemic heart disease: the big picture

DEFINITION

Ischaemic heart disease (IHD) is a group of related syndromes caused by **myocardial ischaemia**: an imbalance between the heart's demand for oxygenated blood and its supply. In more than 90% of cases the cause is reduced coronary flow from **atherosclerosis**; IHD is therefore often called coronary artery disease.

Ischaemia cuts **oxygen** and also **nutrient delivery and waste removal**, so it is worse than hypoxia alone. Other causes: emboli, vasospasm, vasculitis; raised demand (tachycardia, LVH); poor oxygen carriage (anaemia, CO, shock).

FOUR MECHANISMS, ONE DISEASE

- **Fixed stenosis.** Symptoms on exertion when ≥ **70%** of the lumen is narrowed ("critical stenosis"); at rest when ≥ **90%**.
- **Acute plaque change.** Rupture or fissuring, erosion or ulceration, or haemorrhage into the plaque exposes thrombogenic material.
- **Thrombosis** on the disrupted plaque: partial and transient, or total and persistent.
- **Vasoconstriction** (platelet thromboxane, endothelial dysfunction) narrows the lumen further.

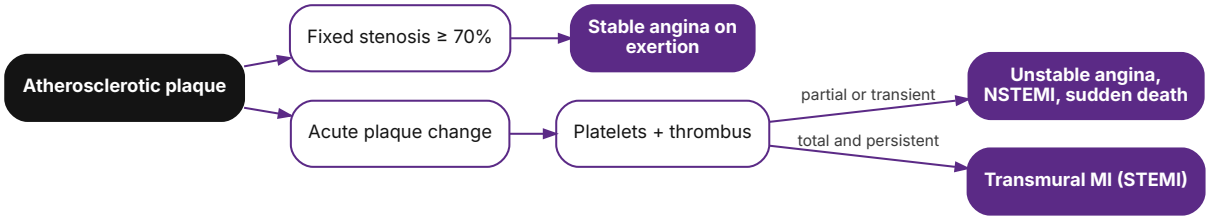


Figure 26.1 · From plaque to syndrome. What happens on the plaque surface decides the clinical picture.

> **CONCEPT CLEAR**
>
> **The dangerous plaque is not the biggest one.** Most plaques that rupture and cause MI were only mildly to moderately stenotic beforehand. What makes a plaque **vulnerable** is a large soft lipid core, a **thin fibrous cap** and many macrophages digesting collagen with metalloproteinases. This is why a man with a "normal" exercise test can still infarct (see Chapter 25 for plaque structure).

THE IHD SYNDROMES

- **Stable angina:** fixed stenosis; exertional pain relieved by rest or nitroglycerin
- **Prinzmetal (variant) angina:** coronary **vasospasm**; pain at rest, ST elevation
- **Unstable (crescendo) angina:** plaque disruption + non-occlusive thrombus; heralds MI
- **Myocardial infarction:** occlusive thrombus → myocyte death (26.2)
- **Chronic IHD:** repeated infarcts → progressive heart failure
- **Sudden cardiac death:** lethal arrhythmia within about 1 hour of symptoms

Acute coronary syndrome = unstable angina, acute MI and sudden cardiac death: all three begin with acute plaque change. IHD is a leading cause of death in Pakistan, where South Asians develop it younger.

26.2 Myocardial infarction

> **DEFINITION**
>
> **Myocardial infarction (MI)** is death (necrosis) of cardiac muscle due to prolonged severe ischaemia, usually from an occlusive thrombus on a disrupted coronary atherosclerotic plaque. Risk rises with age, male sex (women catch up after menopause), hypertension, smoking, diabetes and dyslipidaemia.

Pathogenesis: a race against time

Within **seconds** of occlusion aerobic metabolism stops, anaerobic glycolysis takes over and ATP falls; **contractility is lost within 2 minutes** (so even brief ischaemia can cause heart failure). Lactic acid accumulates. Severe ischaemia for **20–40 minutes** causes **irreversible injury**; microvascular injury follows after about an hour, and the infarct reaches its **full extent in 3–6 hours**.

> **CONCEPT CLEAR**
>
> **Why the subendocardium dies first.** It is the last zone reached by blood from the epicardial coronary arteries and it sits under the highest intramural pressure. Necrosis therefore starts there and spreads outwards as a **wavefront** towards the epicardium over 3–6 hours. This is the scientific basis of "**time is muscle**": reperfusion in the first hours salvages the outer myocardium.

PATTERNS OF INFARCTION

- **Transmural** (STEMI): full or nearly full wall thickness in one coronary territory; occlusive thrombus.
- **Subendocardial** (NSTEMI): inner third to half of the wall; transient or partial thrombus, or shock (circumferential necrosis).
- **Microscopic:** small-vessel occlusion (vasculitis, emboli).

WHICH ARTERY, WHICH WALL

Artery (frequency)	Infarct site
LAD (40–50%)	Anterior LV wall near apex, anterior two-thirds of septum, apex
RCA (30–40%)	Inferior/posterior LV wall, posterior third of septum, posterior RV wall
Left circumflex (15–20%)	Lateral LV wall except the apex

Morphology: dating an infarct

Examiners love this table because the sequence is pure acute inflammation and repair (General Pathology) happening in one organ. **Neutrophils come first, macrophages clear, granulation tissue fills, collagen scars.** An MI less than 12 hours old is invisible to the naked eye, but slices immersed in **triphenyltetrazolium chloride (TTC)** reveal it: viable muscle with intact dehydrogenase activity turns **brick-red**, the infarct stays **pale**.



Figure 26.2 · The infarct diary. Learn the six headline events; the table adds the detail.

EVOLUTION OF MORPHOLOGICAL CHANGES IN MYOCARDIAL INFARCTION

Time	Gross	Light microscopy
0–½ h	None	None (EM: glycogen loss, mitochondrial swelling: reversible)
½–4 h	None	Usually none; variable **waviness of fibres** at the border
4–12 h	Occasional dark mottling	Early **coagulation necrosis**; oedema; haemorrhage
12–24 h	Dark mottling	Ongoing necrosis; nuclear pyknosis; **hypereosinophilic** myocytes; marginal **contraction band** necrosis; early neutrophils
1–3 days	Mottling with **yellow-tan** infarct centre	Coagulation necrosis with **loss of nuclei and striations**; **brisk neutrophilic infiltrate**
3–7 days	**Hyperaemic border**; central yellow-tan softening	Disintegration of dead myofibres, dying neutrophils; early **macrophages** at the border
7–10 days	Maximally yellow-tan and soft; depressed red-tan margins	Well-developed phagocytosis; **granulation tissue** at the margins
10–14 days	Red-grey depressed borders	Well-established granulation tissue with new vessels and collagen
2–8 weeks	Grey-white scar, progressing from border to core	Increased collagen, decreasing cellularity
> 2 months	Scarring complete	Dense collagenous scar

CONCEPT CLEAR

Reperfusion changes the picture. Opening the artery (thrombolysis, PCI) saves muscle but the injured zone becomes **haemorrhagic** (damaged vessels leak) and shows **contraction band necrosis**: calcium floods into dying cells and the myofibrils hypercontract into dense eosinophilic bands. Oxygen free radicals add **reperfusion injury**. Salvaged muscle may stay non-contractile for days: **stunned myocardium**.

Clinical features and laboratory diagnosis

Severe crushing substernal pain radiating to the left arm, jaw or epigastrium, lasting more than 30 minutes and not relieved by nitroglycerin; sweating, nausea, dyspnoea. About 10–15% are "**silent**", especially in diabetics (autonomic neuropathy) and the elderly. Diagnosis rests on symptoms, **ECG** (ST elevation, Q waves) and **cardiac biomarkers** leaked from dead myocytes.

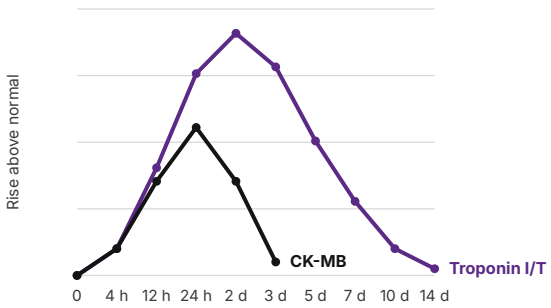


Figure 26.3 · Cardiac markers after MI. Schematic rise above normal; troponin stays up far longer than CK-MB.

LAB VALUES

Troponin I and T cardiac-specific; rise **2–4 h**, peak about **48 h**, stay raised **7–10 days**. The test of choice.

CK-MB

rises 2–4 h, peaks 24–48 h, back to normal by about **72 h**: useful to detect **re-infarction**.

Single negative value

does not exclude MI; repeat over hours (rising pattern).

Variant figures

clinical tables quote rise 3–12 h, peak ~24 h, cTnT up to 14 days. Troponin also rises in myocarditis, PE, sepsis, renal failure.

Complications of MI: think by the calendar

Each complication appears when the infarct is at the stage that allows it. Remember the diary above and the table writes itself.

COMPLICATIONS OF MYOCARDIAL INFARCTION BY TIME

When	Complication	Why then
Minutes–first day	**Arrhythmias** (VF, heart block) → **sudden death**	Ischaemic, electrically unstable myocardium
Hours–days	**Contractile dysfunction**: LV failure, pulmonary oedema; **cardiogenic shock** if > 40% of LV lost; RV infarction with inferior MI	Loss of contracting muscle
2–3 days	**Fibrinous pericarditis** (transmural MI): friction rub, chest pain	Inflammation reaches the epicardium
3–7 days	**Myocardial rupture**: free wall → haemopericardium and **tamponade**; septum → VSD (left-to-right shunt); papillary muscle → acute severe mitral regurgitation	**Softest phase**: macrophages have removed dead muscle, scar not yet laid down
Days–weeks	**Mural thrombus** → systemic emboli (brain, kidney, spleen, limbs)	Akinetic endocardium + stasis
Weeks	**Dressler syndrome**: autoimmune pericarditis	Antibodies to released myocardial antigens
Weeks–months	**Ventricular aneurysm** (large transmural anteroseptal MI); progressive **chronic heart failure**; papillary muscle dysfunction	Thin non-contractile scar bulges outwards

MEMORY HOOK

DARTH VADER

Death · **A**rrhythmia · **R**upture · **T**amponade · **H**eart failure · **V**alve (papillary muscle) disease · **A**neurysm · **D**ressler · **E**mbolism · **R**ecurrence (re-infarction)

EXAM TRAP

Who ruptures? Age > 60, anterior or lateral wall, women, **first MI** (no old scar to buttress the wall) and **no LVH**. A ventricular aneurysm, unlike rupture, is **late** and rarely ruptures because its wall is tough scar.

26.3 Hypertensive heart disease and cor pulmonale

Both are **pressure-overload** diseases. A ventricle pumping against high resistance adds sarcomeres **in parallel**: myocytes get thicker, the wall thickens and the cavity stays normal or small. This is **concentric hypertrophy**, in contrast to the **eccentric hypertrophy** (wall thickening plus dilatation) of volume overload such as regurgitant valves.

Systemic (left-sided) hypertensive heart disease

Minimal criteria: (1) left ventricular hypertrophy, usually concentric, without other cardiovascular disease to explain it, and (2) a history or pathological evidence of hypertension.

GROSS

- LV wall **> 2.0 cm** thick; heart may exceed **500 g**
- Cavity normal or small (concentric); left atrium enlarged

MICROSCOPY

- Broad myocytes with **enlarged, hyperchromatic, rectangular ("boxcar") nuclei**
- Interstitial fibrosis; hyaline arteriolosclerosis

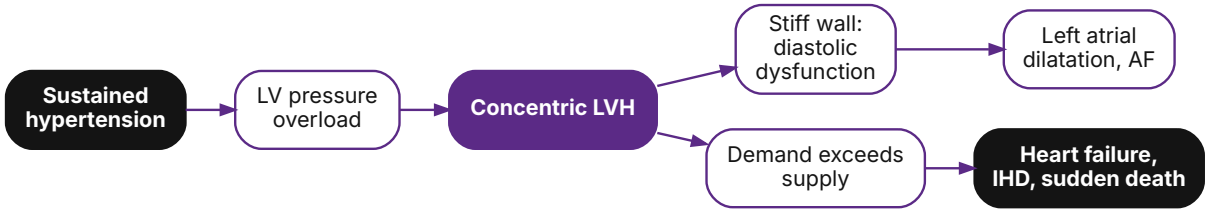


Figure 26.4 · How hypertension damages the heart.

Cor pulmonale (pulmonary hypertensive heart disease)

DEFINITION

Cor pulmonale is right ventricular hypertrophy and/or dilatation caused by **pulmonary hypertension** arising from disorders of the lung parenchyma or pulmonary vasculature. RV changes due to left-sided heart disease or congenital shunts are excluded by definition.

- **Acute:** massive **pulmonary embolism** → sudden RV **dilatation without hypertrophy**; may cause death.
- **Chronic: COPD** (commonest), interstitial fibrosis, recurrent small emboli, primary pulmonary hypertension, sleep apnoea, kyphoscoliosis.
- **Chronic morphology:** RV wall **≥ 1.0 cm** (normal 0.3–0.5 cm), may approach LV thickness.
- Right heart failure: raised JVP, nutmeg liver, ankle oedema.

26.4 Rheumatic fever and rheumatic heart disease

DEFINITION

Rheumatic fever (RF) is an acute, immunologically mediated, multisystem inflammatory disease that follows **group A β-haemolytic streptococcal pharyngitis** after an interval of a few weeks. **Chronic rheumatic heart disease (RHD)** is its main sequel: deforming fibrotic valvular disease, especially **mitral stenosis**.

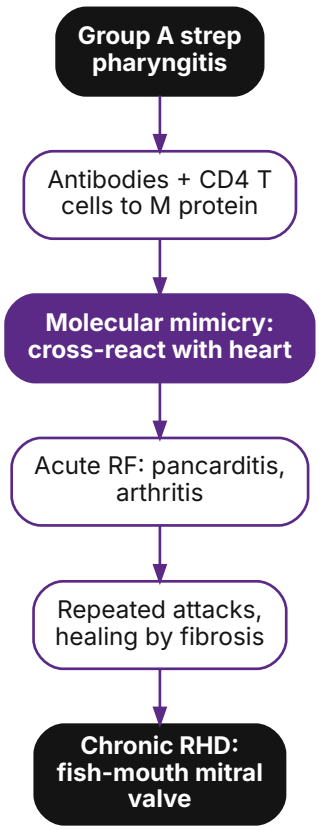


Figure 26.5 · Pathogenesis of rheumatic fever. The bacteria are gone; the immune response does the damage.

CONCEPT CLEAR

Why the heart? Antibodies and CD4+ T cells against streptococcal **M protein** cross-react with cardiac self-antigens (**molecular mimicry**); T-cell cytokines activate macrophages, hence the Aschoff body. Only about **3%** of untreated patients develop RF (genetic susceptibility). Skin infection does not cause RF (it can cause PSGN, Chapter 16).

MEMORY HOOK

J♥NES

Major criteria: **J**oints (migratory polyarthritis of large joints) · ♥ Carditis · **N**odules (subcutaneous) · **E**rythema marginatum · **S**ydenham chorea

LAB VALUES

Jones criteria: evidence of recent group A strep infection (raised ASO titre, positive throat culture or rapid antigen) **plus two major**, or **one major and two minor** criteria. **Minor:** fever, arthralgia, raised ESR/CRP, prolonged PR interval on ECG.

ACUTE RHEUMATIC CARDITIS

- **Pancarditis**: all three layers
- **Pericarditis**: fibrinous or serofibrinous ("bread-and-butter"); resolves
- **Myocarditis**: scattered Aschoff bodies; may cause dilatation and failure (the cause of death in the acute stage)
- **Valvulitis**: fibrinoid necrosis along lines of closure → tiny **1–2 mm warty vegetations (verrucae)**; mitral and aortic
- **MacCallum plaques**: subendocardial thickenings in the left atrium

ASCHOFF BODY (PATHOGNOMONIC)

- Focus of **T lymphocytes**, occasional plasma cells and plump activated macrophages
- **Anitschkow cells**: macrophages with abundant cytoplasm and a central nucleus whose chromatin forms a **slender wavy ribbon** ("**caterpillar cells**"); some fuse into multinucleate **Aschoff giant cells**
- Early lesions: central fibrinoid necrosis
- Later replaced by fibrous scar

Chronic RHD: mitral valve involved in almost all cases (mitral alone 65–70%, mitral + aortic about 25%). Leaflets **thickened and fibrotic**, **commissures fused**, chordae tendineae **short, thick and fused** → a rigid funnel with a **"fish-mouth" or "buttonhole" orifice**; calcification and neovascularisation of leaflets. Aschoff bodies are rarely found now. The **left atrium dilates** and may hold a mural thrombus; lungs show chronic venous congestion; the RV hypertrophies.

Clinical: acute RF peaks at **5–15 years**: fever and migratory polyarthritis (one large joint after another), carditis with friction rub, murmurs, arrhythmia. Arthritis heals without deformity. Chronic RHD appears **decades later** with mitral stenosis and its consequences: **atrial fibrillation**, left atrial thrombus and **embolic stroke**, pulmonary hypertension, right heart failure and **infective endocarditis** on the scarred valve.

PAKISTAN FOCUS

Rheumatic fever and RHD **remain common in Pakistan**, unlike high-income countries where they have almost vanished. The classic local presentation is a **young woman with mitral stenosis**, often first unmasked in **pregnancy**, when the rise in blood volume and heart rate tips her into pulmonary oedema. Prompt penicillin for streptococcal sore throat and secondary prophylaxis prevent most of this disease.

26.5 Infective endocarditis and other valve disease

DEFINITION

Infective endocarditis (IE) is microbial infection of the heart valves or mural endocardium that leads to formation of **vegetations** of thrombotic debris and organisms, often with destruction of the underlying cardiac tissue.

ACUTE VS SUBACUTE INFECTIVE ENDOCARDITIS

Feature	Acute IE	Subacute IE
Organism	Highly virulent: **Staphylococcus aureus**	Low virulence: **viridans streptococci** (commonest overall on damaged valves)
Valve	Often a **previously normal** valve	Previously **damaged** valve (RHD, congenital, MVP)
Onset and course	Rapid, destructive; days to weeks; high mortality	Insidious, weeks to months; most recover with antibiotics
Vegetations	Large, bulky, friable; destroy valve; **ring abscess**	Smaller, less destructive; granulation tissue at the base

ORGANISMS AND PREDISPOSITIONS

- **Viridans streptococci** 50–60%: dental work, damaged valves
- **S. aureus** 10–20%: the major organism in **IV drug users**, in whom the **tricuspid** valve is typical
- **Coagulase-negative staphylococci** (S. epidermidis): **prosthetic valves**
- Enterococci, **HACEK** group (oral commensals); about 10% culture-negative
- Risk: RHD, MVP, calcific or bicuspid valves, prosthetic valves, catheters, immunosuppression, diabetes

MORPHOLOGY

Friable, bulky, potentially destructive vegetations, single or multiple, mostly on **aortic and mitral** valves. They may perforate cusps, rupture chordae or erode into the myocardium as a **ring abscess**. **Micro:** fibrin, neutrophils and **colonies of bacteria** over a necrotic valve.

MEMORY HOOK

FROM JANE

Fever (most consistent sign) · **R**oth spots (retinal haemorrhages) · **O**sler nodes (painful, finger pulp) · **M**urmur (new or changing) · **J**aneway lesions (painless, palms/soles) · **A**naemia · **N**ail-bed splinter haemorrhages · **E**mboli

LAB VALUES

Duke criteria. Major: typical organism in separate blood cultures; echo evidence (vegetation, abscess, new valve dehiscence or regurgitation). **Minor:** predisposition (heart lesion, IV drugs); fever; vascular and immunological phenomena; microbiology short of major. **Definite IE:** 2 major, or 1 major + 3 minor, or 5 minor.

Complications: valve regurgitation, ring abscess, heart failure; **septic emboli** (infarcts and abscesses in brain, spleen, kidney) and **mycotic aneurysms**; **immune-complex glomerulonephritis**.

FOUR KINDS OF VEGETATION: A FAVOURITE COMPARISON

Type	Size and site	Nature	Setting
Rheumatic (verrucae)	Small (1–2 mm), warty, in a row along **lines of closure**	Sterile, fibrinoid; not destructive	Acute rheumatic fever
Infective endocarditis	**Large, irregular, friable**; may extend to chordae	Infected; **destructive**; septic emboli	S. aureus, viridans strep
NBTE (non-bacterial thrombotic)	Small, bland, along lines of closure	Sterile thrombi; can embolise	Hypercoagulable states: mucin-secreting adenocarcinoma, DIC
Libman–Sacks	Small–medium, on **both surfaces** of leaflets	Sterile, with fibrinoid necrosis	SLE, antiphospholipid syndrome

Calcific aortic stenosis

The commonest valve abnormality: age-related "wear and tear" presenting in the **70s–80s**, or in the **50s–60s** on a **congenital bicuspid valve**. Valve cells turn osteoblast-like and calcify the cusps.

- **Gross: heaped-up calcified masses** on the **outflow surfaces** of the cusps, bulging into the sinuses of Valsalva; **no commissural fusion** and a normal mitral valve (rheumatic AS: fused commissures + mitral disease)
- Consequence: concentric LVH → **angina, syncope, heart failure**; prognosis poor once symptoms start unless the valve is replaced

Mitral valve prolapse (myxomatous mitral valve)

Leaflets are **floppy and balloon back into the left atrium** in systole. A few per cent of adults, more women; also in **Marfan syndrome**.

- **Gross:** enlarged, redundant, **hooded** leaflets; elongated, thinned chordae
- **Micro:** thinned fibrosa; **myxomatous degeneration**: spongiosa expanded by mucoid (proteoglycan) material
- Usually silent (**mid-systolic click**); few develop IE, mitral regurgitation, emboli or arrhythmias

26.6 Cardiomyopathies and myocarditis

DEFINITION

Cardiomyopathy is heart disease resulting from a **primary abnormality of the myocardium**, not explained by coronary, valvular, hypertensive or congenital disease. Many are genetic. They are grouped by function into three patterns.

THE THREE FUNCTIONAL PATTERNS OF CARDIOMYOPATHY

Feature	Dilated (DCM)	Hypertrophic (HCM)	Restrictive (RCM)
Basic defect	Impaired contractility: **systolic** dysfunction	Impaired compliance: **diastolic** dysfunction ± outflow obstruction	Impaired compliance: **diastolic** dysfunction
LV ejection fraction	**< 40%**	50–80%	45–90%
Gross	Heavy (2–3 × normal), **flabby, all chambers dilated**; mural thrombi	Massive hypertrophy **without dilatation**; **asymmetric septal** thickening; banana-shaped LV cavity; mitral contact plaque	Ventricles normal size, **firm**; **both atria dilated**
Microscopy	Non-specific: hypertrophied myocytes with large nuclei, interstitial fibrosis	Massive myocyte hypertrophy, **myofibre disarray**, interstitial and replacement fibrosis	According to cause: amyloid, fibrosis, granulomas
Causes	**Genetic** in 20–50% (titin, TTN, commonest); **alcohol**; myocarditis (**Coxsackie B**); doxorubicin; **peripartum**; iron overload	**Autosomal dominant** mutations of sarcomere proteins: **β-myosin heavy chain** (MYH7), myosin-binding protein C, troponin T	**Amyloidosis** (commonest), sarcoidosis, radiation fibrosis, endomyocardial fibrosis, Loeffler endomyocarditis, storage diseases
Classic picture	Progressive heart failure at 20–50 years	**Sudden death in a young athlete**; exertional dyspnoea, systolic murmur	Heart failure mimicking constrictive pericarditis

CONCEPT CLEAR

How HCM obstructs itself and kills. The bulging septum narrows the outflow tract; in systole the anterior mitral leaflet is drawn towards it (systolic anterior motion), so obstruction is **dynamic** and worsens with dehydration, standing or exercise. The thick, stiff ventricle fills poorly (**diastolic failure**), and the disarrayed, scarred myocardium is a substrate for **ventricular fibrillation**. Screen first-degree relatives.

MYOCARDITIS (BRIEF)

Inflammation of the myocardium with myocyte injury. **Viruses** are the commonest cause (**Coxsackie A and B**, other enteroviruses); others: *Trypanosoma cruzi* (Chagas), diphtheria toxin, Lyme disease, drug hypersensitivity (eosinophils), giant cell myocarditis, SLE, transplant rejection.

Micro: interstitial lymphocytic infiltrate with **focal myocyte necrosis**. It may heal, cause sudden death, or progress to **dilated cardiomyopathy**.

EXAM ZONE Past-Paper Pattern SEQs — Solved

CLASSIC **a)** Describe the sequential gross and microscopic changes in myocardial infarction. **b)** Enumerate its complications. **3 + 2 marks**

MODEL ANSWER PLAN

a) Time table (gross + micro): 0–4 h nothing/wavy fibres; 4–24 h coagulation necrosis, mottling; 1–3 d neutrophils, yellow-tan centre; 3–7 d macrophages, hyperaemic border; 7–14 d granulation tissue; 2–8 weeks collagen; scar by about 2 months.

b) Arrhythmias, LV failure/shock, pericarditis (2–3 d), rupture (3–7 d), mural thrombus and emboli, Dressler syndrome, aneurysm, chronic failure.

Score booster: write complications against their day; the examiner sees that you understand *why* rupture happens at 3–7 days.

MODULE PATTERN A 10-year-old boy has fever and painful swelling of the right knee, then the left ankle, three weeks after a sore throat. A new apical murmur is heard. **a)** Diagnosis and the criteria used. **b)** Pathogenesis. **c)** Morphology of the heart in the acute and chronic stages. **1 + 2 + 2 marks**

MODEL ANSWER PLAN

a) Acute rheumatic fever; Jones criteria (2 major, or 1 major + 2 minor, with evidence of strep infection). **b)** Strep pharyngitis → antibodies and CD4 T cells to M protein → molecular mimicry with cardiac antigens → macrophage activation (Fig 26.5). **c)** Acute: pancarditis, Aschoff bodies, Anitschkow cells, verrucae, MacCallum plaques. Chronic: fibrosis, commissural fusion, fused chordae, fish-mouth mitral stenosis, LA thrombus.

PROF PATTERN **a)** Differentiate between acute and subacute infective endocarditis. **b)** Describe the morphology and complications of infective endocarditis. **2 + 3 marks**

MODEL ANSWER PLAN

a) Table: organism, prior valve state, onset/course, destructiveness, mortality. **b)** Friable bulky vegetations (fibrin, neutrophils, bacteria) on aortic/mitral (tricuspid in IV drug users), ring abscess. Complications: regurgitation, septic emboli, mycotic aneurysm, immune-complex GN.

BCQs One-Best-Answer Practice

1 A 60-year-old man dies 2 days after the onset of severe chest pain. Sections of the left ventricle are most likely to show:

A. Dense collagenous scar
B. Granulation tissue with new capillaries
C. Coagulation necrosis with a brisk neutrophilic infiltrate
D. Macrophages laden with haemosiderin and lipofuscin
E. Normal myocardium

2 On day 5 after an anterior MI a woman suddenly develops hypotension, raised JVP and muffled heart sounds. The most likely cause is:

A. Dressler syndrome
B. Ventricular aneurysm
C. Mural thrombus
D. Rupture of the free wall with tamponade
E. Re-infarction

3 An infarct involving the anterior wall of the left ventricle, the apex and the anterior two-thirds of the septum results from occlusion of the:

A. Left anterior descending artery
B. Right coronary artery
C. Left circumflex artery
D. Posterior descending artery
E. Marginal branch of RCA

4 Four days after an MI, chest pain recurs. Which marker best detects re-infarction at this time?

A. Troponin I
B. Troponin T
C. Myoglobin
D. Lactate dehydrogenase
E. CK-MB

5 Plump macrophages with a slender wavy ribbon of nuclear chromatin in a myocardial lesion are called:

A. Anitschkow cells
B. Langhans giant cells
C. Reed–Sternberg cells
D. Touton giant cells
E. Heart failure cells

6 A 25-year-old intravenous drug user has fever and a large vegetation on the tricuspid valve. The most likely organism is:

A. Streptococcus viridans
B. Staphylococcus aureus
C. Staphylococcus epidermidis
D. Enterococcus faecalis
E. Candida albicans

7 A 19-year-old footballer collapses and dies on the field. Autopsy shows a thick septum bulging into the outflow tract and myofibre disarray. The most likely mutated gene encodes:

A. Dystrophin
B. Titin
C. Fibrillin-1
D. Transthyretin
E. β-myosin heavy chain

8 An elderly man has heart failure with normal-sized, stiff ventricles and dilated atria. Endomyocardial biopsy shows material with apple-green birefringence after Congo red. The cardiomyopathy is:

A. Dilated
B. Hypertrophic
C. Restrictive
D. Arrhythmogenic
E. Takotsubo

9 Which feature best distinguishes senile calcific aortic stenosis from rheumatic aortic stenosis?

A. Presence of LVH
B. Absence of commissural fusion
C. Angina on exertion
D. Calcification of cusps
E. Systolic murmur

10 Small sterile vegetations on both surfaces of the mitral leaflets in a young woman with a malar rash are:

A. Rheumatic verrucae
B. Infective vegetations
C. Non-bacterial thrombotic vegetations
D. Libman–Sacks vegetations
E. Myxomatous nodules

11 A massive saddle embolus causes sudden death. The right ventricle at autopsy shows:

A. Dilatation without hypertrophy
B. Concentric hypertrophy
C. Wall thickness of 1.5 cm
D. Old scar
E. Myofibre disarray

12 Histology of a floppy, hooded mitral leaflet with a mid-systolic click shows:

A. Aschoff bodies
B. Bacterial colonies
C. Myxomatous expansion of the spongiosa
D. Fibrinoid necrosis
E. Calcified nodules on the atrial surface

ANSWER KEY WITH ONE-LINE REASONS

1. **C** 1–3 days: loss of nuclei and striations with a neutrophilic infiltrate.
2. **D** Rupture peaks at 3–7 days, the softest phase of the infarct.
3. **A** LAD is the commonest culprit (40–50%) and supplies this territory.
4. **E** Troponins stay raised 7–10 days; CK-MB normalises by 72 h, so a new rise means re-infarction.
5. **A** Anitschkow (caterpillar) cells are part of the Aschoff body of rheumatic carditis.
6. **B** S. aureus causes acute right-sided IE in IV drug users.
7. **E** HCM: autosomal dominant sarcomere mutation, β-MHC most often.
8. **C** Amyloidosis is the commonest cause of restrictive cardiomyopathy.
9. **B** Calcific AS has heaped calcium on the outflow side without commissural fusion.
10. **D** Libman–Sacks endocarditis of SLE involves both leaflet surfaces.
11. **A** Acute cor pulmonale: no time for hypertrophy, only dilatation.
12. **C** MVP: proteoglycan-rich myxomatous degeneration; associated with Marfan syndrome.

VIVA Rapid-Fire Viva Voce

Q What is critical stenosis?	Narrowing of ≥ 70% of a coronary lumen, causing ischaemia on exertion (stable angina).
Q Why does necrosis start in the subendocardium?	It is last to be perfused and under the highest intramural pressure.
Q When does an infarct first become visible grossly?	After about 12 hours as dark mottling; TTC staining shows it earlier as an unstained pale area.
Q What is contraction band necrosis?	Hypercontracted myofibrils forming dense eosinophilic bands, typical of reperfused myocardium.
Q When is rupture most likely and why?	3–7 days, when macrophages have removed necrotic muscle before collagen is laid down.
Q What is Dressler syndrome?	Autoimmune pericarditis appearing weeks after MI.
Q Define cor pulmonale.	RV hypertrophy and/or dilatation due to pulmonary hypertension from lung or pulmonary vascular disease.
Q What is an Aschoff body?	Focal inflammatory lesion of rheumatic carditis: T cells, plasma cells and Anitschkow macrophages, sometimes fibrinoid necrosis.
Q Why is mitral stenosis called fish-mouth?	Fused commissures and thick leaflets make a rigid funnel with a narrow slit-like orifice.
Q Which organism causes endocarditis on prosthetic valves?	Coagulase-negative staphylococci, especially S. epidermidis.
Q What is myofibre disarray?	Haphazard arrangement of hypertrophied myocytes and myofibrils: hallmark of HCM.
Q Commonest viral cause of myocarditis?	Coxsackie A and B viruses (enteroviruses).

OSPE Practice Stations

STATION 1 · SLIDE — 2 minutes

TASK

Myocardium: intensely pink myocytes without nuclei or striations but with preserved outlines; many neutrophils between them. (a) Identify and date the lesion. (b) Type of necrosis? (c) Most feared complication in the next few days?

ANSWER

(a) Acute myocardial infarction, about 1–3 days old. (b) Coagulative necrosis. (c) Myocardial rupture (free wall, septum or papillary muscle), peaking at 3–7 days.

STATION 2 · GROSS SPECIMEN — 1 minute

TASK

Heart opened through a dilated left atrium with thrombus in the appendage. Mitral leaflets thick, rigid, fused at the commissures; chordae short and thick; slit-like orifice. (a) Diagnosis. (b) Two gross features. (c) Two complications.

ANSWER

(a) Chronic rheumatic heart disease with mitral stenosis. (b) Commissural fusion; thickened, fused chordae giving a fish-mouth orifice. (c) Atrial fibrillation with systemic embolism; pulmonary hypertension and right heart failure; infective endocarditis.

STATION 3 · LAB DATA 2 minutes

TASK

Man, 52, chest pain for 2 hours; ST elevation V1–V4. Troponin I normal on arrival, markedly raised 6 hours later; CK-MB raised. (a) Interpret. (b) Likely artery? (c) Why was the first troponin normal?

ANSWER

(a) Acute anterior ST-elevation MI with a rising troponin. (b) Left anterior descending artery. (c) Troponin takes about 2–4 hours to rise; a single early value does not exclude MI, so serial testing is required.

RAPID REVISION The Night-Before Page

15 ONE-LINERS

01 IHD: > 90% coronary atherosclerosis; symptoms at ≥ 70% stenosis.

02 Acute coronary syndromes start with acute plaque change + thrombus.

03 Myocyte injury irreversible after 20–40 min; infarct complete in 3–6 h.

04 LAD 40–50%, RCA 30–40%, LCX 15–20% of infarcts.

05 Neutrophils 1–3 d; macrophages 3–7 d; granulation 1–2 wk; scar by 2 months.

06 TTC: viable muscle brick-red, infarct pale.

07 Troponin: up 2–4 h, peak ~48 h, lasts 7–10 d; CK-MB gone by 72 h.

08 Rupture at 3–7 d; pericarditis 2–3 d; Dressler weeks; aneurysm late.

09 Hypertensive heart: concentric LVH, wall > 2 cm, boxcar nuclei.

10 Cor pulmonale: RVH from lung disease; acute = massive PE.

11 RF: strep throat → M-protein mimicry → Aschoff bodies, pancarditis.

12 Chronic RHD: mitral almost always; fish-mouth stenosis, LA thrombus.

13 Acute IE: S. aureus, normal valve; subacute: viridans strep, damaged valve.

14 Calcific AS: outflow-side calcium, no commissural fusion; MVP: myxomatous.

15 DCM systolic (EF < 40%); HCM β-MHC disarray; RCM amyloid.

MEMORY HOOKS

- **DARTH VADER**: complications of MI
- **J❤NES**: major Jones criteria
- **FROM JANE**: signs of infective endocarditis
- **"Neutrophils first, macrophages next, collagen last"**: dating an MI

TEST YOURSELF: FILL THE GAPS

1. Anitschkow cells are activated _____.
2. Rupture after MI peaks at _____ days.
3. Marker for re-infarction: _____.
4. Vegetations on both leaflet surfaces: _____.
5. Commonest gene in HCM: _____.
6. Prosthetic valve IE organism: _____.

Answers: macrophages · 3–7 · CK-MB · Libman–Sacks · β-myosin heavy chain · S. epidermidis

DRAW FROM MEMORY

The MI timeline: six chevrons from 0 h to 2 months with the key cell at each stage (Figure 26.2).

DRAW FROM MEMORY

The rheumatic fever chain from strep throat to fish-mouth mitral stenosis (Figure 26.5).

SPACED REVISION ☐ Day 1 ☐ Day 3 ☐ Day 7 ☐ Day 14 ☐ Day 30

27

Bone: Metabolic Disease, Osteomyelitis & Bone Tumours

Bone looks like stone but is rebuilt every day by two rival cell teams. Tip the balance and the skeleton thins, softens or overgrows; let a cell escape control and it builds tumour bone in the one place its age and site predict.

BY THE END OF THIS CHAPTER YOU CAN

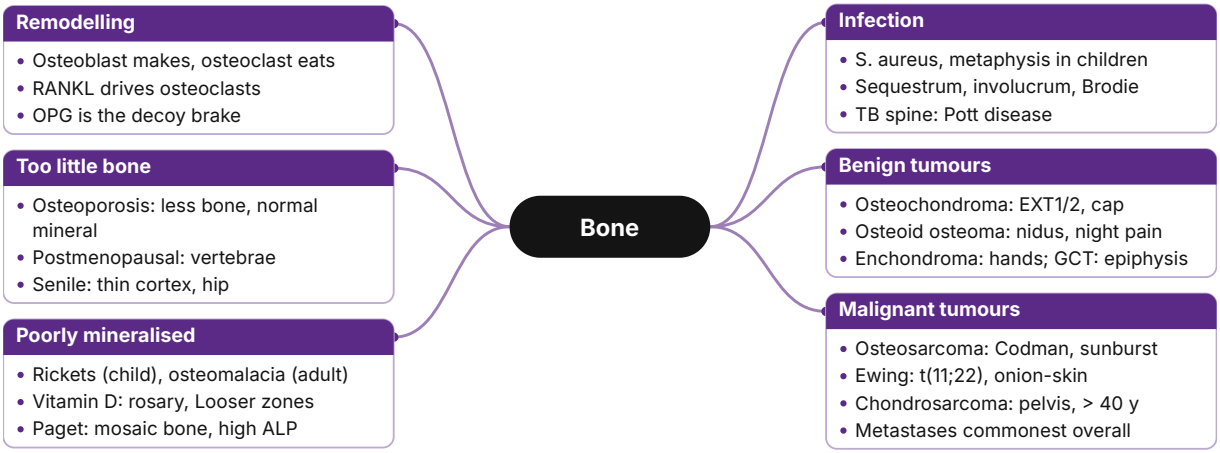
1. Explain bone remodelling and the RANK–RANKL–OPG pathway
2. Describe the causes, pathogenesis and morphology of osteoporosis and contrast it with osteomalacia
3. Describe vitamin D metabolism and the skeletal changes of rickets and osteomalacia; outline Paget disease
4. Describe the pathogenesis and morphology of pyogenic and tuberculous osteomyelitis
5. Classify bone tumours and predict the likely tumour from age, site and radiograph
6. Describe osteochondroma, osteosarcoma, giant cell tumour, Ewing sarcoma and chondrosarcoma for the OSPE

EXAM RADAR

Theory SEQs	★★★
BCQs / MCQs	★★★
Viva voce	★★★
OSPE / slides	★★★

Osteomyelitis, osteoporosis and osteosarcoma are classic long questions; osteochondroma, osteosarcoma and giant cell tumour are the Module 12 lab slides.

ROBBINS LINK Robbins Basic Pathology — *Bones, Joints, and Soft Tissue Tumors*: bone remodelling, osteoporosis, Paget disease, rickets and osteomalacia, osteomyelitis, bone-forming, cartilage-forming and miscellaneous bone tumours, metastases.



Chapter at a glance. Metabolism, infection and tumours: three ways bone fails.

27.1 Bone remodelling: the balance that every disease tips

Adult bone is constantly resorbed by **osteoclasts** (multinucleate, derived from monocyte–macrophage precursors) and rebuilt by **osteoblasts** (mesenchymal), which lay down **osteoid** (type I collagen) that is then **mineralised** with calcium hydroxyapatite. Osteoblasts trapped in matrix become osteocytes. About 10% of the skeleton is remodelled each year.

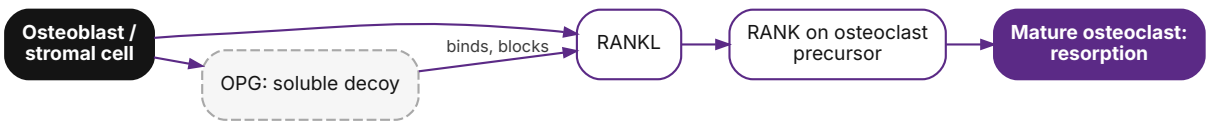


Figure 27.1 · The RANK–RANKL–OPG switch. Osteoblasts control how many osteoclasts exist.

CONCEPT CLEAR

One switch explains many diseases. Anything that **raises RANKL** or **lowers OPG** makes osteoclasts win: PTH, IL-1, TNF and glucocorticoids raise RANKL; **oestrogen** and Wnt/β-catenin signalling raise OPG. That is why menopause, steroids, myeloma, metastases and rheumatoid arthritis all erode bone, and why **denosumab** (anti-RANKL antibody) treats osteoporosis and giant cell tumour.

27.2 Osteoporosis

DEFINITION

Osteoporosis is an acquired condition of **reduced bone mass** (increased porosity) that makes bone fragile and prone to fracture. The bone present is **normal in composition and mineralisation**; there is simply too little of it. WHO definition on DEXA: bone mineral density **T-score ≤ −2.5** (osteopenia: −1 to −2.5).

PRIMARY

- **Postmenopausal** (oestrogen loss): rapid loss, mainly of **trabecular** bone
- **Senile** (age-related): slower loss of trabecular and **cortical** bone in both sexes

SECONDARY

- **Endocrine:** hyperparathyroidism, hyperthyroidism, Cushing syndrome, hypogonadism, type 1 diabetes
- **Drugs: corticosteroids** (commonest secondary cause), heparin, anticonvulsants, alcohol
- **GIT:** malnutrition, malabsorption, liver disease, vitamin C/D deficiency
- **Others: immobilisation**, multiple myeloma, carcinomatosis, anaemia

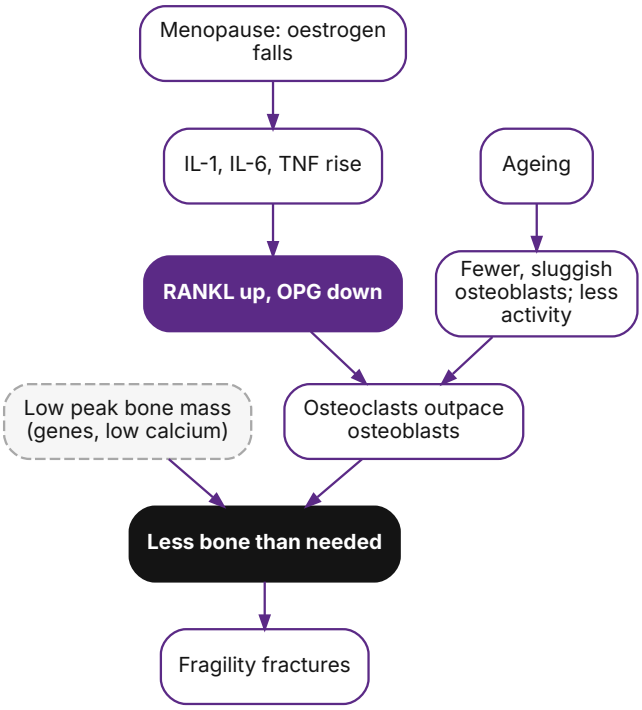


Figure 27.2 · Pathogenesis of osteoporosis. Peak bone mass minus the rate of loss.

GROSS

- Whole skeleton affected, but unevenly
- **Postmenopausal:** vertebral bodies (cancellous) → **crush (compression) fractures**, loss of height, **kyphosis** (dowager's hump)
- **Senile:** thin cortex → **femoral neck** and Colles fractures

MICROSCOPY

- Trabeculae **thinned, perforated** and **lose interconnections**
- Cortex thinned by endosteal and subperiosteal resorption; **widened Haversian canals** (cortex looks cancellous in severe cases)
- **Mineral content normal**; no excess osteoid

Clinical: silent until a fracture. X-rays show osteopenia only after **30–40%** of bone is lost, so **DEXA** is used. Serum **calcium, phosphate and ALP are normal**. Hip fractures bring immobility, pneumonia and pulmonary embolism. Treatment: calcium, vitamin D, exercise, bisphosphonates, denosumab.

OSTEOPOROSIS VS OSTEOMALACIA: BOTH LOOK "THIN" ON X-RAY

Feature	Osteoporosis	Osteomalacia
Defect	**Quantity**: too little normal bone	**Quality**: matrix not mineralised

Feature	Osteoporosis	Osteomalacia
Histology	Thin trabeculae, normal seams	**Wide osteoid seams** (unmineralised)
Serum Ca / PO_4 / ALP	**Normal** / normal / normal	Low or normal / **low** / **raised**; PTH raised
Pain	Only with fractures	Diffuse bone pain, proximal muscle weakness
X-ray clue	Vertebral crush fractures	**Looser zones** (pseudofractures)

27.3 Paget disease of bone (osteitis deformans), in brief

A disorder of **excessive but chaotic remodelling** that gives bone that is **thicker yet weaker**. Middle-aged and elderly, commonest in people of British descent. Familial cases often carry SQSTM1 mutations that increase NF-κB activity in osteoclasts; a slow paramyxovirus infection has been suspected but not proven. About 85% have **polyostotic** disease (skull, pelvis, spine, femur). It runs through **three phases**: **osteolytic** (huge osteoclasts with up to 100 nuclei) → **mixed** osteoclastic–osteoblastic → burnt-out **osteosclerotic** (thick, dense, disordered bone).

Hallmark microscopy: a **mosaic ("jigsaw-puzzle") pattern** of lamellar bone with haphazard **cement lines**. **Labs:** very high **ALP**, raised urinary hydroxyproline, **normal calcium and phosphate**.

Clinical: bone pain, enlarging skull (hats no longer fit), deafness from nerve compression, bowed tibia, chalk-stick fractures, secondary osteoarthritis, high-output heart failure. **Osteosarcoma** develops in about **1%**.

27.4 Rickets and osteomalacia

DEFINITION

Rickets (children) and **osteomalacia** (adults) are disorders of **defective mineralisation of osteoid**, usually from vitamin D deficiency or abnormal vitamin D metabolism. In children the **growth plates** are also affected, which produces the deformities of rickets.

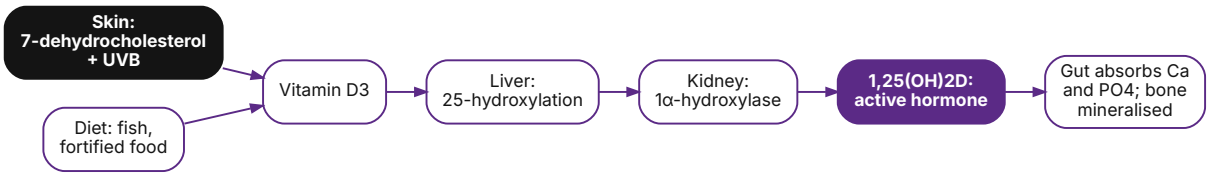


Figure 27.3 · Vitamin D: where it is made and where it fails.

CAUSES (FOLLOW THE PATHWAY)

- **Too little made or eaten:** poor sunlight, covered clothing, dark skin, poor diet
- **Malabsorption:** coeliac disease, biliary obstruction, pancreatic insufficiency
- **Liver disease** (25-hydroxylation); anticonvulsants (faster breakdown)
- **Renal disease** (1α-hydroxylation): renal osteodystrophy; vitamin D-dependent rickets
- **Phosphate wasting:** X-linked hypophosphataemic rickets (PHEX → high FGF23)

CONCEPT CLEAR

Why the bone softens. Low vitamin D → less gut calcium → low serum calcium → **secondary hyperparathyroidism**. PTH restores calcium by resorbing bone but makes the kidney waste phosphate. With a **low calcium × phosphate product**, new osteoid cannot be mineralised. In a child the cartilage of the growth plate also fails to calcify, so it piles up in irregular masses.

RICKETS: WHAT YOU SEE IN THE CHILD

- **Skull:** craniotabes (soft, "ping-pong" occiput), **frontal bossing**, squared head, delayed fontanelle closure and dentition
- **Chest: rachitic rosary** (overgrown cartilage at costochondral junctions), **pigeon breast**, **Harrison sulcus**
- **Limbs:** widened wrists and ankles; **bowed legs** once walking; lumbar lordosis

RICKETS / OSTEOMALACIA: MICROSCOPY

- Growth plate: **overgrown, irregular cartilage** projecting into marrow; disorderly replacement by osteoid
- Osteoid deposited on **unmineralised cartilage** remnants; overgrowth of capillaries and fibroblasts
- Adults: excess **unmineralised osteoid seams** around trabeculae (seen in undecalcified sections)

Osteomalacia: diffuse bone pain, proximal myopathy (waddling gait), fractures of vertebrae and femoral neck, and **Looser zones**: narrow radiolucent bands at right angles to the cortex (pubic rami, ribs, scapula). **Labs:** low/normal calcium, **low phosphate**, **high ALP**, high PTH, **low 25-hydroxyvitamin D**.

PAKISTAN FOCUS

Vitamin D deficiency is very common in Pakistan despite abundant sunshine: covered clothing, indoor lives and poor diets keep skin synthesis and intake low. Rickets is still seen in children, and osteomalacia in women, particularly after repeated pregnancies and long breastfeeding. Bone pain with a raised ALP in a young woman should prompt a vitamin D level before any other diagnosis.

27.5 Osteomyelitis

DEFINITION

Osteomyelitis is inflammation of bone and marrow, almost always due to infection. It may be acute or chronic. Organisms reach bone by the **blood** (commonest), by **spread from a contiguous focus**, or by **direct implantation** (compound fracture, surgery).

PYOGENIC OSTEOMYELITIS: ORGANISM BY SETTING

Setting	Organism
Most cases (80–90% of culture-positive)	**Staphylococcus aureus**
Neonates	H. influenzae, group B streptococci
Sickle cell disease	**Salmonella**
UTI, IV drug users	E. coli, Pseudomonas, Klebsiella
Trauma, open fracture	Mixed bacteria

CONCEPT CLEAR

Why the metaphysis of a child? Metaphyseal capillaries loop back into wide, sluggish **sinusoidal veins** just below the growth plate: bacteria settle where blood slows, and the end-arteries offer little defence. In **neonates** vessels cross the plate, so infection spreads into the epiphysis and **joint** (septic arthritis). In **adults** after plate closure, epiphyses and especially the **vertebrae** are hit.



Figure 27.4 · Natural history of acute haematogenous osteomyelitis. Pus under pressure kills bone; the periosteum walls it in.

GROSS

- **Sequestrum**: pale, dead bone fragment lying free in pus
- **Involucrum**: shell of reactive new bone around it, perforated by openings (cloacae)
- **Draining sinus** to skin; soft-tissue abscess
- **Brodie abscess**: small walled-off intraosseous (often cortical) abscess with sclerotic rim

MICROSCOPY

- Acute: **neutrophils** in marrow spaces, **necrotic bone** with empty osteocyte lacunae
- Chronic: lymphocytes, plasma cells, macrophages; fibrosis of marrow; osteoclastic resorption; **reactive woven bone** at the periphery
- Garré sclerosing osteomyelitis (jaw): exuberant new bone

Clinical: fever, rigors, leukocytosis and throbbing local pain in children; infants may show only fever. X-ray: lytic focus with surrounding sclerosis (after 1–2 weeks). Blood cultures often positive. **Complications** (5–25% become chronic): pathological fracture, septic arthritis, bacteraemia and endocarditis, **secondary amyloidosis**, **squamous cell carcinoma** in the sinus tract, rarely sarcoma.

Tuberculous osteomyelitis and Pott disease

Complicates about **1–3%** of tuberculosis, usually by blood spread in childhood. Favourite sites: **spine** (thoracic and lumbar vertebrae), knees and hips. It is **more destructive and harder to control** than pyogenic infection. **Pott disease**: caseating infection crosses the **intervertebral discs** to involve several vertebrae, causing collapse, angular **kyphosis (gibbus)**, cord compression (paraplegia) and a **cold psoas abscess** tracking to the groin. **Micro:** caseating epithelioid granulomas with Langhans giant cells.

PAKISTAN FOCUS

Pakistan is among the **top five high-TB-burden countries**, with an incidence of roughly **266 per 100,000**. Back pain with fever, weight loss and a gibbus in a young adult is spinal TB until proven otherwise; GeneXpert on aspirated pus confirms it and detects rifampicin resistance.

27.6 Bone tumours

Three facts frame every bone tumour question. (1) **Metastases** are far commoner than primary bone tumours, and among primaries **myeloma** is commonest (Chapter 21). (2) **Benign tumours** outnumber malignant ones, especially under 40; in the elderly a bone tumour is more likely malignant. (3) **Age + site + radiograph** narrow the diagnosis before the biopsy: many tumours arise around the **knee**, where the growth plates are busiest.

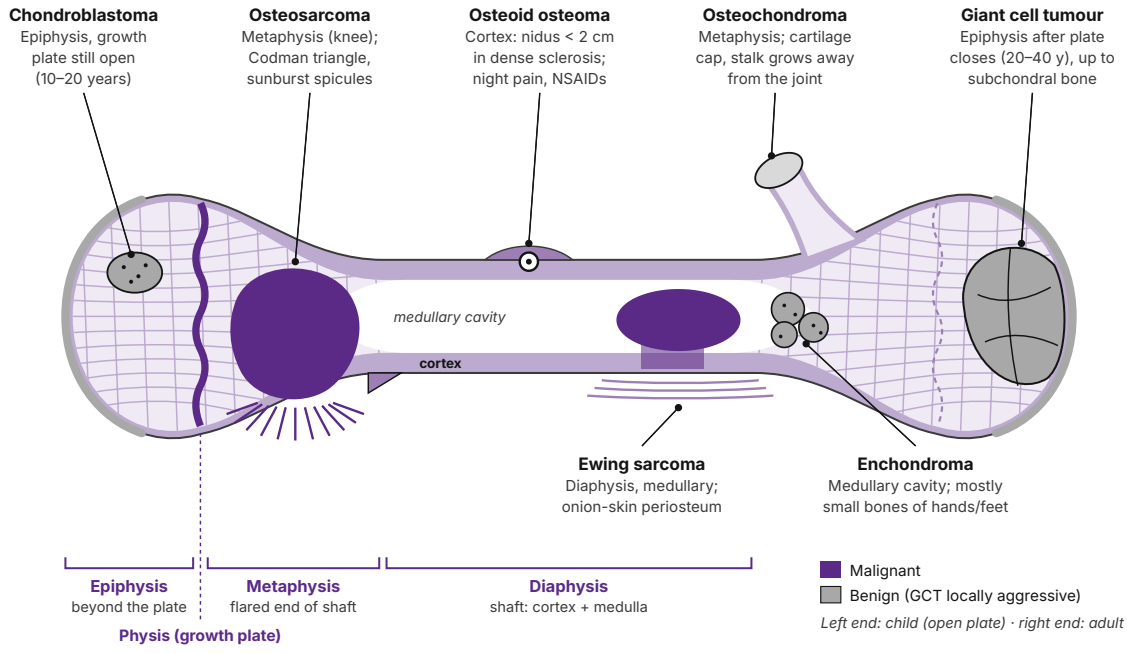


Figure 27.5 · Where bone tumours arise in a long bone. Epiphysis: chondroblastoma (young) and giant cell tumour (adult). Metaphysis: osteosarcoma and osteochondroma. Diaphysis: Ewing sarcoma. Cortex: osteoid osteoma. Medulla: enchondroma.

LAB VALUES

Classification by the matrix the tumour makes (benign | malignant). Bone-forming: osteoid osteoma, osteoblastoma | osteosarcoma. **Cartilage-forming:** osteochondroma, enchondroma, chondroblastoma | chondrosarcoma. **Uncertain origin:** giant cell tumour (locally aggressive), aneurysmal bone cyst | Ewing sarcoma. **Haematopoietic:** myeloma, lymphoma.

Bone-forming tumours

Osteoid osteoma: < 2 cm, in the **cortex** of femur or tibia; teens and twenties, males. Severe pain, **worse at night and relieved by aspirin/NSAIDs** because the tumour makes **PGE_2**. X-ray: small lucent **nidus** in **dense reactive sclerosis**. Treated by radiofrequency ablation.

Osteoblastoma: same histology but **> 2 cm**, in the **posterior elements of vertebrae**; dull ache **not** relieved by aspirin; little sclerosis. **Micro (both):** interlacing trabeculae of **woven bone rimmed by plump osteoblasts** in loose vascular stroma.

Osteosarcoma

DEFINITION

Osteosarcoma is a malignant mesenchymal tumour in which the cancer cells **produce bone matrix (osteoid)**. It is the commonest primary malignant tumour of bone after myeloma and lymphoma.

- **Age:** bimodal. About **75% under 20 years** (growth spurt); a second peak in the **elderly** with Paget disease, bone infarcts or prior radiation. Males slightly more.
- **Site: metaphysis** of long bones; about **50% around the knee** (distal femur, proximal tibia), then proximal humerus.
- **X-ray:** large destructive **mixed lytic–blastic** mass with infiltrative margins; tumour lifts the periosteum to form **Codman triangle**; spicules give a **sunburst** pattern.
- **Labs:** raised ALP.

MOLECULAR KEY

RB loss: children with hereditary retinoblastoma have a vastly increased risk. **TP53**: Li–Fraumeni syndrome; mutated in many sporadic tumours. **CDKN2A** (p16, p14ARF) inactivation. **MDM2 and CDK4 amplification**, typical of low-grade (parosteal) tumours. Most tumours are sporadic.

GROSS

- Bulky, **gritty, grey-white** tumour with **haemorrhage and cystic change**
- Destroys the cortex and forms a **soft-tissue mass**; spreads along the medullary canal
- Rarely crosses the growth plate or enters the joint

MICROSCOPY

- Pleomorphic cells: large **hyperchromatic nuclei**, **bizarre tumour giant cells**, many and atypical mitoses
- **Neoplastic bone**: coarse or fine **lace-like osteoid** made directly by malignant cells (the diagnostic feature)
- Malignant cartilage may be present; vascular invasion, necrosis

Behaviour: painful, enlarging mass; sometimes a pathological fracture. Spreads by blood, chiefly to the **lungs**: **10–20%** have pulmonary metastases at diagnosis. With neoadjuvant chemotherapy and surgery, 5-year survival is about **60–70%** without overt metastases.

Cartilage-forming tumours

OSTEOCHONDROMA (EXOSTOSIS)

A benign **cartilage-capped outgrowth** attached to the underlying bone by a **bony stalk**: the **commonest benign bone tumour**. About **85% are solitary**; the rest occur in **multiple hereditary exostoses** (autosomal dominant, germline EXT1 or EXT2 loss with a second hit; these genes make the enzymes for heparan sulfate). Late adolescence and early adulthood; **men about 3 times** more often. Arise from the **metaphysis near the growth plate** of long bones, especially the knee, and **point away from the joint**. They stop growing when the plate closes. Malignant change to chondrosarcoma is rare, commoner in the hereditary form.

GROSS

- Sessile or pedunculated, 1–20 cm
- Bluish-white **cartilage cap** over a bony stalk
- **Cortex and medulla continuous** with those of the host bone (X-ray key)

MICROSCOPY

- **Cap of benign hyaline cartilage** covered by **perichondrium**
- Deep cartilage undergoes **enchondral ossification**, forming the bony head and stalk
- Marrow space of the stalk merges with host marrow

Enchondroma (chondroma): benign hyaline cartilage nodule inside the **medullary cavity**, usually of the **small tubular bones of the hands and feet**; 20–50 years; often an incidental finding or a pathological fracture. Gross: grey-blue, translucent, < 3 cm; micro: well-circumscribed lobules of hyaline cartilage with bland chondrocytes. X-ray: lucency with ring-like calcification and intact cortex. **IDH1/IDH2** mutations. Multiple: **Ollier disease**; with soft-tissue haemangiomas: **Maffucci syndrome**; both carry a risk of chondrosarcoma.

Chondroblastoma: rare benign cartilage tumour of the **epiphysis** in adolescents, while the growth plate is still open; sheets of chondroblasts with "chicken-wire" calcification.

Chondrosarcoma: malignant tumour that makes **cartilage** (not osteoid). Patients **≥ 40 years**, men more; **axial skeleton**: pelvis, shoulder girdle, ribs (rare in hands and feet, unlike enchondroma). Conventional type about 90%; may arise from enchondroma or osteochondroma. Gross: large, lobulated, **grey-white, translucent, glistening**, myxoid or gelatinous with **spotty calcification**. Micro: malignant cartilage permeating marrow and trapping native trabeculae. **Grade decides prognosis**: grade 1 rarely metastasises; grade 3 often spreads to the lungs.

Ewing sarcoma

A malignant **small round cell tumour** of bone and soft tissue: the **second commonest bone sarcoma in children**; about 80% of patients are under 20. Arises in the **medulla of the diaphysis** of long bones (femur) and in the **pelvis**. Fever, raised ESR, anaemia and a warm, tender swelling **mimic osteomyelitis**. X-ray: lytic, permeative lesion with **onion-skin** layers of periosteal new bone.

MOLECULAR KEY

t(11;22)(q24;q12) in about 85%: **EWSR1–FLI1** fusion, an aberrant transcription factor. Rest: EWSR1–ERG and others. IHC: **CD99** membrane positivity, FLI1.

GROSS

- Soft, tan-white tumour filling the medulla, breaking through cortex into soft tissue
- Haemorrhage and necrosis

MICROSCOPY

- **Sheets of uniform small round cells**, slightly larger than lymphocytes, scant stroma
- Scant **clear cytoplasm rich in glycogen (PAS+)**
- **Homer–Wright rosettes** (neural differentiation)
- Chemotherapy + surgery/radiation: 5-year survival about 75%; chemotherapy-induced necrosis predicts outcome

Giant cell tumour of bone (osteoclastoma)

A **benign but locally aggressive** tumour of adults aged **20–40 years**. The **neoplastic cells are mononuclear osteoblast precursors** that express high levels of **RANKL**; they recruit and activate the non-neoplastic **osteoclast-type giant cells** that destroy bone. Most carry a mutation in the histone gene *H3F3A*. It is the classic **epiphyseal** tumour of adults (after plate closure), around the **knee** and distal radius, extending to the subchondral plate (arthritis-like pain).

X-ray: large, **purely lytic, eccentric** lesion expanding into a thin shell of reactive bone: "**soap-bubble**". **Behaviour:** recurs after curettage in up to 40–60%; rare lung "metastases". Denosumab shrinks it.

GROSS

- Large, **red-brown**, haemorrhagic, often **cystic**
- Thin bony shell; may abut articular cartilage

MICROSCOPY

- Sheets of **uniform oval mononuclear cells** (the tumour) with mitoses
- Evenly scattered **osteoclast-type giant cells with up to 100 or more nuclei**, nuclei resembling those of the stromal cells
- Haemorrhage, haemosiderin, reactive bone

Metastatic tumours

The **commonest malignancy of bone**. Adults: **prostate, breast, kidney, lung**, thyroid; children: neuroblastoma, Wilms tumour. They seek red marrow: **vertebrae, pelvis, ribs, skull, proximal femur**. Most are **lytic** (tumour PTHrP and cytokines raise RANKL); **prostate** is classically **blastic**; breast often mixed. Expect pain, fractures, hypercalcaemia, multiple lesions.

BONE TUMOURS AT A GLANCE: THE TABLE THAT ANSWERS MOST MCQS

Tumour	Age	Site	X-ray	Key micro / marker
Osteochondroma	10–30	Metaphysis, knee	Stalk continuous with host bone	Hyaline cap; EXT1/2
Osteoid osteoma	10–30	Cortex, femur/tibia	Nidus + sclerosis	Woven bone + osteoblasts; PGE_2
Enchondroma	20–50	Medulla, hands/feet	Ring calcification	Bland hyaline cartilage; IDH1/2
Giant cell tumour	20–40	**Epiphysis**, knee	Soap-bubble, eccentric	Osteoclast-type giant cells; RANKL
Osteosarcoma	< 20; elderly	**Metaphysis**, knee	Codman triangle, sunburst	Malignant osteoid; RB, TP53
Ewing sarcoma	< 20	**Diaphysis**, pelvis	Onion-skin	Small round cells, CD99; t(11;22)
Chondrosarcoma	≥ 40	Pelvis, shoulder, ribs	Popcorn calcification, endosteal scalloping	Malignant cartilage; IDH1/2

LAB SPOTTER

Bone slides in 10 seconds. Pink lace-like osteoid between pleomorphic cells = osteosarcoma. **Blue cartilage cap over bone trabeculae with marrow** = osteochondroma. **Giant cells evenly sprinkled among plump oval mononuclear cells** = giant cell tumour. **Blue sheets of small round cells** with clear cytoplasm = Ewing sarcoma. **Lobules of atypical cartilage invading marrow** = chondrosarcoma. **Dead bone with empty lacunae among neutrophils** = osteomyelitis.

EXAM TRAP

Giant cells are not only GCT. They also fill the brown tumour of hyperparathyroidism, aneurysmal bone cyst and many osteosarcomas. Adult age and an **epiphyseal** site point to GCT; serum calcium and PTH exclude a brown tumour.

EXAM ZONE Past-Paper Pattern SEQs — Solved

PROF PATTERN A 15-year-old boy has a painful swelling above the knee for 2 months. X-ray shows a destructive lesion of the distal femoral metaphysis with a sunburst appearance and Codman triangle. **a)** Most likely diagnosis and two risk factors. **b)** Describe its morphology. **2 + 3 marks**

MODEL ANSWER PLAN

a) Osteosarcoma. Risk: hereditary retinoblastoma (RB), Li–Fraumeni (TP53), Paget disease, prior radiation, bone infarcts. **b)** Gross: gritty grey-white metaphyseal mass, haemorrhage, cystic change, cortical destruction, soft-tissue extension, spares the growth plate. Micro: pleomorphic malignant cells, bizarre giant cells, atypical mitoses, **lace-like osteoid formed by tumour cells**. Spread: lungs.

Score booster: state the single diagnostic criterion: "osteoid produced directly by malignant cells".

CLASSIC **a)** Name the organisms causing pyogenic osteomyelitis in different settings. **b)** Describe the pathogenesis and morphology. **c)** List the complications. **2 + 2 + 1 marks**

MODEL ANSWER PLAN

a) S. aureus (most); neonates H. influenzae, group B strep; sickle cell Salmonella; UTI/IV drugs E. coli, Pseudomonas. **b)** Haematogenous seeding of the metaphysis → acute inflammation → necrosis in 48 h → subperiosteal abscess → **sequestrum** → **involucrum**, sinus; Brodie abscess. **c)** Chronicity, fracture, septic arthritis, amyloidosis, SCC in sinus tract, endocarditis.

MODULE PATTERN **a)** Define osteoporosis and list its causes. **b)** Differentiate osteoporosis from osteomalacia. **2 + 3 marks**

MODEL ANSWER PLAN

a) Reduced bone mass with normal mineralisation, causing fragility (T-score ≤ −2.5). Primary: postmenopausal, senile. Secondary: steroids, hyperparathyroidism, hyperthyroidism, immobilisation, malabsorption, myeloma. **b)** Table: quantity vs mineralisation defect; normal vs wide osteoid seams; normal labs vs low PO_4/high ALP/high PTH; crush fractures vs Looser zones.

BCQs One-Best-Answer Practice

1 A 65-year-old woman has back pain and loss of height. Serum calcium, phosphate and ALP are normal; DEXA T-score is −3.0. The basic defect is:

A. Defective mineralisation of osteoid
B. Reduced bone mass with normal mineralisation
C. Mosaic remodelling with cement lines
D. Excess PTH-driven osteitis fibrosa
E. Defective type I collagen synthesis

2 A child with sickle cell anaemia develops osteomyelitis. The organism to be suspected especially is:

A. Salmonella
B. Haemophilus influenzae
C. Pseudomonas aeruginosa
D. Mycobacterium tuberculosis
E. Neisseria gonorrhoeae

3 The sleeve of new bone formed around a sequestrum in chronic osteomyelitis is called:

A. Cloaca
B. Brodie abscess
C. Codman triangle
D. Callus
E. Involucrum

4 A 2-year-old has frontal bossing, a rachitic rosary and widened wrists. Expected laboratory findings are:

A. High calcium, high phosphate, low ALP
B. Normal calcium, phosphate and ALP
C. Low or normal calcium, low phosphate, high ALP
D. High calcium, low phosphate, low PTH
E. Low calcium, high phosphate, low PTH

5 A 17-year-old has severe night pain in the tibia relieved by aspirin. X-ray shows a 1 cm lucent nidus surrounded by dense sclerosis in the cortex. Diagnosis:

A. Osteoblastoma
B. Brodie abscess
C. Osteoid osteoma
D. Ewing sarcoma
E. Enchondroma

6 The single histological criterion for osteosarcoma is:

A. Osteoclast-type giant cells
B. Osteoid produced directly by malignant cells
C. Homer–Wright rosettes
D. Malignant cartilage only
E. Reactive periosteal bone

7 Multiple hereditary exostoses are caused by germline mutations in:

A. IDH1
B. RB1
C. TP53
D. EXT1 or EXT2
E. SQSTM1

8 A 30-year-old has pain near the knee. X-ray: eccentric, purely lytic epiphyseal lesion with a soap-bubble appearance. Microscopy shows uniform mononuclear cells with evenly scattered multinucleate giant cells. The neoplastic cell is the:

A. Multinucleate giant cell
B. Mature osteoclast
C. Macrophage
D. Chondroblast
E. RANKL-expressing osteoblast precursor

9 A 12-year-old has fever, raised ESR and a tender swelling of the femoral shaft. X-ray shows an onion-skin periosteal reaction. Biopsy shows sheets of small round cells. Which test best confirms the diagnosis?

A. Detection of t(11;22) EWSR1–FLI1
B. Blood culture
C. Serum ALP
D. Congo red stain
E. Bone marrow iron stain

10 A 55-year-old man has a large pelvic tumour with lobules of grey-white translucent tissue and spotty calcification. The most likely diagnosis is:

A. Osteosarcoma
B. Enchondroma
C. Giant cell tumour
D. Osteochondroma
E. Chondrosarcoma

11 A mosaic pattern of lamellar bone with irregular cement lines and a very high serum ALP with normal calcium suggests:

A. Osteomalacia
B. Paget disease
C. Hyperparathyroidism
D. Osteoporosis
E. Osteogenesis imperfecta

12 Which metastatic tumour is classically osteoblastic (sclerotic)?

A. Renal cell carcinoma
B. Lung carcinoma
C. Thyroid carcinoma
D. Prostatic adenocarcinoma
E. Melanoma

ANSWER KEY WITH ONE-LINE REASONS

1. **B** Osteoporosis: less bone of normal composition; labs are normal.
2. **A** Salmonella osteomyelitis is characteristic of sickle cell disease (S. aureus still common).
3. **E** Lifted periosteum lays down an involucrum around dead bone.
4. **C** Vitamin D deficiency with secondary hyperparathyroidism: phosphate falls, ALP rises.
5. **C** Cortical nidus < 2 cm with sclerosis and aspirin-responsive night pain (PGE_2).
6. **B** Tumour cells making osteoid defines osteosarcoma.
7. **D** EXT genes encode enzymes for heparan sulfate synthesis.
8. **E** In GCT the mononuclear stromal cells are neoplastic; giant cells are recruited.
9. **A** Ewing sarcoma mimics osteomyelitis; the translocation (and CD99) confirms it.
10. **E** Chondrosarcoma: middle age, axial skeleton, glistening cartilaginous lobules.
11. **B** Jigsaw-puzzle bone of Paget disease; ALP very high, calcium normal.
12. **D** Prostate cancer stimulates osteoblasts; most others are lytic.

VIVA Rapid-Fire Viva Voce

Q What does RANKL do?	Binds RANK on osteoclast precursors and drives their maturation and bone resorption.
Q What is OPG?	Osteoprotegerin: a soluble decoy receptor that binds RANKL and blocks osteoclast formation.
Q Define osteoporosis on DEXA.	Bone mineral density T-score of −2.5 or lower.

Q Where do postmenopausal osteoporotic fractures occur?	Vertebral bodies (crush fractures), also distal radius and femoral neck.
Q What is a Looser zone?	A pseudofracture: radiolucent band of unmineralised osteoid at right angles to the cortex in osteomalacia.
Q What is a rachitic rosary?	Beading of the costochondral junctions from overgrown, unmineralised cartilage and osteoid in rickets.
Q Define sequestrum.	A fragment of dead bone separated from living bone in osteomyelitis.
Q What is a Brodie abscess?	A small walled-off intraosseous abscess, often cortical, surrounded by sclerotic reactive bone.
Q What is Codman triangle?	Triangular reactive bone at the edge where tumour lifts the periosteum; typical of osteosarcoma.
Q Translocation in Ewing sarcoma?	t(11;22)(q24;q12), EWSR1–FLI1 fusion.
Q Which bone tumour is epiphyseal in adults?	Giant cell tumour (osteoclastoma).
Q Why is aspirin effective in osteoid osteoma?	The tumour makes prostaglandin E2; NSAIDs block its production.

OSPE Practice Stations

STATION 1 · SLIDE — 2 minutes

TASK

Bone lesion: sheets of pleomorphic spindle and polygonal cells with large hyperchromatic nuclei, bizarre giant cells and atypical mitoses. Between the cells lie pink, lace-like strands of matrix. (a) Identify. (b) Two features. (c) Commonest site of metastasis.

ANSWER

(a) Osteosarcoma. (b) Malignant pleomorphic cells; osteoid formed directly by tumour cells. (c) Lungs.

STATION 2 · SLIDE — 2 minutes

TASK

Section shows a cap of bluish hyaline cartilage covered by a fibrous layer. Beneath it, cartilage merges into bony trabeculae with fatty marrow. (a) Identify. (b) Name the gene in the familial form. (c) Name the rare complication.

ANSWER

(a) Osteochondroma (exostosis). (b) EXT1 or EXT2 (multiple hereditary exostoses). (c) Transformation to chondrosarcoma.

STATION 3 · SLIDE — 2 minutes

TASK

Sheets of uniform plump oval mononuclear cells with numerous evenly distributed multinucleate giant cells, some with dozens of nuclei; areas of haemorrhage. The lesion was in the distal femoral epiphysis of a 28-year-old. (a) Identify. (b) Typical X-ray sign. (c) Which molecule drives the giant cells?

ANSWER

(a) Giant cell tumour of bone. (b) Eccentric lytic "soap-bubble" lesion. (c) RANKL secreted by the neoplastic mononuclear cells.

RAPID REVISION The Night-Before Page

15 ONE-LINERS

01 RANKL drives osteoclasts; OPG is the decoy; oestrogen raises OPG.

02 Osteoporosis: less bone, normal mineral, normal labs; T-score ≤ −2.5.

03 Postmenopausal: vertebral crush; senile: thin cortex, hip fracture.

04 Steroids are the commonest secondary cause of osteoporosis.

05 Paget: lytic → mixed → sclerotic; mosaic bone; very high ALP.

06 Rickets: rosary, bossing, bowed legs; osteomalacia: Looser zones.

07 Vitamin D deficiency: low PO_4, high ALP, high PTH.

08 Osteomyelitis: S. aureus; children metaphysis; sickle cell Salmonella.

09 Sequestrum = dead bone; involucrum = new sleeve; Brodie = walled abscess.

10 Pott: TB crosses discs, gibbus, cold psoas abscess.

11 Osteochondroma: commonest benign; cap + stalk; EXT1/2.

12 Osteoid osteoma: cortex, nidus < 2 cm, night pain, aspirin.

13 Osteosarcoma: metaphysis, knee, < 20 y; osteoid by tumour cells; lungs.

14 Ewing: diaphysis, t(11;22), CD99, onion-skin, mimics infection.

15 GCT: epiphysis, 20–40 y, RANKL, soap-bubble; recurs locally.

MEMORY HOOKS

- **G-O-E from the joint**: GCT (epiphysis), Osteosarcoma (metaphysis), Ewing (diaphysis)
- **Sequestrum dead, Involucrum new**: "S before I"
- **Prostate builds, kidney eats**: blastic vs lytic metastases

TEST YOURSELF: FILL THE GAPS

1. Decoy receptor for RANKL: _____
2. Pseudofractures of osteomalacia: _____ zones.
3. Osteomyelitis in sickle cell: _____
4. Osteochondroma genes: _____
5. Ewing sarcoma IHC marker: _____.
6. Reactive bone at lifted periosteum: _____ triangle.

Answers: OPG · Looser · Salmonella · EXT1/EXT2 · CD99 · Codman

DRAW FROM MEMORY

A long bone with epiphysis, metaphysis, diaphysis and the seven tumour sites (Figure 27.5).

DRAW FROM MEMORY

Acute osteomyelitis from metaphyseal seeding to sequestrum, involucrum and sinus (Figure 27.4).

SPACED REVISION ☐ Day 1 ☐ Day 3 ☐ Day 7 ☐ Day 14 ☐ Day 30

28

Joints & Skeletal Muscle

A joint can be worn out, attacked by its own immune system, invaded by bacteria or choked with crystals. A muscle fails because its nerve has gone or because the fibre itself is breaking. Ask "which one?" and the slide answers itself.

BY THE END OF THIS CHAPTER YOU CAN

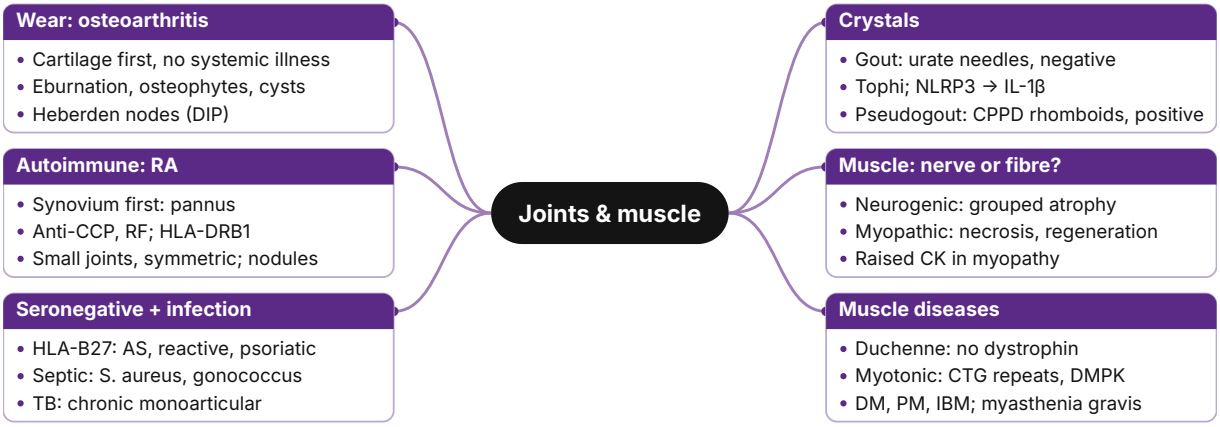
1. Differentiate osteoarthritis from rheumatoid arthritis in pathogenesis, morphology and clinical pattern
2. Describe the pathogenesis and morphology of rheumatoid arthritis, including pannus and the rheumatoid nodule
3. Outline the seronegative spondyloarthropathies and septic and tuberculous arthritis
4. Explain the pathogenesis of gout and compare gout with pseudogout
5. Differentiate neurogenic from myopathic changes in muscle
6. Compare Duchenne and Becker muscular dystrophy; outline myotonic dystrophy, inflammatory myopathies and myasthenia gravis

EXAM RADAR

Theory SEQs	★★★
BCQs / MCQs	★★★
Viva voce	★★☆
OSPE / slides	★★☆

OA vs RA, gout and Duchenne muscular dystrophy are the favourite long questions; crystals, autoantibodies and muscle biopsy patterns are classic BCQs.

ROBBINS LINK Robbins Basic Pathology — *Bones, Joints, and Soft Tissue Tumors* (arthritis, gout) and *Peripheral Nerves and Muscles* (neurogenic vs myopathic change, muscular dystrophies, inflammatory myopathies, neuromuscular junction).



Chapter at a glance. Four ways a joint fails, and two ways a muscle fails.

28.1 Osteoarthritis

DEFINITION

Osteoarthritis (OA), or degenerative joint disease, is the commonest joint disease. Its essential feature is **degeneration of articular cartilage**; the structural and functional failure of the joint follows. Synovial inflammation, if any, is secondary and mild.

- **Primary (idiopathic):** appears with ageing, often in a few joints, without obvious cause. Weight-bearing joints and hands: **knees, hips**, lower lumbar and cervical spine, **DIP and PIP** joints, first carpometacarpal and first MTP joints.
- **Secondary:** at any age in a previously damaged joint: trauma, developmental deformity, **obesity**, diabetes, haemochromatosis, ochronosis, previous inflammatory arthritis.
- **Clinical:** after 50; deep aching pain **worse with use**, short morning stiffness (< 30 min), crepitus, limited movement. **Heberden nodes** (DIP) and Bouchard nodes (PIP) are osteophytes, prominent in women. No systemic features.



Figure 28.1 · Pathogenesis of osteoarthritis. The chondrocyte fails under load.

GROSS

- Cartilage **soft, granular, fibrillated**, then lost in patches
- Exposed subchondral bone polished like ivory: **eburnation**
- **Osteophytes**: mushroom-shaped bony spurs at the joint margins
- **Loose bodies** ("joint mice"): dislodged cartilage and bone fragments

MICROSCOPY

- Early: **clusters (clones) of proliferating chondrocytes**, loss of proteoglycan
- **Fibrillation**: vertical and horizontal clefts in the matrix
- Thickened sclerotic subchondral bone; **subchondral cysts** with fibrous walls (synovial fluid forced through cracks)
- Synovium: mild, secondary congestion and fibrosis

28.2 Rheumatoid arthritis

DEFINITION

Rheumatoid arthritis (RA) is a chronic systemic autoimmune disease that mainly attacks joints, producing a **non-suppurative proliferative and inflammatory synovitis** that often progresses to destruction of articular cartilage and ankylosis. It affects about 1% of people, women **3–5 times** more than men, usually in middle age.

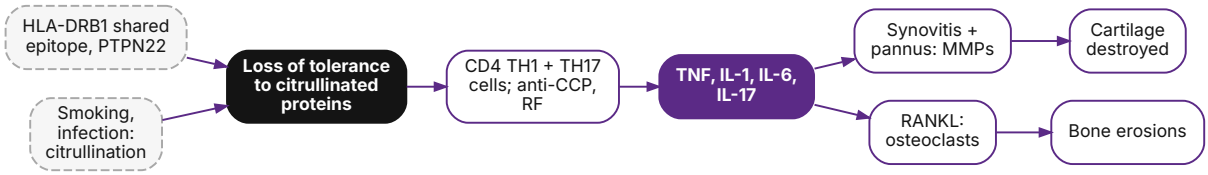


Figure 28.2 · Pathogenesis of rheumatoid arthritis. T cells start it, cytokines sustain it, pannus destroys the joint.

CONCEPT CLEAR

Why anti-TNF works. The inflamed synovium becomes the destroyer: **TNF and IL-1** make synovial cells release **metalloproteinases** that digest cartilage, while **RANKL** on T cells and synovial fibroblasts recruits osteoclasts that erode bone. Blocking TNF slows both arms.

JOINTS

- **Symmetric polyarthritis**: small joints of hands and feet first (**MCP, PIP**), then wrists, ankles, elbows, knees; **DIP usually spared**; cervical spine (atlanto-axial subluxation)
- Synovium swollen, thick, with **villous** projections
- **Pannus**: a mass of inflamed synovium, granulation tissue and fibroblasts that creeps over and erodes the cartilage
- Late: **fibrous then bony ankylosis**; **ulnar deviation** of fingers, radial deviation of wrist, **swan-neck** and **boutonnière** deformities

SYNOVIUM

- **Synovial cell hyperplasia**
- Dense infiltrate of **CD4+ T cells, B cells, plasma cells** and macrophages, often forming **lymphoid follicles**
- **Increased vascularity** (angiogenesis)
- **Fibrin** on the synovial surface; neutrophils in the fluid
- Osteoclastic activity in underlying bone: **marginal erosions**, subchondral cysts, periarticular osteoporosis

Rheumatoid nodule (about a quarter of patients, usually severe, RF-positive disease): firm, non-tender subcutaneous nodule over pressure points (**extensor forearm, elbow**, occiput). Micro: **central fibrinoid necrosis** surrounded by a **palisade of epithelioid macrophages** and an outer rim of lymphocytes, plasma cells and granulation tissue.

Extra-articular disease: lung (interstitial fibrosis, pleuritis, nodules; **Caplan syndrome** with pneumoconiosis), pericarditis, **vasculitis**, scleritis, Sjögren syndrome, **Felty syndrome** (RA + splenomegaly + neutropenia), anaemia of chronic disease, **secondary AA amyloidosis**.

LAB VALUES

Anti-CCP (ACPA): about 70%, **highly specific**, may precede symptoms by years. **Rheumatoid factor:** IgM (or IgA) against the **Fc of IgG**; about 80%, not specific. **X-ray:** effusion, **juxta-articular osteopenia**, marginal erosions, joint-space narrowing.

OSTEOARTHRITIS VS RHEUMATOID ARTHRITIS: THE MOST-ASKED COMPARISON IN JOINT PATHOLOGY

Feature	Osteoarthritis	Rheumatoid arthritis
Nature / target	Degenerative; **articular cartilage**	**Autoimmune, systemic**; **synovium**
Age / sex	Elderly; both sexes	Any adult age, peak middle age; **women** 3–5 ×
Joints	Weight-bearing (hips, knees), **DIP**, PIP; asymmetric	Small joints of hands and feet (**MCP, PIP**), wrists; **symmetric**; DIP spared
Morning stiffness	Brief (< 30 min); pain worse with use	**Prolonged (> 1 hour)**; improves with use
Key lesions	Fibrillation, **eburnation**, **osteophytes**, subchondral cysts	**Pannus**, lymphoid follicles, erosions, **ankylosis**
Nodules	**Heberden** (DIP), Bouchard (PIP): bony	**Rheumatoid nodules**: necrobiotic, subcutaneous
Labs	Normal ESR; RF negative; non-inflammatory fluid	**Raised ESR/CRP**, **anti-CCP**, RF; inflammatory fluid

28.3 Seronegative spondyloarthropathies and infectious arthritis

Spondyloarthropathies are **RF-negative** arthritides in which inflammation starts at the **entheses** (where ligaments and tendons attach to bone) rather than in the synovium. They involve the **sacroiliac joints and spine**, are linked to HLA-B27, and are often triggered by infection.

Disease	Key points
Ankylosing spondylitis	Young men; **sacroiliitis**, spinal fusion (“**bamboo spine**”); uveitis; HLA-B27 in about 90%
Reactive arthritis	Weeks after **GU** (Chlamydia) or **GI** infection (Shigella, Salmonella, Yersinia, Campylobacter); triad of **arthritis, non-gonococcal urethritis or cervicitis, conjunctivitis**
Psoriatic arthritis	DIP joints, dactylitis (“sausage digit”), nail pitting; “pencil-in-cup” erosions
Enteropathic arthritis	With Crohn disease or ulcerative colitis

Suppurative (septic) arthritis

Bacteria reach the joint by blood (commonest), from adjacent osteomyelitis or by direct inoculation. A **medical emergency**: enzymes from neutrophils destroy cartilage within days.

- **Organisms:** *H. influenzae* under 2 years; **S. aureus** in older children and adults; **gonococcus** in adolescents and young adults (complement C5–C9 deficiency predisposes to disseminated disease); **Salmonella** in sickle cell disease
- Usually **one joint**, the **knee** commonest: sudden hot, swollen, painful joint, fever, leukocytosis
- **Joint aspiration** is diagnostic: purulent fluid, organisms

Tuberculous arthritis

A **chronic, progressive, monoarticular** arthritis, from adjacent tuberculous osteomyelitis or blood spread. Weight-bearing joints: **hips, knees**, ankles; spine as Pott disease (Chapter 27).

- Insidious pain, swelling, low-grade fever, weight loss
- **Micro:** synovium with **caseating granulomas**; a pannus of granulation tissue destroys cartilage and bone
- Heals with **fibrous ankylosis** and obliteration of the joint space

LAB VALUES

Synovial fluid at a glance. Normal: clear, viscous, WBC < 200/µL. **Non-inflammatory (OA):** clear yellow, WBC < 2,000/µL. **Inflammatory (RA, gout, pseudogout):** cloudy, low viscosity, WBC 2,000–50,000/µL, mostly neutrophils; look for **crystals**. **Septic:** turbid or purulent, WBC usually > 50,000/µL with > 75% neutrophils, low glucose, Gram stain and culture positive.

28.4 Gout and pseudogout

DEFINITION

Gout is a disease of recurrent attacks of acute arthritis caused by **monosodium urate (MSU) crystals** deposited in and around joints, leading in time to chronic gouty arthritis and to masses of urate called **tophi**. **Hyperuricaemia** (plasma urate > 6.8 mg/dL) is essential but not sufficient: most hyperuricaemic people never develop gout.

CAUSES OF HYPERURICAEMIA

- **Primary gout (about 90%)**: usually **reduced renal excretion** of urate; less often overproduction. Rare enzyme defects: partial **HGPRT** deficiency (complete = **Lesch–Nyhan** syndrome), PRPP synthetase overactivity
- **Secondary (about 10%)**: high cell turnover (**leukaemia**, myeloproliferative disease, **chemotherapy**: tumour lysis), **chronic renal disease**, drugs (**thiazides**, low-dose aspirin), alcohol, lead, von Gierke disease
- Risk of attacks: age (men after 30, women after menopause), alcohol, obesity, purine-rich meals, genes

CONCEPT CLEAR

Why the big toe at night? Urate is less soluble at **lower temperature** and lower pH. The first MTP joint is cool, peripheral and repeatedly traumatised by walking, and at night body temperature dips and the joint fluid is reabsorbed, concentrating urate. Crystals precipitate and the attack begins.



Figure 28.3 · Pathogenesis of acute gout. Crystals are treated as danger signals.

ACUTE AND CHRONIC GOUTY ARTHRITIS

- **Acute:** red, hot, exquisitely painful joint, classically the **first MTP joint (podagra)**: about half of first attacks, most patients eventually
- **Chronic tophaceous:** chalky urate crust on cartilage, thick fibrotic synovium, **pannus**, juxta-articular **punched-out erosions**, ankylosis; on average about 12 years after the first attack
- **Tophi** in ear helix, olecranon bursa, Achilles tendon, fingers; may ulcerate and discharge chalky material
- **Kidney:** urate in medullary interstitium (gouty nephropathy), **uric acid stones**

MICROSCOPY

- Acute: dense **neutrophilic infiltrate** in synovium and fluid; **long, slender, needle-shaped** crystals, many inside neutrophils
- **Tophus (pathognomonic)**: amorphous mass of urate crystals surrounded by **macrophages, lymphocytes and foreign-body giant cells**
- Polarised light: crystals are **strongly negatively birefringent** (yellow when parallel to the compensator axis)
- Fix in **alcohol**: urate dissolves in aqueous formalin

Stages: asymptomatic hyperuricaemia → acute attacks → symptom-free intercritical periods → chronic tophaceous gout. Treat attacks with NSAIDs, colchicine or steroids; prevent with allopurinol or febuxostat.

Pseudogout (calcium pyrophosphate dihydrate, **CPPD**, crystal deposition disease) usually begins after **50 years**. Crystals form first in articular cartilage, menisci and discs (**chondrocalcinosis**) when pyrophosphate accumulates in the matrix; they later shed into the joint and provoke inflammation. It is **sporadic** (age), **hereditary** (*ANKH* mutation)

or **secondary** to joint damage, **hyperparathyroidism, haemochromatosis**, hypomagnesaemia, hypothyroidism or diabetes. On H&E the deposits are oval **blue-purple** aggregates.

GOUT VS PSEUDOGOUT: READ THE CRYSTAL

Feature	Gout	Pseudogout (CPPD)
Crystal	**Monosodium urate**	**Calcium pyrophosphate dihydrate**
Shape	**Long needles**	**Rhomboid** or short rods
Birefringence	**Strongly negative** (yellow parallel)	**Weakly positive** (blue parallel)
Typical patient	Middle-aged man, hyperuricaemia	Elderly (> 50); both sexes
Typical joint	**First MTP** (big toe)	**Knee**, wrist, shoulder
X-ray	Punched-out erosions, tophi	**Chondrocalcinosis** (calcified cartilage)
Associations	Diuretics, alcohol, renal disease, leukaemia	Age, hyperparathyroidism, haemochromatosis, low Mg

28.5 Skeletal muscle: neurogenic or myopathic?

A muscle biopsy answers one question first: is the **nerve** or the **muscle fibre** diseased? Normal muscle shows a **checkerboard** mix of type 1 (slow, oxidative) and type 2 (fast, glycolytic) fibres, because one motor neuron supplies fibres scattered through the fascicle.

NEUROGENIC VS MYOPATHIC CHANGE ON MUSCLE BIOPSY

Feature	Neurogenic (denervation)	Myopathic (primary muscle)
Atrophic fibres	Small, **angular**, both fibre types; **grouped atrophy**	Small **rounded** fibres scattered, with hypertrophic fibres
Fibre types	**Fibre-type grouping** (reinnervation by sprouting axons)	Checkerboard preserved
Necrosis / regeneration	Absent	**Segmental necrosis**, myophagocytosis, **regenerating basophilic fibres**
Other	Target fibres	**Internal nuclei**, fibre splitting, endomysial fibrosis, fat; inflammation in myositis
Clinical	Distal weakness, fasciculations; CK normal or mildly raised	**Proximal weakness**; CK often **raised**
Examples	Motor neuron disease, spinal muscular atrophy, **poliomyelitis**, neuropathies	Muscular dystrophies, myositis, toxic and metabolic myopathies

CONCEPT CLEAR

Why denervation makes groups. When a motor neuron dies, neighbouring axons sprout, adopt its orphaned fibres and convert them to their own type: **type grouping**. If the adopting neuron dies too, the whole patch shrinks: **grouped atrophy**. Disuse and steroids give non-specific **type 2 fibre atrophy**.

PAKISTAN FOCUS

Pakistan and Afghanistan remain the **last countries with endemic wild poliovirus type 1**. Polio destroys anterior horn motor neurons, so the paralysed muscle shows classic **neurogenic atrophy**. Every missed vaccine dose keeps this picture alive.

28.6 Muscle diseases

Muscular dystrophies: Duchenne and Becker

Muscular dystrophies are inherited diseases of progressive **muscle fibre degeneration** with replacement by fibrous and fatty tissue. **Duchenne muscular dystrophy (DMD)** is the commonest and most severe: X-linked, about **1 in 3,500 live male births**. The *DMD* gene at **Xp21** is one of the largest human genes, so new mutations are frequent; **deletions** account for about two-thirds.

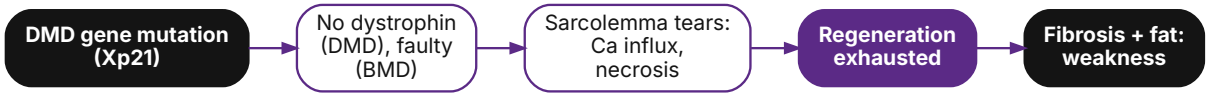


Figure 28.4 · Why dystrophin loss destroys muscle.

CONCEPT CLEAR

Frame decides fate. Dystrophin anchors actin to the **dystroglycan–sarcoglycan complex** and the basement membrane, like a shock absorber. An **out-of-frame** deletion scrambles the code beyond it, so **no dystrophin** is made: Duchenne. An **in-frame** deletion yields a shorter but **partly working** dystrophin: Becker, milder and later.

MICROSCOPY

- **Marked variation in fibre size**: small and hypertrophic rounded fibres, fibre splitting
- **Necrosis, myophagocytosis** and basophilic **regenerating fibres**; internal nuclei
- **Endomysial fibrosis**; late, extensive **fatty replacement**
- **Immunostain: dystrophin absent** (DMD) or reduced/patchy (BMD)

DUCHENNE VS BECKER MUSCULAR DYSTROPHY

Feature	Duchenne	Becker
Dystrophin	**Absent** (out-of-frame mutation)	**Reduced or truncated**, partly functional (in-frame)
Onset	Early childhood: walking late, clumsy, weak **by about 5 years**	Later childhood, adolescence or adult life
Weakness	Pelvic girdle first, then shoulder; **Gowers sign** (climbs up own legs to stand); waddling gait	Same pattern, much milder and slower
Calves	**Pseudohypertrophy**: enlarged by fat and fibrous tissue	Pseudohypertrophy common
Course	**Wheelchair by 10–12 years**; death in early adult life from respiratory failure, pneumonia or cardiomyopathy	Slow progression; walking into adult life; **near-normal lifespan**; cardiomyopathy may dominate
Other	Serum **CK very high** early, falling as muscle is lost; cardiomyopathy; some have cognitive impairment	CK raised

Carrier mothers are usually asymptomatic with a raised CK. Diagnosis: CK, genetic testing for deletions, biopsy with dystrophin immunostaining.

Myotonic dystrophy

The **commonest muscular dystrophy of adults**; autosomal dominant. A **CTG trinucleotide repeat expansion** in the 3′ untranslated region of the DMPK gene (chromosome 19q) makes a toxic RNA that traps splicing regulators (mis-spliced chloride channel → myotonia). Repeats lengthen in successive generations: **anticipation** (congenital form when transmitted by the mother). **Myotonia**: sustained contraction with slow relaxation, "cannot let go of a handshake". Weakness is **distal and facial** ("**hatchet face**", ptosis); **cataracts**, **frontal balding**, testicular atrophy, conduction defects. Biopsy: chains of **internal nuclei**, **ring fibres**, sarcoplasmic masses, type 1 fibre atrophy; little necrosis.

Inflammatory myopathies

Feature	Dermatomyositis	Polymyositis	Inclusion body myositis
Who	**Children** and adults	Adults	**Over 50** (commonest inflammatory myopathy over 65), men
Muscles	Proximal, symmetric	Proximal, symmetric	**Distal**, often asymmetric: quadriceps, finger flexors
Skin	**Heliotrope** (lilac) eyelids, **Gottron papules** on knuckles, shawl rash	None	None
Mechanism	Type I interferon-driven injury of **capillaries**, complement (MAC)	**CD8+ T cells** attack muscle fibres	Degenerative + inflammatory

Feature	Dermatomyositis	Polymyositis	Inclusion body myositis
Biopsy	**Perifascicular atrophy**; perivascular and perimysial inflammation; capillary loss	**Endomysial** lymphocytes invading non-necrotic fibres; MHC class I up	**Rimmed vacuoles**; inclusions of β-amyloid, tau, TDP-43; endomysial T cells
Extras	Anti-Mi-2, anti-Jo-1 (interstitial lung disease); **visceral cancer** risk in adults	Anti-Jo-1 may be present	**Poor response** to steroids

Myasthenia gravis (in brief)

Autoimmune disease of the **neuromuscular junction**: antibodies against the **acetylcholine receptor** (about 85%; others against MuSK or LRP4) block and internalise receptors and fix complement. The thymus is abnormal in most: **thymic hyperplasia** or **thymoma**. **Fatigable weakness** that worsens with use and recovers with rest; **ptosis and diplopia** first, then bulbar and limb muscles. Improves with anticholinesterases and thymectomy; biopsy is usually unremarkable.

⚠ EXAM TRAP

Myasthenia vs Lambert–Eaton. Lambert–Eaton syndrome has antibodies to **presynaptic voltage-gated calcium channels**, is often **paraneoplastic** (small cell lung carcinoma), spares the eyes, has autonomic symptoms and shows **strength that improves** with repeated effort (incremental response), the opposite of myasthenia gravis.

EXAM ZONE Past-Paper Pattern SEQs — Solved

CLASSIC **a)** Differentiate between osteoarthritis and rheumatoid arthritis. **b)** Describe the pathogenesis and morphology of rheumatoid arthritis. **2 + 3 marks**

MODEL ANSWER PLAN

a) Five points: cartilage vs synovium; degenerative vs autoimmune; DIP/weight-bearing vs MCP/PIP symmetric; osteophytes and eburnation vs pannus and ankylosis; normal labs vs anti-CCP, RF, high ESR. **b)** HLA-DRB1 + smoking → citrullinated proteins → CD4 TH1/TH17 cells, anti-CCP → TNF, IL-1, IL-6, IL-17 → synovitis, pannus (MMPs), RANKL-driven erosions. Morphology: villous hyperplastic synovium, lymphoid follicles, plasma cells, angiogenesis, fibrin, pannus, ankylosis; rheumatoid nodule.

Score booster: draw the OA vs RA table and add one labelled deformity (ulnar deviation or swan-neck).

PROF PATTERN A 48-year-old man wakes at night with an excruciatingly painful, red, swollen big toe after a wedding feast. He takes a thiazide diuretic. **a)** Diagnosis. **b)** Pathogenesis. **c)** Morphology of a tophus and findings in joint fluid. **1 + 2 + 2 marks**

MODEL ANSWER PLAN

a) Acute gouty arthritis (podagra). **b)** Hyperuricaemia (thiazide reduces excretion, purine-rich meal, alcohol) → MSU crystals → macrophage NLRP3 inflammasome → IL-1β → neutrophils → acute inflammation. **c)** Tophus: urate crystals surrounded by macrophages, lymphocytes and foreign-body giant cells. Fluid: inflammatory, neutrophils with intracellular needle-shaped, strongly negatively birefringent crystals.

MODULE PATTERN A 5-year-old boy rises from the floor by "climbing up" his legs; his calves are enlarged. **a)** Diagnosis and genetic basis. **b)** Describe the muscle biopsy and compare with Becker dystrophy. **2 + 3 marks**

MODEL ANSWER PLAN

a) Duchenne muscular dystrophy: X-linked, *DMD* gene at Xp21, out-of-frame deletion → no dystrophin. **b)** Fibre size variation, necrosis, regeneration, internal nuclei, endomysial fibrosis, fat replacement, absent dystrophin staining. Becker: in-frame mutation, reduced dystrophin, later onset, slower course, near-normal lifespan.

BCQs One-Best-Answer Practice

1 A 65-year-old woman has knee pain worse on walking, morning stiffness for 10 minutes and bony swellings of the DIP joints. The primary tissue affected is:

A. Synovium
B. Articular cartilage
C. Enthesis
D. Subcutaneous tissue
E. Joint capsule

2 The autoantibody most specific for rheumatoid arthritis is directed against:

A. Citrullinated peptides
B. Fc portion of IgG
C. Double-stranded DNA
D. Centromere proteins
E. Acetylcholine receptor

3 Central fibrinoid necrosis surrounded by palisading epithelioid macrophages in a nodule over the elbow is a:

A. Tophus
B. Caseating granuloma
C. Heberden node
D. Rheumatoid nodule
E. Gumma

4 A 25-year-old man has low back stiffness, sacroiliitis and recurrent anterior uveitis. The most likely genetic association is:

A. HLA-DR3
B. HLA-B27
C. HLA-DRB1 shared epitope
D. HLA-DQ2
E. HLA-B51

5 A sexually active 20-year-old woman has an acutely swollen knee and pustular skin lesions. She has had two episodes of neisserial meningitis. She most likely has deficiency of:

A. C1 inhibitor
B. C3
C. C4
D. C5–C9 (membrane attack complex)
E. C2

6 Synovial fluid shows needle-shaped crystals that are yellow when parallel to the compensator axis. The crystals are:

A. Monosodium urate
B. Calcium pyrophosphate
C. Cholesterol
D. Hydroxyapatite
E. Calcium oxalate

7 The inflammatory mediator central to acute gout, released after crystals activate the NLRP3 inflammasome, is:

A. IL-4
B. Histamine
C. IL-1β
D. Interferon-γ
E. Leukotriene E4

8 A muscle biopsy shows patches of fibres of a single histochemical type and groups of small angular atrophic fibres. The disease is most likely:

A. Duchenne dystrophy
B. Polymyositis
C. Spinal muscular atrophy
D. Myotonic dystrophy
E. Steroid myopathy

9 Perifascicular atrophy on muscle biopsy is characteristic of:

A. Polymyositis
B. Inclusion body myositis
C. Becker dystrophy
D. Myasthenia gravis
E. Dermatomyositis

10 A man has myotonia, frontal balding, cataracts and a hatchet face; his son is more severely affected. The mutation is:

A. CTG repeat expansion in DMPK
B. Out-of-frame deletion of dystrophin
C. CAG repeat in huntingtin
D. Point mutation in SMN1
E. Sarcoglycan mutation

11 Becker muscular dystrophy differs from Duchenne in that:

A. It is autosomal dominant
B. Dystrophin is completely absent
C. Calf pseudohypertrophy never occurs
D. CK is normal
E. Dystrophin is reduced or truncated but partly functional

12 Fatigable ptosis and diplopia that improve with an anticholinesterase are due to antibodies against:

A. Presynaptic calcium channels
B. Postsynaptic acetylcholine receptors
C. Dystrophin
D. Myelin basic protein
E. Ganglioside GM1

ANSWER KEY WITH ONE-LINE REASONS

1. **B** OA begins with degeneration of articular cartilage; Heberden nodes are osteophytes.
2. **A** Anti-CCP (ACPA) is highly specific; RF (anti-Fc IgG) is not.
3. **D** Classic rheumatoid nodule in seropositive RA.
4. **B** Ankylosing spondylitis: HLA-B27 in about 90%.
5. **D** Terminal complement deficiency predisposes to disseminated neisserial infection.
6. **A** Strong negative birefringence = urate (gout).
7. **C** Caspase-1 cleaves pro-IL-1β; IL-1 blockers abort attacks.
8. **C** Fibre-type grouping and grouped atrophy are neurogenic.

9. **E** Capillary injury hits the periphery of fascicles in dermatomyositis.
10. **A** Myotonic dystrophy: CTG expansion with anticipation.
11. **E** In-frame mutations produce a partly working dystrophin.
12. **B** Myasthenia gravis: anti-AChR in about 85%; thymic hyperplasia or thymoma.

VIVA Rapid-Fire Viva Voce

Q What is eburnation?	Polished ivory-like subchondral bone exposed after full-thickness cartilage loss in osteoarthritis.
Q What are Heberden nodes?	Osteophytes at the DIP joints in osteoarthritis.
Q Define pannus.	Inflamed hyperplastic synovium with granulation tissue that grows over and erodes articular cartilage in RA.
Q What is rheumatoid factor?	IgM (or IgA) autoantibody against the Fc portion of IgG.
Q Name the triad of reactive arthritis.	Arthritis, non-gonococcal urethritis or cervicitis, and conjunctivitis.
Q Commonest organism in septic arthritis of adults?	Staphylococcus aureus; gonococcus in sexually active young adults.
Q What is a tophus?	A mass of urate crystals surrounded by macrophages, lymphocytes and foreign-body giant cells.
Q Why fix a gouty specimen in alcohol?	Urate crystals dissolve in aqueous formalin.
Q Crystals of pseudogout?	Calcium pyrophosphate: rhomboid, weakly positively birefringent.
Q What is Gowers sign?	Using the hands to climb up the thighs to stand: proximal (pelvic girdle) weakness, as in Duchenne.
Q What is pseudohypertrophy?	Calf enlargement due to fat and fibrous tissue replacing muscle in Duchenne dystrophy.
Q What is grouped atrophy?	Clusters of atrophic fibres of one type when a reinnervating motor neuron dies: neurogenic disease.

OSPE Practice Stations

STATION 1 · LAB DATA — 2 minutes

TASK

Knee aspirate of a 72-year-old woman with an acute hot knee: cloudy, WBC 24,000/µL (80% neutrophils), Gram stain negative. Under polarised light: short rhomboid crystals, weakly positively birefringent. X-ray: calcified menisci. (a) Diagnosis. (b) Name the crystal. (c) Two associated diseases to exclude.

ANSWER

(a) Acute pseudogout (CPPD deposition disease). (b) Calcium pyrophosphate dihydrate. (c) Hyperparathyroidism, haemochromatosis (also hypomagnesaemia, hypothyroidism).

STATION 2 · SLIDE — 2 minutes

TASK

Synovium with tall villous folds, a hyperplastic lining, dense lymphocytes and plasma cells forming follicles, many new vessels and fibrin on the surface. (a) Identify. (b) Name the destructive tissue it forms over cartilage. (c) Most specific serological test.

ANSWER

(a) Rheumatoid synovitis (rheumatoid arthritis). (b) Pannus. (c) Anti-CCP (anti-citrullinated peptide) antibodies.

STATION 3 · SLIDE 2 minutes

TASK

Muscle biopsy from a boy: marked variation in fibre size with rounded hypertrophic and small fibres, scattered necrotic fibres being phagocytosed, basophilic regenerating fibres, increased endomysial fibrous tissue and fat. Immunostain for dystrophin is negative. (a) Diagnosis. (b) Mode of inheritance. (c) Expected serum enzyme change.

ANSWER

(a) Duchenne muscular dystrophy. (b) X-linked recessive. (c) Markedly raised creatine kinase.

RAPID REVISION The Night-Before Page

15 ONE-LINERS

01 OA: cartilage first; fibrillation, eburnation, osteophytes, subchondral cysts.

02 Heberden nodes (DIP) = OA; RA spares DIP.

03 RA: synovium first; symmetric MCP/PIP; pannus; ankylosis.

04 RA: HLA-DRB1 + smoking → anti-CCP; TNF drives MMPs; RANKL erodes bone.

05 RF = IgM against Fc of IgG; anti-CCP is the specific test.

06 Rheumatoid nodule: fibrinoid necrosis + palisading macrophages.

07 Spondyloarthropathies: RF-negative, entheses, sacroiliac, HLA-B27.

08 Septic arthritis: S. aureus; gonococcus in young adults; knee.

09 TB arthritis: chronic monoarticular hip/knee, fibrous ankylosis.

10 Gout: urate needles, strongly negative; first MTP; NLRP3 → IL-1β.

11 Tophus: urate + foreign-body giant cells; fix in alcohol.

12 Pseudogout: CPPD rhomboids, weakly positive; knee; chondrocalcinosis.

13 Neurogenic: angular fibres, type grouping, grouped atrophy.

14 Duchenne: Xp21, no dystrophin, Gowers, pseudohypertrophy, CK high.

15 DM: perifascicular atrophy; PM: CD8 endomysial; IBM: rimmed vacuoles.

MEMORY HOOKS

- **Needles Negative, Pyrophosphate Positive**: gout vs pseudogout
- **"OA wears, RA tears"**: cartilage vs synovium
- **Frame decides fate**: out-of-frame Duchenne, in-frame Becker

TEST YOURSELF: FILL THE GAPS

1. Tissue that erodes cartilage in RA: _____.
2. Gout crystals are strongly _____ birefringent.
3. Inflammasome in gout: _____.
4. Dystrophin gene locus: _____.
5. Myotonic dystrophy repeat: _____ in DMPK.
6. Biopsy sign of dermatomyositis: _____ atrophy.

Answers: pannus · negatively · NLRP3 · Xp21 · CTG · perifascicular

DRAW FROM MEMORY

The OA vs RA joint side by side: osteophyte, eburnation and cyst vs pannus, erosion and ankylosis.

DRAW FROM MEMORY

The gout chain from hyperuricaemia to NLRP3, IL-1β and neutrophils (Figure 28.3).

SPACED REVISION ☐ Day 1 ☐ Day 3 ☐ Day 7 ☐ Day 14 ☐ Day 30

29

CNS I: Cerebrovascular Disease, Infections, Demyelinating & Degenerative Disease

The brain cannot store fuel, cannot replace its neurons and lives inside a rigid box. That is why four minutes without oxygen, a few millilitres of extra fluid or one misfolded protein can change a person for ever.

BY THE END OF THIS CHAPTER YOU CAN

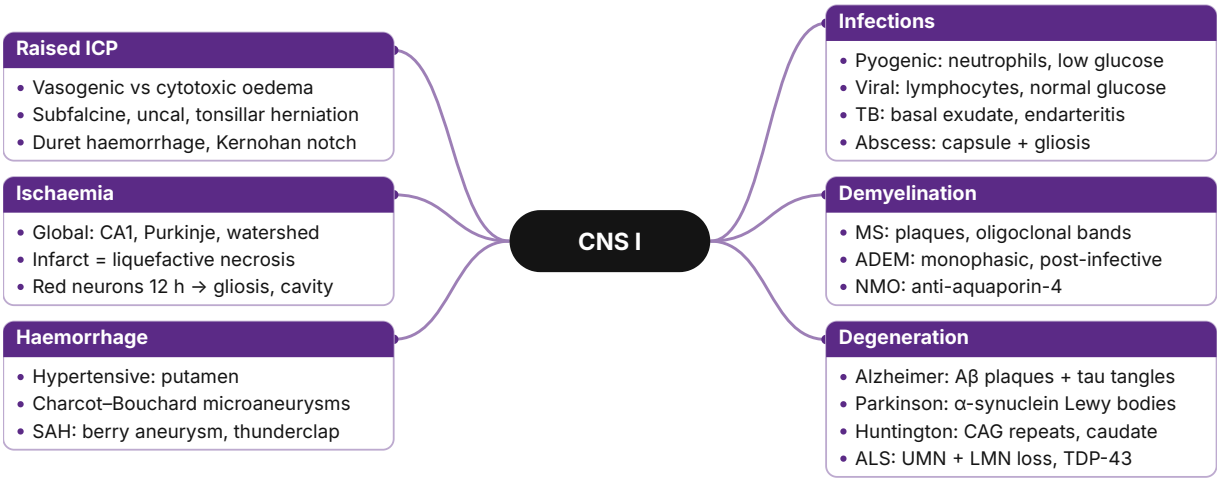
1. Explain cerebral oedema, raised intracranial pressure and the three herniation syndromes
2. Describe global ischaemia and the evolution of a cerebral infarct from hours to months
3. Differentiate hypertensive intracerebral haemorrhage from subarachnoid haemorrhage due to a berry aneurysm
4. Interpret CSF findings to separate bacterial, viral and tuberculous meningitis, and describe brain abscess
5. Describe the pathogenesis and plaque morphology of multiple sclerosis and contrast it with ADEM
6. Name the protein, inclusion and target region in Alzheimer, Parkinson and Huntington disease and ALS

EXAM RADAR

Theory SEQs	★★★
BCQs / MCQs	★★★
Viva voce	★★★
OSPE / slides	★★☆

Meningitis with CSF findings, cerebral infarction and Alzheimer disease are perennial long questions; inclusions, herniations and aneurysm sites are favourite BCQs.

ROBBINS LINK Robbins Basic Pathology — *Central Nervous System*: oedema, herniation and hydrocephalus; cerebrovascular diseases; infections; demyelinating and neurodegenerative diseases.



Chapter at a glance. Six themes cover every non-tumour CNS question.

29.1 Cerebral oedema, raised ICP and herniation

CONCEPT CLEAR

The skull is a closed box. Brain, blood and CSF fill it, and the total volume cannot grow. A little extra volume (oedema, haematoma, tumour, abscess) is absorbed by squeezing out CSF and venous blood; once that reserve is used up, intracranial pressure (ICP) rises steeply. Perfusion then falls, adding ischaemia, and the brain is **pushed past the rigid folds of dura** (falx, tentorium) and through the foramen magnum. This displacement, **herniation**, is what kills.

CEREBRAL OEDEMA

- **Vasogenic:** blood–brain barrier breaks down; **extracellular** fluid around tumours, abscesses,

HYDROCEPHALUS (EXCESS CSF IN VENTRICLES)

- **Non-communicating:** block inside the ventricles (aqueduct stenosis, colloid cyst, posterior fossa

haemorrhage, contusion
- **Cytotoxic:** energy failure stops the Na^+/K^+ pump; **cells swell** (global hypoxia–ischaemia, toxins)
- **Interstitial (hydrocephalic):** CSF forced across the ependyma into periventricular white matter
- Gross: heavy brain, **flattened gyri, narrowed sulci, slit-like ventricles**

tumour)
- **Communicating:** poor resorption at arachnoid villi after meningitis or SAH
- Infants (sutures open): the **head enlarges**
- **Ex vacuo:** passive ventricular enlargement after brain loss (Alzheimer, old infarct)

HERNIATION SYNDROMES: WHAT MOVES, WHAT GETS SQUEEZED

Herniation	What is displaced	Consequences
Subfalcine (cingulate)	Cingulate gyrus under the falx	**Anterior cerebral artery** compressed → infarct (leg weakness)
Transtentorial (uncal)	Uncus / medial temporal lobe over the tentorial edge	**CN III** → **ipsilateral dilated pupil**, impaired eye movement · **posterior cerebral artery** → visual cortex infarct · opposite peduncle pressed on tentorium (**Kernohan notch**) → hemiparesis on the *same* side as the mass (false localising) · **Duret haemorrhages** (linear, flame-shaped) in midbrain and pons
Tonsillar	Cerebellar tonsils through the foramen magnum	**Medulla compressed** → respiratory and cardiac arrest. A lumbar puncture in a patient with a mass lesion can trigger it.

29.2 Hypoxia, ischaemia and cerebral infarction

The brain receives about 15% of resting cardiac output, uses about 20% of the body's oxygen and has almost no energy reserve. It can be starved **globally** (whole-brain hypoperfusion or hypoxia) or **focally** (one artery blocked). A stroke is an acute focal neurological deficit of vascular origin; a TIA is one that resolves, usually within an hour and by definition within 24 hours.

> **DEFINITION**
> **Red neuron** (acute neuronal injury) is the earliest light-microscopic sign of irreversible hypoxic damage, seen **12–24 hours** after the insult: shrunken cell body, intensely **eosinophilic** cytoplasm, loss of Nissl substance and a **pyknotic** nucleus without a nucleolus.

GLOBAL ISCHAEMIA (CARDIAC ARREST, SHOCK)

Mild episodes cause transient confusion; severe ones kill neurons in a **selectively vulnerable** pattern: **hippocampal CA1 pyramidal cells** (Sommer sector), **cerebellar Purkinje cells** and **neocortical pyramidal neurons** of layers 3, 5, 6 (**pseudolaminar necrosis**). The most severe injury ends in **brain death**.

BORDER-ZONE (WATERSHED) INFARCTS

Hypotension starves the most distal parts of arterial territories. The ACA–MCA border gives a **sickle-shaped band** of necrosis over the convexity, a few centimetres lateral to the interhemispheric fissure.

FOCAL ISCHAEMIA: THROMBOTIC VS EMBOLIC INFARCTION

Feature	Thrombotic occlusion	Embolic occlusion (more common)
Source	Atherosclerotic plaque in situ	**Cardiac mural thrombus** (recent MI, atrial fibrillation, valve disease), carotid plaque, paradoxical embolus
Sites	**Carotid bifurcation**, origin of MCA, ends of basilar artery	**MCA territory** (continues the internal carotid); branch points
Onset	Stepwise; often preceded by TIAs	Sudden, maximal at onset
Infarct	**Pale** (non-haemorrhagic)	Often **red** (haemorrhagic)

CONCEPT CLEAR

Why an embolic infarct turns red. An embolus often fragments or lyses. Blood flows back into vessels whose walls have died and leaks out as **petechial haemorrhages** that may merge. A thrombus on a plaque stays put, so its infarct stays pale. This is why CT comes first: **thrombolytics and anticoagulants** are dangerous in a bleeding brain. And unlike other organs, the brain undergoes **liquefactive necrosis**: it has little stroma and much lipid, and enzymes from neutrophils and microglia dissolve it into a cavity.



Figure 29.1 · Evolution of a non-haemorrhagic cerebral infarct. One timeline answers both "describe the morphology" and "date the lesion".

GROSS

- **48 h:** pale, soft, swollen; **grey–white junction blurred**
- **2–10 days:** gelatinous, friable; margin becomes distinct
- **10 days–3 weeks:** liquefies
- **Late:** fluid-filled **cavity** with a grey wall; overlying meninges intact

MICROSCOPY

- **From 12 h:** red neurons, swollen endothelium and astrocytes, disintegrating myelin
- **Up to 48 h: neutrophils** emigrate, then decline
- **48 h to 2–3 weeks: macrophages** stuffed with myelin and lipid (gitter cells); may persist for years
- **From ~1 week:** reactive astrocytes (**gemistocytes**) at the edge → dense **glial scar**

Haemorrhagic infarct: the same sequence plus extravasated red cells and later **haemosiderin-laden macrophages**. **Stroke at the bedside:** BE-FAST (Balance, Eyes, Face, Arm, Speech, Time).

29.3 Intracranial haemorrhage

ISCHAEMIC VS HAEMORRHAGIC STROKE

Feature	Ischaemic stroke (most strokes)	Haemorrhagic stroke
Mechanism	Thrombosis or embolism → infarction	Vessel rupture → blood in brain or subarachnoid space
Main causes	Atherosclerosis, cardiac emboli	**Hypertension** (ICH), **berry aneurysm** (SAH), amyloid angiopathy, AVM
Onset	Sudden or stepwise; often on waking	Sudden, often on exertion; **severe headache, vomiting**, early coma
Early CT	Often normal; hypodense later	**Hyperdense** blood at once
Thrombolysis	May help (within the time window)	**Contraindicated**

Primary (hypertensive) intracerebral haemorrhage

Spontaneous bleeding into the brain substance, peaking around **60 years**; **hypertension** is the leading cause. Sites: **putamen (50–60%)**, thalamus, pons, cerebellum. Basal ganglia and thalamic bleeds are **ganglionic haemorrhages**.

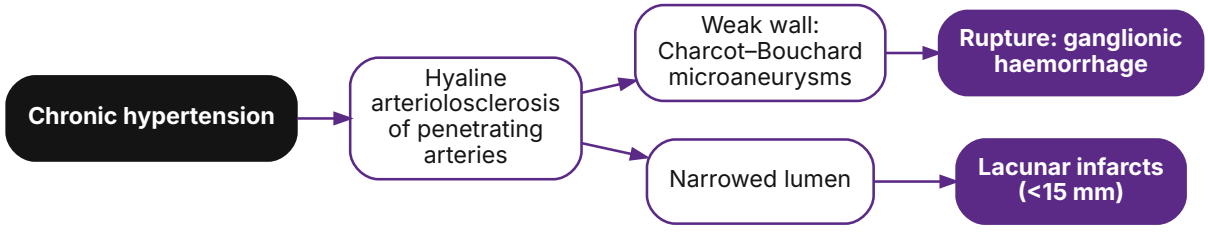


Figure 29.2 · Hypertension and the brain. One vessel lesion, two kinds of stroke.

GROSS

- **Acute:** clot compresses brain; may rupture into ventricles
- **Old: slit-like cavity** with brown haemosiderin lining

- **Lacunes:** small cavities in lenticular nucleus, thalamus, internal capsule, pons; pure motor or sensory strokes. **Slit haemorrhages** come from small penetrating vessels.

- **Cerebral amyloid angiopathy:** Aβ stiffens cortical vessels (Congo red positive) → recurrent **lobar haemorrhages** in the elderly; linked to Alzheimer disease.

Subarachnoid haemorrhage (SAH) and berry aneurysm

The commonest cause of non-traumatic SAH is a ruptured **saccular (berry) aneurysm**; others are AVMs, trauma, extension of an intracerebral bleed, bleeding disorders and tumours.

- Outpouching at a **branch point** of the circle of Willis; about **90% anterior circulation**; multiple in 20–30%
- Sites: **ACA–anterior communicating junction** (commonest) > MCA bifurcation > ICA–posterior communicating junction > basilar tip
- **Not truly congenital:** a gap in the media at the branch point balloons in adult life
- Associations: **ADPKD**, Ehlers–Danlos IV, NF1, Marfan, coarctation; **smoking** and **hypertension**; rupture risk rises sharply above 10 mm
- Wall: hyalinised intima and adventitia; **no smooth muscle or internal elastic lamina**; rupture at the **apex**
- Peak 5th decade: sudden **"worst headache of my life"**, neck stiffness, collapse; 25–50% die with the first bleed
- Complications: **rebleeding**, **vasospasm** → infarcts, **communicating hydrocephalus** (meningeal fibrosis)
- CSF: uniformly bloody, then **xanthochromic**

⚠ EXAM TRAP

SAH is not a traumatic haematoma. Epidural: skull fracture tears the **middle meningeal artery**; biconvex clot; lucid interval then rapid decline. **Subdural:** torn **bridging veins** (elderly, shaken infants); crescent-shaped clot; slow course. **AVM:** tangled vessels without capillaries, young men, seizures and bleeds.

29.4 CNS infections: meningitis and brain abscess

Microbes reach the CNS by four routes: **haematogenous** (commonest), **direct implantation** (trauma, meningomyelocele), **local extension** (sinus, mastoid, dental) and along **peripheral nerves** (rabies, herpes viruses).

DEFINITION

Meningitis is inflammation of the leptomeninges (pia–arachnoid) and the CSF in the subarachnoid space: **acute pyogenic** (bacterial), **aseptic** (usually viral) or **chronic** (tuberculous, spirochaetal, cryptococcal). Spread into the brain is meningoencephalitis.

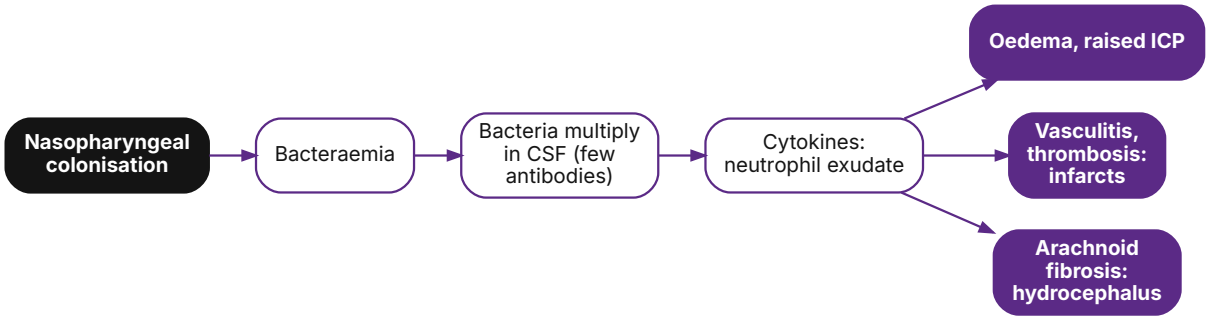


Figure 29.3 · Acute bacterial meningitis. The exudate, not just the germ, injures the brain beneath.

Age	Usual bacteria
Neonates	*E. coli*, **group B streptococci**, *Listeria*
Children	*S. pneumoniae*; *H. influenzae* b (rare if vaccinated)
Young adults	*N. meningitidis* (hostels, camps)

Clinical: fever, headache, photophobia, vomiting, clouding of consciousness and **neck stiffness** (Kernig, Brudzinski). Meningococcaemia adds a **purpuric rash** and may cause **Waterhouse–Friderichsen syndrome** (bilateral adrenal haemorrhage, shock).

Aseptic (viral) meningitis, usually **enteroviral**: similar onset but a milder, **self-limiting** course; only mild lymphocytic leptomeningeal infiltrate.

Age	Usual bacteria
Older adults	*S. pneumoniae*, *L. monocytogenes*

CSF FINDINGS: THE LAB KEY TO EVERY MENINGITIS QUESTION

Feature	Normal	Acute bacterial	Viral (aseptic)	Tuberculous
Appearance	Clear	**Turbid / purulent**	Clear	Clear or hazy; **cobweb clot** on standing
Pressure	Normal	Markedly raised	Normal or mildly raised	Raised
Cells /mm³	0–5 lymphocytes	Hundreds to many thousands, **neutrophils**	10–500 **lymphocytes**	50–500 **lymphocytes** (mixed early)
Protein	15–45 mg/dL	Markedly raised	Normal or mildly raised	**Markedly raised**
Glucose	About ⅔ of blood	**Very low**	**Normal**	**Low**
Test	—	Gram stain, culture	PCR (enterovirus, HSV)	ZN, **GeneXpert**, culture

Glucose rule: low glucose + neutrophils = bacterial; low glucose + lymphocytes = TB or fungal (cryptococcus: India ink, antigen test); normal glucose + lymphocytes = viral.

PYOGENIC MENINGITIS: GROSS

- Congested vessels; **creamy purulent exudate** in the leptomeninges
- **Pneumococcus:** over the **convexities**; **H. influenzae: basal**

MICROSCOPY

- **Neutrophils fill the subarachnoid space**; Gram-stained organisms
- Inflamed vessels → **vasculitis**, venous thrombosis, **haemorrhagic infarcts**

Complications: herniation, infarcts, ventriculitis, abscess, subdural empyema, **hydrocephalus** (pneumococcal capsule → gelatinous exudate → adhesive arachnoiditis), **deafness**, seizures.

Tuberculous meningitis (chronic meningitis)

Spreads from a pulmonary or miliary focus; weeks of headache, malaise, confusion and vomiting, often with **cranial nerve palsies**.

GROSS

- **Gelatinous, fibrinous exudate at the base of the brain**, filling the cisterns and **encasing cranial nerves**
- Small white tubercles; **tuberculoma** (caseous mass) in brain

MICROSCOPY

- Lymphocytes, plasma cells, macrophages; **caseating granulomas** with Langhans giant cells
- **Obliterative endarteritis** → infarcts; arachnoid fibrosis → **hydrocephalus**

PAKISTAN FOCUS

Pakistan is among the **five countries with the highest tuberculosis burden** (WHO estimates incidence around 266 per 100,000), so tuberculous meningitis is a common cause of chronic meningitis, especially in children. Suspect it with weeks of fever, headache and a cranial nerve palsy. **GeneXpert MTB/RIF** on CSF gives a rapid answer and detects rifampicin resistance.

Brain abscess

A localised focus of **liquefactive necrosis** with inflammation, usually bacterial (**streptococci, staphylococci**, anaerobes). Sources: **local extension** (mastoid, sinus, teeth), trauma or surgery, and **haematogenous** spread. Predisposed: **acute bacterial endocarditis** (multiple abscesses), **cyanotic congenital heart disease** (right-to-left shunt bypasses the lung filter), bronchiectasis.

GROSS

- Cavity of **pus** with a **fibrous capsule**; marked surrounding **oedema**

MICROSCOPY

- Centre: necrosis and neutrophils → **granulation tissue** → **collagen capsule** → outer **reactive gliosis** (gemistocytes)

Clinical: progressive focal deficit + raised ICP; fever may be absent. CSF: cells and protein up, **glucose normal**. Risks: herniation, rupture into ventricle, meningitis, venous sinus thrombosis.

29.5 Demyelinating diseases

Demyelinating diseases destroy **normal myelin** with relative sparing of axons; **dysmyelinating** diseases (leukodystrophies) make abnormal myelin from the start because of an inherited metabolic defect. Conduction fails without myelin, so symptoms appear although axons survive.

DEFINITION

Multiple sclerosis (MS) is an autoimmune demyelinating disorder marked by episodes of neurological deficit **separated in time**, caused by white-matter lesions **separated in space**. It is the commonest demyelinating disease; women are affected about twice as often; onset is usually in young adults. Most cases are **relapsing–remitting**, later secondary progressive; about 10% are progressive from the start.

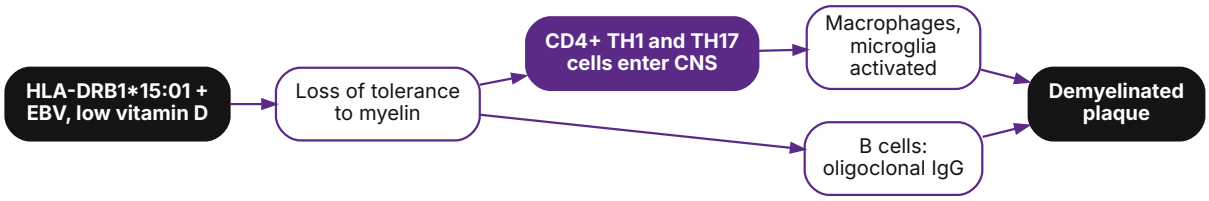


Figure 29.4 · Pathogenesis of multiple sclerosis.

GROSS

- Well-circumscribed, slightly depressed, **glassy grey-tan plaques** in white matter
- Sites: **periventricular**, **optic nerves and chiasm**, brainstem, cerebellum, spinal cord

MICROSCOPY

- **Active:** myelin breakdown with **lipid-laden macrophages**; **perivenular lymphocytic cuffs**; axons relatively preserved
- **Inactive:** no myelin, few oligodendrocytes, dense **gliosis**, axons reduced
- **Shadow plaque:** thin, partly regenerated myelin, ill-defined edge

Clinical: optic neuritis (painful unilateral visual loss, common first event); brainstem signs (diplopia, nystagmus, **internuclear ophthalmoplegia**); cord signs (weakness, spasticity, **bladder dysfunction**). **CSF:** mildly raised protein rich in **IgG**, mild lymphocytosis in a third, **oligoclonal IgG bands** (made by plasma cells inside the CNS, absent from serum) in most patients.

MS VS ADEM VS NEUROMYELITIS OPTICA

Feature	Multiple sclerosis	ADEM	Neuromyelitis optica
Trigger	Autoimmune; genes + environment	1–2 weeks after **viral infection** (rarely vaccination)	Antibodies to **aquaporin-4**
Course	Relapsing, chronic	**Monophasic**, rapid; diffuse (headache, coma); may be fatal	Relapsing; optic neuritis + long cord lesions
Lesions	Plaques of **different ages**	Perivenular demyelination, **all one age**	Optic nerves and cord; necrosis

Also know: osmotic demyelination (central pontine myelinolysis) follows over-rapid correction of hyponatraemia.

29.6 Neurodegenerative diseases

CONCEPT CLEAR

One idea explains them all: misfolded proteins. In each disease a specific protein misfolds, resists degradation and aggregates as **inclusions**, spreading cell to cell in a prion-like way, while a particular group of neurons dies. Ask three questions: **which protein, which inclusion, which neurons**. The neurons lost decide the symptoms: cortex → dementia; basal ganglia → movement disorder; motor neurons → weakness.

NEURODEGENERATIVE DISEASES: PROTEIN, INCLUSION AND AREA

Disease	Protein	Inclusion / lesion	Main area lost	Clinical
Alzheimer	Aβ; tau	**Neuritic plaques**, **tangles**	Hippocampus, association cortex	Dementia, memory first
Parkinson	α-synuclein	**Lewy bodies**	Substantia nigra	Tremor, rigidity, bradykinesia
Huntington	Huntingtin	Intranuclear inclusions	**Caudate**, putamen	Chorea, dementia
ALS	TDP-43, SOD1	Cytoplasmic inclusions	Upper + lower motor neurons	Weakness, wasting, spasticity
Prion disease (CJD)	PrP^{sc}	Spongiform change	Cortex, deep grey matter	Rapid dementia, myoclonus

Alzheimer disease (AD)

The **commonest cause of dementia** in older adults; incidence climbs from about 3% at 65–74 years to nearly half beyond 85. Most cases are sporadic; **5–10% are familial** and early-onset. Clinically: insidious **memory loss** → disorientation, aphasia, behavioural change → mute and bed-bound after 5–10 years; death usually from **pneumonia**.

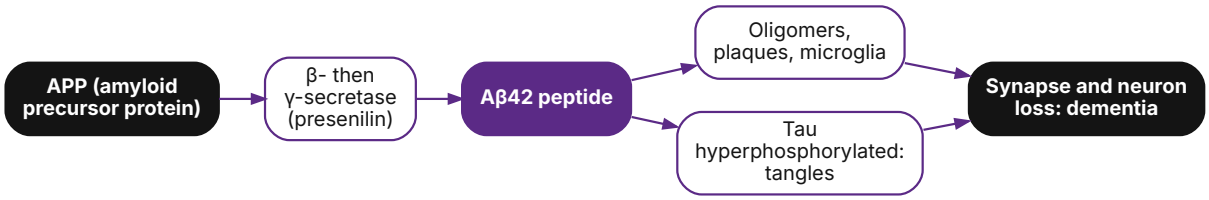


Figure 29.5 · Pathogenesis of Alzheimer disease. Aβ comes first; tau delivers the damage.

GROSS

- **Cortical atrophy**: narrow gyri, **wide sulci** (frontal, temporal, parietal)
- **Hippocampus** and entorhinal cortex shrunken early
- **Hydrocephalus ex vacuo**

MICROSCOPY (SILVER STAIN, Aβ / TAU IHC)

- **Neuritic plaques:** dilated, tortuous **dystrophic neurites** around a central **Aβ amyloid core**, 20–200 μm; diffuse plaques lack neurites
- **Neurofibrillary tangles:** paired helical filaments of hyperphosphorylated **tau**; **flame-shaped** in pyramidal cells; "ghost" tangles outlive the neuron
- **Amyloid angiopathy** almost always

MOLECULAR KEY

Genes in AD. APP (chr 21): an extra copy in **Down syndrome** gives Alzheimer changes by the 40s. **PSEN1** (chr 14), **PSEN2** (chr 1): γ-secretase parts; early-onset familial AD. **APOE ε4** (chr 19): main risk allele in sporadic AD, lowers age of onset; ε2 protects. Normal **α-secretase** cleavage cuts inside Aβ and prevents its formation.

Parkinson disease (PD)

Parkinsonism (resting pill-rolling **tremor, rigidity, bradykinesia**, masked face, stooped festinating gait) results from damage to the **nigrostriatal dopaminergic** pathway. Idiopathic PD is the commonest cause; others are dopamine-blocking drugs and the toxin MPTP. Levodopa helps early; many patients later become demented (cortical Lewy bodies: **dementia with Lewy bodies**).

GROSS	MICROSCOPY
• **Pallor of the substantia nigra** and locus coeruleus	• Loss of pigmented neurons, gliosis • **Lewy bodies:** round **eosinophilic** cytoplasmic inclusions with a **dense core and pale halo** (α-synuclein)

Genes: SNCA (α-synuclein) and **LRRK2** (commonest familial) are dominant; **Parkin, PINK1, DJ-1** are recessive and point to defective mitophagy. Themes: protein aggregation, mitochondrial failure, poor protein clearance.

Huntington disease (HD)

Autosomal dominant chorea and dementia. The **HTT** gene (**chr 4p16.3**) has an expanded **CAG repeat** (normal 11–34; disease alleles usually 40 or more) → polyglutamine huntingtin misfolds (**toxic gain of function**). Longer repeats give earlier onset (**anticipation**), mainly through paternal transmission.

GROSS	MICROSCOPY
• Striking **atrophy of the caudate**, less of the putamen; dilated lateral ventricles	• Loss of **medium spiny striatal neurons**, gliosis; **intranuclear** huntingtin inclusions

Onset in the **40s–50s**; chorea often precedes dementia; depression and suicide; death about 15 years after onset.

Amyotrophic lateral sclerosis (motor neuron disease)

Death of **lower** (anterior horn, brainstem) and **upper** (Betz cell) motor neurons: muscle wasting (**amyotrophy**) plus scarring of the **lateral** corticospinal tracts (**sclerosis**). LMN signs (wasting, **fasciculations**) mix with UMN signs (spasticity, hyperreflexia); **sensation, bladder and eye movements are spared**. Most cases are sporadic; 5–10% familial: **C9orf72** repeat expansion (commonest), **SOD1**, TDP-43, FUS. Morphology: **thin anterior roots**, loss of anterior horn cells, pale lateral columns, neurogenic muscle atrophy. Death from respiratory failure.

PAKISTAN FOCUS

Anterior horn disease in a Pakistani child is polio until proven otherwise. Pakistan and Afghanistan remain the last countries with endemic wild poliovirus type 1. Polio causes acute, asymmetric, **purely lower motor neuron** flaccid paralysis after a febrile illness; ALS is a slowly progressive adult disease with **both upper and lower** motor neuron signs.

LAB SPOTTER

CNS slides in 10 seconds. A **cavity** of foamy macrophages with a gliotic wall: **old infarct**. Pus in the subarachnoid space: **acute meningitis**; granulomas at the base: **TB**. Pus with granulation tissue, collagen capsule and outer gliosis: **abscess**. Pale, sharply bordered area on myelin stain with axons kept: **MS plaque**. Silver stain with plaques and flame-shaped tangles: **Alzheimer**.

EXAM ZONE Past-Paper Pattern SEQs — Solved

PROF PATTERN **a)** Differentiate between ischaemic and haemorrhagic stroke. **b)** Describe the morphological changes in a cerebral infarct with time. **2 + 3 marks**

MODEL ANSWER PLAN

a) Four points: mechanism (occlusion vs rupture), causes (atherosclerosis/emboli vs hypertension/berry aneurysm), early CT (normal vs hyperdense blood), thrombolysis (possible vs contraindicated). **b)** Gross: 48 h pale, soft, swollen, blurred grey–white junction → 2–10 days gelatinous → 10 days–3 weeks liquefaction → cavity. Micro: red neurons from 12 h; neutrophils to 48 h; macrophages 48 h–3 weeks; gemistocytes then glial scar.

Score booster: draw the timeline (Fig 29.1) and write "**liquefactive** necrosis": that word is often a mark.

MODULE PATTERN A 17-year-old hostel student has high fever, severe headache, vomiting, neck stiffness and a purpuric rash on the legs. **a)** Likely diagnosis and organism? **b)** Expected CSF findings? **c)** Four complications. **1 + 2 + 2 marks**

MODEL ANSWER PLAN

a) Acute pyogenic meningitis, ***Neisseria meningitidis***. **b)** Turbid CSF, high pressure, thousands of neutrophils, high protein, very low glucose, Gram-negative diplococci. **c)** Waterhouse–Friderichsen syndrome, herniation, venous thrombosis with infarction, hydrocephalus, deafness, abscess.

Score booster: add a 3-column CSF mini-table (bacterial vs viral vs TB) even if only bacterial is asked.

PROF PATTERN **a)** Enumerate the causes of subarachnoid haemorrhage. **b)** Write a note on berry aneurysm. **2 + 3 marks**

MODEL ANSWER PLAN

a) Ruptured berry aneurysm (commonest), AVM, trauma, extension of intracerebral haemorrhage, bleeding disorders, tumours. **b)** 90% anterior circulation at branch points (ACA–ACoA commonest); medial defect, no internal elastic lamina; ADPKD, Ehlers–Danlos IV, smoking, hypertension; thunderclap headache; rebleeding, vasospasm, hydrocephalus.

CLASSIC **a)** Describe the pathogenesis of multiple sclerosis. **b)** Describe its morphology and CSF findings. **2 + 3 marks**

MODEL ANSWER PLAN

a) HLA-DRB1*15:01 + EBV/low vitamin D → loss of tolerance → CD4 TH1/TH17 cells enter CNS → macrophages/microglia demyelinate; B cells make oligoclonal IgG. **b)** Glassy grey periventricular, optic, brainstem and cord plaques; active (lipid-laden macrophages, perivenular lymphocytes, axons spared), inactive (gliosis), shadow plaques. CSF: oligoclonal bands, raised IgG.

PROF PATTERN **a)** Name four neurodegenerative diseases with the protein that accumulates in each. **b)** Describe the morphology of Alzheimer disease. **2 + 3 marks**

MODEL ANSWER PLAN

a) Alzheimer (Aβ, tau), Parkinson (α-synuclein), Huntington (huntingtin), ALS (TDP-43, SOD1). **b)** Cortical atrophy with wide sulci, hippocampal atrophy, hydrocephalus ex vacuo; neuritic plaques (Aβ core + dystrophic neurites), flame-shaped tau tangles, amyloid angiopathy; silver stain.

BCQs One-Best-Answer Practice

1 A 70-year-old man dies 4 days after a sudden left hemiplegia. Microscopy of the infarcted cortex is most likely to show:

A. Only neutrophils with normal neurons
B. A dense glial scar around a cyst
C. Numerous lipid-laden macrophages, red neurons and oedema
D. Fibrous tissue with collagen deposition
E. Caseating granulomas

2 The commonest site of hypertensive intracerebral haemorrhage is the:

A. Cerebellar hemisphere
B. Pons
C. Occipital cortex
D. Thalamus
E. Putamen

3 A 45-year-old woman develops a sudden 'worst headache of my life'; CT shows blood in the basal cisterns. The underlying lesion most often lies at the:

A. Tip of the basilar artery
B. Junction of the anterior cerebral and anterior communicating arteries
C. Posterior inferior cerebellar artery
D. Middle meningeal artery
E. Vertebral artery

4 CSF: turbid, 2,500 cells/mm³ (mostly neutrophils), protein 250 mg/dL, glucose 15 mg/dL. The most likely diagnosis is:

A. Viral meningitis
B. Tuberculous meningitis
C. Multiple sclerosis
D. Acute bacterial meningitis
E. Subarachnoid haemorrhage

5 The most common cause of bacterial meningitis in a 10-day-old baby is:

A. Group B streptococcus
B. Neisseria meningitidis
C. Streptococcus pneumoniae
D. Haemophilus influenzae type b
E. Mycobacterium tuberculosis

6 At autopsy a child has a thick gelatinous exudate at the base of the brain encasing cranial nerves, obliterative endarteritis and hydrocephalus. The diagnosis is:

A. Pneumococcal meningitis
B. Viral meningitis
C. Tuberculous meningitis
D. Brain abscess
E. Acute disseminated encephalomyelitis

7 Oligoclonal IgG bands in the CSF but not in the serum are most characteristic of:

A. Alzheimer disease
B. Parkinson disease
C. Huntington disease
D. Acute bacterial meningitis
E. Multiple sclerosis

8 A 26-year-old woman had painful visual loss in one eye; two years later she has leg weakness and urinary urgency. Her brain lesions will show:

A. Neuritic plaques with tangles
B. Sharply defined myelin loss with relatively preserved axons
C. Perivenular demyelination all of one age after a viral illness
D. Spongiform change without inflammation
E. Caseating granulomas

9 Lewy bodies are composed mainly of:

A. Tau
B. Aβ amyloid
C. Huntingtin
D. α-synuclein
E. TDP-43

10 A 42-year-old man has chorea, depression and progressive dementia; his father had the same illness. The most striking atrophy will be in the:

A. Substantia nigra
B. Hippocampus
C. Caudate nucleus
D. Anterior horn of the spinal cord
E. Cerebellar vermis

11 In transtentorial (uncal) herniation, compression of the oculomotor nerve produces:

A. A dilated pupil on the opposite side
B. A dilated pupil with impaired eye movement on the same side
C. Bilateral pinpoint pupils
D. Leg weakness from anterior cerebral artery compression
E. Sudden respiratory arrest

12 A 45-year-old man with Down syndrome develops progressive memory loss. The extra gene that explains his early Alzheimer disease encodes:

A. Amyloid precursor protein
B. Presenilin 1
C. Apolipoprotein E
D. Tau
E. α-synuclein

ANSWER KEY WITH ONE-LINE REASONS

1. **C** From 48 h to 2–3 weeks macrophages dominate the infarct.
2. **E** Putamen accounts for 50–60% of hypertensive bleeds.
3. **B** Berry aneurysms are commonest at the ACA–ACoA junction.
4. **D** Neutrophils with very low glucose = pyogenic meningitis.
5. **A** Neonates: E. coli and group B streptococci.
6. **C** Basal exudate with endarteritis and granulomas is typical of TB.
7. **E** Intrathecal IgG from CNS plasma cells is found in most MS patients.
8. **B** Multiple sclerosis: demyelinated plaques with axonal sparing.
9. **D** Lewy bodies are α-synuclein aggregates in nigral neurons.
10. **C** Huntington disease: CAG expansion in HTT; caudate atrophy.
11. **B** CN III is compressed against the tentorium on the side of the mass.
12. **A** APP lies on chromosome 21; three copies increase Aβ production.

VIVA Rapid-Fire Viva Voce

Q What are red neurons?	Neurons 12–24 h after lethal hypoxia: shrunken, eosinophilic, pyknotic nucleus, no nucleolus or Nissl substance.
Q Which neurons are most sensitive to hypoxia?	Hippocampal CA1 pyramidal cells, cerebellar Purkinje cells, neocortical pyramidal neurons.
Q What is a watershed infarct?	Infarct at the border zone between two arterial territories (ACA–MCA) after global hypoperfusion.

Q Why is cerebral infarction liquefactive?	Little stroma, much lipid; enzymes from neutrophils and microglia digest it into a cavity.
Q What are Charcot–Bouchard aneurysms?	Microaneurysms of small penetrating arteries in hypertension; rupture causes ganglionic haemorrhage.
Q What is a Duret haemorrhage?	Linear or flame-shaped bleed in midbrain or pons secondary to transtentorial herniation.
Q CSF in viral meningitis?	Clear; lymphocytes; normal or mildly raised protein; normal glucose.
Q What is Waterhouse–Friderichsen syndrome?	Meningococcal septicaemia with bilateral adrenal haemorrhage, shock and purpura.
Q Layers of a brain abscess wall?	Central necrosis, granulation tissue with collagen capsule, outer reactive gliosis and oedema.
Q What is a shadow plaque?	An MS plaque with thin, partly regenerated myelin and an ill-defined border.
Q Hallmark lesions of Alzheimer disease?	Neuritic plaques with an Aβ core and neurofibrillary tangles of hyperphosphorylated tau.
Q What is a Lewy body?	Round eosinophilic cytoplasmic inclusion with a pale halo, made of α-synuclein, in nigral neurons.

OSPE Practice Stations

STATION 1 · CSF DATA — 2 minutes

TASK

A 6-year-old with 3 weeks of fever, headache and a sixth-nerve palsy. CSF: clear, fine clot on standing; cells 280/mm³, 85% lymphocytes; protein 240 mg/dL; glucose 28 mg/dL (blood 110). (a) Diagnosis? (b) One rapid confirmatory test. (c) Two complications.

ANSWER

(a) Tuberculous meningitis (lymphocytes, very high protein, low glucose, cobweb clot). (b) GeneXpert MTB/RIF on CSF (or ZN stain, culture). (c) Communicating hydrocephalus; infarcts from obliterative endarteritis; cranial nerve palsies.

STATION 2 · GROSS SPECIMEN — 1 minute

TASK

Coronal slice of brain: in the lateral part of one hemisphere a collapsed, irregular cavity with a ragged grey wall; overlying meninges intact. (a) Identify. (b) Two microscopic features. (c) Commonest cause.

ANSWER

(a) Old (remote) cerebral infarct, MCA territory. (b) Lipid-laden macrophages; gliotic wall of reactive astrocytes with new capillaries. (c) Embolism (heart, carotid) or atherosclerotic thrombosis.

STATION 3 · SLIDE — 2 minutes

TASK

Silver-stained cerebral cortex: rounded extracellular structures with a dense central core ringed by swollen, twisted nerve processes; many neurons contain dark flame-shaped fibrillary bundles. (a) Diagnosis. (b) Name the two lesions and their proteins. (c) Which genetic syndrome predisposes to early onset?

ANSWER

(a) Alzheimer disease. (b) Neuritic plaques (Aβ core); neurofibrillary tangles (hyperphosphorylated tau). (c) Down syndrome (extra APP gene, chromosome 21).

RAPID REVISION

The Night-Before Page

15 ONE-LINERS

01 Vasogenic oedema = leaky BBB; cytotoxic = swollen cells from energy failure.

02 Uncal herniation: ipsilateral dilated pupil, PCA infarct, Duret bleeds.

03 Tonsillar herniation compresses the medulla and kills.

04 Red neurons at 12–24 h are the first sign of hypoxic death.

05 Most vulnerable: CA1 (Sommer sector), Purkinje cells, cortical layers 3, 5, 6.

06 Embolic infarcts outnumber thrombotic; MCA territory; often turn red.

07 Infarct: neutrophils to 48 h, macrophages to 3 weeks, gliotic cavity later.

08 Hypertensive ICH: putamen; Charcot–Bouchard microaneurysms.

09 Lobar bleeds in the elderly: cerebral amyloid angiopathy.

10 Berry aneurysm: ACA–ACoA, no elastic lamina, ADPKD, thunderclap headache.

11 Bacterial CSF: neutrophils, high protein, very low glucose.

12 TB meningitis: basal exudate, endarteritis, hydrocephalus, cobweb clot.

13 MS: periventricular plaques, axons spared, oligoclonal bands.

14 Alzheimer: Aβ plaques + tau tangles; APP on chr 21; APOE ε4 risk.

15 PD: Lewy bodies (α-synuclein); HD: CAG repeats, caudate atrophy.

MEMORY HOOKS

- **BE-FAST**: Balance, Eyes, Face, Arm, Speech, Time
- **Glucose rule**: low + neutrophils = bacteria; low + lymphocytes = TB/fungus; normal + lymphocytes = virus
- **NIS** (Charcot triad in MS): Nystagmus, Intention tremor, Scanning speech
- **Protein, inclusion, neurons**: the three questions for every degeneration

TEST YOURSELF: FILL THE GAPS

1. Lacunes are smaller than _____ mm.
2. Commonest berry aneurysm site: _____ junction.
3. Meningitis in young adults: _____.
4. NMO antibodies target _____.
5. Huntington gene lies on chromosome _____.
6. Commonest familial ALS gene: _____.

Answers: 15 · ACA–ACoA · N. meningitidis · aquaporin-4 · 4 (4p16.3) · C9orf72

DRAW FROM MEMORY

A coronal brain showing subfalcine, uncal and tonsillar herniation with what each compresses.

DRAW FROM MEMORY

The infarct timeline: 12 h, 48 h, 2–10 days, 3 weeks, months (Figure 29.1).

SPACED REVISION ☐ Day 1 ☐ Day 3 ☐ Day 7 ☐ Day 14 ☐ Day 30

30

CNS II: Brain Tumours & Nerve Sheath Tumours

In the brain a 'benign' tumour in the wrong place can kill, and a malignant one almost never leaves the skull. The patient's age, the tumour's address and a few molecular markers tell you more than the word benign.

BY THE END OF THIS CHAPTER YOU CAN

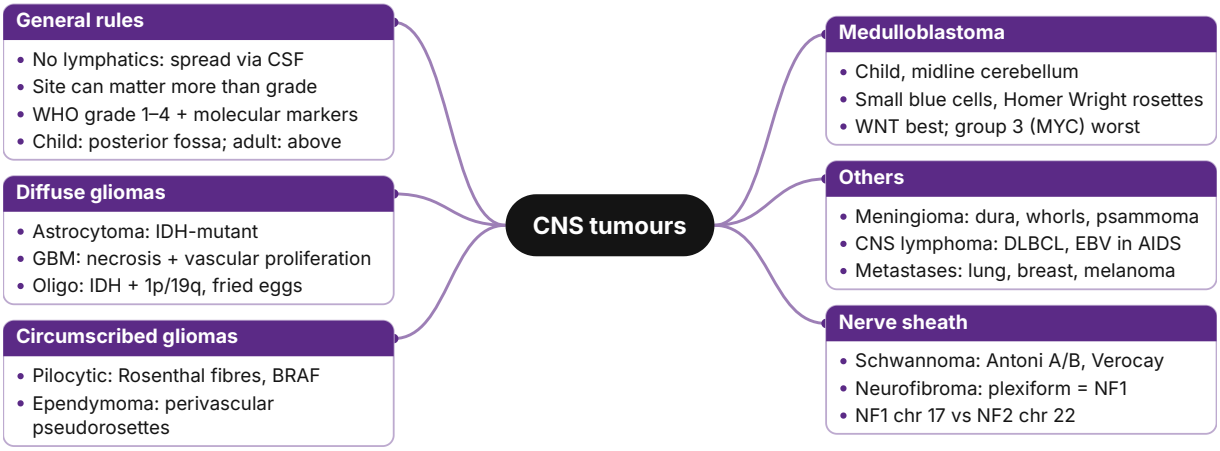
1. Explain why CNS tumours behave differently from tumours elsewhere and how WHO grading works
2. Predict the likely tumour from the patient's age and the tumour's site
3. Describe diffuse astrocytoma, glioblastoma, oligodendroglioma, pilocytic astrocytoma and ependymoma with their markers
4. Describe medulloblastoma and its molecular groups
5. Describe the morphology and grading of meningioma; recognise CNS lymphoma and metastases
6. Differentiate schwannoma from neurofibroma and NF1 from NF2

EXAM RADAR

Theory SEQs	★★☆
BCQs / MCQs	★★★
Viva voce	★★★
OSPE / slides	★★★

Glioblastoma, meningioma and schwannoma are the classic lab slides; age and site, IDH and 1p/19q, rosettes and NF1 vs NF2 are favourite BCQs and viva points.

ROBBINS LINK Robbins Basic Pathology — *Central Nervous System*: tumours (gliomas, embryonal tumours, meningiomas, lymphoma, metastases, familial tumour syndromes); *Peripheral Nerves*: nerve sheath tumours, neurofibromatosis.



Chapter at a glance. Think age, site, histology and marker for every tumour.

30.1 General features of CNS tumours

Intracranial tumours arise in roughly 10–17 per 100,000 people a year (half to three-quarters primary) and make up as many as **20% of childhood cancers**.

BRAIN TUMOURS BY AGE AND SITE: THE FIRST STEP IN EVERY DIAGNOSIS

Site	Adults	Children
Supratentorial (cerebral hemispheres; about 70% of adult tumours)	**Glioblastoma**, diffuse astrocytoma, **oligodendroglioma**; **meningioma** (dura); **metastases** (grey–white junction); CNS lymphoma (deep, periventricular)	Craniopharyngioma (suprasellar); pilocytic astrocytoma of optic pathway or hypothalamus (NF1); supratentorial ependymoma
Infratentorial (posterior fossa; about 70% of childhood tumours)	**Schwannoma** (cerebellopontine angle); **haemangioblastoma** (cerebellum, VHL); metastases	**Pilocytic astrocytoma** (cerebellum), **medulloblastoma** (vermis), **ependymoma** (fourth ventricle), diffuse midline glioma (pons)
Spinal	**Ependymoma** (cord; myxopapillary at filum), meningioma (thoracic, women), schwannoma (roots)	Drop metastases from posterior fossa tumours

CONCEPT CLEAR

Four rules that make brain tumours different. (1) **Benign vs malignant is blurred:** even low-grade gliomas infiltrate widely, cannot be fully removed and slowly progress. (2) **Location decides outcome:** a WHO grade 1 tumour in the brainstem can kill, while a convexity meningioma is cured by surgery. (3) **They almost never spread outside the CNS** (no lymphatics), but some seed through the **CSF** to the spinal cord and cauda equina (**"drop metastases"**: medulloblastoma, ependymoma). (4) **Symptoms** come from raised ICP (morning headache, vomiting, papilloedema), seizures and focal deficits; posterior fossa tumours block CSF and cause **hydrocephalus** early.

WHO GRADING (GRADE 1–4)

- **1:** low proliferation, often cured by resection (pilocytic astrocytoma, most meningiomas, schwannoma)
- **2:** infiltrative, few mitoses, recur and progress (diffuse astrocytoma, oligodendroglioma)
- **3:** anaplasia, brisk mitoses (anaplastic astrocytoma)
- **4:** mitoses, necrosis, rapidly fatal (glioblastoma, medulloblastoma)

EXAM TRAP

Robbins vs the newest WHO book. Robbins (10th ed) uses grades I–IV and the 2016 WHO scheme; the 2021 WHO (CNS5) writes grades 1–4 and gives an **integrated diagnosis** (histology + molecular). It reserves "glioblastoma" for **IDH-wildtype** tumours; the IDH-mutant grade 4 tumour is now "astrocytoma, IDH-mutant, grade 4" (old "secondary GBM"). Either answer earns marks if you name the IDH status.

MEMORY HOOK

Children fall backwards, adults look up

Child, posterior fossa: **P-M-E**: **P**ilocytic (cystic, lateral), **M**edulloblastoma (solid, midline), **E**pendymoma (fourth ventricle floor). Adult, above the tentorium: **G-M-M**: **G**lioblastoma, **M**etastasis, **M**eningioma.

30.2 Diffuse gliomas: astrocytoma, glioblastoma, oligodendroglioma

Gliomas arise from glial cells (astrocytes, oligodendrocytes, ependymal cells). **Diffuse (infiltrating) astrocytomas** make up about **80% of adult primary brain tumours**, usually in the **cerebral hemispheres** of people in their 4th to 6th decades, presenting with seizures, headache or focal deficits. They form a spectrum: **diffuse astrocytoma (grade 2) → anaplastic astrocytoma (grade 3) → glioblastoma (grade 4)**.

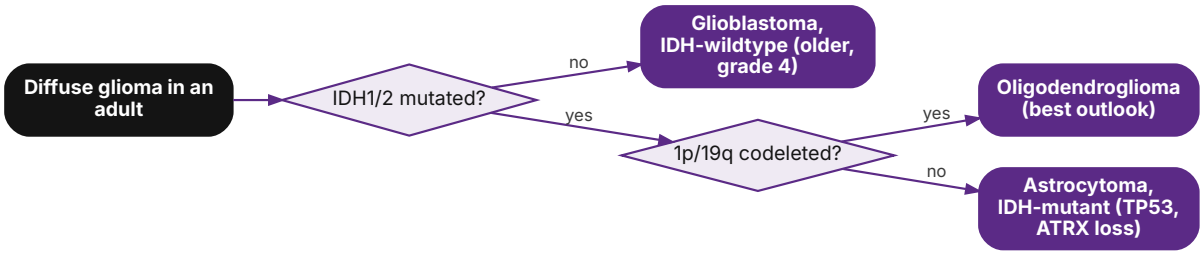


Figure 30.1 · Diagnosing a diffuse glioma in an adult. Two molecular questions sort the three main tumours.

CONCEPT CLEAR

Why IDH matters. Mutant IDH1/2 makes the **oncometabolite 2-hydroxyglutarate**, which blocks DNA and histone demethylases; the hypermethylated cells cannot differentiate. IDH-mutant gliomas strike **younger adults** and do **better** than IDH-wildtype tumours of the same grade.

MOLECULAR KEY

Astrocytoma: IDH + **TP53** + **ATRX** loss.
Oligodendroglioma: IDH + **1p/19q codeletion**. **GBM, IDH-wildtype** (about 90% of GBMs): **EGFR amplification**, **PTEN** loss, **TERT** promoter mutation, CDKN2A deletion. **MGMT promoter methylation** predicts temozolomide response.

Diffuse astrocytoma (grade 2–3)

GROSS

- **Poorly defined**, grey, infiltrative mass that expands and distorts the brain; firm or gelatinous, may be cystic

MICROSCOPY

- More glial nuclei with **atypia** in a **fibrillary GFAP+ background**; no clear edge with normal brain
- **Anaplastic (grade 3):** denser cells, more pleomorphism, **mitoses**

Glioblastoma (GBM, grade 4)

The commonest and most malignant primary brain tumour of adults. It may cross the corpus callosum into both hemispheres (**"butterfly glioma"**). With surgery, radiotherapy and temozolomide, median survival is only about **15 months**.

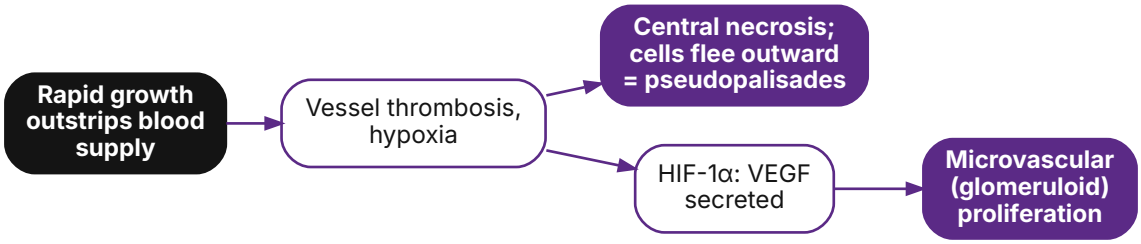


Figure 30.2 · Where the two hallmarks of GBM come from.

GROSS

- **Variegated** cut surface: firm white areas, soft **yellow necrosis**, **haemorrhage**, cysts
- Ill-defined edge; may cross the corpus callosum

MICROSCOPY

- Highly cellular, pleomorphic astrocytic tumour with many mitoses
- **Pseudopalisading necrosis:** tumour nuclei line up around serpiginous necrotic zones
- **Microvascular proliferation:** tufts of proliferating endothelial cells (glomeruloid)
- Either hallmark makes a diffuse astrocytoma grade 4

Oligodendroglioma (grade 2–3)

About **5–15% of gliomas**, mostly in the 4th and 5th decades, in the **white matter of the frontal and temporal lobes**; often a long history of **seizures**. Defined by **IDH mutation + 1p/19q codeletion**, which also predicts **sensitivity to chemotherapy** and a better prognosis than astrocytomas of the same grade.

GROSS

- **Well-circumscribed**, gelatinous, grey mass
- Cysts, focal haemorrhage and **calcification** (visible on imaging)

MICROSCOPY

- Sheets of **uniform round cells** with finely granular chromatin
- **Perinuclear clear halo**: **"fried-egg"** cells (a fixation artefact)
- Delicate branching capillaries: **"chicken-wire"** pattern
- **Microcalcifications**; grade 3 adds mitoses, necrosis, vascular proliferation

30.3 Circumscribed gliomas: pilocytic astrocytoma and ependymoma

Pilocytic astrocytoma (grade 1)

A slow-growing, **well-demarcated** tumour of **children and young adults**, usually in the **cerebellum**; also the third ventricle floor, hypothalamus and **optic nerves (NF1)**. Driven by **BRAF activation**, most often a **KIAA1549–BRAF fusion** (tandem duplication), less often BRAF V600E; **no IDH mutation**. Resection is often curative.

GROSS

- Classic: **cyst with a mural nodule**
- Otherwise a well-circumscribed solid mass

MICROSCOPY

- **Bipolar cells with long, thin, hair-like ("piloid") processes**, GFAP-positive
- Compact areas alternating with **loose microcystic** areas
- **Rosenthal fibres** (bright eosinophilic, corkscrew-shaped) and **eosinophilic granular bodies**
- Hyalinised vessels; mitoses and necrosis rare

Ependymoma

Arises next to the ependyma-lined ventricles and the central canal. In the **first two decades** it is usually in the **fourth ventricle**, where it blocks CSF flow and causes hydrocephalus; in **adults** it favours the **spinal cord** (linked to **NF2**). Supratentorial tumours carry a **ZFTA (C11orf95)–RELA fusion**. Posterior fossa ependymomas do worse because complete removal is difficult; CSF spread can occur.

GROSS

- Solid or papillary mass arising from the **floor of the fourth ventricle**
- Spinal cord: well-demarcated intramedullary mass
- **Myxopapillary ependymoma:** filum terminale of adults

MICROSCOPY

- Uniform round–oval nuclei with granular chromatin; dense fibrillary background
- **Perivascular pseudorosettes** (common): cells around a vessel with a **nucleus-free zone** of processes
- **True ependymal rosettes** (canals with a central lumen): diagnostic but fewer
- GFAP-positive

30.4 Medulloblastoma

DEFINITION

Medulloblastoma is a highly malignant **embryonal** tumour of the **cerebellum**, mainly of children (about 20% of childhood brain tumours). It is **midline (vermis)** in children and more often lateral in adults. The cells are primitive but may show neuronal differentiation (synaptophysin-positive).

GROSS

- Well-circumscribed, **grey, friable** midline mass
- May reach the folial surface and seed the **leptomeninges**; compresses the fourth ventricle

MICROSCOPY

- Very cellular sheets of **small round blue cells**: scant cytoplasm, hyperchromatic, often crescent-shaped nuclei, **many mitoses**
- **Homer Wright rosettes:** tumour cells around a central core of pink fibrillary **neuropil** (no lumen)
- Desmoplastic/nodular pattern in the SHH group

Behaviour: spreads through CSF (**drop metastases** to cord and cauda equina) but is **exquisitely radiosensitive**; with excision, craniospinal radiation and chemotherapy, 5-year survival can reach about 75%.

MOLECULAR GROUPS OF MEDULLOBLASTOMA: PROGNOSIS FOLLOWS THE PATHWAY

Group	Genetic driver and notes	Prognosis
WNT	**CTNNB1** (β-catenin) mutation; monosomy 6; nuclear β-catenin on IHC	**Best**
SHH	PTCH1, SMO, SUFU (Gorlin); infants and adults; desmoplastic	Intermediate (poor if TP53-mutant)
Group 3	**MYC amplification**; young children; often metastatic	**Worst**
Group 4	Isochromosome 17q; commonest group	Intermediate

30.5 Meningioma

A predominantly **benign tumour of adults** arising from **meningothelial (arachnoid cap) cells** and attached to the **dura**. Commoner in **women** (they often carry progesterone receptors and may grow in pregnancy). Sites: **parasagittal** (falx, superior sagittal sinus), cerebral convexity, sphenoid wing, olfactory groove, sella, foramen magnum, thoracic spinal canal. Risk factors: **NF2** (multiple meningiomas), **loss of chromosome 22** / NF2 mutation in about half of sporadic tumours, and **prior radiation**.

GROSS

- Rounded, **well-defined, dura-based** mass that indents but **separates easily from the brain**
- Firm, grey-white, may feel **gritty** (psammoma bodies)
- May invade overlying bone (**hyperostosis**) or spread as a sheet along the dura (en plaque)

MICROSCOPY

- **Meningothelial (syncytial):** lobules of cells with indistinct borders in **whorls**
- **Fibrous:** elongated cells with collagen; **transitional:** both, with many whorls
- **Psammomatous:** numerous **psammoma bodies** (laminated calcified whorls)
- Others: secretory, microcystic; EMA and PR positive

GRADING MENINGIOMAS

Grade	Criteria and behaviour
1 (most)	Low mitotic rate, no brain invasion; cured by complete excision
2 (atypical)	**≥ 4 mitoses/10 HPF** or **brain invasion**, or three of: high cellularity, small cells with high N:C ratio, prominent nucleoli, sheet-like growth, necrosis; recurs more often
3 (anaplastic)	**≥ 20 mitoses/10 HPF** or frank anaplasia resembling carcinoma, melanoma or sarcoma; highly aggressive

30.6 CNS lymphoma, metastases and familial tumour syndromes

PRIMARY CNS LYMPHOMA

- Mostly **diffuse large B-cell lymphoma**
- **Commonest CNS tumour in immunosuppressed** patients (AIDS, transplant), where it is almost always **EBV-positive**; also seen in older immunocompetent adults
- Often **multifocal, deep** (periventricular white matter, basal ganglia, corpus callosum)
- Micro: **angiocentric** cuffs of large CD20+ cells around vessels, infiltrating brain

METASTATIC TUMOURS

- Mostly carcinomas: a quarter to half of intracranial tumours in hospital practice
- Primaries (about 80%): **lung, breast, skin (melanoma), kidney, GI tract**
- Gross: **sharply demarcated, often multiple** masses at the **grey–white junction** with marked **oedema**
- Paraneoplastic: cerebellar degeneration (anti-Yo), limbic encephalitis (anti-Hu, small cell lung cancer)

Familial syndromes (besides NF1/NF2, section 30.7): **tuberous sclerosis** (TSC1/TSC2): cortical tubers, subependymal giant cell astrocytoma, seizures, angiofibromas, renal angiomyolipoma; **von Hippel–Lindau** (VHL, 3p): **cerebellar haemangioblastoma**, retinal angiomas, renal cell carcinoma, phaeochromocytoma, polycythaemia; **Li–Fraumeni** (TP53): gliomas; **Gorlin** (PTCH1): SHH medulloblastoma.

30.7 Peripheral nerve sheath tumours and neurofibromatosis

Schwannoma

A benign, **encapsulated** tumour of neural-crest-derived **Schwann cells** that causes symptoms by compressing the nerve or neighbouring structures. The classic site is the **cerebellopontine angle** on the vestibular branch of CN VIII (**vestibular schwannoma, "acoustic neuroma"**): **tinnitus and sensorineural hearing loss**. Others arise on spinal roots and limb nerves. Sporadic and NF2-associated tumours both lose **merlin** (NF2 gene); **bilateral vestibular schwannomas** mean NF2. Malignant change is extremely rare.

GROSS

- **Well-circumscribed, encapsulated** globular mass attached to the nerve but **separable from it** (nerve pushed to one side)
- Firm grey-tan; yellow, cystic or haemorrhagic areas

MICROSCOPY

- **Antoni A:** dense, cellular fascicles of spindle cells; **nuclear palisading**; **Verocay bodies** (two rows of palisaded nuclei separated by a nucleus-free zone of processes)
- **Antoni B:** loose, hypocellular, **myxoid**, microcystic
- Thick **hyalinised vessels**, haemosiderin; axons largely **excluded**
- **S-100 strongly and diffusely positive**

Neurofibroma

The commonest peripheral nerve sheath tumour. Three forms: **localised cutaneous** (dermis and subcutis; sporadic or NF1), **diffuse**, and **plexiform**, which is **pathognomonic of NF1** and carries a risk of transformation to **malignant peripheral nerve sheath tumour (MPNST)**, a high-grade sarcoma of which about half arise in NF1.

GROSS

- Cutaneous: soft, skin-coloured nodule, often pedunculated; **not encapsulated**
- **Plexiform:** diffuse expansion of a large nerve and its branches, a "**bag of worms**"

MICROSCOPY

- Mixed cells: Schwann cells, perineurial-like cells, fibroblasts, **mast cells**, CD34+ cells
- Wavy, serpentine nuclei in a loose **myxoid** stroma with "**shredded-carrot**" collagen
- **Axons run through** the tumour; S-100 patchy

SCHWANNOMA VS NEUROFIBROMA

Feature	Schwannoma	Neurofibroma
Capsule	**Encapsulated**	Unencapsulated
Relation to nerve	Eccentric: nerve **displaced**, can be spared at surgery	Nerve **incorporated**; axons run through it
Pattern	**Antoni A + B, Verocay bodies**, hyalinised vessels	Loose myxoid, shredded-carrot collagen, wavy nuclei
S-100	Diffuse, strong	Patchy
Syndrome	**NF2** (bilateral vestibular)	**NF1** (multiple, plexiform)
Malignant change	Extremely rare	Plexiform → MPNST

NF1 VS NF2: BOTH AUTOSOMAL DOMINANT, OTHERWISE DIFFERENT DISEASES

Feature	NF1 (von Recklinghausen)	NF2
Gene / chromosome	**NF1**, **17q11.2**; neurofibromin; about 1 in 3,000	**NF2**, **22q12**; merlin
Protein function	GTPase-activating protein that **switches RAS off**	Cytoskeletal protein for **contact inhibition**
Nerve tumours	Multiple **neurofibromas**, **plexiform** neurofibroma, MPNST	**Bilateral vestibular schwannomas** (hallmark)
Other tumours	Optic glioma (pilocytic), phaeochromocytoma, GIST, leukaemia	**Multiple meningiomas**, spinal **ependymomas**
Skin and eyes	**Café-au-lait spots**, axillary/inguinal **freckling**, **Lisch nodules** (iris hamartomas)	Few café-au-lait spots; juvenile posterior subcapsular cataract

MEMORY HOOK

"von Recklinghausen" has 17 letters

So **NF1 lives on chromosome 17**. **NF2** = **2 ears**: bilateral acoustic schwannomas, gene on chromosome **22**.

LAB SPOTTER

Tumour slides in 10 seconds. Necrosis rimmed by palisaded nuclei: **GBM**. Clear halos + chicken-wire capillaries: **oligodendroglioma**. Hair-like cells, microcysts, pink corkscrew fibres: **pilocytic**. Nucleus-free collars round vessels: **ependymoma**. Small blue cells with neuropil rosettes: **medulloblastoma**. Whorls + calcospherites: **meningioma**. Dense and loose areas with Verocay bodies: **schwannoma**.

EXAM ZONE Past-Paper Pattern SEQs — Solved

PROF PATTERN **a)** Classify gliomas. **b)** Describe the gross and microscopic features of glioblastoma. **2 + 3 marks**

MODEL ANSWER PLAN

a) Astrocytic (pilocytic grade 1; diffuse astrocytoma 2; anaplastic 3; glioblastoma 4), oligodendroglioma (2–3), ependymoma (myxopapillary, grade 2–3). Add molecular basis: IDH-mutant astrocytoma, IDH + 1p/19q oligodendroglioma, IDH-wildtype GBM. **b)** Gross: ill-defined, variegated (white, yellow necrosis, haemorrhage, cysts), butterfly across corpus callosum. Micro: pleomorphic, mitotically active astrocytes, **pseudopalisading necrosis**, **microvascular proliferation**; GFAP+.

Score booster: draw Fig 30.2 to explain pseudopalisading and vascular proliferation; quote median survival of about 15 months.

PROF PATTERN **a)** Give the common sites and risk factors of meningioma. **b)** Describe its morphology and grading. **2 + 3 marks**

MODEL ANSWER PLAN

a) Parasagittal, convexity, sphenoid wing, olfactory groove, sella, spinal; women, NF2, chr 22 loss, prior radiation. **b)** Dura-based, well-circumscribed, separable from brain, may cause hyperostosis; whorls, psammoma bodies (meningothelial, fibrous, transitional, psammomatous); grade 1, atypical (≥ 4 mitoses/10 HPF or brain invasion), anaplastic (≥ 20).

CLASSIC **a)** Differentiate between schwannoma and neurofibroma. **b)** Write a note on neurofibromatosis type 1. **3 + 2 marks**

MODEL ANSWER PLAN

a) Table of six points: capsule, relation to nerve, cell types, Antoni A/B + Verocay vs myxoid shredded collagen, S-100 diffuse vs patchy, NF2 vs NF1. **b)** Autosomal dominant, NF1 gene chr 17, neurofibromin (RAS-GAP); neurofibromas (plexiform), café-au-lait spots, freckling, Lisch nodules, optic glioma, MPNST, phaeochromocytoma.

Score booster: one line on molecular function ("loss of neurofibromin leaves RAS switched on") lifts the answer.

BCQs One-Best-Answer Practice

1 A 62-year-old man has a rapidly enlarging frontal mass crossing the corpus callosum. Which microscopic feature defines it as grade 4?

A. Rosenthal fibres
B. Perinuclear halos
C. Psammoma bodies
D. Pseudopalisading necrosis
E. Verocay bodies

2 A 40-year-old has a calcified frontal white-matter tumour composed of round cells with clear halos and branching capillaries. Which finding defines this tumour?

A. IDH mutation with 1p/19q codeletion
B. EGFR amplification
C. KIAA1549–BRAF fusion
D. MYC amplification
E. ZFTA–RELA fusion

3 A cystic cerebellar tumour with a mural nodule in a 9-year-old shows piloid cells and bright eosinophilic corkscrew fibres. The diagnosis is:

A. Medulloblastoma
B. Ependymoma
C. Pilocytic astrocytoma
D. Haemangioblastoma
E. Glioblastoma

4 A midline cerebellar tumour in a 5-year-old consists of sheets of small blue cells with rosettes around fibrillary cores. The rosettes are:

A. Flexner–Wintersteiner rosettes
B. True ependymal rosettes
C. Perivascular pseudorosettes
D. Psammoma bodies
E. Homer Wright rosettes

5 Which medulloblastoma group has the best prognosis?

A. SHH-activated, TP53-mutant
B. WNT-activated
C. Group 3 with MYC amplification
D. Group 4
E. All have equal outcome

6 Perivascular pseudorosettes in a tumour of the fourth ventricle of a child indicate:

A. Medulloblastoma
B. Meningioma
C. Ependymoma
D. Pilocytic astrocytoma
E. Choroid plexus papilloma

7 A dura-based parasagittal mass in a 50-year-old woman shows whorls of cells and laminated calcified bodies. The cell of origin is the:

A. Arachnoid (meningothelial) cap cell
B. Astrocyte
C. Ependymal cell
D. Schwann cell
E. Fibroblast of the pia

8 A 25-year-old with bilateral vestibular schwannomas most likely has a mutation on chromosome:

A. 17
B. 9
C. 3
D. 22
E. 13

9 An encapsulated tumour on a spinal root shows cellular areas with nuclear palisading alternating with loose myxoid areas. The diagnosis is:

A. Neurofibroma
B. Schwannoma
C. MPNST
D. Meningioma
E. Ependymoma

10 A 'bag of worms' thickening of the brachial plexus is pathognomonic of:

A. NF2
B. Tuberous sclerosis
C. von Hippel–Lindau disease
D. Li–Fraumeni syndrome
E. NF1

11 The commonest CNS tumour in a patient with advanced AIDS is:

A. Glioblastoma
B. Meningioma
C. Metastatic carcinoma
D. EBV-positive primary CNS lymphoma
E. Medulloblastoma

12 A 60-year-old smoker has three well-circumscribed masses at the grey–white junction with marked oedema. The most likely diagnosis is:

A. Glioblastoma
B. Oligodendroglioma
C. Metastatic carcinoma
D. Meningioma
E. Pilocytic astrocytoma

ANSWER KEY WITH ONE-LINE REASONS

1. **D** Necrosis with pseudopalisading (or microvascular proliferation) marks glioblastoma.
2. **A** Oligodendroglioma = IDH-mutant and 1p/19q-codeleted.
3. **C** Rosenthal fibres in a cyst-with-nodule tumour = pilocytic astrocytoma.
4. **E** Medulloblastoma: Homer Wright rosettes around neuropil.
5. **B** WNT (CTNNB1-mutant) tumours have the best survival.
6. **C** Cells around vessels with a nucleus-free zone = ependymoma.
7. **A** Meningioma arises from arachnoid cap cells; psammoma bodies are classic.
8. **D** NF2 (merlin) lies on 22q12.
9. **B** Antoni A and B areas with Verocay bodies = schwannoma.
10. **E** Plexiform neurofibroma is diagnostic of NF1.
11. **D** Immunosuppression-related PCNSL is almost always EBV-driven DLBCL.
12. **C** Multiple sharply demarcated grey–white junction lesions = metastases (lung).

VIVA Rapid-Fire Viva Voce

Q Why do brain tumours rarely metastasise outside the CNS?	There are no lymphatics in the brain; spread is mainly local and through the CSF.
Q Commonest site of brain tumours in children and adults?	Children: posterior fossa (about 70%). Adults: supratentorial, cerebral hemispheres.
Q Two hallmarks of glioblastoma?	Pseudopalisading necrosis and microvascular (glomeruloid) proliferation.
Q What are Rosenthal fibres?	Bright eosinophilic, corkscrew-shaped GFAP-rich structures in astrocytic processes; pilocytic astrocytoma and chronic gliosis.

Q Why are oligodendroglioma cells called fried eggs?	Round nuclei surrounded by a clear cytoplasmic halo, an artefact of delayed fixation.
Q Pseudorosette vs true rosette in ependymoma?	Pseudorosette: cells around a vessel with a nucleus-free zone. True rosette: cells around a central lumen.
Q What is a Homer Wright rosette?	Tumour cells around a central core of fibrillary neuropil without a lumen; medulloblastoma, neuroblastoma.
Q What are drop metastases?	CSF seeding of tumour to the spinal cord and cauda equina, e.g. medulloblastoma.
Q Cell of origin of meningioma?	Meningothelial (arachnoid cap) cells.
Q What are Verocay bodies?	Two rows of palisaded nuclei separated by a nucleus-free zone of processes, in Antoni A areas of schwannoma.
Q What does neurofibromin do?	It is a GTPase-activating protein that switches RAS off; lost in NF1.
Q What are Lisch nodules?	Pigmented hamartomas of the iris in NF1.

OSPE Practice Stations

STATION 1 · SLIDE (LAB) — 2 minutes

TASK

Encapsulated spindle-cell tumour: compact cellular areas with nuclei arranged in parallel rows separated by pink nucleus-free zones, alternating with loose, pale, myxoid areas; thick hyalinised vessels. (a) Diagnosis. (b) Name the two areas and the palisaded structure. (c) Commonest intracranial site and symptoms.

ANSWER

(a) Schwannoma. (b) Antoni A and Antoni B areas; Verocay bodies. (c) Cerebellopontine angle on the vestibular nerve (CN VIII): tinnitus and sensorineural hearing loss; bilateral = NF2.

STATION 2 · SLIDE (LAB) — 2 minutes

TASK

Highly cellular glial tumour with marked nuclear pleomorphism and mitoses; irregular necrotic zones rimmed by densely packed tumour nuclei; vessels with multilayered endothelial tufts. (a) Diagnosis and grade. (b) Name the two key features. (c) Two molecular changes.

ANSWER

(a) Glioblastoma, WHO grade 4. (b) Pseudopalisading necrosis; microvascular proliferation. (c) EGFR amplification, PTEN loss (also TERT promoter mutation; IDH wildtype).

STATION 3 · SLIDE (LAB) — 1 minute

TASK

Lobules and whorls of plump cells with indistinct borders and oval nuclei; scattered round, concentrically laminated purple bodies. (a) Diagnosis. (b) Name the purple bodies. (c) One genetic association.

ANSWER

(a) Meningioma (meningothelial/transitional, grade 1). (b) Psammoma bodies. (c) NF2 / loss of chromosome 22.

RAPID REVISION

The Night-Before Page

15 ONE-LINERS

01 Children: 70% posterior fossa; adults: 70% supratentorial.

02 CNS tumours spread via CSF, almost never outside the CNS.

03 Diffuse astrocytomas: about 80% of adult primary brain tumours.

04 IDH-mutant gliomas: younger patients, better prognosis (2-hydroxyglutarate).

05 GBM: pseudopalisading necrosis + microvascular proliferation; about 15 months.

06 IDH-wildtype GBM: EGFR amplification, PTEN loss, TERT promoter.

07 Oligodendroglioma: IDH + 1p/19q, fried eggs, chicken wire, calcification.

08 Pilocytic: child, cerebellar cyst + nodule, Rosenthal fibres, BRAF fusion.

09 Ependymoma: fourth ventricle (child), cord (adult, NF2); pseudorosettes.

10 Medulloblastoma: midline cerebellum, Homer Wright rosettes, radiosensitive.

11 WNT medulloblastoma best; group 3 (MYC) worst.

12 Meningioma: arachnoid cap cells, whorls, psammoma, NF2/chr 22.

13 PCNSL: DLBCL, deep and multifocal, EBV-positive in AIDS.

14 Schwannoma: encapsulated, Antoni A/B, Verocay, S-100 diffuse.

15 Plexiform neurofibroma = NF1; risk of MPNST.

MEMORY HOOKS

- **P-M-E** (child, posterior fossa): Pilocytic, Medulloblastoma, Ependymoma
- **G-M-M** (adult, supratentorial): Glioblastoma, Metastasis, Meningioma
- **von Recklinghausen** = 17 letters = chromosome 17 (NF1)
- **NF2 = 2 ears**: bilateral acoustic schwannomas, chromosome 22

TEST YOURSELF: FILL THE GAPS

1. Oncometabolite of mutant IDH: _____.
2. Methylation of _____ predicts temozolomide response.
3. Pilocytic astrocytoma driver: _____ fusion.
4. Atypical meningioma: ≥ _____ mitoses/10 HPF.
5. Merlin is the product of the _____ gene.
6. Cerebellar haemangioblastoma syndrome: _____.

Answers: 2-hydroxyglutarate · MGMT · KIAA1549–BRAF · 4 · NF2 · von Hippel–Lindau

DRAW FROM MEMORY

A sagittal brain marking where each childhood and adult tumour arises.

DRAW FROM MEMORY

The IDH → 1p/19q decision tree for diffuse gliomas (Figure 30.1).

SPACED REVISION ☐ Day 1 ☐ Day 3 ☐ Day 7 ☐ Day 14 ☐ Day 30

31

Skin: Dermatoses, Blistering Diseases & Skin Tumours

Skin is the one organ every patient shows you. Learn a dozen words for what the pathologist sees, and rashes, blisters and pigmented spots turn into patterns you can name.

BY THE END OF THIS CHAPTER YOU CAN

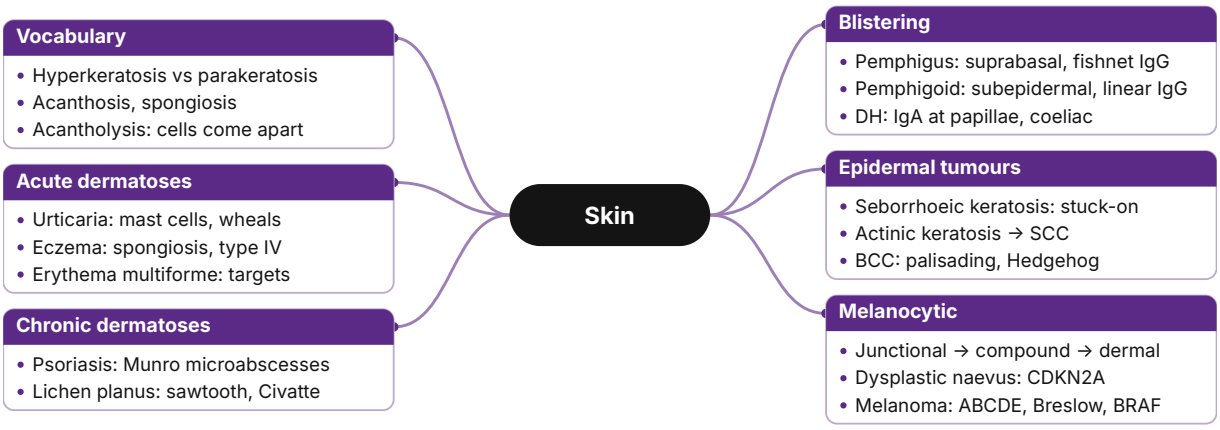
1. Define the key clinical and microscopic terms of dermatopathology
2. Describe urticaria, acute eczematous dermatitis and erythema multiforme with their mechanisms
3. Contrast psoriasis with lichen planus in pathogenesis, morphology and clinical signs
4. Differentiate pemphigus vulgaris, bullous pemphigoid and dermatitis herpetiformis by level of split and immunofluorescence
5. Describe seborrhoeic keratosis, keratoacanthoma, actinic keratosis, squamous cell and basal cell carcinoma
6. Trace the path from naevus to dysplastic naevus to melanoma and list its prognostic factors

EXAM RADAR

Theory SEQs	★★★
BCQs / MCQs	★★★
Viva voce	★★★
OSPE / slides	★★☆

Melanoma (ABCDE, Breslow), psoriasis vs lichen planus and the three bullous diseases are classic long questions; terminology and immunofluorescence patterns are favourite BCQs.

ROBBINS LINK Robbins Basic Pathology — *Skin*: acute and chronic inflammatory dermatoses, blistering diseases, benign and premalignant epithelial lesions, malignant epidermal tumours, melanocytic proliferations.



Chapter at a glance. Six families of skin disease.

31.1 Reading the skin: structure and vocabulary

The **epidermis** is a stratified squamous epithelium of keratinocytes held together by **desmosomes** and anchored to the basement membrane by **hemidesmosomes**. Melanocytes sit in the basal layer and pass pigment to keratinocytes; **Langerhans cells** catch antigens and carry them to lymph nodes, making the skin an immune organ. Below lie the dermis and adnexa. Every skin disease is described with a small set of words: learn them once and every slide becomes readable.

Clinical terms: macule flat coloured spot ≤ 5 mm (patch if larger) · **papule** raised ≤ 5 mm, **nodule** > 5 mm · **plaque** raised, flat-topped > 5 mm · **vesicle** fluid-filled ≤ 5 mm, **bulla** > 5 mm · **pustule** pus-filled · **wheal** itchy, transient dermal oedema · **scale** plate of imperfect keratin · **lichenification** thick, rough skin from rubbing · **excoriation** scratch mark · **onycholysis** nail lifted off its bed.

DERMATOPATHOLOGY GLOSSARY: WHAT THE PATHOLOGIST SEES

Term	Meaning	Example	Term	Meaning	Example
Hyperkeratosis	Thick stratum corneum	Keratoses, lichen planus	**Acantholysis**	**Loss of cohesion** between keratinocytes	**Pemphigus**

Term	Meaning	Example	Term	Meaning	Example
Parakeratosis	**Nuclei retained** in stratum corneum	**Psoriasis**	**Spongiosis**	**Intercellular oedema** of epidermis	**Eczema**
Hypergranulosis	Thick granular layer	Lichen planus	Exocytosis	Inflammatory cells in epidermis	Eczema, psoriasis
Acanthosis	Diffuse epidermal hyperplasia	Psoriasis	Vacuolisation	Vacuoles at basal layer–BM junction	EM, lichen planus
Papillomatosis	Surface bumps over enlarged papillae	Seborrhoeic keratosis	Erosion / ulcer	Partial / full loss of epidermis	Burst blister / tumour
Dyskeratosis	Premature keratinisation below granular layer	SCC	Lentiginous	Single melanocytes along basal layer	Dysplastic naevus

31.2 Acute inflammatory dermatoses

Acute lesions last days to weeks; mononuclear cells, oedema and epidermal or vascular injury dominate.

Urticaria (hives)

Localised **mast cell degranulation** makes dermal venules leaky, producing **itchy, oedematous wheals** that appear and fade within hours (usually under 24 h). Peak age 20–40. **Angioedema** is the same process deeper in the dermis and subcutis.

Causes: IgE-dependent type I hypersensitivity (pollen, foods, drugs, insect venom); **IgE-independent** direct mast cell release (opiates, some antibiotics, radiocontrast); **hereditary angioedema** from inherited **C1 inhibitor** deficiency (uncontrolled bradykinin; laryngeal oedema can kill). **Microscopy** is subtle: superficial dermal oedema separating collagen bundles, sparse perivenular mononuclear cells and a few eosinophils.

Acute eczematous dermatitis

"Eczema" is a clinical term for red, **papulovesicular, oozing, crusted** lesions that, if they persist, become raised scaling plaques. Types: **allergic contact** (nickel, poison ivy, cosmetics), **atopic**, drug-related, photo-eczematous and primary irritant dermatitis.

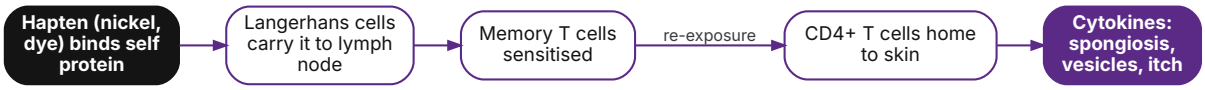


Figure 31.1 · Allergic contact dermatitis: a type IV (delayed) hypersensitivity.

CLINICAL

- Acute: red, itchy, weeping papulovesicles at the site of contact
- Chronic: dry, scaly, **lichenified** plaques

MICROSCOPY

- **Spongiosis**: oedema stretches intercellular bridges → **spongiotic vesicles**
- Superficial perivascular lymphocytes, papillary dermal oedema; eosinophils in drug reactions
- Chronic: **acanthosis** and hyperkeratosis

Erythema multiforme (EM)

A self-limited hypersensitivity reaction to **infections** (**herpes simplex**, mycoplasma) and **drugs** (sulfonamides, penicillin, anticonvulsants such as phenytoin, salicylates, antimalarials). **Cytotoxic CD8+ T cells** attack basal keratinocytes. Lesions are "multiform" (macules, papules, vesicles, bullae); the hallmark is the **target lesion**: a red macule or papule with a pale, vesicular or eroded centre.

Stevens–Johnson syndrome: extensive disease with erosions of lips, mouth, conjunctiva, urethra and genitals, often drug-induced. **Toxic epidermal necrolysis:** full-thickness necrosis and sloughing of skin and mucosa, like a burn. **Microscopy:** lymphocytes along the dermoepidermal junction with **necrotic keratinocytes** (interface dermatitis), dermal oedema → blisters.

31.3 Chronic inflammatory dermatoses

Psoriasis

A common chronic immune-mediated dermatosis (1–2% of people in Western series), sometimes with **arthritis** (see Chapter 28), myopathy or enteropathy. Plaques favour the **elbows, knees, scalp, lumbosacral area**, intergluteal cleft and glans penis.

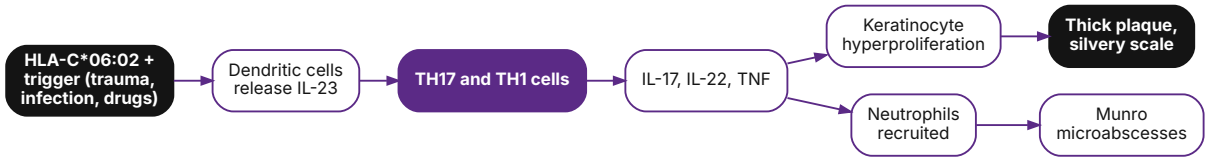


Figure 31.2 · Pathogenesis of psoriasis. T-cell cytokines drive keratinocytes to divide too fast.

CLINICAL

- Well-demarcated pink–salmon **plaques** with loosely adherent **silver-white scale**
- **Auspitz sign**: pinpoint bleeding when the scale is lifted
- **Koebner phenomenon**: lesions at sites of trauma
- **Nails** (about a third): pitting, yellow-brown discolouration, **onycholysis**

MICROSCOPY

- Marked **acanthosis** with **regular elongation of rete ridges** ("test tubes in a rack"); mitoses above the basal layer
- **Thin or absent granular layer**, thick **parakeratotic** scale
- **Thinned suprapapillary plates** over dilated, tortuous papillary capillaries (hence Auspitz)
- Neutrophils: **spongiform pustules of Kogoj** (spinous layer) and **Munro microabscesses** (stratum corneum)

Lichen planus

A self-limited (often 1–2 years) disorder of skin and mucosa. **CD8+ T cells** attack basal keratinocytes that display altered self antigens, so this is an **interface dermatitis**. Lesions favour the **flexor wrists**, elbows and glans penis; **oral lesions** (white, lacy, net-like) occur in about 70% and may persist for years. Healing leaves hyperpigmentation.

CLINICAL

- Itchy, violaceous, flat-topped polygonal papules
- **Wickham striae**: fine white lines on the papules (hypergranulosis)
- Koebner phenomenon

MICROSCOPY

- Dense **band-like lymphocytic infiltrate** hugging the dermoepidermal junction
- Basal cell degeneration (vacuolisation), "squamatisation" of the basal layer
- **Sawtooth rete ridges**; hyperkeratosis, **wedge-shaped hypergranulosis**
- **Civatte (colloid) bodies**: anucleate necrotic basal cells in the papillary dermis

PSORIASIS VS LICHEN PLANUS

Feature	Psoriasis	Lichen planus
Lesion	Salmon plaques, silvery scale; extensor surfaces	Purple polygonal papules; flexor wrists; mouth
Sign	**Auspitz**, Koebner, nail pitting	**Wickham striae**, Koebner
Granular layer	**Thin or absent**	**Thickened** (wedge-shaped)
Stratum corneum	**Parakeratosis** with Munro microabscesses	Orthokeratotic hyperkeratosis
Rete / infiltrate	Regular elongated rete; neutrophils in epidermis	**Sawtooth** rete; **band-like** lymphocytes; Civatte bodies

MEMORY HOOK

Lichen planus: the six Ps

Pruritic, **P**urple, **P**olygonal, **P**lanar (flat-topped) **P**apules and **P**laques. Psoriasis instead is **S**almon, **S**ilvery-**S**caled and on **e**xtensor surfaces.

31.4 Blistering (bullous) diseases

CONCEPT CLEAR

The level of the split decides what you see. Keratinocytes are glued to each other by **desmosomes** (desmogleins) and to the basement membrane by **hemidesmosomes** (BP180, BP230). Antibodies against desmosomes make cells fall apart **within the epidermis** (acantholysis): the roof is thin, so blisters are **flaccid**, rupture easily and leave erosions. Antibodies against hemidesmosomes lift the **whole epidermis** off the dermis: the roof is thick, so blisters are **tense** and stay intact.

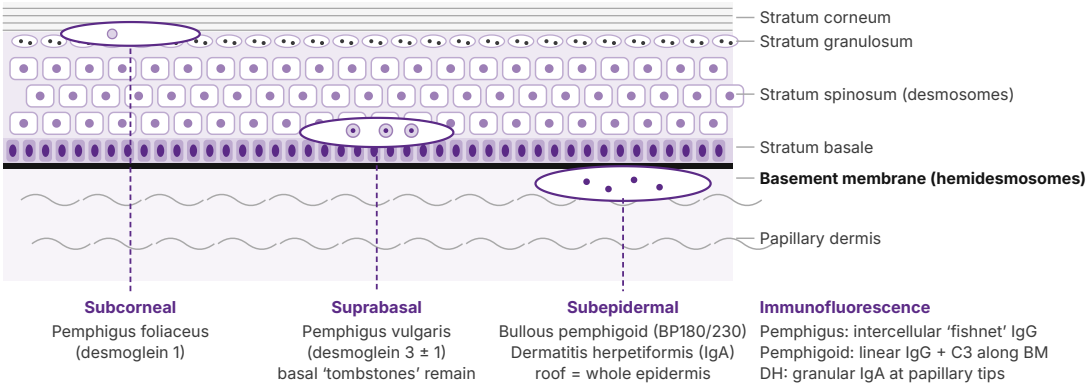


Figure 31.3 · Where the epidermis splits. Read the level, then the immunofluorescence, and the diagnosis follows.

Pemphigus

An uncommon autoimmune disease (usually 40–60 years) in which **IgG autoantibodies against desmogleins** cause **acantholysis** (type II hypersensitivity). **Pemphigus vulgaris** (over 80% of cases) targets **desmoglein 3** (± 1) and involves mucosa and skin: scalp, face, axilla, groin, trunk and pressure points. **Pemphigus foliaceus** targets **desmoglein 1**, gives superficial (subcorneal) blisters on scalp, face, chest and back and rarely affects mucosa. A paraneoplastic form accompanies lymphomas.

CLINICAL

- **Flaccid** vesicles and bullae that rupture easily, leaving painful **erosions** with crust
- Oral erosions often appear first
- **Nikolsky sign positive**

MICROSCOPY

- **Suprabasal acantholytic blister** containing rounded, detached keratinocytes
- Basal cells stay attached to the BM: a "**row of tombstones**"
- **IF:** intercellular IgG around keratinocytes: **"fishnet" / net-like**

Bullous pemphigoid

The commonest autoimmune blistering disease, typically in the **elderly**. IgG autoantibodies bind hemidesmosome proteins **BPAG2 (BP180, type XVII collagen)** and BPAG1 (BP230); complement is activated and neutrophils and **eosinophils** are recruited, splitting the epidermis from the dermis at the lamina lucida.

CLINICAL

- **Tense bullae** with clear fluid on normal or red skin
- Inner thighs, flexor forearms, axillae, groin, lower abdomen
- Oral involvement in only about 10–15%; heals **without scarring**

MICROSCOPY

- **Subepidermal, non-acantholytic** blister; roof is the whole epidermis
- Dermal infiltrate rich in **eosinophils**, some lymphocytes and neutrophils
- **IF: linear** IgG and C3 along the basement membrane

Dermatitis herpetiformis (DH)

Intensely **itchy, grouped vesicles** and urticarial plaques, bilateral and symmetrical on **extensor surfaces**: elbows, knees, upper back, buttocks. Men in the 3rd–4th decades. Closely linked to **coeliac disease** (HLA-DQ2/DQ8); both respond to a **gluten-free diet** (dapsone controls the rash).

Pathogenesis: gluten → IgA against gliadin and transglutaminase → IgA cross-reacts with skin (epidermal transglutaminase, reticulin) and deposits at the **tips of dermal papillae** → complement → neutrophils → microabscesses → blister. **Microscopy: neutrophils and fibrin at the papillary tips**, basal vacuolisation and small **subepidermal** blisters; **IF: granular IgA** at the papillary tips.

PEMPHIGUS VULGARIS VS BULLOUS PEMPHIGOID VS DERMATITIS HERPETIFORMIS

Feature	Pemphigus vulgaris	Bullous pemphigoid	Dermatitis herpetiformis
Antigen	**Desmoglein 3** (± 1)	**BP180** (type XVII collagen), BP230	Transglutaminase (gluten-related)
Blister	**Flaccid**, ruptures; erosions	**Tense**, intact	Small grouped vesicles, very itchy
Mucosa	**Usually involved** (often first)	Uncommon	Rare
Nikolsky sign	**Positive**	Negative	Negative
Level of split	**Suprabasal** (intraepidermal), acantholysis	**Subepidermal**	**Subepidermal**, papillary tips
Inflammation	Sparse	**Eosinophils**	**Neutrophils** (microabscesses)
Immunofluorescence	Intercellular IgG (**fishnet**)	**Linear** IgG + C3 at BM	**Granular IgA** at papillary tips
Association	Other autoimmunity; paraneoplastic form	—	**Coeliac disease**

EXAM TRAP

Nikolsky sign (epidermis shears off with sideways pressure) is positive in **pemphigus vulgaris**, and also in toxic epidermal necrolysis and staphylococcal scalded skin syndrome, all of which split **within** or just below a weak epidermis. It is negative in **bullous pemphigoid**. Do not confuse "pemphigus" (desmosome, young-middle age, mouth) with "pemphig**oid**" (hemidesmosome, elderly, tense).

31.5 Benign and premalignant epithelial lesions

Seborrhoeic keratosis

Very common in middle-aged and older people, mainly on the **trunk**. Activating **FGFR3** mutations in many. The sudden eruption of hundreds of lesions (**Leser–Trélat sign**) is a paraneoplastic marker, usually of **GI adenocarcinoma**.

GROSS

- Round, flat, coin-like, **waxy** tan–brown plaque with a velvety or granular surface; "**stuck-on**" look

MICROSCOPY

- Exophytic, sharply demarcated sheets of small **basaloid** cells; variable melanin
- Hyperkeratosis with **horn cysts** and **pseudo-horn cysts** (keratin-filled invaginations)

- **Acanthosis nigricans**: thick, velvety, **hyperpigmented** skin in flexures (axilla, neck, groin). Usually benign (obesity, **insulin resistance**); in older adults it may signal an occult adenocarcinoma, often gastric.

- **Keratoacanthoma**: a flesh-coloured, dome-shaped nodule with a **central keratin-filled crater** on sun-exposed skin of older men; grows rapidly over weeks and often **regresses spontaneously**. Microscopy: cup-shaped mass of glassy eosinophilic keratinocytes around a keratin plug; many regard it as a well-differentiated SCC variant.

Actinic (solar) keratosis: the precursor of SCC

Caused by chronic **sun exposure** in fair-skinned people; lesions are usually < 1 cm, tan-brown, red or skin-coloured, with a **rough, sandpaper-like** feel; some form a cutaneous horn. **TP53** mutations are common. A minority progress to SCC, so they are treated by cryotherapy or topical agents.

GROSS	MICROSCOPY
• Rough, scaly patch on face, scalp, back of hands	• **Atypia of the lower epidermis** (basal layers), not full thickness • **Parakeratosis**, loss of granular layer • **Solar elastosis**: blue-grey degenerated elastic fibres in the dermis

31.6 Malignant epidermal tumours: SCC and BCC

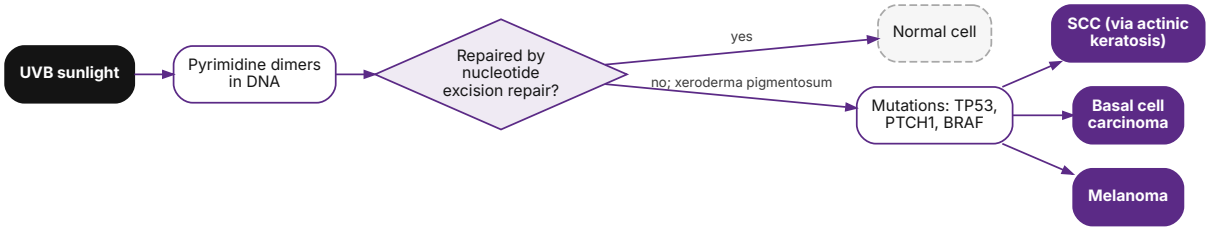


Figure 31.4 · How sunlight causes skin cancer. UVB damages DNA; failed repair fixes the mutations.

Squamous cell carcinoma (SCC)

The second commonest skin cancer, on sun-exposed skin of older people, men more than women. Risk factors: **UV light** (TP53 mutation), **immunosuppression** (transplant recipients; HPV-related), **xeroderma pigmentosum**, arsenic, ionising radiation, and **chronic scars, burns and draining sinuses** (Marjolin ulcer). **SCC in situ (Bowen disease)** is a sharply defined, red, scaly plaque with full-thickness atypia.

GROSS	MICROSCOPY
• Nodular, **hyperkeratotic**, scaly lesion that may **ulcerate**	• Nests and sheets of atypical keratinocytes **invading the dermis** • Well differentiated: polygonal cells, intercellular bridges, **keratin pearls**, dyskeratosis • Poorly differentiated: anaplastic cells, little keratin

Behaviour: grows locally; fewer than 5% of skin SCCs metastasise to regional nodes (higher in mucosa, scars and the immunosuppressed).

Basal cell carcinoma (BCC)

The **commonest invasive cancer in humans**. It grows slowly on sun-exposed skin of fair people and **rarely metastasises**, but untreated it erodes deeply ("**rodent ulcer**") into bone and sinuses. It is driven by the **Hedgehog pathway**: loss of **PTCH1** (germline in **Gorlin / naevoid BCC syndrome**, with multiple BCCs, jaw cysts and medulloblastoma) or activating **SMO** mutations; the SMO inhibitor vismodegib treats advanced cases.

GROSS

- **Pearly papule** with **rolled edges** and **telangiectasia**; some pigmented
- Late: central ulcer with rolled border
- Face, especially around the nose and eyes

MICROSCOPY

- Nests and cords of **basaloid cells** (hyperchromatic, scant cytoplasm) from the epidermis
- **Peripheral palisading** of nuclei at the edge of the nests
- **Retraction clefts** between tumour and fibrotic or mucinous stroma
- Superficial multifocal or nodular patterns

PAKISTAN FOCUS

Consanguinity and childhood skin cancer. Cousin marriage is common in Pakistan and raises the risk of **autosomal recessive** disease. **Xeroderma pigmentosum** is one: defective nucleotide excision repair leaves UV damage unrepaired, so affected children develop freckling, actinic keratoses, SCC, BCC and melanoma at a young age. Strict sun protection and genetic counselling of the family are essential.

31.7 Melanocytic naevi and malignant melanoma

Melanocytic naevi

A **melanocytic naevus** is any benign, congenital or acquired neoplasm of melanocytes. Acquired naevi carry activating **BRAF** (or NRAS) mutations; the cells grow, then stop because the oncogene triggers **senescence** (p16). Clinically they are small (usually < 6 mm), tan–brown, **uniformly pigmented** with round, sharp borders.

LIFE HISTORY OF AN ACQUIRED NAEVUS

Stage	Where the naevus cells are	Clinical look
Junctional	Nests at the dermoepidermal junction	Flat, pigmented macule
Compound	Junctional nests + cords growing into the dermis	Raised, pigmented papule
Intradermal	Dermis only; junctional component lost	Raised, often less pigmented, may be pedunculated

CONCEPT CLEAR

Maturation: the single feature that separates a naevus from melanoma. In a benign naevus, the superficial cells are large, nested and pigmented; deeper cells become smaller with less pigment, form cords and finally spindle cells that resemble nerve ("neurotisation"). This orderly change with depth is **maturation**. Melanoma cells stay large and atypical all the way down: **no maturation**.

Dysplastic naevus

Larger than ordinary naevi (often > 5 mm), flat or slightly raised with a "pebbly" surface, **variegated** pigment and **irregular borders**; may number in the hundreds (**dysplastic naevus syndrome**, familial melanoma with germline **CDKN2A** mutation). Microscopy: compound naevus with **architectural atypia** (enlarged junctional nests that **bridge** adjacent rete ridges, **lentiginous** single-cell spread along the basal layer) and **cytologic atypia** (enlarged, angulated, hyperchromatic nuclei), plus **lamellar fibrosis** around rete ridges and a sparse lymphocytic infiltrate. Most never progress, but they mark a person at risk of melanoma.

Malignant melanoma

Far less common than BCC and SCC but far more deadly. Risk factors: **intermittent intense sun exposure** (childhood sunburns), **fair skin**, many or dysplastic naevi, family history, **xeroderma pigmentosum**. Most arise in the skin; others in mucosa, eye (uvea) and meninges.



Figure 31.5 · From naevus to metastatic melanoma. Each step adds a genetic hit.

MOLECULAR KEY

Genes in melanoma. BRAF (V600E, in about half) or **NRAS** → MAPK pathway; **CDKN2A** (p16/INK4a) loss frees the cell cycle and is the main familial gene; **PTEN** loss → PI3K–AKT; **TERT** promoter mutations → immortality; **KIT** in acral and mucosal melanoma. Therapy follows: **BRAF inhibitors** (vemurafenib) and **PD-1 checkpoint inhibitors**.

MEMORY HOOK

ABCDE of melanoma

Asymmetry · **B**order irregular · **C**olour variegated · **D**iameter > 6 mm · **E**volution (change in size, shape, colour; itching, bleeding). A new pigmented lesion in adult life, or a naevus that changes, needs a biopsy.

CLINICAL TYPES OF CUTANEOUS MELANOMA

Type	Typical site and patient	Growth
Superficial spreading (commonest)	Intermittently sun-exposed skin: trunk, legs	Long radial phase, then vertical
Lentigo maligna	Chronically sun-damaged face of the elderly	Very long radial phase; indolent
Acral / mucosal lentiginous	Palms, soles, nail beds, mucosa; **commonest type in dark-skinned people**; not UV-related (KIT)	Radial, then vertical; often diagnosed late
Nodular	Any site	**Vertical from the start**; worst

GROSS

- Irregular, **variegated** lesion: black, brown, red, blue, grey; white areas of regression
- Notched, irregular border; a **raised nodule** in a flat lesion signals vertical growth
- Ulceration, bleeding

MICROSCOPY

- Large atypical melanocytes with **big nuclei**, irregular contours, chromatin clumped at the membrane and **prominent red (eosinophilic) nucleoli**
- **Radial phase:** single cells and poorly formed nests scattered at all levels of the epidermis (**pagetoid spread**)
- **Vertical phase:** expansile dermal nests or sheets with **no maturation**, mitoses
- Variable melanin; **IHC:** S-100, SOX10, **HMB-45**, Melan-A

PROGNOSTIC FACTORS

Breslow thickness (top of the granular layer to the deepest tumour cell, in mm) is the most important. Also adverse: high **mitotic rate**, **ulceration**, **sentinel lymph node** involvement, male sex and truncal site. Favourable: **tumour-infiltrating lymphocytes**.

EXAM TRAP

Breslow, not Clark. Clark levels (I–V, anatomical layer reached) appear in older books and vivas; staging now uses **Breslow thickness**. Melanoma metastasises **late and widely** (liver, lung, brain), sometimes as an **amelanotic** tumour: use S-100/SOX10/HMB-45.

BCC VS SCC VS MELANOMA

Feature	Basal cell carcinoma	Squamous cell carcinoma	Melanoma
Frequency	**Commonest**	Second	Least common, **most lethal**
Key gene	**PTCH1** (Hedgehog)	**TP53**	**BRAF**, NRAS, CDKN2A
Precursor	None	Actinic keratosis, Bowen disease	Dysplastic naevus (minority)
Microscopy	**Palisading** basaloid nests, **retraction clefts**	**Keratin pearls**, intercellular bridges	Atypical melanocytes, **pagetoid spread**, no maturation
Spread	Local invasion; almost never metastasises	Nodes in < 5%	**Lymph and blood**, widely

LAB SPOTTER

Melanoma slide in 10 seconds. Atypical melanocytes **scattered upward** through the epidermis (pagetoid) instead of neat junctional nests; dermal sheets of large cells with **cherry-red nucleoli** and pigment that do **not get smaller with depth**; mitoses, ulceration. A naevus is symmetrical, nested and matures; a melanoma is not.

EXAM ZONE Past-Paper Pattern SEQs — Solved

PROF PATTERN **a)** Define acanthosis, acantholysis, spongiosis and parakeratosis. **b)** Describe the pathogenesis and microscopic features of psoriasis. **2 + 3 marks**

MODEL ANSWER PLAN

a) Acanthosis: diffuse epidermal hyperplasia (psoriasis). Acantholysis: loss of cohesion between keratinocytes (pemphigus). Spongiosis: intercellular epidermal oedema (eczema). Parakeratosis: nuclei retained in the stratum corneum (psoriasis). **b)** HLA-C*06:02 + trigger → IL-23 → TH17/TH1 → IL-17, IL-22, TNF → keratinocyte hyperproliferation. Micro: acanthosis, regular rete elongation, thin granular layer, parakeratosis, thin suprapapillary plates, dilated capillaries, Kogoj pustules, Munro microabscesses.

Score booster: link the Auspitz sign to the thin suprapapillary plates.

PROF PATTERN **a)** Differentiate pemphigus vulgaris from bullous pemphigoid. **b)** Explain the pathogenesis of dermatitis herpetiformis. **3 + 2 marks**

MODEL ANSWER PLAN

a) Table: age, antigen (desmoglein 3 vs BP180), flaccid vs tense, mucosa, Nikolsky, suprabasal vs subepidermal, fishnet IgG vs linear IgG/C3. **b)** HLA-DQ2/DQ8; gluten → IgA to gliadin/transglutaminase → IgA at papillary tips → complement → neutrophil microabscesses → subepidermal vesicles; coeliac disease.

Score booster: draw the three split levels (Fig 31.3) in one small sketch.

MODULE PATTERN A 62-year-old farmer has a slowly growing pearly nodule beside the nose with rolled edges, dilated vessels and a central ulcer. **a)** Most likely diagnosis? **b)** Describe its microscopy. **c)** Name the pathway involved and the inherited syndrome. **1 + 2 + 2 marks**

MODEL ANSWER PLAN

a) Basal cell carcinoma (rodent ulcer). **b)** Nests of basaloid cells from the epidermis, peripheral palisading, retraction clefts, fibromucinous stroma. **c)** Hedgehog pathway: PTCH1 loss or SMO activation; naevoid BCC (Gorlin) syndrome. Add: locally invasive, almost never metastasises.

CLASSIC **a)** List the risk factors and warning signs of malignant melanoma. **b)** Describe its morphology and prognostic factors. **2 + 3 marks**

MODEL ANSWER PLAN

a) Intermittent intense UV, fair skin, many/dysplastic naevi, CDKN2A family history, xeroderma pigmentosum; ABCDE. **b)** Gross: variegated, irregular, nodule. Micro: large atypical melanocytes with prominent nucleoli, pagetoid spread (radial), dermal nodules without maturation (vertical); S-100/HMB-45. Prognosis: Breslow thickness, mitoses, ulceration, TILs, sentinel node, sex and site.

Score booster: define Breslow thickness exactly (granular layer to deepest cell, mm).

BCQs One-Best-Answer Practice

1 Retention of nuclei in the stratum corneum is called:

A. Hyperkeratosis
B. Parakeratosis
C. Dyskeratosis
D. Acanthosis
E. Spongiosis

2 A biopsy of a silvery-scaled plaque on the elbow shows elongated rete ridges, thin suprapapillary plates and collections of neutrophils within the parakeratotic stratum corneum. These collections are:

A. Pautrier microabscesses
B. Civatte bodies
C. Kamino bodies
D. Munro microabscesses
E. Spongiform pustules of Kogoj

3 Itchy purple flat-topped papules on the wrists show a band-like lymphocytic infiltrate and sawtooth rete ridges. Anucleate eosinophilic bodies in the papillary dermis are:

A. Civatte (colloid) bodies
B. Russell bodies
C. Councilman bodies
D. Psammoma bodies
E. Negri bodies

4 A 78-year-old woman has tense bullae on the inner thighs and groin; the mouth is spared. IF shows linear IgG and C3 along the basement membrane. The target antigen is:

A. Desmoglein 3
B. Desmoglein 1
C. BP180 (type XVII collagen)
D. Tissue transglutaminase
E. Type VII collagen

5 A 45-year-old man has painful oral erosions and flaccid skin blisters; Nikolsky sign is positive and biopsy shows suprabasal acantholysis. Direct IF will show:

A. Linear IgG along the basement membrane
B. Granular IgA at the dermal papillae
C. Intercellular net-like IgG between keratinocytes
D. Perivascular IgA
E. No immune deposits

6 A young man with chronic diarrhoea has intensely itchy grouped vesicles on both elbows and buttocks. Skin IF would most likely show:

A. Linear IgG and C3
B. Intercellular IgG
C. Linear IgA in the epidermis
D. Negative staining
E. Granular IgA at the tips of dermal papillae

7 A waxy, 'stuck-on' brown plaque on the back of a 65-year-old shows basaloid cells with keratin-filled horn cysts. The diagnosis is:

A. Seborrhoeic keratosis
B. Actinic keratosis
C. Basal cell carcinoma
D. Melanoma
E. Keratoacanthoma

8 The sudden appearance of numerous seborrhoeic keratoses (Leser–Trélat sign) should prompt a search for:

A. Melanoma
B. Gastrointestinal adenocarcinoma
C. Lymphoma of skin
D. Hypothyroidism
E. Coeliac disease

9 The naevoid basal cell carcinoma (Gorlin) syndrome results from germline mutation of:

A. BRAF
B. TP53
C. CDKN2A
D. PTCH1
E. KIT

10 The single most important prognostic factor in a localised cutaneous melanoma is:

A. Colour of the lesion
B. Patient's age
C. Clark level
D. Presence of melanin
E. Breslow thickness

11 Which microscopic feature best distinguishes a melanoma from a benign compound naevus?

A. Presence of melanin pigment
B. Lack of maturation of cells with depth
C. Nests at the dermoepidermal junction
D. S-100 positivity
E. Location on sun-exposed skin

12 Recurrent erythema multiforme with target lesions is most often triggered by:

A. Herpes simplex infection
B. Nickel contact
C. Dietary gluten
D. IgE to pollen
E. Ultraviolet light

ANSWER KEY WITH ONE-LINE REASONS

1. **B** Normal on mucosa, abnormal on skin (psoriasis).
2. **D** Munro: stratum corneum; Kogoj: spinous layer.
3. **A** Necrotic basal keratinocytes of lichen planus.
4. **C** Bullous pemphigoid: hemidesmosomal BP180/BP230.
5. **C** Pemphigus vulgaris: anti-desmoglein IgG.
6. **E** Dermatitis herpetiformis, linked to coeliac disease.
7. **A** Horn cysts + stuck-on look.
8. **B** A paraneoplastic sign, most often of GI adenocarcinoma.
9. **D** PTCH1 loss activates Hedgehog signalling.
10. **E** Depth in mm, granular layer to deepest cell.
11. **B** Naevus cells mature with depth; melanoma cells do not.
12. **A** HSV and drugs (sulfonamides) trigger EM.

VIVA Rapid-Fire Viva Voce

Q What is parakeratosis?	Nuclei retained in the stratum corneum, e.g. psoriasis.
Q What is acantholysis?	Keratinocytes lose cohesion as desmosomes fail; pemphigus.
Q What is the Auspitz sign?	Pinpoint bleeding when psoriatic scale is lifted.
Q What is the Koebner phenomenon?	New lesions at sites of trauma; psoriasis, lichen planus.
Q What are Wickham striae?	White lines on lichen planus papules (hypergranulosis).
Q What is the Nikolsky sign?	Epidermis shears off with sideways pressure; pemphigus.
Q What is a 'row of tombstones'?	Basal cells left on the BM in a pemphigus blister floor.
Q What is a target lesion?	Red ring with a pale or blistered centre; erythema multiforme.
Q What is peripheral palisading?	Nuclei lined up at the edge of basal cell carcinoma nests.
Q Three IHC markers of melanoma?	S-100 (or SOX10), HMB-45, Melan-A.
Q What is Breslow thickness?	Granular layer to deepest tumour cell in mm; key prognostic factor.
Q Radial vs vertical growth?	Radial: sideways, cannot metastasise. Vertical: dermal nodule, can.

OSPE Practice Stations

STATION 1 · SLIDE (LAB) — 2 minutes

TASK

Skin: large atypical cells with vesicular nuclei and prominent red nucleoli lie singly and in nests at all levels of the epidermis and form sheets in the dermis that do not get smaller with depth; much brown pigment; several mitoses. (a) Diagnosis. (b) Two features that separate it from a naevus. (c) Most important prognostic measurement.

ANSWER

(a) Malignant melanoma. (b) Pagetoid spread through the epidermis; lack of maturation with depth (also mitoses, nuclear atypia, asymmetry). (c) Breslow thickness in mm.

STATION 2 · SLIDE — 1 minute

TASK

Skin from the face: nests of dark, crowded cells with scant cytoplasm budding from the epidermis into the dermis; nuclei at the edge of each nest line up in a row; clear gaps separate the nests from the stroma. (a) Diagnosis. (b) Name the two features described. (c) Its behaviour.

ANSWER

(a) Basal cell carcinoma. (b) Peripheral palisading; retraction clefts. (c) Locally invasive (rodent ulcer); metastasis is extremely rare.

STATION 3 · IMMUNOFLUORESCENCE — 1 minute

TASK

Direct IF, perilesional skin; 50-year-old with oral erosions and flaccid blisters: IgG outlines every keratinocyte in a net-like pattern. (a) Diagnosis. (b) Target antigen. (c) Level of the blister.

ANSWER

(a) Pemphigus vulgaris. (b) Desmoglein 3 (with or without desmoglein 1). (c) Suprabasal (intraepidermal) with acantholysis.

RAPID REVISION The Night-Before Page

15 ONE-LINERS

01 Parakeratosis = nuclei in stratum corneum; spongiosis = intercellular oedema.

02 Acantholysis = keratinocytes lose cohesion (desmosomes fail).

03 Urticaria: mast cells, wheals within hours; C1 inhibitor deficiency = hereditary angioedema.

04 Allergic contact dermatitis: type IV, Langerhans cells, spongiosis.

05 Erythema multiforme: HSV or drugs, CD8 T cells, target lesions.

06 Psoriasis: TH17/IL-23, parakeratosis, Munro microabscesses, Auspitz.

07 Lichen planus: band-like lymphocytes, sawtooth rete, Civatte bodies.

08 Pemphigus vulgaris: desmoglein 3, suprabasal, fishnet IgG, Nikolsky +.

09 Bullous pemphigoid: BP180, subepidermal, eosinophils, linear IgG/C3.

10 DH: coeliac, granular IgA at papillary tips, neutrophil microabscesses.

11 Seborrhoeic keratosis: stuck-on, horn cysts; many at once = Leser–Trélat.

12 Actinic keratosis: lower-epidermal atypia, solar elastosis → SCC.

13 BCC: commonest; PTCH1; palisading, retraction clefts; rarely metastasises.

14 Naevus matures with depth; melanoma does not.

15 Melanoma: BRAF, CDKN2A; ABCDE; Breslow is the key prognostic factor.

MEMORY HOOKS

- **ABCDE**: Asymmetry, Border, Colour, Diameter > 6 mm, Evolution
- **Six Ps** of lichen planus: pruritic, purple, polygonal, planar papules, plaques
- **Pemphig-US splits Up** in the epidermis; pemphig-OID lifts Off the dermis
- **BCC = 3 Ps**: Pearly, Palisading, PTCH1

TEST YOURSELF: FILL THE GAPS

1. Neutrophils in the spinous layer of psoriasis: _____.
2. Pemphigus foliaceus antigen: _____
3. Seborrhoeic keratosis mutation: _____.
4. Keratin pearls are typical of _____.
5. Main familial melanoma gene: _____.
6. Melanoma type commonest in dark skin: _____.

Answers: Kogoj pustules · desmoglein 1 · FGFR3 · SCC · CDKN2A · acral lentiginous

DRAW FROM MEMORY

The epidermis with the split level of pemphigus foliaceus, pemphigus vulgaris and bullous pemphigoid (Figure 31.3).

DRAW FROM MEMORY

Junctional, compound and intradermal naevus side by side, then a melanoma with pagetoid spread.

SPACED REVISION ☐ Day 1 ☐ Day 3 ☐ Day 7 ☐ Day 14 ☐ Day 30

High-Yield Master Tables

Four tables that examiners raid for one-liners, BCQ stems and viva openers: genes, markers, stains and eponymous bodies, each linked to its chapter. Read every row as a sentence; ***Gen*** *= General Pathology.*

A Genes, mutations & translocations

Disease	Gene / translocation	Product · what goes wrong	Ch
BLOOD & LYMPHOID			
CML; Ph+ B-ALL (adults)	**t(9;22)** Philadelphia chromosome: **BCR–ABL1**	Constitutively active tyrosine kinase; imatinib target; poor prognosis in ALL	13
B-ALL in children	**t(12;21)** ETV6–RUNX1; hyperdiploidy	Disturbed transcription factors; both favourable	13
Acute promyelocytic leukaemia	**t(15;17)** PML–RARA	Maturation block at promyelocyte; ATRA/arsenic response; DIC	13
AML, favourable types	**t(8;21)** RUNX1–RUNX1T1; inv(16) CBFB–MYH11	Core-binding-factor fusions block maturation	13
PV; ET and myelofibrosis	**JAK2 V617F** (almost all PV; about half of ET/PMF); CALR, MPL	JAK–STAT always on → growth-factor-independent proliferation	13
CLL / SLL	del 13q14 (miR-15/16), trisomy 12, del 11q, del 17p	Loss of BCL2-restraining microRNAs; 17p (TP53) loss is worst	13
Burkitt lymphoma	**t(8;14)** MYC–IGH (or t(2;8), t(8;22))	MYC overexpression; Ki-67 near 100%	14
Follicular lymphoma	**t(14;18)** IGH–BCL2	BCL2 overexpressed → apoptosis blocked	14
Mantle cell lymphoma	**t(11;14)** CCND1–IGH	Cyclin D1 overexpressed → G1/S drive	14
Gastric MALT lymphoma	t(11;18) BIRC3–MALT1	NF-κB on; will not regress after H. pylori eradication	2, 14
Sickle cell disease	HBB point mutation: **glutamate → valine** at position 6	HbS polymerises when deoxygenated	12
β-thalassaemia	HBB point mutations (splicing, promoter, chain terminator)	β^0 none or β^+ less β-chain; excess α precipitates	12
α-thalassaemia	HBA1/HBA2 **gene deletions** (1 to 4 genes)	Carrier → trait → HbH disease → hydrops (Hb Barts)	12
Hereditary spherocytosis	Ankyrin (commonest), spectrin, band 3, protein 4.2	Weak membrane skeleton → spherocytes	12
G6PD deficiency	G6PD (X-linked)	Low NADPH → oxidant haemolysis, Heinz bodies	12
Paroxysmal nocturnal haemoglobinuria	PIGA (acquired, X-linked)	No GPI anchor → CD55/CD59 lost → complement lysis	12
Haemophilia A / B	F8 / F9 (X-linked recessive)	Factor VIII / IX deficiency → long aPTT	15
von Willebrand disease	VWF (mostly autosomal dominant)	Poor platelet adhesion + low factor VIII	15
GIT, LIVER & PANCREAS			
Familial adenomatous polyposis	**APC** (5q21), germline	β-catenin not degraded → Wnt signalling on	4
Adenoma–carcinoma sequence	APC → **KRAS** → SMAD4 → **TP53**	Stepwise oncogene gain and suppressor loss	4
Lynch syndrome (HNPCC)	MLH1, MSH2 (MSH6, PMS2)	Mismatch repair fails → microsatellite instability	4
Peutz–Jeghers syndrome	STK11 (LKB1)	Hamartomatous polyps + lip/buccal pigmentation	4
GIST	**KIT** (most) or PDGFRA	Activated receptor kinase; CD117+; imatinib	2
Coeliac disease	HLA-DQ2 / DQ8	Present deamidated gliadin to CD4+ T cells	3
Hereditary haemochromatosis	**HFE C282Y** (6p21)	Low hepcidin → excess iron absorption	6
Wilson disease	ATP7B (13q)	Copper not excreted in bile or loaded on caeruloplasmin	6
α1-antitrypsin deficiency	SERPINA1, PiZZ	Misfolded protein trapped in hepatocyte ER; emphysema	6, 9
Gilbert / Crigler–Najjar	UGT1A1 (reduced / absent activity)	Poor conjugation → unconjugated bilirubin rises	5
Dubin–Johnson syndrome	ABCC2 (MRP2)	Conjugated bilirubin not secreted; black liver	5
Pancreatic adenocarcinoma	**KRAS** (> 90%), CDKN2A, SMAD4, TP53	The most KRAS-driven carcinoma	7
LUNG			
Lung adenocarcinoma	**EGFR** (non-smokers, women, Asians), **ALK**/ROS1 fusions, KRAS (smokers)	Targetable kinases: EGFR and ALK inhibitors	10
Squamous cell carcinoma lung	TP53, CDKN2A (p16) loss	Smoking-driven; few targetable changes	10
Small cell carcinoma	TP53 and **RB1** loss (nearly all)	Neuroendocrine, fast-growing, early spread	10

Disease	Gene / translocation	Product · what goes wrong	Ch
KIDNEY & URINARY TRACT			
Clear cell RCC; VHL syndrome	**VHL** (3p25) loss	HIF not degraded → VEGF, angiogenesis	18
Papillary RCC	**MET** (7q31); trisomy 7, 17	HGF receptor kinase activated	18
Wilms tumour	**WT1** (11p13: WAGR, Denys–Drash); WT2 (11p15: Beckwith–Wiedemann)	Loss of a renal developmental suppressor	18
ADPKD; ARPKD	**PKD1** (16p, polycystin-1, ~85%), PKD2 (4q); ARPKD: PKHD1	Ciliary signalling defect → cysts; ARPKD adds hepatic fibrosis	17
Alport syndrome	**COL4A5** (X-linked)	Abnormal α5 type IV collagen → split GBM, deafness	16
Urothelial carcinoma	FGFR3, HRAS (papillary); TP53, RB (flat CIS)	Two pathways with different behaviour	18
REPRODUCTIVE TRACT & BREAST			
Hereditary breast–ovary cancer	**BRCA1** (17q21), **BRCA2** (13q12)	Homologous-recombination DNA repair lost	21, 22
HER2-positive breast carcinoma	**ERBB2 (HER2)** amplification (17q)	Growth receptor overexpressed; trastuzumab	22
Lobular breast carcinoma; hereditary diffuse gastric cancer	CDH1 loss (germline in the gastric syndrome)	No E-cadherin → single-file or signet-ring cells	2, 22
Cervical carcinoma	HPV 16/18 **E6** and **E7**	E6 degrades p53; E7 inactivates RB	20
Endometrial carcinoma	Type I: **PTEN**, PIK3CA, KRAS, MSI; type II: TP53	Endometrioid (oestrogen-driven) vs serous (aggressive)	20
High-grade serous ovarian carcinoma	TP53 (nearly all); BRCA1/2	Arises mostly from the fallopian tube	21
Hydatidiform mole	Complete: 46,XX **all paternal**; partial: **69,XXY** triploid	Complete: no fetal parts, diffuse trophoblast; partial: fetal parts	21
Testicular germ cell tumours	Isochromosome 12p, **i(12p)**	Present in nearly all GCTs	19
Prostatic adenocarcinoma	TMPRSS2–ERG fusion (about half), PTEN loss	Androgen-driven ETS transcription factor	19
ENDOCRINE			
Papillary carcinoma thyroid	**BRAF V600E**; **RET/PTC** rearrangement (radiation)	MAPK pathway switched on	23
Follicular; anaplastic thyroid	RAS, PAX8–PPARG fusion; anaplastic: TP53	PI3K/AKT and MAPK signalling; TP53 loss = dedifferentiation	23
Medullary carcinoma; MEN2A/2B	**RET** (10q11) point mutation, germline in MEN2	RET kinase on in C cells; MTC + phaeo; 2A parathyroid, 2B neuromas	23
MEN1	MEN1 (menin, 11q13)	Parathyroid, pancreas, pituitary	24
Congenital adrenal hyperplasia	CYP21A2 (21-hydroxylase)	Cortisol ↓, ACTH ↑ → androgens ↑ ± salt loss	24
CARDIOVASCULAR			
Hypertrophic cardiomyopathy	**MYH7** (β-myosin heavy chain), MYBPC3; AD	Myocyte disarray, asymmetric septal hypertrophy	26
Marfan syndrome	**FBN1** (fibrillin-1, 15q21)	Excess TGF-β → cystic medial degeneration, dissection	25
BONE & MUSCLE			
Ewing sarcoma	**t(11;22)(q24;q12)** EWSR1–FLI1	Aberrant transcription factor	27
Osteosarcoma	RB1 (retinoblastoma), TP53 (Li–Fraumeni), CDKN2A; MDM2/CDK4 gain	Cell-cycle checkpoints lost	27
Multiple osteochondromas	**EXT1 / EXT2**	Heparan sulfate defect in growth-plate chondrocytes	27
Enchondroma, chondrosarcoma	IDH1 / IDH2	Oncometabolite 2-hydroxyglutarate	27
Duchenne / Becker dystrophy	**DMD** (Xp21): frameshift vs in-frame deletion	Absent vs reduced, abnormal dystrophin	28
NERVOUS SYSTEM & SKIN			
Diffuse astrocytoma, oligodendroglioma	**IDH1/IDH2** mutation	Defines lower-grade gliomas; better survival	30
Oligodendroglioma	IDH-mutant + **1p/19q co-deletion**	Chemosensitive; best prognosis of diffuse gliomas	30
Glioblastoma (IDH-wild-type)	EGFR amplification, PTEN loss, TERT promoter, CDKN2A	Receptor kinase and PI3K pathways on	30
Pilocytic astrocytoma	**BRAF** (KIAA1549–BRAF fusion)	MAPK activation; grade 1	30
Medulloblastoma	WNT (CTNNB1), SHH (PTCH1), group 3 (MYC)	Molecular group predicts outcome (WNT best)	30
Neurofibromatosis type 1	**NF1** (17q11), neurofibromin	RAS-GAP lost → RAS overactive	30
Neurofibromatosis type 2	**NF2** (22q12), merlin	Schwannomas, meningiomas, ependymomas	30

Disease	Gene / translocation	Product · what goes wrong	Ch
Alzheimer disease	APP, PSEN1, PSEN2 (familial); APOE ε4 (risk)	More Aβ42 → plaques	29
Melanoma	**BRAF V600E**, NRAS; CDKN2A (familial); KIT (acral)	MAPK on; p16 brake lost	31

MEMORY HOOK · TRANSLOCATIONS BY NUMBER

9-22 CML · 15-17 APL · 8-14 Burkitt · 14-18 Follicular · 11-14 Mantle · 11-22 Ewing

Say it as a roll-call: "**Philadelphia, Promyelocytes, Burkitt, BCL2, Cyclin, Ewing**." The chromosome 14 partner is always IGH, the immunoglobulin heavy-chain locus, which is why B-cell tumours dominate the list.

B Tumour markers & immunohistochemistry

Serum markers are for diagnosis support and follow-up, never screening on their own (PSA is the debated exception). IHC answers three questions: what lineage, which organ, which treatment.

Marker	Positive in	Remember	Ch
SERUM (AND URINE) TUMOUR MARKERS			
α-fetoprotein (AFP)	**HCC**; **yolk sac tumour** (testis, ovary)	Also pregnancy, neural tube defects, liver regeneration	6, 19
β-hCG	**Choriocarcinoma** (very high), hydatidiform mole, non-seminomas	Mild rise in seminoma with syncytiotrophoblasts	19, 21
LDH	Germ cell tumour bulk (seminoma), lymphoma	Non-specific; reflects tumour burden	19
CEA	**Colorectal**; also gastric, pancreatic, lung, breast, MTC	Follow-up for recurrence; mild rise in smokers	4
CA 19-9	**Pancreatic** and biliary adenocarcinoma	Also raised in pancreatitis, obstructive jaundice	7
CA-125	**Ovarian serous carcinoma**	Also endometriosis, PID, pregnancy, peritonitis	21
PSA	**Prostate carcinoma**	Organ-specific, not cancer-specific: BPH, prostatitis	19
Calcitonin	**Medullary carcinoma thyroid**	Screen MEN2 families; amyloid in stroma	23
Metanephrines, VMA	**Phaeochromocytoma**; VMA/HVA in neuroblastoma	24-hour urine or plasma metanephrines	24
Urinary 5-HIAA	Carcinoid syndrome (usually liver metastases)	Serotonin metabolite	2, 10
M-protein, Bence Jones	**Multiple myeloma**	Serum/urine electrophoresis; free light chains	14
IMMUNOHISTOCHEMISTRY: LINEAGE AND ORGAN			
Cytokeratin (pan-CK)	Carcinomas, mesothelioma	"Epithelial" lineage	Gen
CK7 / CK20	CK7+ CK20− lung, breast; CK20+ CK7− colorectal	Tracks the origin of a metastasis	4, 10
Vimentin	Sarcomas, melanoma, lymphoma, RCC (with CK), meningioma	"Mesenchymal" lineage	Gen
Desmin; SMA	Desmin: smooth and skeletal muscle tumours; SMA: smooth muscle	Leiomyosarcoma, rhabdomyosarcoma	20
S-100	**Melanoma**, **schwannoma** (strong, diffuse), Langerhans cells	Neural-crest marker	30, 31
HMB-45, Melan-A, SOX10	**Melanoma** (SOX10 also nerve sheath)	Melanocytic lineage	31
GFAP	Astrocytoma, glioblastoma, ependymoma	Glial lineage	30
Chromogranin, synaptophysin	Carcinoid, small cell carcinoma, MTC, phaeochromocytoma	Neuroendocrine granules	2, 10
TTF-1	Lung adenocarcinoma, small cell carcinoma, thyroid	Napsin A also lung adenocarcinoma	10, 23
p40 / p63	**Squamous cell carcinoma**	Separates SCC from adenocarcinoma	10
Calretinin, WT1, D2-40	**Mesothelioma**	Adenocarcinoma: CEA, TTF-1 instead	10
CD117 (KIT), DOG1	**GIST**; CD117 also seminoma, mast cells	Predicts imatinib response	2
CD31, CD34, ERG	Vascular tumours, angiosarcoma	HHV-8 LANA nuclear stain = Kaposi	25
ER, PR	Luminal breast carcinoma; endometrioid carcinoma	Predict hormone therapy response	20, 22
HER2	Breast carcinoma (IHC 3+ or FISH amplified)	Predicts trastuzumab response	22
E-cadherin	**Lost** in lobular breast and diffuse gastric carcinoma	Explains single-file / signet-ring pattern	2, 22

Marker	Positive in	Remember	Ch
OCT3/4, PLAP, CD117; inhibin	**Seminoma** (CD30 = embryonal; AFP = yolk sac); inhibin = sex cord–stromal	Germ cell vs sex cord tumours of testis/ovary	19
p16 (diffuse block)	HPV-driven high-grade cervical lesions	Surrogate for E7 activity	20
MLH1, MSH2, MSH6, PMS2	**Loss** = mismatch-repair deficiency (colon, endometrium)	Screen for Lynch syndrome	4, 20
LEUKAEMIA & LYMPHOMA (CD MARKERS)			
CD45 (LCA)	All leucocytes; lymphoma vs carcinoma	Negative in Reed–Sternberg cells	14
CD19, CD20; CD3	B cells (CD20 = rituximab target); T cells (CD3)	Lineage of a lymphoma or ALL	14
CD5 + CD23	**CLL/SLL**	Mantle cell: CD5+, CD23−, cyclin D1+	13, 14
CD10, BCL6	Germinal-centre lymphomas: follicular, Burkitt	BCL2+ follicular; BCL2− Burkitt	14
CD15, CD30	**Reed–Sternberg cells** of classical Hodgkin	Popcorn cells of NLPHL: CD20+, CD15−, CD30−	14
TdT, CD34	**Lymphoblasts** (ALL, lymphoblastic lymphoma)	CD34 marks any immature blast	13
MPO, CD13, CD33, CD117	**Myeloblasts** (AML)	Auer rods are MPO-positive	13

C Special stains & cytochemistry

Stain	Shows (colour)	Classic use in special pathology	Ch
Congo red	Amyloid: salmon-pink; **apple-green birefringence** in polarised light	MTC stroma, myeloma (AL), restrictive cardiomyopathy, renal amyloid, T2DM islets	16, 23, 26
Perls Prussian blue	Haemosiderin (ferric iron): blue	Haemochromatosis; marrow iron (absent in IDA); ring sideroblasts; heart-failure cells; asbestos bodies	6, 11
PAS	Glycogen, neutral mucin, basement membrane, fungi: magenta	KW nodules and GBM; Whipple macrophages; Ewing glycogen; block positivity in ALL	3, 13, 16
PAS-diastase	Diastase removes glycogen; mucin/glycoprotein stays	**α1-antitrypsin globules** in periportal hepatocytes	6
Ziehl–Neelsen	Acid-fast bacilli: red on blue	Tuberculosis (Fite modification for leprosy); auramine for fluorescence	8
Jones methenamine silver	GBM and mesangial matrix: black	**Spikes** in membranous; **tram-tracks** in MPGN	16
Grocott (GMS) silver	Fungal walls, Pneumocystis: black	Opportunistic pneumonia, fungal sinusitis	8
Warthin–Starry silver	Spiral bacteria: black	H. pylori, spirochaetes	2
Bielschowsky silver	Neurofibrils: black	Neuritic plaques, tangles, Pick bodies	29
Reticulin	Type III collagen fibres: black	Thick plates/lost framework in HCC; marrow fibrosis grading	6, 13
Masson trichrome	Collagen blue-green, muscle red	Staging fibrosis in chronic hepatitis, cirrhosis; old infarct scar	5, 6, 26
Elastic van Gieson	Elastic fibres: black	Broken internal elastic lamina in GCA; medial loss in dissection	25
Oil red O, Sudan III/IV	Neutral lipid (frozen tissue): red/orange	Fatty change, fat embolism, lipid in MCD tubules	6, 16
Mucicarmine	Epithelial mucin: red	Signet-ring cell carcinoma; capsule of Cryptococcus	2, 29
Alcian blue (pH 2.5)	Acid mucin: blue	Goblet cells in **Barrett oesophagus**	1
Giemsa	Bacteria and blood cells	H. pylori in gastric biopsy; blood/marrow films; malaria	2, 11
von Kossa / Alizarin red	Calcium: black / red	Dystrophic and metastatic calcification; psammoma bodies	Gen
Rhodanine; orcein	Copper; HBsAg and copper-binding protein	Wilson disease; **ground-glass hepatocytes** (orcein)	5, 6
MPO / Sudan black B; NSE	Myeloblast granules, Auer rods; NSE: monocytic cells	**AML vs ALL** (blasts positive in AML); NSE for monocytic AML	13
LAP (NAP) score	Neutrophil alkaline phosphatase	**Low in CML**; high in leukaemoid reaction and PV	13

D Eponymous cells, bodies & inclusions

Name	What it is	Seen in	Ch
CELLS			
Reed–Sternberg cell	Large bilobed/binucleate cell, "owl-eye" nucleoli; CD15+ CD30+; lacunar variant in clear space	**Classical Hodgkin lymphoma** (lacunar: nodular sclerosis)	14
Popcorn (LP) cell	RS variant with multilobed "popped-corn" nucleus; CD20+	Nodular lymphocyte-predominant HL	14
Anitschkow (caterpillar) cell	Activated macrophage with wavy central chromatin bar	Aschoff bodies, rheumatic carditis	26

Name	What it is	Seen in	Ch
Langhans giant cell	Nuclei in a **horseshoe** at the periphery	Tuberculous granuloma	8
Hürthle (oncocytic) cell	Large cell, granular pink cytoplasm (mitochondria)	Hashimoto thyroiditis, Hürthle cell tumours	23
Signet-ring cell	Mucin vacuole pushing nucleus to one side	Diffuse gastric carcinoma, Krukenberg tumour	2, 21
Koilocyte	Squamous cell with perinuclear halo, wrinkled nucleus	HPV infection: condyloma, LSIL/CIN 1	20
Heart-failure cell	Haemosiderin-laden alveolar macrophage	Chronic pulmonary congestion (LVF, mitral stenosis)	26
Smudge (basket) cell	Fragile lymphocyte crushed during smearing	CLL	13
Sea-blue histiocyte	Macrophage full of membrane debris	CML (high cell turnover)	13
BODIES, INCLUSIONS & CRYSTALS			
Aschoff body	Fibrinoid focus with T cells, plasma cells, Anitschkow cells	**Rheumatic fever** (pathognomonic)	26
Councilman (acidophil) body	Shrunken eosinophilic apoptotic hepatocyte	Acute viral hepatitis	5
Ground-glass hepatocyte	Finely granular cytoplasm full of HBsAg in ER	Chronic hepatitis B	5
Mallory–Denk body	Ropy pink cytoplasmic clump of keratin 8/18 + ubiquitin	Alcoholic hepatitis; also NASH, Wilson, PBC	6
Psammoma body	Concentric, laminated calcified sphere	**P**apillary thyroid, **S**erous ovary, **M**eningioma, **M**esothelioma	21, 23, 30
Russell body	Pink globule of immunoglobulin in plasma cell	Chronic inflammation, myeloma (Dutcher body if intranuclear)	14
Auer rod; faggot cell	Needle-shaped fused azurophil granules (MPO+); bundles of them in one cell	**AML**; faggot cells in APL	13
Howell–Jolly body	Small round nuclear (DNA) remnant in RBC	Asplenia, sickle cell disease, megaloblastic anaemia	12
Heinz body	Precipitated denatured Hb (supravital stain)	**G6PD deficiency** → bite cells	12
Pappenheimer body	Iron-containing RBC granules (Perls+)	Sideroblastic anaemia, post-splenectomy	11
Basophilic stippling	Aggregated ribosomes in RBC	Lead poisoning, thalassaemia	11, 12
Döhle body	Pale-blue patch of rough ER in a neutrophil	Severe infection, leukaemoid reaction	13
Charcot–Leyden crystals	Crystallised eosinophil protein (galectin-10)	Asthma sputum	9
Curschmann spirals	Whorled mucus plugs with shed epithelium	Asthma	9
Ferruginous (asbestos) body	Golden-brown beaded rod with iron-protein coat	Asbestosis	9
Schaumann / asteroid body	Laminated calcium body / star inside giant cell	Sarcoidosis (not specific)	9
Kimmelstiel–Wilson nodule	PAS+ ball of mesangial matrix	**Diabetic nephropathy**	16, 24
Michaelis–Gutmann body	Laminated calcified body inside macrophages	Malakoplakia (bladder, E. coli)	18
Schiller–Duval body	Glomerulus-like vessel cuffed by tumour cells	**Yolk sac tumour**	19, 21
Call–Exner body	Mini-follicle of eosinophilic fluid amid granulosa cells	**Granulosa cell tumour**	21
Orphan Annie-eye nucleus	Optically clear (ground-glass) nucleus with grooves	**Papillary carcinoma thyroid**	23
Zellballen	Nests of chief cells rimmed by sustentacular cells	Phaeochromocytoma, paraganglioma	24
Civatte (colloid) body	Anucleate dead basal keratinocyte in papillary dermis	Lichen planus	31
Munro microabscess	Neutrophils in parakeratotic stratum corneum	Psoriasis	31
NERVOUS SYSTEM & TUMOUR PATTERNS			
Rosenthal fibre	Thick, pink, corkscrew GFAP-rich astrocytic process	**Pilocytic astrocytoma**; chronic gliosis	30
Homer Wright rosette	Tumour cells round a central fibrillary core (no lumen)	Medulloblastoma, neuroblastoma, Ewing	27, 30
Perivascular pseudorosette	Tumour cells radiating round a vessel	Ependymoma (true rosettes also)	30
Verocay body	Two rows of palisaded nuclei with pink fibrillary zone between	**Schwannoma** (Antoni A)	30
Lewy body	Round pink α-synuclein inclusion with pale halo	**Parkinson disease**, Lewy body dementia	29
Neurofibrillary tangle	Flame-shaped hyperphosphorylated tau	Alzheimer disease	29
Pick body	Smooth round silver-positive tau inclusion	Pick disease (frontotemporal)	29
Negri body	Pink cytoplasmic viral inclusion in Purkinje/hippocampal neurons	Rabies	29
Owl-eye inclusion	Large intranuclear viral inclusion with halo in an enlarged cell	Cytomegalovirus (pneumonia in the immunosuppressed)	8

Pathology Buzzwords A–Z

*The phrases that turn a stem into a diagnosis in five seconds. Each entry reads: the classic description → **diagnosis**, with the chapter number in the small box.*

> ⚠ **EXAM TRAP · A BUZZWORD IS A DOOR, NOT THE ANSWER**
>
> Marks go to the **description behind** the buzzword: "sheets of monotonous medium-sized lymphoid cells, many mitoses, scattered tingible-body macrophages" earns what "starry sky" alone does not. Some buzzwords overlap (onion-skin: Ewing, malignant hypertension, PSC), so always name the organ.

A

- **ABCDE** (asymmetry, border, colour, diameter > 6 mm, evolution) → melanoma [31]
- **Adenoma–carcinoma sequence** (APC → KRAS → TP53) → colorectal carcinoma [4]
- **Antoni A / Antoni B** areas (cellular palisaded / loose myxoid) → schwannoma [30]
- **Apple-core / napkin-ring** lesion → annular carcinoma of left colon [4]
- **Apple-green birefringence** on Congo red in polarised light → amyloid [16, 23, 26]
- **Auer rods**; bundles of them = faggot cells → AML; APL [13]
- **Auspitz sign** (pinpoint bleeding when a scale is lifted) → psoriasis [31]

B

- **Bag of worms** in the scrotum → varicocele [19]
- **Bamboo spine** → ankylosing spondylitis [28]
- **Berry aneurysm** at circle of Willis branch points → subarachnoid haemorrhage; linked to ADPKD [29]
- **Bird's-beak** narrowing on barium swallow → achalasia [1]
- **Bite cells** (spleen has plucked out Heinz bodies) → G6PD deficiency [12]
- **Blue bloater** (cyanosis, oedema, cor pulmonale) → chronic bronchitis [9]
- **Bread-and-butter pericarditis** → fibrinous pericarditis: MI, uraemia, rheumatic fever [26]
- **Brodie abscess** (walled off by sclerotic bone) → subacute osteomyelitis [27]
- **Bronze diabetes** (pigmentation + diabetes + cirrhosis) → haemochromatosis [6]
- **Brown tumour** (osteitis fibrosa cystica) → hyperparathyroidism [23]
- **Bunch of grapes** → hydatidiform mole; in a girl's vagina, sarcoma botryoides [20, 21]
- **Butterfly glioma** crossing the corpus callosum → glioblastoma [30]

C

- **Café-au-lait spots** (six or more) → neurofibromatosis type 1 [30]
- **Caput medusae** → portal hypertension [6]
- **Caseous necrosis** (cheesy, structureless) → tuberculosis [8]
- **Caterpillar (Anitschkow) cells** → rheumatic carditis [26]
- **Charcot–Bouchard microaneurysms** → hypertensive intracerebral haemorrhage (basal ganglia) [29]
- **Charcot–Leyden crystals**, **Curschmann spirals** → asthma [9]
- **Chicken-wire capillaries** → oligodendroglioma [30]
- **Chicken-wire (pericellular) fibrosis** → steatohepatitis, alcoholic or NASH [6]
- **Chocolate cyst** → ovarian endometriosis (endometrioma) [20]
- **Chondromyxoid stroma** with ducts → pleomorphic adenoma [1]
- **Cobblestone mucosa**, **creeping fat** → Crohn disease [3]
- **Cobweb clot** in standing CSF → tuberculous meningitis [29]
- **Codman triangle** (lifted periosteum) → osteosarcoma or any aggressive bone lesion [27]
- **Coin lesion** → solitary pulmonary nodule: granuloma, hamartoma, carcinoma [10]
- **Coffee-bean nuclei** (longitudinal grooves) → granulosa cell or Brenner tumour [21]
- **Comedo necrosis** in ducts → high-grade DCIS [22]
- **Contraction bands** in myocytes → reperfused infarct [26]
- **Councilman bodies** → acute viral hepatitis [5]
- **Courvoisier sign** (palpable gallbladder + jaundice) → carcinoma head of pancreas [7]
- **Crab-like, gritty** stellate breast mass → invasive ductal carcinoma [22]
- **Crescents** in Bowman space → RPGN [16]
- **Crew-cut (hair-on-end) skull** → β-thalassaemia major, sickle cell [12]
- **Currant-jelly sputum** → Klebsiella pneumonia [8]

D

- **Dawson fingers** (plaques at right angles to ventricles) → multiple sclerosis [29]
- **Double-barrelled aorta** → aortic dissection [25]
- **Dressler syndrome** (pericarditis weeks after MI) → autoimmune post-infarct pericarditis [26]
- **Dry tap** on marrow aspiration → myelofibrosis [13]

E

- **Egg-shell calcification** of hilar nodes → silicosis [9]
- **Erythroplakia** (red velvety oral patch) → high-risk premalignant lesion [1]

F

- **Fatty streak** of foam cells → earliest atherosclerosis [25]
- **Ferruginous bodies**, pleural plaques → asbestos exposure [9]
- **Fish-mouth (button-hole) mitral valve** → chronic rheumatic mitral stenosis [26]
- **Flask-shaped ulcers** → amoebic colitis [3]
- **Flea-bitten kidney** (cortical petechiae) → malignant hypertension; also infective endocarditis [25, 26]
- **Foamy PAS-positive macrophages** in villi → Whipple disease [3]

- **Fried-egg cells** (perinuclear halo) → oligodendroglioma; clear seminoma cells look alike [30, 19]

G

- **Ghon complex** (subpleural focus + caseous hilar nodes) → primary tuberculosis [8]
- **Golden-yellow, variegated** renal mass → clear cell RCC [18]
- **Gower sign**, calf pseudohypertrophy → Duchenne muscular dystrophy [28]
- **Grey hepatisation** (dry, grey-brown, firm lobe) → lobar pneumonia, stage 3 [8]
- **Ground-glass hepatocytes** → chronic hepatitis B (HBsAg) [5]

H

- **Heaped-up, rolled edges** to a gastric ulcer → malignant ulcer [2]
- **Heart-failure cells** in sputum or alveoli → chronic pulmonary congestion [26]
- **Heberden / Bouchard nodes** (DIP / PIP) → osteoarthritis [28]
- **Heliotrope rash**, Gottron papules → dermatomyositis [28]
- **Hobnail liver** (small regular nodules) → micronodular (alcoholic) cirrhosis [6]
- **Honeycomb lung** → end-stage fibrosis, e.g. IPF (UIP) [9]
- **Hürthle cells** + lymphoid follicles → Hashimoto thyroiditis [23]
- **Hyaline membranes** lining alveoli → diffuse alveolar damage (ARDS) [9]
- **Hypersegmented neutrophils** (≥ 5 lobes) → megaloblastic anaemia [11]

I

- **Indian-file** (single-file) cells → invasive lobular carcinoma [22]
- **Insulitis / islet amyloid** → type 1 / type 2 diabetes [24]
- **Interface hepatitis** (limiting plate eroded) → chronic hepatitis [5]
- **Involucrum** (new bone sheath round a sequestrum) → chronic osteomyelitis [27]

J–K

- **Janeway lesions**, Osler nodes, Roth spots, splinter haemorrhages → infective endocarditis [26]
- **Jaw claudication**, tender temporal artery → giant cell arteritis [25]
- **Kayser–Fleischer ring** → Wilson disease [6]
- **Keratin pearls**, intercellular bridges → squamous cell carcinoma [10, 31]
- **Kimmelstiel–Wilson nodules** → diabetic nephropathy [16, 24]
- **Koilocytes** → HPV infection [20]
- **Krukenberg tumour** (bilateral ovarian signet-ring deposits) → metastatic gastric carcinoma [2, 21]

L

- **Lead-pipe colon** (no haustra) → long-standing ulcerative colitis [3]
- **Leather-bottle stomach** → linitis plastica (diffuse gastric carcinoma) [2]
- **Lepidic growth** along alveolar walls → adenocarcinoma in situ of lung [10]
- **Leukoplakia** (white patch that will not rub off) → oral premalignant lesion; naswar, gutka [1]
- **Lisch nodules** (iris hamartomas) → NF1 [30]
- **Looser zones** (pseudofractures) → osteomalacia [27]
- **Lumpy-bumpy** (granular) immunofluorescence → immune-complex GN [16]
- **Lymphoid aggregates** + steatosis in portal tracts → chronic hepatitis C [5]

M

- **MacCallum plaque** (left atrial endocardium) → rheumatic heart disease [26]
- **Mallory–Denk bodies** → alcoholic hepatitis [6]
- **Maltese crosses** / oval fat bodies in urine → nephrotic syndrome [16]
- **Millet-seed nodules** in many organs → miliary tuberculosis [8]
- **Moon face**, buffalo hump, purple striae → Cushing syndrome [24]
- **Mosaic pattern** of cement lines → Paget disease of bone [27]
- **Moth-eaten** (permeative) bone destruction → Ewing sarcoma, lymphoma, myeloma [27]
- **Muddy-brown granular casts** → acute tubular necrosis [17]
- **Munro microabscesses** → psoriasis [31]
- **Mural nodule in a cyst** (child's cerebellum) → pilocytic astrocytoma [30]

N

- **Needle-shaped, negatively birefringent** crystals → gout; rhomboid, positive → pseudogout [28]
- **Negri bodies** → rabies [29]
- **Nikolsky sign** (skin shears with rubbing) → pemphigus vulgaris [31]
- **Nuclear grooves**, pseudoinclusions → papillary carcinoma thyroid [23]
- **Nutmeg liver** (red centrilobular, pale periportal) → chronic passive congestion [26]

O

- **Oat cells**, nuclear moulding, crush artefact → small cell carcinoma [10]
- **Oligoclonal bands** in CSF → multiple sclerosis [29]
- **Onion-skin arterioles** ± fibrinoid necrosis → malignant hypertension [25]
- **Onion-skin periductal fibrosis**, beaded ducts → primary sclerosing cholangitis [3, 6]
- **Onion-skin periosteum** → Ewing sarcoma [27]
- **Orphan Annie-eye nuclei** → papillary carcinoma thyroid [23]
- **Owl-eye nucleoli** in a binucleate giant cell → Reed–Sternberg cell [14]

P

- **Paget disease of the nipple** (eczematous nipple) → underlying DCIS or carcinoma [22]
- **Palisading** of peripheral nuclei, retraction clefts → basal cell carcinoma [31]
- **Pannus** eroding cartilage → rheumatoid arthritis [28]
- **Pearly papule** with telangiectasia; **rodent ulcer** → basal cell carcinoma [31]
- **Peau d'orange** → breast carcinoma in dermal lymphatics [22]
- **Pencil cells** + microcytic hypochromic RBCs → iron deficiency anaemia [11]
- **Perifascicular atrophy** → dermatomyositis [28]
- **Pink puffer**, barrel chest → emphysema [9]
- **Popcorn cells** → nodular lymphocyte-predominant Hodgkin [14]
- **Porcelain gallbladder** → calcified chronic cholecystitis; carcinoma risk [7]

- **Psammoma bodies** → papillary thyroid, serous ovary, meningioma, mesothelioma [21, 23, 30]
- **Pseudomembranes** ("volcano" eruptions) → C. difficile colitis [3]
- **Pseudopalisading necrosis**, vascular proliferation → glioblastoma [30]
- **Pseudopolyps**, continuous disease from rectum → ulcerative colitis [3]
- **Pulseless disease** (young Asian woman) → Takayasu arteritis [25]
- **Punched-out lytic lesions** of skull → multiple myeloma [14]
- **Punched-out ulcer**, clean base, lesser curve → benign peptic ulcer [2]

R

- **Rachitic rosary**, bowed legs → rickets [27]
- **Red hepatisation** (red, firm, liver-like lobe) → lobar pneumonia, stage 2 [8]
- **Red neurons** (12–24 h after ischaemia) → acute irreversible neuronal injury [29]
- **Reid index > 0.4** → chronic bronchitis [9]
- **Ring sideroblasts** (Perls stain) → sideroblastic anaemia [11]
- **Rokitansky protuberance** (hair, sebum, teeth) → mature cystic teratoma [21]
- **Rokitansky–Aschoff sinuses** → chronic cholecystitis [7]
- **Rosenthal fibres** → pilocytic astrocytoma [30]
- **Rouleaux** + high ESR → multiple myeloma [14]
- **Rule of 10s** → phaeochromocytoma [24]

S

- **Saddle nose**, sinusitis, haematuria → granulomatosis with polyangiitis [25]
- **Salmon-pink (red, velvety) tongues** above the GOJ → Barrett oesophagus [1]
- **Salt-and-pepper chromatin** → neuroendocrine tumour: carcinoid, SCLC, MTC [2, 10]
- **Sawtooth rete ridges**, band of lymphocytes → lichen planus [31]
- **Saw-toothed (serrated) crypts** → hyperplastic polyp / sessile serrated lesion [4]
- **Scalloped colloid**, tall cells → Graves disease [23]
- **Schistocytes** (helmet cells) → microangiopathic haemolysis: DIC, TTP, HUS [15]
- **Sea-blue histiocytes** in marrow → CML [13]
- **Sequestrum** (dead bone fragment) → chronic osteomyelitis [27]
- **Signet-ring cells** → diffuse gastric carcinoma [2]
- **Sister Mary Joseph nodule** → umbilical metastasis (stomach, ovary) [2]
- **Skip lesions**, transmural granulomas → Crohn disease [3]
- **Smoky / cola-coloured urine** with RBC casts → nephritic syndrome [16]
- **Smudge cells** → CLL [13]
- **Snowstorm** on ultrasound, no fetus → complete hydatidiform mole [21]
- **Soap-bubble** lytic epiphyseal lesion → giant cell tumour of bone [27]
- **Spherocytes**, Coombs-negative, family history → hereditary spherocytosis [12]
- **Spike and dome** on silver stain → membranous nephropathy [16]
- **Spongiosis** (intercellular oedema) → eczema [31]
- **Staghorn calculus** → struvite stone (urease-positive Proteus) [18]
- **Starry sky** → Burkitt lymphoma [14]
- **Stones, bones, groans, moans** → hyperparathyroidism [23]
- **Strawberry gallbladder** → cholesterolosis [7]
- **Strawberry naevus** → capillary haemangioma of infancy [25]
- **String sign** on barium → Crohn stricture of terminal ileum [3]
- **Stuck-on** greasy plaque with horn cysts → seborrhoeic keratosis [31]
- **Sunburst** periosteal reaction → osteosarcoma [27]
- **Swiss-cheese endometrium** (cystic glands) → endometrial hyperplasia [20]

T

- **Target cells** → thalassaemia, liver disease, HbC, post-splenectomy [12]
- **Target (iris) skin lesions** → erythema multiforme [31]
- **Tear-drop cells** → myelofibrosis [13]
- **Thick trabeculae** with lost reticulin → hepatocellular carcinoma [6]
- **Thunderclap headache** → subarachnoid haemorrhage [29]
- **Thyroidisation** of tubules, U-shaped scars over blunted calyces → chronic pyelonephritis [17]
- **Tombstone** basal row after suprabasal split → pemphigus vulgaris [31]
- **Tophi** (chalky urate deposits) → chronic gout [28]
- **Tram-track** (double-contour) GBM → MPGN [16]
- **Transverse (girdle) ulcers** of ileum → intestinal tuberculosis [3]
- **Tree-bark** wrinkling of aortic intima → syphilitic aortitis [25]
- **Trousseau sign** (migratory thrombophlebitis) → pancreatic carcinoma [7]

V–Z

- **Vegetations**: small warty on closure lines → rheumatic; large friable → infective; both valve surfaces → Libman–Sacks [26]
- **Verocay bodies** → schwannoma [30]
- **Villous atrophy** with intraepithelial lymphocytes → coeliac disease [3]
- **Virchow node** (left supraclavicular) → gastric carcinoma [2]
- **Waterhouse–Friderichsen** → adrenal haemorrhage in meningococcal sepsis [24]
- **Watershed infarcts** → global hypotension (border zones) [29]
- **Wavy fibres** at the infarct border (½–4 h) → earliest microscopic sign of MI [26]
- **Whorled** grey-white cut surface in the uterus → leiomyoma [20]
- **Wickham striae** (white lacy lines on violaceous papules) → lichen planus [31]
- **Wire loops** (thick capillary walls) → lupus nephritis, class IV [16]
- **Woody, stony-hard thyroid** fixed to neck → Riedel thyroiditis [23]
- **Zellballen** (cell balls) → phaeochromocytoma, paraganglioma [24]

Lab OSPE Atlas

Every slide, smear and specimen of the Special Pathology lab list, described the way you will see it and the way the examiner wants to hear it.

THE 60-SECOND SLIDE ROUTINE

1. Organ (what normal tissue is left at the edge?) → **2. Process** (inflammation, hyperplasia or tumour?) → **3. Benign or malignant** (circumscribed vs infiltrative; atypia, mitoses) → **4. Name it** + **two features** you can point at → **5. Answer the add-on question** (cause, marker, complication). In each card below, **LP** = low power, **HP** = high power, **ID** = the three points to write.

M10 Module 10 · GIT, liver, lung, blood

H. pylori gastritis — CH 2 · SLIDE

LP Antral mucosa with a dense band of lymphocytes and plasma cells in the superficial lamina propria; lymphoid follicles with germinal centres.

HP Neutrophils inside gland and pit epithelium; small curved rods in the surface mucus.

ID **1** lymphoplasmacytic infiltrate + follicles **2** intraepithelial neutrophils (activity) **3** organisms on Giemsa or silver stain

EXAMINER ASKS Two serious complications? → Peptic ulcer; gastric adenocarcinoma; MALT lymphoma.

Chronic viral hepatitis — CH 5 · SLIDE

LP Portal tracts expanded by lymphocytes (dense lymphoid aggregates in HCV); inflammation spills into the lobule; fibrous septa later.

HP Lymphocytes eroding the limiting plate (interface hepatitis); spotty necrosis; ground-glass hepatocytes (HBV) or steatosis (HCV).

ID **1** portal lymphocytic infiltrate **2** interface hepatitis **3** portal or bridging fibrosis

EXAMINER ASKS Grade vs stage? → Grade = activity (inflammation, necrosis); stage = fibrosis, ending in cirrhosis.

Hepatocellular carcinoma — CH 6 · SLIDE

LP Thick trabeculae and sheets of liver-like cells separated by sinusoids; cirrhotic liver alongside.

HP Polygonal cells, granular pink cytoplasm, high N:C ratio, prominent nucleoli; bile plugs; vascular invasion.

ID **1** trabeculae more than 3 cells thick, reticulin lost **2** atypical hepatocytes **3** bile production

EXAMINER ASKS Marker and local risk factors? → AFP; HCV (leading in Pakistan) and HBV cirrhosis, aflatoxin.

Pulmonary tuberculosis — CH 8 · SLIDE

LP Many rounded granulomas, some fusing, with pink centres; fibrosis around older lesions.

HP Amorphous pink caseation; rim of epithelioid histiocytes, Langhans giant cells (horseshoe nuclei), lymphocytes.

ID **1** caseous necrosis **2** epithelioid cells + Langhans cells **3** lymphocyte cuff and fibrosis

EXAMINER ASKS Stain and mechanism? → Ziehl–Neelsen for AFB; type IV hypersensitivity (Th1, IFN-γ activates macrophages).

Chronic pancreatitis — CH 7 · SLIDE

LP Lobules replaced by dense fibrous tissue; few atrophic acini; islets spared and crowded together.

HP Dilated ducts containing pink protein plugs or calcified concretions; scattered chronic inflammatory cells.

ID **1** fibrosis with acinar loss **2** dilated ducts with protein plugs **3** preserved islets trapped in scar

EXAMINER ASKS Commonest cause and two complications? → Long-term alcohol; pseudocyst, steatorrhoea, diabetes.

Cirrhosis — CH 6 · SLIDE

LP Normal lobules lost; whole section made of regenerative nodules of hepatocytes ringed by fibrous septa (trichrome blue).

HP Septa carry lymphocytes and proliferating bile ductules; nodules lack a central vein; fatty change or Mallory–Denk bodies if alcoholic.

ID **1** diffuse process **2** bridging fibrous septa **3** parenchymal nodules without normal architecture

EXAMINER ASKS Three effects of portal hypertension? → Ascites, varices (oesophageal, rectal, caput medusae), splenomegaly, encephalopathy.

Lobar pneumonia — CH 8 · SLIDE

LP Every alveolus in the field filled with exudate; alveolar walls intact, so the outline of the lung is preserved.

HP Red hepatisation: neutrophils, red cells, fibrin. Grey hepatisation: fibrin mesh with dying neutrophils, red cells lysed.

ID **1** diffuse alveolar exudate **2** septa intact **3** stage-specific contents

EXAMINER ASKS Name the stages and the usual organism. → Congestion, red and grey hepatisation, resolution; Streptococcus pneumoniae.

Squamous cell carcinoma, lung — CH 10 · SLIDE

LP Nests and sheets of malignant squamous cells infiltrating a desmoplastic stroma near a bronchus.

HP Keratin pearls, single-cell keratinisation, intercellular bridges; pleomorphic nuclei, mitoses.

ID **1** invasive squamous nests **2** keratin pearls **3** intercellular bridges

EXAMINER ASKS Typical paraneoplastic syndrome? → Hypercalcaemia from PTHrP; strongest risk factor smoking.

PERIPHERAL SMEARS AND MARROW (ANAEMIAS AND LEUKAEMIAS)

Smear	What you see	Clinch it with	Examiner asks → answer
Iron deficiency anaemia	Small pale RBCs with wide central pallor, anisopoikilocytosis, pencil cells	MCV, MCH low; ferritin low; TIBC high; RDW high	Commonest cause in an adult man? → chronic GI blood loss (ulcer, carcinoma, hookworm)
Megaloblastic anaemia	Macro-ovalocytes, hypersegmented neutrophils (≥ 5 lobes); marrow megaloblasts, giant metamyelocytes	MCV > 100 fL; low B12 or folate	Which deficiency damages the spinal cord? → B12 (subacute combined degeneration)
β-thalassaemia major	Marked microcytosis, hypochromia, target cells, nucleated RBCs, basophilic stippling	Hb electrophoresis: HbF very high, HbA absent or low	Why iron overload? → transfusions + more gut absorption (ineffective erythropoiesis, low hepcidin)
Sickle cell disease	Sickle cells, target cells, Howell–Jolly bodies (autosplenectomy)	Sickling test; HbS on electrophoresis	The mutation? → glutamate → valine at position 6 of β-globin
Hereditary spherocytosis	Small dense spherocytes without central pallor; polychromasia	MCHC high; osmotic fragility up; Coombs negative	Treatment that corrects the anaemia? → splenectomy
Acute myeloid leukaemia	Large blasts (≥ 20%), fine chromatin, nucleoli, Auer rods; faggot cells in APL	MPO+, CD13, CD33, CD117	Which subtype causes DIC? → APL, t(15;17); treat with ATRA
Acute lymphoblastic leukaemia	Blasts with scant agranular cytoplasm, condensed chromatin, small nucleoli	TdT+; CD19 (B) or CD3 (T)	Peak age and good-risk genetics? → 2–5 years; t(12;21), hyperdiploidy
Chronic myeloid leukaemia	WBC often > 100 × 10^9/L, all stages of granulocytes, basophilia, eosinophilia, few blasts; marrow hypercellular	Ph chromosome (BCR–ABL); low LAP score; sea-blue histiocytes	How does it differ from a leukaemoid reaction? → basophilia, splenomegaly, low LAP, BCR–ABL
Chronic lymphocytic leukaemia	Sheets of small mature lymphocytes; smudge cells	CD5+, CD23+, CD20 dim	Two complications? → warm AIHA; hypogammaglobulinaemia with infections; Richter transformation

GROSS SPECIMENS · MODULE 10

Specimen	Describe it (organ → lesion → cut surface)	Examiner asks → answer
Chronic peptic ulcer	Stomach (lesser curve): round punched-out ulcer usually < 2 cm, straight walls, flat margins, clean base, folds radiating from it	Complications? → haemorrhage, perforation, pyloric obstruction
Gastric carcinoma	Fungating or ulcerated mass with heaped-up, rolled edges; or a diffusely thick rigid wall (linitis plastica)	Ovarian spread called? → Krukenberg tumour
Colorectal carcinoma	Right colon: bulky polypoid fungating mass. Left colon: annular napkin-ring stricture with dilated bowel above	Follow-up marker? → CEA; stage by TNM
Cirrhotic liver	Shrunken, nodular surface; cut surface of nodules (micro < 3 mm, macro > 3 mm) in grey-white scar; yellow or green tint	Leading cause in Pakistan? → hepatitis C
Hepatocellular carcinoma	Large soft mass or many nodules, variegated yellow-green and haemorrhagic, in a cirrhotic liver; portal vein invasion	Tumour marker? → AFP
Chronic cholecystitis with stones	Thick fibrotic gallbladder wall, flattened mucosa, many faceted yellow (cholesterol) or mixed stones	Name the risk group? → female, fertile, fat, forty (and family)
Lobar pneumonia	Whole lobe solid and airless: red-brown (red hepatisation) or grey-brown, dry (grey); fibrin on pleura	Two complications? → abscess, empyema, bacteraemia
Tuberculous lung	Apical cavity with ragged caseous lining and fibrous wall; or millet-sized yellow-white nodules (miliary)	Primary lesion with nodes called? → Ghon complex
Bronchogenic carcinoma	Firm grey-white hilar mass encircling and narrowing a bronchus, spreading into lung; enlarged black-grey nodes	Central tumours in smokers? → squamous and small cell carcinoma
Emphysematous lung	Voluminous pale lungs with enlarged airspaces and apical bullae	Type with α1-AT deficiency? → panacinar, lower lobes

M11 Module 11 · Kidney, reproductive tract, breast, thyroid

Renal cell carcinoma, clear cell — CH 18 · SLIDE

LP Nests and sheets of cells separated by a delicate branching network of thin-walled vessels; haemorrhage; pseudocapsule.

HP Cells with water-clear cytoplasm (glycogen, lipid), crisp cell membranes, small round nuclei.

ID **1** clear cells with sharp borders **2** rich delicate vasculature **3** haemorrhage, necrosis

EXAMINER ASKS Gene and paraneoplastic effects? → VHL (3p); polycythaemia (EPO), hypercalcaemia (PTHrP).

Urothelial (transitional cell) carcinoma — CH 18 · SLIDE

LP Papillary fronds with fibrovascular cores covered by thickened urothelium; stromal invasion in high-grade tumours.

HP Loss of polarity, nuclear pleomorphism, hyperchromasia and mitoses at all levels.

ID **1** papillae with fibrovascular cores **2** thick disordered urothelium **3** atypia ± invasion

EXAMINER ASKS Risk factors? → Smoking, aniline dyes, cyclophosphamide, phenacetin; Schistosoma → squamous carcinoma.

Seminoma
CH 19 · SLIDE

LP Sheets and lobules of uniform cells divided by thin fibrous septa full of lymphocytes; granulomas may be present.

HP Large round cells, clear glycogen-rich cytoplasm, distinct borders, central nucleus with a prominent nucleolus.

ID **1** uniform clear cells **2** lymphocyte-rich septa **3** granulomatous reaction

EXAMINER ASKS Markers and prognosis? → AFP normal; hCG mildly up in some; PLAP, OCT3/4+; radiosensitive, excellent outlook.

Benign prostatic hyperplasia
CH 19 · SLIDE

LP Nodules of glands and fibromuscular stroma; glands of varying size, some cystic with papillary infoldings.

HP Two cell layers: inner tall columnar secretory cells and outer flat basal cells; corpora amylacea in lumens.

ID **1** nodular gland + stroma hyperplasia **2** basal cell layer preserved **3** infoldings, corpora amylacea

EXAMINER ASKS Hormone and zone? → Dihydrotestosterone; transition zone, so it obstructs the urethra early. Not premalignant.

Adenomyosis
CH 20 · SLIDE

LP Islands of endometrium deep in the myometrium, well below the endometrial–myometrial junction, with thick muscle around them.

HP Benign endometrial glands surrounded by endometrial stroma; no atypia.

ID **1** glands **and** stroma **2** within myometrium **3** reactive muscle hypertrophy

EXAMINER ASKS Difference from endometriosis? → Adenomyosis is inside the uterine wall; endometriosis is outside the uterus (ovary, pouch of Douglas).

Endometrial carcinoma (endometrioid)
CH 20 · SLIDE

LP Crowded back-to-back glands with little or no stroma, cribriform areas, invading myometrium.

HP Stratified atypical columnar cells, mitoses; squamous morules may be present.

ID **1** back-to-back glands **2** cytological atypia **3** myometrial invasion

EXAMINER ASKS Type I vs type II? → I: oestrogen, hyperplasia, PTEN, good prognosis. II: serous, atrophic endometrium, TP53, aggressive.

Mucinous cystadenoma, ovary
CH 21 · SLIDE

LP Multiple cyst spaces with fibrous walls, lined by a single layer of epithelium.

HP Tall columnar cells full of apical mucin with small basal nuclei; no stratification, atypia or invasion.

ID **1** multilocular cysts **2** single layer of tall mucinous cells **3** bland basal nuclei

EXAMINER ASKS How does it differ from serous? → Mucinous: larger, multilocular, rarely bilateral, mucin. Serous: often bilateral, ciliated cells, psammoma bodies.

Complete hydatidiform mole
CH 21 · SLIDE

LP Large swollen chorionic villi with central fluid-filled cisterns; no fetal vessels.

HP Circumferential proliferation of cytotrophoblast and syncytiotrophoblast around villi, with atypia.

ID **1** hydropic villi with cisterns **2** diffuse trophoblast proliferation **3** no fetal parts

EXAMINER ASKS Karyotype and follow-up? → 46,XX all paternal; serial hCG, as some progress to invasive mole or choriocarcinoma.

Choriocarcinoma
CH 21 · SLIDE

LP Haemorrhagic, necrotic tumour; **no chorionic villi**.

HP Sheets of malignant cytotrophoblast mixed with multinucleate syncytiotrophoblast, invading vessels.

ID **1** absent villi **2** biphasic trophoblast **3** extensive haemorrhage and necrosis

EXAMINER ASKS Marker and spread? → Very high hCG; early blood spread to lungs, vagina, brain; very chemosensitive.

Fibroadenoma, breast
CH 22 · SLIDE

LP Well-circumscribed nodule; loose stroma compresses ducts into slits (intracanalicular) or surrounds round ducts (pericanalicular).

HP Ducts with epithelial and myoepithelial layers; bland stroma without atypia or mitoses.

ID **1** circumscribed biphasic lesion **2** slit-like ducts, two cell layers **3** bland stroma

EXAMINER ASKS Differentiate from phyllodes tumour? → Phyllodes: older women, leaf-like clefts, cellular atypical stroma, stromal overgrowth, can recur.

Invasive ductal carcinoma (NST)
CH 22 · SLIDE

LP Irregular cords, nests and glands of tumour cells invading fat in a dense desmoplastic stroma; DCIS nearby.

HP Pleomorphic hyperchromatic nuclei, nucleoli, mitoses; no myoepithelial layer; lymphovascular invasion.

ID **1** infiltrative edge **2** desmoplasia **3** malignant nests without myoepithelium

EXAMINER ASKS Prognostic factors? → Node status (most important), size, grade, LVI, ER/PR, HER2, Ki-67.

Hashimoto thyroiditis
CH 23 · SLIDE

LP Thyroid replaced by dense lymphoplasmacytic infiltrate with lymphoid follicles and germinal centres; small atrophic follicles.

HP Hürthle (oncocytic) cells with abundant granular pink cytoplasm lining follicles; scant colloid.

ID **1** lymphoid follicles **2** Hürthle cells **3** follicular atrophy, little colloid

EXAMINER ASKS Antibodies and tumour risk? → Anti-TPO, anti-thyroglobulin; B-cell (marginal zone) lymphoma risk raised.

Papillary carcinoma, thyroid
CH 23 · SLIDE

LP Branching papillae with fibrovascular cores; scattered laminated psammoma bodies.

HP Clear "Orphan Annie-eye" nuclei, overlapping, with grooves and intranuclear pseudoinclusions.

ID **1** papillae with cores **2** diagnostic nuclear features **3** psammoma bodies

EXAMINER ASKS Spread and risk factor? → Lymphatics to cervical nodes; childhood radiation; BRAF, RET/PTC.

Leiomyoma, uterus
CH 20 · SLIDE

LP Well-demarcated nodule of interlacing smooth-muscle bundles in whorls.

HP Uniform spindle cells with cigar-shaped nuclei, pink cytoplasm, very few mitoses, no necrosis.

ID **1** circumscribed **2** whorled fascicles **3** bland cigar-shaped nuclei

EXAMINER ASKS Features of leiomyosarcoma? → Tumour necrosis, nuclear atypia, high mitotic count.

GROSS SPECIMENS · MODULE 11

Specimen	Describe it (organ → lesion → cut surface)	Examiner asks → answer
Renal cell carcinoma	Spherical mass at one pole; golden-yellow with grey-white, red haemorrhagic and cystic areas; may grow into renal vein	Classic triad? → haematuria, flank pain, mass
Adult polycystic kidney	Both kidneys hugely enlarged, replaced by cysts (up to several cm) with clear, brown or haemorrhagic fluid	Extrarenal lesions? → liver cysts, berry aneurysm, MVP
Hydronephrosis	Dilated pelvis and calyces, thin rim of compressed cortex; stone or stricture at the outlet	Causes? → stone, BPH, tumour, stricture; PUJ obstruction in children
Papillary bladder carcinoma	Frond-like exophytic papillary growth from mucosa (lateral walls, trigone); wall thick if invasive	Main risk factor? → smoking
Seminoma testis	Enlarged testis; homogeneous grey-white lobulated cut surface, no haemorrhage or necrosis	Variegated, haemorrhagic tumour suggests? → non-seminoma
Nodular hyperplasia, prostate	Enlarged prostate; cut surface of many well-defined solid and cystic nodules; urethra compressed into a slit	Bladder changes? → trabeculation, diverticula, hydronephrosis
Leiomyoma uterus	Firm grey-white whorled nodules, well circumscribed: submucosal, intramural, subserosal	Commonest symptom? → menorrhagia
Mucinous cystadenoma	Large multiloculated cyst, smooth outer surface, sticky gelatinous contents	Usually unilateral or bilateral? → unilateral
Mature cystic teratoma	Unilocular cyst with hair, greasy sebum and sometimes teeth; Rokitansky protuberance	Malignant change? → squamous carcinoma (rare)
Hydatidiform mole	Mass of translucent grape-like vesicles, no fetus	Serum marker? → very high hCG
Breast carcinoma	Hard grey-white stellate mass, gritty to cut, chalky streaks; nipple retraction, peau d'orange skin	Commonest quadrant? → upper outer
Multinodular goitre	Asymmetric, enlarged thyroid; colloid-filled nodules with haemorrhage, cysts and calcification	Two complications? → tracheal compression; toxic nodule

M12 Module 12 · Bone, skin, nervous system, heart

Osteochondroma — CH 27 · SLIDE

LP Cap of hyaline cartilage covered by perichondrium over a stalk of trabecular bone with marrow.

HP Cap chondrocytes like a disorganised growth plate; enchondral ossification at the base of the cap.

ID **1** cartilage cap + perichondrium **2** enchondral ossification **3** cortex and marrow continuous with host bone

EXAMINER ASKS Site and genetics? → Metaphysis near growth plate (knee); multiple hereditary exostoses: EXT1/EXT2.

Osteosarcoma — CH 27 · SLIDE

LP Destructive sheet of malignant cells laying down lace-like osteoid; host trabeculae destroyed.

HP Bizarre pleomorphic cells, atypical mitoses, tumour giant cells; osteoid made directly by tumour cells.

ID **1** malignant osteoid from tumour cells **2** marked pleomorphism **3** bone destruction

EXAMINER ASKS Age, site and X-ray? → Under 20, knee metaphysis; Codman triangle, sunburst; lung metastases.

Giant cell tumour of bone — CH 27 · SLIDE

LP Sheets of mononuclear cells with evenly scattered osteoclast-like giant cells; haemorrhage.

HP Giant cells with up to 100 nuclei that look like the nuclei of the mononuclear cells.

ID **1** many uniformly spread giant cells **2** mononuclear stromal cells **3** epiphyseal site (clinical)

EXAMINER ASKS Driving molecule? → RANKL from neoplastic stromal cells; age 20–40, soap-bubble X-ray, recurs locally.

Ewing sarcoma — CH 27 · SLIDE

LP Sheets of small round blue cells divided by fibrous strands; necrosis.

HP Uniform cells a little larger than lymphocytes, scant clear glycogen-rich cytoplasm; Homer Wright rosettes.

ID **1** small round blue cells **2** PAS+ glycogen, CD99+ **3** Homer Wright rosettes

EXAMINER ASKS Translocation and X-ray? → t(11;22) EWSR1–FLI1; diaphysis, onion-skin periosteum.

Malignant melanoma — CH 31 · SLIDE

LP Asymmetric lesion; atypical melanocytes in nests and singly at all levels of the epidermis (pagetoid) and invading dermis; no maturation with depth.

HP Large cells, pleomorphic nuclei, cherry-red nucleoli, mitoses, variable melanin.

ID **1** pagetoid spread **2** no maturation **3** atypia, mitoses, melanin; S-100, HMB-45+

EXAMINER ASKS Single best prognostic factor? → Breslow thickness; also ulceration, mitotic rate, nodes.

Schwannoma — CH 30 · SLIDE

LP Encapsulated spindle-cell tumour with cellular Antoni A and loose myxoid Antoni B areas; thick hyalinised vessels.

HP Palisaded nuclei in two rows with pink fibrillary zone between them: Verocay bodies.

ID **1** capsule **2** Antoni A + B **3** Verocay bodies; S-100 diffusely +

EXAMINER ASKS Commonest intracranial site? → Vestibular nerve at cerebellopontine angle; bilateral = NF2.

Glioblastoma — CH 30 · SLIDE

LP Very cellular infiltrating glioma with necrosis; tumour cells line up around necrotic zones.

HP Pleomorphic astrocytic cells, mitoses, glomeruloid microvascular proliferation.

ID **1** pseudopalisading necrosis **2** microvascular proliferation **3** pleomorphic GFAP+ cells

EXAMINER ASKS Grade and outlook? → WHO grade 4; butterfly lesion; median survival about 15 months.

Meningioma — CH 30 · SLIDE

LP Lobules and concentric whorls of meningothelial cells; laminated psammoma bodies.

HP Syncytial cells with indistinct borders, oval nuclei, occasional pseudoinclusions.

ID **1** whorls **2** psammoma bodies **3** bland syncytial cells

EXAMINER ASKS Origin and genetics? → Arachnoid cap cells; NF2 / chromosome 22 loss; prior radiation.

Pilocytic astrocytoma — CH 30 · SLIDE

LP Biphasic: compact areas of bipolar hair-like cells and loose microcystic areas.

HP Bright pink corkscrew Rosenthal fibres and eosinophilic granular bodies; few mitoses.

ID **1** biphasic pattern **2** Rosenthal fibres **3** granular bodies, bland cells

EXAMINER ASKS Age, site, behaviour? → Child's cerebellum, cyst with mural nodule; BRAF; grade 1, good outlook.

Medulloblastoma — CH 30 · SLIDE

LP Densely cellular sheets of small round blue cells in the cerebellar midline.

HP Hyperchromatic carrot-shaped nuclei, scant cytoplasm, many mitoses; Homer Wright rosettes.

ID **1** small round blue cells **2** Homer Wright rosettes **3** high mitotic rate

EXAMINER ASKS How does it spread? → Through CSF (drop metastases); radiosensitive; WNT group best.

OTHER BONE AND BRAIN TUMOURS ON THE LAB LIST (ONE-LINE SLIDES)

Tumour	What you see	Examiner asks → answer
Chondrosarcoma	Lobules of atypical hyaline or myxoid cartilage permeating marrow spaces and trapping trabeculae; binucleate chondrocytes	Age and site? → over 40; pelvis, shoulder, ribs
Ependymoma	Perivascular pseudorosettes (cells round vessels with a nucleus-free zone), true ependymal rosettes; GFAP+	Site in children? → fourth ventricle
Oligodendroglioma	Sheets of round cells with perinuclear halos (fried-egg), chicken-wire capillaries, calcification	Molecular definition? → IDH-mutant + 1p/19q co-deleted
Chronic osteomyelitis	Dead bone with empty lacunae (sequestrum), chronic inflammation, reactive new bone (involucrum)	Commonest organism? → S. aureus; Salmonella in sickle cell

GROSS SPECIMENS · MODULE 12 (WITH THE OSPE FAVOURITES FROM THE HEART AND BRAIN)

Specimen	Describe it (organ → lesion → cut surface)	Examiner asks → answer
Osteosarcoma	Large grey-white gritty haemorrhagic mass in the distal femoral metaphysis, breaking the cortex, lifting periosteum, into soft tissue	Where does it spread? → lungs
Osteochondroma	Mushroom-shaped bony outgrowth near the metaphysis with a bluish-white cartilage cap, pointing away from the joint	Malignant change? → rare chondrosarcoma, more in multiple exostoses
Giant cell tumour	Expanded bone end; red-brown, haemorrhagic, cystic tumour within a thin shell of bone, reaching the joint surface	Commonest site? → around the knee
Chronic osteomyelitis	Thickened bone with dead sequestrum, sheath of new bone (involucrum) and draining sinus tracts	Late complications? → fracture, amyloidosis, SCC in sinus
Atherosclerotic aorta	Raised yellow-white plaques, worst in the abdominal aorta and around ostia; ulceration, calcification, thrombus	Earliest lesion? → fatty streak
Myocardial infarct	Left ventricle (LAD territory): yellow, soft infarct with red border (3–7 days) or grey-white scar (weeks)	Rupture is most likely when? → 3–7 days
Rheumatic mitral stenosis	Thick fibrotic calcified leaflets, fused commissures, short thick chordae: fish-mouth orifice; dilated left atrium	Pathognomonic lesion of carditis? → Aschoff body
Cerebral infarct	MCA territory: recent soft swollen pale (or haemorrhagic, if embolic) area; old infarct a cystic cavity with gliosis	Neurons most vulnerable? → CA1 hippocampus, Purkinje cells
Meningioma	Rounded rubbery grey-white dura-based mass compressing, not invading, the brain	Calcified bodies seen? → psammoma bodies
Glioblastoma	Ill-defined hemispheric mass, variegated: yellow necrosis, red haemorrhage, cysts; may cross corpus callosum	Microscopic hallmark? → pseudopalisading necrosis

Top 60 Most-Repeated SEQ Themes

The themes that keep returning in BUMDC and UHS-pattern papers, in the words students meet them, with what a full-mark answer must contain. They are recurring themes, not a forecast of any paper: aim to write each as define or classify (1) + mechanism with a flowchart (2) + morphology or a comparison table (2) in about 12 minutes.

#	Recurring theme → what your answer must include	Ch	✓
GASTROINTESTINAL TRACT			
1	**Oral leukoplakia, erythroplakia & oral SCC** → naswar, gutka, paan; dysplasia; SCC morphology; nodal spread	1	☐
2	**Pleomorphic adenoma** → classify salivary tumours; epithelium + chondromyxoid stroma; recurrence; malignant change	1	☐
3	**Barrett oesophagus; oesophageal carcinoma** → goblet-cell metaplasia → dysplasia → adenocarcinoma; SCC vs adeno table	1	☐
4	**H. pylori gastritis & peptic ulcer** → urease, CagA; antral gastritis; acid vs defence; ulcer zones; complications	2	☐
5	**Gastric carcinoma** → risk factors; intestinal vs diffuse; signet-ring cells, linitis plastica; Krukenberg, Virchow node	2	☐
6	**Coeliac disease (malabsorption)** → gluten, HLA-DQ2/8, anti-tTG; villous atrophy, crypt hyperplasia, IELs; lymphoma risk	3	☐
7	**Crohn disease vs ulcerative colitis** → table: site, continuity, depth, granulomas, fistulae, pseudopolyps, cancer risk	3	☐
8	**Polyps & colorectal carcinoma** → classify polyps; APC → KRAS → TP53; FAP, Lynch; right vs left; TNM, CEA	4	☐
LIVER, GALLBLADDER & PANCREAS			
9	**Jaundice** → bilirubin pathway; pre-hepatic / hepatic / post-hepatic lab table; Gilbert, Dubin–Johnson	5	☐
10	**Hepatitis B** → serology graph; carrier vs chronic; acute vs chronic morphology; ground-glass cells	5	☐
11	**Cirrhosis & portal hypertension** → definition; stellate cells; micro vs macronodular; ascites, varices, splenomegaly	6	☐
12	**Haemochromatosis** → HFE C282Y, low hepcidin; liver, pancreas, heart, skin; Perls stain; HCC risk	6	☐
13	**Hepatocellular carcinoma** → HCV, HBV, aflatoxin; gross patterns; trabecular histology; AFP; vein invasion	6	☐
14	**Gallstones & cholecystitis** → cholesterol vs pigment; risk factors; acute vs chronic cholecystitis; complications	7	☐
15	**Acute & chronic pancreatitis** → gallstones, alcohol; trypsin flowchart; fat necrosis; lipase; pseudocyst	7	☐
LUNG			
16	**Lobar pneumonia vs bronchopneumonia** → four stages, gross and micro of each; patchy pattern; complications	8	☐
17	**Tuberculosis** → type IV hypersensitivity, granuloma; Ghon complex; primary vs secondary; cavity, miliary	8	☐
18	**COPD & asthma** → centriacinar vs panacinar; protease–antiprotease; Reid index; Th2/IgE; Curschmann spirals	9	☐
19	**Lung carcinoma** → smoking; types; EGFR, ALK; central vs peripheral; paraneoplastic table; Pancoast, SVC	10	☐
BLOOD & LYMPHOID SYSTEM			
20	**Anaemia; IDA vs chronic disease** → morphological + aetiological classes; IDA smear; iron studies table; hepcidin	11	☐
21	**Megaloblastic anaemia** → B12 vs folate; macro-ovalocytes, hypersegmented neutrophils; marrow; neurology	11	☐
22	**β-thalassaemia major** → genetics; ineffective erythropoiesis; iron overload; smear; HbF; crew-cut skull	12	☐
23	**AML vs ALL** → ≥ 20% blasts; cytochemistry and CD markers; Auer rods; APL t(15;17) and DIC	13	☐
24	**CML vs leukaemoid reaction** → t(9;22) BCR–ABL; phases; smear; LAP, basophils, spleen; imatinib	13	☐
25	**Lymphomas: Hodgkin vs non-Hodgkin** → classify; Reed–Sternberg cell, CD15/CD30; Ann Arbor; t(14;18), t(8;14)	14	☐
26	**Multiple myeloma** → CRAB; M-spike, Bence Jones protein; punched-out lesions; marrow plasma cells; kidney	14	☐
27	**DIC & bleeding tests** → triggers; consumption flowchart; PT, aPTT, platelets, D-dimer; haemophilia vs vWD	15	☐
KIDNEY & URINARY TRACT			
28	**Nephritic vs nephrotic; PSGN** → comparison table; PSGN pathogenesis; LM, IF, EM humps; low C3, ASO	16	☐
29	**Nephrotic diseases: MCD, membranous, FSGS** → LM, IF, EM of each; child vs adult; complications	16	☐
30	**Acute tubular necrosis** → ischaemic vs nephrotoxic; pathogenesis; morphology; three phases; casts	17	☐
31	**Acute & chronic pyelonephritis** → ascending route, reflux; abscesses, papillary necrosis; U-shaped scars	17	☐

#	Recurring theme → what your answer must include	Ch	✓
32	**Renal cell carcinoma** → types and genes (VHL, MET); clear cells; triad; paraneoplastic; renal vein spread	18	☐
33	**Urothelial carcinoma** → smoking, aniline dyes; papillary vs flat CIS; low vs high grade; invasion	18	☐
34	**Urolithiasis** → stone types table (oxalate, struvite, uric acid, cystine); causes; complications	18	☐
REPRODUCTIVE TRACT & BREAST			
35	**Testicular tumours** → classify; cryptorchidism; seminoma vs non-seminoma table; AFP, hCG, LDH	19	☐
36	**BPH vs prostatic carcinoma** → zone; DHT; basal cells lost in cancer; PSA; Gleason; osteoblastic metastases	19	☐
37	**Carcinoma cervix & CIN** → HPV E6–p53, E7–RB; CIN / SIL grades; Pap smear; HPV vaccine	20	☐
38	**Endometriosis & adenomyosis** → definitions; sites; theories; chocolate cyst; infertility	20	☐
39	**Endometrial carcinoma, type I vs II** → unopposed oestrogen, hyperplasia, PTEN vs TP53; morphology; prognosis	20	☐
40	**Ovarian tumours** → classify; markers; serous vs mucinous table; mature teratoma; Krukenberg	21	☐
41	**Hydatidiform mole; choriocarcinoma** → complete vs partial table (karyotype, villi, fetus, hCG); follow-up	21	☐
42	**Breast carcinoma** → risk factors, BRCA; DCIS vs invasive; IDC vs ILC; ER/PR/HER2; prognostic factors	22	☐
ENDOCRINE			
43	**Graves disease vs Hashimoto thyroiditis** → TSH-receptor vs TPO antibodies; scalloped colloid vs follicles + Hürthle cells	23	☐
44	**Thyroid carcinomas** → papillary nuclei vs follicular invasion; spread; medullary: calcitonin, MEN2	23	☐
45	**Hyperparathyroidism** → primary, secondary, tertiary; stones, bones, groans, moans; osteitis fibrosa	23	☐
46	**Diabetes mellitus** → type 1 vs 2 pathogenesis; criteria; AGEs, polyol; nephropathy (KW), vascular disease	24	☐
47	**Cushing syndrome & adrenal disorders** → causes; diagnostic flowchart; Addison disease; phaeochromocytoma	24	☐
CARDIOVASCULAR SYSTEM			
48	**Atherosclerosis** → risk factors; response-to-injury flowchart; fatty streak → plaque; complications	25	☐
49	**Aneurysms & aortic dissection** → AAA, syphilitic, berry; Marfan, cystic medial degeneration; Stanford A/B	25	☐
50	**Myocardial infarction** → pathogenesis; timeline of morphology; troponin curve; complications by day	26	☐
51	**Rheumatic heart disease** → molecular mimicry; Jones criteria; Aschoff body; pancarditis; mitral stenosis	26	☐
52	**Infective endocarditis** → acute vs subacute: organisms, valves, vegetations; Duke criteria; emboli	26	☐
BONE, JOINTS & MUSCLE			
53	**Osteomyelitis** → organisms by group; metaphyseal spread; sequestrum, involucrum; complications	27	☐
54	**Bone tumours** → osteosarcoma vs Ewing vs GCT table: age, site, X-ray sign, histology, genetics	27	☐
55	**RA vs OA; gout** → pannus, RF / anti-CCP vs cartilage wear, osteophytes; urate needles, tophi	28	☐
NERVOUS SYSTEM & SKIN			
56	**Meningitis** → CSF table (bacterial, viral, TB); organisms by age; morphology; complications	29	☐
57	**Stroke: infarction vs haemorrhage** → thrombotic vs embolic; timeline of changes; hypertensive bleed; berry aneurysm	29	☐
58	**Brain tumours** → glioblastoma, meningioma, medulloblastoma: age, site, grade, hallmark histology	30	☐
59	**Melanoma; BCC vs SCC** → UV, BRAF, CDKN2A; ABCDE; growth phases; Breslow; palisading vs keratin pearls	31	☐
60	**Psoriasis vs lichen planus; bullous diseases** → histology table; pemphigus vs pemphigoid: blister level, IF	31	☐

SCORE BOOSTERS THAT WORK ON ANY OF THE 60

- **Open with a definition or classification** in one or two lines; the first mark is usually there.
- **Draw one labelled flowchart** for pathogenesis and **one table** for any "compare" or "differentiate": both are quicker to mark than paragraphs.
- **Morphology always in two halves**: gross (size, colour, consistency, cut surface) then microscopy (architecture → cells → special features).
- **Finish with complications or prognosis**, and one marker, gene or Pakistan-relevant risk factor where it fits.

Grand Mock Paper

One full Special Pathology paper across all three modules, written to the BUMDC/UHS pattern: 45 one-best-answer BCQs and 9 structured SEQs, with an answer key and a marking scheme.

PAPER FORMAT AND TIME ALLOCATION

Part 1 · 45 BCQs, 1 mark each, no negative marking: **45 minutes** (one minute per question); cover the key until you finish. **Part 2 · 9 SEQs**, 5 marks each: about **2 hours** (12–13 minutes per question) plus 10 minutes to review. In the marking scheme, half marks go to each correct, specific point; a labelled diagram or table earns the same marks as a paragraph. Timings in your own college may differ slightly.

PART 1 45 One-Best-Answer BCQs · 45 minutes

1 A 50-year-old man with long-standing heartburn has red, velvety tongues of mucosa extending 3 cm above the gastro-oesophageal junction. Which biopsy finding is required to confirm Barrett oesophagus?

A. Eosinophils more than 15 per high-power field
B. Basal zone hyperplasia with elongated papillae
C. Columnar epithelium with goblet cells
D. Parietal and chief cells in the mucosa
E. Koilocytes in the squamous epithelium

2 Antral-predominant Helicobacter pylori gastritis predisposes to duodenal ulcer mainly because it:

A. Increases gastrin and acid secretion
B. Destroys parietal cells, causing achlorhydria
C. Induces antibodies to intrinsic factor
D. Causes bile reflux into the stomach
E. Blocks prostaglandin synthesis in the duodenum

3 A 42-year-old woman has a diffusely thickened, rigid stomach wall and bilateral ovarian masses. A biopsy of the gastric wall will most likely show:

A. Spindle cells positive for CD117
B. Small lymphocytes forming lymphoepithelial lesions
C. Well-formed glands arising in intestinal metaplasia
D. Nests of cells with salt-and-pepper chromatin
E. Discohesive signet-ring cells

4 Which finding in a colectomy specimen favours Crohn disease over ulcerative colitis?

A. Continuous disease starting in the rectum
B. Transmural inflammation with non-caseating granulomas
C. Crypt abscesses
D. Inflammatory pseudopolyps
E. Toxic megacolon

5 A 38-year-old man has a right-sided colon carcinoma that shows microsatellite instability; his mother had endometrial carcinoma at 45. His germline defect most likely involves:

A. APC
B. STK11
C. SMAD4
D. A mismatch-repair gene such as MLH1
E. TP53

6 A health worker's serology shows HBsAg negative, anti-HBs positive and total anti-HBc positive. This indicates:

A. Past infection with recovery and immunity
B. Vaccination only
C. Acute hepatitis B in the window period
D. Chronic hepatitis B carrier state
E. Early acute hepatitis B

7 A 21-year-old student turns mildly yellow while fasting during exams. Bilirubin is 2.8 mg/dL, almost all unconjugated; liver enzymes, haemoglobin and reticulocytes are normal. The diagnosis is:

A. Crigler–Najjar syndrome type I
B. Dubin–Johnson syndrome
C. Gilbert syndrome
D. Hereditary spherocytosis
E. Rotor syndrome

8 The principal source of the excess collagen deposited in cirrhosis is the:

A. Kupffer cell
B. Sinusoidal endothelial cell
C. Cholangiocyte
D. Hepatocyte
E. Hepatic stellate (Ito) cell

9 A 52-year-old man has diabetes, bronze skin and hepatomegaly; transferrin saturation is 85%. Liver biopsy shows golden-brown hepatocyte pigment that is blue on Perls stain. The basic defect is:

A. Excess hepcidin blocking iron export
B. HFE mutation causing inappropriately low hepcidin
C. ATP7B mutation impairing copper excretion
D. Misfolded α1-antitrypsin retained in the ER
E. Repeated blood transfusions

10 Chalky white spots in the peripancreatic fat of a patient who died of acute pancreatitis represent:

A. Caseous necrosis
B. Coagulative necrosis
C. Calcified lymph nodes
D. Enzymatic fat necrosis with calcium soaps
E. Fibrinoid necrosis

11 At autopsy a lung lobe is firm, dry and grey-brown. Alveoli are packed with fibrin and disintegrating neutrophils; red cells have lysed. The stage of lobar pneumonia is:

A. Congestion
B. Red hepatisation
C. Resolution
D. Organisation
E. Grey hepatisation

12 After primary tuberculosis, a child's radiograph shows a subpleural focus near the interlobar fissure with enlarged caseous hilar nodes. This combination is called:

A. Simon focus
B. Ghon complex
C. Assmann focus
D. Miliary pattern
E. Pott lesion

13 A 36-year-old non-smoker has emphysema and raised liver enzymes. The expected pattern of emphysema is:

A. Panacinar, worst in the lower lobes
B. Centriacinar, worst in the upper lobes
C. Distal acinar, beneath the pleura
D. Irregular, around scars
E. Bullous, confined to the apex

14 A 60-year-old smoker has a central hilar mass and serum calcium of 12.8 mg/dL without bone metastases. Biopsy shows keratin pearls. The hypercalcaemia is caused by:

A. Ectopic ACTH
B. Calcitonin deficiency
C. Inappropriate ADH secretion
D. PTH-related protein secreted by the tumour
E. Vitamin D toxicity

15 A 58-year-old woman with rheumatoid arthritis has Hb 9.6 g/dL, MCV 79 fL, low serum iron, low TIBC and raised ferritin. The key mediator of her anaemia is:

A. Excess erythropoietin
B. Vitamin B12 deficiency
C. Hepcidin induced by IL-6
D. Transferrin receptor mutation
E. Lead inhibiting haem synthesis

16 A 60-year-old with macrocytic anaemia and hypersegmented neutrophils develops loss of vibration sense and ataxia. The deficient vitamin and its likely cause are:

A. Folate; poor diet
B. Vitamin B12; autoimmune gastritis
C. Iron; chronic blood loss
D. Vitamin B6; isoniazid
E. Folate; methotrexate

17 A 10-month-old child of first-cousin parents has Hb 5 g/dL, MCV 58 fL, hepatosplenomegaly, target cells and nucleated red cells. Haemoglobin electrophoresis is expected to show:

A. HbS about 90%
B. Raised HbA2 with normal HbA
C. HbH inclusions
D. Hb Barts
E. Very high HbF with little or no HbA

18 A 12-year-old with jaundice, splenomegaly and spherocytes has a negative direct Coombs test and increased osmotic fragility; his father had a splenectomy. The defective protein is most often:

A. Glucose-6-phosphate dehydrogenase
B. PIGA
C. Ankyrin
D. β-globin
E. Pyruvate kinase

19 A 32-year-old has gum bleeding, low platelets, low fibrinogen and raised D-dimer. Marrow shows hypergranular promyelocytes with bundles of Auer rods. The genetic lesion is:

A. t(15;17) PML–RARA
B. t(8;21) RUNX1–RUNX1T1
C. t(9;22) BCR–ABL1
D. t(12;21) ETV6–RUNX1
E. inv(16) CBFB–MYH11

20 A lymph node shows nodules separated by broad collagen bands; large cells lie in clear spaces and are CD15+ and CD30+. The subtype is:

A. Mixed cellularity
B. Lymphocyte-rich
C. Lymphocyte-depleted
D. Nodular sclerosis
E. Nodular lymphocyte-predominant

21 A 19-year-old woman has had heavy periods and gum bleeding since menarche; her father bruises easily. Platelet count and PT are normal, bleeding time is prolonged and aPTT mildly prolonged. The most likely diagnosis is:

A. Haemophilia A
B. Immune thrombocytopenic purpura
C. Glanzmann thrombasthenia
D. von Willebrand disease
E. Vitamin K deficiency

22 A 55-year-old man has nephrotic syndrome. Biopsy shows diffusely thickened capillary walls, granular IgG along the GBM and subepithelial deposits with spikes on silver stain. The diagnosis is:

A. Membranous nephropathy
B. Minimal change disease
C. Focal segmental glomerulosclerosis
D. IgA nephropathy
E. Post-streptococcal glomerulonephritis

23 After repair of a ruptured aortic aneurysm with prolonged hypotension, a man becomes oliguric. Urine shows muddy-brown granular casts and FENa above 2%. The diagnosis is:

A. Prerenal azotaemia
B. Acute pyelonephritis
C. Post-streptococcal glomerulonephritis
D. Urinary tract obstruction
E. Ischaemic acute tubular necrosis

24 A 62-year-old smoker has haematuria, a flank mass and Hb 18.5 g/dL. Nephrectomy shows a golden-yellow tumour of cells with clear cytoplasm. The gene most often inactivated in this tumour is:

A. WT1
B. MET
C. VHL
D. PKD1
E. TSC2

25 A woman with recurrent Proteus urinary infections develops a staghorn calculus. Its composition is most likely:

A. Calcium oxalate
B. Magnesium ammonium phosphate
C. Uric acid
D. Cystine
E. Pure calcium phosphate

26 A 30-year-old man has a testicular mass and serum AFP of 2,000 ng/mL. Which component is most likely present in the tumour?

A. Classic seminoma
B. Spermatocytic tumour
C. Leydig cell tumour
D. Choriocarcinoma
E. Yolk sac tumour

27 Which statement about prostatic adenocarcinoma is correct?

A. Most arise in the transition zone
B. Bone metastases are typically osteolytic
C. The Gleason score is based on nuclear pleomorphism
D. Malignant glands lack a basal cell layer
E. PSA is raised only in carcinoma

28 The HPV E7 oncoprotein promotes cervical carcinogenesis mainly by binding and inactivating:

A. RB
B. p53
C. BRCA1
D. APC
E. PTEN

29 A 24-year-old has vaginal bleeding, a uterus large for dates and very high hCG. Evacuated tissue shows swollen villi with diffuse trophoblast proliferation and no fetal parts. The karyotype is most likely:

A. 69,XXY
B. 47,XXY
C. 46,XX with all chromosomes paternal
D. 45,X
E. 46,XY with one maternal and one paternal set

30 A breast carcinoma infiltrates as single files of small, uniform cells and forms no tubules. Loss of which protein explains this pattern?

A. HER2
B. E-cadherin
C. Oestrogen receptor
D. BRCA1
E. Cyclin D1

31 The diagnosis of papillary carcinoma of the thyroid rests mainly on:

A. Characteristic nuclear features
B. Capsular invasion
C. Vascular invasion
D. Amyloid in the stroma
E. Raised serum calcitonin

32 A 40-year-old hypothyroid woman has a firm, diffusely enlarged, painless thyroid. The expected microscopy is:

A. Tall hyperplastic follicles with scalloped colloid
B. Granulomas with giant cells around colloid
C. Lymphoid follicles with germinal centres and Hürthle cells
D. Dense fibrosis extending into neck muscles
E. Papillae with psammoma bodies

33 In long-standing type 2 diabetes the pancreatic islets characteristically show:

A. Insulitis with T lymphocytes
B. Amyloid deposition
C. Increased β-cell mass
D. Eosinophilic infiltrate
E. Granulomas

34 A 35-year-old woman has central obesity, purple striae and hypertension. ACTH is high and suppresses with high-dose dexamethasone. The cause is most likely:

A. Adrenal cortical adenoma
B. Adrenal carcinoma
C. Ectopic ACTH from small cell carcinoma
D. Exogenous corticosteroids
E. Pituitary corticotroph adenoma

35 The foam cells of a fatty streak are derived mainly from:

A. Endothelial cells taking up native LDL
B. Neutrophils engulfing cholesterol crystals
C. Platelets releasing lipid granules
D. Macrophages taking up oxidised LDL via scavenger receptors
E. Fibroblasts storing triglyceride

36 A man dies suddenly 5 days after an anterior myocardial infarction; the pericardial sac is full of blood. Which feature of the infarct at this stage explains the event?

A. Wavy fibres at the border
B. Maximal softening as macrophages remove dead muscle
C. Dense collagenous scar
D. Contraction band necrosis from reperfusion
E. Early coagulative necrosis without inflammation

37 A 26-year-old pregnant woman has dyspnoea and a mid-diastolic murmur; she had fever with flitting joint pains in childhood. The valve lesion and its classic gross appearance are:

A. Aortic stenosis; calcified nodules on the cusps
B. Mitral regurgitation; floppy ballooning leaflets
C. Tricuspid regurgitation; large friable vegetations
D. Pulmonary stenosis; fused cusps
E. Mitral stenosis; fish-mouth orifice with fused commissures

38 An intravenous drug user has fever and septic pulmonary emboli; echocardiography shows large friable vegetations on the tricuspid valve. The most likely organism is:

A. Staphylococcus aureus
B. Streptococcus viridans
C. Enterococcus faecalis
D. Coxiella burnetii
E. Candida albicans

39 A 30-year-old has a lytic soap-bubble lesion in the distal femoral epiphysis reaching the joint surface; biopsy shows numerous osteoclast-like giant cells. The neoplastic cells recruit the giant cells by producing:

A. Parathyroid hormone
B. Osteoprotegerin
C. Alkaline phosphatase
D. RANKL
E. TGF-β

40 A 12-year-old has a diaphyseal femoral tumour with onion-skin periosteal reaction. Biopsy shows small round blue cells with PAS-positive glycogen that are CD99-positive. The characteristic translocation is:

A. t(9;22)
B. t(X;18)
C. t(11;22)
D. t(2;13)
E. t(12;16)

41 A 5-year-old boy climbs up his own legs to stand and has bulky calves; CK is very high and muscle shows no dystrophin. The genetic lesion is:

A. CTG repeat expansion in DMPK
B. In-frame DMD deletion with reduced dystrophin
C. Frameshift deletion in the DMD gene
D. Antibodies against the acetylcholine receptor
E. Mitochondrial DNA mutation

42 CSF from a 25-year-old with three weeks of headache and a sixth-nerve palsy is clear but forms a cobweb clot; 250 cells/µL, mostly lymphocytes; protein markedly raised; glucose low. The diagnosis is:

A. Acute bacterial meningitis
B. Viral meningitis
C. Subarachnoid haemorrhage
D. Multiple sclerosis
E. Tuberculous meningitis

43 A 7-year-old has a cerebellar cyst with a mural nodule. Microscopy shows bipolar hair-like cells, microcysts and Rosenthal fibres. The WHO grade and usual outcome are:

A. Grade 4; median survival about 15 months
B. Grade 1; excellent outcome after resection
C. Grade 2; inevitable progression to glioblastoma
D. Grade 3; spreads through the CSF
E. Grade 4; radiosensitive embryonal tumour

44 A 45-year-old has painful oral erosions and flaccid skin blisters that spread with rubbing. Biopsy shows suprabasal acantholysis with a tombstone row of basal cells; IF shows net-like intercellular IgG. The target antigen is:

A. Desmoglein 3
B. BP180 (type XVII collagen)
C. Type VII collagen
D. Tissue transglutaminase
E. Laminin-332

45 The single most important prognostic factor for a localised cutaneous melanoma is:

A. Diameter of the lesion
B. Degree of pigmentation
C. Age of the patient
D. Breslow thickness
E. BRAF mutation status

PART 1 · ANSWER KEY WITH ONE-LINE REASONS

1. **C** Barrett oesophagus needs intestinal metaplasia, i.e. goblet cells, in columnar mucosa.
2. **A** Antral inflammation lowers somatostatin, so gastrin and acid rise and injure the duodenum.
3. **E** Linitis plastica with Krukenberg tumours = diffuse-type gastric carcinoma.
4. **B** Transmural disease and granulomas mark Crohn; crypt abscesses occur in both.
5. **D** Lynch syndrome: mismatch-repair failure, MSI, colon and endometrial cancers.
6. **A** Anti-HBc proves infection; anti-HBs without HBsAg proves recovery. Vaccination gives anti-HBs alone.
7. **C** Mild, fluctuating unconjugated hyperbilirubinaemia with stress or fasting and no haemolysis.
8. **E** Activated stellate cells become myofibroblasts that lay down collagen in the space of Disse.
9. **B** Hereditary haemochromatosis: HFE C282Y → low hepcidin → unchecked iron absorption.
10. **D** Lipase releases fatty acids that combine with calcium (saponification).
11. **E** Grey hepatisation follows red hepatisation as red cells break down in the fibrin mesh.
12. **B** Ghon focus + draining hilar nodes = Ghon complex (Ranke complex once calcified).
13. **A** α1-antitrypsin deficiency causes panacinar emphysema of the lower zones.
14. **D** Squamous cell carcinoma secretes PTHrP: humoral hypercalcaemia of malignancy.

15. **C** Anaemia of chronic disease: IL-6 → hepcidin → iron trapped in macrophages.
16. **B** Neurological damage (subacute combined degeneration) points to B12; pernicious anaemia is the classic cause.
17. **E** β-thalassaemia major: β-chains absent or scarce, so HbF dominates.
18. **C** Hereditary spherocytosis: ankyrin is the commonest mutated membrane-skeleton protein.
19. **A** Acute promyelocytic leukaemia with DIC; treated with all-trans retinoic acid.
20. **D** Collagen bands + lacunar cells = nodular sclerosis classical Hodgkin lymphoma.
21. **D** Platelet-type bleeding + low factor VIII (long aPTT), autosomal dominant: von Willebrand disease.
22. **A** Spike-and-dome subepithelial deposits are the signature of membranous nephropathy.
23. **E** Granular casts and high FENa show tubular damage, not a prerenal state.
24. **C** Clear cell RCC: VHL (3p) loss; EPO secretion explains the polycythaemia.
25. **B** Urease-producing organisms alkalinise urine → struvite staghorn stones.
26. **E** A high AFP signals yolk sac elements; pure seminoma does not make AFP.
27. **D** Loss of basal cells separates carcinoma from benign glands; most tumours arise peripherally.
28. **A** E7 binds RB, releasing E2F; E6 degrades p53.
29. **C** Complete mole: an empty ovum fertilised by sperm, so the genome is androgenetic.
30. **B** Invasive lobular carcinoma: loss of E-cadherin makes the cells discohesive.
31. **A** Clear nuclei, grooves and pseudoinclusions are diagnostic even without papillae.
32. **C** Hashimoto thyroiditis: lymphoid follicles and oncocytic (Hürthle) metaplasia.
33. **B** Islet amyloid (amylin) is typical of type 2; insulitis is type 1.
34. **E** Cushing disease: ACTH-dependent and suppressible; adrenal causes have low ACTH.
35. **D** Monocyte-derived macrophages ingest oxidised LDL and become foam cells.
36. **B** Free wall rupture is commonest at 3–7 days, when the infarct is softest.
37. **E** Chronic rheumatic heart disease: fibrotic, fused mitral leaflets.
38. **A** Acute right-sided endocarditis in drug users is usually S. aureus.
39. **D** Giant cell tumour: stromal cells express RANKL; denosumab blocks it.
40. **C** Ewing sarcoma: EWSR1–FLI1 fusion from t(11;22).
41. **C** Duchenne: frameshift deletion → no dystrophin; in-frame deletions cause Becker.
42. **E** Subacute course, lymphocytes, very high protein and low glucose: tuberculous meningitis.
43. **B** Pilocytic astrocytoma is grade 1 and often cured by excision.
44. **A** Pemphigus vulgaris: IgG against desmoglein 3 (with desmoglein 1 in mucocutaneous disease).
45. **D** Depth of invasion (Breslow thickness) best predicts metastasis and survival.

PART 2 Structured SEQs · 9 × 5 marks

SEQ 1 A 48-year-old man has burning epigastric pain relieved by food and has passed black stools. **1 + 2 + 2 marks**
Endoscopy shows a 1.5-cm punched-out ulcer in the first part of the duodenum; the rapid urease test on an antral biopsy is positive. **a)** Name the organism and two of its virulence factors. **b)** Explain how this organism leads to duodenal ulceration. **c)** Describe the gross and microscopic features of a chronic peptic ulcer and name two complications.

SEQ 2 A 45-year-old woman, anti-HCV positive for 12 years, presents with abdominal distension, **1 + 2 + 2 marks**
splenomegaly and an episode of haematemesis. **a)** Define cirrhosis. **b)** Explain the pathogenesis of fibrosis in cirrhosis. **c)** Enumerate the consequences of portal hypertension in this patient.

SEQ 3 A 24-year-old man has had cough for six weeks with evening fever, weight loss and blood-streaked **1 + 2 + 2 marks**
sputum. Chest X-ray shows a right apical cavity. **a)** Give the most likely diagnosis and two tests to confirm it. **b)** Describe how the tuberculous granuloma forms. **c)** Differentiate primary from secondary tuberculosis.

SEQ 4 A 2-year-old child of a first-cousin marriage has been pale since six months of age and has frontal **1 + 1 + 2 + 1 marks**
bossing and hepatosplenomegaly. Hb 5.8 g/dL, MCV 58 fL; the smear shows target cells and nucleated red cells. **a)** What is the most likely diagnosis? **b)** Name the confirmatory test and its expected result. **c)** Explain the pathogenesis of the anaemia. **d)** Why does iron overload develop?

SEQ 5 A 7-year-old boy passes smoky urine and has puffy eyes two weeks after a sore throat. BP is 140/90 **1 + 2 + 2 marks**
mmHg; urine shows RBC casts and 1+ protein. **a)** Name the condition and two blood tests that support it. **b)** Describe its pathogenesis and the light, immunofluorescence and electron microscopic findings. **c)** Differentiate nephritic from nephrotic syndrome (four points).

SEQ 6 A 46-year-old woman has a hard, painless lump in the upper outer quadrant of the left breast with skin dimpling and a palpable axillary node. Core biopsy shows invasive carcinoma of no special type. **a)** List four risk factors for this disease. **b)** Describe its gross and microscopic features. **c)** Name the prognostic and predictive factors. — 1 + 2 + 2 marks

SEQ 7 A 28-year-old woman has a solitary cold thyroid nodule and a firm lymph node in the neck. FNA shows papillary clusters of cells with optically clear, grooved nuclei and intranuclear pseudoinclusions. **a)** Give the diagnosis and two risk factors. **b)** Describe its morphology. **c)** Differentiate it from follicular carcinoma. — 1 + 2 + 2 marks

SEQ 8 A 58-year-old diabetic smoker has had crushing central chest pain for six hours; the ECG shows ST elevation in the anterior leads. **a)** Explain the pathogenesis of the atherosclerotic plaque. **b)** Which cardiac marker is most useful, and why? **c)** List the complications of myocardial infarction with their usual timing. — 2 + 1 + 2 marks

SEQ 9 A 15-year-old boy has had pain and swelling above the left knee for two months. X-ray shows a destructive lesion in the distal femoral metaphysis with a Codman triangle and a sunburst pattern. **a)** Give the most likely diagnosis and two predisposing factors. **b)** Describe its gross and microscopic features. **c)** Tabulate how it differs from Ewing sarcoma and giant cell tumour. — 1 + 2 + 2 marks

MARKING Part 2 · Marking scheme

SEQ 1 H. pylori and peptic ulcer — Marking scheme · 5 marks

1. 1 *Helicobacter pylori* (0.5); any two of urease, flagella, adhesins, CagA, VacA (0.25 each).
2. 2 Antral-predominant gastritis (0.5) → less somatostatin, more gastrin → acid hypersecretion (0.5) → gastric metaplasia in the duodenum, which the organism colonises (0.5) → inflammation, ammonia and cytokines weaken mucosal defence, so acid and pepsin ulcerate it (0.5).
3. 2 Gross: round to oval, sharply punched-out, usually < 2 cm, straight walls, flat margins, clean base, radiating folds (1). Microscopy: necrotic fibrinoid debris, neutrophilic layer, granulation tissue, fibrous scar (0.5). Complications: bleeding, perforation, pyloric obstruction (0.5).

SEQ 2 Cirrhosis and portal hypertension — Marking scheme · 5 marks

1. 1 A diffuse process (0.25) with bridging fibrous septa (0.25) and parenchymal regenerative nodules (0.25) that destroys normal lobular architecture (0.25).
2. 2 Chronic injury (HCV) → Kupffer cells and lymphocytes release TGF-β, PDGF, TNF (0.5) → stellate cells activate into myofibroblasts (0.5) → collagen I and III in the space of Disse; sinusoids lose fenestrae (capillarisation) (0.5) → septa link portal tracts and central veins; surviving hepatocytes form nodules (0.5).
3. 2 Any four, 0.5 each: ascites; portosystemic shunts (oesophageal varices causing haematemesis, rectal varices, caput medusae); congestive splenomegaly with hypersplenism; hepatic encephalopathy.

SEQ 3 Pulmonary tuberculosis — Marking scheme · 5 marks

1. 1 Secondary (post-primary) pulmonary tuberculosis (0.5); sputum ZN smear for AFB, GeneXpert MTB/RIF, culture (any two, 0.25 each).
2. 2 Bacilli are taken up by alveolar macrophages and survive by blocking phagolysosome fusion (0.5) → after about 3 weeks antigens are presented and Th1 cells develop under IL-12 (0.5) → IFN-γ activates macrophages into epithelioid cells and Langhans giant cells (0.5) → TNF recruits monocytes; caseous necrosis reflects type IV hypersensitivity (0.5).
3. 2 Four rows, 0.5 each: unsensitised vs sensitised host; subpleural mid-lung vs apical site; Ghon complex vs cavitating fibrocaseous lesion; prominent vs minimal nodes; healing and calcification vs progression.

SEQ 4 β-thalassaemia major

Marking scheme · 5 marks

1. 1 β-thalassaemia major (homozygous β-thalassaemia).
2. 1 Haemoglobin electrophoresis or HPLC (0.5): HbF markedly raised, HbA absent or very low (0.5).
3. 2 β-globin mutations reduce or abolish β-chains (0.5) → less HbA, so cells are microcytic and hypochromic (0.5) → unpaired α-chains precipitate, damaging precursors in the marrow (ineffective erythropoiesis) and red cells in the spleen (haemolysis) (0.5) → anaemia drives erythropoietin, marrow expansion (frontal bossing, crew-cut skull) and extramedullary haematopoiesis (0.5).
4. 1 Repeated transfusions (0.5) and increased gut absorption because ineffective erythropoiesis suppresses hepcidin (0.5).

SEQ 5 Post-streptococcal glomerulonephritis

Marking scheme · 5 marks

1. 1 Acute post-streptococcal glomerulonephritis (0.5); raised ASO or anti-DNase B and low C3 (0.5).
2. 2 Nephritogenic group A streptococci → latent period of 1–4 weeks → immune complexes (e.g. SpeB) in glomeruli → complement activation → neutrophil and monocyte influx (1). LM: enlarged hypercellular glomeruli with neutrophils; IF: granular IgG and C3; EM: subepithelial humps (1).
3. 2 Any four, 0.5 each: haematuria with RBC casts vs none; proteinuria < 3.5 vs > 3.5 g/day; periorbital vs generalised oedema; hypertension vs normal BP; hyperlipidaemia and lipiduria only in nephrotic.

SEQ 6 Invasive breast carcinoma

Marking scheme · 5 marks

1. 1 Any four, 0.25 each: age; early menarche or late menopause; nulliparity or late first pregnancy; family history or BRCA1/2; postmenopausal obesity; hormone therapy; atypical hyperplasia; chest radiation.
2. 2 Gross: hard grey-white stellate mass with infiltrative edges, gritty on cutting, chalky streaks; skin or nipple retraction (1). Microscopy: cords, nests and glands of pleomorphic cells with mitoses in a desmoplastic stroma, invading fat, no myoepithelial layer; graded on tubules, pleomorphism and mitoses (1).
3. 2 Prognostic: nodal status (most important), size, distant metastasis, grade, lymphovascular invasion, histological type (1). Predictive: ER and PR for hormone therapy, HER2 for trastuzumab; Ki-67 as proliferation index (1).

SEQ 7 Papillary carcinoma of the thyroid

Marking scheme · 5 marks

1. 1 Papillary thyroid carcinoma (0.5); ionising radiation in childhood and family history, with BRAF or RET/PTC as the molecular basis (0.5).
2. 2 Gross: solitary or multifocal, grey-white, may be cystic or calcified (0.5). Microscopy: branching papillae with fibrovascular cores (0.5); optically clear Orphan Annie-eye nuclei, grooves and pseudoinclusions (0.5); psammoma bodies and lymphatic invasion (0.5).
3. 2 Four rows, 0.5 each: diagnosis by nuclear features vs by capsular or vascular invasion; spread to nodes vs blood (lung, bone); BRAF, RET/PTC vs RAS, PAX8–PPARG; psammoma bodies present vs absent.

SEQ 8 Atherosclerosis and myocardial infarction

Marking scheme · 5 marks

1. 2 Endothelial dysfunction from hypertension, smoking, hyperlipidaemia, diabetes (0.5) → LDL enters the intima and is oxidised; monocytes adhere (VCAM-1) and become macrophages that take up oxidised LDL → foam cells, fatty streak (0.5) → cytokines and PDGF draw in and multiply smooth muscle cells that make matrix (0.5) → fibrous cap over a lipid–necrotic core: the fibrofatty plaque; rupture of a thin cap triggers thrombosis (0.5).
2. 1 Cardiac troponin I or T (0.5): most specific and sensitive for myocardial necrosis, rises within hours and stays raised for 1–2 weeks (0.5).
3. 2 Any four with timing, 0.5 each: arrhythmias (first hours to days); cardiogenic shock or pump failure (early); pericarditis (2–3 days; Dressler weeks later); rupture of free wall, septum or papillary muscle (3–7 days); mural thrombus with embolism; ventricular aneurysm (weeks to months).

SEQ 9 Osteosarcoma

Marking scheme · 5 marks

1. 1 Osteosarcoma (0.5); any two of hereditary retinoblastoma (RB1), Li–Fraumeni (TP53), Paget disease, prior radiation, bone infarcts (0.5).
2. 2 Gross: large grey-white gritty tumour with haemorrhage and cystic change, destroying cortex, lifting periosteum into soft tissue, usually stopping at the growth plate (1). Microscopy: pleomorphic malignant cells with atypical mitoses and giant cells that lay down lace-like neoplastic osteoid directly; cartilage may be present (1).
3. 2 Four rows, 0.5 each: age (osteosarcoma and Ewing mostly under 20; GCT 20–40 years); site (metaphysis vs diaphysis vs epiphysis); X-ray (sunburst, Codman vs onion-skin vs soap-bubble); microscopy (malignant osteoid vs small round blue CD99+ cells vs osteoclast-like giant cells).

Last-Day Checklist

Forty facts you must be able to say aloud the evening before the paper. Tick each one only if you can explain it without looking.

GIT & LIVER

- [ ] **Barrett oesophagus** = columnar mucosa with goblet cells; precursor of adenocarcinoma, not SCC. [1]
- [ ] **H. pylori**: antral gastritis → duodenal ulcer, gastric carcinoma, MALT lymphoma. Autoimmune gastritis → B12 deficiency. [2]
- [ ] **Diffuse gastric cancer**: signet-ring cells, linitis plastica, CDH1; ovarian spread = Krukenberg. [2]
- [ ] **Crohn**: skip lesions, transmural, granulomas, fistulae. **UC**: from rectum, continuous, mucosal, pseudopolyps. [3]
- [ ] **Adenoma–carcinoma**: APC → KRAS → TP53; large, villous, high-grade adenomas carry most risk. [4]
- [ ] **HBV**: anti-HBc IgM = acute; HBsAg beyond 6 months = chronic; anti-HBs alone = vaccinated. [5]
- [ ] **Cirrhosis** = diffuse fibrosis + regenerative nodules (stellate cells); **HCC**: HCV in Pakistan, AFP. [6]
- [ ] **Haemochromatosis**: HFE C282Y, low hepcidin, Perls-positive liver; bronze diabetes. [6]
- [ ] **Acute pancreatitis**: gallstones, alcohol; lipase; enzymatic fat necrosis; pseudocyst. [7]

LUNG

- [ ] **Lobar pneumonia**: congestion → red → grey hepatisation → resolution; S. pneumoniae. [8]
- [ ] **TB granuloma**: caseation, epithelioid cells, Langhans cells; type IV hypersensitivity; ZN stain. [8]
- [ ] **Emphysema**: centriacinar (smokers, upper lobes) vs panacinar (α1-AT, lower lobes). [9]
- [ ] **Lung cancer**: SCC → PTHrP; small cell → ACTH, ADH, TP53 + RB1; adenocarcinoma → EGFR, ALK. [10]

BLOOD

- [ ] **IDA**: ferritin low, TIBC high. **Chronic disease**: ferritin high, TIBC low, hepcidin up. [11]
- [ ] **B12** deficiency = megaloblastic anaemia + nerve damage; folate spares the nerves. [11]
- [ ] **β-thal major**: HbF high, HbA absent or low, target cells, crew-cut skull, iron overload. [12]
- [ ] **AML** ≥ 20% blasts, MPO+, Auer rods; **APL** t(15;17) → DIC, ATRA. **ALL**: TdT+, children. [13]
- [ ] **CML**: t(9;22) BCR–ABL, basophilia, low LAP, big spleen; imatinib. [13]
- [ ] **Translocations**: 8;14 Burkitt · 14;18 follicular · 11;14 mantle cell · 11;22 Ewing. [14]
- [ ] **Reed–Sternberg**: CD15+ CD30+, owl-eye nucleoli; popcorn cells (NLPHL) CD20+. [14]
- [ ] **DIC**: low platelets and fibrinogen, long PT and aPTT, high D-dimer, schistocytes. [15]

KIDNEY

- [ ] **Nephritic**: haematuria, RBC casts, hypertension. **Nephrotic**: protein > 3.5 g/day, albumin < 3 g/dL, lipids. [16]
- [ ] **PSGN**: subepithelial humps, low C3, ASO up. **MCD**: normal LM, foot-process effacement. [16]
- [ ] **Clear cell RCC**: VHL (3p), golden-yellow, renal vein spread, polycythaemia. [18]

REPRODUCTIVE & BREAST

- [ ] **Seminoma**: clear cells, lymphocytic septa, AFP normal. **Yolk sac**: AFP, Schiller–Duval bodies. [19]
- [ ] **Prostate cancer**: peripheral zone, no basal cells, Gleason (architecture), osteoblastic metastases. [19]
- [ ] **HPV 16/18**: E6 → p53, E7 → RB; CIN 1 = LSIL, CIN 2–3 = HSIL. [20]
- [ ] **Moles**: complete 46,XX paternal, no fetus; partial 69,XXY with fetal parts. [21]
- [ ] **Breast**: IDC commonest; ILC loses E-cadherin; nodal status = strongest prognostic factor. [22]

ENDOCRINE

- [ ] **PTC**: clear grooved nuclei, psammoma, node spread. **Follicular**: capsular/vascular invasion, blood spread. [23]
- [ ] **MTC**: C cells, calcitonin, amyloid, RET. **Hashimoto**: lymphoid follicles, Hürthle cells, anti-TPO. [23]
- [ ] **Diabetes**: type 1 insulitis vs type 2 islet amyloid; KW nodules; hyaline arteriolosclerosis. [24]

HEART & VESSELS

- [ ] **Atheroma**: fatty streak → fibrofatty plaque; thin-cap rupture → thrombosis → MI. [25]
- [ ] **MI**: neutrophils days 1–3; macrophages days 3–7 (rupture risk); granulation from day 7; scar by 2 months. [26]
- [ ] **Rheumatic heart**: Aschoff bodies, Anitschkow cells; mitral stenosis with fish-mouth orifice. [26]

BONE, JOINTS, CNS, SKIN

- [ ] **Bone tumour sites**: epiphysis GCT; metaphysis osteosarcoma, osteochondroma; diaphysis Ewing (t(11;22)). [27]
- [ ] **Gout**: negatively birefringent needles. **Pseudogout**: positive rhomboids. **RA**: pannus, anti-CCP. [28]
- [ ] **CSF**: bacterial = neutrophils, low glucose; viral = lymphocytes, normal glucose; TB = lymphocytes, very high protein. [29]
- [ ] **Brain tumours**: GBM pseudopalisading; pilocytic Rosenthal fibres; meningioma psammoma; schwannoma Verocay. [30]
- [ ] **Skin**: pemphigus suprabasal, Dsg3; pemphigoid subepidermal; melanoma prognosis = Breslow thickness. [31]

EXAM-DAY TIPS

- **First 5 minutes:** read the whole SEQ paper and start with the question you know best.
- **Budget time:** one minute per BCQ; 12–13 minutes per 5-mark SEQ. Never leave a part blank.
- **BCQs:** read the last line first, underline age, sex and site, eliminate, and change an answer only for a clear reason.
- **SEQs:** definition → flowchart → gross then microscopy → complications; use a table for every "compare".
- **Slides:** organ → process → benign or malignant → name + two features; know the stain or marker.
- **Viva:** open with the definition, classify, then steer to what you know well; say so if you do not know.

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12 + 12
BCQs and viva questions in every chapter

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night-before page per chapter

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